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LAPIS
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FEUL610Q418-01

ML610Q418/ML610Q418C
User's Manual

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Preface

This manual describes the operation of the hardware of the 8-bit microcontroller ML610Q418/ML610Q418C.

The following manuals are also available. Read them as necessary.

- nX-U8/100 Core Instruction Manual
  Description on the basic architecture and the each instruction of the nX-U8/100 Core.

- MACU8 Assembler Package User's Manual
  Description on the method of operating the relocatable assembler, the linker, the librarian, and the object converter and also on the specifications of the assembler language.

- CCU8 User's Manual
  Description on the method of operating the compiler.

- CCU8 Programming Guide
  Description on the method of programming.

- CCU8 Language Reference
  Description on the language specifications.

- DTU8 Debugger User's Manual
  Description on the method of operating the debugger DTU8.

- IDEU8 User's Manual
  Description on the integrated development environment IDEU8.

- uEASE User's Manual
  Description on the on-chip debug tool uEASE.

- uEASE connection Manual
  Description about the connection between uEASE and ML610Qxxx.

- FWuEASE Flash Writer Host Program User's Manual
  Description on the Flash Writer host program.

Notation

Classification	Notation	Description
♦ Numeric value	xxh, xxH	Indicates a hexadecimal number. x: Any value in the range of 0 to F
	xxb	Indicates a binary number; "b" may be omitted. x: A value 0 or 1
♦ Unit	word, W	1 word = 16 bits
	byte, B	1 byte = 8 bits
	nibble, N	1 nibble = 4 bits
	mega-, M	10^6
	kilo-, K	$2^{10} = 1024$
	kilo-, k	$10^3 = 1000$
	milli-, m	10^{-3}
	micro-, µ	10^{-6}
	nano-, n	10^{-9}
	second, s (lower case)	second
♦ Terminology	"H" level, "1" level	Indicates high voltage signal levels V_{IH} and V_{OH} as specified by the electrical characteristics.
	"L" level, "0" level	Indicates low voltage signal levels V_{IL} and V_{OL} as specified by the electrical characteristics.

♦ Register description

R/W: Indicates that Read/Write attribute. "R" indicates that data can be read and "W" indicates that data can be written. "R/W" indicates that data can be read or written.

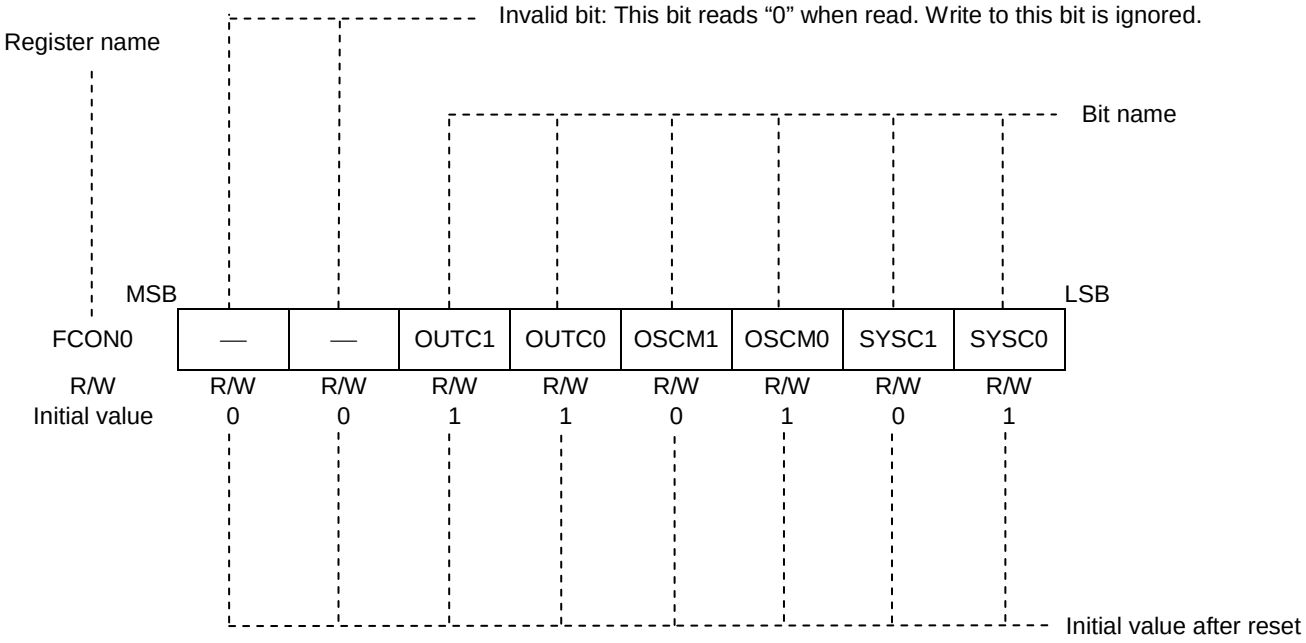


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Chapter 1

Overview

1. Overview

1.1 Features

This LSI is a high-performance 8-bit CMOS microcontroller into which rich peripheral circuits, such as synchronous serial port, UART, I2C bus interface (master), melody driver, battery level detect circuit, RC oscillation type A/D converter, and LCD driver, are incorporated around 8-bit CPU nX-U8/100.
The CPU nX-U8/100 is capable of efficient instruction execution in 1-instruction 1-clock mode by 3-stage pipe line architecture parallel processing. The Flash ROM that is installed as program memory achieves low-voltage low-power consumption operation (read operation) equivalent to mask ROM and is most suitable for battery-driven applications.
The on-chip debug function that is installed enables program debugging and programming.

- CPU
  - 8-bit RISC CPU (CPU name: nX-U8/100)
  - Instruction system: 16-bit instructions
  - Instruction set: Transfer, arithmetic operations, comparison, logic operations, multiplication/division, bit manipulations, bit logic operations, jump, conditional jump, call return stack manipulations, arithmetic shift, and so on
  - On-Chip debug function
  - Minimum instruction execution time
    30.5 μs (@32.768 kHz system clock)
    0.24 4μs (@4.096 MHz system clock)

- Internal memory
  - Internal 128KByte Flash ROM (64K×16 bits) (including unusable 1KByte + 2Byte TEST area)
  - Internal 4KByte Data Flash (2K×16 bits)
  - Internal 4KByte Data RAM (4096×8 bits), 240×9bit Display Allocation RAM

- Interrupt controller
  - 1 non-maskable interrupt sources (Internal source: 1)
  - 22 maskable interrupt sources (Internal sources: 17, External sources: 5)

- Time base counter
  - Low-speed time base counter ×1 channel
    Frequency compensation (Compensation range: Approx. –488ppm to +488ppm. Compensation accuracy: Approx. 0.48ppm)
  - High-speed time base counter ×1 channel

- Watchdog timer
  - Non-maskable interrupt and reset
  - Free running
  - Overflow period: 4 types selectable (125ms, 500ms, 2s, and 8s)

- Timers
  - 8 bits × 4 channels (Timer0-3: 16-bit x 2 configuration available by using Timer0-1 or Timer2-3)
  - Clock frequency measurement mode (in one channel of 16-bit configuration using Timer2-3)

- Capture
  – Time base capture × 2 channels (4096 Hz to 32 Hz)

- PWM
  – Resolution 16 bits × 1 channel

- Synchronous serial port
  – Master/slave selectable × 2 channel
  – LSB first/MSB first selectable
  – 8-bit length/16-bit length selectable

- UART
  – Half-duplex × 2 channel or full-duplex × 1 channel
  – Bit length, parity/no parity, odd parity/even parity, 1 stop bit/2 stop bits
  – Positive logic/negative logic selectable
  – Built-in baud rate generator

- I^2C bus interface
  – Master function only
  – Fast mode (400 kbps@4MHz), standard mode (100 kbps@4MHz, 50kbps@500kHz)

- Melody driver
  – Scale: 29 types (Melody sound frequency: 508 Hz to 32.768 kHz)
  – Tone length: 63 types
  – Tempo: 15 types
  – Buzzer output mode (4 output modes, 8 frequencies, 16 duty levels)

- RC oscillation type A/D converter
  – 24-bit counter
  – Time division × 2 channels

- Successive approximation type A/D converter (SA-ADC)
  – 12-bit A/D converter
  – Input × 4 channels

- General-purpose ports
  – Input-only port × 6 channels (including secondary functions)
  – Output-only port × 3 channels (including secondary functions)
  – Input/output port
    ML610Q418 : 18 channels (including secondary functions)
    ML610Q418C : 26 channels (including secondary functions)

- LCD driver
  – Dot matrix can be supported.
    ML610Q418 : 192 dots max. (48 seg × 4 com)
    ML610Q418C : 160 dots max. (40 seg × 4 com)
  – 1/1 to 1/4 duty
  – 1/2, 1/3 bias (built-in bias generation circuit)
  – Frame frequency selectable: approx. 64Hz, 73Hz, 85Hz, and 102Hz
  – Bias voltage multiplying clock selectable (8 types)
  – LCD drive stop mode, LCD display mode, all LCDs on mode, and all LCDs off mode selectable
  – Programmable display allocation function

- Reset
  - Reset by the RESET_N pin
  - Reset by power-on detection
  - Reset when oscillation stop of the low-speed clock is detected.
  - Reset by low level detection (LLD)
    The voltage which is released from reset is selectable by the code-option: 1.1V, 1.8V (Max.)
  - Reset by the watchdog timer (WDT) 2nd overflow

- Power supply voltage detect function
  - Judgment voltages: One of 16 levels
  - Judgment accuracy: ±2% (Typ.)

- Clock
  - Low-speed clock: (This LSI can not guarantee the operation without low-speed clock)
    Crystal oscillation (32.768 kHz)
  - High-speed clock:
    Built-in RC oscillation (500kHz)
    Built-in PLL oscillation (8.192 MHz ±2.5%), crystal/ceramic oscillation (4.096 MHz), external clock
  - Selection of high-speed clock mode by software:
    Built-in RC oscillation, built-in PLL oscillation, crystal/ceramic oscillation, external clock

- Power management
  - HALT mode: Instruction execution by CPU is suspended (peripheral circuits are in operating states).
  - STOP mode: Stop of low-speed oscillation and high-speed oscillation (Operations of CPU and peripheral circuits are stopped.)
  - Clock gear: The frequency of high-speed system clock can be changed by software (1/1, 1/2, 1/4, or 1/8 of the oscillation clock)
  - Block Control Function: Power down (reset registers and stop clock supply) the circuits of unused peripherals.

- Guaranteed operating range
  - Operating temperature: –20°C to 70°C
  - Operating voltage: V_{DD} = 1.1V to 3.6V

• Product name – Supported Function

The line-up of the ML610Q418 is below.

- Chip (Die) -	ROM type	Operating temperature	LCD driver	Product availability
ML610Q418-xxxWA	Flash ROM	-20°C to +70°C	192 dots max. (48 seg x 4 com)	Under construction
ML610Q418C-xxxWA	Flash ROM	-20°C to +70°C	160 dots max. (40 seg x 4 com)	Under construction

-100-pin plastic TQFP -	ROM type	Operating temperature	LCD driver	Product availability
ML610Q418-xxxTB	Flash ROM	-20°C to +70°C	192 dots max. (48 seg x 4 com)	Yes
ML610Q418C-xxxTB	Flash ROM	-20°C to +70°C	160 dots max. (40 seg x 4 com)	Under construction

xxx: ROM code number (xxx of the blank product is NNN)
Q:Flash ROM version
WA: Chip
TB: TQFP

1.2 Configuration of Functional Blocks

1.2.1 Block Diagram of ML610Q418

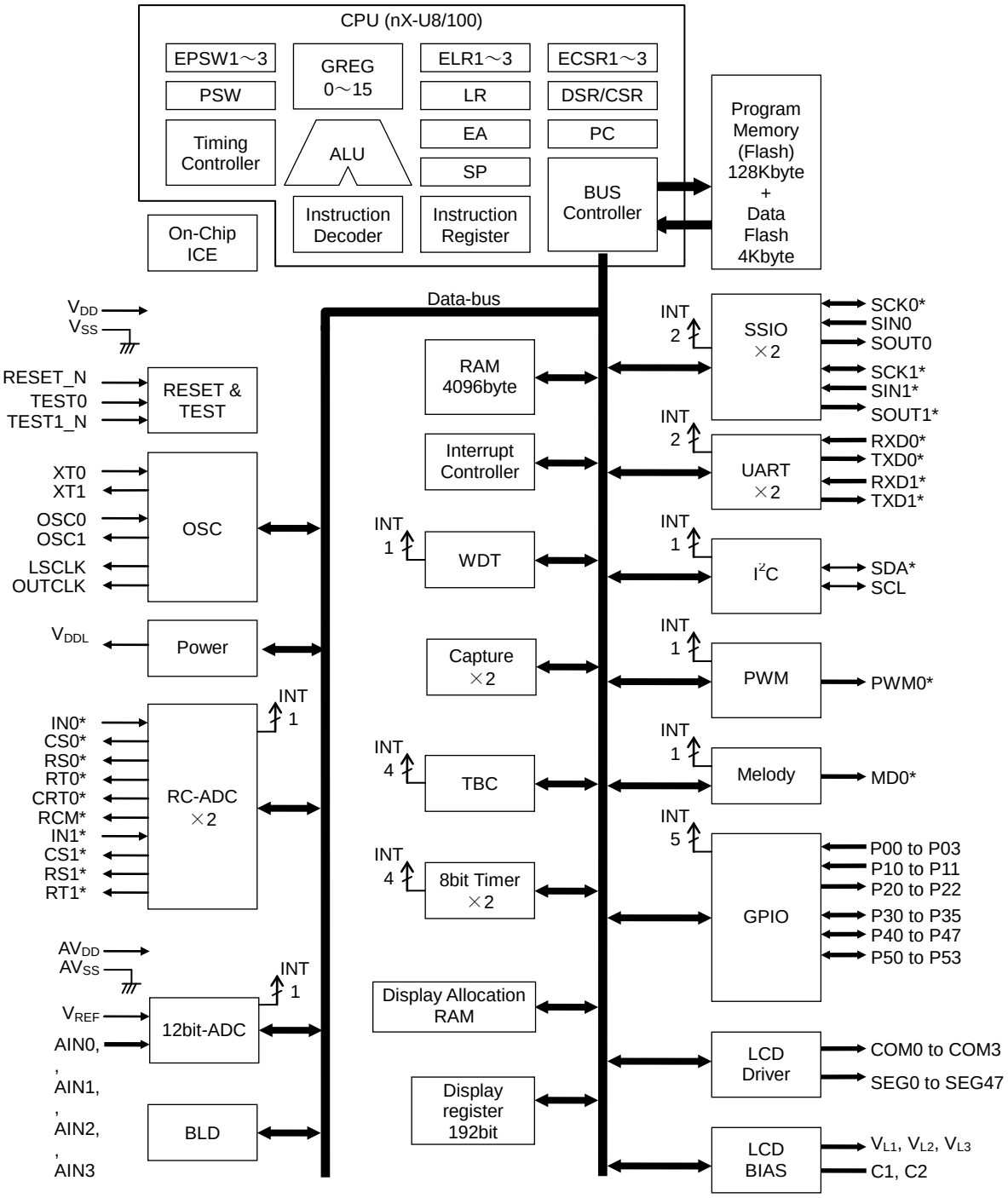


Figure 1-1 Block Diagram of ML610Q418

1.2.2 Block Diagram of ML610Q418C

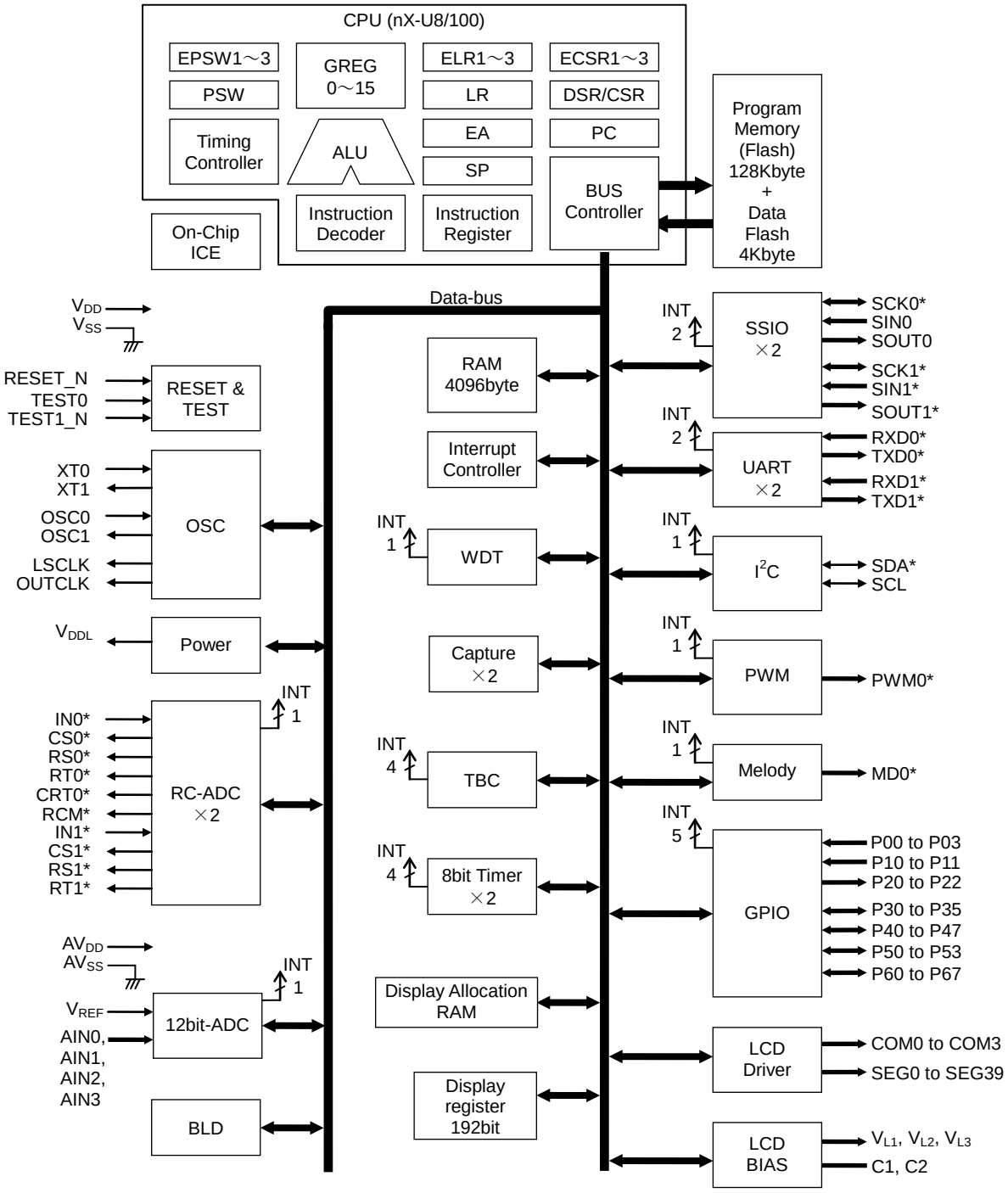


Figure 1-2 Block Diagram of ML610Q418C

1.3 Pins

1.3.1 Pin Layout

1.3.1.1 Pin Layout of ML610Q418 TQFP Package

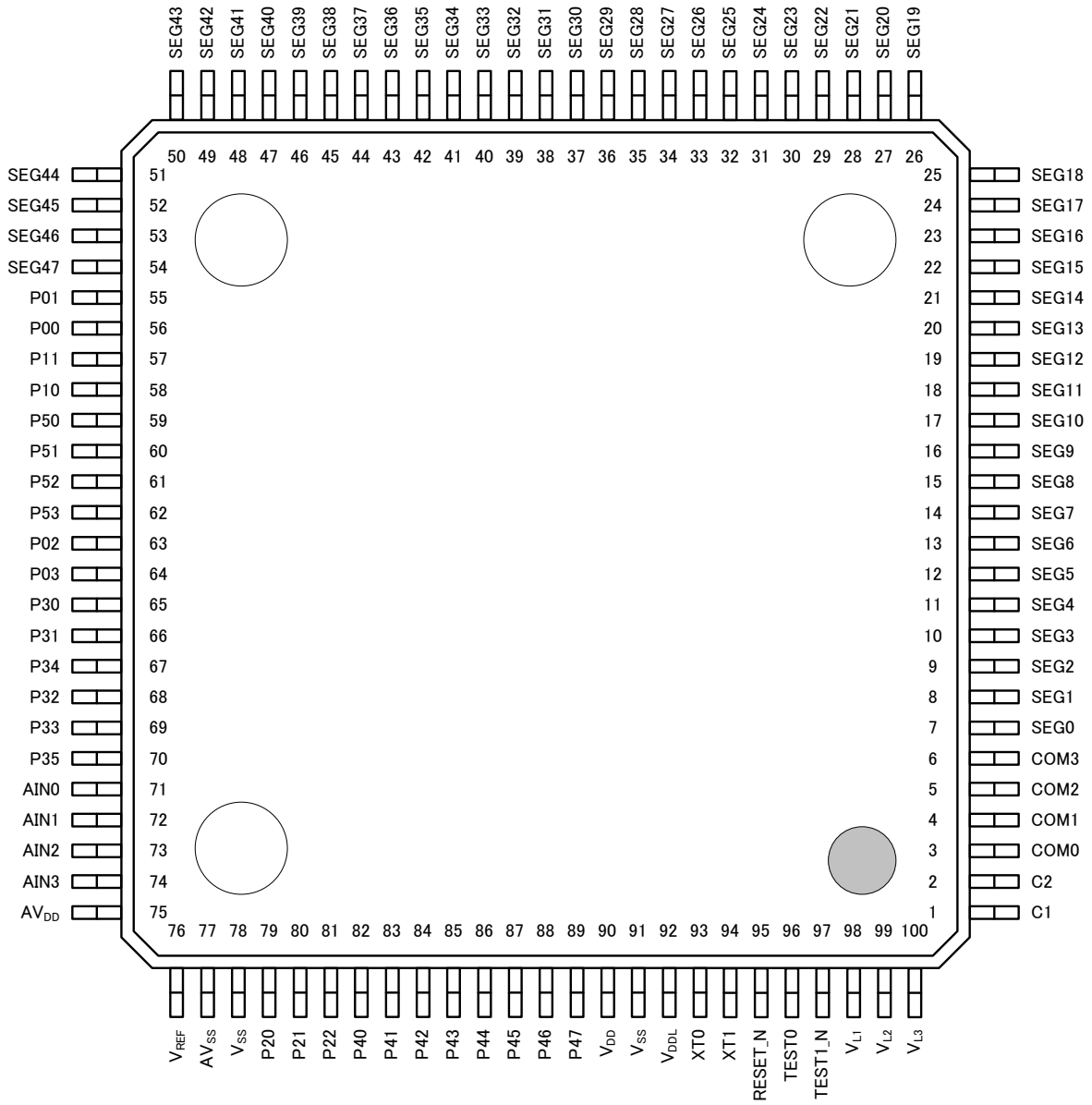


Figure 1-3 Pin Layout of ML610Q418 Package

1.3.1.2 Pin Layout of ML610Q418C TQFP Package

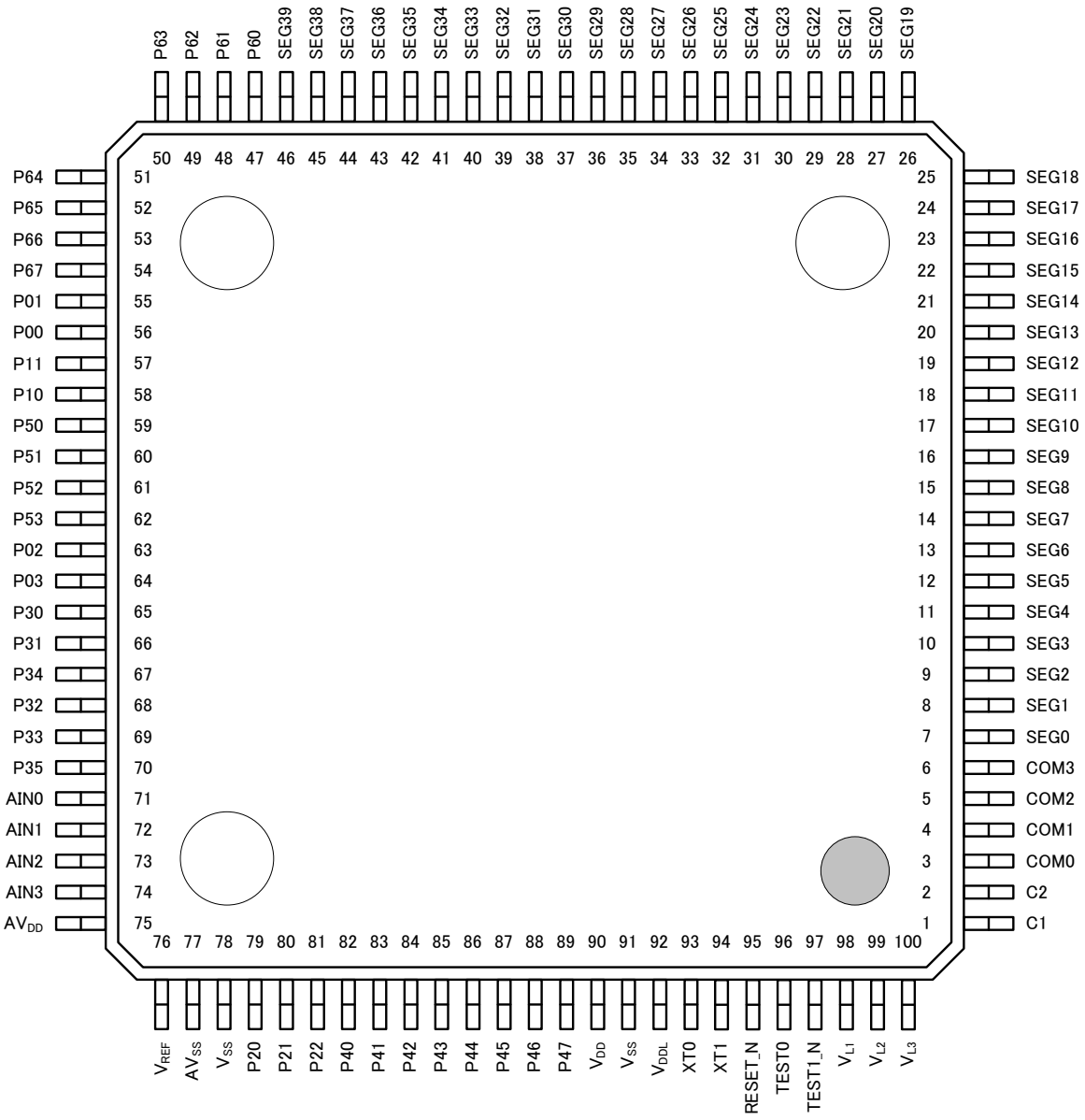
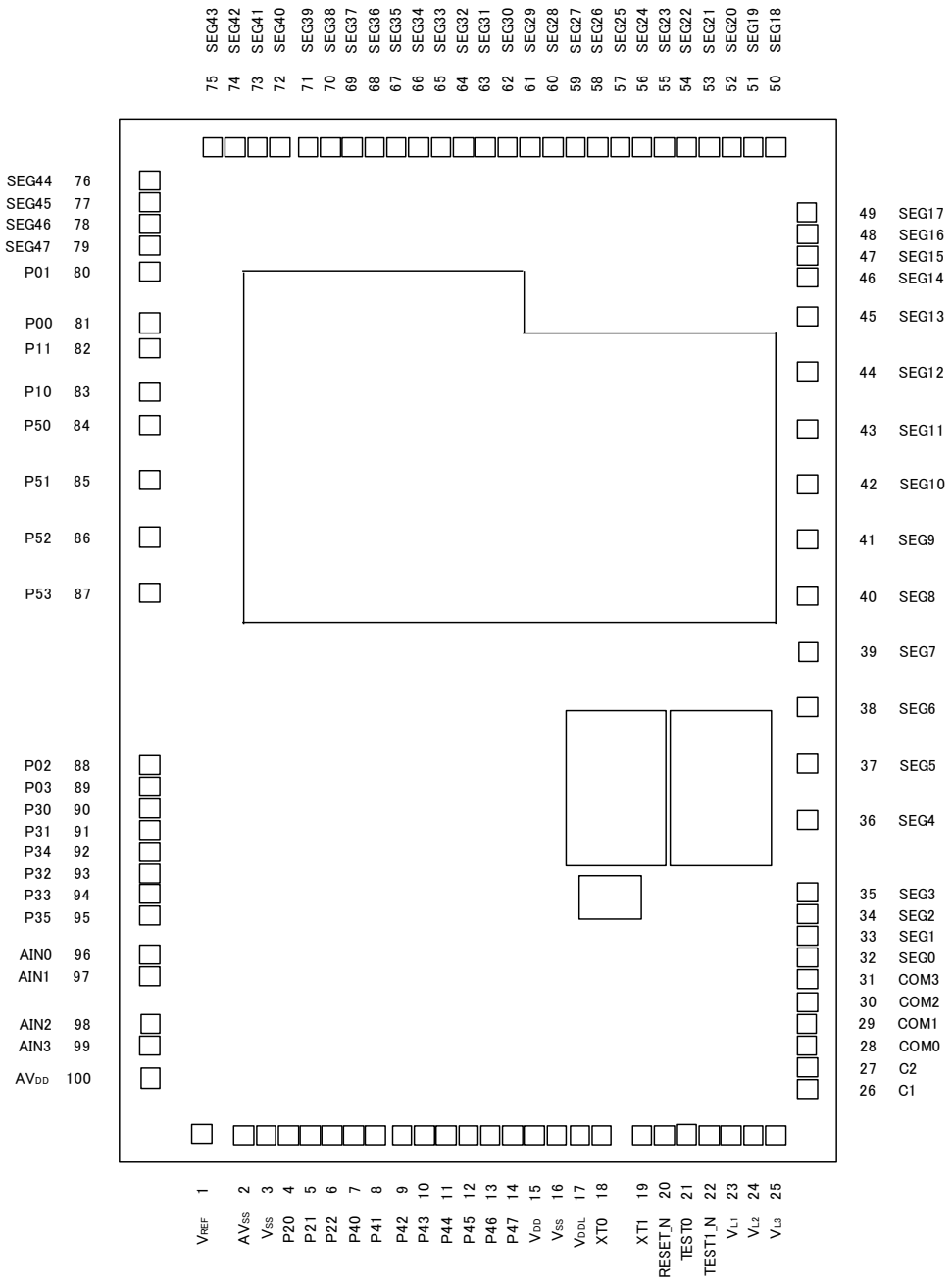


Figure 1-4 Pin Layout of ML610Q418C Package

1.3.1.3 Pin Layout of ML610Q418 Chip

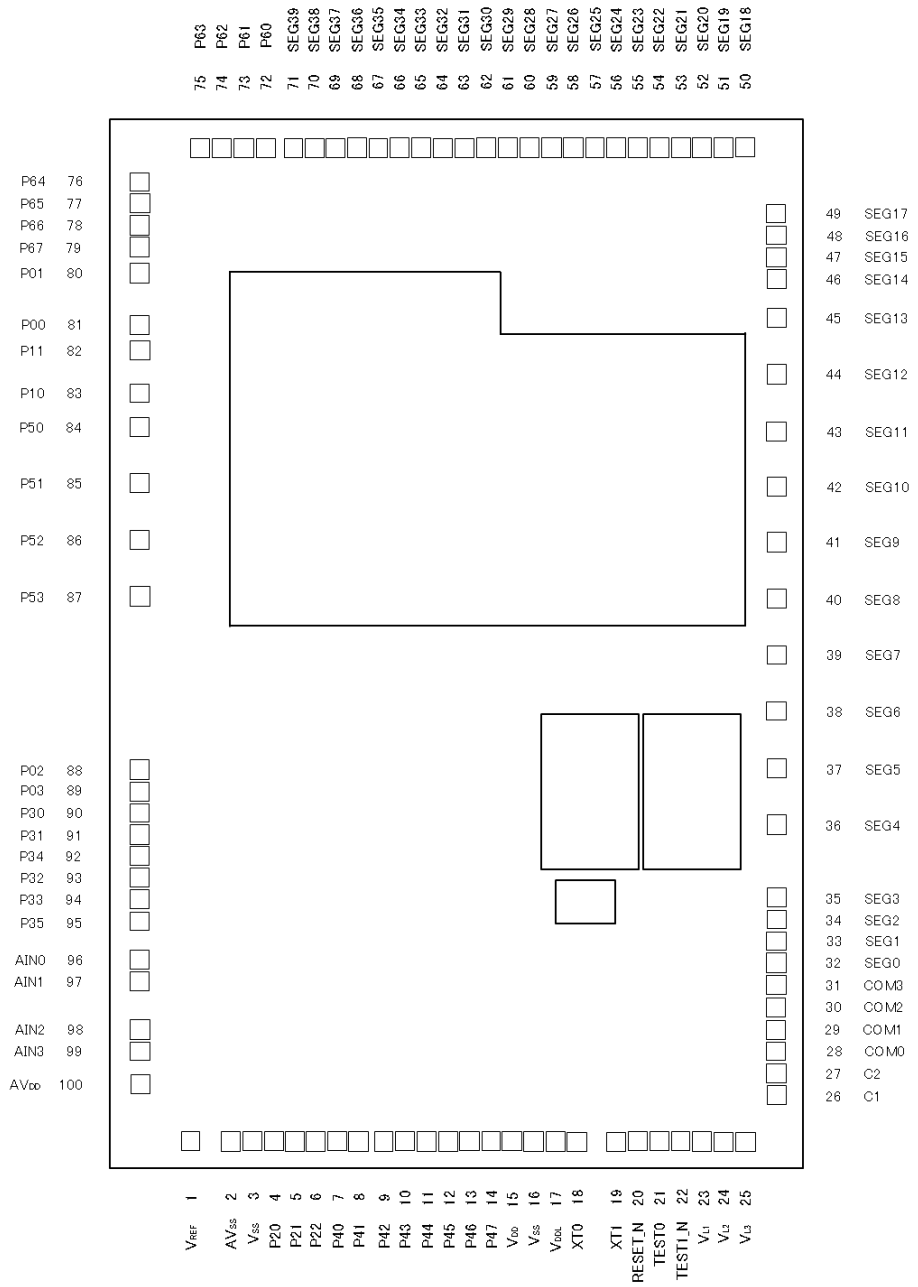


Chip size:	2.64mm×3.84mm
PAD count:	100 pins
Minimum PAD pitch:	80 μm
PAD aperture:	70 μm × 70 μm
Chip thickness:	350 μm
Voltage of the rear side of chip:	V_{SS} level

Figure 1-5 Dimensions of ML610Q418 Chip

Note:
Figure 1-5 is an image figure of the order of PAD, and it differs from an actual image. Refer to the PAD coordinate for detailed arrangement.
A chip angle can be checked by the distinguishing mark of three figures.

1.3.1.4 Pin Layout of ML610Q418C Chip



Chip size:	2.64mm×3.84mm
PAD count:	100 pins
Minimum PAD pitch:	80 μm
PAD aperture:	70 μm × 70 μm
Chip thickness:	350 μm
Voltage of the rear side of chip:	V_{SS} level

Figure 1-6 Dimensions of ML610Q418C Chip

Note:
Figure 1-6 is an image figure of the order of PAD, and it differs from an actual image. Refer to the PAD coordinate for detailed arrangement.
A chip angle can be checked by the distinguishing mark of three figures.

1.3.1.5 Pad Coordinates of ML610Q418 Chip

Table 1-1 Pad Coordinates of ML610Q418

Chip Center: X=0,Y=0

PAD No.	Pad Name	X (μm)	Y (μm)	PAD No.	Pad Name	X (μm)	Y (μm)	PAD No.	Pad Name	X (μm)	Y (μm)
1	V_{ref}	-1020	-1814	36	SEG4	1214	-658	71	SEG39	-620	1814
2	AV_{SS}	-860	-1814	37	SEG5	1214	-452	72	SEG40	-730	1814
3	V_{SS}	-780	-1814	38	SEG6	1214	-246	73	SEG41	-810	1814
4	P20	-700	-1814	39	SEG7	1214	-40	74	SEG42	-890	1814
5	P21	-620	-1814	40	SEG8	1214	166	75	SEG43	-970	1814
6	P22	-540	-1814	41	SEG9	1214	372	76	SEG44	-1214	1700
7	P40	-460	-1814	42	SEG10	1214	578	77	SEG45	-1214	1620
8	P41	-380	-1814	43	SEG11	1214	784	78	SEG46	-1214	1540
9	P42	-280	-1814	44	SEG12	1214	990	79	SEG47	-1214	1460
10	P43	-200	-1814	45	SEG13	1214	1196	80	P01	-1214	1360
11	P44	-120	-1814	46	SEG14	1214	1335	81	P00	-1214	1170
12	P45	-40	-1814	47	SEG15	1214	1415	82	P11	-1214	1080
13	P46	40	-1814	48	SEG16	1214	1495	83	P10	-1214	920
14	P47	120	-1814	49	SEG17	1214	1575	84	P50	-1214	796
15	V_{DD}	204	-1814	50	SEG18	1060	1814	85	P51	-1214	590
16	V_{SS}	284	-1814	51	SEG19	980	1814	86	P52	-1214	384
17	V_{DDL}	364	-1814	52	SEG20	900	1814	87	P53	-1214	178
18	XT0	452	-1814	53	SEG21	820	1814	88	P02	-1214	-452
19	XT1	612	-1814	54	SEG22	740	1814	89	P03	-1214	-532
20	RESET_N	692	-1814	55	SEG23	660	1814	90	P30	-1214	-612
21	TEST0	772	-1814	56	SEG24	580	1814	91	P31	-1214	-692
22	TEST1_N	852	-1814	57	SEG25	500	1814	92	P34	-1214	-772
23	V_{L1}	932	-1814	58	SEG26	420	1814	93	P32	-1214	-852
24	V_{L2}	1012	-1814	59	SEG27	340	1814	94	P33	-1214	-932
25	V_{L3}	1092	-1814	60	SEG28	260	1814	95	P35	-1214	-1012
26	C1	1214	-1645	61	SEG29	180	1814	96	AIN0	-1214	-1153
27	C2	1214	-1565	62	SEG30	100	1814	97	AIN1	-1214	-1233
28	COM0	1214	-1485	63	SEG31	20	1814	98	AIN2	-1214	-1405
29	COM1	1214	-1405	64	SEG32	-60	1814	99	AIN3	-1214	-1485
30	COM2	1214	-1325	65	SEG33	-140	1814	100	AV_{DD}	-1214	-1611
31	COM3	1214	-1245	66	SEG34	-220	1814				
32	SEG0	1214	-1165	67	SEG35	-300	1814				
33	SEG1	1214	-1085	68	SEG36	-380	1814				
34	SEG2	1214	-1005	69	SEG37	-460	1814				
35	SEG3	1214	-925	70	SEG38	-540	1814				

1.3.1.6 Pad Coordinates of ML610Q418C Chip

Table 1-2 Pad Coordinates of ML610Q418C

Chip Center: X=0,Y=0

PAD No.	Pad Name	X (μm)	Y (μm)	PAD No.	Pad Name	X (μm)	Y (μm)	PAD No.	Pad Name	X (μm)	Y (μm)
1	V_{ref}	-1020	-1814	36	SEG4	1214	-658	71	SEG39	-620	1814
2	AV_{SS}	-860	-1814	37	SEG5	1214	-452	72	P60	-730	1814
3	V_{SS}	-780	-1814	38	SEG6	1214	-246	73	P61	-810	1814
4	P20	-700	-1814	39	SEG7	1214	-40	74	P62	-890	1814
5	P21	-620	-1814	40	SEG8	1214	166	75	P63	-970	1814
6	P22	-540	-1814	41	SEG9	1214	372	76	P64	-1214	1700
7	P40	-460	-1814	42	SEG10	1214	578	77	P65	-1214	1620
8	P41	-380	-1814	43	SEG11	1214	784	78	P66	-1214	1540
9	P42	-280	-1814	44	SEG12	1214	990	79	P67	-1214	1460
10	P43	-200	-1814	45	SEG13	1214	1196	80	P01	-1214	1360
11	P44	-120	-1814	46	SEG14	1214	1335	81	P00	-1214	1170
12	P45	-40	-1814	47	SEG15	1214	1415	82	P11	-1214	1080
13	P46	40	-1814	48	SEG16	1214	1495	83	P10	-1214	920
14	P47	120	-1814	49	SEG17	1214	1575	84	P50	-1214	796
15	V_{DD}	204	-1814	50	SEG18	1060	1814	85	P51	-1214	590
16	V_{SS}	284	-1814	51	SEG19	980	1814	86	P52	-1214	384
17	V_{DDL}	364	-1814	52	SEG20	900	1814	87	P53	-1214	178
18	XT0	452	-1814	53	SEG21	820	1814	88	P02	-1214	-452
19	XT1	612	-1814	54	SEG22	740	1814	89	P03	-1214	-532
20	RESET_N	692	-1814	55	SEG23	660	1814	90	P30	-1214	-612
21	TEST0	772	-1814	56	SEG24	580	1814	91	P31	-1214	-692
22	TEST1_N	852	-1814	57	SEG25	500	1814	92	P34	-1214	-772
23	V_{L1}	932	-1814	58	SEG26	420	1814	93	P32	-1214	-852
24	V_{L2}	1012	-1814	59	SEG27	340	1814	94	P33	-1214	-932
25	V_{L3}	1092	-1814	60	SEG28	260	1814	95	P35	-1214	-1012
26	C1	1214	-1645	61	SEG29	180	1814	96	AIN0	-1214	-1153
27	C2	1214	-1565	62	SEG30	100	1814	97	AIN1	-1214	-1233
28	COM0	1214	-1485	63	SEG31	20	1814	98	AIN2	-1214	-1405
29	COM1	1214	-1405	64	SEG32	-60	1814	99	AIN3	-1214	-1485
30	COM2	1214	-1325	65	SEG33	-140	1814	100	AV_{DD}	-1214	-1611
31	COM3	1214	-1245	66	SEG34	-220	1814				
32	SEG0	1214	-1165	67	SEG35	-300	1814				
33	SEG1	1214	-1085	68	SEG36	-380	1814				
34	SEG2	1214	-1005	69	SEG37	-460	1814				
35	SEG3	1214	-925	70	SEG38	-540	1814				

1.3.2 List of Pins

1.3.2.1 Pad list of ML610Q418/ML610Q418C Chip

PAD No.		Primary function			Secondary function			Tertiary function			Quaternary function		
Q418	Q418C	Pin name	I/O	Function	Pin name	I/O	Function	Pin name	I/O	Function	Pin name	I/O	Function
3, 16	3. 16	Vss	—	Negative power supply pin	—	—	—	—	—	—	—	—	—
15	15	V_{DD}	—	Positive power supply pin	—	—	—	—	—	—	—	—	—
17	17	V_{DDL}	—	Power supply pin for internal logic (internally generated)	—	—	—	—	—	—	—	—	—
2	2	AV_{SS}	—	Negative power supply pin for successive approximation type ADC	—	—	—	—	—	—	—	—	—
100	100	AV_{DD}	—	Positive power supply pin for successive approximation type ADC	—	—	—	—	—	—	—	—	—
1	1	V_{REF}	—	Reference power supply pin for successive approximation type ADC	—	—	—	—	—	—	—	—	—
96	96	AIN0	—	Successive approximation type ADC input	—	—	—	—	—	—	—	—	—
97	97	AIN1	—	Successive approximation type ADC input	—	—	—	—	—	—	—	—	—
98	98	AIN2	—	Successive approximation type ADC input	—	—	—	—	—	—	—	—	—
99	99	AIN3	—	Successive approximation type ADC input	—	—	—	—	—	—	—	—	—
23	23	V_{L1}	—	Power supply pin for LCD bias (internally generated)	—	—	—	—	—	—	—	—	—
24	24	V_{L2}	—	Power supply pin for LCD bias (internally generated)	—	—	—	—	—	—	—	—	—
25	25	V_{L3}	—	Power supply pin for LCD bias (internally generated)	—	—	—	—	—	—	—	—	—
26	26	C1	—	Capacitor connection pin for LCD bias generation	—	—	—	—	—	—	—	—	—
27	27	C2	—	Capacitor connection pin for LCD bias generation	—	—	—	—	—	—	—	—	—
21	21	TEST0	I/O	Test pin	—	—	—	—	—	—	—	—	—
22	22	TEST1_N	I	Test pin	—	—	—	—	—	—	—	—	—
20	20	RESET_N	I	Reset input pin	—	—	—	—	—	—	—	—	—

PAD No.		Primary function			Secondary function			Tertiary function			Quaternary function		
Q418	Q418C	Pin name	I/O	Function	Pin name	I/O	Function	Pin name	I/O	Function	Pin name	I/O	Function
18	18	XT0	I	Low-speed clock oscillation pin	—	—	—	—	—	—	—	—	—
19	19	XT1	O	Low-speed clock oscillation pin	—	—	—	—	—	—	—	—	—
81	81	P00/EXI0/ CAP0	I	Input port, External interrupt, Capture 0 input	—	—	—	—	—	—	—	—	—
80	80	P01/EXI1/ CAP1	I	Input port, External interrupt, Capture 1 input	—	—	—	—	—	—	—	—	—
88	88	P02/EXI2 /RXD0	I	Input port, External interrupt, UART0 received data	—	—	—	—	—	—	—	—	—
89	89	P03/EXI3 /RXD1	I	Input port, External interrupt UART1 received data	—	—	—	—	—	—	—	—	—
83	83	P10	I	Input port	OSC0	I	High-speed oscillation	—	—	—	—	—	—
82	82	P11	I	Input port	OSC1	O	High-speed oscillation	—	—	—	—	—	—
4	4	P20/LED0	O	Output port	LSCLK	O	Low-speed clock output	—	—	—	—	—	—
5	5	P21/LED1	O	Output port	OUTCLK	O	High-speed clock output	—	—	—	—	—	—
6	6	P22/LED2	O	Output port	MD0	O	Melody 0 output	—	—	—	—	—	—
90	90	P30	I/O	Input/output port	IN0	I	RC type ADC0 oscillation input pin	—	—	—	—	—	—
91	91	P31	I/O	Input/output port	CS0	O	RC type ADC0 reference capacitor connection pin	—	—	—	—	—	—
93	93	P32	I/O	Input/output port	RS0	O	RC type ADC0 reference resistor connection pin	—	—	—	—	—	—
94	94	P33	I/O	Input/output port	RT0	O	RC type ADC0 measurement resistor sensor connection pin	—	—	—	—	—	—
92	92	P34	I/O	Input/output port	RCT0	O	RC type ADC0 resistor/capacitor sensor connection pin	PWM0	O	PWM output			
95	95	P35	I/O	Input/output port	RCM	O	RC type ADC oscillation monitor	—	—	—	—	—	—
7	7	P40	I/O	Input/output port	SDA	I/O	I2C data input/output	SIN0	I	SSIO0 data input	—	—	—
8	8	P41	I/O	Input/output port	SCL	I/O	I2C clock input/output	SCK0	I/O	SSIO0 synchronous clock	—	—	—
9	9	P42	I/O	Input/output port	RXD0	I	UART0 data input	SOUT0	O	SSIO0 data output	—	—	—
10	10	P43	I/O	Input/output port	TXD0	O	UART0 data output	PWM0	O	PWM output	TXD1	O	UART1 data output
11	11	P44/T02P	I/O	Input/output port, Timer 0/Timer	IN1	I	RC type ADC1 oscillation	SIN0	I	SSIO0 data input	—	—	—

PAD No.		Primary function			Secondary function			Tertiary function			Quaternary function		
Q418	Q418C	Pin name	I/O	Function	Pin name	I/O	Function	Pin name	I/O	Function	Pin name	I/O	Function
		0CK		2/PWM0 external clock input			input pin						
12	12	P45/T13CK	I/O	Input/output port, Timer 1/Timer 3 external clock input	CS1	O	RC type ADC1 reference capacitor connection pin	SCK0	I/O	SSIO0 synchronous clock	—	—	—
13	13	P46	I/O	Input/output port	RS1	O	RC type ADC1 reference resistor connection pin	SOUT0	O	SSIO0 data output	—	—	—
14	14	P47	I/O	Input/output port	RT1	O	RC type ADC1 measurement resistor sensor connection pin	—	—	—	—	—	—
84	84	P50/EXI8	I/O	Input/output port, External interrupt	MD0	O	Melody 0 output	SIN1	I	SSIO1 data input	—	—	—
85	85	P51/EXI8	I/O	Input/output port, External interrupt	—	—	—	SCK1	I/O	SSIO1 synchronous clock input/output	—	—	—
86	86	P52/EXI8	I/O	Input/output port, External interrupt	RXD1	I	UART1 received data	SOUT1	O	SSIO1 data output	—	—	—
87	87	P53/EXI8	I/O	Input/output port, External interrupt	TXD1	O	UART1 data output	—	—	—	—	—	—
28	28	COM0	O	LCD common pin	—	—	—	—	—	—	—	—	—
29	29	COM1	O	LCD common pin	—	—	—	—	—	—	—	—	—
30	30	COM2	O	LCD common pin	—	—	—	—	—	—	—	—	—
31	31	COM3	O	LCD common pin	—	—	—	—	—	—	—	—	—
32	32	SEG0	O	LCD segment pin	—	—	—	—	—	—	—	—	—
33	33	SEG1	O	LCD segment pin	—	—	—	—	—	—	—	—	—
34	34	SEG2	O	LCD segment pin	—	—	—	—	—	—	—	—	—
35	35	SEG3	O	LCD segment pin	—	—	—	—	—	—	—	—	—
36	36	SEG4	O	LCD segment pin	—	—	—	—	—	—	—	—	—
37	37	SEG5	O	LCD segment pin	—	—	—	—	—	—	—	—	—
38	38	SEG6	O	LCD segment pin	—	—	—	—	—	—	—	—	—
39	39	SEG7	O	LCD segment pin	—	—	—	—	—	—	—	—	—
40	40	SEG8	O	LCD segment pin	—	—	—	—	—	—	—	—	—
41	41	SEG9	O	LCD segment pin	—	—	—	—	—	—	—	—	—
42	42	SEG10	O	LCD segment pin	—	—	—	—	—	—	—	—	—
43	43	SEG11	O	LCD segment pin	—	—	—	—	—	—	—	—	—
44	44	SEG12	O	LCD segment pin	—	—	—	—	—	—	—	—	—
45	45	SEG13	O	LCD segment pin	—	—	—	—	—	—	—	—	—
46	46	SEG14	O	LCD segment pin	—	—	—	—	—	—	—	—	—
47	47	SEG15	O	LCD segment pin	—	—	—	—	—	—	—	—	—
48	48	SEG16	O	LCD segment pin	—	—	—	—	—	—	—	—	—
49	49	SEG17	O	LCD segment pin	—	—	—	—	—	—	—	—	—
50	50	SEG18	O	LCD segment pin	—	—	—	—	—	—	—	—	—
51	51	SEG19	O	LCD segment pin	—	—	—	—	—	—	—	—	—
52	52	SEG20	O	LCD segment pin	—	—	—	—	—	—	—	—	—
53	53	SEG21	O	LCD segment pin	—	—	—	—	—	—	—	—	—
54	54	SEG22	O	LCD segment pin	—	—	—	—	—	—	—	—	—
55	55	SEG23	O	LCD segment pin	—	—	—	—	—	—	—	—	—
56	56	SEG24	O	LCD segment pin	—	—	—	—	—	—	—	—	—
57	57	SEG25	O	LCD segment pin	—	—	—	—	—	—	—	—	—
58	58	SEG26	O	LCD segment pin	—	—	—	—	—	—	—	—	—

PAD No.		Primary function			Secondary function			Tertiary function			Quaternary function		
Q418	Q418C	Pin name	I/O	Function	Pin name	I/O	Function	Pin name	I/O	Function	Pin name	I/O	Function
59	59	SEG27	O	LCD segment pin	—	—	—	—	—	—	—	—	—
60	60	SEG28	O	LCD segment pin	—	—	—	—	—	—	—	—	—
61	61	SEG29	O	LCD segment pin	—	—	—	—	—	—	—	—	—
62	62	SEG30	O	LCD segment pin	—	—	—	—	—	—	—	—	—
63	63	SEG31	O	LCD segment pin	—	—	—	—	—	—	—	—	—
64	64	SEG32	O	LCD segment pin	—	—	—	—	—	—	—	—	—
65	65	SEG33	O	LCD segment pin	—	—	—	—	—	—	—	—	—
66	66	SEG34	O	LCD segment pin	—	—	—	—	—	—	—	—	—
67	67	SEG35	O	LCD segment pin	—	—	—	—	—	—	—	—	—
68	68	SEG36	O	LCD segment pin	—	—	—	—	—	—	—	—	—
69	69	SEG37	O	LCD segment pin	—	—	—	—	—	—	—	—	—
70	70	SEG38	O	LCD segment pin	—	—	—	—	—	—	—	—	—
71	71	SEG39	O	LCD segment pin	—	—	—	—	—	—	—	—	—
72	—	SEG40	O	LCD segment pin	—	—	—	—	—	—	—	—	—
73	—	SEG41	O	LCD segment pin	—	—	—	—	—	—	—	—	—
74	—	SEG42	O	LCD segment pin	—	—	—	—	—	—	—	—	—
75	—	SEG43	O	LCD segment pin	—	—	—	—	—	—	—	—	—
76	—	SEG44	O	LCD segment pin	—	—	—	—	—	—	—	—	—
77	—	SEG45	O	LCD segment pin	—	—	—	—	—	—	—	—	—
78	—	SEG46	O	LCD segment pin	—	—	—	—	—	—	—	—	—
79	—	SEG47	O	LCD segment pin	—	—	—	—	—	—	—	—	—
—	72	P60	I/O	Input/output port	—	—	—	—	—	—	—	—	—
—	73	P61	I/O	Input/output port	—	—	—	—	—	—	—	—	—
—	74	P62	I/O	Input/output port	—	—	—	—	—	—	—	—	—
—	75	P63	I/O	Input/output port	—	—	—	—	—	—	—	—	—
—	76	P64	I/O	Input/output port	—	—	—	—	—	—	—	—	—
—	77	P65	I/O	Input/output port	—	—	—	—	—	—	—	—	—
—	78	P66	I/O	Input/output port	—	—	—	—	—	—	—	—	—
—	79	P67	I/O	Input/output port	—	—	—	—	—	—	—	—	—

1.3.2.2 Pin list of ML610Q418/ML610Q418C

Pin No.		Primary function			Secondary function			Tertiary function			Quaternary function		
Q418	Q418C	Pin name	I/O	Function	Pin name	I/O	Function	Pin name	I/O	Function	Pin name	I/O	Function
78,91		Vss	—	Negative power supply pin	—	—	—	—	—	—	—	—	—
90		V_{DD}	—	Positive power supply pin	—	—	—	—	—	—	—	—	—
92		V_{DDL}	—	Power supply pin for internal logic (internally generated)	—	—	—	—	—	—	—	—	—
77		AV_{SS}	—	Negative power supply pin for successive approximation type ADC	—	—	—	—	—	—	—	—	—
75		AV_{DD}	—	Positive power supply pin for successive approximation type ADC	—	—	—	—	—	—	—	—	—
76		V_{REF}	—	Reference power supply pin for successive approximation type ADC	—	—	—	—	—	—	—	—	—
71		AIN0	—	Successive approximation type ADC input	—	—	—	—	—	—	—	—	—
72		AIN1	—	Successive approximation type ADC input	—	—	—	—	—	—	—	—	—
73		AIN2	—	Successive approximation type ADC input	—	—	—	—	—	—	—	—	—
74		AIN3	—	Successive approximation type ADC input	—	—	—	—	—	—	—	—	—
98		V_{L1}	—	Power supply pin for LCD bias (internally generated)	—	—	—	—	—	—	—	—	—
99		V_{L2}	—	Power supply pin for LCD bias (internally generated)	—	—	—	—	—	—	—	—	—
100		V_{L3}	—	Power supply pin for LCD bias (internally generated)	—	—	—	—	—	—	—	—	—
1		C1	—	Capacitor connection pin for LCD bias generation	—	—	—	—	—	—	—	—	—
2		C2	—	Capacitor connection pin for LCD bias generation	—	—	—	—	—	—	—	—	—
96		TEST0	I/O	Test pin	—	—	—	—	—	—	—	—	—
97		TEST1_N	I	Test pin	—	—	—	—	—	—	—	—	—
95		RESET_N	I	Reset input pin	—	—	—	—	—	—	—	—	—
93		XT0	I	Low-speed clock oscillation pin	—	—	—	—	—	—	—	—	—

Pin No.		Primary function			Secondary function			Tertiary function			Quaternary function		
Q418	Q418C	Pin name	I/O	Function	Pin name	I/O	Function	Pin name	I/O	Function	Pin name	I/O	Function
94		XT1	O	Low-speed clock oscillation pin	—	—	—	—	—	—	—	—	—
56		P00/EXI0/ CAP0	I	Input port, External interrupt, Capture 0 input	—	—	—	—	—	—	—	—	—
55		P01/EXI1/ CAP1	I	Input port, External interrupt, Capture 1 input	—	—	—	—	—	—	—	—	—
63		P02/EXI2 /RXD0	I	Input port, External interrupt, UART0 received data	—	—	—	—	—	—	—	—	—
64		P03/EXI3 /RXD1	I	Input port, External interrupt UART1 received data	—	—	—	—	—	—	—	—	—
58		P10	I	Input port	OSC0	I	High-speed oscillation	—	—	—	—	—	—
57		P11	I	Input port	OSC1	O	High-speed oscillation	—	—	—	—	—	—
79		P20/LED0	O	Output port	LSCLK	O	Low-speed clock output	—	—	—	—	—	—
80		P21/LED1	O	Output port	OUTCLK	O	High-speed clock output	—	—	—	—	—	—
81		P22/LED2	O	Output port	MD0	O	Melody 0 output	—	—	—	—	—	—
65		P30	I/O	Input/output port	IN0	I	RC type ADC0 oscillation input pin	—	—	—	—	—	—
66		P31	I/O	Input/output port	CS0	O	RC type ADC0 reference capacitor connection pin	—	—	—	—	—	—
68		P32	I/O	Input/output port	RS0	O	RC type ADC0 reference resistor connection pin	—	—	—	—	—	—
69		P33	I/O	Input/output port	RT0	O	RC type ADC0 measurement resistor sensor connection pin	—	—	—	—	—	—
67		P34	I/O	Input/output port	RCT0	O	RC type ADC0 resistor/capacitor sensor connection pin	PWM0	O	PWM output			
70		P35	I/O	Input/output port	RCM	O	RC type ADC oscillation monitor	—	—	—	—	—	—
82		P40	I/O	Input/output port	SDA	I/O	I2C data input/output	SIN0	I	SSIO0 data input	—	—	—
83		P41	I/O	Input/output port	SCL	I/O	I2C clock input/output	SCK0	I/O	SSIO0 synchronous clock	—	—	—
84		P42	I/O	Input/output port	RXD0	I	UART0 data input	SOUT0	O	SSIO0 data output	—	—	—
85		P43	I/O	Input/output port	TXD0	O	UART0 data output	PWM0	O	PWM output	TXD1	O	UART1 data output
86		P44/T02P 0CK	I/O	Input/output port, Timer 0/Timer 2/PWM0 external clock input	IN1	I	RC type ADC1 oscillation input pin	SIN0	I	SSIO0 data input	—	—	—

Pin No.		Primary function			Secondary function			Tertiary function			Quaternary function		
Q418	Q418C	Pin name	I/O	Function	Pin name	I/O	Function	Pin name	I/O	Function	Pin name	I/O	Function
87		P45/T13CK	I/O	Input/output port, Timer 1/Timer 3 external clock input	CS1	O	RC type ADC1 reference capacitor connection pin	SCK0	I/O	SSIO0 synchronous clock	—	—	—
88		P46	I/O	Input/output port	RS1	O	RC type ADC1 reference resistor connection pin	SOUT0	O	SSIO0 data output	—	—	—
89		P47	I/O	Input/output port	RT1	O	RC type ADC1 measurement resistor sensor connection pin	—	—	—	—	—	—
59		P50/EXI8	I/O	Input/output port, External interrupt	MD0	O	Melody 0 output	SIN1	I	SSIO1 data input	—	—	—
60		P51/EXI8	I/O	Input/output port, External interrupt	—	—	—	SCK1	I/O	SSIO1 synchronous clock input/output	—	—	—
61		P52/EXI8	I/O	Input/output port, External interrupt	RXD1	I	UART1 received data	SOUT1	O	SSIO1 data output	—	—	—
62		P53/EXI8	I/O	Input/output port, External interrupt	TXD1	O	UART1 data output	—	—	—	—	—	—
3		COM0	O	LCD common pin	—	—	—	—	—	—	—	—	—
4		COM1	O	LCD common pin	—	—	—	—	—	—	—	—	—
5		COM2	O	LCD common pin	—	—	—	—	—	—	—	—	—
6		COM3	O	LCD common pin	—	—	—	—	—	—	—	—	—
7		SEG0	O	LCD segment pin	—	—	—	—	—	—	—	—	—
8		SEG1	O	LCD segment pin	—	—	—	—	—	—	—	—	—
9		SEG2	O	LCD segment pin	—	—	—	—	—	—	—	—	—
10		SEG3	O	LCD segment pin	—	—	—	—	—	—	—	—	—
11		SEG4	O	LCD segment pin	—	—	—	—	—	—	—	—	—
12		SEG5	O	LCD segment pin	—	—	—	—	—	—	—	—	—
13		SEG6	O	LCD segment pin	—	—	—	—	—	—	—	—	—
14		SEG7	O	LCD segment pin	—	—	—	—	—	—	—	—	—
15		SEG8	O	LCD segment pin	—	—	—	—	—	—	—	—	—
16		SEG9	O	LCD segment pin	—	—	—	—	—	—	—	—	—
17		SEG10	O	LCD segment pin	—	—	—	—	—	—	—	—	—
18		SEG11	O	LCD segment pin	—	—	—	—	—	—	—	—	—
19		SEG12	O	LCD segment pin	—	—	—	—	—	—	—	—	—
20		SEG13	O	LCD segment pin	—	—	—	—	—	—	—	—	—
21		SEG14	O	LCD segment pin	—	—	—	—	—	—	—	—	—
22		SEG15	O	LCD segment pin	—	—	—	—	—	—	—	—	—
23		SEG16	O	LCD segment pin	—	—	—	—	—	—	—	—	—
24		SEG17	O	LCD segment pin	—	—	—	—	—	—	—	—	—
25		SEG18	O	LCD segment pin	—	—	—	—	—	—	—	—	—
26		SEG19	O	LCD segment pin	—	—	—	—	—	—	—	—	—
27		SEG20	O	LCD segment pin	—	—	—	—	—	—	—	—	—
28		SEG21	O	LCD segment pin	—	—	—	—	—	—	—	—	—
29		SEG22	O	LCD segment pin	—	—	—	—	—	—	—	—	—
30		SEG23	O	LCD segment pin	—	—	—	—	—	—	—	—	—
31		SEG24	O	LCD segment pin	—	—	—	—	—	—	—	—	—
32		SEG25	O	LCD segment pin	—	—	—	—	—	—	—	—	—
33		SEG26	O	LCD segment pin	—	—	—	—	—	—	—	—	—
34		SEG27	O	LCD segment pin	—	—	—	—	—	—	—	—	—

Pin No.		Primary function			Secondary function			Tertiary function			Quaternary function		
Q418	Q418C	Pin name	I/O	Function	Pin name	I/O	Function	Pin name	I/O	Function	Pin name	I/O	Function
35		SEG28	O	LCD segment pin	—	—	—	—	—	—	—	—	—
36		SEG29	O	LCD segment pin	—	—	—	—	—	—	—	—	—
37		SEG30	O	LCD segment pin	—	—	—	—	—	—	—	—	—
38		SEG31	O	LCD segment pin	—	—	—	—	—	—	—	—	—
39		SEG32	O	LCD segment pin	—	—	—	—	—	—	—	—	—
40		SEG33	O	LCD segment pin	—	—	—	—	—	—	—	—	—
41		SEG34	O	LCD segment pin	—	—	—	—	—	—	—	—	—
42		SEG35	O	LCD segment pin	—	—	—	—	—	—	—	—	—
43		SEG36	O	LCD segment pin	—	—	—	—	—	—	—	—	—
44		SEG37	O	LCD segment pin	—	—	—	—	—	—	—	—	—
45		SEG38	O	LCD segment pin	—	—	—	—	—	—	—	—	—
46		SEG39	O	LCD segment pin	—	—	—	—	—	—	—	—	—
47	—	SEG40	O	LCD segment pin	—	—	—	—	—	—	—	—	—
48	—	SEG41	O	LCD segment pin	—	—	—	—	—	—	—	—	—
49	—	SEG42	O	LCD segment pin	—	—	—	—	—	—	—	—	—
50	—	SEG43	O	LCD segment pin	—	—	—	—	—	—	—	—	—
51	—	SEG44	O	LCD segment pin	—	—	—	—	—	—	—	—	—
52	—	SEG45	O	LCD segment pin	—	—	—	—	—	—	—	—	—
53	—	SEG46	O	LCD segment pin	—	—	—	—	—	—	—	—	—
54	—	SEG47	O	LCD segment pin	—	—	—	—	—	—	—	—	—
—	47	P60	I/O	Input/output port	—	—	—	—	—	—	—	—	—
—	48	P61	I/O	Input/output port	—	—	—	—	—	—	—	—	—
—	49	P62	I/O	Input/output port	—	—	—	—	—	—	—	—	—
—	50	P63	I/O	Input/output port	—	—	—	—	—	—	—	—	—
—	51	P64	I/O	Input/output port	—	—	—	—	—	—	—	—	—
—	52	P65	I/O	Input/output port	—	—	—	—	—	—	—	—	—
—	53	P66	I/O	Input/output port	—	—	—	—	—	—	—	—	—
—	54	P67	I/O	Input/output port	—	—	—	—	—	—	—	—	—

1.3.3 Description of Pins

Table 1-3 Description of Pins

Pin name	I/O	Description	Primary/ Secondary/ Tertiary	Logic
System				
RESET_N	I	Reset input pin. When this pin is set to a "L" level, system reset mode is set and the internal section is initialized. When this pin is set to a "H" level subsequently, program execution starts. A pull-up resistor is internally connected.	—	Negative
XT0	I	Crystal connection pin for low-speed clock.	—	—
XT1	O	A 32.768 kHz crystal oscillator (see measuring circuit 1) is connected to this pin. Capacitors CDL and CGL are connected across this pin and V_{SS} as required.	—	—
OSC0	I	Crystal/ceramic connection pin for high-speed clock.	Secondary	—
OSC1	O	A crystal or ceramic is connected to this pin (4.1 MHz max.). Capacitors CDH and CGH (see measuring circuit 1) are connected across this pin and V_{SS}. This pin is used as the secondary function of the P10 pin(OSC0) and P11 pin(OSC1).	Secondary	—
LSCLK	O	Low-speed clock output pin. This pin is used as the secondary function of the P20 pin.	Secondary	—
OUTCLK	O	High-speed clock output pin. This pin is used as the secondary function of the P21 pin.	Secondary	—
General-purpose input port				
P00-P03	I	General-purpose input port. Since these pins have secondary functions, the pins cannot be used as a port when the secondary functions are used.	Primary	Positive
P10-P11	I	General-purpose input port. Since these pins have secondary functions, the pins cannot be used as a port when the secondary functions are used.	Primary	Positive
General-purpose output port				
P20-P22	O	General-purpose output port. Since these pins have secondary functions, the pins cannot be used as a port when the secondary functions are used.	Primary	Positive
General-purpose input/output port				
P30-P35	I/O	General-purpose input/output port. Since these pins have secondary functions, the pins cannot be used as a port when the secondary functions are used.	Primary	Positive
P40-P47	I/O	General-purpose input/output port. Since these pins have secondary functions, the pins cannot be used as a port when the secondary functions are used.	Primary	Positive
P50-P53	I/O	General-purpose input/output port. Since these pins have secondary functions, the pins cannot be used as a port when the secondary functions are used.	Primary	Positive
P60-P67	I/O	General-purpose input/output port. These pins are for the ML610Q418C, but are not provided in the ML610Q418.	Primary	Positive

Pin name	I/O	Description	Primary/ Secondary/ Tertiary	Logic
UART				
TXD0	O	UART0 data output pin. This pin is used as the secondary function of the P43 pin.	Secondary	Positive
RXD0	I	UART0 data input pin. This pin is used as the secondary function of the P42 or the primary function of the P02 pin.	Primary/Secondary	Positive
TXD1	O	UART1 data output pin. This pin is used as the quaternary function of the P43 pin or the secondary function of the P53.	Secondary/ Quaternary	Positive
RXD1	I	UART1 data input pin. This pin is used as the primary and secondary function of the P52 or the primary function of the P03 pin.	Primary/Secondary	Positive
I^2C bus interface				
SDA	I/O	I^2C data input/output pin. This pin is used as the secondary function of the P40 pin. This pin has an NMOS open drain output. When using this pin as a function of the I^2C, externally connect a pull-up resistor.	Secondary	Positive
SCL	O	I^2C clock output pin. This pin is used as the secondary function of the P41 pin. This pin has an NMOS open drain output. When using this pin as a function of the I^2C, externally connect a pull-up resistor.	Secondary	Positive
Synchronous serial (SSIO)				
SCK0	I/O	Synchronous serial clock input/output pin. This pin is used as the tertiary function of the P41 or P45 pin.	Tertiary	—
SIN0	I	Synchronous serial data input pin. This pin is used as the tertiary function of the P40 or P44 pin.	Tertiary	Positive
SOUT0	O	Synchronous serial data output pin. This pin is used as the tertiary function of the P42 or P46 pin.	Tertiary	Positive
SCK1	I/O	Synchronous serial clock input/output pin. This pin is used as the tertiary function of the P51 pin.	Tertiary	—
SIN1	I	Synchronous serial data input pin. This pin is used as the tertiary function of the P50 pin.	Tertiary	Positive
SOUT1	O	Synchronous serial data output pin. This pin is used as the tertiary function of the P52 pin.	Tertiary	Positive
PWM				
PWM0	O	PWM0 output pin. This pin is used as the tertiary function of the P43 or P34 pin.	Tertiary	Positive
T0P0CK	I	PWM0 external clock input pin. This pin is used as the primary function of the P44 pin.	Primary	—
External interrupt				
EXI0-3	I	External maskable interrupt input pins. Interrupt enable and edge selection can be performed for each bit by software. These pins are used as the primary functions of the P00-P03 pins.	Primary	Positive/ negative
EXI8	I	External maskable interrupt input pins. Interrupt enable and edge selection can be performed for each bit by software. These pins are used as the primary functions of the P50-P53 pins.	Primary	Positive/ negative
Capture				
CAP0	I	Capture trigger input pins. The value of the time base counter is captured in the register synchronously with the interrupt edge selected by software. These pins are used as the primary functions of the P00 pin(CAP0) and P01 pin(CAP1).	Primary	—
CAP1	I		Primary	—
Timer				
T02P0CK	I	External clock input pin used for Timer 0. This pin is used as the primary function of the P44 pin.	Primary	—
T13CK	I	External clock input pin used for Timer 1. This pin is used as the primary function of the P45 pin.	Primary	—
Melody				
MD0	O	Melody/buzzer signal output pin. This pin is used as the secondary function of the P22 pin.	Secondary	Positive/ negative

Pin name	I/O	Description	Primary/ Secondary/ Tertiary	Logic
LED drive				
LED0-2	O	Nch open drain output pins to drive LED.	Primary	Positive/ negative
RC oscillation type A/D converter				
IN0	I	Channel 0 oscillation input pin. This pin is used as the secondary function of the P30 pin.	Secondary	—
CS0	O	Channel 0 reference capacitor connection pin. This pin is used as the secondary function of the P31 pin.	Secondary	—
RS0	O	This pin is used as the secondary function of the P32 pin which is the reference resistor connection pin of Channel 0.	Secondary	—
RT0	O	Resistor sensor connection pin of Channel 0 for measurement. This pin is used as the secondary function of the P34 pin.	Secondary	—
CRT0	O	Resistor/capacitor sensor connection pin of Channel 0 for measurement. This pin is used as the secondary function of the P33 pin.	Secondary	—
RCM	O	RC oscillation monitor pin. This pin is used as the secondary function of the P35 pin.	Secondary	—
IN1	I	Oscillation input pin of Channel 1. This pin is used as the secondary function of the P44 pin.	Secondary	—
CS1	O	Reference capacitor connection pin of Channel 1. This pin is used as the secondary function of the P45 pin.	Secondary	—
RS1	O	Reference resistor connection pin of Channel 1. This pin is used as the secondary function of the P46 pin.	Secondary	—
RT1	O	Resistor sensor connection pin for measurement of Channel 1. This pin is used as the secondary function of the P47 pin.	Secondary	—
Successive approximation type A/D converter				
AV_{SS}	—	Negative power supply pin for successive approximation type A/D converter.	—	—
AV_{DD}	—	Positive power supply pin for successive approximation type A/D converter.	—	—
V_{REF}	—	Reference power supply pin for successive approximation type A/D converter.	—	—
AIN0	I	Channel 0 analog input for successive approximation type A/D converter.	—	—
AIN1	I	Channel 1 analog input for successive approximation type A/D converter.	—	—
AIN2	I	Channel 2 analog input for successive approximation type A/D converter.	—	—
AIN3	I	Channel 3 analog input for successive approximation type A/D converter.	—	—
LCD drive signal				
COM0-3	O	Common output pins.	—	—
SEG0-39	O	Segment output pins.	—	—
SEG40-47	O	Segment output pins. These pins are for the ML610Q418, but are not provided in the ML610Q418C.	—	—
LCD driver power supply				
V_{L1}	—	Power supply pins for LCD bias (internally generated). Capacitors Ca, Cb, and Cc (see measuring circuit 1) are connected between V_{SS} and V_{L1}, V_{L2}, and V_{L3} respectively.	—	—
V_{L2}	—		—	—
V_{L3}	—		—	—
C1	—	Power supply pins for LCD bias (internally generated). Capacitors C12 is connected between C1 and C2.	—	—
C2	—		—	—
For testing				
TEST0	I/O	Input/output pin for testing. A pull-down resistor is internally connected.	—	—
TEST1_N	I	Input/output pin for testing. A pull-up resistor is internally connected.	—	—

Pin name	I/O	Description	Primary/ Secondary/ Tertiary	Logic
Power supply				
V_{SS}	—	Negative power supply pin.	—	—
V_{DD}	—	Positive power supply pin.	—	—
V_{DDL}	—	Positive power supply pin (internally generated) for internal logic. Capacitors CL0 and CL1 (see measuring circuit 1) are connected between this pin and V_{SS}.	—	—

1.3.4 Termination of Unused Pins

Table 1-3 shows methods of terminating the unused pins.

Table 1-4 Termination of Unused Pins

Pin	Recommended pin termination
AV_{DD}	V_{SS}
AV_{SS}	V_{SS}
V_{REF}	V_{SS}
AIN0, AIN1, AIN2, AIN3	Open
V_{L1}, V_{L2}, V_{L3}	Open
C1, C2	Open
RESET_N	Open
TEST0	Open
TEST1_N	Open
P00 to P03	V_{DD} or V_{SS}
P10 to P11	V_{DD}
P20 to P22	Open
P30 to P35	Open
P40 to P47	Open
P50 to P53	Open
P60 to P67	Open
COM0 to 3	Open
SEG0 to 47	Open

Note:
– It is recommended to set the unused input ports and input/output ports to the inputs with pull-down resistors/pull-up resistors or the output mode since the supply current may become excessively large if the pins are left open in the high impedance input setting.

1.3.5 The main difference points of ML610Q418 and ML610Q418C

Table 1-5 The main difference points of ML610Q418 and ML610Q418C.

Function	ML610Q418	ML610Q418C
LCD SEG	SEG47 to SEG0	SEG39 to SEG0
PORT6	—	P60～P67

Chapter 2

CPU and Memory Space

2. CPU and Memory Space

2.1 Overview

This LSI includes 8-bit CPU nX-U8/100 and the memory model is "LARGE model" .
For details of the CPU nX-U8/100, see "nX-U8/100 Core Instruction Manual".

2.2 Program Memory Space

The program memory space is used to store program codes, table data (ROM window), or vector tables.
The program codes have a length of 16 bits and are specified by a 16-bit program counter (PC).
The ROM window area data has a length of 8 bits and can be used as table data.
The vector table, which has 16-bit long data, can be used as reset vectors, hardware interrupt vectors, and software interrupt vectors.
The program memory space consists of 1 segment and has 128-Kbyte (64-Kword) capacity.
Figure 2-1 shows the configuration of the program memory space.

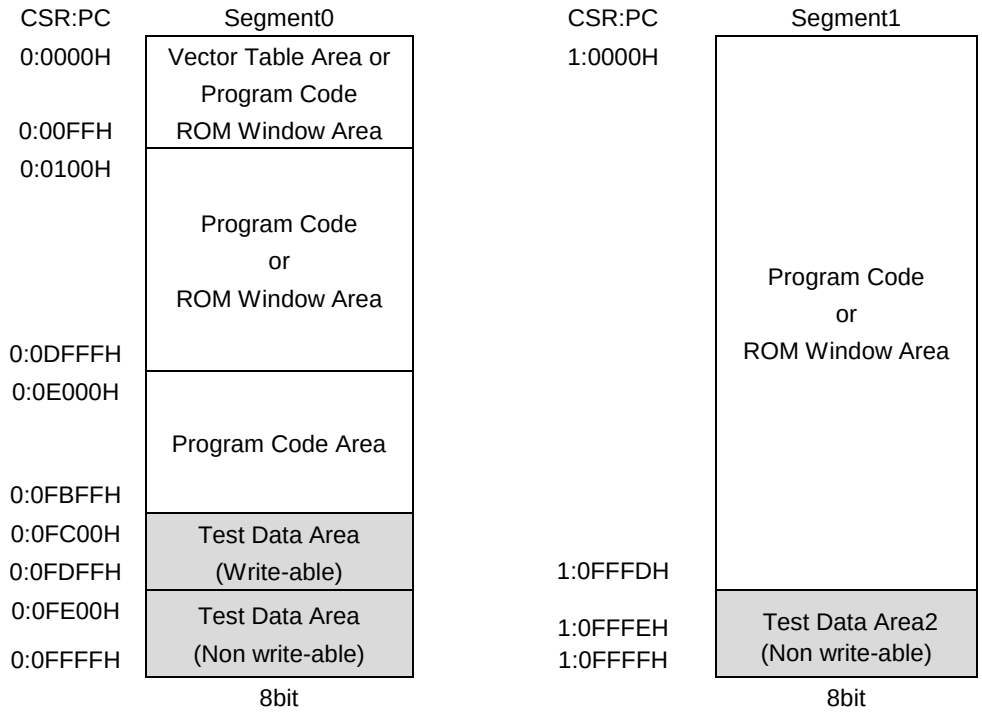


Figure 2-1 Configuration of Program Memory Space

Notes:
Since test program data is stored in the 1024Byte (512Word) test data area (0:FC00H to 0:FFFFH) of Segment 0 and in the 2Byte of test data area (1:0FFFEH～1:0FFFFH) of segment 1, this area cannot be used as a program code area.
The address "0: FC00H to 0:FDFFH" in the test area is write-able and erase-able. Fill the area with "0FFH". If data in the area is uncertain or other data (i.e. not 0FFH), operating with the code cannot be guaranteed.
Set "0FFH" data (BRK instruction) in the unused area of the program memory space.
The address in the specified range at DTU8 "Save the program" execution should exclude "01FFEH and 01FFFH".

2.3 Data Memory Space

The data memory space of this LSI consists of the ROM window area, 4KByte RAM area and SFR area of the Segment 0 , the ROM reference area of the Segment 1,the 4KByte Data Flash Area of the Segment 2 and the ROM reference area of the Segment 8 and 9.
The data memory stores 8-bit data and is specified by 20 bits consisting of higher 4 bits as DSR and lower 16 bits as addressing specified by each instruction.
Figure 2-2 shows the configuration of the data memory space.

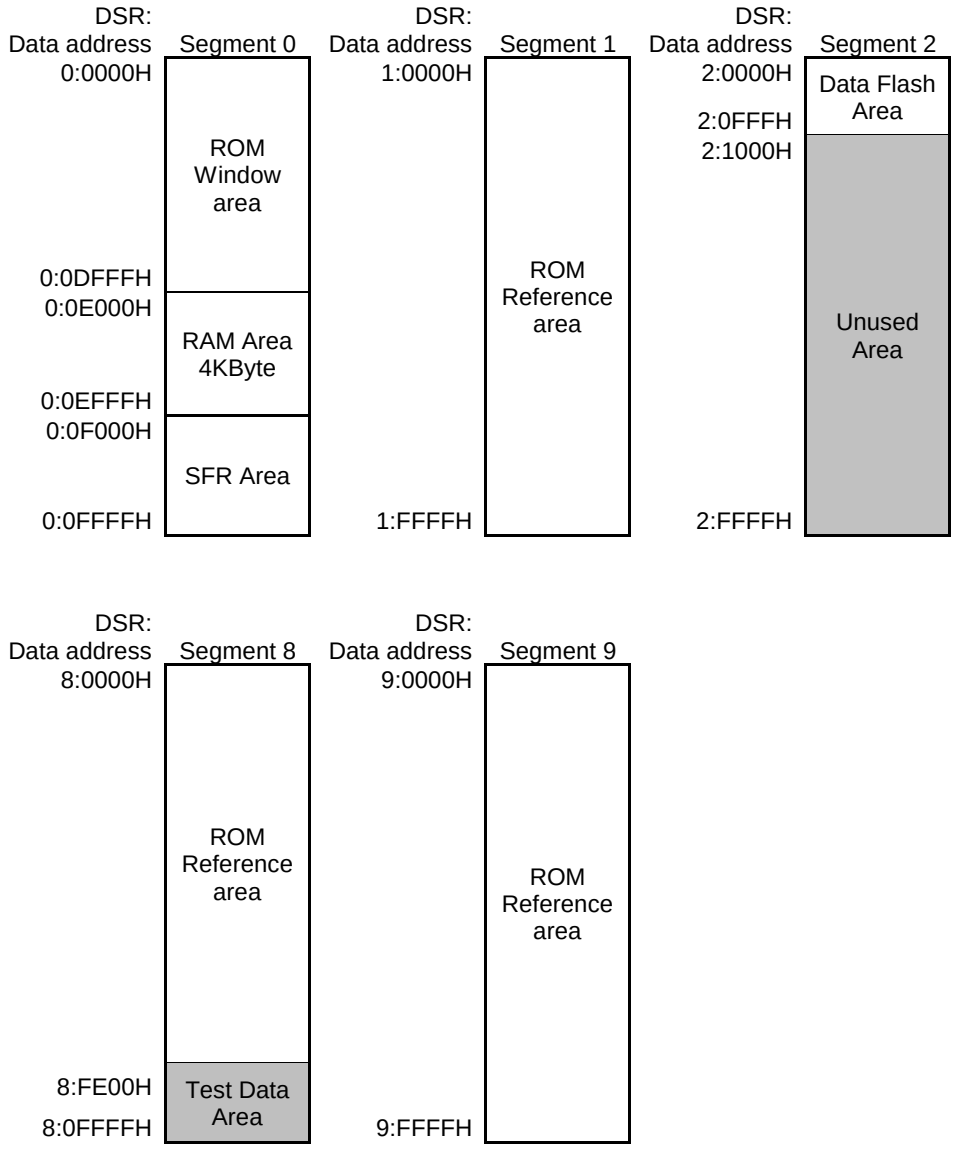


Figure 2-2 Configuration of Data Memory Space

Notes:
The contents of the 4-Kbyte RAM area are undefined at system reset. Initialize this area by software.
The contents of Segment 0 of the program memory space is read from the ROM reference area of the Segment 8.
The contents of Segment 1 of the program memory space is read from the ROM reference area of the Segment 9.
The Data Flash area consists of Flash memory and can be rewritten by the user program. For detail the Data Flash area, see Chapter 28, "Flash Memory Programming".

2.4 Instruction Length

The length of a instruction is 16 bits.

2.5 Data Type

The data types supported include byte (8 bits) and word (16 bits).

2.6 Description of Registers

2.6.1 List of Registers

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F000H	Data segment register	DSR	—	R/W	8	00H

2.6.2 Data Segment Register (DSR)

Address: 0F000H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
DSR	—	—	—	—	DSR3	DSR2	DSR1	DSR0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

DSR is a special function register (SFR) to retain a data segment. For details of DSR, see "nX-U8/100 Core Instruction Manual".

[Description of Bits]

- **DSR3-DSR0** (bits 3-0)

DSR3	DSR2	DSR1	DSR0	Description
0	0	0	0	Data segment 0 (initial value)
0	0	0	1	Data segment 1
0	0	1	0	Data segment 2
0	0	1	1	Prohibit
0	1	0	0	
0	1	0	1	
0	1	1	0	
0	1	1	1	
1	0	0	0	Data segment 8
1	0	0	1	Data segment 9
1	0	1	0	Prohibit
1	0	1	1	
1	1	0	0	
1	1	0	1	
1	1	1	0	
1	1	1	1	

Chapter 3

Reset Function

3. Reset Function

3.1 Overview

This LSI has the five reset functions shown below. If any of the five reset conditions is satisfied, this LSI enters system reset mode.

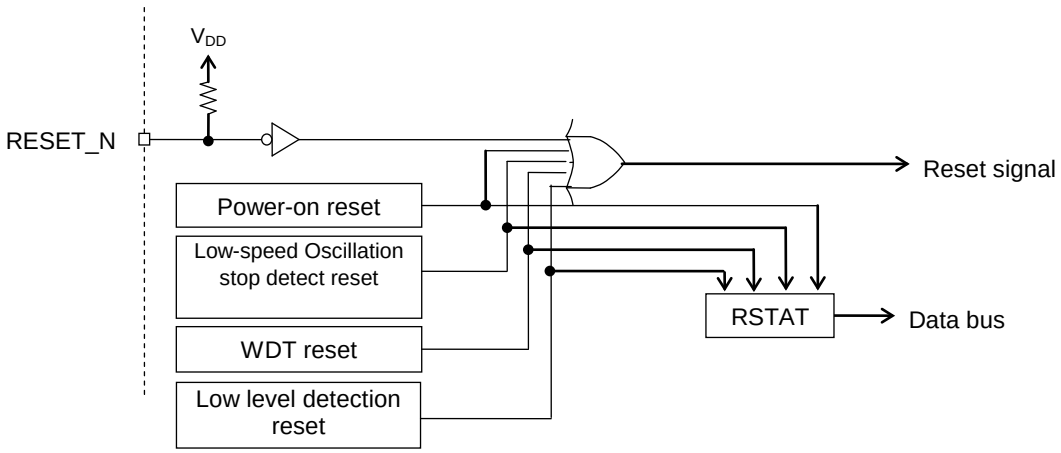
- Reset by the RESET_N pin
- Reset by power-on detection
- Reset by low level detection
- Reset by the low-speed oscillation stop detection
- Reset by the 2nd watchdog timer (WDT) overflow
- Software reset by execution of the BRK instruction

3.1.1 Features

- The RESER_N pin has an internal pull-up resistor
- The reset by low level detection occurs, when the voltage of V_{DD} is less than 1.1V (max.) or 1.8V(max.). The voltage level, 1.1V(max.) or 1.8V(max.), can be selected as the voltage which releases the reset mode by the code-option. For the details of the code-option, refer to Chapter 28, "Code Option"
- The low-speed oscillation stop detection time is 3 ms (typ.)
- 250 ms, 1 sec, 4 sec, or 16 sec can be selected as the watchdog timer (WDT) overflow period
- Built-in reset status register (RSTAT) indicating the reset generation causes
- Only the CPU is reset by the BRK instruction (neither the RAM area nor the SFR area are reset).

3.1.2 Configuration

Figure 3-1 shows the configuration of the reset generation circuit.



RSTAT: Reset status register

Figure 3-1 Configuration of Reset Generation Circuit

3.1.3 List of Pin

Pin name	I/O	Description
RESET_N	I	Reset input pin

3.2 Description of Registers

3.2.1 List of Registers

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F001H	Reset status register	RSTAT	—	R/W	8	—

3.2.2 Reset Status Register (RSTAT)

Address: 0F001H
Access: R/W
Access size: 8 bits
Initial value: Undefined

	7	6	5	4	3	2	1	0
RSTAT	—	RSTR	—	LLDR	—	WDTR	XSTR	POR
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	x	0	x	0	0	x	1

RSTAT is a special function register (SFR) that indicates the causes by which the reset is generated.
At the occurrence of reset, the contents of RSTAT are not initialized, while the bit indicating the cause of the reset is set to "1". When checking the reset cause using this function, perform write operation to RSTAT in advance and initialize the contents of RSTAT to "00H".

[Description of Bits]

- **POR** (bit 0)
  The POR bit is a flag that indicates that the power-on reset is generated. This bit is set to "1" when powered on.

POR	Description
0	Power-on reset not generated
1	Power-on reset generated

- **XSTR** (bit 1)
  The XSTR bit is a flag that indicates the generation of low-speed oscillation stop detect reset. When low-speed oscillation stops for the period specified by the low-speed oscillation stop detection time (TSTOP) or more, this bit is set to "1".

XSTR	Description
0	Low-speed oscillation stop detect reset not occurred
1	Low-speed oscillation stop detect reset occurred

- **WDTR** (bit 2)
  The WSDTR is a flag that indicates that the watchdog timer reset is generated. This bit is set to "1" when the reset by overflow of the watchdog timer is generated.

WDTR	Description
0	Watchdog timer reset not occurred
1	Watchdog timer reset occurred

- **LLDR** (bit 4)
  The LLDR is a flag that indicates that the low level detection reset is generated. This bit is set to "1" when the voltage of V_{DD} is less than 1.8V(max.) or 1.1V (max.). The voltage level can be selected by the code option. For the details of the code option, refer to Chapter 28, "Code-Option"

LLDR	Description
0	Low level detection reset not occurred
1	Low level detection reset occurred

- **RSTR** (bit 6)
  The RSTR is a flag that indicates that the occurrence of reset by the RESET_N pin. This bit is set to "1" when the reset by the RESET_N pin is generated.

RSTR	Description
0	RESET_N pin reset not occurred
1	RESET_N pin reset occurred

3.3 Description of Operation

3.3.1 Operation of System Reset Mode

System reset has the highest priority among all the processing and any other processing being executed up to then is cancelled.
The system reset mode is set by any of the following causes.

- Reset by the RESET_N pin
- Reset by power-on detection
- Reset by low level detection
- Reset by low-speed oscillation stop detection
- Reset by watchdog timer (WDT) overflow
- Software reset by the BRK instruction (only the CPU is reset)

In system reset mode, the following processing is performed.

(1) The power circuit is initialized, but not initialized by the reset by the BRK instruction execution. For the details of the power circuit, refer to Chapter 27, "Power Circuit".
(2) All the special function registers (SFRs) whose initial value is not undefined are initialized. However, the initialization is not performed by software reset due to execution of the BRK instruction. See Appendix A "Registers" for the initial values of the SFRs.
(3) CPU is initialized.
   - All the registers in CPU are initialized.
   - The contents of addresses 0000H and 0001H in the program memory are set to the stack pointer (SP).
   - The contents of addresses 0002H and 0003H in the program memory are set to the program counter (PC). However, when the interrupt level (ELEEVL) of the program status word (PSW) at reset by the BRK instruction is 1 or lower, the contents of addresses 0004H and 0005H of the program memory are set in the program counter (PC). For the BRK instruction, see "nX-U8/100 Core Instruction Manual".

Note:
In system reset mode, the contents of data memory and those of any SFR whose initial value is undefined are not initialized and are undefined. Initialize them by software.
In system reset mode by the BRK instruction, no special function register (SFR) that has a fixed initial value is initialized either. Therefore initialize such an SFR by software.

Chapter 4

MCU Control Function

4. MCU Control Function

4.1 Overview

The operating states of this LSI are classified into the following 4 modes including system reset mode:

System reset mode
Program run mode
HALT mode
STOP mode

For system reset mode, see Chapter 3, "Reset Function".

This LSI has a block control function, which power downs the circuits of unused peripherals (reset registers and stop clock supplies) to make even more reducing the current consumption.

4.1.1 Features

- HALT mode, where the CPU stops operating and only the peripheral circuit is operating
- STOP mode, where both low-speed oscillation and high-speed oscillation stop
- Stop code acceptor function, which controls transition to STOP mode
- Block control function, which power downs the circuits of unused peripherals (reset registers and stop clock supplies).

4.1.2 Configuration

Figure 4-1 shows an operating state transition diagram.

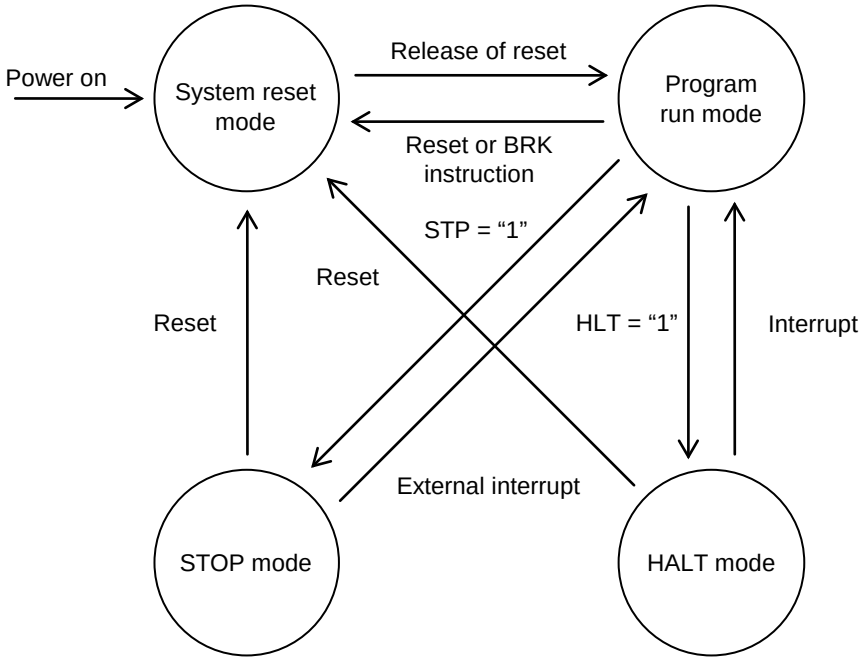


Figure 4-1 Operating State Transition Diagram

4.2 Description of Registers

4.2.1 List of Registers

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F008H	Stop code acceptor	STPACP	—	W	8	—
0F009H	Standby control register	SBYCON	—	W	8	00H
0F028H	Block control register 0	BLKCON0	—	R/W	8	00H
0F029H	Block control register 1	BLKCON1	—	R/W	8	00H
0F02AH	Block control register 2	BLKCON2	—	R/W	8	00H
0F02BH	Block control register 3	BLKCON3	—	R/W	8	00H
0F02CH	Block control register 4	BLKCON4	—	R/W	8	00H

4.2.2 Stop Code Acceptor (STPACP)

Address: 0F008H
Access: W
Access size: 8 bits
Initial value: — (Undefined)

	7	6	5	4	3	2	1	0
STPACP	—	—	—	—	—	—	—	—
W	W	W	W	W	W	W	W	W
Initial value	-	-	-	-	-	-	-	-

STPACP is a write-only special function register (SFR) that is used for setting a STOP mode.
When STPACP is read, "00H" is read.
When data is written to STPACP in the order of "5nH"(n: an arbitrary value) and "0AnH"(n: an arbitrary value), the stop code acceptor is enabled. When the STP bit of the standby control register (SBYCON) is set to "1" in this state, the mode is changed to the STOP mode. When the STOP mode is set, the STOP code acceptor is disabled.
When another instruction is executed between the instruction that writes "5nH" to STPACP and the instruction that writes "0AnH", the stop code acceptor is enabled after "0AnH" is written. However, if data other than "0AnH" is written to STPACP after "5nH" is written, the "5nH" write processing becomes invalid so that data must be written again starting from "5nH".
During a system reset, the stop code acceptor is disabled.

Note:
The STOP code acceptor can not be enabled on the condition of that both any interrupt enable flag and the corresponding interrupt request flag are "1"(An interrupt request occurrence with resetting MIE flag will have the condition).

4.2.3 Standby Control Register (SBYCON)

Address: 0F009H
Access: W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
SBYCON	—	—	—	—	—	—	STP	HLT
W	W	W	W	W	W	W	W	W
Initial value	0	0	0	0	0	0	0	0

SBYCON is a special function register (SFR) to control operating mode of MCU.

[Description of Bits]

- **STP** (bit 1)
  The STP bit is used for setting the STOP mode. When the STP bit is set to "1" with the stop code adapter enabled by using STPACP, the mode is changed to the STOP mode. When the NMI interrupt request or the P00–P03 interrupt request enabled by the interrupt enable register 1 (IE1) is issued, the STP bit is set to "0" and the LSI returns to the program run mode.

- **HLT** (bit 0)
  The HALT bit is used for setting a HALT mode. When the HALT bit is set to "1", the mode is changed to the HALT mode. When the NMI interrupt request, WDT interrupt request, or enabled (the interrupt enable flag is "1") interrupt request is issued, the HALT bit is set to "1" and the mode is returned to program run mode.

STP	HLT	Description
0	0	Program run mode (initial value)
0	1	HALT mode
1	0	STOP mode
1	1	Prohibited

Note:
The mode can not be changed to HALT mode or STOP mode on the condition of that both any interrupt enable flag and the corresponding interrupt request flag are "1"(An interrupt request occurrence with resetting MIE flag will have the condition).
When a maskable interrupt source (interrupt with enable bit) occurs while the MIE flag of the program status word (PSW) in the nX-U8/100 core is "0", the STOP mode and the HALT mode are simply released and interrupt processing is not performed. Refer to the "nX-U8/100 Core Instruction Manual" for details of PSW.

4.2.4 Block Control Register 0(BLKCON0)

Address: 0F028H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
BLKCON0	—	—	—	—	DTM3	DTM2	DTM1	DTM0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

BLKCON0 is a special function register (SFR) to make even more reducing current consumption by turning unused peripherals off.

[Description of Bits]

- **DTM3** (bit 3)
  The DTM3 bit is used to control Timer3 operation. When the DTM3 bit is set to "1", the circuits related to Timer 3 are reset and turned off.

DTM3	Description
0	Enable operating Timer 3 (initial value)
1	Disable operating Timer 3

- **DTM2** (bit 2)
  The DTM2 bit is used to control Timer2 operation. When the DTM2 bit is set to "1", the circuits related to Timer 2 are reset and turned off.

DTM2	Description
0	Enable operating Timer 2 (initial value)
1	Disable operating Timer 2

- **DTM1** (bit 1)
  The DTM1 bit is used to control Timer1 operation. When the DTM1 bit is set to "1", the circuits related to Timer 1 are reset and turned off.

DTM1	Description
0	Enable operating Timer 1 (initial value)
1	Disable operating Timer 1

- **DTM0** (bit 0)
  The DTM0 bit is used to control Timer3 operation. When the DTM0 bit is set to "1", the circuits related to Timer 3 are reset and turned off.

DTM0	Description
0	Enable operating Timer 3 (initial value)
1	Disable operating Timer 3

Note:
When certain bits of block control registers are set to "1", corresponding peripherals are reset (all registers are reset) and operating clocks for the peripherals stop. Writing to every SFR (special function register) in the corresponding peripherals is not valid while the bits of block control registers are set to "1" and returns the initial value for read. Ensure the bits are reset to "0" before using the peripherals to enable the operation.

See Chapter 9, "Timers" for detail about operation of Timer 0, Timer 1, Timer 2 and Timer 3.

4.2.5 Block Control Register 1(BLKCON1)

Address: 0F029H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
BLKCON1	—	DCAPR	—	—	—	—	—	DPW0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

BLKCON1 is a special function register (SFR) to make even more reducing current consumption by turning unused peripherals off.

[Description of Bits]

- **DCAPR** (bit 6)
  The DCAPR bit is used to control Capture operation. When the DCAPR bit is set to "1", the circuits related to Capture are reset and turned off.

DCAPR	Description
0	Enable operating Capture (initial value)
1	Disable operating Capture

- **DPW0** (bit 0)
  The DPW0 bit is used to control PWM0 operation. When the DPW0 bit is set to "1", the circuits related to PWM0 are reset and turned off.

DPW0	Description
0	Enable operating PWM0 (initial value)
1	Disable operating PWM0

Note:
When certain bits of block control registers are set to "1", corresponding peripherals are reset (all registers are reset) and operating clocks for the peripherals stop. Writing to every SFR (special function register) in the corresponding peripherals is not valid while the bits of block control registers are set to "1" and returns the initial value for read. Ensure the bits are reset to "0" before using the peripherals to enable the operation.

See Chapter 8, "Capture" for detail about operation of Capture.
See Chapter 10, "PWM" for detail about operation of PWM.

4.2.6 Block Control Register 2(BLKCON2)

Address: 0F02AH
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
BLKCON2	DI2C0	—	—	—	DUA1	DUA0	DSIO1	DSIO0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

BLKCON2 is a special function register (SFR) to make even more reducing current consumption by turning unused peripherals off.

[Description of Bits]

- **DI2C0** (bit 7)
  The DI2C0 bit is used to control I2C bus interface operation. When the DI2C0 bit is set to "1", the circuits related to I2C bus interface are reset and turned off.

DI2C0	Description
0	Enable operating I2C (initial value)
1	Disable operating I2C

- **DUA1** (bit 3)
  The DUA1 bit is used to control UART1 operation. When the DUA1 bit is set to "1", the circuits related to UART1 are reset and turned off.

DUA1	Description
0	Enable operating UART1 (initial value)
1	Disable operating UART1

- **DUA0** (bit 2)
  The DUA0 bit is used to control UART0 operation. When the DUA0 bit is set to "1", the circuits related to UART0 are reset and turned off.

DUA0	Description
0	Enable operating UART0 (initial value)
1	Disable operating UART0

- **DSIO1** (bit 1)
  The DSIO1 bit is used to control SSIO1 operation. When the DSIO1 bit is set to "1", the circuits related to SSIO1are reset and turned off.

DSIO1	Description
0	Enable operating SSIO1 (initial value)
1	Disable operating SSIO1

- **DSIO0** (bit 0)
  The DSIO0 bit is used to control SSIO0 operation. When the DSIO0 bit is set to "1", the circuits related to SSIO0 are reset and turned off.

DSIO0	Description
0	Enable operating SSIO0 (initial value)
1	Disable operating SSIO0

Note:
When certain bits of block control registers are set to "1", corresponding peripherals are reset (all registers are reset) and operating clocks for the peripherals stop. Writing to every SFR (special function register) in the corresponding peripherals is not valid while the bits of block control registers are set to "1" and returns the initial value for read. Ensure the bits are reset to "0" before using the peripherals to enable the operation.

See Chapter 14, "I2C Bus Interface" for detail about operation of I2C Bus Interface.
See Chapter 13, "UART" for detail about operation of UART.
See Chapter 12, "Synchronous Serial Port" for detail about operation of SSIO.

4.2.7 Block Control Register 3(BLKCON3)

Address: 0F02BH
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
BLKCON3	—	—	—	—	—	—	—	DMD0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

BLKCON3 is a special function register (SFR) to make even more reducing current consumption by turning unused peripherals off.

[Description of Bits]

- **DMD0** (bit 0)
  The DMD0 bit is used to control Melody/Buzzer operation. When the DMD0 bit is set to "1", the circuits related to Melody/Buzzer are reset and turned off.

DMD0	Description
0	Enable operating Melody/Buzzer (initial value)
1	Disable operating Melody/Buzzer

Note:
When certain bits of block control registers are set to "1", corresponding peripherals are reset (all registers are reset) and operating clocks for the peripherals stop. Writing to every SFR (special function register) in the corresponding peripherals is not valid while the bits of block control registers are set to "1" and returns the initial value for read. Ensure the bits are reset to "0" before using the peripherals to enable the operation.

See Chapter 22, "Melody Driver" for detail about operation of Melody/Buzzer.

4.2.8 Block Control Register 4(BLKCON4)

Address: 0F02CH
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
BLKCON4	—	DLCD	DBLD	DXTSP	—	—	DRAD	DSAD
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

BLKCON4 is a special function register (SFR) to make even more reducing current consumption by turning unused peripherals off.

[Description of Bits]

- **DLCD** (bit 6)
  The DLCD bit is used to control LCD driver operation. When the DLCD bit is set to "1", the circuits related to LCD driver are reset and turned off.

DLCD	Description
0	Enable operating LCD driver (initial value)
1	Disable operating LCD driver

- **DBLD** (bit 5)
  The DBLD bit is used to control BLD (Battery Level Detector) operation. When the DBLD bit is set to "1", the circuits related to BLD are reset and turned off.

DBLD	Description
0	Enable operating BLD (initial value)
1	Disable operating BLD driver

- **DXTSP** (bit 4)
  The DXTSP bit is used to control 32kHz oscillation stop detect operation. Only during HALT mode, When the DXTSP bit is set to "1", the circuits related to 32kHz oscillation stop detect are reset and turned off. When the operating mode is not in HALT, the 32kHz oscillation stop detect is always working regardless the condition of this bit.

DXTSP	Description
0	Enable operating 32kHz oscillation stop detect (initial value)
1	Disable operating 32kHz oscillation stop detect in HALT mode

- **DRAD** (bit 1)
  The DRAD bit is used to control RC type A/D converter operation. When the DRAD bit is set to "1", the circuits related to RC type A/D converter are reset and turned off.

DRAD	Description
0	Enable operating RC type A/D converter (initial value)
1	Disable operating RC type A/D converter

- **DSAD** (bit 0)
  The DSAD bit is used to control SA type A/D converter operation. When the DSAD bit is set to "1", the circuits related to SA type A/D converter are reset and turned off.

DSAD	Description
0	Enable operating SA type A/D converter (initial value)
1	Disable operating SA type A/D converter

Note:
When certain bits of block control registers are set to "1", corresponding peripherals are reset (all registers are reset) and operating clocks for the peripherals stop. Writing to every SFR (special function register) in the corresponding peripherals is not valid while the bits of block control registers are set to "1" and returns the initial value for read. Ensure the bits are reset to "0" before using the peripherals to enable the operation.

See Chapter 25, "LCD Driver" for detail about operation of LCD driver.
See Chapter 26, "Battery Level Detector" for detail about operation of BLD.
See Chapter 3, "Reset Function" for detail about operation of 32kHz oscillation stop detector.
See Chapter 23, "RC Oscillation Type A/D Converter" for detail about operation of RC oscillation type A/D converter.
See Chapter 24, "Successive Approximation" for detail about operation of SA type A/D converter.

4.3 Description of Operation

4.3.1 Program Run Mode

The program run mode is the state where the CPU executes instructions sequentially.
At power-on reset, RESET_N pin reset, low-speed oscillation stop detect reset, or WDT overflow reset, the CPU executes instructions from the addresses that are set in addresses 0002H and 0003H of program memory (ROM) after the system reset mode is released.
At reset by the BRK instruction, the CPU executes instructions from the addresses that are set in the addresses 0004H and 0005H of the program memory after the system reset mode is released. However, when the value of the interrupt level bit (ELEVEL) of the program status word (PSW) is 02H or higher at execution of the BRK instruction (after the occurrence of the WDT interrupt or NMI interrupt), the CPU executes instructions from the addresses that are set in the addresses 0002H and 0003H.
For details of the BRK instruction and PSW, see the "nX-U8/100 Core Instruction Manual" and for the reset function, see Chapter 3, "Reset Function".

4.3.2 HALT Mode

The HALT mode is the state where the CPU interrupts execution of instructions and only the peripheral circuits are running.
When the HLT bit of the standby control register (SBYCON) is set to "1", the HALT mode is set.
When a NMI interrupt request, a WDT interrupt request, or an interrupt request enabled by an interrupt enable register (IE1–IE7) is issued, the HLT bit is set to "0" on the falling edge of the next system clock (SYSCLK) and the HALT mode is returned to the program run mode released.
Figure 4-2 shows the operation waveforms in HALT mode.

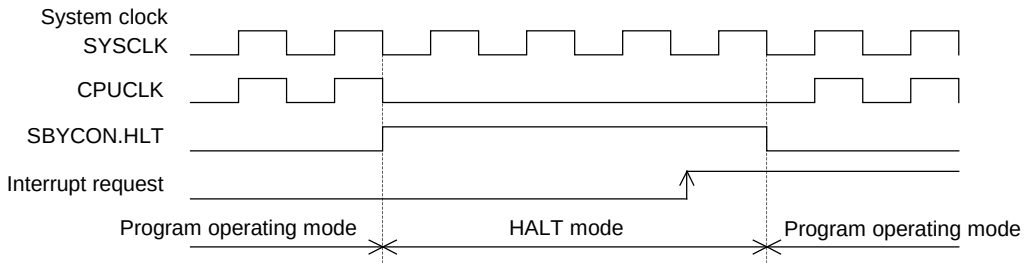


Figure 4-2 Operation Waveforms in HALT Mode

Note:
Since up to two instructions are executed during the period between HALT mode release and a transition to interrupt processing, place two NOP instructions next to the instruction that sets the HLT bit to "1".

4.3.3 STOP Mode

The STOP mode is the state where low-speed oscillation and high-speed oscillation stop and the CPU and peripheral circuits stop the operation.
When the stop code acceptor is enabled by writing "5nH"(n: an arbitrary value) and "0AnH"(n: an arbitrary value) to the stop code acceptor (STPACP) sequentially and the STP bit of the standby control register (SBYCON) is set to "1", the STOP mode is entered. When the STOP mode is set, the stop code acceptor is disabled.
When a NMI interrupt request or an interrupt-enabled (the interrupt enable flag is "1") P00 to P003 interrupt request is issued, the STP bit is set to "0", the STOP mode is released, and the mode is returned to the program run mode.

4.3.3.1 STOP Mode When CPU Operates with Low-Speed Clock

When the stop code acceptor is in the enabled state and the STP bit of SBYCON is set to "1", the STOP mode is entered, stopping low-speed oscillation and high-speed oscillation.
When the NMI interrupt request or the interrupt-enabled (the interrupt enable flag is "1") P00 to P03 interrupt request is issued, the STP bit is set to "0" and low-speed oscillation restarts. If the high-speed clock was oscillating before the STOP mode is entered, the high-speed oscillation restarts. When the high-speed clock was not oscillating before the STOP mode is entered, high-speed oscillation does not start.
When an interrupt request occurs, the STOP mode is released after the elapse of the low-speed oscillation start time (TXTL) and the low-speed clock (LSCLK) oscillation settling time (8192-pulse count), the mode is returned to the program mode, and the low-speed clock (LSCLK) restarts supply to the peripheral circuits. If the high-speed clock already started oscillation at this time, the high-speed clocks (OSCLK and HSCLK) also restart supply to the peripheral circuits.
For the low-speed oscillation start time (TXTL), see the "Electrical Characteristics" Section in Appendix C.
Figure 4-3 shows the operation waveforms in STOP mode when CPU operates with the low-speed clock.

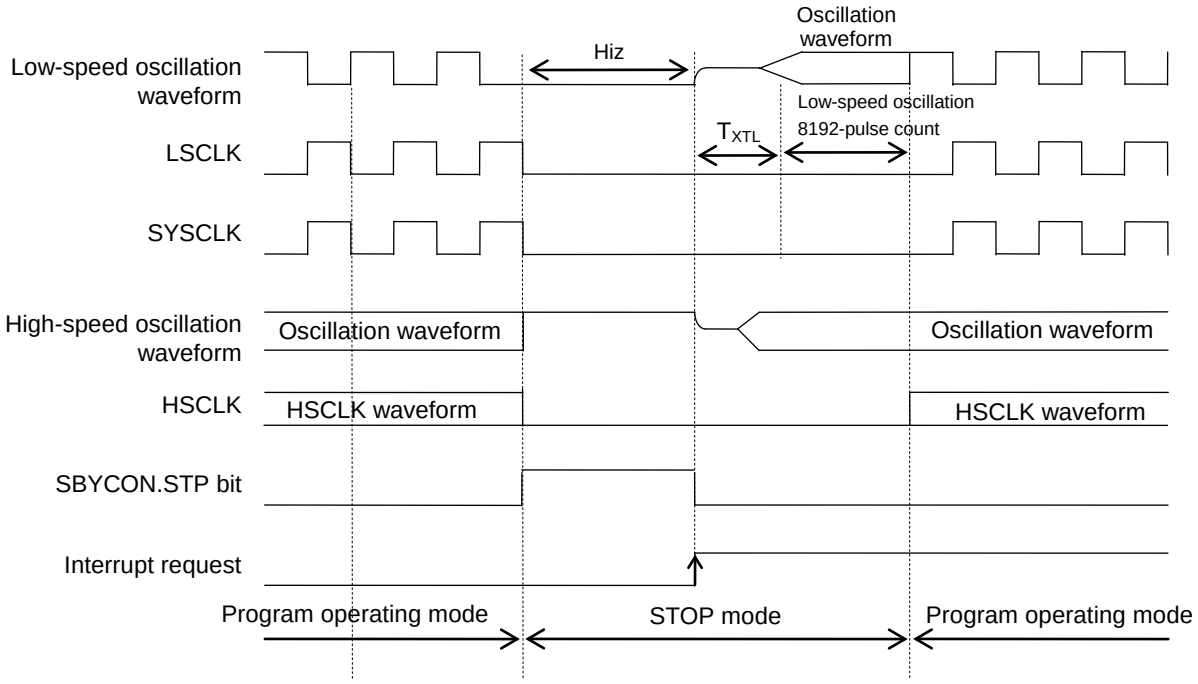


Figure 4-3 Operation Waveforms in STOP Mode When CPU Operates with Low-Speed Clock

4.3.3.2 STOP Mode When CPU Operates with High-Speed Clock

When the CPU is operating with a high-speed clock and the STP bit of SBYCON is set to "1" with the stop code acceptor enabled, the STOP mode is entered and high-speed oscillation and low-speed oscillation stop.
When the NMI interrupt request or the interrupt-enabled (the interrupt enable flag is "1") P00 to P03 interrupt request is issued, the STP bit is set to "0" and the low-speed and high-speed oscillation restart.
When an interrupt request is issued, the STOP mode is released after the elapse of the high-speed oscillation start time (TXTH/TRC) and the high-speed clock (OSCLK) oscillation stabilization time (8192-pulse count), the mode is returned to the program run mode, and the high-speed clocks (OSCLK and HSCLK) restart supply to the peripheral circuits.
The low-speed clock (LSCLK) restarts supply to the peripheral circuits after the elapse of the low-speed oscillation start time (TXTL) and low-speed clock (LSCLK) oscillation settling time (8192 count).
For the high-speed oscillation start time (TXTH) and low-speed oscillation start time (TXTL), see the "Electrical Characteristics" Section in Appendix C.
Figure 4-4 shows the operation waveforms in STOP mode when CPU operates with the high-speed clock.

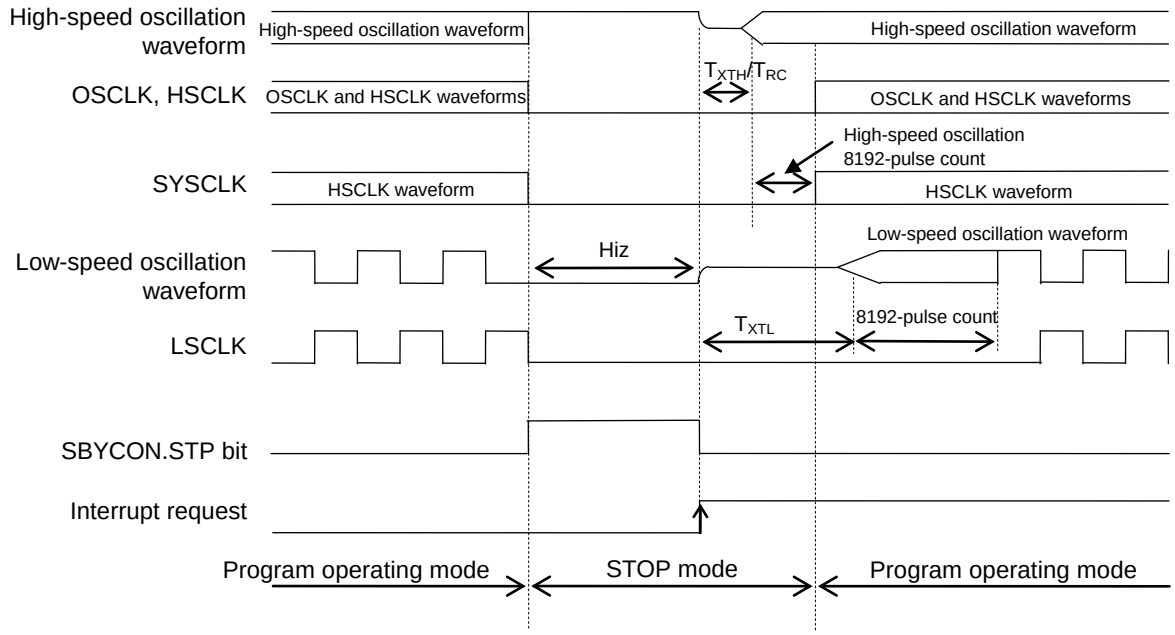


Figure 4-4 Operation Waveforms in STOP Mode When CPU Operates with High-Speed Clock

Note:
The STOP mode is entered two cycles after the instruction that sets the STP bit to "1" and up to two instructions are executed during the period between STOP mode release and a transition to interrupt processing. Therefore, place two NOP instructions next to the instruction that set the STP bit to "1".

4.3.3.3 Note on Return Operation from STOP/HALT Mode

The operation of returning from the STOP mode and HALT mode varies according to the interrupt level (ELEVEL) of the program status word (PSW), master interrupt enable flag (MIE), the contents of the interrupt enable register (IE0 to IE3), and whether the interrupt is a non-maskable interrupt or a maskable interrupt.
For details of PSW and the IE and IRQ registers, see "nX-U8/100 Core Instruction Manual" and Chapter 5, "Interrupt", respectively.

Table 4-1 and Table 4-2 show the return operations from STOP/HALT mode.

Table 4-1 Return Operation from STOP/HALT Mode (Non-Maskable Interrupt)

ELEVEL	MIE	IEn.m	IRQn.m	Return operation from STOP/HALT mode
*	*	–	0	Not returned from STOP/HALT mode.
3	*	–	1	After the mode is returned from STOP/HALT mode, the program operation restarts from the instruction following the instruction that sets the STP/HLT bit to "1". The program operation does not go to the interrupt routine.
0, 1, 2	*	–	1	After the mode is returned from the STOP/HALT mode, program operation restarts from the instruction following the instruction that sets the STP/HLT bit to "1", then goes to the interrupt routine.

Table 4-2 Return Operation from STOP/HALT Mode (Maskable Interrupt)

ELEVEL	MIE	IEn.m	IRQn.m	Return operation from STOP/HALT mode
*	*	*	0	Not returned from STOP/HALT mode.
*	*	0	1	
*	0	1	1	After the mode is returned from STOP/HALT mode, the program operation restarts from the instruction following the instruction that sets the STP/HLT bit to "1". The program operation does not go to the interrupt routine.
2,3	1	1	1	
0, 1	1	1	1	After the mode is returned from the STOP/HALT mode, program operation restarts from the instruction following the instruction that sets the STP/HLT bit to "1", then goes to the interrupt routine.

Notes:
- If the ELEVEL bit is 0H, it indicates that the CPU is performing neither non-maskable interrupt processing nor maskable interrupt processing nor software interrupt processing.
- If the ELEVEL bit is 1H, it indicates that the CPU is performing maskable interrupt processing or software interrupt processing. (ELEVEL is set during interrupt transition cycle.)
- If the ELEVEL bit is 2H, it indicates that the CPU is performing non-maskable interrupt processing. (ELEVEL is set during interrupt transition cycle.)
- If the ELEVEL bit is 3H, it indicates that the CPU is performing interrupt processing specific to the emulator. This setting is not allowed in normal applications.

4.3.4 Block Control Function

This LSI has a block control function, which resets and completely turns operating circuits of unused peripherals off to make even more reducing current consumption.

When certain bits of block control registers are set to "1", corresponding peripherals are reset (all registers are reset) and operating clocks for the peripherals stop. Writing to every SFR (special function register) in the corresponding peripherals is not valid while the bits of block control registers are set to "1" and returns the initial value for read. Ensure the bits are reset to "0" before using the peripherals to enable the operation.

BLKCON0 register controls(disables/enables) operation of Timer 0, Timer 1, Timer 2 and Timer 3.
BLKCON1 register controls(disables/enables) operation of Capture, 1kHz Timer and PWM.
BLKCON2 register controls(disables/enables) operation of I2C, UART and SSIO.
BLKCON3 register controls(disables/enables) operation of Melody/Buzzer.
BLKCON4 register controls(disables/enables) operation of LCD driver, Battery Level Detector, 32kHz oscillation stop detector, RC type A/D converter and SAR type A/D converter.

Note:
DXTSP bit (bit 4) of BLKCON4 register disables the operation of 32kHz oscillation stop detector in HALT mode only.
See the each chapter for detail about the operation of each peripheral and relevant notes.

Chapter 5

Interrupts (INTs)

5. Interrupts (INTs)

5.1 Overview

This LSI has 23 interrupt sources (External interrupts: 5 sources, Internal interrupts: 18 sources) and a software interrupt (SWI).
For details of each interrupt, see the following chapters:

Chapter 7, "Time Base Counter"
Chapter 9, "Timer"
Chapter 10, "PWM"
Chapter 11, "Watchdog Timer"
Chapter 12, "Synchronous Serial Port"
Chapter 13, "UART"
Chapter 14, "I^2C Bus Interface"
Chapter 15, "Port0"
Chapter 20, "Port5"
Chapter 22, "Melody Driver"
Chapter 23, "RC Oscillation Type A/D Converter"
Chapter 24, "Successive Approximation Type A/D Converter"

5.1.1 Features

- 1 non-maskable interrupt sources (Internal source: 1)
- 22 maskable interrupt sources (Internal sources: 17, External sources: 5)
- Software interrupt (SWI): 64 sources max.
- External interrupts allow edge selection and sampling selection.

5.2 Description of Registers

5.2.1 List of Registers

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F011H	Interrupt enable register 1	IE1	—	R/W	8	00H
0F012H	Interrupt enable register 2	IE2	—	R/W	8	00H
0F013H	Interrupt enable register 3	IE3	—	R/W	8	00H
0F014H	Interrupt enable register 4	IE4	—	R/W	8	00H
0F015H	Interrupt enable register 5	IE5	—	R/W	8	00H
0F016H	Interrupt enable register 6	IE6	—	R/W	8	00H
0F017H	Interrupt enable register 7	IE7	—	R/W	8	00H
0F018H	Interrupt request register 0	IRQ0	—	R/W	8	00H
0F019H	Interrupt request register 1	IRQ1	—	R/W	8	00H
0F01AH	Interrupt request register 2	IRQ2	—	R/W	8	00H
0F01BH	Interrupt request register 3	IRQ3	—	R/W	8	00H
0F01CH	Interrupt request register 4	IRQ4	—	R/W	8	00H
0F01DH	Interrupt request register 5	IRQ5	—	R/W	8	00H
0F01EH	Interrupt request register 6	IRQ6	—	R/W	8	00H
0F01FH	Interrupt request register 7	IRQ7	—	R/W	8	00H

5.2.2 Interrupt Enable Register 1 (IE1)

Address: 0F011H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
IE1	—	—	—	—	EP03	EP02	EP01	EP00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

IE1 is a special function register (SFR) to control enable/disable for each interrupt request.
When an interrupt is accepted, the master interrupt enable flag (MIE) is set to "0", but the corresponding flag of IE1 is not reset.

[Description of Bits]

- **EP00** (bit 0)
  EP00 is the enable flag for the input port P00 pin interrupt (P00INT).

EP00	Description
0	Disabled (initial value)
1	Enabled

- **EP01** (bit 1)
  EP01 is the enable flag for the input port P01 pin interrupt (P01INT).

EP01	Description
0	Disabled (initial value)
1	Enabled

- **EP02** (bit 2)
  EP02 is the enable flag for the input port P02 pin interrupt (P02INT).

EP02	Description
0	Disabled (initial value)
1	Enabled

- **EP03** (bit 3)
  EP03 is the enable flag for the input port P03 pin interrupt (P03INT).

EP03	Description
0	Disabled (initial value)
1	Enabled

5.2.3 Interrupt Enable Register 2 (IE2)

Address: 0F012H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
IE2	EI2CM	—	—	—	EP5	ESAD	ESIO1	ESIO0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

IE2 is a special function register (SFR) to control enable/disable for each interrupt request.
When an interrupt is accepted, the master interrupt enable flag (MIE) is set to "0", but the corresponding flag of IE2 is not reset.

[Description of Bits]

- **ESIO0** (bit 0)
  ESIO0 is the enable flag for the synchronous serial port 0 interrupt (SIO0INT).

ESIO0	Description
0	Disabled (initial value)
1	Enabled

- **ESIO1** (bit 1)
  ESIO1 is the enable flag for the synchronous serial port 1 interrupt (SIO1INT).

ESIO1	Description
0	Disabled (initial value)
1	Enabled

- **ESAD** (bit 2)
  ESAD is the enable flag for the successive approximation type A/D converter interrupt (SADINT).

ESAD	Description
0	Disabled (initial value)
1	Enabled

- **EP5** (bit 3)
  EP5 is the enable flag for the external 8 interrupt (P5INT).

EP5	Description
0	Disabled (initial value)
1	Enabled

- **EI2CM** (bit 7)
  EI2CM is the enable flag for the I2C bus 0 interrupt (I2C0INT).

EI2CM	Description
0	Disabled (initial value)
1	Enabled

5.2.4 Interrupt Enable Register 3 (IE3)

Address: 0F013H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
IE3	—	—	—	—	—	—	ETM1	ETM0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

IE3 is a special function register (SFR) to control enable/disable for each interrupt request.
When an interrupt is accepted, the master interrupt enable flag (MIE) is set to "0", but the corresponding flag of IE3 is not reset.

[Description of Bits]

- **ETM0** (bit 0)
  ETM0 is the enable flag for the timer 0 interrupt (TM0INT).

ETM0	Description
0	Disabled (initial value)
1	Enabled

- **ETM1** (bit 1)
  ETM1 is the enable flag for the timer 1 interrupt (TM1INT).

ETM1	Description
0	Disabled (initial value)
1	Enabled

5.2.5 Interrupt Enable Register 4 (IE4)

Address: 0F014H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
IE4	—	—	ERAD	—	—	EMD0	EUA1	EUA0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

IE4 is a special function register (SFR) to control enable/disable for each interrupt request.
When an interrupt is accepted, the master interrupt enable flag (MIE) is set to "0", but the corresponding flag of IE4 is not reset.

[Description of Bits]

- **EUA0** (bit 0)
  EUA0 is the enable flag for the UART0 interrupt (UA0INT).

EUA0	Description
0	Disabled (initial value)
1	Enabled

- **EUA1** (bit 1)
  EUA1 is the enable flag for the UART1 interrupt (UA1INT).

EUA1	Description
0	Disabled (initial value)
1	Enabled

- **EMD0** (bit 2)
  EMD0 is the enable flag for the melody 0 interrupt (MD0INT).

EMD0	Description
0	Disabled (initial value)
1	Enabled

- **ERAD** (bit 5)
  ERAD is the enable flag for the RC oscillation type A/D converter interrupt (RADINT).

ERAD	Description
0	Disabled (initial value)
1	Enabled

5.2.6 Interrupt Enable Register 5 (IE5)

Address: 0F015H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
IE5	—	—	ETM3	ETM2	—	—	—	—
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

IE5 is a special function register (SFR) to control enable/disable for each interrupt request.
When an interrupt is accepted, the master interrupt enable flag (MIE) is set to "0", but the corresponding flag of IE5 is not reset.

[Description of Bits]

- **ETM2** (bit 4)
  ETM2 the enable flag for the timer 2 interrupt (TM2INT).

ETM2	Description
0	Disabled (initial value)
1	Enabled

- **ETM3** (bit 5)
  ETM3 the enable flag for the timer 3 interrupt (TM3INT)

ETM3	Description
0	Disabled (initial value)
1	Enabled

5.2.7 Interrupt Enable Register 6 (IE6)

Address: 0F016H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
IE6	E32H	—	E128H	—	—	—	—	EPW0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

IE6 is a special function register (SFR) to control enable/disable for each interrupt request.
When an interrupt is accepted, the master interrupt enable flag (MIE) is set to "0", but the corresponding flag of IE6 is not reset.

[Description of Bits]

- **EPW0** (bit 0)
  EPW0 is the enable flag for the PWM0 interrupt (PW0INT)

EPW0	Description
0	Disabled (initial value)
1	Enabled

- **E128H** (bit 5)
  E128H is the enable flag for the time base counter 128 Hz interrupt (T128HINT).

E128H	Description
0	Disabled (initial value)
1	Enabled

- **E32H** (bit 7)
  E32H is the enable flag for the time base counter 32 Hz interrupt (T32HINT).

E32H	Description
0	Disabled (initial value)
1	Enabled

5.2.8 Interrupt Enable Register 7 (IE7)

Address: 0F017H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
IE7	—	—	—	—	E2H	—	—	E16H
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

IE7 is a special function register (SFR) to control enable/disable for each interrupt request.
When an interrupt is accepted, the master interrupt enable flag (MIE) is set to "0", but the corresponding flag of IE7 is not reset.

[Description of Bits]

- **E16H** (bit 0)
  E16H is the enable flag for the time base counter 16 Hz interrupt (T16HINT).

E16H	Description
0	Disabled (initial value)
1	Enabled

- **E2H** (bit 3)
  E2H is the enable flag for the time base counter 2 Hz interrupt (T2HINT).

E2H	Description
0	Disabled (initial value)
1	Enabled

5.2.9 Interrupt Request Register 0 (IRQ0)

Address: 0F018H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
IRQ0	—	—	—	—	—	—	—	QWDT
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

IRQ0 is a special function register (SFR) to request an interrupt for each interrupt source.
The watchdog timer interrupt is non-maskable interrupts that do not depend on MIE. In this case, an interrupt is requested to the CPU regardless of the value of the Mask Interrupt Enable flag (MIE).
Each IRQ0 request flag is set to "1" regardless of the MIE value when an interrupt is generated. By setting the IRQ0 request flag to "1" by software, an interrupt can be generated.
The corresponding flag of IRQ0 is set to "0" by hardware when the interrupt request is accepted by the CPU.

[Description of Bits]

- **QWDT** (bit 0)
  QWDT is the request flag for the watchdog timer interrupt (WDTINT).

QWDT	Description
0	No request (initial value)
1	Request

Note:
When an interrupt is generated by the write instruction to the interrupt request register (IRQ0), the interrupt shift cycle starts after the next 1 instruction is executed.

5.2.10 Interrupt Request Register 1 (IRQ1)

Address: 0F019H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
IRQ1	—	—	—	—	QP03	QP02	QP01	QP00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

IRQ1 is a special function register (SFR) to request an interrupt for each interrupt source.
Each IRQ1 request flag is set to "1" regardless of the IE1 and MIE values when an interrupt is generated. In this case, an interrupt is requested to the CPU when the related flag of the interrupt enable register (IE1) is set to "1" and the master interrupt enable flag (MIE) is set to "1".
By setting the IRQ1 request flag to "1" by software, an interrupt can be generated.
The corresponding flag of IRQ1 is set to "0" by hardware when the interrupt request is accepted by the CPU.

[Description of Bits]

- **QP00** (bit 0)
  QP00 is the request flag for the input port P00 pin interrupt (P00INT).

QP00	Description
0	No request (initial value)
1	Request

- **QP01** (bit 1)
  QP01 is the request flag for the input port P01 pin interrupt (P01INT).

QP01	Description
0	No request (initial value)
1	Request

- **QP02** (bit 2)
  QP02 is the request flag for the input port P02 pin interrupt (P02INT).

QP02	Description
0	No request (initial value)
1	Request

- **QP03** (bit 3)
  QP03 is the request flag for the input port P03 pin interrupt (P03INT).

QP03	Description
0	No request (initial value)
1	Request

Note:
When an interrupt is generated by the write instruction to the interrupt request register (IRQ1) or to the interrupt enable register (IE1), the interrupt shift cycle starts after the next 1 instruction is executed.

5.2.11 Interrupt Request Register 2 (IRQ2)

Address: 0F01AH
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
IRQ2	QI2C0	—	—	—	QP5	QSAD	QSIO1	QSIO0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

IRQ2 is a special function register (SFR) to request an interrupt for each interrupt source.
Each IRQ2 request flag is set to "1" regardless of the IE2 and MIE values when an interrupt is generated. In this case, an interrupt is requested to the CPU when the related flag of the interrupt enable register (IE2) is set to "1" and the master interrupt enable flag (MIE) is set to "1".
By setting the IRQ2 request flag to "1" by software, an interrupt can be generated.
The corresponding flag of IRQ2 is set to "0" by hardware when the interrupt request is accepted by the CPU.

[Description of Bits]

- **QSIO0** (bit 0)
  QSIO0 is the request flag for the synchronous serial port 0 interrupt (SIO0INT).

QSIO0	Description
0	No request (initial value)
1	Request

- **QSIO1** (bit 1)
  QSIO1 is the request flag for the synchronous serial port 1 interrupt (SIO1INT).

QSIO1	Description
0	No request (initial value)
1	Request

- **QSAD** (bit 2)
  QSAD is the request flag for the successive approximation type A/D converter interrupt (SADINT)

QSAD	Description
0	No request (initial value)
1	Request

- **QP5** (bit 3)
  QP5 is the request flag for the external 8 interrupt (P5INT).

EP5	Description
0	No request (initial value)
1	Request

- **QI2C0** (bit 7)
  QI2C0 is the request flag for the I2C bus 0 interrupt (I2C0INT).

QI2C0	Description
0	No request (initial value)
1	Request

Note:
When an interrupt is generated by the write instruction to the interrupt request register (IRQ2) or to the interrupt enable register (IE2), the interrupt shift cycle starts after the next 1 instruction is executed.

5.2.12 Interrupt Request Register 3 (IRQ3)

Address: 0F01BH
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
IRQ3	—	—	—	—	—	—	QTM1	QTM0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

IRQ3 is a special function register (SFR) to request an interrupt for each interrupt source.
Each IRQ3 request flag is set to "1" regardless of the IE3 and MIE values when an interrupt is generated. In this case, an interrupt is requested to the CPU when the related flag of the interrupt enable register (IE3) is set to "1" and the master interrupt enable flag (MIE) is set to "1".
By setting the IRQ3 request flag to "1" by software, an interrupt can be generated.
The corresponding flag of IRQ3 is set to "0" by hardware when the interrupt request is accepted by the CPU.

[Description of Bits]

- **QTM0** (bit 0)
  QTM0 is the request flag for the timer 0 interrupt (TM0INT).

QTM0	Description
0	No request (initial value)
1	Request

- **QTM1** (bit 1)
  QTM1 is the request flag for the timer 1 interrupt (TM1INT).

QTM1	Description
0	No request (initial value)
1	Request

Note:
When an interrupt is generated by the write instruction to the interrupt request register (IRQ3) or to the interrupt enable register (IE3), the interrupt shift cycle starts after the next 1 instruction is executed.

5.2.13 Interrupt Request Register 4 (IRQ4)

Address: 0F01CH
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
IRQ4	—	—	QRAD	—	—	QMD0	QUA1	QUA0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

IRQ4 is a special function register (SFR) to request an interrupt for each interrupt source.
Each IRQ4 request flag is set to "1" regardless of the IE4 and MIE values when an interrupt is generated. In this case, an interrupt is requested to the CPU when the related flag of the interrupt enable register (IE4) is set to "1" and the master interrupt enable flag (MIE) is set to "1".
By setting the IRQ4 request flag to "1" by software, an interrupt can be generated.
The corresponding flag of IRQ4 is set to "0" by hardware when the interrupt request is accepted by the CPU.

[Description of Bits]

- **QUA0** (bit 0)
  QUA0 is the request flag for the UART0 interrupt (UA0INT).

QUA0	Description
0	No request (initial value)
1	Request

- **QUA1** (bit 1)
  QUA1 is the request flag for the UART1 interrupt (UA1INT).

QUA1	Description
0	No request (initial value)
1	Request

- **QMD0** (bit 2)
  QMD0 is the request flag for the melody 0 interrupt (MD0INT).

QMD0	Description
0	No request (initial value)
1	Request

- **QRAD** (bit 5)
  QRAD is the request flag for the RC oscillation type A/D converter interrupt (RADINT).

QRAD	Description
0	No request (initial value)
1	Request

Note:
When an interrupt is generated by the write instruction to the interrupt request register (IRQ4) or to the interrupt enable register (IE4), the interrupt shift cycle starts after the next 1 instruction is executed.

5.2.14 Interrupt Request Register 5 (IRQ5)

Address: 0F01DH
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
IRQ5	—	QTM3	QTM2	—	—	—	—	—
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

IRQ5 is a special function register (SFR) to request an interrupt for each interrupt source.
Each IRQ5 request flag is set to "1" regardless of the IE5 and MIE values when an interrupt is generated. In this case, an interrupt is requested to the CPU when the related flag of the interrupt enable register (IE5) is set to "1" and the master interrupt enable flag (MIE) is set to "1".
By setting the IRQ5 request flag to "1" by software, an interrupt can be generated.
The corresponding flag of IRQ5 is set to "0" by hardware when the interrupt request is accepted by the CPU.

[Description of Bits]

- **QTM2** (bit 5)
  QTM2 is the request flag for the timer 2 interrupt (TM2INT).

QTM2	Description
0	No request (initial value)
1	Request

- **QTM3** (bit 6)
  QTM3 is the request flag for the timer 3 interrupt (TM3INT).

QTM3	Description
0	No request (initial value)
1	Request

Note:
When an interrupt is generated by the write instruction to the interrupt request register (IRQ5) or to the interrupt enable register (IE5), the interrupt shift cycle starts after the next 1 instruction is executed.

5.2.15 Interrupt Request Register 6 (IRQ6)

Address: 0F01EH
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
IRQ6	Q32H	—	Q128H	—	—	—	—	QPW0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

IRQ6 is a special function register (SFR) to request an interrupt for each interrupt source.
Each IRQ6 request flag is set to "1" regardless of the IE6 and MIE values when an interrupt is generated. In this case, an interrupt is requested to the CPU when the related flag of the interrupt enable register (IE6) is set to "1" and the master interrupt enable flag (MIE) is set to "1".
By setting the IRQ6 request flag to "1" by software, an interrupt can be generated.
The corresponding flag of IRQ6 is set to "0" by hardware when the interrupt request is accepted by the CPU.

[Description of Bits]

- **QPW0** (bit 0)
  QPW0 is the request flag for the PWM0 interrupt (PW0INT).

QPW0	Description
0	No request (initial value)
1	Request

- **Q128H** (bit 5)
  Q128H is the request flag for the time base counter 128 Hz interrupt (T128HINT).

Q128H	Description
0	No request (initial value)
1	Request

- **Q32H** (bit 7)
  Q32H is the request flag for the time base counter 32 Hz interrupt (T32HINT).

Q32H	Description
0	No request (initial value)
1	Request

Note:
When an interrupt is generated by the write instruction to the interrupt request register (IRQ6) or to the interrupt enable register (IE6), the interrupt shift cycle starts after the next 1 instruction is executed.

5.2.16 Interrupt Request Register 7 (IRQ7)

Address: 0F01FH
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
IRQ7	—	—	—	—	Q2H	—	—	Q16H
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

IRQ7 is a special function register (SFR) to request an interrupt for each interrupt source.
Each IRQ7 request flag is set to "1" regardless of the IE7 and MIE values when an interrupt is generated. In this case, an interrupt is requested to the CPU when the related flag of the interrupt enable register (IE7) is set to "1" and the master interrupt enable flag (MIE) is set to "1".
By setting the IRQ7 request flag to "1" by software, an interrupt can be generated.
The corresponding flag of IRQ7 is set to "0" by hardware when the interrupt request is accepted by the CPU.

[Description of Bits]

- **Q16H** (bit 0)
  Q16H is the request flag for the time base counter 8 Hz interrupt (T8HINT).

Q16H	Description
0	No request (initial value)
1	Request

- **Q2H** (bit 3)
  Q2H is the request flag for the time base counter 2 Hz interrupt (T2HINT).

Q2H	Description
0	No request (initial value)
1	Request

Note:
When an interrupt is generated by the instruction to write to the interrupt request register (IRQ7) or to the interrupt enable register (IE7), the the interrupt shift cycle starts after the next 1 instruction is executed.

5.3 Description of Operation

With the exception of the watchdog timer interrupt (WDTINT) and the NMI interrupt (NMINT), interrupt enable/disable for 23 sources is controlled by the master interrupt enable flag (MIE) and the individual interrupt enable registers (IE1 to 7). WDTINT and NMIINT are non-maskable interrupts.
When the interrupt conditions are satisfied, the CPU calls a branching destination address from the vector table determined for each interrupt source and the interrupt shift cycle starts to branch to the interrupt processing routine.
Table 5-1 lists the interrupt sources.

Table 5-1 Interrupt Sources

Priority	Interrupt source	Symbol	Vector table address
1	Watchdog timer interrupt	WDTINT	0008H
2	P00 interrupt	P00INT	0010H
3	P01 interrupt	P01INT	0012H
4	P02 interrupt	P02INT	0014H
5	P03 interrupt	P03INT	0016H
6	Synchronous serial port 0 interrupt	SIO0INT	0020H
7	Synchronous serial port 1 interrupt	SIO1INT	0022H
8	Successive approximation type A/D converter interrupt	SADINT	0024H
9	External 8 interrupt	P5INT	0026H
10	I^2C bus 0 interrupt	I2C0INT	002EH
11	Timer 0 interrupt	TM0INT	0030H
12	Timer 1 interrupt	TM1INT	0032H
13	UART 0 interrupt	UA0INT	0040H
14	UART 1 interrupt	UA1INT	0042H
15	Melody 0 interrupt	MD0INT	0044H
16	RC oscillation type A/D converter interrupt	RADINT	004AH
17	Timer 2 interrupt	TM2INT	0058H
18	Timer 3 interrupt	TM3INT	005AH
19	PWM0 interrupt	PW0INT	0060H
20	TBC128Hz interrupt	T128HINT	006AH
21	TBC32Hz interrupt	T32HINT	006EH
22	TBC16Hz interrupt	T16HINT	0070H
23	TBC2Hz interrupt	T2HINT	0076H

Note:
When multiple interrupts are generated concurrently, the interrupts are serviced according to this priority and processing of low-priority interrupts is pending.
Please define vector tables for all unused interrupts for fail safe.

5.3.1 Maskable Interrupt Processing

When an interrupt is generated with the MIE flag set to "1", the following processing is executed by hardware and the processing of program shifts to the interrupt destination.

(1) Transfer the program counter (PC) to ELR1.
(2) Transfer CSR to ECSR1.
(3) Transfer PSW toEPSW1.
(4) Set the MIE flag to "0".
(5) Set the ELEVEL field to"1".
(6) Load the interrupt start address into PC.

5.3.2 Non-Maskable Interrupt Processing

When an interrupt is generated regardless of the state of MIE flag, the following processing is performed by hardware and the processing of program shifts to the interrupt destination.

(1) Transfer PC to ELR2.
(2) Transfer CSR to ECSR2.
(3) Transfer PSW to EPSW2.
(4) Set the ELEVEL field to "2".
(5) Load the interrupt start address into PC.

5.3.3 Software Interrupt Processing

A software interrupt is generated as required within an application program. When the SWI instruction is performed within the program, a software interrupt is generated, the following processing is performed by hardware, and the processing program shifts to the interrupt destination. The vector table is specified by the SWI instruction.

(1) Transfer PC to ELR1.
(2) Transfer CSR to ECSR1.
(3) Transfer PSW to EPSW1.
(4) Set the MIE flag to "0".
(5) Set the ELEVEL field to "1".
(6) Load the interrupt start address into PC.

Reference:
For the MIE flag, Program Counter (PC), CSR, PSW, and ELEVEL, see "nX-U8/100 Core Instruction Manual".

5.3.4 Notes on Interrupt Routine

Notes are different in programming depending on whether a subroutine is called or not by the program in executing an interrupt routine, whether multiple interrupts are enabled or disabled, and whether such interrupts are maskable or non-maskable.

State A: Maskable interrupt is being processed

A-1: When a subroutine is not called by the program in executing an interrupt routine

A-1-1: When multiple interrupts are disabled

- Processing immediately after the start of interrupt routine execution
  No specific notes.
- Processing at the end of interrupt routine execution
  Specify the RTI instruction to return the contents of the ELR register to the PC and those of the EPSW register to PSW.

A-1-2: When multiple interrupts are enabled

- Processing immediately after the start of interrupt routine execution
  Specify "PUSH ELR, EPSW" to save the interrupt return address and the PSW status in the stack.
- Processing at the end of interrupt routine execution
  Specify "POP PC, PSW" instead of the RTI instruction to return the contents of the stack to PC and PSW.

Example of description: State A-1-1

```
Intrpt_A-1-1;          ; A-1-1 state
   DI                  ; Disable interrupt
    :
    :
    :
   RTI                 ; Return PC from ELR
                       ; Return PSW form EPSW
                       ; End
```

Example of description: State A-1-2

```
Intrpt_A-1-2;          ; Start
   PUSH ELR, EPSW      ; Save ELR and EPSW at the
                         beginning
   EI                  ; Enable interrupt
    :
    :
    :
    :
    :
   POP PC, PSW         ; Return PC from the stack
                       ; Return PSW from the stack
                       ; End
```

A-2: When a subroutine is called by the program in executing an interrupt routine

A-2-1: When multiple interrupts are disabled

- Processing immediately after the start of interrupt routine execution
  Specify the "PUSH LR" instruction to save the subroutine return address in the stack.
- Processing at the end of interrupt routine execution
  Specify "POP LR" immediately before the RTI instruction to return from the interrupt processing after returning the subroutine return address to LR.

A-2-2: When multiple interrupts are enabled

- Processing immediately after the start of interrupt routine execution
  Specify "PUSH LR, ELR, EPSW" to save the interrupt return address, the subroutine return address, and the EPSW status in the stack.
- Processing at the end of interrupt routine execution
  Specify "POP PC, PSW, LR" instead of the RTI instruction to return the saved data of the interrupt return address to PC, the saved data of EPSW to PSW, and the saved data of LR to LR.

Example of description: A-2-2

```
Intrpt_A-2-2;           ; Start
   PUSH ELR, EPSW,      ; Save ELR, EPSW, LR at
   LR                     the beginning

   EI                   ; Enable interrupt
    :
    :
    :

   BL Sub_1             ; Call subroutine Sub_1
    :
   POP PC, PSW, LR      ; Return PC from the stack
                        ; Return PSW from the stack
                        ; Return LR from the stack
                        ; End
```

```
Sub_1;          ;
DI              ; Disable interrupt
                    :
                    :
                    :
RT              ; Return PC from LR
                ; End of subroutine
```

State B: Non-maskable interrupt is being processed

B-1: When no instruction is executed in an interrupt routine
- Processing immediately after the start of interrupt routine execution
  Specify the RTI instruction to return the contents of the ELR register to the PC and those of the EPSW register to PSW.

B-2: When one or more instructions are executed in an interrupt routine

B-2-1: When a subroutine is not called by the program in executing an interrupt routine
- Processing immediately after the start of interrupt routine execution
  Specify "PUSH ELR, EPSW" to save the interrupt return address and the PSW status in the stack.
- Processing at the end of interrupt routine execution
  Specify "POP PC, PSW" instead of the RTI instruction to return the contents of the stack to PC and PSW.

B-2-2: When a subroutine is called by the program in executing an interrupt routine
- Processing immediately after the start of interrupt routine execution
  Specify "PUSH LR, ELR, EPSW" to save the interrupt return address, the subroutine return address, and the EPSW status in the stack.
- Processing at the end of interrupt routine execution

Specify "POP PC, PSW, LR" instead of the RTI instruction to return the saved data of the interrupt return address to PC, the saved data of EPSW to PSW, and the saved data of LR to LR..

Example of description: B-1

```
Intrpt_B-1:          ; B-1 state
  RTI                ; Return PC from ELR
                     ; Return PSW form EPSW
                     ; End
```

Example of description: B-2-1

```
Intrpt_B-2-1:        ; Start
  PUSH  ELR, EPSW    ; Save ELR, EPSW at the
                       beginning
  :
  :
  :
  POP  PC, PSW       ; Return PC from the stack
                     ; Return PSW from the stack
                     ; End
```

Example of description: B-2-2

```
Intrpt_B-2-2:        ; Start
  PUSH  ELR,EPSW,LR  ; Save ELR, EPSW, LR at the
                       beginning
  :
  :
  :
  BL Sub_1           ; Call subroutine Sub_1
  :
  POP  PC, PSW, LR   ; Return PC from the stack
                     ; Return PSW from the stack
                     ; Return LR from the stack
                     ; End
```

```
Sub_1:               ;
  :
  :
  :
  :
  RT                 ; Return PC from LR
                     ; End of subroutine
```

5.3.5 Interrupt Disable State

Even if the interrupt conditions are satisfied, an interrupt may not be accepted depending on the operating state. This is called an interrupt disabled state. See below for the interrupt disabled state and the handling of interrupts in this state.

Interrupt disabled state 1: Between the interrupt shift cycle and the instruction at the beginning of the interrupt routine

When the interrupt conditions are satisfied in this section, an interrupt is generated immediately following the execution of the instruction at the beginning of the interrupt routine corresponding to the interrupt that has already been enabled.

Interrupt disabled state 2: Between the DSR prefix instruction and the next instruction

When the interrupt conditions are satisfied in this section, an interrupt is generated immediately after execution of the instruction following the DSR prefix instruction.

Reference:
For the DSR prefix instruction, see "nX-U8/100 Core Instruction Manual".

Chapter 6

Clock Generation Circuit

6. Clock Generation Circuit

6.1 Overview

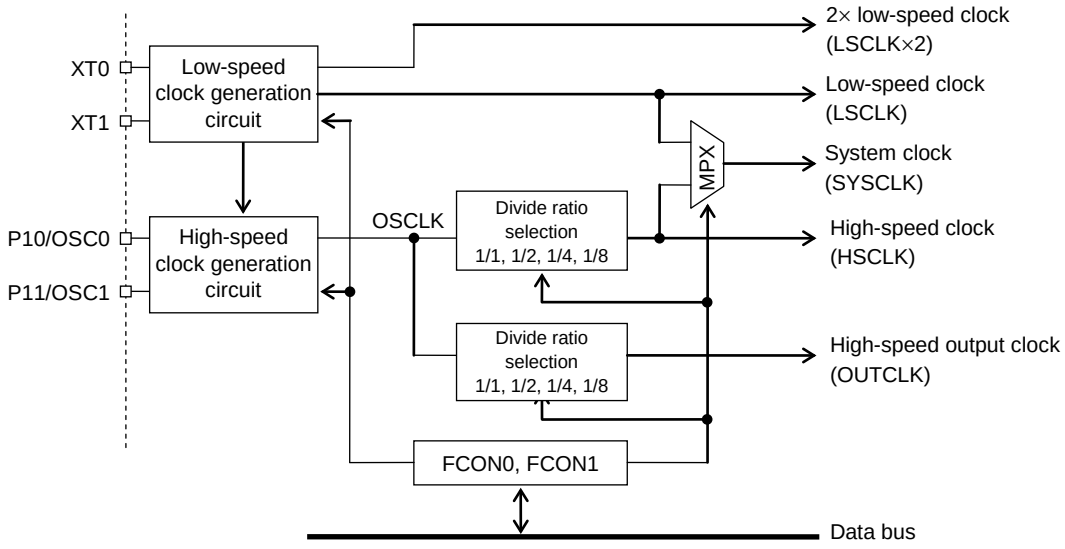
The clock generation circuit generates and provides a low-speed clock (LSCLK), 2× low-speed clock (LSCLK×2), a high-speed clock (HSCLK), a system clock (SYSCLK), and a high-speed output clock (OUTCLK). LSCLK, LSCLK×2, and HSCLK are time base clocks for the peripheral circuits, SYSCLK is a basic operation clock of CPU, and OUTCLK is a clock that is output from a port.
For the OUTCLK output port, see Chapter 17, "Port 2".
Additionally, for the STOP mode described in this chapter, see Chapter 4, "MCU Control Function", and for BLD, see Chapter 26, "Battery Level Detection Circuit".

6.1.1 Features

- Low-speed clock: 32.768 kHz crystal oscillation mode
  – Capable of generating LSCLK × 2 (64 kHz) to be used for some peripherals.
- High-speed clock: Software selection
  – 500 kHz RC oscillation mode
  – Crystal/ceramic oscillation mode
  – Built-in PLL oscillation mode
  – External clock input mode

6.1.2 Configuration

Figure 6-1 shows the configuration of the clock generation circuit.



FCON0 : Frequency control register 0
FCON1 : Frequency control register 1

Figure 6-1 Configuration of Clock Generation Circuit

Note:
After power on, or after system reset, this LSI starts operating by 32.768kHz low speed clock. At initialization by software, set the FCON0 or FCON1 register to switch the clock to a required one. Operation of this LSI is not guaranteed under a condition where a low-speed clock is not supplied.

6.1.3 List of Pins

Pin name	I/O	Description
XT0	I	Pin for connecting a crystal for low-speed clock
XT1	O	Pin for connecting a crystal for low-speed clock
P10/OSC0	I	Pin for connecting a crystal/ceramic resonator for high-speed clock Used for the secondary function of the P10 pin
P11/OSC1	O	Pin for connecting a crystal/ceramic resonator for high-speed clock Used for the secondary function of the P11 pin

6.2 Description of Registers

6.2.1 List of Registers

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F002H	Frequency control register 0	FCON0	FCON	R/W	8/16	33H
0F003H	Frequency control register 1	FCON1		R/W	8	00H

6.2.2 Frequency Control Register 0 (FCON0)

Address: 0F002H
Access: R/W
Access size: 8/16 bits
Initial value: 33H

	7	6	5	4	3	2	1	0
FCON0	—	—	OUTC1	OUTC0	OSCM1	OSCM0	SYSC1	SYSC0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	1	1	0	0	1	1

FCON0 is a special function register (SFR) to control the high-speed clock generation circuit and to select system clock.

[Description of Bits]

- **SYSC1, SYSC0** (bits 1, 0)
  The SYSC1 and SYSC0 bits are used to select the frequency of the high-speed clock (HSCLK) used for system clock and periphera1 circuits (including high-speed time base counter). OSCLK, 1/2OSCLK, 1/4OSCLK, or 1/8OSCLK can be selected. The maximum operating frequency guaranteed for the system clock (SYSCLK) of this LSI is 4.2 MHz.
  At system reset, 1/8OSCLK is selected.

SYSC1	SYSC0	Description
0	0	OSCLK (1/2OSCLK in built-in PLL oscillation mode)
0	1	1/2OSCLK
1	0	1/4OSCLK
1	1	1/8OSCLK (initial value)

- **OSCM1, OSCM0** (bits 3, 2)
  The OSCM1 and OSCM0 bits are used to select the mode of the high-speed clock generation circuit. RC oscillation mode, crystal/ceramic oscillation mode, PLL oscillation mode, or external clock input mode can be selected.
  The setting of OSCM1 and OSCM0 can be changed only when high-speed oscillation is being stopped (ENOSC bit of FCON1 is "0"). At system reset, RC oscillation mode is selected.
  When switching the high-speed oscillation mode, please first switch back to low speed clock before switching to other high-speed clock (set the ENOSC bit and SYSCLK bit of FCON1 to "0").

OSCM1	OSCM0	Description
0	0	RC oscillation mode (initial value)
0	1	Crystal/ceramic oscillation mode
1	0	Built-in PLL oscillation mode
1	1	External clock input mode

- **OUTC1, OUTC0** (bits 5, 4)
  The OUTC1 and OUTC0 bits are used to select the frequency of the high-speed output clock which is output when the secondary function of the port is used.
  OSCLK, 1/2OSCLK, 1/4OSCLK, or 1/8OSCLK can be selected.
  At system reset, 1/8OSCLK is selected.

OUTC1	OUTC0	Description
0	0	OSCLK
0	1	1/2OSCLK
1	0	1/4OSCLK
1	1	1/8OSCLK (initial value)

Note:
To switch the mode of the high-speed clock generation circuit using the OSCM1 and OSCM0 bits, stop the high-speed oscillation and set the system clock to the low-speed clock (set the ENOSC bit and SYSCLK of FCON1 to "0").
The oscillators that are connected to the P10/OSC0 and P11/OSC1 pins must not exceed 4.2 MHz. In external clock mode, input a clock that does not exceed 4.2 MHz. When a built-in PLL oscillation mode is selected (OSCM1 = "1", OSCM0 = "0"), 1/2OSCLK (about 4.096 MHz) is output as HSCLK even if OSCLK (SYSC0 = "0", SYSC1 = "1") is selected.
When built-in PLL (about 8.192 MHz) oscillation mode is selected (OSCM1 = "1", OSCM0 = "0"), 1/2OSCLK (about 4.096 MHz) is output as HSCLK even if OSCLK (SYSC0 = "0", SYSC1 = "1") is selected.

6.2.3 Frequency Control Register 1 (FCON1)

Address: 0F003H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
FCON1	LPLL	—	—	—	—	ENMLT	ENOSC	SYSCLK
R/W	R	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

FCON1 is a special function register (SFR) to control the high-speed clock generation circuit and to select system clock.

[Description of Bits]

- **SYSCLK** (bit 0)
  The SYSCLK bit is used to select system clock. It allows selection of the low-speed clock (LSCLK) or HSCLK (1/nOSCLK: n = 1, 2, 4, 8) selected by using the high-speed clock frequency select bit (SYSC1, 0) of FCON0.
  When the oscillation of high-speed clock is stopped (ENOSC bit = "0"), the SYSCLK bit is fixed to "0" and the low-speed clock (LSCLK) is selected for system clock.

SYSCLK	Description
0	LSCLK (initial value)
1	HSCLK

- **ENOSC** (bit 1)
  The ENOSC bit is used to select enable/disable of the oscillation of the high-speed clock oscillator.

ENOSC	Description
0	Disables high-speed oscillation (initial value)
1	Enables high-speed oscillation

- **ENMLT** (bit 2)
  The ENMLT bit is used to select enable/disable of the operation of the 2× low-speed clock (LSCLK×2).

ENMLT	Description
0	Disables 2× low-speed clock operation (initial value)
1	Enables 2× low-speed clock operation

- **LPLL** (bit 7)
  The LPLL bit is used as a flag to indicate the oscillation state of PLL oscillation.
  When the LPLL bit is set to "1", this indicates that the PLL oscillation frequency is locked within 8.192 MHz±2.5%.
  When the LPLL bit is set to "0", this indicates that the PLL oscillation is inactive or the PLL oscillation frequency is not within 8.192 MHz±2.5%.
  LPLL is a read-only bit.

LPLL	Description
0	Disables the use of PLL oscillation (initial value)
1	Enables the use of PLL oscillation

6.3 Description of Operation

6.3.1 Low-Speed Clock

6.3.1.1 Low-Speed Clock Generation Circuit

Figure 6-2 shows the configuration of the low-speed clock generation circuit.
A low-speed clock generation circuit is provided with an external 32.768 kHz crystal. To match the oscillation frequency by using a trimmer capacitor, connect external capacitors (C_{GL} and C_{DL}) as required.
In STOP mode, V_{DDL} is powered off to stop low-speed oscillation, and the XT0 and XT1 pins become Hiz (Hi Impedance state).
When the ENMLT bit of FCON1 is set to "1", the 2× low-speed clock circuit starts to generate the LSCLK×2(64kHz)

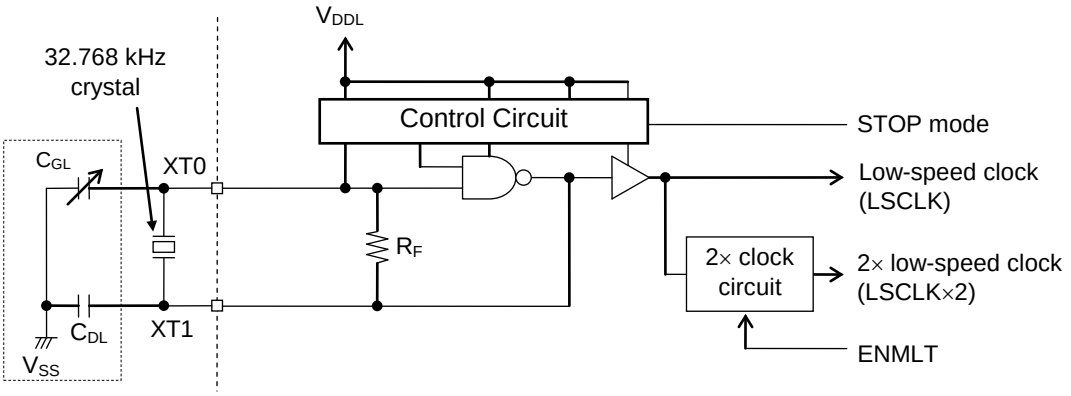


Figure 6-2 Circuit Configuration of 32.768 kHz Crystal Oscillation Mode

Notes:
Install a crystal as close to the LSI as possible and make sure that signals causing noise and power supply wiring are not near the crystal and its wiring.
Note that oscillation may stop due to condensation.

6.3.1.2 Operation of Low-Speed Clock Generation Circuit

The low-speed clock generation circuit is activated by the occurrence of power ON reset.
A low-speed clock (LSCLK) is supplied to the peripheral circuits after the elapse of the low-speed oscillation start period (T_{XTL}) and oscillation stabilization period (8192 counts) after powered on.
The low-speed clock generation circuit stops the oscillation in STOP mode. When oscillation is resumed by releasing of the STOP mode by external interrupt, LSCLK is supplied to the peripheral circuits after the elapse of the low-speed oscillation start period (T_{XTL}) and low-speed clock (LSCLK) oscillation stabilization period. For STOP mode, see Chapter 4, "MCU Control Function".
Figure 6-3 shows the waveforms of the low-speed clock generation circuit. For the low-speed oscillation start time (T_{XTL}), see Appendix C, "Electrical Characteristics".

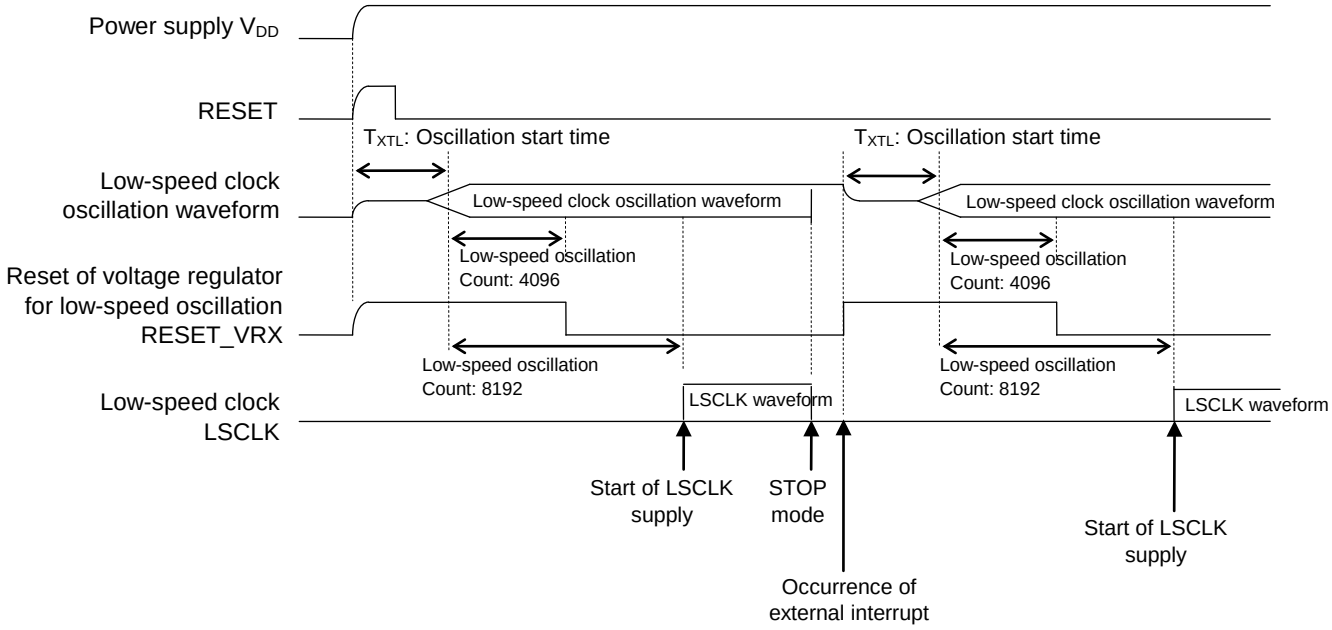


Figure 6-3 Operation of Low-Speed Clock Generation Circuit

Note:
After the power supply is turned on, CPU starts operation with a high-speed clock (500 kHz RC oscillation). It is recommended to switch to the low-speed clock after confirming that the low-speed clock is oscillating by checking that the 128 Hz interrupt request bit (Q128H) of the low-speed time base counter is "1". If the clock is switched before the low-speed clock oscillates, the CPU stops operation until oscillation of the low-speed clock starts.

6.3.2 High-Speed Clock

Setting of the OSCM1 and OSCM0 bits of the frequency control register 0 (FCON0) allows selection of the 500 kHz RC oscillation mode, crystal/ceramic oscillation mode, built-in PLL (Phase Locked Loop) oscillation mode, or external clock input mode for the high-speed clock generation circuit.

6.3.2.1 500 kHz RC Oscillation

In RC oscillation mode (OSCM0 = "0", OSCM1 = "0"), supply of OSCLK (high-speed oscillation clock) is started when RC oscillation clock pulse count reaches 128 after oscillation is enabled (ENOSC is set to "1").
In 500 kHz RC oscillation mode, both the P10/OSC0 pin and the P11/OSC1 pin can be used as general-purpose input ports.
Figure 6-4 shows the circuit configuration in RC oscillation mode.

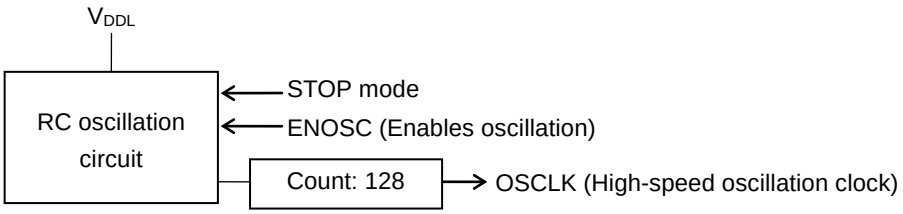


Figure 6-4 Circuit Configuration in RC Oscillation Mode

Notes:
The RC oscillation mode is allowed within the range of V_{DD} = 1.3 V to 3.6 V.
After system reset mode is released, supply of OSCLK starts after the RC oscillation clock pulse count reaches 8192.
After release of a STOP mode, supply of OSCLK starts.

6.3.2.2 Crystal/Ceramic Oscillation Mode

In crystal/ceramic oscillation mode, both the P10/OSC0 pin and the P11/OSC1 pin are used for crystal ceramic oscillation.
In crystal/ceramic oscillation mode, a crystal or a ceramic resonator is externally connected to the P10/OSC0 and P11/OSC1 pins. If the high-speed oscillation clock pulse count reaches 4096 after oscillation enable, the clock is output to OSCLK (high-speed oscillation clock).
Figure 6-5 shows the circuit configuration in crystal/ceramic oscillation mode.

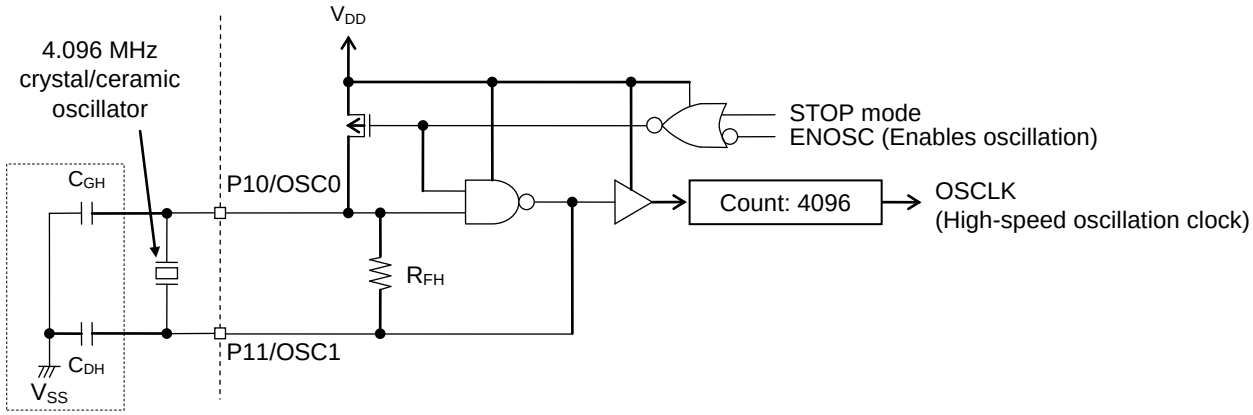


Figure 6-5 Circuit Configuration in Crystal/Ceramic Oscillation Mode

Notes:
The crystal/ceramic oscillation mode can be used within a V_{DD} range of 1.8 V to 3.6 V. Select a frequency according to the operating voltage range by using the power supply voltage detection circuit (BLD). See Chapter 26, "Battery Level Detection Circuit" for details of BLD.
Install a crystal or a ceramic resonator as close to the LSI as possible and make sure that signals causing noise and power supply wiring are not near the crystal or the ceramic resonator and their wiring.
Note that oscillation may stop due to condensation.
The crystal or the ceramic resonator connected to the P10/OSC0 and P11/OSC1 pins should not exceed the guaranteed maximum operation frequency of 4.2 MHz of the system clock (SYSCLK) of this LSI.

6.3.2.3 Built-in PLL Oscillation Mode

The PLL oscillation circuit generates a clock of 8.192 MHz (= 32.768 kHz × 250) ±2.5%. When the PLL oscillation clock (OSCLK) reaches within 8.192 MHz±2.5%, the LPLL flag of FCON1 is set.
In built-in PLL oscillation mode (OSCM0 = "0", OSCM1 = "1"), supply of OSCLK (high-speed oscillation clock) is started when PLL oscillation clock pulse count reaches 8192 after oscillation is enabled (ENOSC is set to "1").
In PLL oscillation mode, both the P10/OSC0 pin and the P11/OSC1 pin can be used as general-purpose input ports.
Figure 6-6 shows the circuit configuration in PLL oscillation mode.

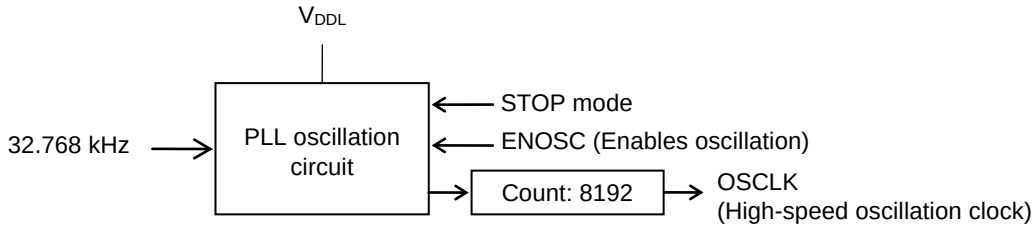


Figure 6-6 Circuit Configuration in PLL Oscillation Mode

Note:
The PLL oscillation mode can be used within a V_{DD} range of 1.8 V to 3.6 V. Select a frequency according to the operating voltage range by using the power supply voltage detection circuit (BLD).
When OSCLK is selected through SYSC1 or SYSC0 of FCON0 in PLL oscillation mode, about 4.096MHz, which is the same as 1/2OSCLK, is selected.
To use a PLL oscillation mode, a frequency of low-speed crystal oscillation 32.768kHz is necessary. The frequency of 32.768kHz is not adjusted by the frequency adjustment circuit of the time base counter.

6.3.2.4 External Clock Input Mode

In external clock input mode, external clock is input from the P10/OSC0 pin. The P11/OSC1 pin can be used as a general-purpose input port.
Figure 6-7 shows the circuit configuration in external clock input mode.

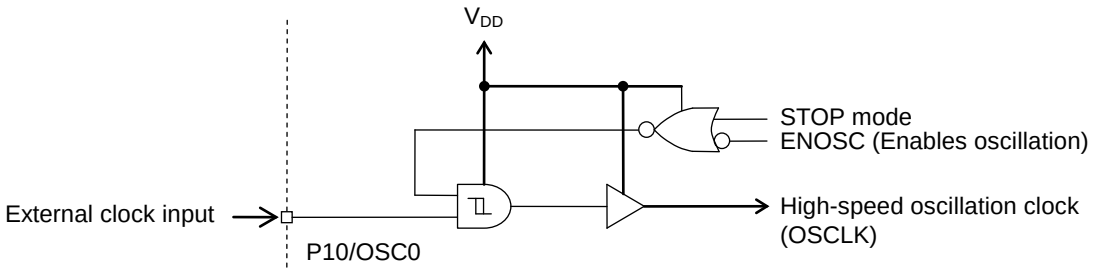


Figure 6-7 Circuit Configuration in External Clock Input Mode

Notes:
The external clock input mode can be used within a VDD range of 1.8 V to 3.6 V. Select a frequency according to the operation voltage range by using the power supply voltage detection circuit (BLD).
Since the diodes are included between the P10/OSC0 pin and VDD and between the P10/OSC0 pin and VSS, do not apply voltages higher than VDD and lower than VSS to the P10/OSC0 pin.
If the P10/OSC0 pin is left open in external clock input mode, excessive current can flow. Therefore, make sure that the "H" level (VDD) or the "L" level (VSS) is input.
The clock that is input should not exceed the guaranteed maximum operating frequency 4.2 MHz of the system clock (SYSCLK) of this LSI.

6.3.2.5 Operation of High-Speed Clock Generation Circuit

The high-speed clock generation circuit is activated in 500Hz RC oscillation mode by power-on reset generation.
As a result of the occurrence of power-on reset, the circuit goes into system reset mode and then shifts to program operating mode after the elapse of the high-speed RC oscillation start time (T_{RC}) and the oscillation stabilization time (Count: 8192) of the high-speed oscillation clock (OSCLK) and at the same time, a high-speed clock (HSCLK) is supplied to the peripheral circuits.
Figure 6-8 shows the waveforms of the high-speed clock generation circuit at power on. For the high-speed RC oscillation start time (T_{RC}), see Appendix C, "Electrical Characteristics".

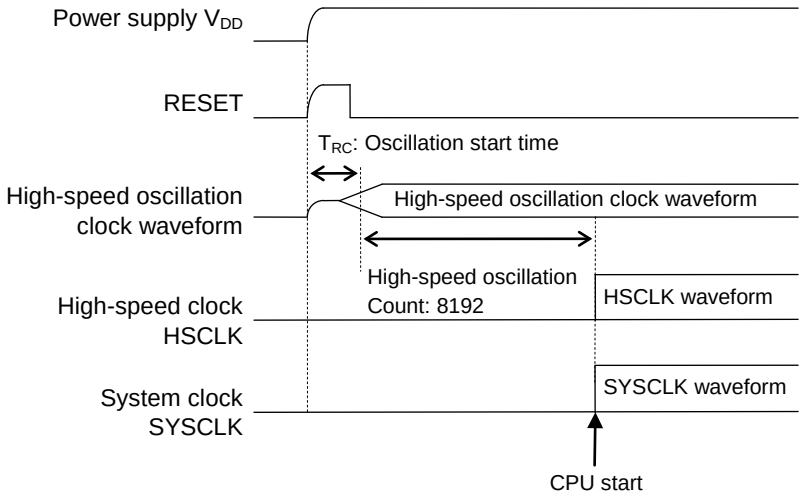


Figure 6-8 Operation of High-Speed Clock Generation Circuit at Power-On

The high-speed clock generation circuit allows selection of an oscillation mode and start/stop of oscillation by using the frequency control registers 0 and 1 (FCON0 and FCON1).
Oscillation can be started by setting the ENOSC bit to "1" after selecting a high-speed oscillation mode in FCON0 and a high-speed oscillation frequency. After the start of oscillation, HSCLK starts supply of a clock to the peripheral circuits following the elapse of the high-speed oscillation start period (T_{RC}/T_{XTH}/T_{PLL}) in each mode and the oscillation stabilization period of the high-speed oscillation clock (OSCLK).
The high-speed clock generation circuit stops oscillation when it shifts to a STOP mode by the software. When the STOP mode is released by external interrupt, HSCLK supplies clocks to peripheral circuits following the elapse of the high-speed oscillation start period (T_{RC}/T_{XTH}/T_{PLL}) in each mode and the oscillation stabilization period of the high-speed clock (OSCLK). The oscillation stabilization period is the duration of 128 clock pulses in 500 kHz RC oscillation mode and external clock input mode and the duration of 4096 clock pulses in the crystal/ceramic oscillation mode and PLL oscillation mode.

Figure 6-9 shows the waveforms of the high-speed clock generation circuit in crystal/ceramic oscillation mode.

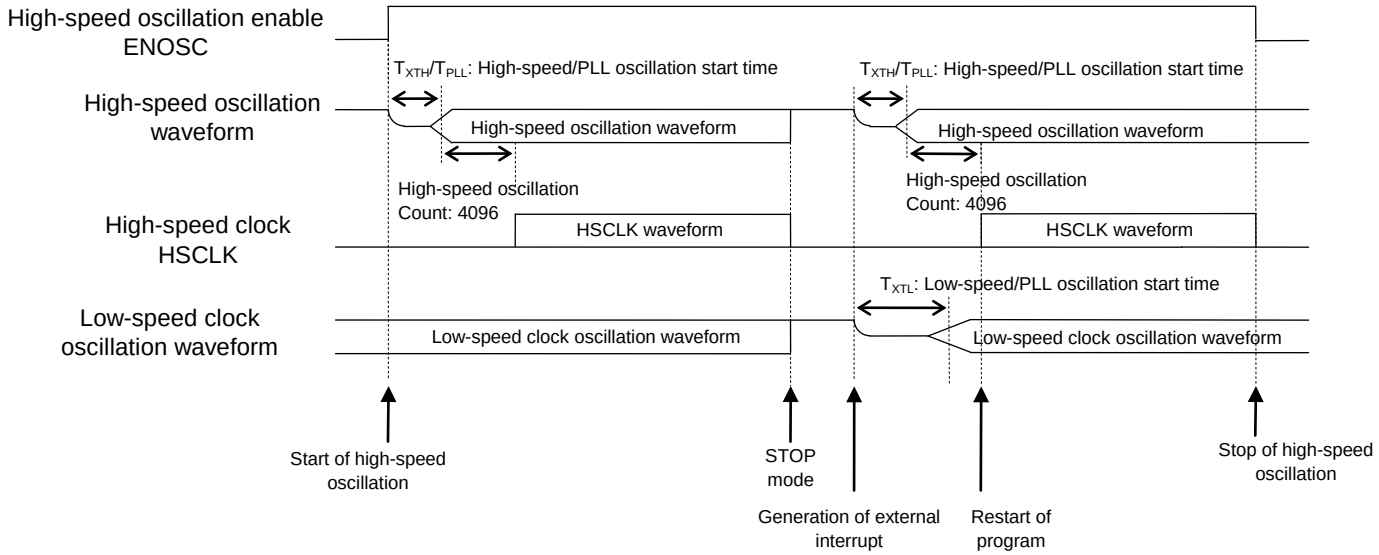


Figure 6-9 Operation of High-Speed Clock Generation Circuit in Crystal/Ceramic Oscillation Mode

6.3.3 Switching of System Clock

The system clock can be switched between high-speed clock (HSCLK) and low-speed clock (LSCLK) by using the frequency control registers (FCON0, FCON1).
Figure 6-10 shows a flow of system clock switching processing (HSCLK→LSCLK) and Figure 6-11 shows a flow of system clock switching processing (LSCLK→HSCLK).

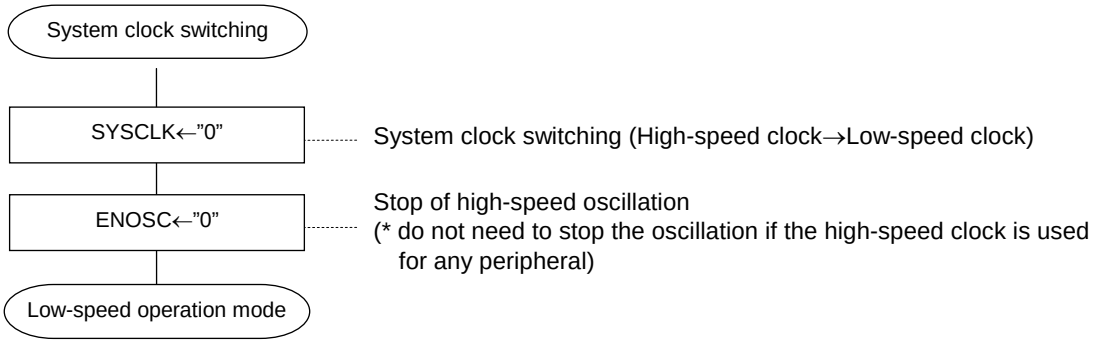


Figure 6-10 Flow of System Clock Switching Processing (HSCLK→LSCLK)

Note:
After the power is turned on or if the system clock is switched from HSCLK to LSCLK immediately following return from the STOP mode, the CPU becomes inactive until LSCLK starts clock supply to the peripheral circuits. Therefore, It is recommended to switch to LSCLK after confirming that the LSCLK is oscillating by checking that the time base counter interrupt request bit (Q128H) is "1".

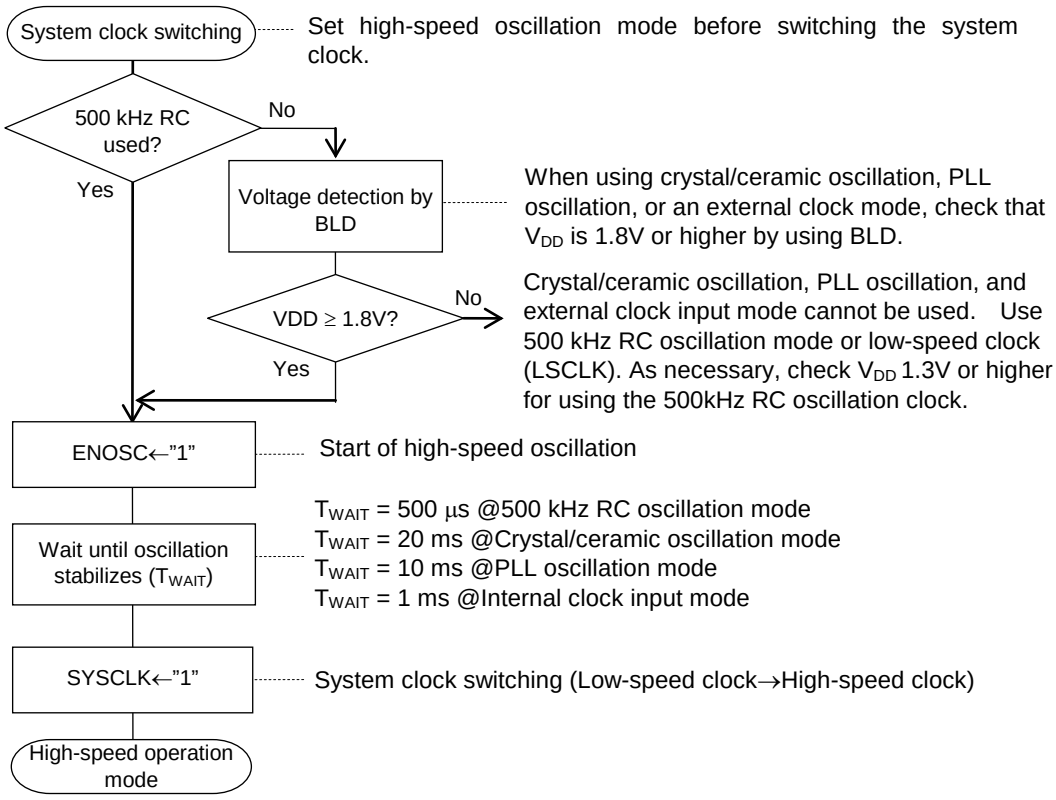


Figure 6-11 Flow of System Clock Switching Processing (LSCLK→HSCLK)

Note:
If the system clock is switched from a low-speed clock to a high-speed clock before the high-speed clock (HSCLK) starts oscillation, the CPU becomes inactive until HSCLK starts clock supply to the peripheral circuits.

6.4 Specifying port registers

When you want to make sure clock output functions are working, please check related port registers are specified. See Chapter 17, "Port2" for detail about the port registers.

6.4.1 Functioning P21 (OUTCLK) as the high speed clock output

Set P21MD bit (bit1 of P2MOD register) to "1" for specifying the high speed clock output as the secondary function of P21.

Reg. name	P2MOD register (Address: 0F214H)							
Bit	7	6	5	4	3	2	**1**	0
Bit name	-	-	-	-	-	P22MD	**P21MD**	P20MD
Data	-	-	-	-	-	*	**1**	*

Set P21C1 bit (bit1 of P2CON1 register) to "1" and set P21C0 bit(bit1 of P2CON0 register) to "1", for specifying the P21 as CMOS output.

Reg. name	P2CON1 register (Address: 0F213H)							
Bit	7	6	5	4	3	2	**1**	0
Bit name	-	-	-	-	-	P22C1	**P21C1**	P20C1
Data	-	-	-	-	-	*	**1**	*

Reg. name	P2CON0 register (Address: 0F212H)							
Bit	7	6	5	4	3	2	**1**	0
Bit name	-	-	-	-	-	P22C0	**P21C0**	P20C0
Data	-	-	-	-	-	*	**1**	*

Data of P21D bit (bit1 of P2D register) does not affect to the high speed clock output function, so don't care the data for the function.

Reg. name	P2D register (Address: 0F210H)							
Bit	7	6	5	4	3	2	**1**	0
Bit name	-	-	-	-	-	P22D	**P21D**	P20D
Data	-	-	-	-	-	*	******	*

- : Bit does not exist.
* : Bit not related to the high speed clock function
** : Don't care the data.

Note:
P21(Port2) is an output-only port, does not have an register to select the data direction(input or output).

6.4.2 Functioning P20 (LSCLK) as the low speed clock output

Set P20MD bit (bit0 of P2MOD register) to "1" for specifying the low speed clock output as the secondary function of P22.

Reg. name	P2MOD register (Address: 0F214H)							
Bit	7	6	5	4	3	2	1	**0**
Bit name	-	-	-	-	-	P22MD	P21MD	**P20MD**
Data	-	-	-	-	-	*	*	**1**

Set P20C1 bit (bit0 of P2CON1 register) to "1" and P20C0 bit (bit0 of P2CON0 register), for specifying P20 as CMOS output.

Reg. name	P2CON1 register (Address: 0F213H)							
Bit	7	6	5	4	3	2	1	**0**
Bit name	-	-	-	-	-	P22C1	P21C1	**P20C1**
Data	-	-	-	-	-	*	*	**1**

Reg. name	P2CON0 register (Address: 0F212H)							
Bit	7	6	5	4	3	2	1	**0**
Bit name	-	-	-	-	-	P22C0	P21C0	**P20C0**
Data	-	-	-	-	-	*	*	**1**

Data of P20D bit (bit0 of P2D register) does not affect to the low speed clock output function, so don't care the data for the function.

Reg. name	P2D register (Address: 0F210H)							
Bit	7	6	5	4	3	2	1	**0**
Bit name	-	-	-	-	-	P22D	P21D	**P20D**
Data	-	-	-	-	-	*	*	******

- : Bit does not exist.
* : Bit not related to the low speed clock function
** : Don't care the data.

Note:
P20(Port2) is an output-only port, does not have an register to select the data direction(i.e. input or output).

Chapter 7

Time Base Counter

7. Time Base Counter

7.1 Overview

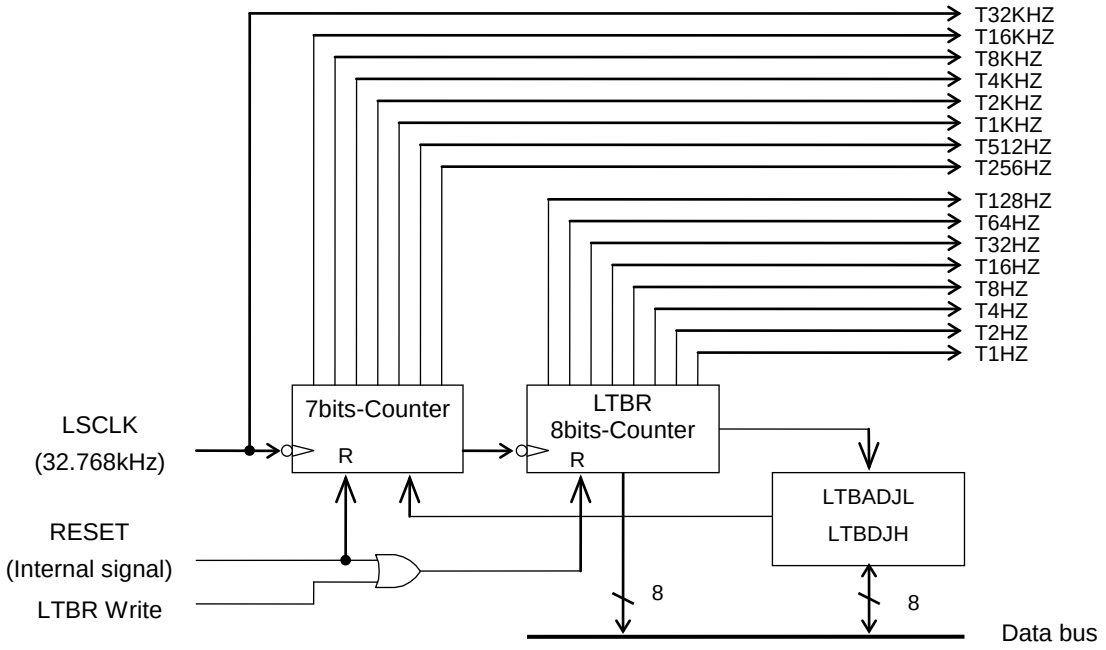
This LSI includes a low-speed time base counter (LTBC) and a high-speed time base counter (HTBC) that generate base clocks for peripheral circuits. By using the time base counter, it is possible to generate events periodically.
For input clocks, see Chapter 6, "Clock Generation Circuit". For interrupt permission, interrupt request flags, etc., described in this chapter, see Chapter 5, "Interrupt".

7.1.1 Features

- LTBC generates T32KHZ to T1HZ signals by dividing the low-speed clock (LSCLK) frequency.
- LTBC allows frequency adjustment (Adjustment range: Approx. -488ppm to +488ppm. Adjustment accuracy: Approx. 0.48ppm) by using the low-speed time base counter frequency adjustment registers (LTBADJH and LTBADJL).
- HTBC generates HTB1 to HTB32 signals by dividing the high-speed clock (HSCLK) frequency.
- Capable of generating 128Hz , 32Hz , 16Hz , and 2Hz interrupts.

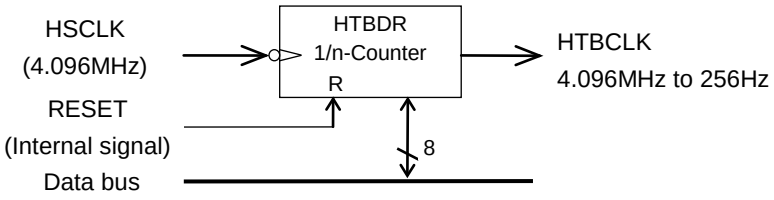
7.1.2 Configuration

Figure 7-1 and Figure 7-2 show the configuration of a low-speed time base counter and a high-speed time base counter, respectively.



LTBR : Low-speed time base counter register
LTBADJL : Low-speed time base counter frequency adjustment register
LTBADJH : Low-speed time base counter frequency adjustment register

Figure 7-1 Configuration of Low-Speed Time Base Counter (LTBC)



HTBDR : High-speed time base counter frequency divide register

Figure 7-2 Configuration of High-Speed Time Base Counter

Note:
The frequency of HSCLK changes according to specified data in SYSC1 bit and SYSC0 bit of Frequency control register 0 (FON0)

7.2 Description of Registers

7.2.1 List of Registers

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F00AH	Low-speed time base counter register	LTBR	−	R/W	8	00H
0F00BH	High-speed time base counter frequency divide register	HTBDR	−	R/W	8	00H
0F00CH	Low-speed time base counter frequency adjustment register L	LTBADJL	LTBADJ	R/W	8/16	00H
0F00DH	Low-speed time base counter frequency adjustment register H	LTBADJH		R/W	8	00H

7.2.2 Low-Speed Time Base Counter Register (LTBR)

Address: 0F00AH
Access: R/W
Access size: 8 bit
Initial value: 00H

	7	6	5	4	3	2	1	0
LTBR	T1HZ	T2HZ	T4HZ	T8HZ	T16HZ	T32HZ	T64HZ	T128HZ
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

LTBR is a special function register (SFR) to read the T128HZ to T1HZ outputs of the low-speed time base counter.
The T128HZ-T1HZ outputs are set to “0” when write operation is performed for LTBR. Write data is invalid.

Note:
A TBC interrupt (128Hz interrupt, 32Hz interrupt, 16Hz interrupt, or 2Hz interrupt) may occur depending on the LTBR write timing. Take this into consideration when programming your software by referring to "Figure 7-4 Interrupt Timing and Reset Timing by Writing to LTBR."

7.2.3 High-Speed Time Base Counter Divide Register (HTBDR)

Address: 0F00BH
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
HTBDR	—	—	—	—	HTD3	HTD2	HTD1	HTD0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

HTBDR is a special function register (SFR) to set the dividing ratio of the 4-bit, 1/n counter.

[Description of Bits]

- **HTD3-HTD0** (bits 3-0)
  The HTD3-HTD0 bits are used to set the dividing ratio of the 4-bit, 1/n counter. The frequency divide ratios selectable include 1/1 to 1/16.

HTD3	HTD2	HTD1	HTD0	Description	
				Dividing ratio	Frequency of HTBCLK (*1)
0	0	0	0	1/16 (initial value)	256kHz
0	0	0	1	1/15	273kHz
0	0	1	0	1/14	293kHz
0	0	1	1	1/13	315kHz
0	1	0	0	1/12	341kHz
0	1	0	1	1/11	372kHz
0	1	1	0	1/10	410kHz
0	1	1	1	1/9	455kHz
1	0	0	0	1/8	512kHz
1	0	0	1	1/7	585kHz
1	0	1	0	1/6	683kHz
1	0	1	1	1/5	819kHz
1	1	0	0	1/4	1024kHz
1	1	0	1	1/3	1365kHz
1	1	1	0	1/2	2048kHz
1	1	1	1	1/1	4096kHz

*1: Indicates the frequency when the high-speed oscillation clock “HSCLK” is 4096kHz.

7.2.4 Low-Speed Time Base Counter Frequency Adjustment Registers L and H (LTBADJL, LTBADJH)

Address: 0F00CH
Access: R/W
Access size: 8/16 bit
Initial value: 00H

	7	6	5	4	3	2	1	0
LTBADJL	LADJ7	LADJ6	LADJ5	LADJ4	LADJ3	LADJ2	LADJ1	LADJ0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

Address: 0F00DH
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
LTBADJH	—	—	—	—	—	LADJS	LADJ9	LADJ8
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

LTBADJL and LTBADJH are special function registers (SFRs) to set the frequency adjustment values of the low-speed time base clock.

[Description of Bits]

- **LADJS, LADJ9 to LADJ8** (bits 2 to 0) **LADJ7-LADJ0** (bits 7 to 0)
  The LADJS and LADJ9 to LADJ0 bits are used to adjust frequency.
  Adjustment range: approximately -488ppm to +488ppm, Adjustment accuracy: approximately 0.48ppm is possible. See Section 7.3.3, “Low-Speed Time Base Counter Frequency Adjustment Function” for the correspondence between the frequency adjustment values (LTBADJH, LTBADJL) and adjustment ratio.

7.3 Description of Operation

7.3.1 Low-speed Time Base Counter

The low-speed time base counter (LTBC) starts counting from 0000H on the LSCLK falling edge after system reset.
The T128HZ, T32HZ, T16HZ, and T2HZ outputs of LTBC are used as time base interrupts and an interrupt is requested on the falling edge of each output. Each of LTBC outputs is also used as an operation clock for peripheral circuits.
The output data of T128HZ to T1HZ of LTBC can be read from the low-speed time base counter register (LTBR). When reading the data, read LTBR twice and check that the two values coincide to prevent reading of undefined data during counting.
Figure 7-3 shows an example of program to read LTBR.

```
        LEA     offset LTBR     ; EA←LTBR address
MARK:
        L       R0,     [EA]    ; 1st read
        L       R1,     [EA]    ; 2nd read
;
        CMP     R0,     R1      ; Comparison for LTBR
        BNE     MARK            ; To MARK when the values do not coincide
;
```

Figure 7-3 Programming Example for Reading LTBR

LTBR is reset when write operation is performed and the T128HZ to T1HZ outputs are set to “0”. Write data is invalid. Since an interrupt occurs if a falling edge occurs in the T128Hz to T2HZ outputs during writing to LTBR, take care in software programming.
Figure 7-4 shows interrupt generation timing and reset timing of the time base counter output by writing to LTBR.

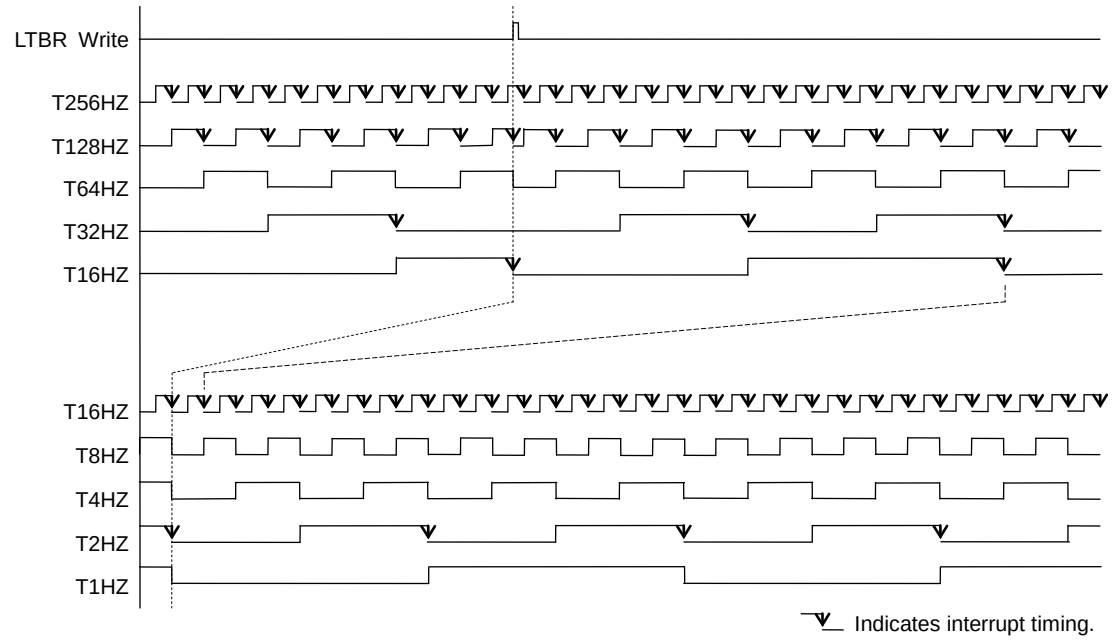


Figure 7-4 Interrupt Timing and Reset Timing by Writing to LTBR

7.3.2 High-Speed Time Base Counter

The high-speed time base counter is configured as a 4-bit 1/n counter (n = 1 to 16).
In the 4-bit 1/n counter, the divided clock (1/16 x HSCLK to 1/1 x HSCLK) selected by the high-speed time base counter divide register (HTBDR) is generated as HTBCLK. HTBCLK is used as a timer and also as an operation clock of PWM.
Figure 7-5 shows the output waveform of HTBCLK.

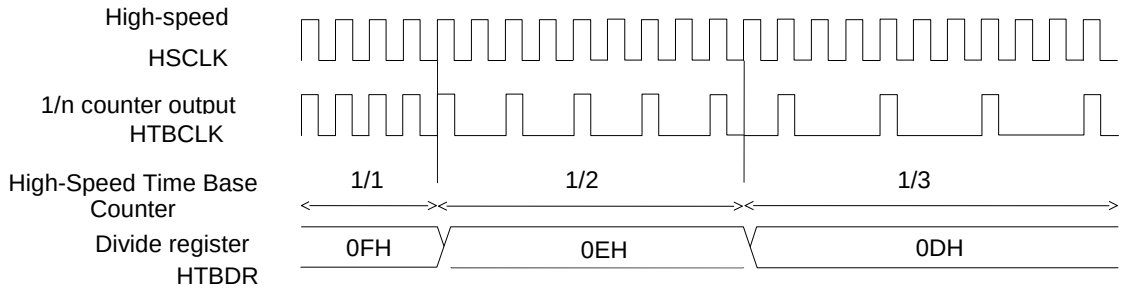


Figure 7-5 Output Waveform of HTBCLK

7.3.3 Low-Speed Time Base Counter Frequency Adjustment Function

Frequency adjustment (Adjustment range: Approx. -488ppm to +488ppm. Adjustment accuracy: Approx. 0.48ppm) is possible for outputs of T8KHZ to T1HZ of LTBC by using the low-speed time base counter frequency adjust registers (LTBADJH and LTBADJL).
Table7-1 shows correspondence between the frequency adjustment values (LTBADJH, LTBADJL) and adjustment ratio.

Table 7-1 Correspondence between Frequency Adjustment Values (LTBADJH, LTBADJL)and Adjustment Ratio

LADJ10～0											Hexadecimal	Frequency adjustment ratio (ppm)
0	1	1	1	1	1	1	1	1	1	1	3FFH	+487.80
0	1	1	1	1	1	1	1	1	1	0	3FEH	+487.33
:	:	:	:	:	:	:	:	:	:	:		:
0	0	0	0	0	0	0	0	0	1	1	003H	+1.43
0	0	0	0	0	0	0	0	0	1	0	002H	+0.95
0	0	0	0	0	0	0	0	0	0	1	001H	+0.48
0	0	0	0	0	0	0	0	0	0	0	000H	0
1	1	1	1	1	1	1	1	1	1	1	7FFH	-0.48
1	1	1	1	1	1	1	1	1	1	0	7FEH	-0.95
:	:	:	:	:	:	:	:	:	:	:		:
1	0	0	0	0	0	0	0	0	0	1	401H	-487.80
1	0	0	0	0	0	0	0	0	0	0	400H	-488.28

The adjustment values (LADJ10 to LADJ0) to be set in LTBADJH and LTBADJL can be obtained by using the following equations:

Adjustment value = Frequency adjustment ratio x 2097152 (decimal)
= Frequency adjustment ratio x 200000h (hexadecimal)

Example 1: When adjusting +15.0ppm (when the clock loses)
Adjustment value = +15.0ppm x 2097152 (decimal)
= +15.0×10-6×2097152
= +31.45728 (decimal)
≒ 01Fh (hexadecimal)

Example 2: When adjusting –25.5ppm (when the clock gains)
Adjustment value = –25.5ppm x 2097152 (decimal)
= –25.5×10-6×2097152
= –53.477376 (decimal)
≒ 7CCh (hexadecimal)

Note:
The low-speed clock (LSCLK) and the outputs of T32KHZ and T16KHZ of LTBC are not adjusted by the frequency adjust function.
The frequency adjustment accuracy does not guarantee the accuracy including the frequency variation of the crystal oscillation (32.768kHz) due to temperature variations.

Chapter 8

Capture

8. Capture

8.1 Overview

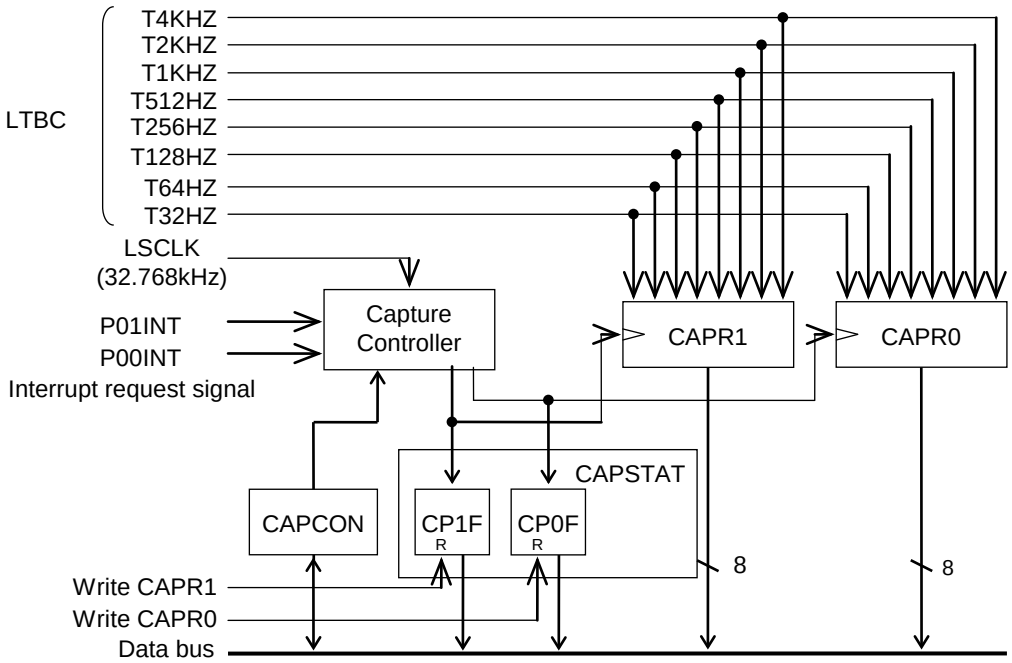
This LSI has two channels of capture circuits that capture the T4KHZ to T32HZ signals of the low-speed time base counter (LTBC) to the capture data register at the occurrence of P00 and P01 interrupts. The circuits capture timings at which each interrupt occurred, based on the time from the time base counter. It can also read out the T4KHZ to T32HZ signals of the low-speed time base counter (LTBC) at any timing.
For the external interrupt (P00INT, P01INT) from the P00 or P01 pin, see Chapter 5, "Interrupt" and Chapter 14, "Port 0".

8.1.1 Features

- Time base capture×2ch (4096Hz to 32Hz)

8.1.2 Configuration

Figure 8-1 shows the configuration of the capture circuit.



CAPCON : Capture control register
CAPSTAT : Capture status register
CAPR0 : Capture data register 0
CAPR1 : Capture data register 1
CAPTB : Capture time base data register

Figure 8-1 Configuration of Capture Circuit

8.1.3 List of Pins

Pin name	I/O	Description
P00/CAP0	I	Capture 0 input pin Used as the secondary function of the P00 pin.
P01/CAP1	I	Capture 1 input pin Used as the secondary function of the P01 pin.

8.2 Description of Registers

8.2.1 List of Registers

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F090H	Capture control register	CAPCON	—	R/W	8	00H
0F091H	Capture status register	CAPSTAT	—	R/W	8	00H
0F092H	Capture data register 0	CAPR0	—	R/W	8	00H
0F093H	Capture data register 1	CAPR1	—	R/W	8	00H
0F094H	Capture time base data register	CAPTB	—	R	8	Undefined

8.2.2 Capture Control Register (CAPCON)

Address: 0F090H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
CAPCON	—	—	—	—	—	—	ECAP1	ECAP0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

CAPCON is a special function register (SFR) to control the capture circuit.

[Description of Bits]

- **ECAP0** (bit 0)
  The ECAP0 bit is used to start or stop the operation of capture 0.

ECAP0	Description
0	Stops the capture 0 operation. (initial value)
1	Starts the capture 0 operation.

- **ECAP1** (bit 1)
  The ECAP1 bit is used to start or stop the operation of capture 1.

ECAP1	Description
0	Stops the capture 1 operation. (initial value)
1	Starts the capture 1 operation.

8.2.3 Capture Status Register (CAPSTAT)

Address: 0F091H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
CAPSTAT	—	—	—	—	—	—	CAPF1	CAPF0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

CAPSTAT is a special function register (SFR) to indicate a state of the capture circuit.

[Description of Bits]

- **CAPF0** (bit 0)
  The CAPF0 bit is the flag to indicate whether data is captured in capture data register 0 (CARP0) or not.
  When the CAPF0 bit is set to "1", it indicates that data is captured in capture data register 0 (CAPR0).
  When the CAPF0 bit is set to "1", the next capture operation is stopped. So perform the write operation to capture data register 0 (CAPR0) to clear the CAPF0 bit to "0".

CAPF0	Description
0	No capture 0 latch (initial value)
1	Capture 0 latch

- **CAPF1** (bit 1)
  The CAPF1 bit is the flag to indicate whether data is captured in capture data register 0 (CARP1) or not.
  When the CAPF0 bit is set to "1", it indicates that data is captured in capture data register 0 (CAPR1).
  When the CAPF1 bit is set to "1", the next capture operation is stopped. So perform the write operation to capture data register 1 (CAPR1) to clear theCAPF0 bit to "0".

CAPF1	Description
0	No capture 1 latch (initial value)
1	Capture 1 latch

8.2.4 Capture Data Register 0 (CAPR0)

Address: 0F092H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
CAPR0	CP07	CP06	CP05	CP04	CP03	CP02	CP01	CP00
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

CAPR0 is a register in which capture data is stored.
The T4KHZ to T32HZ signals of the low-speed time base counter (LTBC) are captured when the P00 interrupt request is generated with the CAPF0 flag (bit 0 of the CAPSTAT register) set to "0".
Writing to CAPR0 sets the CAPF0 flag of CAPSTAT to "0". The value of CAPR0 does not change even if data is written to it.

8.2.5 Capture Data Register 1 (CAPR1)

Address: 0F093H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
CAPR1	CP17	CP16	CP15	CP14	CP13	CP12	CP11	CP10
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

CAPR1 is a register in which capture data is stored.
The T4KHZ to T32HZ signals of the low-speed time base counter (LTBC) are captured when the P01 interrupt request is generated with the CAPF1 flag (bit 1 of the CAPSTAT register) set to "0".
Writing to CAPR1 sets the CAPF1 flag of CAPSTAT to "0". The value of CAPR1 does not change even if data is written to it.

8.2.6 Capture Time Base Data Register (CAPTB)

Address: 0F094H
Access: R
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
CAPTB	CPTB7	CPTB6	CPTB5	CPTB4	CPTB3	CPTB2	CPTB1	CPTB0
R/W	R	R	R	R	R	R	R	R
Initial value	X	X	X	X	X	X	X	X

CAPTB is a special function register (SFR) to read the T4KHZ to T32HZ outputs of the low-speed time base counter (LTBC). The initial value varies depending on the state of the T4KHZ to T32HZ outputs of the low-speed time base counter (LTBC) at the timing when this register is read. When reading the data, read this register twice and check that the two values coincide to prevent reading of undefined data during counting.

8.3 Description of Operation

The capture circuit starts the capture operation by setting the ECAP0 or ECAP1 bit of the capture control register (CAPCON).
When the input trigger from the P00 or P01 pin selected by the external interrupt control register 0 or 1 (EXICON0 or EXICON1) is generated and the P00 or P01 interrupt request flag (QP00 or QP01) is set to "1", the T4KHZ to T32HZ signals of the low-speed time base counter (LTBC) are captured in the capture register 0 or 1 (CAPR0 or CAPR1) on the next low-speed clock (LSCLK) falling edge and the at the same time, the capture flag (CAPF0 or CAPF1) of the capture status register (CAPSTAT) is set to "1".
When the capture flag (CAPF0, CAPF1) is "1", the following capture operation stops.
After reading the value captured in the capture register 0 or 1 (CAPR0, CAPR1), perform write operation (write data is meaningless) for the capture register 0 or 1 (CAPR0, CAPR1), clear the capture flag (CAPF0, CAPF1) to "0", and wait for the next P00 or P01 interrupt.
Figure 8-2 shows the timing of the capture operation.

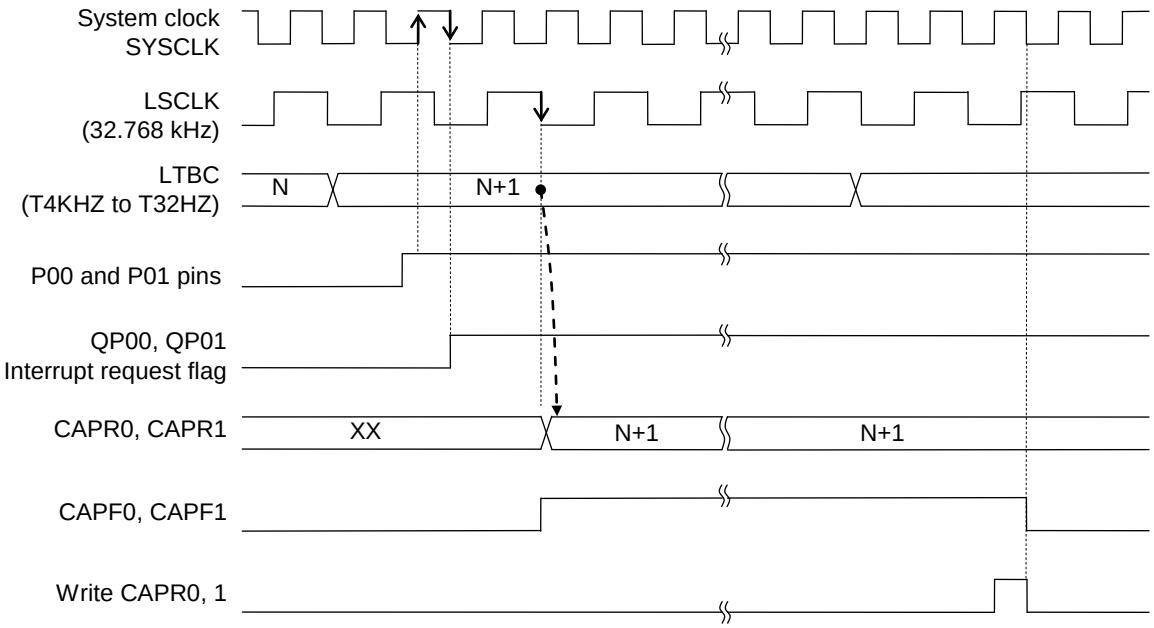


Figure 8-2 Timing Diagram of Capture Operation

Note:
When CPU is operating at the high speed (HSCLK), check that the capture flag (CAPF0, CAPF1) is set to "1" after the P00 or P01 interrupt request is generated and then read capture data register 0 or 1 (CAPR0, CAPR1).

Chapter 9

Timers

9. Timers

9.1 Overview

This LSI includes 4 channels of 8-bit timers.

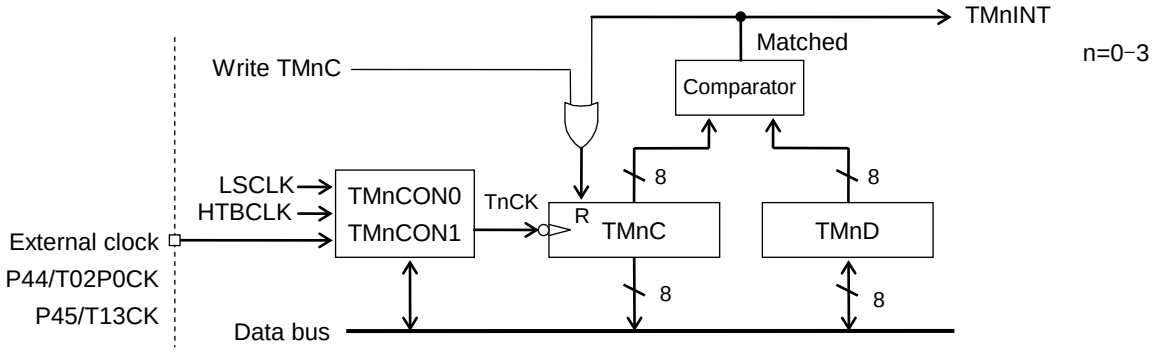
9.1.1 Features

- The timer interrupt (TMnINT) is generated when the values of timer counter register (TMnC, n=0 to 3) and timer data register (TMnD) coincide.
- A timer configured by combining timer 0 and timer 1 or timer 2 and timer 3 can be used as a 16-bit timer.
- For the timer clock, the low-speed clock (LSCLK), high-speed time base clock (HTBCLK), or external clock can be selected.
- A 16bit-timer 2 & 3 has clock frequency measurement mode, which can count HTBCLK and generates the timer interrupt (TM3INT) when the count ends.
  Using the count data to know the frequency by software can determine more accurate baud-rate.

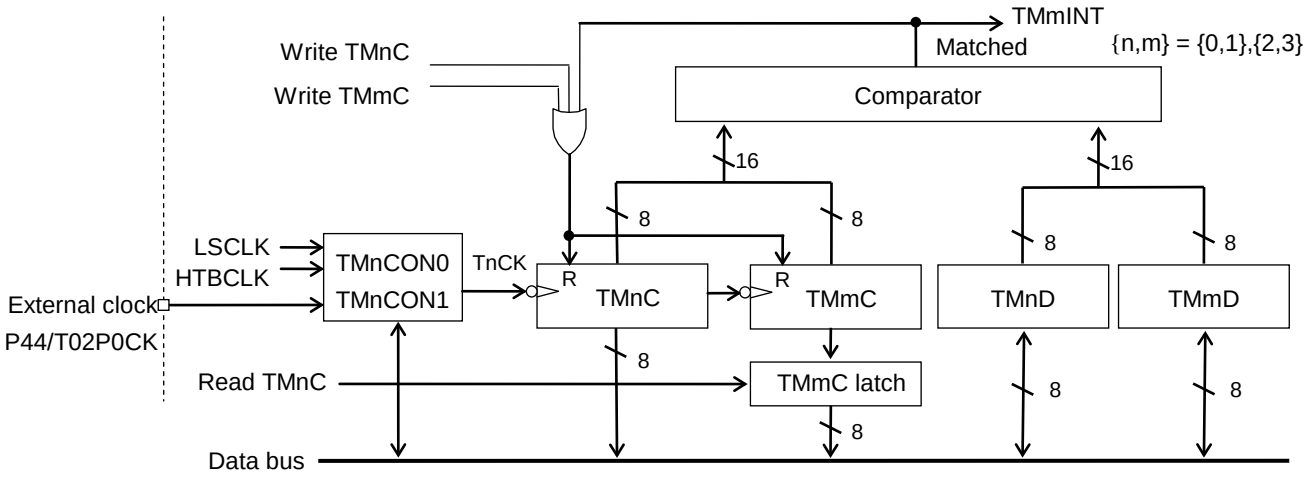
9.1.2 Configuration

Figure 9-1 shows the configuration of the timers.

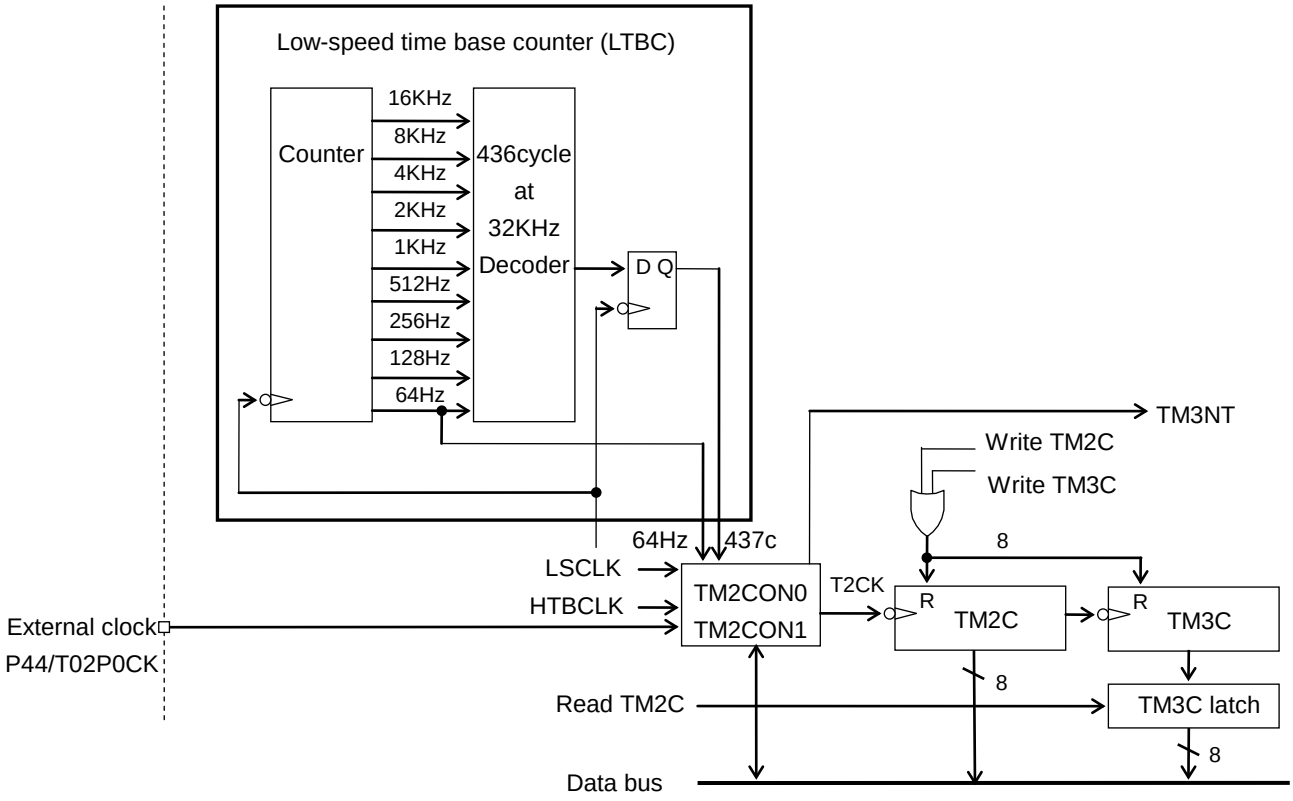
TMnCON0 : Timer control register 0
TMnCON1 : Timer control register 1
TMmD, TMnD : Timer data register
TMmC, TMnC : Timer counter register



(a) In 8-bit Timer Mode (Timers 0 to 3)



(b) In 16-bit Timer Mode (Timers 0 to 3)



(c) Frequency measurement mode with 16 bit timer(Timer2 to 3)

Figure 9-1 Timer Configuration

9.1.3 List of Pins

Pin name	I/O	Function
P44/T02P0CK	I	External clock input pin 8bit timer mode : used for timer0 or timer2 16bit timer mode : used for timer0 to timer1 or timer2 to timer3.
P45/T13CK	I	External clock input pin 8bit timer mode : used for timer1 or timer3

9.2 Description of Registers

9.2.1 List of Registers

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F030H	Timer 0 data register	TM0D	TM0DC	R/W	8/16	0FFH
0F031H	Timer 0 counter register	TM0C		R/W	8	00H
0F032H	Timer 0 control register 0	TM0CON0	TM0CON	R/W	8/16	00H
0F033H	Timer 0 control register 1	TM0CON1		R/W	8	00H
0F034H	Timer 1 data register	TM1D	TM1DC	R/W	8/16	0FFH
0F035H	Timer 1 counter register	TM1C		R/W	8	00H
0F036H	Timer 1 control register 0	TM1CON0	TM1CON	R/W	8/16	00H
0F037H	Timer 1 control register 1	TM1CON1		R/W	8	00H
0F038H	Timer 2 data register	TM2D	TM2DC	R/W	8/16	0FFH
0F039H	Timer 2 counter register	TM2C		R/W	8	00H
0F03AH	Timer 2 control register 0	TM2CON0	TM2CON	R/W	8/16	0A0H
0F03BH	Timer 2 control register 1	TM2CON1		R/W	8	00H
0F03CH	Timer 3 data register	TM3D	TM3DC	R/W	8/16	0FFH
0F03DH	Timer 3 counter register	TM3C		R/W	8	00H
0F03EH	Timer 3 control register 0	TM3CON0	TM3CON	R/W	8/16	00H
0F03FH	Timer 3 control register 1	TM3CON1		R/W	8	00H

9.2.2 Timer 0 Data Register (TM0D)

Address: 0F030H
Access: R/W
Access size: 8-bit
Initial value: 0FFH

	7	6	5	4	3	2	1	0
TM0D	T0D7	T0D6	T0D5	T0D4	T0D3	T0D2	T0D1	T0D0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	1	1	1	1	1	1	1	1

TM0D is a special function register (SFR) to set the value to be compared with the timer 0 counter register (TM0C) value.

Note:
Set TM0D when the timer stops(When T0STAT bit of TM0CON1 register is “0”).
When “00H” is written in TM0D, TM0D is set to “01H”.

9.2.3 Timer 1 Data Register (TM1D)

Address: 0F034H
Access: R/W
Access size: 8-bit
Initial value: 0FFH

	7	6	5	4	3	2	1	0
TM1D	T1D7	T1D6	T1D5	T1D4	T1D3	T1D2	T1D1	T1D0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	1	1	1	1	1	1	1	1

TM1D is a special function register (SFR) to set the value to be compared with the value of the timer 1 counter register (TM1C).

Note:
Set TM1D when the timer stops(When T1STAT bit of TM1CON1 register is “0”).
When “00H” is written in TM1D, TM1D is set to “01H”.

9.2.4 Timer 2 Data Register (TM2D)

Address: 0F038H
Access: R/W
Access size: 8-bit
Initial value: 0FFH

	7	6	5	4	3	2	1	0
TM2D	T2D7	T2D6	T2D5	T2D4	T2D3	T2D2	T2D1	T2D0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	1	1	1	1	1	1	1	1

TM2D is a special function register (SFR) to set the value to be compared with the value of the timer 2 counter register (TM2C).

Note:
Set TM2D when the timer stops(When T2STAT bit of TM2CON1 register is “0”).
When “00H” is written in TM2D, TM2D is set to “01H”.

9.2.5 Timer 3 Data Register (TM3D)

Address: 0F03CH
Access: R/W
Access size: 8-bit
Initial value: 0FFH

	7	6	5	4	3	2	1	0
TM3D	T3D7	T3D6	T3D5	T3D4	T3D3	T3D2	T3D1	T3D0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	1	1	1	1	1	1	1	1

TM3D is a special function register (SFR) to set the value to be compared with the value of the timer 3 counter register (TM3C).

Note:
Set TM3D when the timer stops(When T3STAT bit of TM3CON1 register is “0”).
When “00H” is written in TM3D, TM3D is set to “01H”.

9.2.6 Timer 0 Counter Register (TM0C)

Address: 0F031H
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
TM0C	T0C7	T0C6	T0C5	T0C4	T0C3	T0C2	T0C1	T0C0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

TM0C is a special function register (SFR) that functions as an 8-bit binary counter.
When write operation to TM0C is performed, TM0C is set to "00H". The data that is written is meaningless.
In 16-bit timer mode and 16-bit timer frequency measurement mode, if write operation is performed to either the low-order TM0C or high-order TM1C, both the low-order and the high-order are set to "0000H".

During timer operation, the TM0C content may not be read depending on the conditions of the timer clock and the system clock.
Table 9-1 shows whether a TM0C read is enabled or disabled during timer operation for each condition of the timer clock and system clock.

Table 9-1 TM0C Read Enable/Disable during Timer Operation

Timer clock T0CK	System clock SYSCLK	TM0C read enable/disable
LSCLK	LSCLK	Read enabled.
LSCLK	HSCLK	Read enabled. However, to prevent the reading of undefined data during incremental counting, read TM0C twice and check that the results match.
HTBCLK	LSCLK	Read disabled.
HTBCLK	HSCLK	Read enabled.
External clock	LSCLK	Read disabled.
	HSCLK	

9.2.7 Timer 1 Counter Register (TM1C)

Address: 0F035H
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
TM1C	T1C7	T1C6	T1C5	T1C4	T1C3	T1C2	T1C1	T1C0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

TM1C is a special function register (SFR) that functions as an 8-bit binary counter.
When write operation to TM1C is performed, TM1C is set to "00H". The data that is written is meaningless.
In 16-bit timer mode, if write operation is performed to either the low-order TM0C or high-order TM1C, both the low order and the high order are set to "0000H".
When reading TM1C in 16-bit timer mode, be sure to read TM0C first since the count value of TM1C is stored in the TM1C latch when TM0C is read.

During timer operation, the TM1C content may not be read depending on the conditions of the timer clock and the system clock.
Table 9-2 shows whether a TM1C read is enabled or disabled during timer operation for each condition of the timer clock and system clock.

Table 9-2 TM1C Read Enable/Disable during Timer Operation

Timer clock T1CK	System clock SYSCLK	TM1C read enable/disable
LSCLK	LSCLK	Read enabled.
LSCLK	HSCLK	Read enabled. However, to prevent the reading of undefined data during incremental counting, read TM1C twice and check that the results match.
HTBCLK	LSCLK	Read disabled.
HTBCLK	HSCLK	Read enabled.
External clock	LSCLK	Read disabled.
	HSCLK	

9.2.8 Timer 2 Counter Register (TM2C)

Address: 0F039H
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
TM2C	T2C7	T2C6	T2C5	T2C4	T2C3	T2C2	T2C1	T2C0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

TM2C is a special function register (SFR) that functions as an 8-bit binary counter.
When write operation to TM2C is performed, TM2C is set to "00H". The data that is written is meaningless.
In 16-bit timer mode and 16-bit timer frequency measurement mode, if write operation is performed to either the low-order TM2C or high-order TM3C, both the low order and the high order are set to "0000H".

During timer operation, the TM2C content may not be read depending on the conditions of the timer clock and the system clock.
Table 9-3 shows whether a TM2C read is enabled or disabled during timer operation for each condition of the timer clock and system clock.

Table 9-3 TM2C Read Enable/Disable during Timer Operation

Timer clock T2CK	System clock SYSCLK	TM2C read enable/disable
LSCLK	LSCLK	Read enabled.
LSCLK	HSCLK	Read enabled. However, to prevent the reading of undefined data during incremental counting, read TM2C twice and check that the results match.
HTBCLK	LSCLK	Read disabled.
HTBCLK	HSCLK	Read enabled.
External clock	LSCLK	Read disabled.
	HSCLK	

9.2.9 Timer 3 Counter Register (TM3C)

Address: 0F03DH
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
TM3C	T3C7	T3C6	T3C5	T3C4	T3C3	T3C2	T3C1	T3C0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

TM3C is a special function register (SFR) that functions as an 8-bit binary counter.
When write operation to TM3C is performed, TM3C is set to "00H". The data that is written is meaningless.
In 16-bit timer mode and 16-bit timer frequency measurement mode, if write operation is performed to either the low-order (TM2C) or high-order (TM3C), both the low order and the high order are set to "0000H".
When reading TM3C in 16-bit timer mode, be sure to read TM2C first since the count value of TM3C is stored in the TM3C latch when TM2C is read.

During timer operation, the TM3C content may not be read depending on the conditions of the timer clock and the system clock.
Table 9-4 shows whether a TM3C read is enabled or disabled during timer operation for each condition of the timer clock and system clock.

Table 9-4 TM3C Read Enable/Disable during Timer Operation

Timer clock T3CK	System clock SYSCLK	TM3C read enable/disable
LSCLK	LSCLK	Read enabled.
LSCLK	HSCLK	Read enabled. However, to prevent the reading of undefined data during incremental counting, read TM3C twice and check that the results match.
HTBCLK	LSCLK	Read disabled.
HTBCLK	HSCLK	Read enabled.
External clock	LSCLK	Read disabled.
	HSCLK	

9.2.10 Timer 0 Control Register 0 (TM0CON0)

Address: 0F032H
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
TM0CON0	—	—	—	—	—	T01M16	T0CS1	T0CS0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

TM0CON0 is a special function register (SFR) to control the Timer 0.
Rewrite TM0CON0 while the timer 0 is stopped (T0STAT of the TM0CON1 register is "0").

[Description of Bits]

- **T0CS1, T0CS0** (bits 1, 0)
  The T0CS1 and T0CS0 bits are used for selecting the operation clock of timer 0. LSCLK, HTBCLK, or the external clock (P44/T02P0CK) can be selected by these bits.

T0CS1	T0CS0	Description
0	0	LSCLK (initial value)
0	1	HTBCLK
1	0	Prohibited
1	1	External clock (P44/T02P0CK)

- **T01M16** (bit 2)
  The T01M16 bit is used for selecting the operating mode of timer 0 and timer 1.
  In 8-bit timer mode, each of timer 0 and timer 1 operates independently as a 8-bit timer.
  In 16-bit timer mode, timer 0 and timer 1 are connected and they operate as a 16-bit timer.
  In 16-bit timer mode, timer 1 is incremented by a timer 0 overflow signal.
  A timer 0 interrupt (TM0INT) is not generated.

T01M16	Description
0	8-bit timer mode (initial value)
1	16-bit timer mode

9.2.11 Timer 1 Control Register 0 (TM1CON0)

Address: 0F036H
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
TM1CON0	—	—	—	—	—	—	T1CS1	T1CS0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

TM1CON0 is a special function register (SFR) to control the Timer 1.
Rewrite TM1CON0 while the timer 1 is stopped (T1STAT of the TM1CON1 register is “0”).

[Description of Bits]

- **T1CS1, T1CS0** (bits 1, 0)
  The T1CS1 and T1CS0 bits are used for selecting the operation clock of timer 1. LSCLK, HTBCLK, or the external clock (P45/T13CK) can be selected by these bits.
  In cases where the 16-bit timer mode has been selected by setting T01M16 of TM0CON to “1”, the values of T1CS1 and T1CS0 are invalid.

T1CS1	T1CS0	Description
0	0	LSCLK (initial value)
0	1	HTBCLK
1	0	Prohibited
1	1	External clock (P45/T13CK)

9.2.12 Timer 2 Control Register 0 (TM2CON0)

Address: 0F03AH
Access: R/W
Access size: 8-bit
Initial value: A0H

	7	6	5	4	3	2	1	0
TM2CON0	T2FMA7	T2FMA6	T2FMA5	T2FMA4	T23MFM	T23M16	T2CS1	T2CS0
R/W	R	R	R	R	R/W	R/W	R/W	R/W
Initial value	1	0	1	0	0	0	0	0

TM2CON0 is a special function register (SFR) to control the Timer 2.
Rewrite TM2CON0 while the timer 2 is stopped (T2STAT of the TM2CON1 register is “0”).

[Description of Bits]

- **T2CS1, T2CS0** (bits 1, 0)
  The T2CS1 and T2CS0 bits are used for selecting the operation clock of timer 2. LSCLK, HTBCLK, or the external clock (P44/T02P0CK) can be selected by these bits.

T2CS1	T2CS0	Description
0	0	LSCLK (initial value)
0	1	HTBCLK
1	0	Prohibited
1	1	External clock (P44/T02P0CK)

- **T23MFM, T23M16** (bit 3, 2)
  The T23MFM bit and T23M16 bit is used for selecting the operating mode of timer 2 and timer 3..
  In 8-bit timer mode, each of timer 0 and timer 1 operates independently as a 8-bit timer.
  In 16-bit timer mode, timer 2 and timer 3 are connected and they operate as a 16-bit timer.
  In 16-bit timer mode, timer 3 is incremented by a timer 2 overflow signal.
  A timer 2 interrupt (TM2INT) is not generated.
  In 16-bit timer frequency measurement mode, timer 2 and timer 3 are connected and they operate as a 16-bit clock counter to measure the frequency.
  A timer 2 interrupt (TM2INT) is not generated.
  When this LSI does not have the frequency measurement function (T2FMA7 to T2FMA5 bit values are not "1010"), the T23MFM bit is read-only and the readout is always "0" regardless of the write access value.

T23MFM	T23M16	Description
0	0	8-bit timer mode (initial value)
0	1	16-bit timer mode
1	0	Prohibited
1	1	16-bit timer frequency measurement mode

- **T2FMA7 to T2FMA4** (bit 7 to 4)
  T2FMA7 to T2FMA4 are read-only registers that show whether the frequency measurement function is provided.
  The readout is always "0A".

T2FMA7	T2FMA6	T2FMA5	T2FMA4	Description
1	0	1	0	The frequency measurement mode is available

9.2.13 Timer 3 Control Register 0 (TM3CON0)

Address: 0F03EH
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
TM3CON0	—	—	—	—	—	—	T3CS1	T3CS0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

TM3CON0 is a special function register (SFR) to control the Timer 3.
Rewrite TM3CON0 while the timer 3 is stopped (T3STAT of the TM3CON1 register is “0”).

[Description of Bits]

- **T3CS1, T3CS0** (bits 1, 0)
  The T3CS1 and T3CS0 bits are used for selecting the operation clock of timer 3. LSCLK, HTBCLK, or the external clock (P44/T13CK) can be selected by these bits.
  In cases where the 16-bit timer mode has been selected by setting T23M16 of TM2CON to “1”, the values of T3CS1 and T3CS0 are invalid.

T3CS1	T3CS0	Description
0	0	LSCLK (initial value)
0	1	HTBCLK
1	0	Prohibited
1	1	External clock (P45/T13CK)

9.2.14 Timer 0 Control Register 1 (TM0CON1)

Address: 0F033
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
TM0CON1	T0STAT	—	—	—	—	—	—	T0RUN
R/W	R	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

TM0CON1 is a special function register (SFR) to control the Timer 0.

[Description of Bits]

- **T0RUN** (bit 0)
  The T0RUN bit is used for controlling stop/start of timer 0.

T0RUN	Description
0	Stops counting.
1	Starts counting.

- **T0STAT** (bit 7)
  The T0STAT bit is used for indicating “counting stopped”/”counting in progress” of timer 0.

T0STAT	Description
0	Counting stopped.
1	Counting in progress.

9.2.15 Timer 1 Control Register 1 (TM1CON1)

Address: 0F037H
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
TM1CON1	T1STAT	—	—	—	—	—	—	T1RUN
R/W	R	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

TM1CON1 is a special function register (SFR) to control the Timer 1.

[Description of Bits]

- **T1RUN** (bit 0)
  The T1RUN bit is used for controlling count stop/start of timer 1.
  In 16-bit timer mode, be sure to set this bit to "0". Regardless of the T1RUN value, the Timer 1 counts up by the Timer 0 overflow signal.

T1RUN	Description
0	Stops counting.
1	Starts counting.

- T1STAT (bit 7)
  The T1STAT bit is used for indicating "counting stopped"/"counting in progress" of timer 1.
  In 16-bit timer mode, this bit will read "0".

T1STAT	Description
0	Counting stopped.
1	Counting in progress.

9.2.16 Timer 2 Control Register 1 (TM2CON1)

Address: 0F03BH
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
TM2CON1	T2STAT	—	—	—	—	—	—	T2RUN
R/W	R	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

TM2CON1 is a special function register (SFR) to control the Timer 2.

[Description of Bits]

- **T2RUN** (bit 0)
  The T2RUN bit is used for controlling stop/start of timer 2.
  Setting the T2RUN bit can force cancel the counting in the 16-bit timer frequency measurement mode.
  In that case, TM3INT does not occur.

T2RUN	Description
0	In timer mode: Stops counting. In frequency measurement mode: Stops measurement
1	In timer mode: Starts counting. In frequency measurement mode: Starts measurement

- **T2STAT** (bit 7)
  The T2STAT bit is used for indicating “counting stopped”/”counting in progress” of timer 2.

T2STAT	Description
0	In timer mode: Count halted. In frequency measurement mode: Measurement halted
1	In timer mode: Counting in progress. In frequency measurement mode: Measuring

9.2.17 Timer 3 Control Register 1 (TM3CON1)

Address: 0F03FH
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
TM3CON1	T3STAT	—	—	—	—	—	—	T3RUN
R/W	R	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

TM3CON1 is a special function register (SFR) to control the Timer 3.

[Description of Bits]

- **T3RUN** (bit 0)
  The T3RUN bit is used for controlling stop/start of timer 3.
  In 16-bit timer mode and 16-bit timer frequency measurement mode, be sure to set this bit to "0". Regardless of the T3RUN value, the Timer 3 counts up by the Timer 2 overflow signal.
  In 16-bit timer frequency measurement mode, be sure to set this bit to "0" also.

T3RUN	Description
0	Stops counting.
1	Starts counting.

- **T3STAT** (bit 7)
  The T3STAT bit is used for indicating "counting stopped"/"counting in progress" of timer 3.
  In 16-bit timer mode and 16-bit timer frequency measurement mode, this bit will return "0".

T3STAT	Description
0	Counting stopped.
1	Counting in progress.

9.3 Description of Operation

9.3.1 Timer mode operation

The timer counters (TMnC) are set to an operating state (TnSTAT are set to "1") on the first falling edge of the timer clocks (TnCK) that are selected by the Timer 0 to 3 control register 0 (TMnCON0) when the TnRUN bits of timer 0 to 3 control register 1 (TMnCON1) are set to "1" and increment the count value on the 2nd falling.

When the count value of TM0 to TM3C and the timer 0 to 3 data register (TMnD) coincide, timer 0 to 3 interrupt (TMnINT) occurs on the next timer clock falling edge, TMnC are reset to "00H" and incremental counting continues.

When the TnRUN bits are set to "0", TMnC stop counting after counting once the falling of the timer clock (TnCK). Confirm that TMnC has been stopped by checking that the TnSTAT bit of the Timer 0–3 control register 1 (TMnCON1) is "0".
When the TnRUN bits are set to "1" again, TMn restart incremental counting from the previous values.
To initialize TMnC to "00H", perform write operation in TMnC.

The timer interrupt period (TTMI) is expressed by the following equation.

$$T_{TMI} = \frac{TMnD + 1}{TnCK\ (Hz)} \quad (n=0 \text{ to } 3)$$

TMnD: Timer 0 to 3 data register (TMnD) setting value (01H to 0FFH)
TnCK: Clock frequency selected by the Timer 0 to 3 control register 0 (TMnCON0)

After the TnRUN bits are set to "1", timers are synchronized by the timer clock and counting starts so that an error of a maximum of 1 clock period occurs until the first timer interrupt. The timer interrupt periods from the second time are constant.
Figure 9-2 shows the operation timing diagram of Timer 0 to 3.

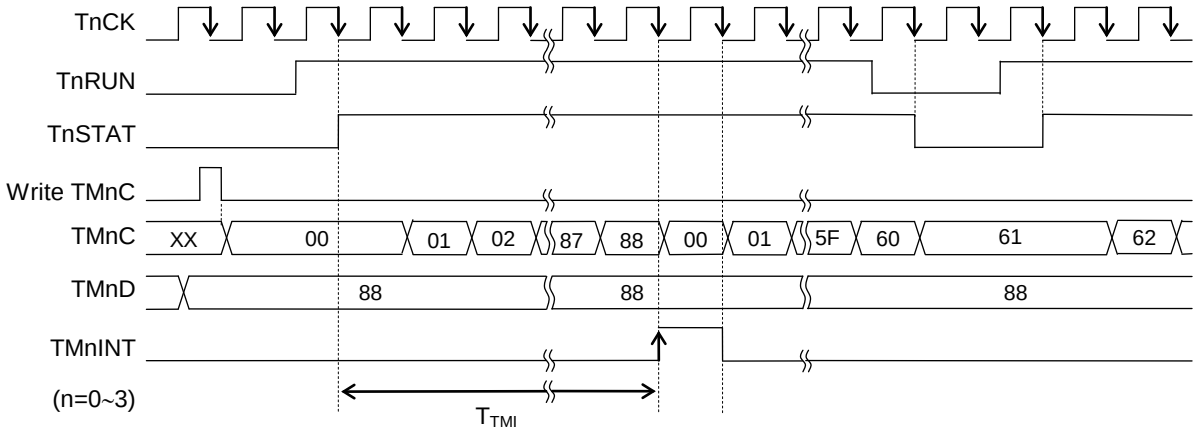


Figure 9-2 Operation Timing Diagram of Timer 0 to 3

Note:
Even if "0" is written to the TnRUN bits, counting operation continues up to the falling edge (the timer 0 to 3 status flag (TnSTA) is in a "1" state) of the next timer clock pulse. Therefore, the timer 0 to 3 interrupt (TMnINT) may occur.

9.3.2 16-bit timer frequency measurement mode operation

The frequency measurement mode in 16-bit timer 2&3, is used to count the frequency of 500kHz RC oscillation clock which typically has temperature variation and production tolerance.

Using the frequency measurement mode can make better accuracy for uart baud-rate clock or timer function.
(1) Reading the count data, calculating and setting it to uart communication baud-rate registers, can make more accurate baud-rate clock.
(2) Reading the count data, calculating and setting it to a timer data register, can make more accurate timing in normal timer mode.

Figure 9-3 shows the operation timing in frequency measurement mode.

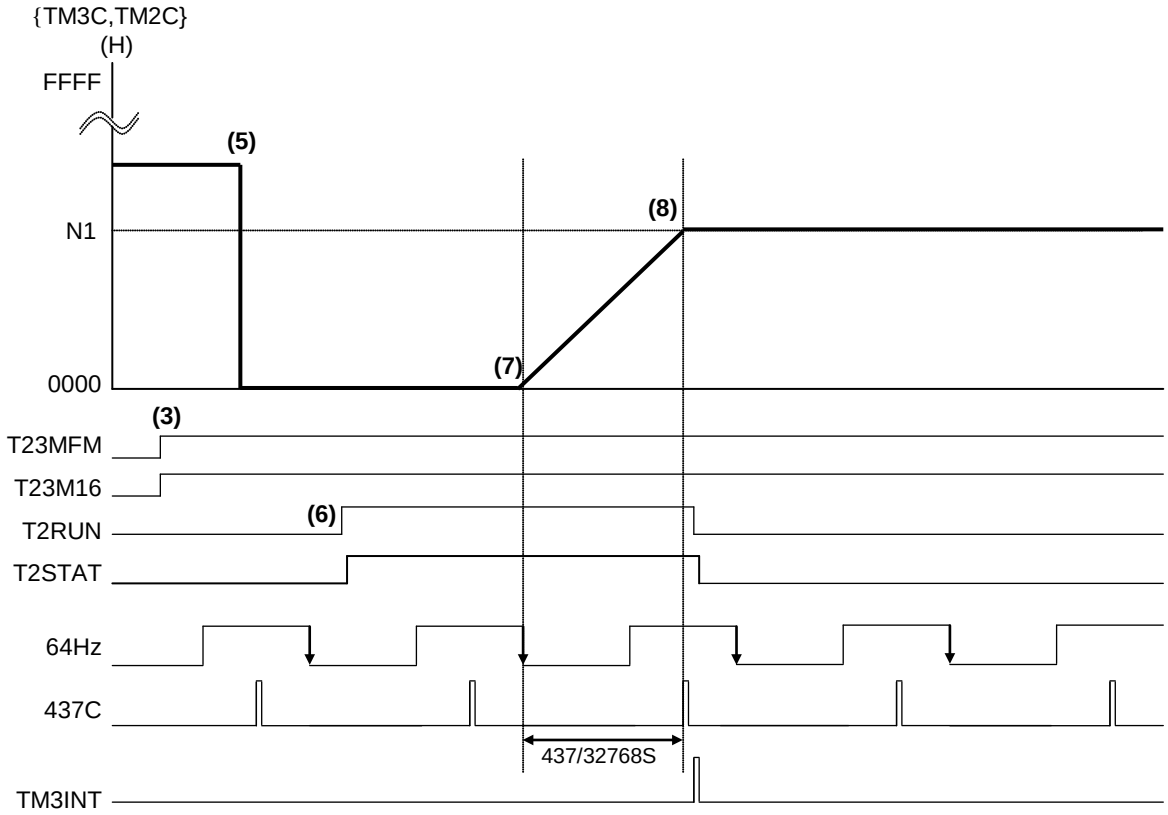


Figure 9-3 Operation Timing in frequency measurement mode

(1) High-speed clock (HSCLK, HTBCLK) has to be in oscillating state by controlling with FCONn registers. And also select 1/1 divide ratio of the high-speed time base counter by setting HTBDR (High-speed Time Base counter Divide Register) register to 0FH.

(2) Reset both T2RUN bit (bit0 of TM2CON1 register) and T3RUN bit (bit0 of TM3CON1 register) to "0" to stop the timer. And then, check both T2STAT bit (bit7 of TM2CON1 register) and T3STAT bit (bit7 of TM3CON1 register) are "0" for making certain the timer stops.

(3) Set T23MFM bit (bit3 of TM2CON0 register) to "1" (Frequency measurement mode), set T23M16 bit (bit2 of TM2CON0 register) to "1" (16bit mode) and set T2CS1-0 bits(bit1/0 of TM2CON0 register) to "01"(HTBCLK mode).

(4) Set "FFH" to both TM2D register and TM3D register.

(5) Clear both TM2C register and TM3C register to “00H”.

(6) Set T2RUN bit (bit0 of TM2CON1 register) to “1” to start counting the timer.

(7) When (T23MFM bit == "1") and (TM23M16 bit == "1") and (T2RUN bit == "1"),
the count up starts with the falling of the 64Hz clock signal.

(8) The count-up stops at the falling edge of the next timer clock (HTBCLK) after 437C signal becomes “1”.
Also, at the same time, T2RUN bit and T2STAT bit become “0” and the interrupt signal TM3INT activates.
The 437C signal is the pulse signal that rises in 437/32768 seconds after the falling of the 64Hz signal.

(9) After checking T2STAT bit or TM3INT interrupt occurs, read out the data (N1) of TM2C register and TM3C register.

For example of utilizing N1, 9600Hz timer interrupt is generated.

Assuming a low-accuracy high-speed clock (HTBCLK) is exactly 600kHz, then the count value N1 is:

N1 = 600000 × 437 / 32768
= 8001 (Decimal)
= 1F41 (Hexadecimal)
= 0001 1111 0100 0001 (Binary)

Because 437/32768 seconds are equivalent to 128 clocks at 9600 Hz (more precisely, 9598 Hz),
dividing the count value NI by 128 provides the frequency ratio (N2) between the HTBCLK and 9600 Hz.
Because 128 = 2^7, this calculation can be solved by truncating the right-hand seven digits of N1 (Binary), that is:

N2 = 8001(Decimal) / 128 (Decimal)
=0001 1111 0 (Binary)
=3E (Hexadecimal)
=62 (Decimal)

This indicates that 9600Hz is about 62 times the cycle of HTBCLK.
Therefore, for an operation in the timer mode where the timer data register is set to
3EH – 1 = 3DH to overflow the counter every 62 counts of the HTBCLK clock,
the TMnINT interrupt signal cycle tTMnINT is:

tTMnINT = 1 / 600000 × 62 = 0.10333ms(9677Hz)

9.3.3 16-bit timer frequency measurement mode application for setting uart baud-rate

For example, when the target baud-rate is 9600bps and the clock is HSCLK(500kHz), the UART0 baud-rate register (UA0BRTH, UA0BRTL) should be set as:

UA0BRTH, UA0BRTL = 500000/9600 – 1 = 51 (decimal) = 33 (hexadecimal) (See Section 14.3.2.)

However, actual 500kHz RC oscillation clock has temperature variation and production tolerance, the calculation by using the fixed value of 500kHz cannot make accurate baud-rate. To compensate it, count the frequency in the frequency measurement mode to set the baud-rate again before operating UART communication.

After finishing the clock count in the frequency measurement mode, assuming HTBCLK is 451kHz, data of TM2C register and TM3C register will be:

N1 = 451000 × 437 / 32768
= 6014 (Decimal)
= 177E (Hexadecimal)
= 101110<u>1111110 (Binary)</u>

As (437 / 32768) sec is equivalent to 128 clocks at 9600Hz (more precisely, 9598Hz), a division of the count (N1) by 128 equals frequency ratio (N2) between the frequency of HTBCLK and 9600Hz. For the calculation, the accuracy of baud-rate depends on truncating (1) or rounding (2) the data.

UART0 baud rate registers H and L are:

UA0BRTH, UA0BRTL = (frequency ratio between HTBCLK clock and baud rate) - 1 = (N1/128)-1 = N2 -1 (See Section 14.3.2.)

① Round data in calculation
N1 = 101110<u>1111110 (binary)</u>
N2 = 101111 (binary) = 47 (decimal) = 2F (hexadecimal)
Set N2-1 (= 2E) to UA0BRTH and UA0BRTL registers. In this case, the actual baud-rate will be 9595.744681.. [bps], so the accuracy = ((9595.744681/9600) -1)* 100= -0.04..[%].

② Truncate data in calculation (the accuracy of baud-rate becomes worse)
N1 = 101110<u>1111110 (</u>binary)
N2 = 101110 (binary) = 46 (decimal) = 2E (hexadecimal)
Set N2-1 (= 2D) to UA0BRTH and UA0BRTL registers. In this case, the actual baud-rate will be 9804.347826.. [bps], so the accuracy = ((9804.347826/9600) -1)* 100= 2.12..[%].

Table 9-5 shows the baud rate and accuracy (theoretical) when the baud rate clock is set to 500KHz. Table 9-6 shows the baud rate and accuracy (theoretical) when the baud rate clock is set to 2MHz.

Table 9-5 Baud Rate and Accuracy (theoretical) for Baud Rate Clock Set to 500KHz

Baud-rate[bps]	Data setting to UA0BRTH register and UA0BRTH register	Theoretical accuracy
300	Round off {N1/4 (2bit right-shift) } - (minus) 1.	~ ±2%
600	Round off {N1/8 (3bit right-shift) } - (minus) 1.	
1200	Round off {N1/16 (4bit right-shift) } - (minus) 1.	
2400	Round off {N1/32 (5bit right-shift) } - (minus) 1.	
4800	Round off {N1/64 (6bit right-shift) } - (minus) 1.	
9600	Round off {N1/128 (7bit right-shift) } - (minus) 1.	
19200	Round off {N1/256 (8bit right-shift) } - (minus) 1.	±2% ~ 2.5%
38400	Round off {N1/512 (9bit right-shift) } - (minus) 1.	±2.5% ~

Table 9-6 Baud Rate and Accuracy (theoretical) for Baud Rate Clock Set to 2MHz

Baud-rate[bps]	Data setting to UA0BRTH register and UA0BRTH register	Theoretical accuracy
300	Round off {N1/4 (2bit right-shift) } - (minus) 1.	~ ±2%
600	Round off {N1/8 (3bit right-shift) } - (minus) 1.	
1200	Round off {N1/16 (4bit right-shift) } - (minus) 1.	
2400	Round off {N1/32 (5bit right-shift) } - (minus) 1.	
4800	Round off {N1/64 (6bit right-shift) } - (minus) 1.	
9600	Round off {N1/128 (7bit right-shift) } - (minus) 1.	
19200	Round off {N1/256 (8bit right-shift) } - (minus) 1.	
38400	Round off {N1/512 (9bit right-shift) } - (minus) 1.	

9.4 Operating Timers by External Clock Inputs

When the external clock is selected as the operation clock for the Timer 0 (8-bit timer mode), operate it by inputting the clock with the P44 set to the 1st function. When the external clock is selected as the operation clock for the Timer 1 (8-bit timer mode), operate it by inputting the clock with the P45 set to the 1st function. When the external clock is selected as the operation clock for the Timer 2 (8-bit timer mode), operate it by inputting the clock with the P44 set to the 1st function. When the external clock is selected as the operation clock for the Timer 3 (8-bit timer mode), operate it by inputting the clock with the P45 set to the 1st function.
When the external clock is selected as the operation clock for the Timers 0 and 1 set to the 16-bit timer mode, operate them by inputting the clock with the P44 set to the 1st function. When the external clock is selected as the operation clock for the Timers 2 and 3 set to the 16-bit timer mode, operate them by inputting the clock with the P44 set to the 1st function.
Set the external clock frequency to a value below the operating frequency (fop) in "Appendix C Electrical Characteristics."

9.4.1 Operating Timer 0 (8-Bit Timer Mode) by External Clock (P44/T02P0CK)

Set the P44MD1 bit (P4MOD1 register's bit 4) to "0" and the P44MD0 bit (P4MOD0 register's bit 4) to "0" for specifying the P44 to the 1st function.

Register name	P4MOD1 register (Address: 0F225H)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	P47MD1	P46MD1	P45MD1	**P44MD1**	P43MD1	P42MD1	P41MD1	P40MD1
Setting value	*	*	*	**0**	*	*	*	*

Register name	P4MOD0 register (Address: 0F224H)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	P47MD0	P46MD0	P45MD0	**P44MD0**	P43MD0	P42MD0	P41MD0	P40MD0
Setting value	*	*	*	**0**	*	*	*	*

Set the P43DIR bit (P4DIR register's bit 3) to "1" for specifying the state mode of the P44 pin to input.

Register name	P4DIR register (Address: 0F221H)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	P47DIR	P46DIR	P45DIR	**P44DIR**	P43DIR	P42DIR	P41DIR	P40DIR
Setting value	*	*	*	**1**	*	*	*	*

Select the external clock(P44/T02P0CK) and timer0 (8-bit mode) in the TM0CON0 register, and input the operation clock for the Timer 0 from the P44 pin.

9.4.2 Operating Timer 1 (8-Bit Timer Mode) by External Clock (P45/T13CK)

Set the P45MD1 bit (P4MOD1 register's bit 5) to "0" and the P45MD0 bit (P4MOD0 register's bit 5) to "0" for specifying the P45 to the 1st function.

Register name	P4MOD1 register (Address: 0F225H)							
Bit	7	6	**5**	4	**3**	2	1	0
Bit name	P47MD1	P46MD1	**P45MD1**	P44MD1	P43MD1	P42MD1	P41MD1	P40MD1
Setting value	*	*	**0**	*	*	*	*	*

Register name	P4MOD0 register (Address: 0F224H)							
Bit	7	6	**5**	4	**3**	2	1	0
Bit name	P47MD0	P46MD0	**P45MD0**	P44MD0	P43MD0	P42MD0	P41MD0	P40MD0
Setting value	*	*	**0**	*	*	*	*	*

Set the P45DIR bit (P4DIR register's bit 5) to "1" for specifying the state mode of the P45 pin to input.

Register name	P4DIR register (Address: 0F221H)							
Bit	7	6	**5**	4	**3**	2	1	0
Bit name	P47DIR	P46DIR	**P45DIR**	P44DIR	P43DIR	P42DIR	P41DIR	P40DIR
Setting value	*	*	**1**	*	*	*	*	*

Select the external clock(P45/T13P0CK) and timer1 (8-bit mode) in the TM0CON0 and TM1CON0 register, and input the operation clock for the Timer 1 from the P45 pin.

9.4.3 Operating Timer 2 (8-Bit Timer Mode) by External Clock (P44/T02P0CK)

Set the P44MD1 bit (P4MOD1 register's bit 4) to "0" and the P44MD0 bit (P4MOD0 register's bit 4) to "0" for specifying the P44 to the 1st function.

Register name	P4MOD1 register (Address: 0F225H)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	P47MD1	P46MD1	P45MD1	**P44MD1**	P43MD1	P42MD1	P41MD1	P40MD1
Setting value	*	*	*	**0**	*	*	*	*

Register name	P4MOD0 register (Address: 0F224H)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	P47MD0	P46MD0	P45MD0	**P44MD0**	P43MD0	P42MD0	P41MD0	P40MD0
Setting value	*	*	*	**0**	*	*	*	*

Set the P44DIR bit (P4DIR register's bit 4) to "1" for specifying the state mode of the P44 pin to input.

Register name	P4DIR register (Address: 0F221H)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	P47DIR	P46DIR	P45DIR	**P44DIR**	P43DIR	P42DIR	P41DIR	P40DIR
Setting value	*	*	*	**1**	*	*	*	*

Select the external clock(P44/T02P0CK) and timer2 (8-bit mode) in the TM2CON0 register, and input the operation clock for the Timer 2 from the P44 pin.

9.4.4 Operating Timer 3 (8-Bit Timer Mode) by External Clock (P45/T13CK)

Set the P45MD1 bit (P4MOD1 register's bit 5) to "0" and the P45MD0 bit (P4MOD0 register's bit 5) to "0" for specifying the P45 to the 1st function.

Register name	P4MOD1 register (Address: 0F225H)							
Bit	7	6	**5**	4	**3**	2	1	0
Bit name	P47MD1	P46MD1	**P45MD1**	P44MD1	P43MD1	P42MD1	P41MD1	P40MD1
Setting value	*	*	**0**	*	*	*	*	*

Register name	P4MOD0 register (Address: 0F224H)							
Bit	7	6	**5**	4	**3**	2	1	0
Bit name	P47MD0	P46MD0	**P45MD0**	P44MD0	P43MD0	P42MD0	P41MD0	P40MD0
Setting value	*	*	**0**	*	*	*	*	*

Set the the P45DIR bit (P4DIR register's bit 5) to "1" for specifying the state mode of the P45 pin to input.

Register name	P4DIR register (Address: 0F221H)							
Bit	7	6	**5**	4	**3**	2	1	0
Bit name	P47DIR	P46DIR	**P45DIR**	P44DIR	P43DIR	P42DIR	P41DIR	P40DIR
Setting value	*	*	**1**	*	*	*	*	*

Select the external clock(P45/T13CK) and timer3 (8-bit mode) in the TM3CON0 register, and input the operation clock for the Timer 3 from the P45 pin.

9.4.5 Operating Timer 0 and Timer 1 (16-Bit Timer Mode) by External Clock (P44/T02P0CK)

Set the P44MD1 bit (P4MOD1 register's bit 4) to "0" and the P44MD0 bit (P4MOD0 register's bit 4) to "0" for specifying the P44 to the 1st function.

Register name	P4MOD1 register (Address: 0F225H)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	P47MD1	P46MD1	P45MD1	**P44MD1**	P43MD1	P42MD1	P41MD1	P40MD1
Setting value	*	*	*	**0**	*	*	*	*

Register name	P4MOD0 register (Address: 0F224H)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	P47MD0	P46MD0	P45MD0	**P44MD0**	P43MD0	P42MD0	P41MD0	P40MD0
Setting value	*	*	*	**0**	*	*	*	*

Set the P43DIR bit (P4DIR register's bit 3) to "1" for specifying the state mode of the P44 pin to input.

Register name	P4DIR register (Address: 0F221H)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	P47DIR	P46DIR	P45DIR	**P44DIR**	P43DIR	P42DIR	P41DIR	P40DIR
Setting value	*	*	*	**1**	*	*	*	*

Select the external clock(P44/T02P0CK) and Timers 0 and 1 (16-bit mode) in the TM0CON0 register, and input the operation clock for the Timers 0 and 1 (16-bit mode) from the P44 pin.

9.4.6 Operating Timer 2 and Timer 3 (16-Bit Timer Mode) by External Clock (P44/T02P0CK)

Set the P44MD1 bit (P4MOD1 register's bit 4) to "0" and the P44MD0 bit (P4MOD0 register's bit 4) to "0" for specifying the P44 to the 1st function.

Register name	P4MOD1 register (Address: 0F225H)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	P47MD1	P46MD1	P45MD1	**P44MD1**	P43MD1	P42MD1	P41MD1	P40MD1
Setting value	*	*	*	**0**	*	*	*	*

Register name	P4MOD0 register (Address: 0F224H)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	P47MD0	P46MD0	P45MD0	**P44MD0**	P43MD0	P42MD0	P41MD0	P40MD0
Setting value	*	*	*	**0**	*	*	*	*

Set the P44DIR bit (P4DIR register's bit 4) to "1" for specifying the state mode of the P44 pin to input.

Register name	P4DIR register (Address: 0F221H)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	P47DIR	P46DIR	P45DIR	**P44DIR**	P43DIR	P42DIR	P41DIR	P40DIR
Setting value	*	*	*	**1**	*	*	*	*

Select the external clock(P44/T02P0CK) and Timers 2 and 3 (16-bit mode) in the TM2CON0 register, and input the operation clock for the Timers 2 and 3 (16-bit mode) from the P44 pin.

Chapter 10

PWM

10. PWM

10.1 Overview

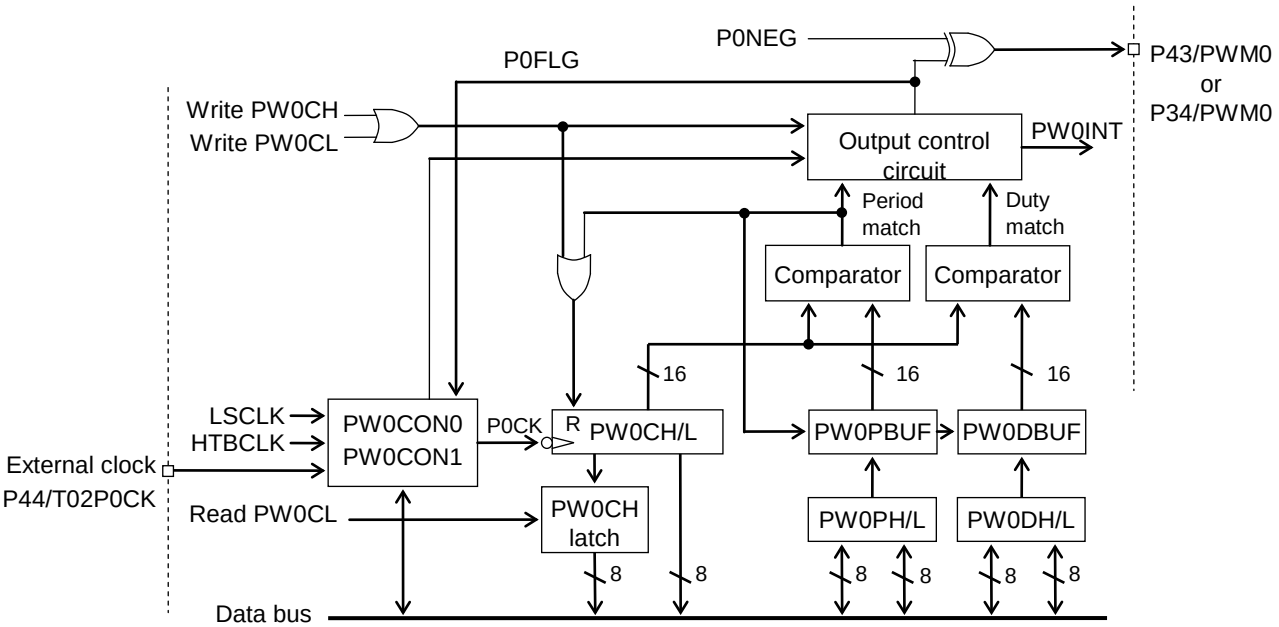
This LSI includes one channel of 16-bit PWM (Pulse Width Modulation).
The PWM output (PWM0) function is assigned to P43(Port 4) and P34(Port 3) as the tertiary function. For the functions of port 4 and port3, see Chapter 19, "Port 4" and Chapter 18, "Port 3".

10.1.1 Features

- The PWM signals with the periods of approximately 488 ns (HTBCLK=4.096MHz) to 2s (@LSCLK=32.768kHz) can be generated and output outside of the LSI.
- The output logic of the PWM signal can be switched to the positive or negative logic.
- At the coincidence of PWM signal period, duties, and period & duty, a PWM interrupt (PW0INT) occurs.
- For the PWM clock, a low-speed clock (LSCLK), a high-speed time base clock (HTBCLK), and an external clock are available.

10.1.2 Configuration

Figure 10 - 1 shows the configuration of the PWM circuit.



PW0PL: PWM0 period register L
PW0PH: PWM0 period register H
PW0PBUF: PWM0 period buffer
PW0DL: PWM0 duty register L
PW0DH: PWM0 duty register H
PW0DBUF: PWM0 duty buffer
PW0CL: PWM0 counter register L
PW0CH: PWM0 counter register H
PW0CON0: PWM0 control register 0
PW0CON1: PWM0 control register 1

Figure 10-1 Configuration of PWM Circuit

10.1.3 List of Pins

Pin name	I/O	Description
P43/PWM0	O	PWM0 output pin Used for the secondary function of the P43 pin.
P34/PWM0	O	PWM0 output pin Used for the secondary function of the P34 pin.

10.2 Description of Registers

10.2.1 List of Registers

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F0A0H	PWM0 period register L	PW0PL	PW0P	R/W	8/16	0FFH
0F0A1H	PWM0 period register H	PW0PH		R/W	8	0FFH
0F0A2H	PWM0 duty register L	PW0DL	PW0D	R/W	8/16	00H
0F0A3H	PWM0 duty register H	PW0DH		R/W	8	00H
0F0A4H	PWM0 counter register L	PW0CL	PW0C	R/W	8/16	00H
0F0A5H	PWM0 counter register H	PW0CH		R/W	8	00H
0F0A6H	PWM0 control register 0	PW0CON0	PW0CON	R/W	8/16	00H
0F0A7H	PWM0 control register 1	PW0CON1		R/W	8	40H

10.2.2 PWM0 Period Registers (PW0PL, PW0PH)

Address: 0F0A0H
Access: R/W
Access size: 8 bits
Initial value: 0FFH

	7	6	5	4	3	2	1	0
PW0PL	P0P7	P0P6	P0P5	P0P4	P0P3	P0P2	P0P1	P0P0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
At reset	1	1	1	1	1	1	1	1

Address: 0F0A1H
Access: R/W
Access size: 8 bits
Initial value: 0FFH

	7	6	5	4	3	2	1	0
PW0PH	P0P15	P0P14	P0P13	P0P12	P0P11	P0P10	P0P9	P0P8
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
At reset	1	1	1	1	1	1	1	1

PW0PH and PW0PL are special function registers (SFRs) to set the PWM0 periods.

Note:
When PW0PH or PW0PL is set to "0000H", the PWM0 period buffer (PW0PBUF) is set to "0001H".

10.2.3 PWM0 Duty Registers (PW0DL, PW0DH)

Address: 0F0A2H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
PW0DL	P0D7	P0D6	P0D5	P0D4	P0D3	P0D2	P0D1	P0D0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
At reset	0	0	0	0	0	0	0	0

Address: 0F0A3H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
PW0DH	P0D15	P0D14	P0D13	P0D12	P0D11	P0D10	P0D9	P0D8
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
At reset	0	0	0	0	0	0	0	0

PW0DH and PW0DL are special function registers (SFRs) to set the duties of PWM0.

Note:
Set PW0DH and PW0DL to values smaller than those to which PW0PH and PW0PL are set.

10.2.4 PWM0 Counter Registers (PW0CH, PW0CL)

Address: 0F0A4H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
PW0CL	P0C7	P0C6	P0C5	P0C4	P0C3	P0C2	P0C1	P0C0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
At reset	0	0	0	0	0	0	0	0

Address: 0F0A5H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
PW0DH	P0C15	P0C14	P0C13	P0C12	P0C11	P0C10	P0C9	P0C8
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
At reset	0	0	0	0	0	0	0	0

PW0CL and PW0CH are special function registers (SFRs) that function as 16-bit binary counters.
When data is written to either PW0CL or PW0CH, PW0CL and PW0CH is set to "0000H". The data that is written is meaningless.
When data is read from PW0CL, the value of PW0CH is latched. When reading PW0CH and PW0CL, use a word type instruction or pre-read PW0CL.

The contents of PW0CH and PW0CL during PWM operation cannot be read depending on the combination of the PWM clock and system clock. Table 10-1 shows PW0CH and PW0CL read enable/disable for each combination of the PWM clock and system clock.

Table 10-1 PW0CH and PW0CL Read Enable/Disable during PWM0 Operation

PWM clock P0CK	System clock SYSCLK	PW0CH and PW0CL read enable/disable
LSCLK	LSCLK	Read enabled
LSCLK	HSCLK	Read enabled. However, to prevent the reading of undefined data during counting, read consecutively PW0CH or PW0CL twice until the last data coincides the previous data.
HTBCLK	LSCLK	Read disabled
HTBCLK	HSCLK	Read enabled
External clock	LSCLK	Read disabled
	HSCLK	

10.2.5 PWM0 Control Register 0 (PW0CON0)

Address: 0F0A6H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
PW0CON0	—	—	—	P0NEG	P0IS1	P0IS0	P0CS1	P0CS0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
At reset	0	0	0	0	0	0	0	0

PW0CON0 is a special function register (SFR) to control PWM.

[Description of Bits]

- **P0CS1, P0CS0** (bits 1, 0)
  The P0CS1 and P0CS0 bits are used to select the PWM0 operation clocks. LSCLK, HTBCLK, or the external clock (P44/T02P0CK) can be selected.

P0CS1	P0CS0	Description
0	0	LSCLK (initial value)
0	1	HTBCLK
1	0	Prohibited (the PWM circuit does not operate)
1	1	External clock (P44/T02P0CK)

- **P0IS1, P0IS0** (bits 3, 2)
  The P0IS1 and P0IS0 bits are used to select the point at which the PWM0 interrupt occurs. "When the periods coincide", "when the duties coincide", or "when the periods and duties coincide" can be selected.

P0IS1	P0IS0	Description
0	0	When the periods coincide. (Initial value)
0	1	When the duties coincide.
1	*	When the periods and duties coincide.

- **P0NEG** (bit 4)
  The P0NEG bit is used to select the output logic. When the positive logic is selected, the initial value of PWM0 output is "1", and when the negative logic is selected, the initial value of PWM0 output is "0".

P0NEG	Description
0	Positive logic (initial value)
1	Negative logic

10.2.6 PWM0 Control Register 1 (PW0CON1)

Address: 0F0A7H
Access: R/W
Access size: 8 bits
Initial value: 40H

	7	6	5	4	3	2	1	0
PW0CON1	P0STAT	P0FLG	—	—	—	—	—	P0RUN
R/W	R	R/W	R/W	R/W	R/W	R/W	R/W	R/W
At reset	0	1	0	0	0	0	0	0

PW0CON1 is a special function register (SFR) to control PWM0.

[Description of Bits]

- **P0RUN** (bit 0)
  The P0RUN bit is used to control count stop/start of PWM0.

P0RUN	Description
0	Stops counting. (Initial value)
1	Starts counting.

- **P0FLG** (bit 6)
  The P0FLG bit is used to read the output flag of PWM0.
  This bit is set to "1" when write operation to PW0CH or PW0CL is performed,

P0FLG	Description
0	PWM0 output flag = "0"
1	PWM0 output flag = "1" (initial value)

- **P0STAT** (bit 7)
  The P0STAT bit indicates "counting stopped or "counting in progress" of PWM0.

P0STAT	Description
0	Counting stopped. (Initial value)
1	Counting in progress.

10.3 Description of Operation

The PWM0 counter registers (PW0CH, PW0CL) are set to an operating state (P0STAT is set to "1") on the first falling edge of the PWM clock (P0CK) that are selected by the PWM0 control register 0 (PW0CON0) when the P0RUN bit of PWM0 control register 1 (PW0CON1) is set to "1" and increment the count value on the 2nd falling edge.

When the count value of PWM0 counter registers and the value of the PWM0 duty buffer (PW0DBUF) coincide, the PWM flag (P0FLG) is set to "0" on the next timer clock falling edge of P0CK.
When the count value of PWM0 counter registers and the value of the PWM0 period buffer (PW0PBUF) coincide, the PWM flag (P0FLG) is set to "1" on the next falling edge of P0CK and PWM0 counter registers is set to "0000H" and incremental counting continues. At the same time, the value of the PWM0 duty register (PW0DH, PW0DL) is transferred to the PWM0 duty buffer (PW0DBUF) and the value of PWM0 period register (PW0PH, PW0PL) to the PWM0 period buffer (PW0PBUF).

When the P0RUN bit is set to "0", PWM0 counter registers stop counting after counting once the falling of the PWM clock (P0CK). Confirm that PW0CH and PW0CL are stopped by checking that the PnSTAT bit of the PWM0 control register 1 (PW0CON1) is "0". When the P0RUN bit is set to "1" again, PWM0 counter registers restarts incremental counting from the previous value on the falling edge of P0CK.

To initialize PWM0 counter registers to "0000H", perform write operation in either of PW0CH or PW0CL. At that time, P0FLG is also set to "1". When data is written in the PWM0 duty register (PW0DH, PW0DL) during count stop (P0RUN is in a "1" state), the data is transferred to the PWM0 duty buffer (PW0DBUF) and when data is written in the PWM0 period register (PW0PH, PW0PL), the data is transferred to the PWM0 period buffer (PW0PBUF).

The PWM clock, the point at which an interrupt of PWM0 occurs, and the logic of the PWM output are selected by PWM0 control register 0 (PW0CN0).

The period of the PWM0 signal (TPWP) and the first half duration (TPWD) of the duty are expressed by the following equations.

$$T_{PWP} = \frac{PW0P + 1}{P0CK\ (Hz)}$$

$$T_{PWD} = \frac{PW0D + 1}{P0CK\ (Hz)}$$

PW0P: PWM0 period registers (PW0PH, PW0PL) setting value (0001H to 0FFFFH)
PW0D: PWM0 duty registers (PW0DH, PW0DL) setting value (0000H to 0FFFEH)
P0CK: Clock frequency selected by the PWM0 control register 0 (PW0CON0)

After the P0RUN bit is set to "1", counting starts in synchronization with the PWM clock. This causes an error of up to 1 clock pulse to the time the first PWM interrupt is issued. The PWM interrupt period from the second time is fixed.
Figure 10-2 shows the operation timing of PWM0.

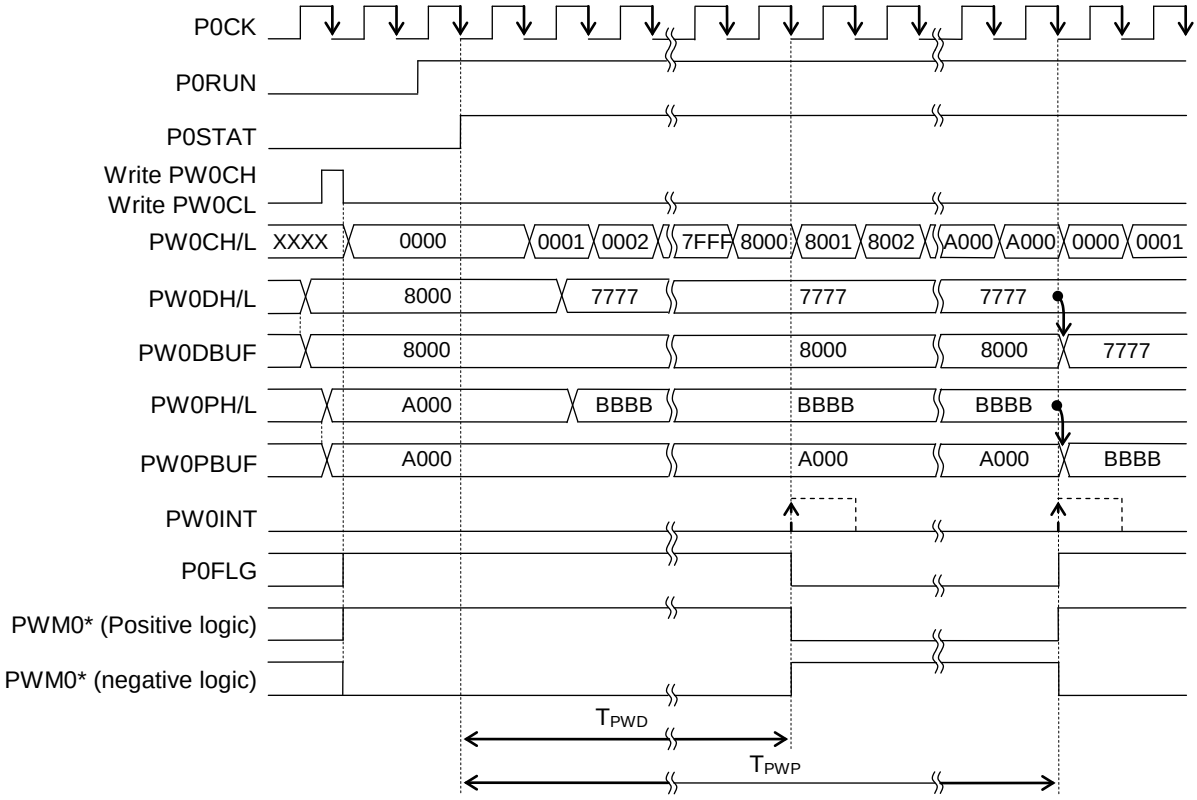


Figure 10-2 (1/2) Operation Timing Diagram of PWM0

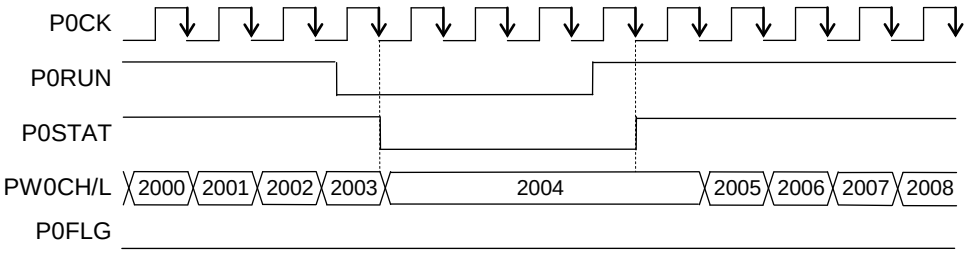


Figure 10-2 (2/2) Operation Timing Diagram of PWM0

Note:
Even if "0" is written to the P0RUN bit, counting operation continues up to the falling edge (the PWM0 status flag (P0STAT) is in a "1" state) of the next PWM clock pulse. Therefore, the PWM0 interrupt (PW0INT) may occur.

10.4 Specifying port registers

When you want to make sure the PWM function is working, please check related port registers are specified. See Chapter 19, "Port 4" and Chapter 18, "Port 3" for detail about the port registers.

10.4.1 Functioning P43 (PWM0) as the PWM output

Set P43MD1 bit (bit3 of P4MOD1 register) to "1" and set P43MD0 bit (bit3 of P4MOD0 register) to "0", for specifying the PWM output as the tertiary function of P43.

Reg. name	P4MOD1 register (Address: 0F225H)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	P47MD1	P46MD1	P45MD1	P44MD1	**P43MD1**	P42MD1	P41MD1	P40MD1
Data	*	*	*	*	**1**	*	*	*

Reg. name	P4MOD0 register (Address: 0F224H)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	P47MD0	P46MD0	P45MD0	P44MD0	**P43MD0**	P42MD0	P41MD0	P40MD0
Data	*	*	*	*	**0**	*	*	*

Set P43C1 bit (bit3 of P4CON1 register) to "1", set P43C0 bit(bit3 of P4CON0 register) to "1" and set P43DIR bit(bit3 of P4DIR register) to "0", for specifying the P43 as CMOS output.

Reg. name	P4CON1 register (Address: 0F223H)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	P47C1	P46C1	P45C1	P44C1	**P43C1**	P42C1	P41C1	P40C1
Data	*	*	*	*	**1**	*	*	*

Reg. name	P4CON0 register (Address: 0F222H)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	P47C0	P46C0	P45C0	P44C0	**P43C0**	P42C0	P41C0	P40C0
Data	*	*	*	*	**1**	*	*	*

Reg. name	P4DIR register (Address: 0F221H)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	P47DIR	P46DIR	P45DIR	P44DIR	**P43DIR**	P42DIR	P41DIR	P40DIR
Data	*	*	*	*	**0**	*	*	*

Data of P43D bit (bit3 of P4D register) does not affect to the PWM output function, so don't care the data for the function.

Reg. name	P4D register (Address: 0F220H)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	P47D	P46D	P45D	P44D	**P43D**	P42D	P41D	P40D
Data	*	*	*	*	******	*	*	*

* : Bit not related to the PWM function
** : Don't care the data.

10.4.2 Functioning P34 (PWM0) as the PWM output

Set P34MD1 bit (bit4 of P3MOD1 register) to "1" and set P34MD0 bit (bit4 of P3MOD0 register) to "0", for specifying the PWM output as the tertiary function of P34.

Reg. name	P3MOD1 register (Address: 0F21DH)							
Bit	7	6	5	**4**	3	2	1	0
Bit name	-	-	P35MD1	**P34MD1**	P33MD1	P32MD1	P31MD1	P30MD1
Data	-	-	*	**1**	*	*	*	*

Reg. name	P3MOD0 register (Address: 0F21CH)							
Bit	7	6	5	**4**	3	2	1	0
Bit name	-	-	P35MD0	**P34MD0**	P33MD0	P32MD0	P31MD0	P30MD0
Data	-	-	*	**0**	*	*	*	*

Set P34C1 bit (bit4 of P3CON1 register) to "1", set P34C0 bit(bit4 of P3CON0 register) to "1" and set P34DIR bit(bit4 of P3DIR register) to "0", for specifying the P34 as CMOS output.

Reg. name	P3CON1 register (Address: 0F21BH)							
Bit	7	6	5	**4**	3	2	1	0
Bit name	-	-	P35C1	**P34C1**	P33C1	P32C1	P31C1	P30C1
Data	-	-	*	**1**	*	*	*	*

Reg. name	P3CON0 register (Address: 0F21AH)							
Bit	7	6	5	**4**	3	2	1	0
Bit name	-	-	P35C0	**P34C0**	P33C0	P32C0	P31C0	P30C0
Data	-	-	*	**1**	*	*	*	*

Reg. name	P3DIR register (Address: 0F219H)							
Bit	7	6	5	**4**	3	2	1	0
Bit name	-	-	P35DIR	**P34DIR**	P33DIR	P32DIR	P31DIR	P30DIR
Data	-	-	*	**0**	*	*	*	*

Data of P34D bit (bit4 of P3D register) does not affect to the PWM output function, so don't care the data for the function.

Reg. name	P3D register (Address: 0F218H)							
Bit	7	6	5	**4**	3	2	1	0
Bit name	-	-	P35D	**P34D**	P33D	P32D	P31D	P30D
Data	-	-	*	******	*	*	*	*

- : Bit does not exist.
* : Bit not related to the PWM function
** : Don't care the data.

Chapter 11

Watchdog Timer

11. Watchdog Timer

11.1 Overview

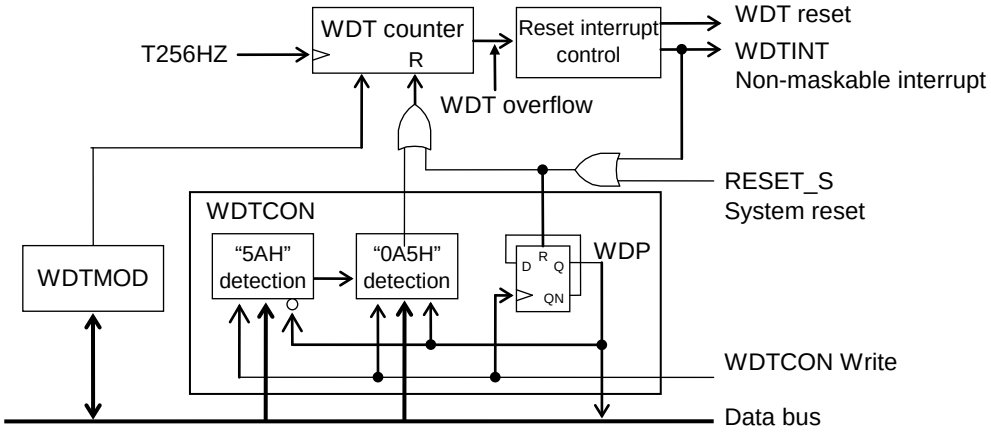
This LSI incorporates a watchdog timer (WDT) that operates at a system reset unconditionally (free-run operation) in order to detect an undefined state of the MCU and return from that state.
If the WDT counter overflows due to the failure of clearing of the WDT counter within the WDT overflow period, the watchdog timer requests a WDT interrupt (non-maskable interrupt). When the second overflow occurs, the watchdog timer generates a WDT reset signal and shifts the mode to a system reset mode.
For interrupts see Chapter 5, "Interrupts," and for WDT interrupt see Chapter 3, "Reset Function".

11.1.1 Features

- Free running (cannot be stopped)
- One of four types of overflow periods (125ms, 500ms, 2s, and 8s) selectable by software
- Non-maskable interrupt by the first overflow
- Reset generated by the second overflow

11.1.2 Configuration

Figure 11-1 shows the configuration of the watchdog timer.



WDTCON : Watchdog timer control register
WDTMOD : Watchdog timer mode register

Figure 11-1 Configuration of Watchdog Timer

11.2 Description of Registers

11.2.1 List of Registers

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F00EH	Watchdog timer control register	WDTCON	—	R/W	8	00H
0F00FH	Watchdog timer mode register	WDTMOD	—	R/W	8	02H

11.2.2 Watchdog Timer Control Register (WDTCON)

Address: 0F00EH
Access: W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
WDTCON	d7	d6	d5	d4	d3	d2	d1	WDP/d0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

WDTCON is a special function register (SFR) to clear the WDT counter.
When WDTCON is read, the value of the internal pointer (WDP) is read from bit 0.

[Description of Bits]

- **WDP/d0** (bit 0)
  The value of the internal pointer (WDP) is read from this bit. The WDP is reset to "0" at the system reset or Watch Dog Timer overflow and is inverted every writing to WDTCON.

- **d7-d0** (bits 7-0)
  This bit is used to write data to clear the WDT counter. Write "5AH" on the condition of WDP is "0" and write "0A5H" on the condition of WDP is "1".

Note:
When the WDT interrupt (WDTINT) occurs by the first WDT counter overflow, the counter and the internal pointer (WDP) are initialized for a half cycle of low speed clock (about 15us). During the time period that they are initialized, writing to WDTCON is disable and the logic of WDP does not change. Therefore, in the case of that you have program codes handle to clear the WDT when the first overflow WDT interrupt occurs and also the codes run at high-speed system clock, please check the WDP gets reversed after writing to WDTCON to see if the writing was surely successful. For example of the program code, see Section 11.3.1, "Handling example when you do not want to use the watch dog timer".

11.2.3 Watchdog Timer Mode Register (WDTMOD)

Address: 0F00FH
Access: W
Access size: 8 bits
Initial value: 02H

	7	6	5	4	3	2	1	0
WDTMOD	—	—	—	—	—	—	WDT1	WDT0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	1	0

WDTMOD is a special function register to set the overflow period of the watchdog timer.

[Description of Bits]

- **WDT1-0** (bits 1-0)
  These bits are used to select an overflow period of the watchdog timer.
  The WDT1 and WDT0 bits set a overflow period (TWOV) of the WDT counter. One of 125ms, 500ms, 2s, and 8s can be selected.

WDT1	WDT0	Description
0	0	125 ms
0	1	500 ms
1	0	2 s (initial value)
1	1	8 s

11.3 Description of Operation

The WDT counter starts counting after the system reset has been released and the low-speed clock oscillation start..
Write "5AH" when the internal pointer (WDP) is "0"and then the WDT counter is cleared by writing "0A5H" when WDP is "1".
WDP is reset to "0" at the time of system reset or when the WDT counter overflows and is inverted whenever data is written to WDTCON.
When the WDT counter cannot be cleared within the WDT counter overflow period (T_{WOV}), a watchdog timer interrupt (WDTINT) occurs. If the WDT counter is not cleared even by the software processing performed following the watchdog timer interrupt and overflow occurs again, WDT reset occurs and the mode shifts to a system reset mode.
For the overflow period (TWOV) of the WDT counter, one of 125ms, 500ms, 2s, and 8s can be selected by the watchdog mode register (WDTMOD).
Clear the WDT counter within the clear period of the WDT counter shown in Table 11-1.

Table 11-1 Clear Period of WDT Counter

WDT1	WDT0	T_{WOV}	T_{WCL}
0	0	125 ms	Approx. 121 ms
0	1	500 ms	Approx. 496 ms
1	0	2000 ms	Approx. 1996 ms
1	1	8000 ms	Approx. 7996 ms

Figure 11-2 shows an example of watchdog timer operation.

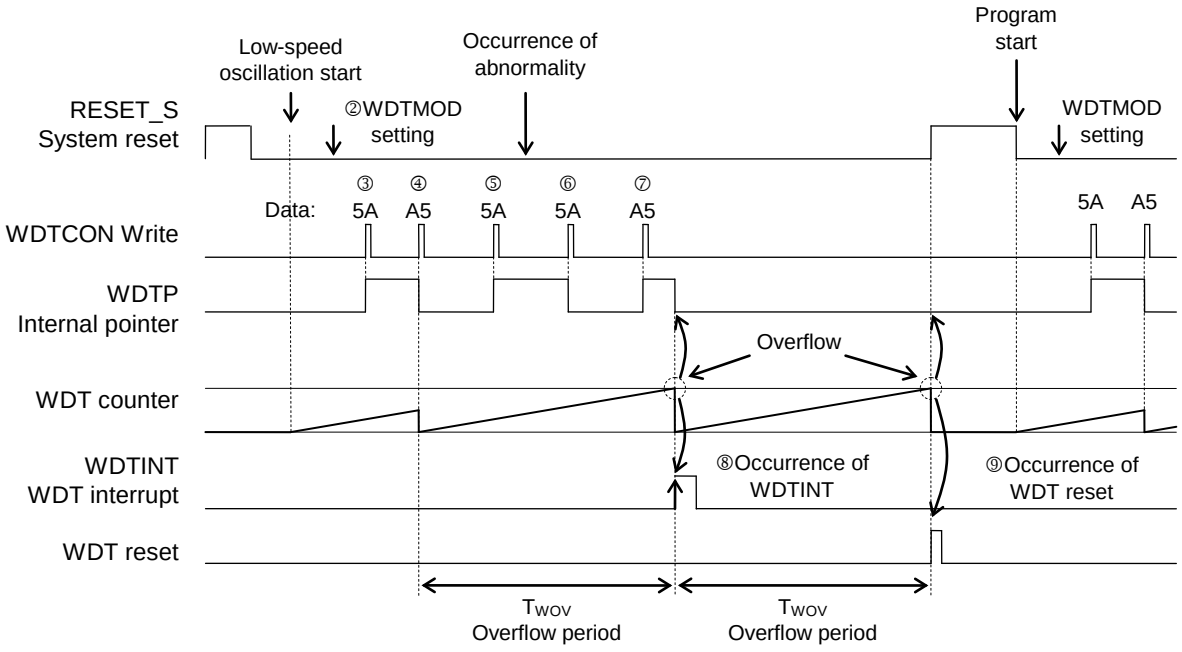


Figure 11-2 Example of Watchdog Timer Operation

① The WDT counter starts counting after the system reset has been released and the low-speed clock oscillation start.
② The overflow period of the WDT counter (TWOV) is set to WDTMOD.
③ "5AH" is written to WDTCON. (Internal pointer 0→1)
④ "0A5H" is written to WDTCON and the WDT counter is cleared. (Internal pointer 1→0)
⑤ "5AH" is written o WDTCON. (Internal pointer 0→1)
⑥ When "5AH" is written to WDTCON after the occurrence of abnormality, it cannot be accepted as the internal pointer is set to "1". (Internal pointer 1→0)
⑦ Although "0A5H" is written to WDTCON, the WDT counter is not cleared since the internal pointer is "0" and the writing of "5AH" is not accepted in ⑥. (Internal pointer 0→1)
⑧ The WDT counter overflows and a watchdog timer interrupt request (WDTINT) is generated. In this case,, the WDT counter and the internal pointer (WDP) are initialized for a half cycle of low speed clock (about 15.26us).
⑨ If the WDT counter is not cleared even by the software processing performed following a watchdog timer interrupt and the WDT counter overflows again, WDT reset occurs and the mode is shifted to a system reset mode.

Note:
- In STOP mode, the watchdog timer operation also stops.
- In HALT mode, the watchdog timer operation does not stop. When the WDT interrupt occurs, the HALT mode is released.
- The watchdog timer cannot detect all the abnormal operations. Even if the CPU loses control, the watchdog timer cannot detect the abnormality in the operation state in which the WDT counter is cleared.

11.3.1 Handling example when you do not want to use the watch dog timer

WDT counter is a free-run counter that starts count-up automatically after the system reset released and the low-speed clock (LSCLK) starts oscillating. If the WDT counter gets overflow, the WDT non-maskable interrupt occurs and then a system reset occurs. Therefore, it is needed to clear the WDT counter even if you do not want to use the WDT as a fale-safe function.
See following example programming codes to clear the WDT counter in the interrupt routine.

Example programming code:

```
__DI();                    // Disable multi-interrupts
do
{
   WDTCON = 0x5a;
} while(WDP != 1)
WDTCON = 0xa5;
__EI();
```

Chapter 12

Synchronous Serial Port

12. Synchronous Serial Port

12.1 Overview

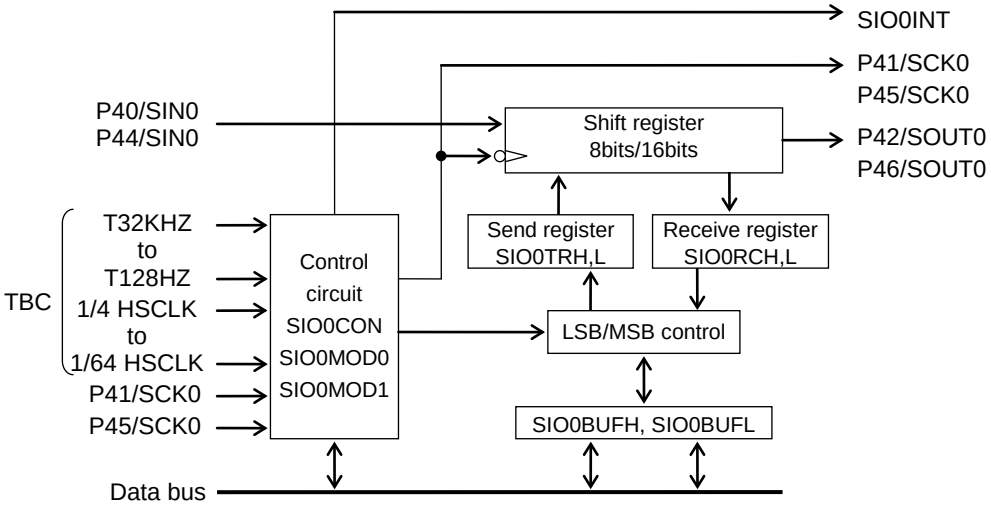
This LSI includes two channels of 8/16-bit synchronous serial ports (SSIO). It can also be used to control the device incorporated with the SPI interface by using one GPIO as the chip enable pin.
When the synchronous serial port is used, the tertiary functions of Port 4 or the tertiary functions of Port 5 must be set. For the tertiary function setting of Port 4, see Chapter 17, "Port 4." For the tertiary function setting of Port 5, see Chapter 18, "Port 5."

12.1.1 Features

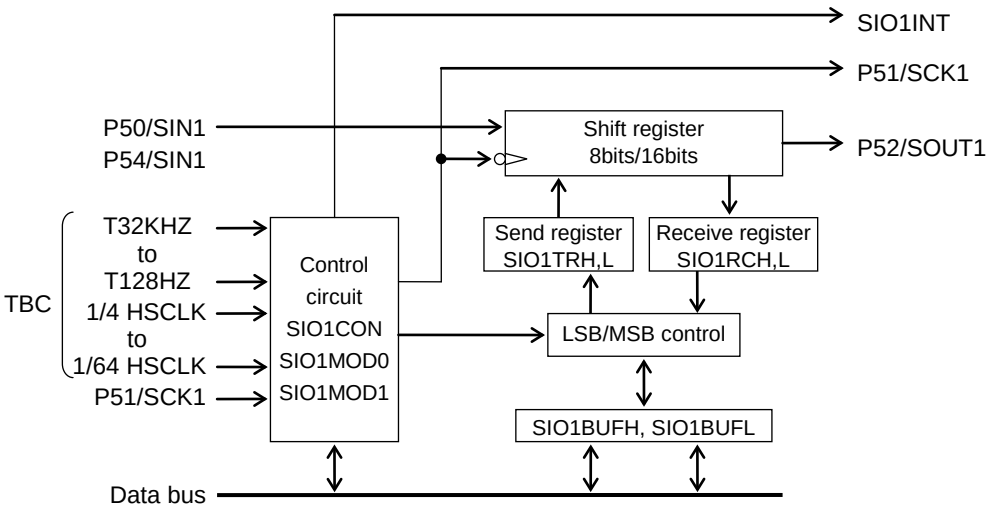
- Master or slave selectable
- MSB first or LSB first selectable
- 8-bit length or 16-bit length selectable fro the data length

12.1.2 Configuration

Figure 12-1 shows the configuration of the synchronous serial port.



Synchronous Serial Port 0 (SSIO0)



Synchronous Serial Port 1(SSIO1)

SIO0BUFL, SIO1BUFL : Serial port transmit/receive buffer L
SIO0BUFH, SIO1BUFH : Serial port transmit/receive buffer H
SIO0CON, SIO1CON : Serial port control register
SIO0MOD0, SIO1MOD0 : Serial port mode register 0
SIO0MOD1, SIO1MOD1 : Serial port mode register 1

Figure 12-1 Configuration of Synchronous Serial Port

12.1.3 List of Pins

Pin name	I/O	Function
P40/SIN0 P44/SIN0	I	Received data input. Used for the tertiary function of the P40 and P44 pins.
P41/SCK0 P45/SCK0	I/O	Synchronous clock input/output. Used for the tertiary function of the P41 and P45 pins.
P42/SOUT0 P46/SOUT0	O	Transmitted data output. Used for the tertiary function of the P42 and P46 pins.
P50/SIN1	I	Received data input. Used for the secondary function of the P50 pin.
P51/SCK1	I/O	Synchronous clock input/output. Used for the secondary function of the P51 pin.
P52/SOUT1	O	Transmitted data output. Used for the secondary function of the P52 pin.

12.2 Description of Registers

12.2.1 List of Registers

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F280H	Serial port 0 transmit/receive buffer L	SIO0BUFL	SIO0BUF	R/W	8/16	00H
0F281H	Serial port 0 transmit/receive buffer H	SIO0BUFH		R/W	8	00H
0F282H	Serial port 0 control register	SIO0CON	—	R/W	8	00H
0F284H	Serial port 0 mode register 0	SIO0MOD0	SIO0MOD	R/W	8/16	00H
0F285H	Serial port 0 mode register 1	SIO0MOD1		R/W	8	00H
0F288H	Serial port 1 transmit/receive buffer L	SIO1BUFL	SIO1BUF	R/W	8/16	00H
0F289H	Serial port 1 transmit/receive buffer H	SIO1BUFH		R/W	8	00H
0F28AH	Serial port 1 control register	SIO1CON	—	R/W	8	00H
0F28CH	Serial port 1 mode register 0	SIO1MOD0	SIO1MOD	R/W	8/16	00H
0F28DH	Serial port 1 mode register 1	SIO1MOD1		R/W	8	00H

12.2.2 Serial Port 0 Transmit/Receive Buffers (SIO0BUFL and SIO0BUFH)

Address: 0F280H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
SIO0BUFL	S0B7	S0B6	S0B5	S0B4	S0B3	S0B2	S0B1	S0B0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

Address: 0F281H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
SIO0BUFH	S0B15	S0B14	S0B13	S0B12	S0B11	S0B10	S0B9	S0B8
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

SIO0BUFL and SIO0BUFH are special function registers (SFRs) to write transmitted data and to read received data of the synchronous serial port 0.
When data is written in SIO0BUFL and SIO0BUFH, the data is written in the transmit registers (SIO0TRL and SIO0TRH). When data is read from SIO0BUFL and SIO0BUFH, the contents of the receive registers (SIO0RCL and SIO0RCH) are read.

12.2.3 Serial Port 1 Transmit/Receive Buffers (SIO1BUFL and SIO1BUFH)

Address: 0F288H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
SIO1BUFL	S1B7	S1B6	S1B5	S1B4	S1B3	S1B2	S1B1	S1B0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

Address: 0F289H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
SIO1BUFH	S1B15	S1B14	S1B13	S1B12	S1B11	S1B10	S1B9	S1B8
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

SIO1BUFL and SIO1BUFH are special function registers (SFRs) to write transmitted data and to read received data of the synchronous serial port 1.
When data is written in SIO1BUFL and SIO1BUFH, the data is written in the transmit registers (SIO1TRL and SIO1TRH). When data is read from SIO1BUFL and SIO1BUFH, the contents of the receive registers (SIO1RCL and SIO1RCH) are read.

12.2.4 Serial Port 0 Control Register (SIO0CON)

Address: 0F282H
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
SIO0CON	—	—	—	—	—	—	—	S0EN
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

SIO0CON is a special function register (SFR) to control the synchronous serial port 0.

[Description of Bits]

- **S0EN** (bit 0)
  The S0EN bit is used to specify start of synchronous serial communication. Writing a "1" to S0EN starts 8-/16-bit data communication. The S0EN bit is set to "0" automatically when 8-/16-bit data communication is terminated.
  The S0EN bit is set to "0" at a system reset.

S0EN	Description
0	Stops communication. (Initial value)
1	Starts communication

12.2.5 Serial port 1 control register (SIO1CON)

Address: 0F28AH
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
SIO1CON	—	—	—	—	—	—	—	S1EN
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

SIO1CON is a special function register (SFR) to control the synchronous serial port 1.

[Description of Bits]

- **S1EN** (bit 0)
  The S1EN bit is used to specify start of synchronous serial communication. Writing a "1" to S1EN starts 8-/16-bit data communication. The S1EN bit is set to "0" automatically when 8-/16-bit data communication is terminated.
  The S1EN bit is set to "0" at a system reset.

S1EN	Description
0	Stops communication. (Initial value)
1	Starts communication

12.2.6 Serial Port 0 Mode Register 0 (SIO0MOD0)

Address: 0F284H
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
SIO0MOD0	—	—	—	—	S0LG	S0MD1	S0MD0	S0DIR
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

SIO0MOD0 is a special function register (SFR) to set mode of the synchronous serial port 0.

[Description of Bits]

- **S0DIR** (bit 0)
  S0DIR is the bit for selecting LSB first or MSB first.

S0DIR	Description
0	LSB first (initial value)
1	MSB first

- **S0MD1 and S0MD0** (bits 2 and 1)
  The S0MD1 and S0MD0 bits are used to select the transmit/receive mode of the synchronous serial port 0. The Receive mode, Transmit mode, or Transmit/Receive mode is selectable.

S0MD1	S0MD0	Description
0	0	Stops transmission/reception (initial value)
0	1	Receive mode
1	0	Transmit mode
1	1	Transmit/receive mode

- **S0LG** (bit 3)
  S0LG is the bit that specifies the transmit/receive buffer bit length. Either 8-bit length or 16-bit length can be selected. The S0LG bit is set to "0" at a system reset.

S0LG	Description
0	8-bit length (initial value)
1	16-bit length

Note:
Do not change any of the SIO0MOD0 register settings during transmission/reception.
When the synchronous serial port 0 is used, the tertiary functions of Port 4 must be set. For the tertiary functions of Port 4, see Chapter 19, "Port 4".

12.2.7 Serial Port 1 Mode Register 0 (SIO1MOD0)

Address: 0F28CH
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
SIO1MOD0	—	—	—	—	S1LG	S1MD1	S1MD0	S1DIR
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

SIO1MOD0 is a special function register (SFR) to set mode of the synchronous serial port 1.

[Description of Bits]

- **S1DIR** (bit 0)
  S1DIR is the bit for selecting LSB first or MSB first.

S1DIR	Description
0	LSB first (initial value)
1	MSB first

- **S1MD1 and S1MD0** (bits 2 and 1)
  The S1MD1 and S1MD0 bits are used to select the transmit/receive mode of the synchronous serial port 1. The Receive mode, Transmit mode, or Transmit/Receive mode is selectable.

S1MD1	S1MD0	Description
0	0	Stops transmission/reception (initial value)
0	1	Receive mode
1	0	Transmit mode
1	1	Transmit/receive mode

- **S1LG** (bit 3)
  S1LG is the bit that specifies the transmit/receive buffer bit length. Either 8-bit length or 16-bit length can be selected. The S1LG bit is set to "0" at a system reset.

S1LG	Description
0	8-bit length (initial value)
1	16-bit length

Note:
•Do not change any of the SIO1MOD0 register settings during transmission/reception.
•When the synchronous serial port 1 is used, the tertiary functions of Port 5 must be set. For the tertiary functions of Port 5, see Chapter 18, "Port 5".

12.2.8 Serial Port 0 Mode Register 1 (SIO0MOD1)

Address: 0F285H
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
SIO0MOD1	—	—	S0NEG	S0CKT	—	S0CK2	S0CK1	S0CK0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

SIO0MOD1 is a special function register (SFR) to set mode of the synchronous serial port 0.

[Description of Bits]

- **S0CK2 to S0CK0** (bits 2 to 0)
  The S0CK2 to S0CK0 bits are used to select the transfer clock of the synchronous serial port. When the internal clock is selected, this LSI is set to master mode and when the external clock is selected, it is set to slave mode.

S0CK2	S0CK1	S0CK0	Description
0	0	0	32 KHz (initial value)
0	0	1	16KHz
0	1	0	1/4 HSCLK
0	1	1	1/8 HSCLK
1	0	0	1/16 HSCLK
1	0	1	1/32 HSCLK
1	1	0	External clock 0 (P41/SCK0)
1	1	1	External clock 0 (P45/SCK0)

- **S0CKT** (bit 4)
  The S0CKT bit is used to select the phase of the transfer clock output.

S0CKT	Description
0	Clock type 0: Output at the “H” level by default (initial value).
1	Clock type 1: Output at the “L” level by default.

- **S0NEG** (bit 5)
  The S0NEG bit is used to select the positive or negative logic of the transfer clock output.

S0NEG	Description
0	Positive logic (initial value)
1	Negative logic

12.2.9 Serial Port 1 Mode Register 1 (SIO1MOD1)

Address: 0F28DH
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
SIO1MOD1	—	—	S1NEG	S1CKT	—	S1CK2	S1CK1	S1CK0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

SIO1MOD1 is a special function register (SFR) to set mode of the synchronous serial port 1.

[Description of Bits]

- **S1CK2 to S1CK0** (bits 2 to 0)
  The S1CK2 to S1CK0 bits are used to select the transfer clock of the synchronous serial port. When the internal clock is selected, this LSI is set to master mode and when the external clock is selected, it is set to slave mode.

S1CK2	S1CK1	S1CK0	Description
0	0	0	32 KHz (initial value)
0	0	1	16KHz
0	1	0	1/4 HSCLK
0	1	1	1/8 HSCLK
1	0	0	1/16 HSCLK
1	0	1	1/32 HSCLK
1	1	0	External clock 1 (P51/SCK1)
1	1	1	Prohibited

- **S1CKT** (bit 4)
  The S1CKT bit is used to select the phase of the transfer clock output.

S1CKT	Description
0	Clock type 0: Output at the “H” level by default (initial value).
1	Clock type 1: Output at the “L” level by default.

- **S1NEG** (bit 5)
  The S1NEG bit is used to select the positive or negative logic of the transfer clock output.

S1NEG	Description
0	Positive logic (initial value)
1	Negative logic

12.3 Description of Operation

12.3.1 Transmit Operation

When “1” is written to the SnMD1 bit and "0" is written to the SnMD0 bit of the serial port mode register (SIOnMOD0), this LSI is set to the transmit mode.
When transmitted data is written to the serial port transmit/receive buffer (SIOnBUFL, "H") and the SnEN bit of the serial port control register (SIOnCON) is set to “1”, transmission starts. When transmission of 8/16-bit data terminates, a synchronous serial port interrupt (SIOnINT) occurs and the SnEN bit is set to “0”.
The transmitted data is output from the Port 4's tertiary function (P42/SOUT0, P46/SOUT0) or from the Port 5's tertiary function (P52/SOUT1, P56/SOUT1).
When an internal clock is selected in the serial port mode register (SIOnMOD1), the LSI is set to a master mode and when an external clock (SCKn) is selected, the LSI is set to a slave mode.
The serial port mode register (SIOnMOD0) enables selection of MSB first/LSB first.
The transmitted data output pin (SOUTn) and the transfer clock input/output pin (SCKn) need to be set to the tertiary function for the Port 4 or to the tertiary function for the Port 5.
The transmission operation waveforms of the synchronous serial port (8-bit length, LSB first) are shown in Figures 12-2 to 12-5, for the clock type 0 (positive logic), clock type 0 (negative logic), clock type 1 (positive logic), and clock type 1 (negative logic), respectively.

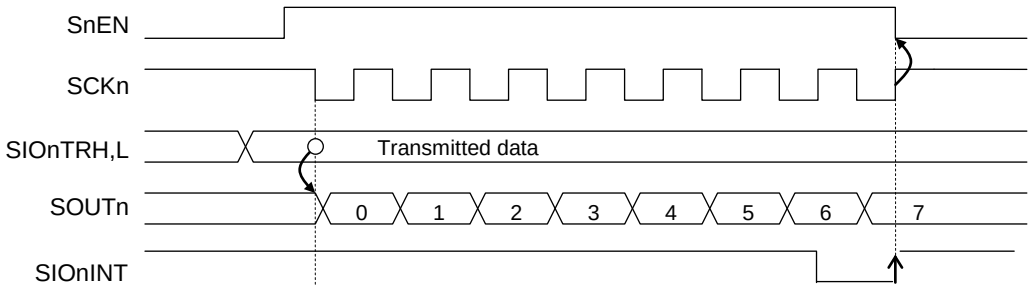


Figure 12-2 Transmit Operation Waveforms of Synchronous Serial Port for Clock Type 0 (Positive Logic)

(8-bit length, LSB first, n = 0, 1)

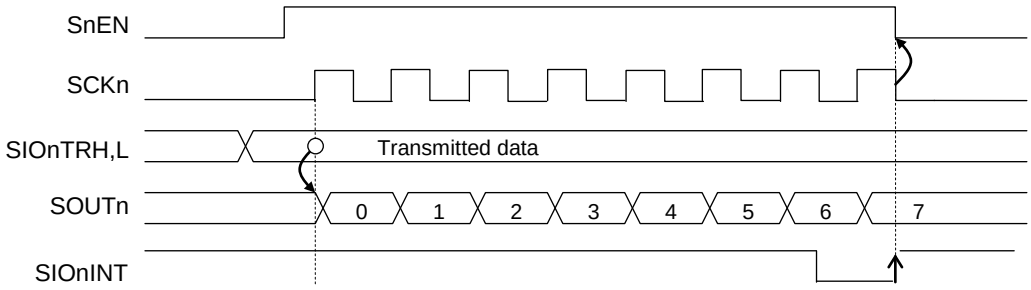


Figure 12-3 Transmit Operation Waveforms of Synchronous Serial Port for Clock Type 0 (Negative Logic)

(8-bit length, LSB first, n = 0, 1)

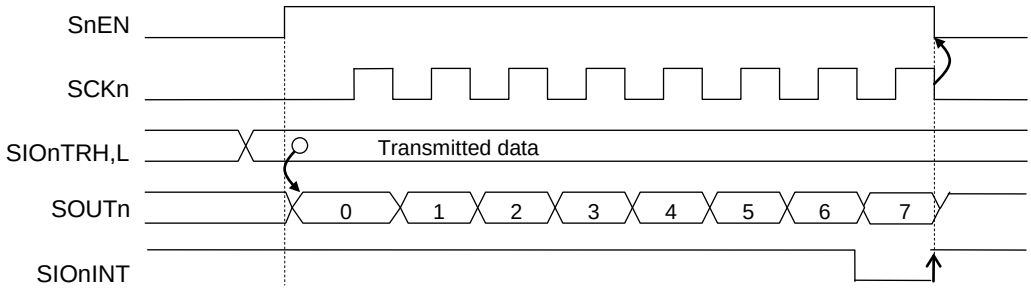


Figure 12-4 Transmit Operation Waveforms of Synchronous Serial Port for Clock Type 1 (Positive Logic)

(8-bit length, LSB first, n = 0, 1)

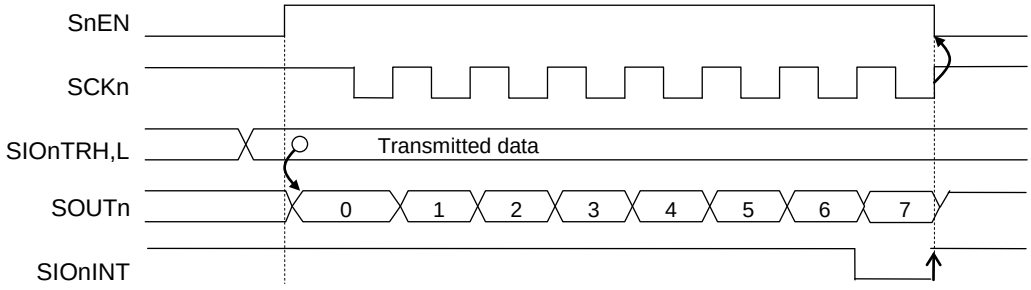


Figure 12-5 Transmit Operation Waveforms of Synchronous Serial Port for Clock Type 1 (Negative Logic)

(8-bit length, LSB first, n = 0, 1)

12.3.2 Receive Operation

When "0" is written to the SnMD1 bit and "1" is written to the SnMD0 bit of the serial port mode register (SIOnMOD0), this LSI is set to a receive mode.
When the SnEN bit of the serial port control register (SIOnCON) is set to "1", reception starts. When reception of 8/16-bit data terminates, a synchronous serial port interrupt (SIOnINT) occurs and the SnEN bit is set to "0".
The received data is input from the tertiary function pins (P40/SIN0 or P44/SIN0) or the tertiary function pin (P50/SIN1, P54SIN1) of GPIO.
When an internal clock is selected in the serial port mode register (SIOnMOD1), the LSI is set to a master mode and when an external clock (SCKn) is selected, the LSI is set to a slave mode.
The serial port mode register (SIOnMOD0) enables selection of MSB first/LSB first.
The received data input pin (SINn) and the transfer clock input/output pin (SCKn) need to be set to the tertiary function for the Port 4 or to the secondary function for the Port 5.
The receive operation waveforms of the synchronous serial port (8-bit length, MSB first) are shown in Figures 12-6 to 12-9, for the clock type 0 (positive logic), clock type 0 (negative logic), clock type 1 (positive logic), and clock type 1 (negative logic), respectively.

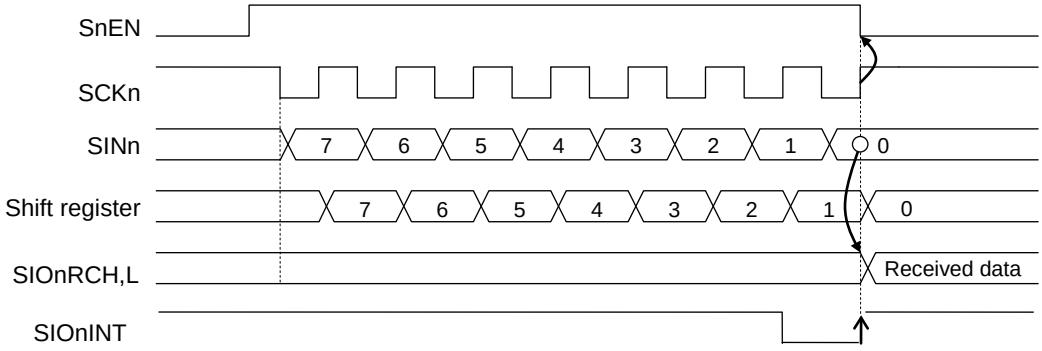


Figure 12-6 Receive Operation Waveforms of Synchronous Serial Port for Clock Type 0 (Positive Logic)

(8-bit length, MSB first)

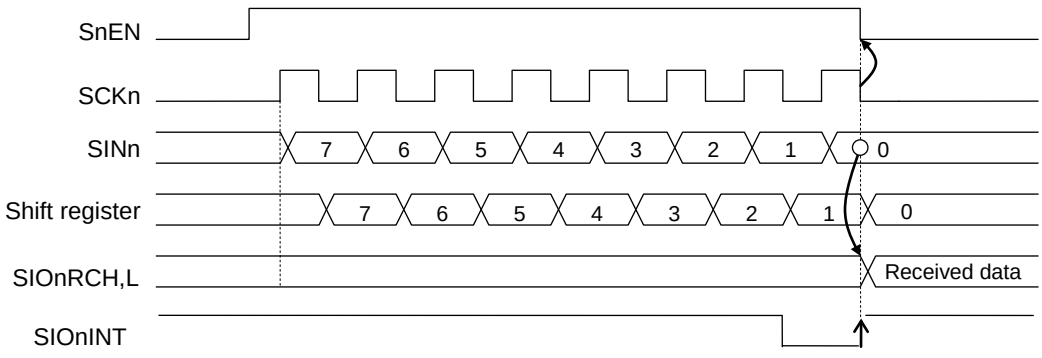


Figure 12-7 Receive Operation Waveforms of Synchronous Serial Port for Clock Type 0 (Negative Logic)

(8-bit length, MSB first)

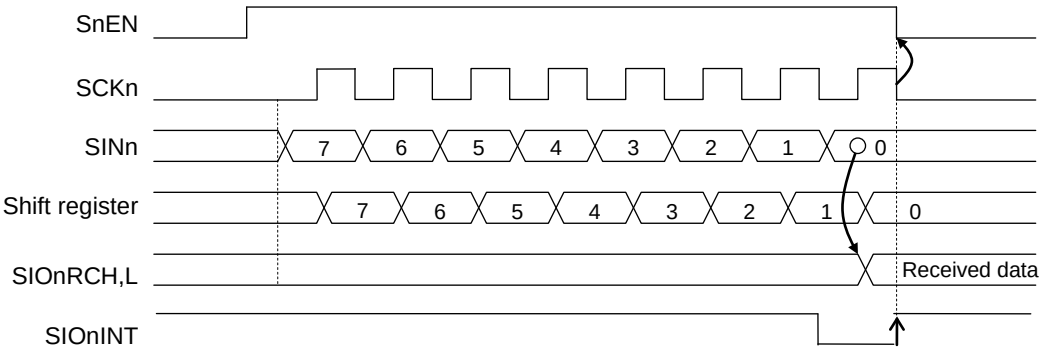


Figure 12-8 Receive Operation Waveforms of Synchronous Serial Port for Clock Type 1 (Positive Logic)

(8-bit length, MSB first)

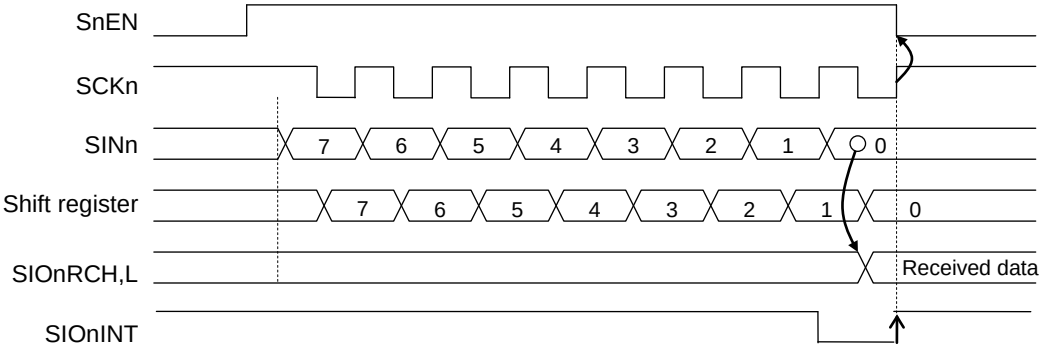


Figure 12-9 Receive Operation Waveforms of Synchronous Serial Port for Clock Type 1 (Negative Logic)

(8-bit length, MSB first)

Note:
When the SOUT0 pin is set to the tertiary function output in receive mode, a “H” level is output from the SOUT0 pin.
When the SOUT1 pin is set to the tertiary function output in receive mode, a “H” level is output from the SOUT1 pin.

12.3.3 Transmit/Receive Operation

When “1” is written to the SnMD1 bit and "1" is written to the SnMD0 bit of the serial port mode register (SIOnMOD0), this LSI is set to the transmit/receive mode.
When the SnEN bit of the serial port control register (SIOnCON) is set to “1”, transmission/reception starts. When transmission/reception of 8/16-bit data terminates, a synchronous serial port interrupt (SIOnINT) occurs and the SnEN bit is set to “0”.
The received data is input from the tertiary function pins (P40/SIN0 or P44/SIN0) or the tertiary function pin (P50/SIN1, P54SIN1) of GPIO, and the transmitted data is output from the tertiary function pins (P42/SOUT0 or P46/SOUT0) or the tertiary function pin (P52/SOUT1) of GPIO.
When an internal clock is selected in the serial port mode register (SIOnMOD1), the LSI is set to a master mode and when an external clock (SCKn) is selected, the LSI is set to a slave mode.
The serial port mode register (SIOnMOD0) enables selection of MSB first/LSB first.
The received data input pin (SINn), the transmitted data output pin (SOUTn), and the transfer clock input/output pin (SCKn) need to be set to the tertiary function for the Port 4 or to the secondary function for the Port 5.
Figure 12-10 shows the transmit/receive operation waveforms of the synchronous serial port (16-bit length, LSB first, clock types 0, positive logic).

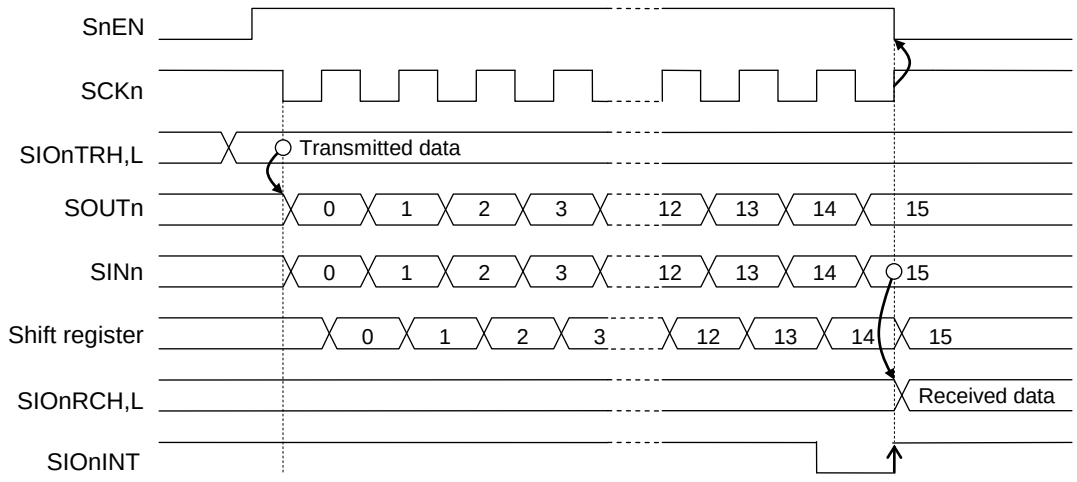


Figure 12-10 Transmit/Receive Operation Waveforms of Synchronous Serial Port

(16-bit Length, LSB First, Clock Type 0, Positive Logic))

12.4 Specifying port registers

To enable the SSIO(SSIO0, SSIO1) function, the applicable bit of each related port register needs to be set. See Chapter 17, "Port 4" and Chapter 18 "Port 5" for detail about the port registers.

12.4.1 Functioning P42 (SOUT0: Output), P41 (SCK0: Input/output), and P40 (SIN0: Input) as the SSIO0/ "Master mode"

Set the P42MD1 to P40MD1 bits (P4MOD1 register bits 2 to 0) to "1" and the P42MD0 to P40MD0 bits (P4MOD0 register bits 2 to 0) to "0" for selecting the SSIO as the tertiary function of the P42, P41 and P40.

Register name	P4MOD1 register (Address: 0F225H)							
Bit	7	6	5	4	3	**2**	**1**	**0**
Bit name	P47MD1	P46MD1	P45MD1	P44MD1	P43MD1	**P42MD1**	**P41MD1**	**P40MD1**
Setting value	*	*	*	*	*	**1**	**1**	**1**

Register name	P4MOD0 register (Address: 0F224H)							
Bit	7	6	5	4	3	**2**	**1**	**0**
Bit name	P47MD0	P46MD0	P45MD0	P44MD0	P43MD0	**P42MD0**	**P41MD0**	**P40MD0**
Setting value	*	*	*	*	*	**0**	**0**	**0**

Set P42C1-P41MC1 bits(bit2-bit1 of P4CON1 register) to "1", set P42C0-P41C0 bits(bit2-bit1 of P4CON0 register) to "1", and set P42DIR-P41DIR bits(bit2-bit1 of P4DIR register) to "0" for selecting the state mode of the P42 and P41 pins to CMOS output. Set the P40DIR bit (P4DIR register bit 0) to "1" for selecting the P40 as an input pin.
The set value ($) is arbitrary for the P40C1 and P40C0 bits. Select an arbitrary state mode depending on the state of the external circuit to which the P40 pin is connected.

Register name	P4CON1 register (Address: 0F223H)							
Bit	7	6	5	4	3	**2**	**1**	**0**
Bit name	P47C1	P46C1	P45C1	P44C1	P43C1	**P42C1**	**P41C1**	**P40C1**
Setting value	*	*	*	*	*	**1**	**1**	**$**

Register name	P4CON0 register (Address: 0F222H)							
Bit	7	6	5	4	3	**2**	**1**	**0**
Bit name	P47C0	P46C0	P45C0	P44C0	P43C0	**P42C0**	**P41C0**	**P40C0**
Setting value	*	*	*	*	*	**1**	**1**	**$**

Register name	P4DIR register (Address: 0F221H)							
Bit	7	6	5	4	3	**2**	**1**	**0**
Bit name	P47DIR	P46DIR	P45DIR	P44DIR	P43DIR	**P42DIR**	**P41DIR**	**P40DIR**
Setting value	*	*	*	*	*	**0**	**0**	**1**

The data of the P42D to P40D bits (P4D register bits 2 to 0) can either be "0" or "1".

Register name	P4D register (Address: 0F220H)							
Bit	7	6	5	4	3	**2**	**1**	**0**
Bit name	P47D	P46D	P45D	P44D	P43D	**P42D**	**P41D**	**P40D**
Setting value	*	*	*	*	*	******	******	******

* : Bit not related to the SSIO0 function
** : Don't care $: Optional

12.4.2 Functioning P42 (SOUT0: Output), P41 (SCK0: Input/output), and P40 (SIN0: Input) as the SSIO0/ "Slave mode"

Set the P42MD1 to P40MD1 bits (P4MOD1 register bits 2 to 0) to "1" and the P42MD0 to P40MD0 bits (P4MOD0 register bits 2 to 0) to "0" for selecting the SSIO as the tertiary function of the P42, P41 and P40. They are the same setting as those in the case of master mode.

Register name	P4MOD1 register (Address: 0F225H)							
Bit	7	6	5	4	3	**2**	**1**	**0**
Bit name	P47MD1	P46MD1	P45MD1	P44MD1	P43MD1	**P42MD1**	**P41MD1**	**P40MD1**
Setting value	*	*	*	*	*	**1**	**1**	**1**

Register name	P4MOD0 register (Address: 0F224H)							
Bit	7	6	5	4	3	**2**	**1**	**0**
Bit name	P47MD0	P46MD0	P45MD0	P44MD0	P43MD0	**P42MD0**	**P41MD0**	**P40MD0**
Setting value	*	*	*	*	*	**0**	**0**	**0**

Set the P42C1 bit (P4CON1 register bit 2) to "1", the P42C0 bit (P4CON0 register bit 2) to "1", and the P42DIR bit (P4DIR register bit 2) to "0" for selecting the P42 pin state mode to CMOS output.
Set P41DIR to P40DIR bits (P4DIR register bit 1 to 0) to "1" for specifying the P41 and P40 as input pins.
The set value ($) is arbitrary for the P41C1 to P40C1 bits and for the P41C0 to P40C0 bits. Select an arbitrary input mode depending on the state of the external circuit to which the P41 or P40 pin is connected.

Register name	P4CON1 register (Address: 0F223H)							
Bit	7	6	5	4	3	**2**	**1**	**0**
Bit name	P47C1	P46C1	P45C1	P44C1	P43C1	**P42C1**	**P41C1**	**P40C1**
Setting value	*	*	*	*	*	**1**	**$**	**$**

Register name	P4CON0 register (Address: 0F222H)							
Bit	7	6	5	4	3	**2**	**1**	**0**
Bit name	P47C0	P46C0	P45C0	P44C0	P43C0	**P42C0**	**P41C0**	**P40C0**
Setting value	*	*	*	*	*	**1**	**$**	**$**

Register name	P4DIR register (Address: 0F221H)							
Bit	7	6	5	4	3	**2**	**1**	**0**
Bit name	P47DIR	P46DIR	P45DIR	P44DIR	P43DIR	**P42DIR**	**P41DIR**	**P40DIR**
Setting value	*	*	*	*	*	**0**	**1**	**1**

The data of the P42D to P40D bits (P4D register bits 2 to 0) can either be "0" or "1".

Register name	P4D register (Address: 0F220H)							
Bit	7	6	5	4	3	**2**	**1**	**0**
Bit name	P47D	P46D	P45D	P44D	P43D	**P42D**	**P41D**	**P40D**
Setting value	*	*	*	*	*	******	******	******

* : Bit not related to the SSIO0 function
** : Don't care $: Optional

12.4.3 Functioning P46 (SOUT0: Output), P45 (SCK0: Input/output) and P44 (SIN0: Input) as the SSIO0/ "Master mode"

Set P46MD1-P44MD1 bits(bit6-bit4 of P4MOD1 register) to "1" and set P46MD0-P44MD0(bit6-bit4 of P4MOD0 register) to "0", for specifying the SSIO0 as the tertiary function of P46, P45 and P44. Set P40MD1 bit(bit0 of P4MOD1) to "0", and set P40 to be primary or secondary function as not to be SSIO function.

Register name	P4MOD1 register (Address: 0F225H)							
Bit	7	**6**	**5**	**4**	3	2	1	**0**
Bit name	P47MD1	**P46MD1**	**P45MD1**	**P44MD1**	P43MD1	P42MD1	P41MD1	**P40MD1**
Setting value	*	**1**	**1**	**1**	*	*	*	**0**

Register name	P4MOD0 register (Address: 0F224H)							
Bit	7	**6**	**5**	**4**	3	2	1	0
Bit name	P47MD0	**P46MD0**	**P45MD0**	**P44MD0**	P43MD0	P42MD0	P41MD0	P40MD0
Setting value	*	**0**	**0**	**0**	*	*	*	*

Set P46C1-P45MC1 bits(bit6-bit5 of P4CON1 register) to "1", set P46C0-P45C0 bits(bit6-bit5 of P4CON0 register) to "1", and set P46DIR-P45DIR bits(bit6-bit5 of P4DIR register) to "0" for selecting the state mode of the P46 and P45 pins to CMOS output. Set P44DIR bit(bit4 of P4DIR register) to "1" for specifying the P44 as an input pin.
The set value ($) is arbitrary for the P44C1 and P44C0 bits. Select an arbitrary state mode depending on the state of the external circuit to which the P44 pin is connected.

Register name	P4CON1 register (Address: 0F223H)							
Bit	7	**6**	**5**	**4**	3	2	1	0
Bit name	P47C1	**P46C1**	**P45C1**	**P44C1**	P43C1	P42C1	P41C1	P40C1
Setting value	*	**1**	**1**	**$**	*	*	*	*

Register name	P4CON0 register (Address: 0F222H)							
Bit	7	**6**	**5**	**4**	3	2	1	0
Bit name	P47C0	**P46C0**	**P45C0**	**P44C0**	P43C0	P42C0	P41C0	P40C0
Setting value	*	**1**	**1**	**$**	*	*	*	*

Register name	P4DIR register (Address: 0F221H)							
Bit	7	**6**	**5**	**4**	3	2	1	0
Bit name	P47DIR	**P46DIR**	**P45DIR**	**P44DIR**	P43DIR	P42DIR	P41DIR	P40DIR
Setting value	*	**0**	**0**	**1**	*	*	*	*

The P46D to P44D bits (P4D register bits 6 to 4) data can either be "0" or "1" (not need to be set).

Register name	P4D register (Address: 0F220H)							
Bit	7	**6**	**5**	**4**	3	2	1	0
Bit name	P47D	**P46D**	**P45D**	**P44D**	P43D	P42D	P41D	P40D
Setting value	*	******	******	******	*	*	*	*

* : Bit not related to the SSIO0 function
** : Don't care $: Optional

12.4.4 Functioning P46 (SOUT0: Output), P45 (SCK0: Input/output) and P44 (SIN0: Input) as the SSIO0/ "Slave mode"

Set P46MD1-P44MD1 bits(bit6-bit4 of P4MOD1 register) to "1" and set P46MD0-P44MD0(bit6-bit4 of P4MOD0 register) to "0", for specifying the SSIO0 as the tertiary function of P46, P45 and P44. Set P40MD1 bit(bit0 of P4MOD1) to "0", and set P40 to be primary or secondary function as not to be SSIO function. They are the same setting as those in the case of master mode.

Register name	P4MOD1 register (Address: 0F225H)							
Bit	7	**6**	**5**	**4**	3	2	1	0
Bit name	P47MD1	**P46MD1**	**P45MD1**	**P44MD1**	P43MD1	P42MD1	P41MD1	**P40MD1**
Setting value	*	**1**	**1**	**1**	*	*	*	**0**

Register name	P4MOD0 register (Address: 0F224H)							
Bit	7	**6**	**5**	**4**	3	2	1	0
Bit name	P47MD0	**P46MD0**	**P45MD0**	**P44MD0**	P43MD0	P42MD0	P41MD0	P40MD0
Setting value	*	**0**	**0**	**0**	*	*	*	*

Set P46C1 bit(bit6 of P4CON1 register) to "1", set P46C0 bit(bit6 of P4CON0 register) to "1", and set P46DIR bit(bit6 of P4DIR register) to "0" for selecting the P46 pin state mode to CMOS output.
Set P45DIR-P44DIR bits(bit5-4 of P4DIR register) to "1" for specifying the P45 and P44 as input pins.
The set value ($) is arbitrary for the P45C1 to P44C1 bits and for the P45C0 to P44C0 bits. Select an arbitrary input mode depending on the state of the external circuit to which the P45 or P44 pin is connected.

Register name	P4CON1 register (Address: 0F223H)							
Bit	7	**6**	**5**	**4**	3	2	1	0
Bit name	P47C1	**P46C1**	**P45C1**	**P44C1**	P43C1	P42C1	P41C1	P40C1
Setting value	*	**1**	**$**	**$**	*	*	*	*

Register name	P4CON0 register (Address: 0F222H)							
Bit	7	**6**	**5**	**4**	3	2	1	0
Bit name	P47C0	**P46C0**	**P45C0**	**P44C0**	P43C0	P42C0	P41C0	P40C0
Setting value	*	**1**	**$**	**$**	*	*	*	*

Register name	P4DIR register (Address: 0F221H)							
Bit	7	**6**	**5**	**4**	3	2	1	0
Bit name	P47DIR	**P46DIR**	**P45DIR**	**P44DIR**	P43DIR	P42DIR	P41DIR	P40DIR
Setting value	*	**0**	**1**	**1**	*	*	*	*

The P46D to P44D bits (P4D register bits 6 to 4) data can either be "0" or "1" (not need to be set).

Register name	P4D register (Address: 0F220H)							
Bit	7	**6**	**5**	**4**	3	2	1	0
Bit name	P47D	**P46D**	**P45D**	**P44D**	P43D	P42D	P41D	P40D
Setting value	*	******	******	******	*	*	*	*

* : Bit not related to the SSIO0 function
** : Don't care $: Optional

12.4.5 Functioning P52 (SOUT1: Output), P51 (SCK1: Input/output), and P50 (SIN1: Input) as the SSIO1/ “Master mode”

Set the P52MD1 to P50MD1 bits (P5MOD1 register bits 2 to 0) to "1", and set the P52MD0 to P50MD0 bits (P5MOD0 register bits 2 to 0) to "0" for specifying the SSIO1 as the secondary function of the P52, P51, and P50.

Register name	P5MOD1 register (address: 0F22DH)							
Bit	7	6	5	4	3	**2**	**1**	**0**
Bit name	—	P56MD1	P55MD1	P54MD1	—	**P52MD1**	**P51MD1**	**P50MD1**
Setting value	*	*	*	*	*	**1**	**1**	**1**

Register name	P5MOD0 register (address: 0F22CH)							
Bit	7	6	5	4	3	**2**	**1**	**0**
Bit name	—	P56MD0	P55MD0	P54MD0	—	**P52MD0**	**P51MD0**	**P50MD0**
Setting value	*	*	*	*	*	**0**	**0**	**0**

Set the P52C0 to P51C0 bits (P5CON0 register bits 2 to 1) to "1", and the P52DIR to P51DIR bits (P5DIR register bits 2 to 1) to "0" for selecting the P52 and P51 pin state mode to CMOS output. Set the P50DIR bit (P5DIR register bit 0) to “1” for selecting the P50 as an input pin.
The set value ($) is arbitrary for the P5UD and P50C0 bits. Select an arbitrary state mode depending on the state of the external circuit to which the P50 pin is connected.

Register name	P5CON1 register (address: 0F22BH)							
Bit	7	6	5	4	3	2	1	**0**
Bit name	—	—	—	—	—	—	—	**P5UD**
Setting value	*	*	*	*	*	*	*	**$**

Register name	P5CON0 register (address: 0F22AH)							
Bit	7	6	5	4	3	**2**	**1**	**0**
Bit name	P57C0	P56C0	P55C0	P54C0	P53C0	**P52C0**	**P51C0**	**P50C0**
Setting value	*	*	*	*	*	**1**	**1**	**$**

Register name	P5DIR register (address: 0F229H)							
Bit	7	6	5	4	3	**2**	**1**	**0**
Bit name	P57DIR	P56DIR	P55DIR	P54DIR	P53DIR	**P52DIR**	**P51DIR**	**P50DIR**
Setting value	*	*	*	*	*	**0**	**0**	**1**

The P52D to P50D bits (P5D register bits 2 to 0) data can either be "0" or "1" (not need to be set).

Register name	P5D register (address: 0F228H)							
Bit	7	6	5	4	3	**2**	**1**	**0**
Bit name	P57D	P56D	P55D	P54D	P53D	**P52D**	**P51D**	**P50D**
Setting value	*	*	*	*	*	******	******	******

* : Bit not related to the SSIO1 function
** : Don’t care $: Optional

12.4.6 Functioning P52 (SOUT1: Output), P51 (SCK1: Input/output), and P50 (SIN1: Input) as the SSIO1/ "Slave mode"

Set the P52MD1 to P50MD1 bits (P5MOD1 register bits 2 to 0) to "1", and set the P52MD0 to P50MD0 bits (P5MOD0 register bits 2 to 0) to "0" for specifying the SSIO as the tertiary function of the P52, P51, and P50. They are the same setting as those in the case of master mode.

Register name	P5MOD1 register (address: 0F22DH)							
Bit	7	6	5	4	3	**2**	**1**	**0**
Bit name	—	P56MD1	P55MD1	P54MD1	—	**P52MD1**	**P51MD1**	**P50MD1**
Setting value	*	*	*	*	*	**1**	**1**	**1**

Register name	P5MOD0 register (address: 0F22CH)							
Bit	7	6	5	4	3	**2**	**1**	**0**
Bit name	—	P56MD0	P55MD0	P54MD0	—	**P52MD0**	**P51MD0**	**P50MD0**
Setting value	*	*	*	*	*	**0**	**0**	**0**

Set the P52C0 bit (P5CON0 register bit 2) to "1", and the P52DIR bit (P5DIR register bit 2) to "0" for selecting the P52 pin state mode to CMOS output.
Set P51DIR to P50DIR bits (P5DIR register bits 1 to 0) to "1" for selecting the P51 and P50 as input pins.
The set value ($) is arbitrary for the P5UD bits and for the P51C0 to P50C0 bits. Select an arbitrary input mode depending on the state of the external circuit to which the P51 or P50 pin is connected.

Register name	P5CON1 register (address: 0F22BH)							
Bit	7	6	5	4	3	**2**	**1**	**0**
Bit name	—	—	—	—	—	—	—	**P5UD**
Setting value	*	*	*	*	*	*	*	**$**

Register name	P5CON0 register (address: 0F22AH)							
Bit	7	6	5	4	3	**2**	**1**	**0**
Bit name	P57C0	P56C0	P55C0	P54C0	P53C0	**P52C0**	**P51C0**	**P50C0**
Setting value	*	*	*	*	*	**1**	**$**	**$**

Register name	P5DIR register (address: 0F229H)							
Bit	7	6	5	4	3	**2**	**1**	**0**
Bit name	P57DIR	P56DIR	P55DIR	P54DIR	P53DIR	**P52DIR**	**P51DIR**	**P50DIR**
Setting value	*	*	*	*	*	**0**	**1**	**1**

The P52D to P50D bits (P5D register bits 2 to 0) data can either be "0" or "1" (not need to be set).

Register name	P5D register (address: 0F228H)							
Bit	7	6	5	4	3	**2**	**1**	**0**
Bit name	P57D	P56D	P55D	P54D	P53D	**P52D**	**P51D**	**P50D**
Setting value	*	*	*	*	*	******	******	******

* : Bit not related to the SSIO1 function
** : Don't care $: Optional

Chapter 13

UART

13. UART

13.1 Overview

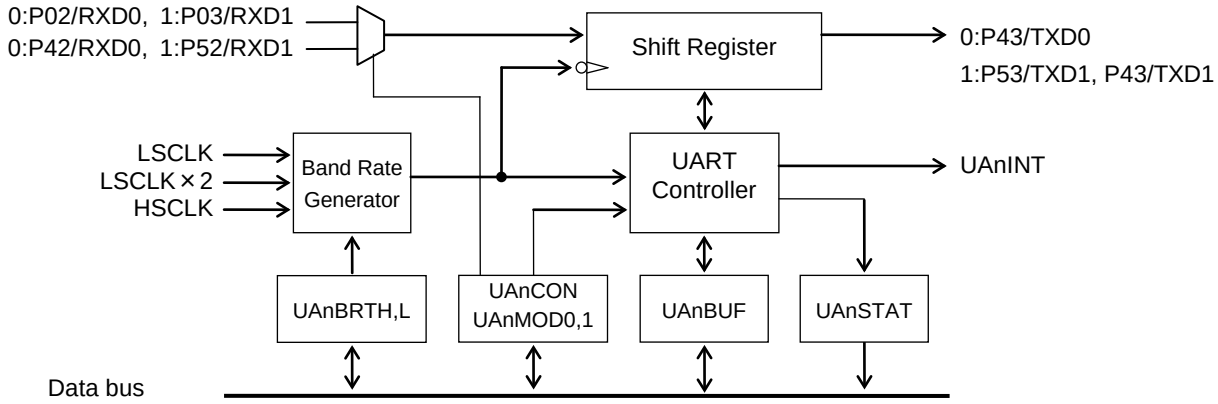
This LSI includes 2 channels of UART (Universal Asynchronous Receiver Transmitter) which are asynchronous serial interface (A full-duplex is also possible by using 2 channels).
For the input clock, see Chapter 6, "Clock Generation Circuit".
The use of UART requires setting of the secondary or quaternary functions of Port 4 and setting of the secondary functions of Port 5. For the secondary or quaternary functions of Port 4, see Chapter 19, "Port 4". For the secondary functions of Port 5, see Chapter 20, "Port 5".

13.1.1 Features

- 5-bit/6-bit/7-bit/8-bit data length selectable.
- Odd parity, even parity, or no parity selectable.
- 1 stop bit or 2 stop bits selectable.
- Provided with parity error flag, overrun error flag, framing error flag, and transmit buffer status flag.
- Positive logic or negative logic selectable as communication logic.
- LSB first or MSB first selectable as a communication direction.
- Communication speed: Settable within the range of 200bps to 38400bps.
- Built-in baud rate generator.

13.1.2 Configuration

Figure 13-1 shows the configuration of the UART.



UAnBUF : UARTn transmit/receive buffer
UAnBRTH,L : UARTn baud rate H and L are:
UAnCON : UARTn control register
UAnMOD0,1 : UARTn mode registers 0 and 1
UAnSTAT : UARTn status register
*n=0,1

Figure 13-1 Configuration of UART

13.1.3 List of Pins

Pin name	I/O	Function
P02/RXD0	I	UART0 data input pin Used for the primary function of the P02 pin.
P42/RXD0	I	UART0 data input pin Used for the secondary function of the P42 pin.
P43/TXD0	O	UART0 data output pin Used for the secondary function of the P43 pin.
P03/RXD1	I	UART1 data input pin Used for the primary function of the P03 pin.
P52/RXD1	I	UART1 data input pin Used for the secondary function of the P52 pin.
P53/TXD1	O	UART1 data output pin Used for the secondary function of the P53 pin.
P43/TXD1	O	UART1 data output pin Used for the quaternary function of the P43 pin.

13.2 Description of Registers

13.2.1 List of Registers

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F290H	UART0 transmit/receive buffer	UA0BUF	−	R/W	8	00H
0F291H	UART0 control register	UA0CON	−	R/W	8	00H
0F292H	UART0 mode register 0	UA0MOD0	UA0MOD	R/W	8/16	00H
0F293H	UART0 mode register 1	UA0MOD1		R/W	8	00H
0F294H	UART0 baud rate register L	UA0BRTL	UA0BRT	R/W	8/16	0FFH
0F295H	UART0 baud rate register H	UA0BRTH		R/W	8	0FH
0F296H	UART0 status register	UA0STAT	−	R/W	8	00H
0F298H	UART1 transmit/receive buffer	UA1BUF	−	R/W	8	00H
0F299H	UART1 control register	UA1CON	−	R/W	8	00H
0F29AH	UART1 mode register 0	UA1MOD0	UA1MOD	R/W	8/16	00H
0F29BH	UART1 mode register 1	UA1MOD1		R/W	8	00H
0F29CH	UART1 baud rate register L	UA1BRTL	UA1BRT	R/W	8/16	0FFH
0F29DH	UART1 baud rate register H	UA1BRTH		R/W	8	0FH
0F29EH	UART1 status register	UA1STAT	−	R/W	8	00H

13.2.2 UART0 Transmit/Receive Buffer (UA0BUF)

Address: 0F290H
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
UA0BUF	U0B7	U0B6	U0B5	U0B4	U0B3	U0B2	U0B1	U0B0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

UA0BUF is a special function register (SFR) to store the transmitted/received data of the UART.
In transmit mode, write transmission data to UA0BUF. To transmit the data consecutively, confirm the U0FUL flag of the UART0 status register (UA0STAT) becomes "0", then write the next transmitted data to the UA0BUF. Any value written to UA0BUF can be read.
In receive mode, since data received at termination of reception is stored in UA0BUF, read the contents of UABUF using the UART0 interrupt at termination of reception. At continuous reception, BA0BUF is updated whenever reception terminates. Any write to BA0BUF is disabled in receive mode.
The bits, which are not required when any of the 5- to 7-bit data length is selected, become invalid in transmit mode and are set to “0” in receive mode.

Note:
For operation in transmit mode, be sure to set the transmit mode (UA0MOD0 and UA0MOD1) before setting the transmitted data in UAOBUF.

13.2.3 UART0 Control Register (UA0CON)

Address: 0F291H
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
UA0CON	—	—	—	—	—	—	—	U0EN
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

UA0CON is a special function register (SFR) to start/stop communication of the UART.

[Description of Bits]

- **U0EN** (bit 0)
  The U0EN bit is used to specify the UART communication operation start. When U0EN is set to "1", UART communication starts. In transmit mode, this bit is automatically set to "0" at termination of transmission. In receive mode, receive operation is continued. To terminate reception, set the bit to "0" by software.

U0EN	Description
0	Stops communication. (Initial value)
1	Starts communication

13.2.4 UART0 Mode Register 0 (UA0MOD0)

Address: 0F292H
Access: R/W
Access size: 8/16 bit
Initial value: 00H

	7	6	5	4	3	2	1	0
UA0MOD0	−	−	U0RSS	U0RSEL	−	U0CK1	U0CK0	U0IO
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

UA0MOD0 is a special function register (SFR) to set the transfer mode of the UART.

[Description of Bits]

- **U0IO** (bit 0)
  The U0IO bit is used to select transmit or receive mode.

U0IO	Description
0	Transmit mode (initial value)
1	Receive mode

- **U0CK1, U0CK0** (bits 2, 1)
  The U0CK1 and U0CK0 bits are used to select the clock to be input to the baud rate generator of the UART0.

U0CK1	U0CK0	Description
0	0	LSCLK (initial value)
0	1	LSCLK×2
1	*	HSCLK

- **U0RSEL** (bit 4)
  The U0RSEL bit is used to select the received data input pin for the UART0.

U0RSEL	Description
0	Selects the P02 pin. (Initial value)
1	Selects the P42 pin.

- **U0RSS** (bit 5)
  U0RSS is the bit that selects the UART0 received data input sampling timing.

U0RSS	Description
0	Value set in the UA0BRTH and UA0BRTL registers/2 (initial value)
1	Value set in the UA0BRTH and UA0BRTL registers/2-1

Note:
Always set the UA0MOD0 register while communication is stopped, and do not rewrite it during communication.
When specifying LSCLK X 2 for the clock, enable the operation of the low-speed double clock by setting bit 2 (ENMLT) of the frequency control register 1 (FCON1) to "1".
When selecting the P42 pin as the received data input pin, it is necessary to configure settings for the Port 4 secondary functions. For the secondary functions of Port 4, see Chapter 17, "Port 4".

13.2.5 UART0 Mode Register 1 (UA0MOD1)

Address: 0F293H
Access: R/W
Access size: 8/16 bit
Initial value: 00H

	7	6	5	4	3	2	1	0
UA0MOD1	—	U0DIR	U0NEG	U0STP	U0PT1	U0PT0	U0LG1	U0LG0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

UA0MOD1 is a special function register (SFR) to set the transfer mode of the UART.

[Description of Bits]

- **U0LG1, U0LG0** (bits 1, 0)
  The U0LG1 and U0LG0 bits are used to specify the data length in the communication of the UART.

U0LG1	U0LG0	Description
0	0	8-bit length (initial value)
0	1	7-bit length
1	0	6-bit length
1	1	5-bit length

- **U0PT1, U0PT0** (bits 3, 2)
  The U0PT1 and U0PT0 bits are used to select “even parity”, odd parity”, or “no parity” in the communication of the UART.

U0PT1	U0PT0	Description
0	0	Even parity (initial value)
0	1	Odd parity
1	*	No parity bit

- **U0STP** (bit 4)
  The U0STP bit is used to select the stop bit length in the communication of the UART.

U0STP	Description
0	1 stop bit (initial value)
1	2 stop bits

- **U0NEG** (bit 5)
  The U0NEG bit is used to select positive logic or negative logic in the communication of the UART.

U0NEG	Description
0	Positive logic (initial value)
1	Negative logic

- **U0DIR** (bit 6)
  The U0DIR bit is used to select LSB first or MSB first in the communication of the UART.

U0DIR	Description
0	LSB first (initial value)
1	MSB first

Note:
Always set the UA0MOD1 register while communication is stopped, and do not rewrite it during communication.

13.2.6 UART0 Baud Rate Registers L, H (UA0BRTL, UA0BRTH)

Address: 0F294H
Access: R/W
Access size: 8/16 bit
Initial value: 0FFH

	7	6	5	4	3	2	1	0
UA0BRTL	U0BR7	U0BR6	U0BR5	U0BR4	U0BR3	U0BR2	U0BR1	U0BR0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	1	1	1	1	1	1	1	1

Address: 0F295H
Access: R/W
Access size: 8-bit
Initial value: 0FH

	7	6	5	4	3	2	1	0
UA0BRTH	—	—	—	—	U0BR11	U0BR10	U0BR9	U0BR8
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	1	1	1	1

UA0BRTL and UA0BRTH are special function registers (SFRs) to set the count value of the baud rate generator which generates baud rate clocks.
For the relationship between the count value of the baud rate generator and baud rate, see Section 13.3.2, “Baud Rate”.

Note:
Always set the UA0BRTL and UA0BRTH registers while communication is stopped, and do not rewrite them during communication.

13.2.7 UART0 Status Register (UA0STAT)

Address: 0F296H
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
UA0STAT	—	—	—	—	U0FUL	U0PER	U0OER	U0FER
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

UA0STAT is a special function register (SFR) to indicate the state of transmit or receive operation of the UART.
When any data is written to UA0STAT, all the flags are initialized to "0".

[Description of Bits]

- **U0FER** (bit 0)
  The U0FER bit is used to indicate occurrence of a framing error of the UART.
  When an error occurs in the start or stop bit, the U0FER bit is set to "1". This bit is updated each time reception is completed.
  The U0FER bit is fixed to "0" in transmit mode.

U0FER	Description
0	No framing error (initial value)
1	With framing error

- **U0OER** (bit 1)
  The U0OER bit is used to indicate occurrence of an overrun error of the UART.
  If the received data in the transmit/receive buffer (UA0BUF) is received again before it is read, this bit is set to "1". Even if reception is stopped by the U0EN bit and then reception is restarted, this bit is set to "1" if the previously received data is not read. Therefore, make sure that data is always read from the transmit/receive buffer even if the data is not required.
  The U0OER bit is fixed to "0" in transmit mode.

U0OER	Description
0	No overrun error (initial value)
1	Overrun error

- **U0PER** (bit 2)
  The U0PER bit is used to indicate occurrence of a parity error of the UART.
  When the parity of the received data and the parity bit attached to the data do not coincide, this bit is set to "1". U0PER is updated whenever data is received.
  The U0PER bit is fixed to "0" in transmit mode.

U0PER	Description
0	No parity error (initial value)
1	Parity error

- **U0FUL** (bit 3)
  The U0FUL bit is used to indicate the state of the transmit/receive buffer of the UART.
  When the transmitted data is written in UA0BUF in transmit mode, this bit is set to “1” and when this transmitted data is transferred to the shift register, this bit is set to “0”. To transmit the data consecutively, confirm the U0FUL flag becomes "0", then write the next transmitted data to the UA0BUF.
  The U0FUL bit is fixed to “0” in receive mode.

U0FUL	Description
0	There is no data in the transmit/receive buffer. (Initial value)
1	There is data in the transmit/receive buffer.

13.2.8 UART1 Transmit/Receive Buffer (UA1BUF)

Address: 0F298H
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
UA0BUF	U1B7	U1B6	U1B5	U1B4	U1B3	U1B2	U1B1	U1B0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

UA1BUF is a special function register (SFR) to store the transmitted/received data of the UART1.
In transmit mode, write transmission data to UA1BUF. To transmit the data consecutively, confirm the U1FUL flag of the UART0 status register (UA1STAT) becomes "0", then write the next transmitted data to the UA1BUF. Any value written to UA1BUF can be read.
In receive mode, since data received at termination of reception is stored in UA1BUF, read the contents of UABUF using the UART1 interrupt at termination of reception. At continuous reception, BA1BUF is updated whenever reception terminates. Any write to BA1BUF is disabled in receive mode.
The bits, which are not required when any of the 5- to 7-bit data length is selected, become invalid in transmit mode and are set to “0” in receive mode.

Note:
For operation in transmit mode, be sure to set the transmit mode (UA1MOD0 and UA1MOD1) before setting the transmitted data in UAOBUF.

13.2.9 UART1 Control Register (UA1CON)

Address: 0F299H
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
UA0CON	—	—	—	—	—	—	—	U1EN
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

UA1CON is a special function register (SFR) to start/stop communication of the UART.

[Description of Bits]

- **U1EN** (bit 0)
  The U1EN bit is used to specify the UART communication operation start. When U1EN is set to "1", UART communication starts. In transmit mode, this bit is automatically set to "0" at termination of transmission. In receive mode, receive operation is continued. To terminate reception, set the bit to "0" by software.

U1EN	Description
0	Stops communication. (Initial value)
1	Starts communication

13.2.10 UART1 Mode Register 1 (UA1MOD0)

Address: 0F29AH
Access: R/W
Access size: 8/16 bit
Initial value: 00H

	7	6	5	4	3	2	1	0
UA0MOD0	−	−	U1RSS	U1RSEL	−	U1CK1	U1CK0	U1IO
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

UA1MOD0 is a special function register (SFR) to set the transfer mode of the UART1.

[Description of Bits]

- **U1IO** (bit 0)
  The U1IO bit is used to select transmit or receive mode.

U1IO	Description
0	Transmit mode (initial value)
1	Receive mode

- **U1CK1, U1CK0** (bits 2, 1)
  The U1CK1 and U0CK0 bits are used to select the clock to be input to the baud rate generator of the UART1.

U1CK1	U1CK0	Description
0	0	LSCLK (initial value)
0	1	LSCLK×2
1	*	HSCLK

- **U1RSEL** (bit 4)
  The U1RSEL bit is used to select the received data input pin for the UART1.

U1RSEL	Description
0	Selects the P03 pin. (Initial value)
1	Selects the P52 pin.

- **U1RSS** (bit 5)
  U1RSS is the bit that selects the UART0 received data input sampling timing.

U1RSS	Description
0	Value set in the UA1BRTH and UA1BRTL registers/2 (initial value)
1	Value set in the UA1BRTH and UA1BRTL registers/2-1

Note:
Always set the UA1MOD0 register while communication is stopped, and do not rewrite it during communication.
When specifying LSCLK X 2 for the clock, enable the operation of the low-speed double clock by setting bit 2 (ENMLT) of the frequency control register 1 (FCON1) to "1".
When selecting the P52 pin as the received data input pin, it is necessary to configure settings for the Port 5 secondary functions. For the secondary functions of Port 5, see Chapter 20, "Port 5".

13.2.11 UART1 Mode Register 1 (UA1MOD1)

Address: 0F29BH
Access: R/W
Access size: 8/16 bit
Initial value: 00H

	7	6	5	4	3	2	1	0
UA1MOD1	—	U1DIR	U1NEG	U1STP	U1PT1	U1PT0	U1LG1	U1LG0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

UA1MOD1 is a special function register (SFR) to set the transfer mode of the UART1.

[Description of Bits]

- **U1LG1, U1LG0** (bits 1, 0)
  The U1LG1 and U1LG0 bits are used to specify the data length in the communication of the UART1.

U1LG1	U1LG0	Description
0	0	8-bit length (initial value)
0	1	7-bit length
1	0	6-bit length
1	1	5-bit length

- **U1PT1, U1PT0** (bits 3, 2)
  The U1PT1 and U1PT0 bits are used to select “even parity”, odd parity”, or “no parity” in the communication of the UART1.

U1PT1	U1PT0	Description
0	0	Even parity (initial value)
0	1	Odd parity
1	*	No parity bit

- **U1STP** (bit 4)
  The U1STP bit is used to select the stop bit length in the communication of the UART1.

U1STP	Description
0	1 stop bit (initial value)
1	2 stop bits

- **U1NEG** (bit 5)
  The U1NEG bit is used to select positive logic or negative logic in the communication of the UART1.

U1NEG	Description
0	Positive logic (initial value)
1	Negative logic

- **U1DIR** (bit 6)
  The U0DIR bit is used to select LSB first or MSB first in the communication of the UART1.

U1DIR	Description
0	LSB first (initial value)
1	MSB first

Note:
Always set the UA1MOD1 register while communication is stopped, and do not rewrite it during communication.

13.2.12 UART1 Baud Rate Registers L, H (UA1BRTL, UA1BRTH)

Address: 0F29CH
Access: R/W
Access size: 8/16 bit
Initial value: 0FFH

	7	6	5	4	3	2	1	0
UA1BRTL	U1BR7	U1BR6	U1BR5	U1BR4	U1BR3	U1BR2	U1BR1	U1BR0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	1	1	1	1	1	1	1	1

Address: 0F29DH
Access: R/W
Access size: 8-bit
Initial value: 0FH

	7	6	5	4	3	2	1	0
UA1BRTH	—	—	—	—	U1BR11	U1BR10	U1BR9	U1BR8
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	1	1	1	1

UA1BRTL and UA1BRTH are special function registers (SFRs) to set the count value of the baud rate generator which generates baud rate clocks.
For the relationship between the count value of the baud rate generator and baud rate, see Section 13.3.2, “Baud Rate”.

Note:
Always set the UA1BRTL and UA1BRTH registers while communication is stopped, and do not rewrite them during communication.

13.2.13 UART1 Status Register (UA1STAT)

Address: 0F29EH
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
UA1STAT	—	—	—	—	U1FUL	U1PER	U1OER	U1FER
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

UA1STAT is a special function register (SFR) to indicate the state of transmit or receive operation of the UART1.
When any data is written to UA1STAT, all the flags are initialized to "0".

[Description of Bits]

- **U1FER** (bit 0)
  The U1FER bit is used to indicate occurrence of a framing error of the UART1.
  When an error occurs in the start or stop bit, the U1FER bit is set to "1". This bit is updated each time reception is completed.
  The U1FER bit is fixed to "0" in transmit mode.

U1FER	Description
0	No framing error (initial value)
1	With framing error

- **U1OER** (bit 1)
  The U1OER bit is used to indicate occurrence of an overrun error of the UART1.
  If the received data in the transmit/receive buffer (UA1BUF) is received again before it is read, this bit is set to "1". Even if reception is stopped by the U0EN bit and then reception is restarted, this bit is set to "1" if the previously received data is not read. Therefore, make sure that data is always read from the transmit/receive buffer even if the data is not required.
  The U1OER bit is fixed to "0" in transmit mode.

U1OER	Description
0	No overrun error (initial value)
1	Overrun error

- **U1PER** (bit 2)
  The U1PER bit is used to indicate occurrence of a parity error of the UART1.
  When the parity of the received data and the parity bit attached to the data do not coincide, this bit is set to "1". U1PER is updated whenever data is received.
  The U1PER bit is fixed to "0" in transmit mode.

U1PER	Description
0	No parity error (initial value)
1	Parity error

- **U1FUL** (bit 3)
  The U1FUL bit is used to indicate the state of the transmit/receive buffer of the UART1.
  When the transmitted data is written in UA1BUF in transmit mode, this bit is set to “1” and when this transmitted data is transferred to the shift register, this bit is set to “0”. To transmit the data consecutively, confirm the U1FUL flag becomes "0", then write the next transmitted data to the UA0BUF.
  The U1FUL bit is fixed to “0” in receive mode.

U1FUL	Description
0	There is no data in the transmit/receive buffer. (Initial value)
1	There is data in the transmit/receive buffer.

13.3 Description of Operation

13.3.1 Transfer Data Format

In the transfer data format, one frame contains a start bit, a data bit, a parity bit, and a stop bit. In this format, 5 to 8 bits can be selected as data bit. For the parity bit, “with parity bit”, “without parity bit”, “even parity”, or “odd parity” can be selected. For the stop bit, “1 stop bit” or “2 stop bits” are available and for the transfer direction, “LSB first” or “MSB first” are available for selection. For serial input/output logic, positive logic or negative logic can be selected.
All these options are set with the UART0 mode register (UA0MOD1).
Figure 13-2 and Figure 13-3 show the positive logic input/output format and negative logic input/output format, respectively.

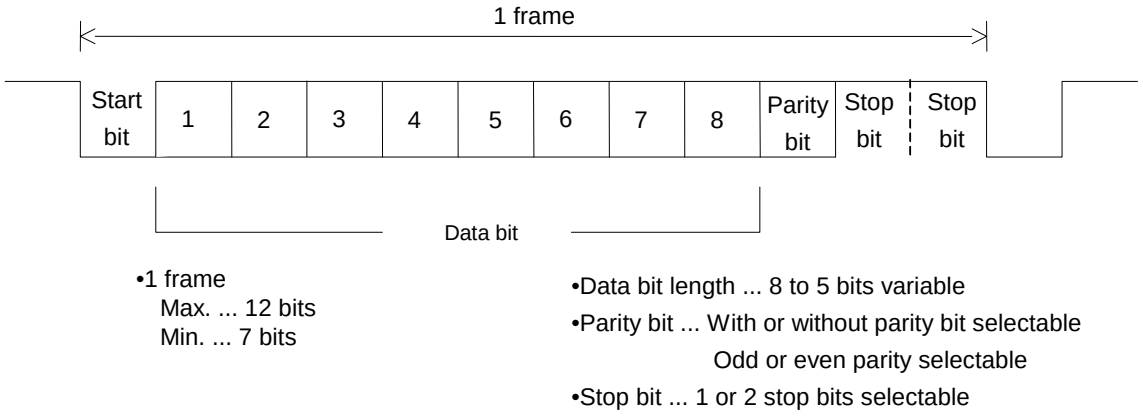


Figure 13-2 Positive Logic Input/Output Format

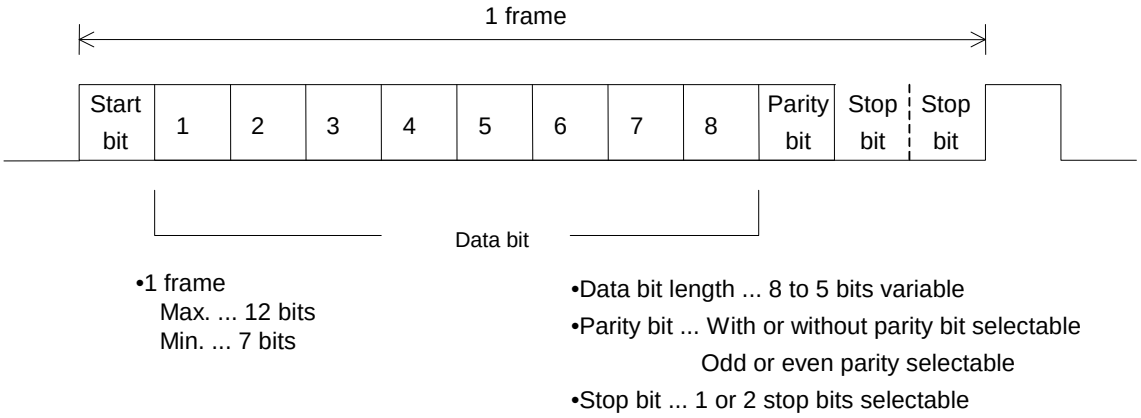


Figure 13-3 Negative Logic Input/Output Format

13.3.2 Baud rate

Baud rates are generated by the baud generator.
The baud rate generator generates a baud rate by counting the clock selected by the baud rate clock selection bits (U0CK1, U0CK0) of the UART0 mode register 0 (UA0MOD0). The count value of the baud rate generator can be set by writing it in the UART0 baud rate register H or L (UA0BRTH, UA0BRTL). The maximum count is 4096.
The setting values of UA0BRTH and UA0BRTL are expressed by the following equation.

$$\text{UA0BRTH, L} = \frac{\text{Clock frequency (Hz)}}{\text{Baud rate (bps)}} - 1$$

Table 13-2 lists the count values for typical baud rates.

Table 13-2 Count Values for Typical Baud Rates

Baud rate	Baud rate generator Clock selection			Baud rate generator counter value				Error [%]
	Baud rate Clock	U0CK1	U0CK0	Count value	Period of one bit	UA0BRTH	UA0BRTL	
1200bps	32.768kHz	0	0	27	Approximately 824us	00H	1AH	1.1
2400bps	32.768kHz	0	0	14	Approximately 427us	00H	0DH	-2.5
	65.536kHz	0	1	27	Approximately 412us	00H	1AH	1.1
4800bps	32.768kHz	0	0	7	Approximately 214us	00H	06H	-2.5
	65.536kHz	0	1	14	Approximately 214us	00H	0DH	-2.5
	500kHz	1	*	104	Approximately 208us	00H	67H	0.2
	2MHz			417	Approximately 208.5us	01H	A0H	-0.1
9600bps	65.536kHz	0	1	7	Approximately 107us	00H	06H	-2.5
	500kHz	1	*	52	Approximately 104us	00H	33H	0.2
	2MHz			208	Approximately 104us	00H	CFH	0.2
19200bps	2MHz	1	*	104	Approximately 52us	00H	67H	0.2
38400bps	2MHz	1	*	52	Approximately 26us	00H	33H	0.2

Note:
When using 65.536KHz (LSCLK X 2) for the baud rate generator input clock, enable the operation of the low-speed double clock by setting bit 2 (ENMLT) of the frequency control register 1 (FCON1) to “1”.

When the baud rate clock generator input clock selection is set to 500kHz or 2MHz, an error of 500kHz±25% or 2MHz±25% may occur. To set the baud rate with accuracy, set the baud rate generator counter value by referring to the frequency measurement modes for Timers 2 and 3 (Sections 9.3.3).

Internal logic voltage (VDDL) is changed by OSCM2 bit. VDDL becomes Typ.1.2V when OSCM2 is set to “0” and 500kHz oscillation is selected. VDDL becomes Typ.1.5V when OSCM2 is set to “1” and 2MHz oscillation is selected. . Ensure to write this bit when the high-speed clock oscillator circuit stops oscillating (During FCON1 register's ENOSC bit is "0").

13.3.3 Transmitted Data Direction

Figure 13-4 shows the relationship between the transmit/receive buffer and the transmitted/received data.

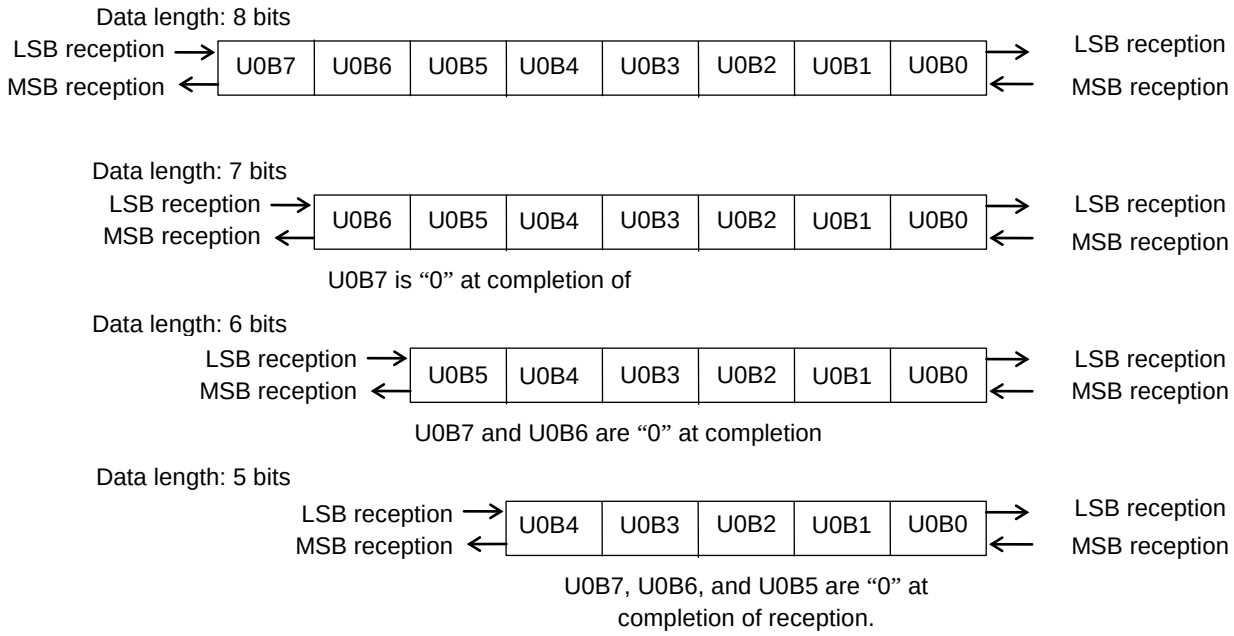


Figure 13-4 Relationship between Transmit/Receive Buffer and Transmitted/Received Data

Note:
When the TXD0 pin is set to serve the secondary function output in receive mode, "H" level is output from the TXD0 output.

13.3.4 Transmit Operation

Transmission is started by setting the U0IO bit of the UART0 mode register 0 (UA0MOD0) to "0" to select transmit mode and setting the U0EN bit of the UART0 control register (UA0CON) to "1".
Figure 13-5 shows the operation timing for transmission.

When the U0EN bit is set to "1" (①), the baud rate generator generates an internal transfer clock of the baud rate set and starts transmission.
The start bit is output to the TXD0 pin by the falling edge of the internal transfer clock (②). Subsequently, the transmitted data, a parity bit, and a stop bit are output.
When the start bit is output (②), a UART0 interrupt is requested. In the UART0 interrupt routine, the next data to be transmitted is written to the transmit/receive buffer (UA0BUF).
When the next data to be transmitted is written to the transmit/receive buffer (UA0OBUF), the transmit buffer status flag (U0FUL) is set to "1" (③) and a UART0 interrupt is requested on the falling edge of the internal transfer clock (④) after transmission of the stop bit. If the UART0 interrupt routine is terminated without writing the next data to the transmit/receive buffer, the U0FUL bit is not set to "1" (⑤) and transmission continues up to the transmission of the stop bit, then the U0EN bit is reset to "0" and a UART0 interrupt is requested.
The valid period for the next transmit data to be written to the transmit/receive buffer is from the generation of an interrupt to the termination of stop bit transmission. (⑥)

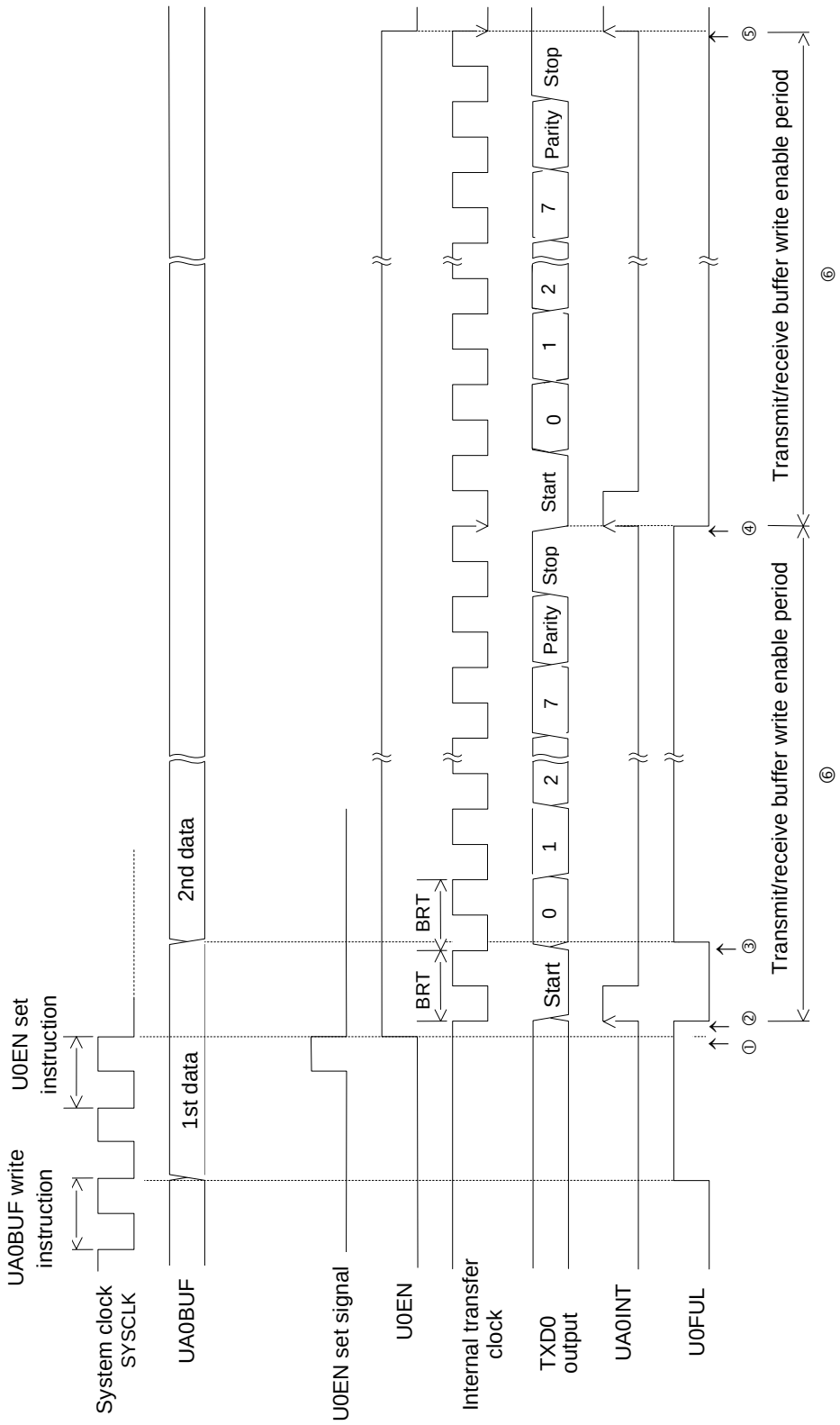


Figure 13-5 Operation Timing in Transmission

13.3.5 Receive Operation

Select the received data pin using the U0RSEL bit of the UART0UART0 mode register 0 (UA0MOD0). Select the receive mode by setting the U0IO bit of the UART0 mode register 0 (UA0MOD0) to "1". Then, set the U0EN bit of the UART0 control register (UA0CON) to "1" to start receiving data.
Figure 13-6 shows the operation timing for reception.

When receive operation starts, the LSI checks the data sent to the input pin RXD0 and waits for the arrival of a start bit.
When detecting a start bit (①), the LSI generates the internal transfer clock of the baud rate set with the start bit detect point as a reference and performs receive operation.
The shift register shifts in the data input to RXD on the rising edge of the internal transfer clock. The data and parity bit are shifted into the shift register and 5- to 8- bit received data is transferred to the transmit/receive buffer (UA0BUF) concurrently with the fall of the internal transfer clock of ③.
The LSI requests a UART0 interrupt on the rising edge of the internal transfer clock subsequent to the internal transfer clock by which the received data was fetched (④) and checks for a stop bit error and a parity bit error. When an error is detected, the LSI sets the corresponding bit of the UART0 status register (UA0STAT) to “1”.

Parity error : S0PER ＝“1”
Overrun error : S0OER ＝“1”
Framing error : S0FER ＝“1”

As shown in Figure 13-6, the rise of the internal transfer clock is set so that it may fall into the middle of the bit interval of the received data.
Reception continues until the U0EN bit is reset to “0” by the program. When the U0EN bit is reset to “0” during reception, the received data may be destroyed. When the U0EN bit is reset to “0” during the “U0EN reset enable period” in Figure 13-6, the received data is protected.

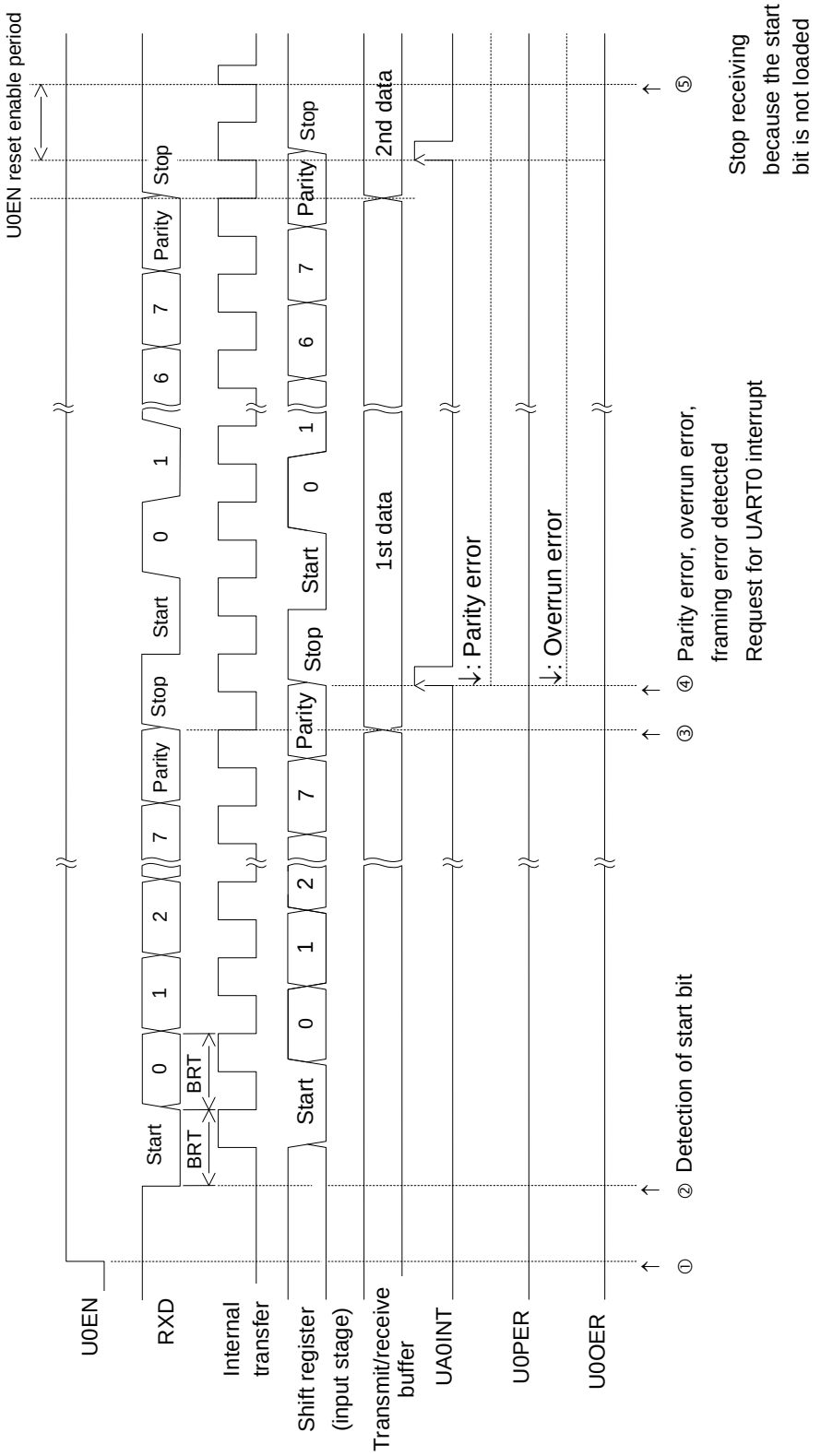


Figure 13-6 Operation Timing in Reception

13.3.5.1 Detection of Start Bit

The Start bit is sampled using the baud rate generator clock (LSCLK, LSCLK x 2, HSCLK) selected by the U0CK1 and U0CK0 bits of the UARTn mode register 0 (UA0MOD0). Therefore, the start bit detection may be delayed for one cycle of the baud rate generate clock at the maximum.
Figure 13-7 shows the start bit detection timing.

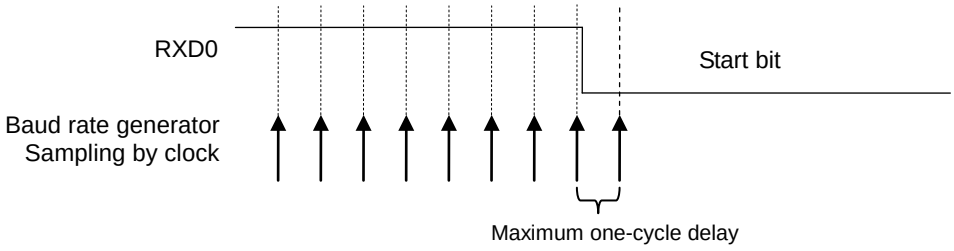
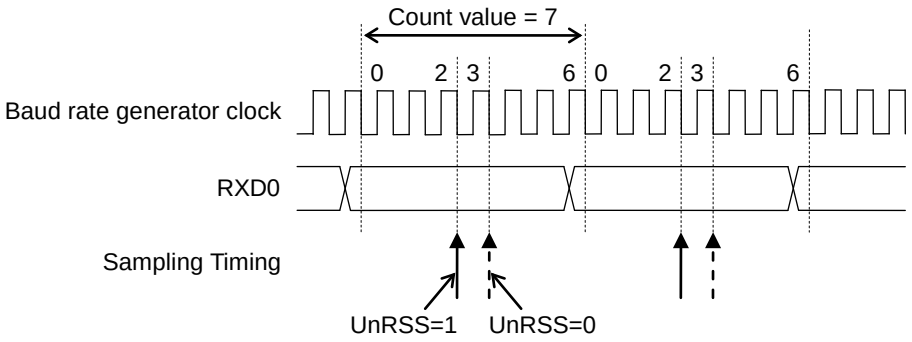


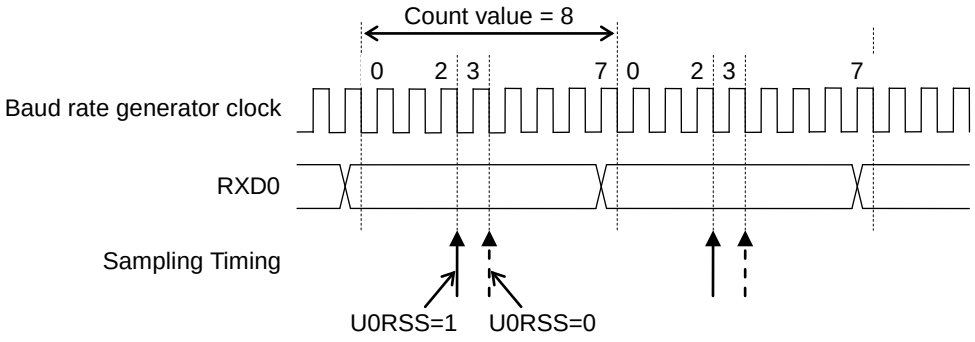
Figure 13-7 Start Bit Detection Timing (Positive Logic)

13.3.5.2 Sampling Timing

When the Start bit is detected, the received data that was input to the RXD0 is sampled almost at the center of the baud rate, then loaded to the shift register.
The loading sampling timing of this shift register can be adjusted for one clock of the baud rate generator clock, using the U0RSS bit of the UART0 mode register 0 (UA0MOD0).
Figure 13-8 shows the relationship between the U0RSS bit and the sampling timing.



(1) When the baud rate generator count value is "7" (odd)



(2) When the baud rate generator count value is "8" (even)

Figure 13-8 Relationship between UORSS Bit and Sampling Timing

13.3.5.3 Receive Margin

If there is an error between the sender baud rate and the baud rate generated by the baud rate generator of this LSI, the error accumulates until the last stop bit loading in one frame, decreasing the receive margin. This receive margin needs to be fully considered, particularly when the baud rate generator clock uses a lower frequency such as LSCLK and LSCLK x 2 to realize a higher bit rate (e.g., 4800bps, 9600bps).
Figure 13-9 shows the baud rate errors and receive margin waveforms.

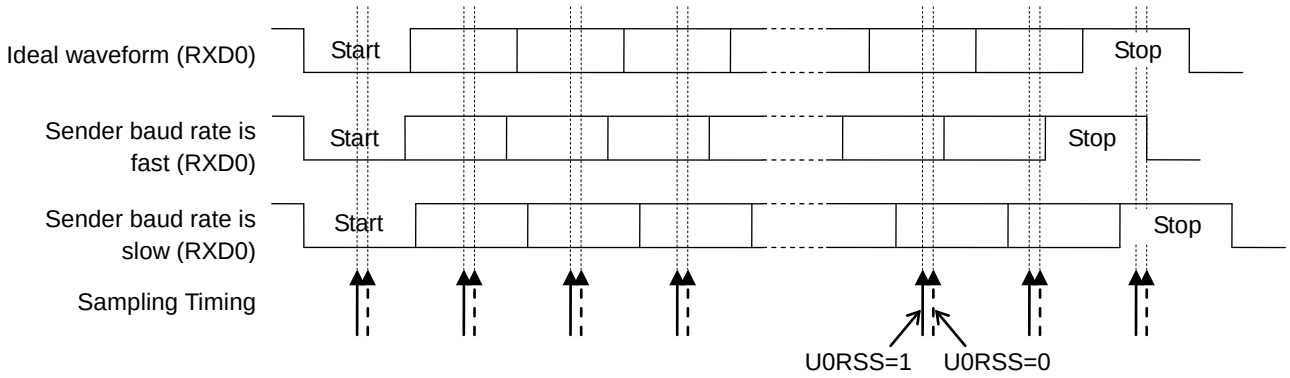


Figure 13-9 Baud Rate Error and Receive Margin

Note:
In system designing, ensure a sufficient receive margin considering the baud rate difference between sender and receiver, start bit detection delay, distortion in receive data, and influence of noise.

13.4 Specifying port registers

To enable the UART function, the applicable bit of each related port register needs to be set. See Chapter 21, “Port 4” and Chapter 18, “Port 0” for detail about the port registers.

13.4.1 Functioning P43(TXD0) and P42(RXD0) as the UART

Set P43MD1-P42MD1 bits(bit3-bit2 of P4MOD1 register) to “0” and set P43MD0-P42MD0(bit3-bit2 of P4MOD0 register) to “1”, for specifying the UART as the secondary function of P43 and P42.

Register name	P4MOD1 register (Address: 0F225H)							
Bit	7	6	5	4	**3**	**2**	1	0
Bit name	P47MD1	P46MD1	P45MD1	P44MD1	**P43MD1**	**P42MD1**	P41MD1	P40MD1
Setting value	*	*	*	*	**0**	**0**	*	*

Register name	P4MOD0 register (Address: 0F224H)							
Bit	7	6	5	4	**3**	**2**	1	0
Bit name	P47MD0	P46MD0	P45MD0	P44MD0	**P43MD0**	**P42MD0**	P41MD0	P40MD0
Setting value	*	*	*	*	**1**	**1**	*	*

Set the P43C1 bit (P4CON1 register's bit 3) to "1", the P43C0 bit (P4CON0 register's bit 3) to "1", and the P43DIR bit (P4DIR register's bit 3) to "0" for specifying the state mode of the P43 pin to CMOS output.
Set P42DIR bit (bit2 of P4DIR register) to “1” for specifying the P42 as an input pin.
The set value ($) is arbitrary for the P42C1 and P42C0 bits. Select an arbitrary input mode depending on the state of the external circuit to which the P42 pin is connected.

Register name	P4CON1 register (Address: 0F223H)							
Bit	7	6	5	4	**3**	**2**	1	0
Bit name	P47C1	P46C1	P45C1	P44C1	**P43C1**	**P42C1**	P41C1	P40C1
Setting value	*	*	*	*	**1**	**$**	*	*

Register name	P4CON0 register (Address: 0F222H)							
Bit	7	6	5	4	**3**	**2**	1	0
Bit name	P47C0	P46C0	P45C0	P44C0	**P43C0**	**P42C0**	P41C0	P40C0
Setting value	*	*	*	*	**1**	**$**	*	*

Register name	P4DIR register (Address: 0F221H)							
Bit	7	6	5	4	**3**	**2**	1	0
Bit name	P47DIR	P46DIR	P45DIR	P44DIR	**P43DIR**	**P42DIR**	P41DIR	P40DIR
Setting value	*	*	*	*	**0**	**1**	*	*

The P43D to P42D bits (P4D register bits 3 to 2) data can either be "0" or "1" (not need to be set).

Register name	P4D register (Address: 0F220H)							
Bit	7	6	5	4	**3**	**2**	1	0
Bit name	P47D	P46D	P45D	P44D	**P43D**	**P42D**	P41D	P40D
Setting value	*	*	*	*	**	**	*	*

* : Bit not related to the UART function
** : Don't care $: Optional

[Note:]
The receive pin (RXD) is selected by U0RSEL bit (bit4 of UA0MOD0 register). The initial value "0" selects the P02 and the value "1" selects the P43.

13.4.2 Functioning P43(TXD0) and P02(RXD0) as the UART

Set P43MD1 bit (bit3 of P4MOD1 register) to “0” and set P43MD0(bit3 of P4MOD0 register) to “1”, for specifying the UART as the secondary function of P43.

Register name	P4MOD1 register (Address: 0F225H)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	P47MD1	P46MD1	P45MD1	P44MD1	**P43MD1**	P42MD1	P41MD1	P40MD1
Setting value	*	*	*	*	**0**	$	*	*

Register name	P4MOD0 register (Address: 0F224H)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	P47MD0	P46MD0	P45MD0	P44MD0	**P43MD0**	P42MD0	P41MD0	P40MD0
Setting value	*	*	*	*	**1**	$	*	*

Set the P43C1 bit (P4CON1 register's bit 3) to "1", the P43C0 bit (P4CON0 register's bit 3) to "1", and the P43DIR bit (P4DIR register's bit 3) to "0" for specifying the state mode of the P43 pin to CMOS output.

Register name	P4CON1 register (Address: 0F223H)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	P47C1	P46C1	P45C1	P44C1	**P43C1**	P42C1	P41C1	P40C1
Setting value	*	*	*	*	**1**	*	*	*

Register name	P4CON0 register (Address: 0F222H)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	P47C0	P46C0	P45C0	P44C0	**P43C0**	P42C0	P41C0	P40C0
Setting value	*	*	*	*	**1**	*	*	*

Register name	P4DIR register (Address: 0F221H)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	P47DIR	P46DIR	P45DIR	P44DIR	**P43DIR**	P42DIR	P41DIR	P40DIR
Setting value	*	*	*	*	**0**	*	*	*

The P43D bit (P4D register bit 3) data can either be "0" or "1" (not need to be set).

Register name	P4D register (Address: 0F220H)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	P47D	P46D	P45D	P44D	**P43D**	P42D	P41D	P40D
Setting value	*	*	*	*	**	*	*	*

The P02 pin is an input-only pin and does not need input/output selection by the register. The set value ($) is arbitrary for the P02C1 and P02C0 bits. Select an arbitrary input mode depending on the state of the external circuit to which the P02 pin is connected.

Register name	P0CON1 register (Address: 0F207H)							
Bit	7	6	5	4	3	**2**	1	0
Bit name	-	-	-	-	P03C1	**P02C1**	P01C1	P00C1
Setting value	-	-	-	-	*	**$**	*	*

Register name	P0CON0 register (Address: 0F206H)							
Bit	7	6	5	4	3	**2**	1	0
Bit name	-	-	-	-	P03C0	**P02C0**	P01C0	P00C0
Setting value	-	-	-	-	*	**$**	*	*

The P02D bit (P0D register bit 2) data can either be "0" or "1" (not need to be set).

Register name	P0D register (Address: 0F204H)							
Bit	7	6	5	4	3	**2**	1	0
Bit name	-	-	-	-	P03D	**P02D**	P01D	P00D
Setting value	-	-	-	-	*	******	*	*

- : Bit that does not exist
* : Bit not related to the UART function
** : Don’t care $: Optional

Note:
•The receive pin (RXD) is selected by U0RSEL bit (bit4 of UA0MOD0 register). The initial value "0" selects the P02 and the value "1" selects the P43.
•Even if the P42 pin is selected as RXD0 by the P42MD1, P42MD0, P42C1, P42C0, and P42IDR bits, the P02 pin will be selected as RXD0 when the U0RSEL bit of the UA0MOD0 register is "0".
•P02(Port 0) is an input-only port, does not have registers that can select data direction(input or output) or mode(primary or secondary function).

13.4.3 Functioning P53(TXD1) and P52(RXD1) as the UART1

Set P53MD1 to P52MD1 bit (bit3 to bit2 of P5MOD1 register) to “0” and set P53MD0 to P52MD0(bit3 to bit2 of P5MOD0 register) to “1”, for specifying the UART1 as the secondary function of P53 and P52.

Register name	P5MOD1 register (Address: 0F22DH)							
Bit	7	6	5	4	**3**	**2**	1	0
Bit name	P57MD1	P56MD1	P55MD1	P54MD1	**P53MD1**	**P52MD1**	P51MD1	P50MD1
Setting value	*	*	*	*	**0**	**0**	*	*

Register name	P5MOD0 register (Address:0F22CH)							
Bit	7	6	5	4	**3**	**2**	1	0
Bit name	P57MD0	P56MD0	P55MD0	P54MD0	**P53MD0**	**P52MD0**	P51MD0	P50MD0
Setting value	*	*	*	*	**1**	**1**	*	*

Set the P53C0 bit(P5CON0 register’s bit3) to "1",and the P53DIR bit (P5DIR register’s bit3) to "0" for for specifying the state mode of the P53 pin to CMOS output.
Set the P52DIR bit(P5DIR register’s bit2) to "1" for specifying the P52 as an input pin.
The set value ($) is arbitrary for the P52C0 bits. Select an arbitrary input mode depending on the state of the external circuit to which the P52 pin is connected.

Register name	P5CON0 register (Address:0F22AH)							
Bit	7	6	5	4	**3**	**2**	1	0
Bit name	P57C0	P56C0	P55C0	P54C0	**P53C0**	**P52C0**	P51C0	P50C0
Setting value	*	*	*	*	**1**	**$**	*	*

Register name	P5DIR register (Address:0F229H)							
Bit	7	6	5	4	**3**	**2**	1	0
Bit name	P57DIR	P56DIR	P55DIR	P54DIR	**P53DIR**	**P52DIR**	P51DIR	P50DIR
Setting value	*	*	*	*	**0**	**1**	*	*

The P53D to P52D bits (P5D register bits 3 to 2) data can either be "0" or "1" (not need to be set).

Register name	P5D register (Address:0F228H)							
Bit	7	6	5	4	**3**	**2**	1	0
Bit name	P57D	P56D	P55D	P54D	**P53D**	**P52D**	P51D	P50D
Setting value	*	*	*	*	******	******	*	*

* : Bit not related to the UART function
** : Don’t care $: Optional

[Note]
The receive pin（RXD1）is selected by U1RSEL bit (bit4 of UA1MOD0 register). The initial value "0" selects the P03 and the value "1" selects the P52.

13.4.4 Functioning P43(TXD1) and P02(RXD0) as the UART1

Set P53MD1 bit (bit3 of P5MOD1 register) to “0” and set P53MD0(bit3 of P5MOD0 register) to “1”, for specifying the UART1 as the secondary function of P53.

Register name	P5MOD1 register (Address: 0F22DH)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	P57MD1	P56MD1	P55MD1	P54MD1	**P53MD1**	P52MD1	P51MD1	P50MD1
Setting value	*	*	*	*	**0**	$	*	*

Register name	P5MOD0 register (Address: 0F22CH)							
Bit	7	6	5	4	3	2	1	0
Bit name	P57MD0	P56MD0	P55MD0	P54MD0	P53MD0	P52MD0	P51MD0	P50MD0
Setting value	*	*	*	*	1	$	*	*

Set the P53C0 bit (P5CON register’s bit3) to “1”, the P53DIR bit (P5DIR register’s bit3) to “0” for specifying the state mode of the P53 pin to CMOS output.

Register name	P5CON0 register (Address: 0F22AH)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	P57C0	P56C0	P55C0	P54C0	**P53C0**	P52C0	P51C0	P50C0
Setting value	*	*	*	*	**1**	*	*	*

Register name	P5DIR register (Address: 0F229H)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	P57DIR	P56DIR	P55DIR	P54DIR	**P53DIR**	P52DIR	P51DIR	P50DIR
Setting value	*	*	*	*	**0**	*	*	*

The P53D bit(P5D register bit3) data can either be “0” or “1”(not need to be set)

Register name	P5D register (Address: 0F228H)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	P57D	P56D	P55D	P54D	**P53D**	P52D	P51D	P50D
Setting value	*	*	*	*	**	*	*	*

The P03 pin is an input-only pin and does not need input/output selection by the register. The set value($) is arbitrary for the P03C1 and p03C0 bits. Select an arbitrary input mode depending on the state of the external circuit to which the P03 pin is connected.

Register name	P0CON1 register (Address: 0F207H)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	-	-	-	-	**P03C1**	P02C1	P01C1	P00C1
Setting value	-	-	-	-	**$**	*	*	*

Register name	P0CON0 register (Address: 0F206H)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	-	-	-	-	**P03C0**	P02C0	P01C0	P00C0
Setting value	-	-	-	-	**$**	*	*	*

The P03D bit(P0D register bit 3) data can either be “0” or “1”(not need to be set)

Register name	P0D register (Address: 0F204H)							
Bit	7	6	5	4	**3**	2	1	0
Bit name	-	-	-	-	**P03D**	P02D	P01D	P00D
Setting value	-	-	-	-	******	*	*	*

- : Bit that does not exist
* : Bit not related to the UART function
** : Don’t care $: Optional

Note:
•The receive pin (RXD1) is selected by U1RSEL bit (bit4 of UA1MOD0 register). The initial value "0" selects the P03 and the value "1" selects the P53.
•Even if the P52 pin is selected as RXD1 by the P52MD1, P52MD0, P52C0, and P52DIR bits, the P03 pin will be selected as RXD1 when the U1RSEL bit of the UA1MOD0 register is "0".
•P03(Port 0) is an input-only port, does not have registers that can select data direction(input or output) or mode(primary or secondary function).

13.4.5 Functioning P43(TXD1) and P42(RXD0) as the UART0/UART1

Set P43MD1 to P42MD1 bits(bit3-bit2 of P4MOD1 register) to “10” and set P43MD0 to M42MD0(bit3-bit2 of P4MOD0 register) to “11”, for specifying the UART as the secondary function of P42 and as the quaternary function of P43.

Register name	P4MOD1 register (Address: 0F225H)							
Bit	7	6	5	4	**3**	**2**	1	0
Bit name	P47MD1	P46MD1	P45MD1	P44MD1	**P43MD1**	**P42MD1**	P41MD1	P40MD1
Setting value	*	*	*	*	**1**	**0**	*	*

Register name	P4MOD0 register (Address:0F224H)							
Bit	7	6	5	4	**3**	**2**	1	0
Bit name	P47MD0	P46MD0	P45MD0	P44MD0	**P43MD0**	**P42MD0**	P41MD0	P40MD0
Setting value	*	*	*	*	**1**	**1**	*	*

Set the P43C1 bit (P4CON1 register's bit 3) to "1", the P43C0 bit (P4CON0 register’s bit 3) to “1”, and the P43DIR bit(P4DIR register’s bit3) to “0” for specifying the state mode of the P43 pin to CMOS outut.
Set P42DIR bit(bit2 of P4DIR register) to “1” for specifying the P42 as an input pin.
The set value($) is arbitrary for the P42C1 and P42C0 bits. Select an arbitrary input mode depending on the state of the external circuit to which the P42 pin is connected.

Register name	P4CON1 register (Address: 0F223H)							
Bit	7	6	5	4	**3**	**2**	1	0
Bit name	P47C1	P46C1	P45C1	P44C1	**P43C1**	**P42C1**	P41C1	P40C1
Setting value	*	*	*	*	**1**	**$**	*	*

Register name	P4CON0 register (Address: 0F222H)							
Bit	7	6	5	4	**3**	**2**	1	0
Bit name	P47C0	P46C0	P45C0	P44C0	**P43C0**	**P42C0**	P41C0	P40C0
Setting value	*	*	*	*	**1**	**$**	*	*

Register name	P4DIR register (Address: 0F221H)							
Bit	7	6	5	4	**3**	**2**	1	0
Bit name	P47DIR	P46DIR	P45DIR	P44DIR	**P43DIR**	**P42DIR**	P41DIR	P40DIR
Setting value	*	*	*	*	**0**	**1**	*	*

The P43D to P42D bits(P4D register bits 3 to 2) data can either be “0” or “1”(not need to be set).

Register name	P4D register (Address: 0F220H)							
Bit	7	6	5	4	**3**	**2**	1	0
Bit name	P47D	P46D	P45D	P44D	**P43D**	**P42D**	P41D	P40D
Setting value	*	*	*	*	**	**	*	*

* : Bit not related to the UART function
** : Don’t care $: Optional

[Note:]
The receive pin (RXD) is selected by U0RSEL bit (bit4 of UA0MOD0 register). The initial value "0" selects the P02 and the value "1" selects the P43.

Chapter 14

I^2C Bus Interface

14 I2C Bus Interface

14.1 Overview

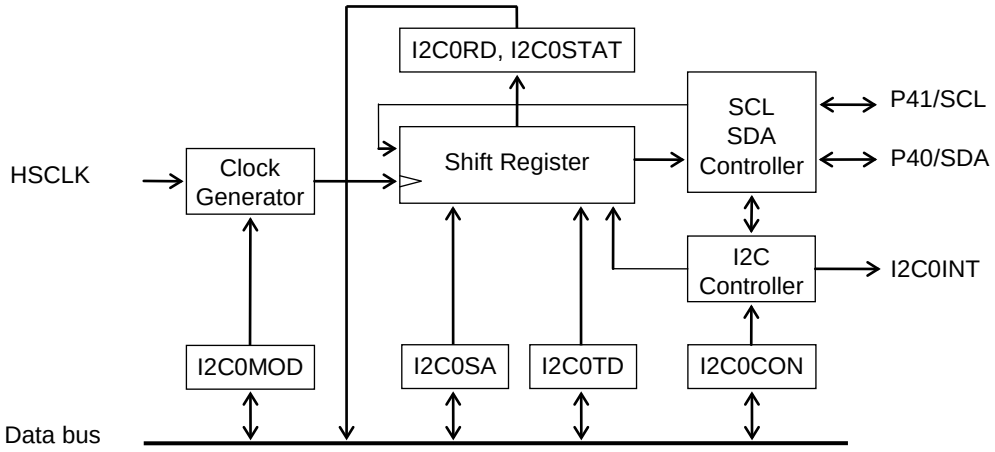
This LSI includes 1 channel of I^2C bus interface (master).
The secondary functions of Port 4 are assigned to the I^2C bus interface data input/output pin and the I^2C bus interface clock input/output pin. For Port4, see Chapter 19, "Port 4".

14.1.1 Features

- Master function
- Communication speeds supported include standard mode (100 kbps@4MHz HSCLK, 50kbps@500kHz HSCLK) and fast mode (400kbps@4MHz HSCLK).
- 7-bit address format (10-bit address can be supported)

14.1.2 Configuration

Figure 14-1 shows the configuration of the I^2C bus interface.



I2C0RD : I^2C bus 0 receive register
I2C0SA : I^2C bus 0 slave address register
I2C0TD : I^2C bus 0 transmit data register
I2C0CON : I^2C bus 0 control register
I2C0MOD : I^2C bus 0 mode register
I2C0STAT : I^2C bus 0 status register

Figure 14-1 Configuration of I^2C Bus Interface

14.1.3 List of Pins

Pin name	I/O	Description
P40/SDA	I/O	I^2C bus interface data input/output pin. Used for the secondary function of the P40 pin.
P41/SCL	I/O	I^2C bus interface clock input/output pin. Used for the secondary function of the P41 pin.

14.2 Description of Registers

14.2.1 List of Registers

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F2A0H	I^2C bus 0 receive register	I2C0RD	—	R	8	00H
0F2A1H	I^2C bus 0 slave address register	I2C0SA	—	R/W	8	00H
0F2A2H	I^2C bus 0 transmit data register	I2C0TD	—	R/W	8	00H
0F2A3H	I^2C bus 0 control register	I2C0CON	—	R/W	8	00H
0F2A4H	I^2C bus 0 mode register	I2C0MOD	—	R/W	8	00H
0F2A5H	I^2C bus 0 status register	I2C0STAT	—	R	8	00H

14.2.2 I^2C Bus 0 Receive Register (I2C0RD)

Address: 0F2A0H
Access: R
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
I2C0RD	I20R7	I20R6	I20R5	I20R4	I20R3	I20R2	I20R1	I20R0
R	R	R	R	R	R	R	R	R
Initial value	0	0	0	0	0	0	0	0

I2C0RD is a read-only special function register (SFR) to store receive data.
I2C0RD is updated after completion of each reception.

[Description of Bits]

- **I20R7-I20R0** (bits 7-0)
  The I20R7 to I20R0 bits are used to store receive data. The signal input to the SDA pin is received at transmission of a slave address and at data transmission/reception in sync with the rising edge of the signal on the SCL pin. Since data that has been output to the SDA and SCL pins is received not only at data reception but also at slave address data transmission and data transmission, it is possible to check whether transmit data has certainly been transmitted.

14.2.3 I^2C Bus 0 Slave Address Register (I2C0SA)

Address: 0F2A1H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
I2C0SA	I20A6	I20A5	I20A4	I20A3	I20A2	I20A1	I20A0	I20RW
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

I2C0SA is a special function register (SFR) to set the address and the transmit/receive mode of the slave device.

[Description of Bits]

- **I20RW** (bit 0)
  The I20RW bit is used to select the data transmit mode (write) or data receive mode (read).

I20RW	Description
0	Data transmit mode (initial value)
1	Data receive mode

- **I20A6-I20A0** (bits 7-1)
  The I20A6 to I20A0 bits are used to set the address of the communication destination.

14.2.4 I^2C Bus 0 Transmit Data Register (I2C0TD)

Address: 0F2A2H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
I2C0TD	I20T7	I20T6	I20T5	I20T4	I20T3	I20T2	I20T1	I20T0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

I2C0TD is a special function register (SFR) to set transmit data.

[Description of Bits]

- **I20T7-0** (bits 7-0)
  The I20T7 to 0 bits are used to set transmit data.

14.2.5 I^2C Bus 0 Control Register (I2C0CON)

Address: 0F2A3H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
I2C0CON	I20ACT	—	—	—	—	I20RS	I20SP	I20ST
R/W	R/W	R/W	R/W	R/W	R/W	W	W	R/W
Initial value	0	0	0	0	0	0	0	0

I2C0CON is a special function register (SFR) to control transmit and receive operations.

[Description of Bits]

- **I20ST** (bit 0)
  The I20ST bit is used to control the communication operation of the I^2C bus interface. When the I20ST bit is set to "1", communication starts. When "1" is overwritten to the I20ST bit in a control register setting wait state after transmission/reception of acknowledgment, communication starts again. When the I20ST bit is set to "0", communication is stopped forcibly.
  The I20ST bit can be set to "1" only when the I2C bus interface is in an operation enable state (I20EN = "1").
  When the I20SP bit is set to "1", the I20ST bit is set to "0".

I20ST	Description
0	Stops communication (initial value)
1	Starts communication

- **I20SP** (bit 1)
  The I20SP bit is a write-only bit used to request a stop condition. When the I20SP bit is set to "1", the I^2C bus shifts to the stop condition and communication stops. When the I20SP bit is read, "0" is always read.

I20SP	Description
0	No stop condition request (initial value)
1	Stop condition request

- **I20RS** (bit 2)
  The I20RS bit is a write-only bit used to request a repeated start. When this bit is set to "1" during data communication, the I^2C bus shifts to the repeated start condition and communication restarts from the slave address. I20RS can be set to "1" only while communication is active (I20ST ="1"). When the I20RS bit is read, "0" is always read.

I20RS	Description
0	No repeated start request (initial value)
1	Repeated start request

- **I20ACT** (bit 7)
  The I20ACT bit is used to set the acknowledge signal to be output at completion of reception.

I20ACT	Description
0	Acknowledgment data "0" (initial value)
1	Acknowledgment data "1"

14.2.6 I²C Bus 0 Mode Register (I2C0MOD)

Address: 0F2A4H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
I2C0MOD	—	—	—	I20SYN	I20DW1	I20DW0	I20MD	I20EN
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

I2C0MOD is a special function register (SFR) to set operating mode.

[Description of Bits]

- **I20EN** (bit 0)
  The I20EN bit is used to enable the operation of the I²C bus interface. Only when the I20EN bit is set to "1", the I20ST bit can be set and the I20BB flag starts operation. When the I20EN bit is set to "0", all the SFRs related to the I²C bus 0 are initialized.

I20EN	Description
0	Stops I²C operation. (Initial value)
1	Enables I²C operation.

- **I20MD** (bit 1)
  The I20MD bit is used to set the communication speed of the I²C bus interface. Standard mode or fast mode can be selected.

I20MD	Description
0	Standard mode (initial value)/ 100kbps@4MHz HSCLK
1	Fast mode / Max. 400kbps@4MHz HSCLK

- **I20DW1, I20DW0** (bits 3, 2)
  The I20DW1 and I20DW0 bits are used to set the communication speed reduction rate of the I²C bus interface. Set this bit so that the communication speed does not exceed 100kpbs/400kpbs.

I20DW1	I20DW0	Description
0	0	No communication speed reduction (initial value)
0	1	10% communication speed reduction
1	0	20% communication speed reduction
1	1	30% communication speed reduction

- **I20SYN** (bit 4)
  Always set The I20SYN bit to 0.

Note:
The I²C bus is set so that the communication speed may become 100kbps/400kbps when HSCLK is 4 MHz. Therefore, when using PLL oscillation (approx. 8.192 MHz) for high-speed oscillation, select 1/2HSCLK at selection of the HSCLK frequency of FCON0 and select 10% communication speed reduction at selection of I2C0MOD communication speed reduction. When 500 kHz RC oscillation is used, communication in standard mode (50kbps) is available with the fast mode by setting I20MD bit to "1".

14.2.7 I^2C Bus 0 Status Register (I2C0STAT)

Address: 0F2A5H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
I2C0STAT	—	—	—	—	—	I20ER	I20ACR	I20BB
R	R	R	R	R	R	R	R	R
Initial value:	0	0	0	0	0	0	0	0

I2C0STAT is a read-only special function register (SFR) to indicate the state of the I^2C bus interface.

[Description of Bits]

- **I20BB** (bit 0)
  The I20BB bit is used to indicate the state of use of the I^2C bus interface. When the start condition is generated on the I^2C bus, this bit is set to "1" and when the stop condition is generated, the bit is set to "0". The I20BB bit is set to "0" when the I20EN bit of I2C0MOD is "0".

I20BB	Description
0	I^2C bus-free state (Initial value)
1	I^2C bus-busy state

- **I20ACR** (bit 1)
  The I20ACR bit is used to store the acknowledgment signal received. Acknowledgment signals are received each time the slave address is received and data transmission or reception is completed. The I20ACR bit is set to "0" when the I20EN bit of I2C0MOD is "0".

I20ACR	Description
0	Receives acknowledgment "0". (Initial value)
1	Receives acknowledgment "1".

- **I20ER** (bit 2)
  The I20ER bit is a flag to indicate a transmit error. When the value of the bit transmitted and the value of the SDA pin do not coincide, this bit is set to "1". The SDA remains the output until the subsequent byte data communication terminates even if I20ER is set to "1".
  The I20ER bit is set to "0" when a write operation to I2C0CON is performed. The I20ER bit is set to "0" when the I20EN bit of I2C0MOD is set to "0".

I20ER	Description
0	No transmit error (initial value)
1	Transmit error

14.3 Description of Operation

14.3.1 Communication Operating Mode

Communication is started when communication mode is selected by using the I^2C bus 0 mode register (I2C0MOD), the I^2C function is enabled by using the I20EN bit, a slave address and a data communication direction are set in the I^2C bus 0 slave address register, and "1" is written to the I20ST bit of the I2C bus 0 control register (I2C0CON).

14.3.1.1 Start Condition

When "1" is written to the I20ST bit of the I^2C bus 0 control register ((I2C0CON) while communication is stopped (the I20ST bit is "0"), communication is started and the start condition waveform is output to the SDA and SCL pins.
After execution of the start condition, the LSI shifts to slave address transmit mode.

14.3.1.2 Repeated Start Condition

When "1" is written to the I20RS and I20ST bits of the I^2C bus 0 control register ((I2C0CON) during communication (the I20ST bit is "0"), the repeated start condition waveform is output to the SDA and SCL pins. If you want to continue the communication without making the stop condition, executing the repeated start condition enables to specify another slave addresses or change the data direction (transmit or receive) See Figure 14-4.

After execution of the repeated start condition, the LSI shifts to slave address transmit mode.

14.3.1.3 Slave Address Transmit Mode

In slave address transmit mode, the values (slave address and data communication direction) of the I^2C bus 0 slave address register (I2C0SA) are transmitted in MSB first, and finally, the acknowledgment signal is received in the I20ACR bit of the I^2C bus 0 status register (I2CSTAT). The I20ACR bit is reset to "0" when receiving an acknowledgment "0" and set to "1" when receiving an acknowledgment "1".
At completion of acknowledgment reception, the LSI shifts to the I^2C bus 0 control register (I2C0CON) setting wait state (control register setting wait state).
The value of I2C0SA output from the SDA pin is stored in I2C0RD during after mentioned Control Register Setting Wait State.

14.3.1.4 Data Transmit Mode

In data transmit mode, the value of I2C0TD is transmitted in MSB first, and finally, the acknowledgment signal is received in the I20ACR bit of the I^2C bus 0 status register (I2CSTAT). The I20ACR bit is reset to "0" when receiving an acknowledgment "0" and set to "1" when receiving an acknowledgment "1".
At completion of acknowledgment reception, the LSI shifts to the I^2C bus 0 control register (I2C0CON) setting wait state (control register setting wait state).
The value of I2C0TD output from the SDA pin is stored in I2C0RD.

14.3.1.5 Data Receive Mode

In data receive mode, the value input in the SDA pin is received synchronously with the rising edge of the serial clock output to the SCL pin, and finally, the value of the I20ACT bit of the I2C bus 0 control register (I2C0CON) is output as an acknowledge signal. For example, as shown in Figure 14-3 and Figure 14-4, the acknowledge slot after I20ACT bit is reset to "0" (I2CON="01H") will have the acknowledge "0" (Shown as "A" in the Figure). In the same way, the acknowledge slot after I20ACT bit is set to "1" (I2CON="81H") will have the non-acknowledge "1" (Shown as "A" in the Figure).
At completion of acknowledgment transmission, the LSI shifts to the I^2C bus 0 control register (I2C0CON) setting wait state (control register setting wait state).
The data received is stored in I2C0RD after the acknowledgment signal is output. The acknowledgment signal output is received in the I20ACR bit of the I^2C bus 0 status register (I2CSTAT).

14.3.1.6 Control Register Setting Wait State

When the LSI shifts to the control register setting wait state, an I^2C bus interface interrupt (I2C0INT) is generated.
In the control register setting wait state, the transmit flag (I20ER) of the I^2C bus 0 status register (I2C0STAT) and acknowledgment receive data (I20ACR) are confirmed and at data reception, the contents of I2C0RD are read in the CPU and the next operation mode is selected.

When "1" is written to the I20ST bit in the control register setting wait state, the LSI shifts to the data transmit or receive mode. When "1" is written to the I20SP bit, the LSI shifts to the stop condition. When "1" is written to the I20RS bit and I20ST bit, the operation shifts to the repeated start condition.

14.3.1.7 Stop Condition

In the stop condition, the stop condition waveform is output to the SDA and SCL pins. After the stop condition waveform is output, an I^2C bus interface interrupt (I2C0INT) is generated.

14.3.2 Communication Operation Timing

Figures 14-2 to 14-4 show the operation timing and control method for each communication mode.

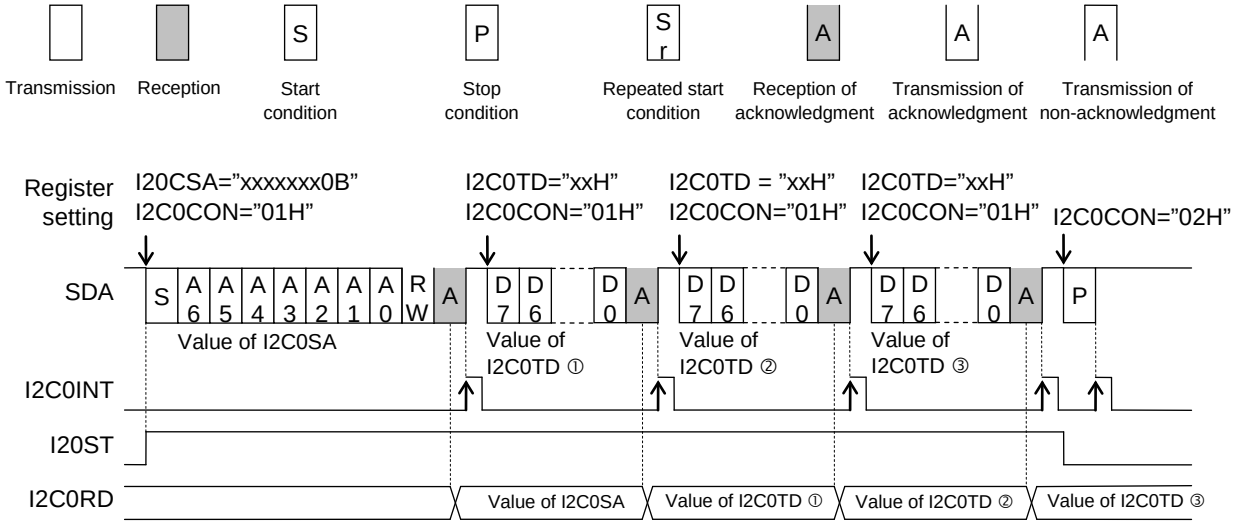


Figure 14-2 Operation Timing in Data Transmit (Write)

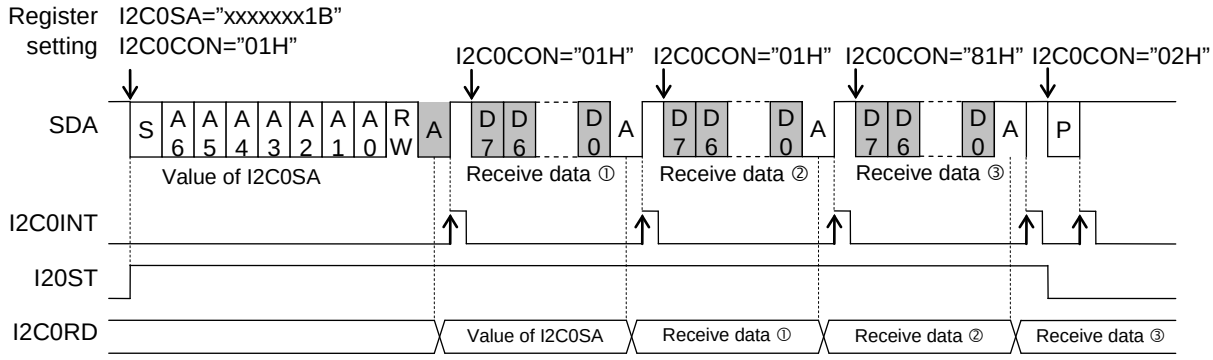


Figure 14-3 Operation Timing in Data Receive (Read)

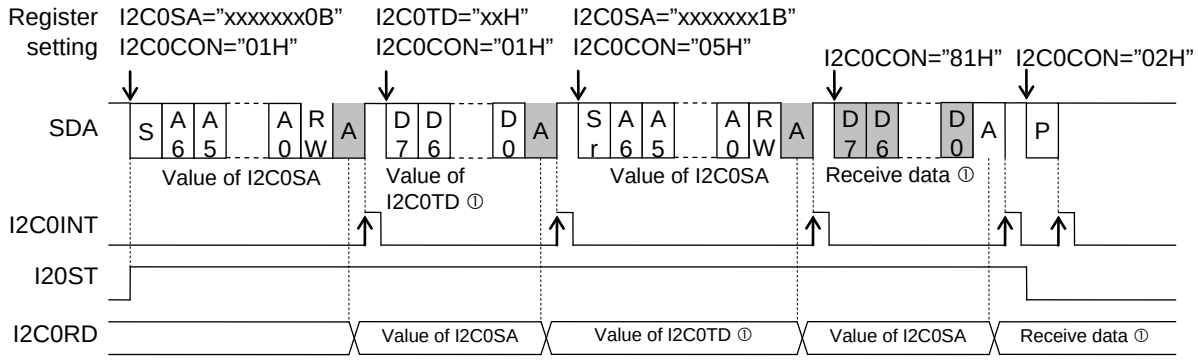


Figure 14-4 Operation Timing at Data Transmit/Receive (Write/Read) Switching

Figure 14-5 shows the operation timing and control method when an acknowledgment error occurs.

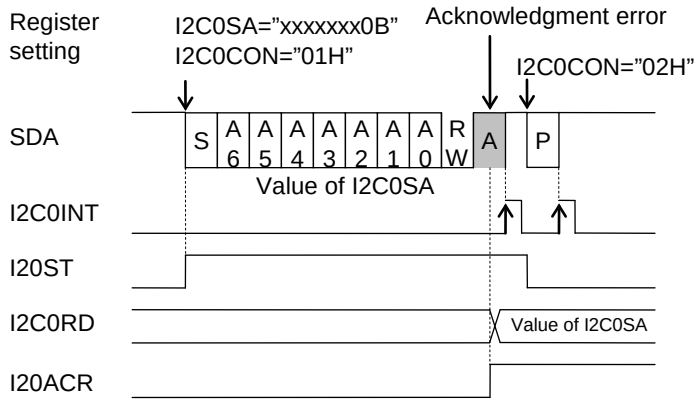


Figure 14-5 Operation Suspend Timing at Occurrence of Acknowledgment Error

When the values of the transmitted bit and the SDA pin do not coincide, the I20ER bit of the I2C bus 0 status register (I2C0STAT) is set to "1" and SDA pin remains the output until termination of the subsequent byte data communication. I20ER bit is initialized to "0" by writing I^2C Bus 0 Control Register (I2C0COCON).
Figure 14-6 shows the operation timing and control method when transmission fails.

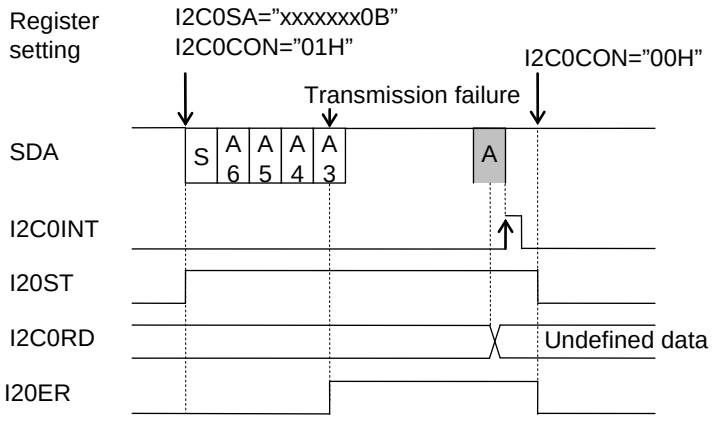


Figure 14-6 Operation Timing When Transmission Fails

14.3.3 Operation Waveforms

Figure 14-7 shows the operation waveforms of the SDA and SCL signals and the I20BB flag. Table 14-1 shows the relationship between communication speeds and HSCLK clock counts.

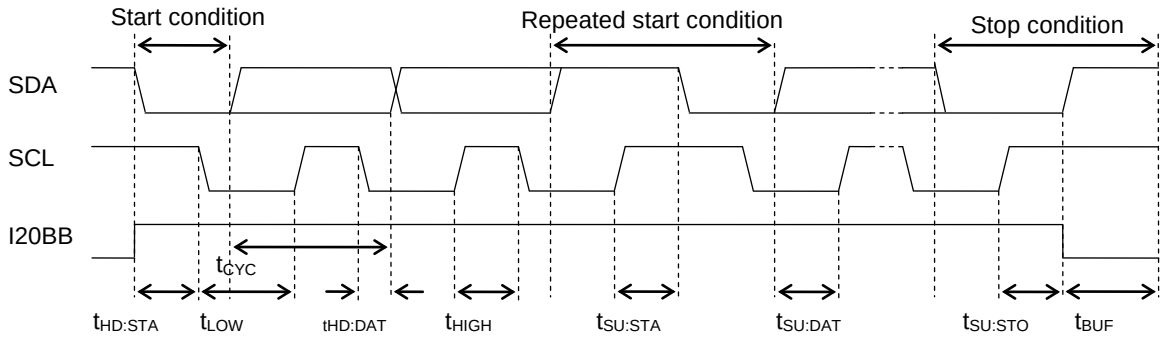


Figure 14-7 Operation Waveforms of SDA and SCL Signals and I20BB Flag

Table 14-1 Relationship between Communication Speeds and HSCLK Clock Counts

Communication speed (I20SP)	Speed reduction (I20DW1, 0)	t_{CYC}	$t_{HD:STA}$	t_{LOW}	$t_{HD:DAT}$	t_{HIGH}	$t_{SU:STA}$	$t_{SU:DAT}$	$t_{SU:STO}$	t_{BUF}
Standard mode 100 kbps	No reduction	40ϕ	18ϕ	22ϕ	4ϕ	18ϕ	22ϕ	18ϕ	18ϕ	22ϕ
	10% reduction	44ϕ	20ϕ	24ϕ	4ϕ	20ϕ	24ϕ	20ϕ	20ϕ	24ϕ
	20% reduction	48ϕ	22ϕ	26ϕ	4ϕ	22ϕ	26ϕ	22ϕ	22ϕ	26ϕ
	30% reduction	52ϕ	24ϕ	28ϕ	4ϕ	24ϕ	28ϕ	24ϕ	24ϕ	28ϕ
Fast mode 400 kbps	No reduction	10ϕ	4ϕ	6ϕ	2ϕ	4ϕ	6ϕ	4ϕ	4ϕ	6ϕ
	10% reduction	11ϕ	4ϕ	7ϕ	2ϕ	4ϕ	7ϕ	5ϕ	4ϕ	7ϕ
	20% reduction	12ϕ	5ϕ	7ϕ	2ϕ	5ϕ	7ϕ	5ϕ	5ϕ	7ϕ
	30% reduction	13ϕ	5ϕ	8ϕ	2ϕ	5ϕ	8ϕ	6ϕ	5ϕ	8ϕ

ϕ: Period of high-speed clock (HSCLK)

Note
The HSCLK clock count is set so that the communication speed may be set to 100kbps/400kbps when HSCLK is 4 MHz. When the high-speed clock frequency is not 4 MHz, select an I2C0MOD communication speed reduction rate and an FCON0 HSCLK frequency so that the communication speed may not exceed 100kbps/400kbps.

14.4 Specifying port registers

When you want to make sure the I2C bus interface function is working, please check related port registers are specified. See Chapter 21, "Port 4" for detail about the port registers.

14.4.1 Functioning P41(SCL) and P40(SDA) as the I2C

Set P41MD1-P40MD1 bits(bit1-bit0 of P4MOD1 register) to "0" and set P41MD0-P40MD0(bit1-bit0 of P4MOD0 register) to "1", for specifying the I2C as the secondary function of P41 and P40.

Reg. name	P4MOD1 register (Address: 0F225H)							
Bit	7	6	5	4	**3**	2	**1**	**0**
Bit name	P47MD1	P46MD1	P45MD1	P44MD1	P43MD1	P42MD1	**P41MD1**	**P40MD1**
Data	*	*	*	*	*	*	**0**	**0**

Reg. name	P4MOD0 register (Address: 0F224H)							
Bit	7	6	5	4	3	2	**1**	**0**
Bit name	P47MD0	P46MD0	P45MD0	P44MD0	P43MD0	P42MD0	**P41MD0**	**P40MD0**
Data	*	*	*	*	*	*	**1**	**1**

Set P41C1-P40C1 bit(bit1-0 of P4CON1 register) to "1", set P41C0-P40C0 bit(bit1-0 of P4CON0 register) to "0", and set P41DIR-P40DIR bit(bit1-0 of P4DIR register) to "0", for specifying the P41 and P40 as Nch open-drain output. The open-drain/open-collector outputs are required on the I2C bus line to avoid collision between H level and L level.

Reg. name	P4CON1 register (Address: 0F223H)							
Bit	7	6	5	4	3	2	**1**	**0**
Bit name	P47C1	P46C1	P45C1	P44C1	P43C1	P42C1	**P41C1**	**P40C1**
Data	*	*	*	*	*	*	**1**	**1**

Reg. name	P4CON0 register (Address: 0F222H)							
Bit	7	6	5	4	3	2	**1**	**0**
Bit name	P47C0	P46C0	P45C0	P44C0	P43C0	P42C0	**P41C0**	**P40C0**
Data	*	*	*	*	*	*	**0**	**0**

Reg. name	P4DIR register (Address: 0F221H)							
Bit	7	6	5	4	3	2	**1**	**0**
Bit name	P47DIR	P46DIR	P45DIR	P44DIR	P43DIR	P42DIR	**P41DIR**	**P40DIR**
Data	*	*	*	*	*	*	**0**	**0**

Data of P41D-P40D bits (bit1-0 of P4D register) do not affect to the I2C function, so don't care the data for the function.

Reg. name	P4D register (Address: 0F220H)							
Bit	7	6	5	4	3	2	**1**	**0**
Bit name	P47D	P46D	P45D	P44D	P43D	P42D	**P41D**	**P40D**
Data	*	*	*	*	*	*	******	******

* : Bit not related to the I2C bus interface function
** : Don't care the data

Chapter 15

Port 0

15. Port 0

15.1 Overview

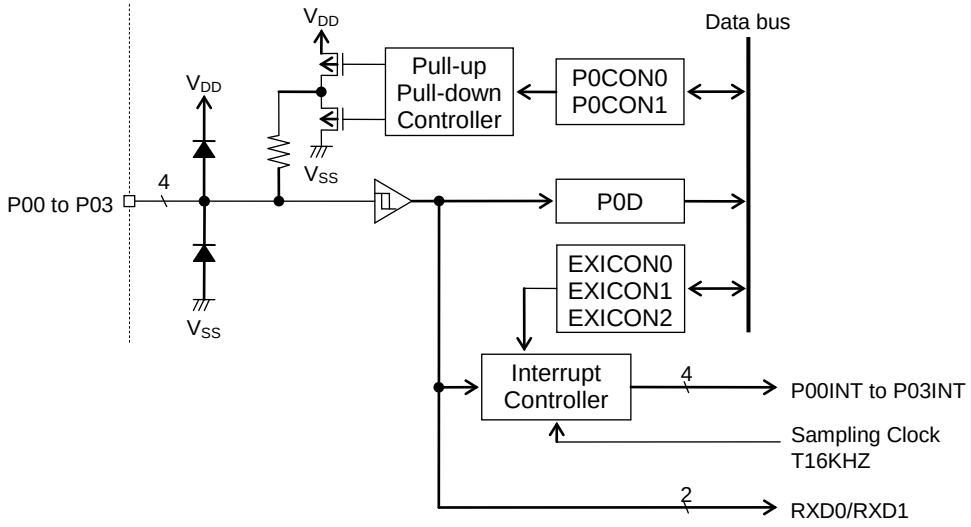
This LSI includes the 4-bit, input-only Port 0 (P00 to P03).

15.1.1 Features

- All bits support a maskable interrupt function.
- Allows selection of interrupt disabled mode, falling-edge interrupt mode, rising-edge interrupt mode, or both-edge interrupt mode for each bit.
- Allows selection of with/without interrupt sampling for each bit. (Sampling frequency: T16KHZ)
- Allows selection of high-impedance input mode, input mode with a pull-down resistor, or input mode with a pull-up resistor for each bit.
- The P00 and P01 pins can be used as the trigger input pins of the capture circuit and the P02 pin can be used as the RXD0 input pin of UART0. The P03 pin can be used as the RXD1 input pin of UART1.

15.1.2 Configuration

Figure 15-1 shows the configuration of Port 0.



P0D : Port 0 data register
P0CON0 : Port 0 control register 0
P0CON1 : Port 0 control register 1
EXICON0 : External interrupt control register 0
EXICON1 : External interrupt control register 1
EXICON2 : External interrupt control register 2

Figure 15-1 Configuration of Port 0

15.1.3 List of Pins

Pin name	I/O	Function
P00/EXI0/CAP0	I	Input port, External 0 interrupt, Capture 0 trigger signal input
P01/EXI1/CAP1	I	Input port, External 1 interrupt, Capture 1 trigger signal input
P02/EXI2/RXD0	I	Input port, External 2 interrupt, UART0 data input (RXD0)
P03/EXI3/RXD1	I	Input port, External 3 interrupt, UART1 data input(RXD1)

15.2 Description of Registers

15.2.1 List of Registers

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F204H	Port 0 data register	P0D	—	R	8	Depends on pin state
0F206H	Port 0 control register 0	P0CON0	P0CON	R/W	8/16	00H
0F207H	Port 0 control register 1	P0CON1		R/W	8	00H
0F020H	External interrupt control register 0	EXICON0	—	R/W	8	00H
0F021H	External interrupt control register 1	EXICON1	—	R/W	8	00H
0F022H	External interrupt control register 2	EXICON2	—	R/W	8	00H

15.2.2 Port 0 Data Register (P0D)

Address: 0F204H
Access: R
Access size: 8-bit
Initial value: Depends on pin state

	7	6	5	4	3	2	1	0
P0D	—	—	—	—	P03D	P02D	P01D	P00D
R	R	R	R	R	R	R	R	R
Initial value	0	0	0	0	x	x	x	x

P0D is a special function register (SFR) to only read the pin level of Port 0.

[Description of Bits]

- **P03D to P00D** (bits 3 to 0)
  The P03D to P00D bits are used to read the pin level of Port 0.

P03D	Description
0	P03 pin input: “L” level
1	P03 pin input: “H” level

P02D	Description
0	P02 pin input: “L” level
1	P02 pin input: “H” level

P01D	Description
0	P01 pin input: “L” level
1	P01 pin input: “H” level

P00D	Description
0	P00 pin input: “L” level
1	P00 pin input: “H” level

15.2.3 Port 0 Control Registers 0, 1 (P0CON0, P0CON1)

Address: 0F206H
Access: R/W
Access size: 8/16 bit
Initial value: 00H

	7	6	5	4	3	2	1	0
P0CON0	—	—	—	—	P03C0	P02C0	P01C0	P00C0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

Address: 0F207H
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
P0CON1	—	—	—	—	P03C1	P02C1	P01C1	P00C1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

P0CON0 and P0CON1 are special function registers (SFRs) to select the input mode of Port 0.

[Description of Bits]

- **P03C0 to P00C0, P03C1 to P00C1** (bits 3 to 0)
  The P03C0 to P00C0 bits and the P03C1 to P00C1 bits are used to select high-impedance input mode, input mode with a pull-down resistor, or input mode with a pull-up resistor. The P0nC0 bit and the P0nC1 bit determine the input mode of P0n (Example: When P02C0 = “0” and P02C1 = “1”, P02 is in input mode with a pull-up resistor).

P03C1-P00C1	P03C0-P00C0	Description
0	0	High-impedance input mode (initial value)
0	1	Input mode with a pull-down resistor
1	0	Input mode with a pull-up resistor
1	1	High-impedance input mode

15.2.4 External Interrupt Control Registers 0, 1 (EXICON0, EXICON1)

Address: 0F020H
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
EXICON0	—	—	—	—	P03E0	P02E0	P01E0	P00E0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

Address: 0F021H
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
EXICON1	—	—	—	—	P03E1	P02E1	P01E1	P00E1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

EXICON0 and EXICON1 are special function registers (SFRs) to select an interrupt edge of Port 0.

[Description of Bits]

- **P03E0 to P00E0**, **P03E1 to P00E1** (bits 3 to 0)
  The P03E0 to P00E0 bits and the P03E1 to P00E1 bits are used to select interrupt disabled mode, falling-edge interrupt mode, rising-edge interrupt mode, or both-edge interrupt mode. The P0nE0 bit and the P0nE1 bit determine the interrupt mode of P0n (Example: When P02E0 = "0" and P02E1 = "1", P02 is in rising-edge interrupt mode).

P03E1-P00E1	P03E0-P00E0	Description
0	0	Interrupt disabled (initial value)
0	1	Falling-edge interrupt mode
1	0	Rising-edge interrupt mode
1	1	Both-edge interrupt mode

15.2.5 External Interrupt Control Register 2 (EXICON2)

Address: 0F022H
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
EXICON2	—	—	—	—	P03SM	P02SM	P01SM	P00SM
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

EXICON2 is a special function register (SFR) to select whether the Port 0 interrupt is with sampling or without sampling.

[Description of Bits]

- **P03SM to P00SM** (bits 3 to 0)
  The P03SM to P00SM bits are used to select detection of signal edge for Port 0 interrupts with or without sampling. The sampling clock is T16KHZ of the low-speed time base counter (LTBC).

P03SM	Description
0	Detects the input signal edge for a P03 interrupt without sampling (initial value).
1	Detects the input signal edge for a P00 interrupt with sampling.

P02SM	Description
0	Detects the input signal edge for a P02 interrupt without sampling (initial value).
1	Detects the input signal edge for a P00 interrupt with sampling.

P01SM	Description
0	Detects the input signal edge for a P01 interrupt without sampling (initial value).
1	Detects the input signal edge for a P00 interrupt with sampling.

P00SM	Description
0	Detects the input signal edge for a P00 interrupt without sampling (initial value).
1	Detects the input signal edge for a P00 interrupt with sampling.

Note:
In STOP mode, since the 16 kHz sampling clock stops, no sampling is performed regardless of the values set in P00SM to P03SM.

15.3 Description of Operation

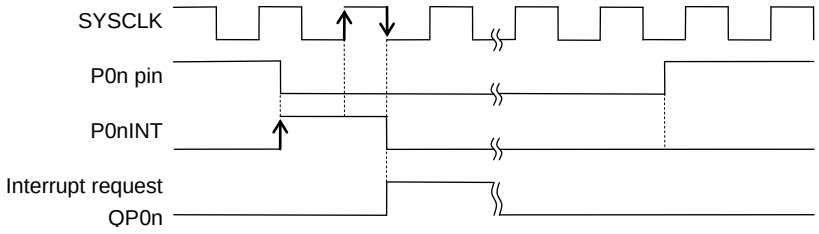
For each pin of Port 0, the setting of the Port 0 control registers 0 and 1 (P0CON0 and P0CON1) allows selection of high-impedance input mode, input mode with a pull-down resistor, or input mode with a pull-up resistor.
High-impedance input mode is selected at system reset.
The pin level of Port 0 can be read by reading the Port 0 data register (P0D).

15.3.1 External Interrupt / Secondary Function

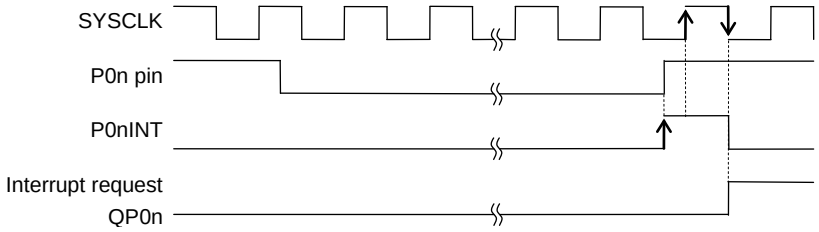
The Port 0 pins (P00, P01, P02, P03) can be used for P00 to P03 interrupts (P00INT to P03INT). The P00 to P03 interrupts are maskable and interrupt enable or disable can be selected. For interrupts, see Chapter 5, "Interrupt."
The P00 and P01 pins can be used as the trigger input to the capture circuit, the P02 pin as the RXD0 input to the UART0 and the P03 pin as the RXD1 input to the UART1.
For the capture function, see Chapter 8, "Capture." For the UART function, see Chapter 13, "UART."

15.3.2 Interrupt Request

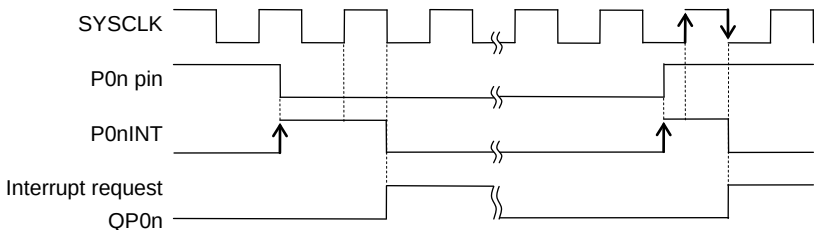
When an interrupt edge selected with the external interrupt control register 0, 1, or 2 (EXICON0, EXICON1, or EXICON2) occurs at a Port 0 pin, any of the maskable P00 to P03 interrupts (P00INT to P03INT) occurs.
Figure 15-2 shows the P00 to P03 interrupt generation timing in rising-edge interrupt mode, in falling-edge interrupt mode, and in both-edge interrupt mode without sampling and the P00 to P03 interrupt generation timing in rising-edge interrupt mode with sampling.



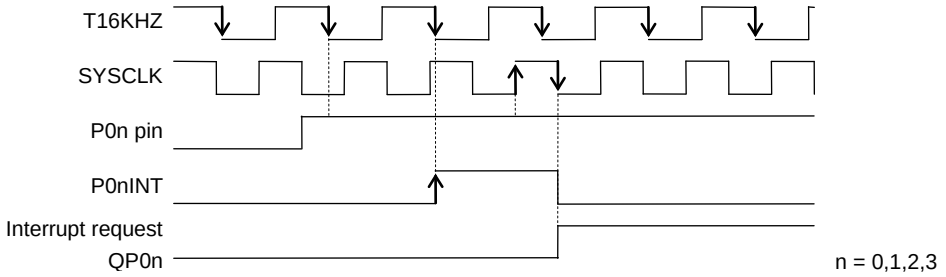
(a) When Falling-Edge Interrupt Mode without Sampling is Selected



(b) When Rising-Edge Interrupt Mode without Sampling is Selected



(c) When Both-Edge Interrupt Mode without Sampling is Selected



(d) When Rising-Edge Interrupt Mode with Sampling is Selected

Figure 15-2 P00 to P03 Interrupts Generation Timing

Chapter 16

Port 1

16. Port 1

16.1 Overview

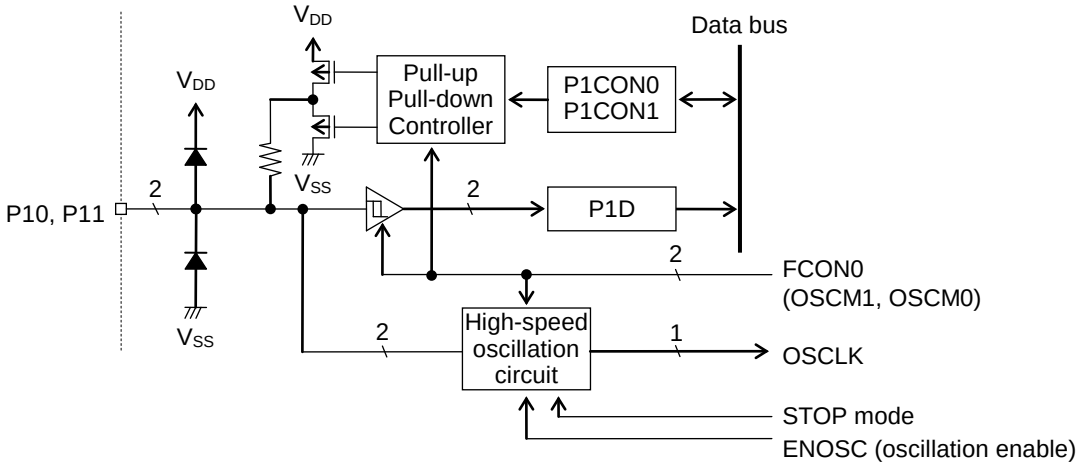
This LSI incorporates a 2-bit input port, Port 1 (P10, P11).
Port 1 can have a high-speed oscillation pin or an external clock input pin as a secondary function. When the port is used as a high-speed oscillation pin, the P11 pin functions as an output pin if crystal/ceramic oscillation mode is selected with the OSCM1–0 bits of the FCON0 register.
For high-speed oscillation and external clock input, see Chapter 6, "Clock Generation Circuit".

16.1.1 Features

- Allows selection of high-impedance input mode, input mode with a pull-down resistor, or input mode with a pull-up resistor for each bit.
- Allows selection of a high-speed crystal/ceramic resonator pin or an external clock input pin as a secondary function.

16.1.2 Configuration

Figure 16-1 shows the configuration of Port 1.



P1D : Port 1 data register
P1CON0 : Port 1 control register 0
P1CON1 : Port 1 control register 1

Figure 16-1 Configuration of Port 1

16.1.3 List of Pins

Pin name	I/O	Primary function	Secondary function
P10/OSC0	I	Input port	High-speed crystal/ceramic oscillation pin, external clock input pin
P11/OSC1	I/O	Input port	High-speed crystal/ceramic oscillation pin

16.2 Description of Registers

16.2.1 List of Registers

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F208H	Port 1 data register	P1D	—	R	8	Depends on pin status
0F20AH	Port 1 control register 0	P1CON0	P1CON	R/W	8/16	00H
0F20BH	Port 1 control register 1	P1CON1		R/W	8	00H

16.2.2 Port 1 Data Register (P1D)

Address: 0F208H
Access: R
Access size: 8 bits
Initial value: Depends on pin status

	7	6	5	4	3	2	1	0
P1D	—	—	—	—	—	—	P11D	P10D
R	R	R	R	R	R	R	R	R
Initial value	0	0	0	0	0	0	x	x

P1D is a special function register (SFR) dedicated to read the input level of the Port 1 pin.

[Description of Bits]

- **P11D, P10D** (bits 1, 0)
  The P11D and P10D bits are used to read the input level of the Port 1 pin.

P11D	Description
0	Input level of the P11 pin: "L"
1	Input level of the P11 pin: "H"

P10D	Description
0	Input level of the P10 pin: "L"
1	Input level of the P10 pin: "H"

16.2.3 Port 1 Control Registers 0, 1 (P1CON0, P1CON1)

Address: 0F20AH
Access: R/W
Access size: 8/16 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
P1CON0	—	—	—	—	—	—	P11C0	P10C0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

Address: 0F20BH
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
P0CON1	—	—	—	—	—	—	P11C1	P10C1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

P1CON0 and P1CON1 are special function registers (SFRs) to select the input mode of Port 1.

[Description of Bits]

- **P11C0, P10C0, P11C1, P00C1** (bits 1-0)
  The P11C0, P10C0, P11C1 and P00C1 bits are used to select high-impedance input mode, input mode with a pull-down resistor, or input mode with a pull-up resistor.

P11C1	P11C0	Description
0	0	P11 pin: high-impedance input mode (initial value)
0	1	P11 pin: input mode with a pull-down resistor
1	0	P11 pin: input mode with a pull-up resistor
1	1	P11 pin: high-impedance input mode

P10C1	P10C0	Description
0	0	P10 pin: high-impedance input mode (initial value)
0	1	P10 pin: input mode with a pull-down resistor
1	0	P10 pin: input mode with a pull-up resistor
1	1	P10 pin: high-impedance input mode

Note:
When using P10 and P11 as crystal/ceramic oscillation pins, be sure to set the P10 and P11 pins to high-impedance input mode.
When using the P10 pin as an external clock input pin, set the P10 pin to high-impedance input mode so that the pull-up or pull-down resistor will not carry current.

16.3 Description of Operation

16.3.1 Input Port Function

For each pin of Port 1, one of high-impedance input mode, input mode with a pull-down resistor, and input mode with a pull-up resistor can be selected by setting the Port 1 control registers 0 and 1 (P1CON0 and P1CON1). At system reset, high-impedance input mode is selected as the initial state.
The input level of the Port 1 pin can be read by reading the Port 1 data register (P1D).

16.3.2 Secondary Function

A high-speed crystal/ceramic oscillation pin or an external clock input pin is assigned to Port 1 as a secondary function. Select high-speed crystal/ceramic oscillation mode or external clock input mode by using the high-speed clock mode select function of the OSCM1 and 0 bits of the frequency control register 0 (FCON0). In crystal/ceramic oscillation mode, both P10 and P11 pins are used as the pins for crystal/ceramic oscillation.
In external clock input mode, the P10 pin is used as the input pin of external clock and the P11 pin can be used as a general-purpose input port.

Note:
No port mode register is provided for switching between the primary function and the secondary function of Port 1. When using the Port 1 pins as high-speed oscillation pins, pin mode is switched according to the values set in the OSCM1 and OSCM0 bits of the FCON0 register.
See Chapter 6, "Clock Generation Circuit," for the details of the FCON0 register, high-speed oscillation, and external clock input.

Chapter 17

Port 2

17. Port 2

17.1 Overview

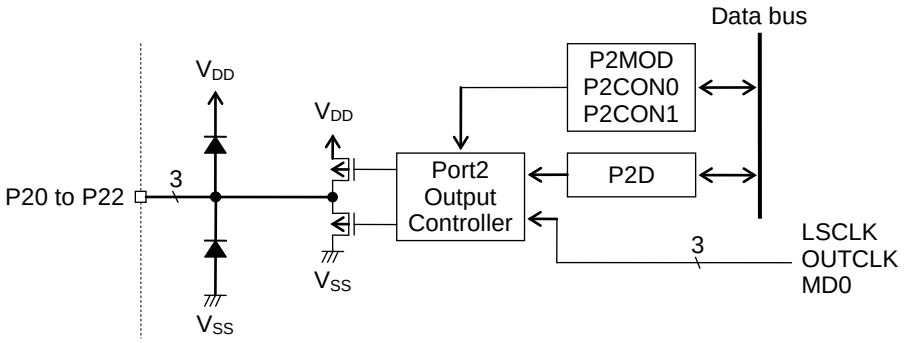
This LSI includes 3-bit Port 2 (P20 to P22) dedicated to output.
Port 2 can output low-speed clock (LSCLK), high-speed output clock (OUTCLK), and melody as a secondary function. For clock output and melody 0 (MD0) output, see Chapter 6, "Clock Generation Circuit" and Chapter 22, "Melody Driver", respectively.

17.1.1 Features

- Allows direct LED drive.
- Allows selection of high-impedance output mode, P-channel open drain output mode, N-channel open drain output mode, or CMOS output mode for each bit.
- Allows output of low-speed clock (LSCLK), high-speed clock (OUTCLK), or melody 0 (MD0) as a secondary function.

17.1.2 Configuration

Figure 17-1 shows the configuration of Port 2.



P2D : Port 2 data register
P2CON0 : Port 2 control register 0
P2CON1 : Port 2 control register 1
P2MOD : Port 2 mode register

Figure 17-1 Configuration of Port 2

17.1.3 List of Pins

Pin name	I/O	Primary function	Secondary function
P20/LED0/LSCLK	O	Output port, Direct LED drive	Low-speed clock output (LSCLK)
P21/LED1/OUTCLK	O	Output port, Direct LED drive	High-speed clock output (OUTCLK)
P22/LED2/MD0	O	Output port, Direct LED drive	Melody 0 output (MD0)

17.2 Description of Registers

17.2.1 List of Registers

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F210H	Port 2 data register	P2D	—	R/W	8	00H
0F212H	Port 2 control register 0	P2CON0	P2CON	R/W	8/16	00H
0F213H	Port 2 control register 1	P2CON1		R/W	8	00H
0F214H	Port 2 mode register	P2MOD	—	R/W	8	00H

17.2.2 Port 2 Data Register (P2D)

Address: 0F210H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
P2D	—	—	—	—	—	P22D	P21D	P20D
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

P2D is a special function register (SFR) to set the output value of Port 2. The value of this register is output to Port 2. The value written to P2D is readable.

[Description of Bits]

- **P22D-P20D** (bits 2-0)
  The P22D to P20D bits are used to set the output value of the Port 2 pin.

P22D	Description
0	Output level of the P22 pin: "L"
1	Output level of the P22 pin: "H"

P21D	Description
0	Output level of the P21 pin: "L"
1	Output level of the P21 pin: "H"

P20D	Description
0	Output level of the P20 pin: "L"
1	Output level of the P20 pin: "H"

17.2.3 Port 2 control registers 0, 1 (P2CON0, P2CON1)

Address: 0F212H
Access: R/W
Access size: 8/16 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
P2CON0	—	—	—	—	—	P22C0	P21C0	P20C0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

Address: 0F213H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
P2CON1	—	—	—	—	—	P22C1	P21C1	P20C1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

P2CON0 and P2CON1 are special function registers (SFRs) to select the output state of the output pin Port 2.

[Description of Bits]

- **P22C0-P20C0, P22C1-P20C1** (bits 2-0)
  The P22C0 to P20C0 and P22C1 to P20C1 bits are used to select high-impedance output mode, P-channel open drain output mode, N-channel open drain output mode, or CMOS output mode.
  To directly drive LEDs, select N-channel open drain output mode.

P22C1	P22C0	Description
0	0	P22 pin: In high-impedance output mode (initial value)
0	1	P22 pin: In P-channel open drain output mode
1	0	P22 pin: In N-channel open drain output mode
1	1	P22 pin: In CMOS output mode

P21C1	P21C0	Description
0	0	P21 pin: In high-impedance output mode (initial value)
0	1	P21 pin: In P-channel open drain output mode
1	0	P21 pin: In N-channel open drain output mode
1	1	P21 pin: In CMOS output mode

P20C1	P20C0	Description
0	0	P20 pin: In high-impedance output mode (initial value)
0	1	P20 pin: In P-channel open drain output mode
1	0	P20 pin: In N-channel open drain output mode
1	1	P20 pin: In CMOS output mode

17.2.4 Port 2 Mode Register (P2MOD)

Address: 0F214H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
P2MOD	—	—	—	—	—	P22MD	P21MD	P20MD
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

P2MOD is a special function register (SFR) to select the primary function or the secondary function of Port 2

[Description of Bits]

- **P22MD** (bit 2)
  The P22MD bit is used to select the primary function or the secondary function of the P22 pin.

P22MD	Description
0	General-purpose output port function (initial value)
1	Melody 0 (MD0) output function

- **P21MD** (bit 1)
  The P21MD bit is used to select the primary function or the secondary function of the P21 pin.

P21MD	Description
0	General-purpose output port function (initial value)
1	High-speed output clock (OUTCLK) output function

- **P20MD** (bit 0)
  The P20MD bit is used to select the primary function or the secondary function of the P20 pin.

P20MD	Description
0	General-purpose output port function (initial value)
1	Low-speed clock (LSCLK) output function

Note:
The output characteristics of port2(P20, P21 and P22) corresponds to VOL1 and VOH1 when each bit (P20MD, P21MD, P22MD) is "1" , and corresponds to VOL2 and VOH2 when the bit is "0", which are shown in Appendix C, "Electrical Characteristics".

17.3 Description of Operation

17.3.1 Output Port Function

For each pin of Port 2, any one of high-impedance output mode, P-channel open drain output mode, N-channel open drain output mode, and CMOS output mode can be selected by setting the Port 2 control registers 0 and 1 (P2CON0 and P2CON1). At a system reset, high-impedance output mode is selected as the initial state.
Depending of the value set in the Port 2 data register (P2D), a "L" level or "H" level signal is output to each pin of Port 2.

17.3.2 Secondary Function

Low-speed clock (LSCLK) output, High-speed output clock (OUTCLK) output, or melody 0 (MD0) output is assigned to Port 2 as a secondary function. The secondary function can be used by setting the P22MD to P20MD bits of the Port 2 mode register (P2MOD) to "1".

Chapter 18

Port 3

18. Port 3

18.1 Overview

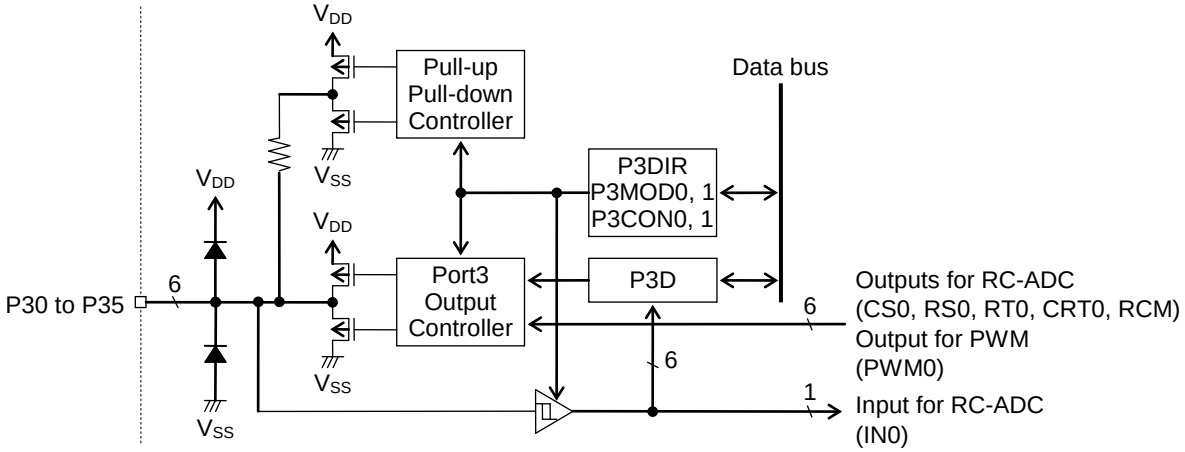
This LSI includes Port 3 (P30 to P35), which is a 6-bit input/output port.
This port can also be used as the RC-ADC (channel 0) oscillation pins (IN0, CS0, RS0, RT0, CRT0, RCM) and the PWM output pin in secondary and tertiary modes.
For RC-ADC and PWM, see Chapter 23, "RC Oscillation Type A/D converter", and Chapter 10, "PWM".

18.1.1 Features

- Allows selection of high-impedance output, P-channel open drain output, N-channel open drain output, or CMOS output in output mode for each bit.
- Allows selection of high-impedance input, input with a pull-down resistor, or input with a pull-up resistor in input mode for each bit.
- The RC-ADC (channel 0) oscillation pins (IN0, CS0, RS0, RT0, CRT0, RCM) and the PWM output pin (PWM0) can be used as the secondary functions.

18.1.2 Configuration

Figure 18-1 shows the configuration of Port 3.



P3D : Port 3 data register
P3DIR : Port 3 direction register
P3CON0 : Port 3 control register 0
P3CON1 : Port 3 control register 1
P3MOD0 : Port 3 mode register 0
P3MOD1 : Port 3 mode register 1

Figure 18-1 Configuration of Port 3

18.1.3 List of Pins

Pin name	I/O	Primary function	Secondary function	Tertiary function
P30/IN0	I/O	Input/output port	Oscillation waveform input pin for RC-ADC	—
P31/CS0	I/O	Input/output port	Reference capacitor connection pin for RC-ADC	—
P32/RS0	I/O	Input/output port	Reference resistor connection pin for RC-ADC	—
P33/RT0	I/O	Input/output port	Resistor sensor connection pin for measurement for RC-ADC	—
P34/CRT0/PWM0	I/O	Input/output port	Resistor/capacitor sensor connection pin for measurement for RC-ADC	PWM0 output pin
P35/RCM	I/O	Input/output port	RC oscillation monitor pin for RC-ADC	—

18.2 Description of Registers

18.2.1 List of Registers

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F218H	Port 3 data register	P3D	—	R/W	8	00H
0F219H	Port 3 direction register	P3DIR	—	R/W	8	00H
0F21AH	Port 3 control register 0	P3CON0	P3CON	R/W	8/16	00H
0F21BH	Port 3 control register 1	P3CON1		R/W	8	00H
0F21CH	Port 3 mode register 0	P3MOD0	P3MOD	R/W	8/16	00H
0F21DH	Port 3 mode register 1	P3MOD1		R/W	8	00H

18.2.2 Port 3 data register (P3D)

Address: 0F218H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
P3D	—	—	P35D	P34D	P33D	P32D	P31D	P30D
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

P3D is a special function register (SFR) to set the value to be output to the Port 3 pin or to read the input level of the Port 3. In output mode, the value of this register is output to the Port 3 pin. The value written to P3D is readable. In input mode, the input level of the Port 3 pin is read when P3D is read. Output mode or input mode is selected by using the port direction register (P3DIR) described later.

[Description of Bits]

- **P35D-P30D** (bits 5-0)
  The P35D to P30D bits are used to set the output value of the Port 3 pin in output mode and to read the pin level of the Port 3 pin in input mode.

P35D	Description
0	Output or input level of the P35 pin: "L"
1	Output or input level of the P35 pin: "H"

P34D	Description
0	Output or input level of the P34 pin: "L"
1	Output or input level of the P34 pin: "H"

P33D	Description
0	Output or input level of the P33 pin: "L"
1	Output or input level of the P33 pin: "H"

P32D	Description
0	Output or input level of the P32 pin: "L"
1	Output or input level of the P32 pin: "H"

P31D	Description
0	Output or input level of the P31 pin: "L"
1	Output or input level of the P31 pin: "H"

P30D	Description
0	Output or input level of the P30 pin: "L"
1	Output or input level of the P30 pin: "H"

18.2.3 Port 3 Direction Register (P3DIR)

Address: 0F219H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
P3DIR	—	—	P35DIR	P34DIR	P33DIR	P32DIR	P31DIR	P30DIR
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

P3DIR is a special function register (SFR) to select the input/output mode of Port 3.

[Description of Bits]

- **P35DIR-P30DIR** (bits 5-0)
  The P35DIR to P30DIR pins are used to set the input/output direction of the Port 3 pin.

P35DIR	Description
0	P35 pin: Output (initial value)
1	P35 pin: Input

P34DIR	Description
0	P34 pin: Output (initial value)
1	P34 pin: Input

P33DIR	Description
0	P33 pin: Output (initial value)
1	P33 pin: Input

P32DIR	Description
0	P32 pin: Output (initial value)
1	P32 pin: Input

P31DIR	Description
0	P31 pin: Output (initial value)
1	P31 pin: Input

P30DIR	Description
0	P30 pin: Output (initial value)
1	P30 pin: Input

18.2.4 Port 3 control registers 0, 1 (P3CON0, P3CON1)

Address: 0F21AH
Access: R/W
Access size: 8/16 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
P3CON0	—	—	P35C0	P34C0	P33C0	P32C0	P31C0	P30C0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

Address: 0F21BH
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
P3CON1	—	—	P35C1	P34C1	P33C1	P32C1	P31C1	P30C1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

P3CON0 and P3CON1 are special function registers (SFRs) to select input/output state of the Port 3 pin. The input/output state is different between input mode and output mode. Input or output is selected by using the P3DIR register.

[Description of Bits]

- **P35C1-P30C1, P35C0-P30C0** (bits 5-0)
  The P35C1 to P30C1 pins and the P35C0 to P30C0 pins are used to select high-impedance output, P-channel open drain output, N-channel open drain output, or CMOS output in output mode and to select high-impedance input, input with a pull-down resistor, or input with a pull-up resistor in input mode.

		When output mode is selected (P35DIR bit = "0")	When input mode is selected (P35DIR bit = "1")
P35C1	**P35C0**	Description	
0	0	P35 pin: High-impedance output (initial value)	P35 pin: High-impedance input
0	1	P35 pin: P-channel open drain output	P35 pin: Input with a pull-down resistor
1	0	P35 pin: N-channel open drain output	P35 pin: Input with a pull-up resistor
1	1	P35 pin: CMOS output	P35 pin: High-impedance input

		When output mode is selected (P34DIR bit = "0")	When input mode is selected (P34DIR bit = "1")
P34C1	**P34C0**	Description	
0	0	P34 pin: High-impedance output (initial value)	P34 pin: High-impedance input
0	1	P34 pin: P-channel open drain output	P34 pin: Input with a pull-down resistor
1	0	P34 pin: N-channel open drain output	P34 pin: Input with a pull-up resistor
1	1	P34 pin: CMOS output	P34 pin: High-impedance input

		When output mode is selected (P33DIR bit = "0")	When input mode is selected (P33DIR bit = "1")
P33C1	**P33C0**	Description	
0	0	P33 pin: High-impedance output (initial value)	P33 pin: High-impedance input
0	1	P33 pin: P-channel open drain output	P33 pin: Input with a pull-down resistor
1	0	P33 pin: N-channel open drain output	P33 pin: Input with a pull-up resistor
1	1	P33 pin: CMOS output	P33 pin: High-impedance input

		When output mode is selected (P32DIR bit = "0")	When input mode is selected (P32DIR bit = "1")
P32C1	**P32C0**	Description	
0	0	P32 pin: High-impedance output (initial value)	P32 pin: High-impedance input
0	1	P32 pin: P-channel open drain output	P32 pin: Input with a pull-down resistor
1	0	P32 pin: N-channel open drain output	P32 pin: Input with a pull-up resistor
1	1	P32 pin: CMOS output	P32 pin: High-impedance input

		When output mode is selected (P31DIR bit = "0")	When input mode is selected (P31DIR bit = "1")
P31C1	**P31C0**	Description	
0	0	P31 pin: High-impedance output (initial value)	P31 pin: High-impedance input
0	1	P31 pin: P-channel open drain output	P31 pin: Input with a pull-down resistor
1	0	P31 pin: N-channel open drain output	P31 pin: Input with a pull-up resistor
1	1	P31 pin: CMOS output	P31 pin: High-impedance input

		When output mode is selected (P30DIR bit = "0")	When input mode is selected (P30DIR bit = "1")
P30C1	**P30C0**	Description	
0	0	P30 pin: High-impedance output (initial value)	P30 pin: High-impedance input
0	1	P30 pin: P-channel open drain output	P30 pin: Input with a pull-down resistor
1	0	P30 pin: N-channel open drain output	P30 pin: Input with a pull-up resistor
1	1	P30 pin: CMOS output	P30 pin: High-impedance input

18.2.5 Port 3 mode registers 0, 1 (P3MOD0, P3MOD1)

Address: 0F21CH
Access: R/W
Access size: 8/16 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
P3MOD0	—	—	P35MD0	P34MD0	P33MD0	P32MD0	P31MD0	P30MD0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

Address: 0F21DH
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
P3MOD1	—	—	P35MD1	P34MD1	P33MD1	P32MD1	P31MD1	P30MD1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

P3MOD0 and P3MOD1 are special function registers (SFRs) to select the primary, secondary, or tertiary function of Port 3.

[Description of Bits]

- **P35MD1, P35MD0** (bit 5)
  The P35MD1 and P35MD0 bits are used to select the primary or secondary function of the P35 pin.

P35MD1	P35MD0	Description
0	0	General-purpose input/output mode (initial value)
0	1	RC oscillation monitor pin for RC-ADC
1	0	Prohibited
1	1	Prohibited

- **P34MD1, P34MD0** (bit 4)
  The P34MD1 and P34MD0 bits are used to select the primary, secondary, or tertiary function of the P34 pin.

P34MD1	P34MD0	Description
0	0	General-purpose input/output mode (initial value)
0	1	Resistor/capacitor sensor connection pin for measurement for RC-ADC (channel 0)
1	0	PWM0 output pin
1	1	Prohibited

- **P33MD1, P33MD0** (bit 3)
  The P33MD1 and P33MD0 bits are used to select the primary or secondary function of the P33 pin.

P33MD1	P33MD0	Description
0	0	General-purpose input/output mode (initial value)
0	1	Resistor/capacitor sensor connection pin for measurement for RC-ADC (channel 0)
1	0	Prohibited
1	1	Prohibited

- **P32MD1, P32MD0** (bit 2)
  The P32MD1 and P32MD0 bits are used to select the primary or secondary function of the P32 pin.

P32MD1	P32MD0	Description
0	0	General-purpose input/output mode (initial value)
0	1	Reference resistor connection pin for RC-ADC (channel 0)
1	0	Prohibited
1	1	Prohibited

- **P31MD1, P31MD0** (bit 1)
  The P31MD1 and P31MD0 bits are used to select the primary or secondary function of the P31 pin.

P31MD1	P31MD0	Description
0	0	General-purpose input/output mode (initial value)
0	1	Reference capacitor connection pin for RC-ADC (channel 0)
1	0	Prohibited
1	1	Prohibited

- **P30MD1, P30MD0** (bit 0)
  The P30MD1 and P30MD0 bits are used to select the primary or secondary function of the P30 pin.

P30MD1	P30MD0	Description
0	0	General-purpose input/output mode (initial value)
0	1	RC oscillation waveform input pin for RC-ADC (channel 0)
1	0	Prohibited
1	1	Prohibited

Note:
If any bit combination out of the above is set to "Prohibited" and the corresponding bit of the port 3 is specified to output mode (selected in port3 control register), status of corresponding pin is fixed, regardless the contents of Port3 register (P3D)

High-impedance output mode: High-impedance
P-channel open drain output mode: High-impedance
N-channel open drain output mode: Fixed to "L"
CMOS output mode: High-impedance: Fixed to "L"

When using RC-ADC as the secondary function, specify each pin be "High-impedance input" even the RC oscillation monitor pin. Pull-up or Pull-down input makes drawing the current.

18.3 Description of Operation

18.3.1 Input/Output Port Functions

For each pin of Port 3, either output or input is selected by setting the Port 3 direction register (P3DIR).
In output mode, high-impedance output mode, P-channel open drain output mode, N-channel open drain output mode, or CMOS output mode can be selected by setting the Port 3 control registers 0 and 1 (P3CON0 and P3CON1).
In input mode, high-impedance input mode, input mode with a pull-down resistor, or input mode with a pull-up resistor can be selected by setting the Port 3 control registers 0 and 1 (P3CON0 and P3CON1).
At a system reset, high-impedance output mode is selected as the initial state.
In output mode, "L" or "H" level is output to each pin of Port 3 depending on the value set by the Port 3 data register (P3D).
In input mode, the input level of each pin of Port 3 can be read from the Port 3 data register (P3D).

18.3.2 Secondary and Tertiary Functions

Secondary and tertiary functions are assigned to Port 3 as the RC-ADC (channel 0) oscillation pins (IN0, CS0, RS0, RT0, CRT0, RCM) and the PWM output pin (PWM0). These pins can be used in a secondary or tertiary function mode by setting the P35MD0 to P30MD0 bits and the P35MD1 to P30MD1 bits of the Port 3 mode registers (P3MOD0, P3MOD1).

Note:
All the port 3 pins except P35/RCM are configured as pins dedicated to the RC-ADC function during A/D conversion. Therefore, if there is any unused pin, that pin cannot be used as its primary function during A/D conversion. For the RC-ADC, see Chapter 23, "RC Oscillation Type A/D Converter".

Chapter 19

Port 4

19. Port 4

19.1 Overview

This LSI includes Port 4 (P40 to P47) which is an 8-bit input/output port.
This port can have the I2C bus, RC-ADC, synchronous serial port, and PWM output functions as secondary and tertiary and quaternary functions.
See the following chapters for reference:

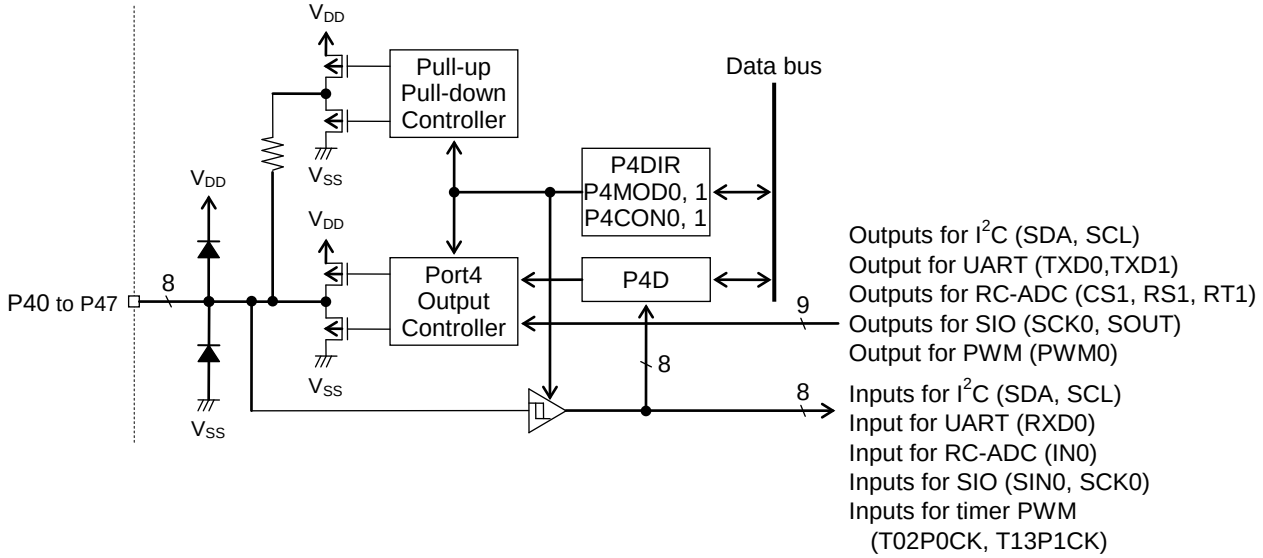
I2C bus:	Chapter 14 "I2C Bus Interface"
UART:	Chapter 13 "UART"
RC-ADC:	Chapter 23 "RC Oscillation Type A/D Converter"
Synchronous serial port:	Chapter 12 "Synchronous Serial Port"
PWM:	Chapter 10 "PWM"

19.1.1 Features

- Allows selection of high-impedance output, P-channel open drain output, N-channel open drain output, or CMOS output for each bit in output mode.
- Allows selection of high-impedance input, input with a pull-down resistor, or input with a pull-up resistor for each bit in input mode.
- The P44 and P45 pins can be used as external clock input pins for the timer and PWM.
- The I2C bus interface pins (SDA, SCL), UART pins (RXD0, TXD0, TXD1), RC-ADC (channel 1) oscillation pins (IN1, CS1, RS1, RT1), synchronous serial port pins (SIN0, SCK0, SOUT0), and PWM output pin (PWM0) can be used as the secondary functions.

19.1.2 Configuration

Figure 19-1 shows the configuration of Port 4.



P4D	: Port 4 data register
P4DIR	: Port 4 direction register
P4CON0	: Port 4 control register 0
P4CON1	: Port 4 control register 1
P4MOD0	: Port 4 mode register 0
P4MOD1	: Port 4 mode register 1

Figure 19-1 Configuration of Port 4

19.1.3 List of Pins

Pin name	I/O	Primary function	Secondary function	Tertiary function	Quaternary function
P40/SDA/SIN0	I/O	Input/output port	I^2C bus 0 data input/output pin	SSIO0 data input pin	—
P41/SCL/SCK0	I/O	Input/output port	I^2C bus 0 clock input/output pin	SSIO0 clock input/output pin	—
P42/RXD0/SOUT0	I/O	Input/output port	UART0 data input pin	SSIO0 data output pin	—
P43/TXD0/PWM0/ TXD1	I/O	Input/output port	UART0 data output pin	PWM0 output pin	UART1 data output pin
P44/ T02P0CK /IN1/SIN0/	I/O	Input/output port, Timer 0/Timer 2/ PWM0 external clock	RC oscillation waveform input pin for RC-ADC	SSIO0 data input pin	—
P45/ T13P1CK /CS1/SCK0/	I/O	Input/output port, Timer 1/Timer 3 external clock	Reference capacitor connection pin for RC-ADC	SSIO0 clock input/output pin	—
P46/RS1/SOUT0	I/O	Input/output port	Reference resistor connection pin for RC-ADC	SSIO0 data output pin	—
P47/RT1	I/O	Input/output port	Resistor sensor connection pin for measurement for RC-ADC	—	—

19.2 Description of Registers

19.2.1 List of Registers

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F220H	Port 4 data register	P4D	—	R/W	8	00H
0F221H	Port 4 direction register	P4DIR	—	R/W	8	00H
0F222H	Port 4 control register 0	P4CON0	P4CON	R/W	8/16	00H
0F223H	Port 4 control register 1	P4CON1		R/W	8	00H
0F224H	Port 4 mode register 0	P4MOD0	P4MOD	R/W	8/16	00H
0F225H	Port 4 mode register 1	P4MOD1		R/W	8	00H

19.2.2 Port 4 Data Register (P4D)

Address: 0F220H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
P4D	P47D	P46D	P45D	P44D	P43D	P42D	P41D	P40D
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

P4D is a special function register (SFR) to set the value to be output to the Port 4 pin or to read the input level of the Port 4. In output mode, the value of this register is output to the Port 4 pin. The value written to P4D is readable. In input mode, the input level of the Port 4 pin is read when P4D is read. Output mode or input mode is selected by using the port direction register (P4DIR) described later.

[Description of Bits]

- **P47D-P40D** (bits 7-0)
  The P47D to P40D bits are used to set the output value of the Port 4 pin in output mode and to read the pin level of the Port 4 pin in input mode.

P47D	Description
0	Output or input level of the P47 pin: "L"
1	Output or input level of the P47 pin: "H"

P46D	Description
0	Output or input level of the P46 pin: "L"
1	Output or input level of the P46 pin: "H"

P45D	Description
0	Output or input level of the P45 pin: "L"
1	Output or input level of the P45 pin: "H"

P44D	Description
0	Output or input level of the P44 pin: "L"
1	Output or input level of the P44 pin: "H"

P43D	Description
0	Output or input level of the P43 pin: "L"
1	Output or input level of the P43 pin: "H"

P42D	Description
0	Output or input level of the P42 pin: "L"
1	Output or input level of the P42 pin: "H"

P41D	Description
0	Output or input level of the P41 pin: "L"
1	Output or input level of the P41 pin: "H"

P40D	Description
0	Output or input level of the P40 pin: "L"
1	Output or input level of the P40 pin: "H"

19.2.3 Port 4 Direction Register (P4DIR)

Address: 0F221H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
P4DIR	P47DIR	P46DIR	P45DIR	P44DIR	P43DIR	P42DIR	P41DIR	P40DIR
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

P4DIR is a special function register (SFR) to select the input/output mode of Port 4.

[Description of Bits]

- **P47DIR-P40DIR** (bits 7-0)
  The P47DIR to P40DIR pins are used to set the input/output direction of the Port 4 pin.

P47DIR	Description
0	P47 pin: Output (initial value)
1	P47 pin: Input

P46DIR	Description
0	P46 pin: Output (initial value)
1	P46 pin: Input

P45DIR	Description
0	P45 pin: Output (initial value)
1	P45 pin: Input

P44DIR	Description
0	P44 pin: Output (initial value)
1	P44 pin: Input

P43DIR	Description
0	P43 pin: Output (initial value)
1	P43 pin: Input

P42DIR	Description
0	P42 pin: Output (initial value)
1	P42 pin: Input

P41DIR	Description
0	P41 pin: Output (initial value)
1	P41 pin: Input

P40DIR	Description
0	P40 pin: Output (initial value)
1	P40 pin: Input

19.2.4 Port 4 Control Registers 0, 1 (P4CON0, P4CON1)

Address: 0F222H
Access: R/W
Access size: 8/16 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
P4CON0	P47C0	P46C0	P45C0	P44C0	P43C0	P42C0	P41C0	P40C0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

Address: 0F223H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
P4CON1	P47C1	P46C1	P45C1	P44C1	P43C1	P42C1	P41C1	P40C1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

P4CON0 and P4CON1 are special function registers (SFRs) to select input/output state of the Port 4 pin. The input/output state is different between input mode and output mode. Input or output is selected by using the P4DIR register.

[Description of Bits]

- **P47C1-P40C1, P47C0-P40C0** (bits 7-0)
  The P47C1 to P40C1 pins and the P47C0 to P40C0 pins are used to select high-impedance output, P-channel open drain output, N-channel open drain output, or CMOS output in output mode and to select high-impedance input, input with a pull-down resistor, or input with a pull-up resistor in input mode.

Setting of P47 pin		When output mode is selected (P47DIR bit = "0")	When input mode is selected (P47DIR bit = "1")
P47C1	**P47C0**	Description	
0	0	High-impedance output (initial value)	High-impedance input
0	1	P-channel open drain output	Input with a pull-down resistor
1	0	N-channel open drain output	Input with a pull-up resistor
1	1	CMOS output	High-impedance input

Setting of P46 pin		When output mode is selected (P46DIR bit = "0")	When input mode is selected (P46DIR bit = "1")
P46C1	**P46C0**	Description	
0	0	High-impedance output (initial value)	High-impedance input
0	1	P-channel open drain output	Input with a pull-down resistor
1	0	N-channel open drain output	Input with a pull-up resistor
1	1	CMOS output	High-impedance input

Setting of P45 pin		When output mode is selected (P45DIR bit = "0")	When input mode is selected (P45DIR bit = "1")
P45C1	**P45C0**	Description	
0	0	High-impedance output (initial value)	High-impedance input
0	1	P-channel open drain output	Input with a pull-down resistor
1	0	N-channel open drain output	Input with a pull-up resistor
1	1	CMOS output	High-impedance input

Setting of P44 pin		When output mode is selected (P44DIR bit = "0")	When input mode is selected (P44DIR bit = "1")
P44C1	**P44C0**	Description	
0	0	High-impedance output (initial value)	High-impedance input
0	1	P-channel open drain output	Input with a pull-down resistor
1	0	N-channel open drain output	Input with a pull-up resistor
1	1	CMOS output	High-impedance input

Setting of P43 pin		When output mode is selected (P43DIR bit = "0")	When input mode is selected (P43DIR bit = "1")
P43C1	**P43C0**	Description	
0	0	High-impedance output (initial value)	High-impedance input
0	1	P-channel open drain output	Input with a pull-down resistor
1	0	N-channel open drain output	Input with a pull-up resistor
1	1	CMOS output	High-impedance input

Setting of P42 pin		When output mode is selected (P42DIR bit = "0")	When input mode is selected (P42DIR bit = "1")
P42C1	**P42C0**	Description	
0	0	High-impedance output (initial value)	High-impedance input
0	1	P-channel open drain output	Input with a pull-down resistor
1	0	N-channel open drain output	Input with a pull-up resistor
1	1	CMOS output	High-impedance input

Setting of P41 pin		When output mode is selected (P41DIR bit = "0")	When input mode is selected (P41DIR bit = "1")
P41C1	**P41C0**	Description	
0	0	High-impedance output (initial value)	High-impedance input
0	1	P-channel open drain output	Input with a pull-down resistor
1	0	N-channel open drain output	Input with a pull-up resistor
1	1	CMOS output	High-impedance input

Setting of P40 pin		When output mode is selected (P40DIR bit = "0")	When input mode is selected (P40DIR bit = "1")
P40C1	**P40C0**	Description	
0	0	High-impedance output (initial value)	High-impedance input
0	1	P-channel open drain output	Input with a pull-down resistor
1	0	N-channel open drain output	Input with a pull-up resistor
1	1	CMOS output	High-impedance input

19.2.5 Port 4 Mode Registers 0, 1 (P4MOD0, P4MOD1)

Address: 0F224H
Access: R/W
Access size: 8/16 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
P4MOD0	P47MD0	P46MD0	P45MD0	P44MD0	P43MD0	P42MD0	P41MD0	P40MD0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

Address: 0F225H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
P4MOD1	P47MD1	P47MD1	P45MD1	P44MD1	P43MD1	P42MD1	P41MD1	P40MD1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

P4MOD0 and P4MOD1 are special function registers (SFRs) to select the primary, secondary, or tertiary function of Port 4.

[Description of Bits]

- **P47MD1, P47MD0** (bit 7)
  The P47MD1 and P47MD0 bits are used to select the primary or secondary function of the P47 pin.

P47MD1	P47MD0	Description
0	0	General-purpose input/output mode (initial value)
0	1	Resistor sensor connection pin for measurement for RC-ADC (channel 1)
1	0	Prohibited
1	1	Prohibited

- **P46MD1, P46MD0** (bit 6)
  The P46MD1 and P46MD0 bits are used to select the primary, secondary, or tertiary function of the P46 pin.

P46MD1	P46MD0	Description
0	0	General-purpose input/output mode (initial value)
0	1	Reference resistor connection pin for RC-ADC (channel 1)
1	0	SIO0 data output pin
1	1	Prohibited

- **P45MD1, P45MD0** (bit 5)
  The P45MD1 and P45MD0 bits are used to select the primary, secondary, or tertiary function of the P45 pin.

P45MD1	P45MD0	Description
0	0	General-purpose input/output mode (initial value)
0	1	Reference capacitor connection pin for RC-ADC (channel 1)
1	0	SIO0 clock input/output pin
1	1	Prohibited

- **P44MD1, P44MD0** (bit 4)
  The P44MD1 and P44MD0 bits are used to select the primary, secondary, or tertiary function of the P44 pin.

P44MD1	P44MD0	Description
0	0	General-purpose input/output mode (initial value)
0	1	RC oscillation waveform input pin for RC-AD (channel 1)
1	0	SIO0 data input pin
1	1	Prohibited

- **P43MD1, P43MD0** (bit 3)
  The P43MD1 and P43MD0 bits are used to select the primary, secondary, or tertiary, or quaternary function of the P43 pin.

P43MD1	P43MD0	Description
0	0	General-purpose input/output mode (initial value)
0	1	UART0 data output pin
1	0	PWM0 output pin
1	1	UART1 data output pin

- **P42MD1, P42MD0** (bit 2)
  The P42MD1 and P42MD0 bits are used to select the primary, secondary, or tertiary function of the P42 pin.

P42MD1	P42MD0	Description
0	0	General-purpose input/output mode (initial value)
0	1	UART0 input pin
1	0	SIO0 data output pin
1	1	Prohibited

- **P41MD1, P41MD0** (bit 1)
  The P41MD1 and P41MD0 bits are used to select the primary, secondary, or tertiary function of the P41 pin.

P41MD1	P41MD0	Description
0	0	General-purpose input/output mode (initial value)
0	1	I^2C bus 0 clock input/output pin
1	0	SIO0 clock input/output pin
1	1	Prohibited

- **P40MD1, P40MD0** (bit 0)
  The P40MD1 and P40MD0 bits are used to select the primary, secondary, or tertiary function of the P40 pin.

P40MD1	P40MD0	Description
0	0	General-purpose input/output mode (initial value)
0	1	I^2C bus 0 data input/output pin
1	0	SIO0 data input pin
1	1	Prohibited

Note:
If any bit combination out of the above is set to "Prohibited" and the corresponding bit of the port 4 is specified to output mode (selected in port4 control register), status of corresponding pin is fixed, regardless the contents of Port4 register (P4D)

High-impedance output mode: High-impedance
P-channel open drain output mode: High-impedance
N-channel open drain output mode: Fixed to "L"
CMOS output mode: High-impedance: Fixed to "L"

When using RC-ADC as the secondary function, specify each pin be "High-impedance input" even the RC oscillation monitor pin. Pull-up or Pull-down input makes drawing the current.

19.3 Description of Operation

19.3.1 Input/Output Port Functions

For each pin of Port 4, either output or input is selected by setting the Port 4 direction register (P4DIR).
In output mode, high-impedance output mode, P-channel open drain output mode, N-channel open drain output mode, or CMOS output mode can be selected by setting the Port 4 control registers 0 and 1 (P4CON0 and P4CON1).
In input mode, high-impedance input mode, input mode with a pull-down resistor, or input mode with a pull-up resistor can be selected by setting the Port 4 control registers 0 and 1 (P4CON0 and P4CON1).
At a system reset, high-impedance output mode is selected as the initial state.
In output mode, "L" or "H" level is output to each pin of Port 4 depending on the value set by the Port 4 data register (P4D).
In input mode, the input level of each pin of Port 4 can be read from the Port 4 data register (P4D).

19.3.2 Secondary and Tertiary and Quaternary Functions

Secondary and tertiary and quaternary functions are assigned to Port 4 as the I2C bus 0 pins (SDA, SCL), UART 0 pins (RXD0, TXD0), UART1 pin(TXD1), RC-ADC (channel 1) oscillation pins (IN1, CS1, RS1, RT1), synchronous serial port 0 pins (SIN0, SCK0, SOUT), and the PWM output pin (PWM0). These pins can be used in a secondary or tertiary or quaternary function mode by setting the P47MD0 to P40MD0 bits and the P47MD1 to P40MD1 bits of the Port 4 mode registers (P4MOD0, P4MOD1).

Note:
The P44 to P47 pins of port 4 are configured as pins dedicated to the RC-ADC function during A/D conversion. Therefore, if there is any unused pin, that pin cannot be used during A/D conversion. For the RC-ADC, see Chapter 23, "RC Oscillation Type A/D Converter".

Chapter 20

Port 5

20. Port 5

20.1 Overview

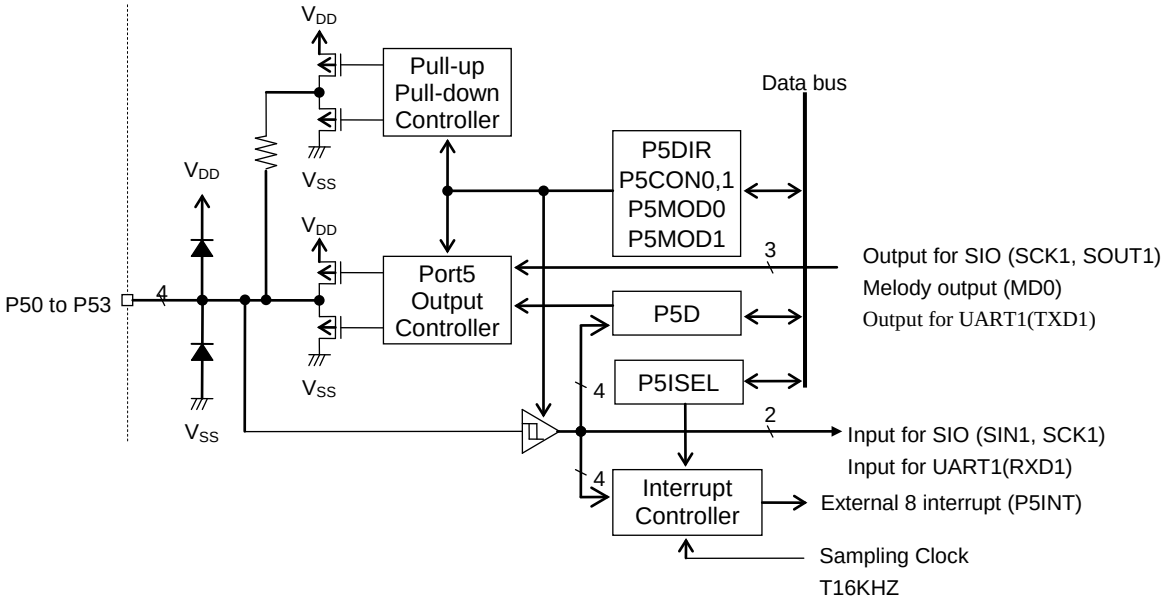
This LSI includes Port 5 (P50 to P53) which is an 4-bit input/output port.

20.1.1 Features

- Allows selection of N-channel open drain output or CMOS output for each bit in the output mode.
- Allows selection of high-impedance input or input with a pull-down/pull-up resistor for each bit in the input mode.
- Allows selection of interrupt disabled or interrupt enabled for each bit in the input mode.
- The Melody 0 (MD0) and UART1(RXD1,TXD1) are available as the secondary function.
- The synchronous serial port pins (SIN1, SCK1, SOUT1) are available as the tertiary function.

20.1.2 Configuration

Figure 20-1 shows the configuration of Port 5.



P5D : Port 5 data register
P5DIR : Port 5 direction register
P5CON0 : Port 5 control register 0
P5CON1 : Port 5 control register 1
P5MOD0 : Port 5 mode register 0
P5MOD1 : Port 5 mode register 1
P5ISEL : Port 5 interrupt mode register

Figure 20-1 Configuration of Port 5

20.1.3 List of Pins

Pin name	Input/output	Primary function	Secondary Function	Tertiary Function
P50/EXI8/MD0/SIN1	I/O	Input/output port, External 8 interrupt	Melody 0 output	SSIO1 data input
P51/EXI8/SCK1	I/O	Input/output port, External 8 interrupt	—	SSIO1 synchronous clock input/output
P52/EXI8/SOUT1 /RXD1	I/O	Input/output port, External 8 interrupt	UART1 data input	SSIO1 data output
P53/EXI8/TXD1	I/O	Input/output port, External 8 interrupt	UART1 data output	—

20.2 Description of Registers

20.2.1 List of Registers

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F228H	Port 5 data register	P5D	—	R/W	8	0FFH
0F229H	Port 5 direction register	P5DIR	—	R/W	8	00H
0F22AH	Port 5 control register 0	P5CON0	P5CON	R/W	8/16	00H
0F22BH	Port 5 control register 1	P5CON1		R/W	8	00H
0F22CH	Port 5 mode register 0	P5MOD0	P5MOD	R/W	8/16	00H
0F22DH	Port 5 mode register 1	P5MOD1		R/W	8	00H
0F22EH	Port 5 interrupt mode register	P5ISEL	—	R/W	8	00H

20.2.2 Port 5 Data Register (P5D)

Address: 0F228H
Access: R/W
Access size: 8-bit
Initial value: 0FFH

	7	6	5	4	3	2	1	0
P5D	—	—	—	—	P53D	P52D	P51D	P50D
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	1	1	1	1	1	1	1	1

P5D is a special function register (SFR) to set the value to be output to the Port 5 pin or to read the input level of the Port 5. In output mode, the value of this register is output to the Port 5 pin. The value written to P5D is readable.
In input mode, the input level of the Port 5 pin is read when P5D is read. Output mode or input mode is selected by using the port direction register (P5DIR) described later.

[Description of Bits]

- **P53D to P50D** (bit 3 to 0)
  The P53D to P50D bits are used to set the output value of the Port 5 pin in output mode and to read the pin level of the Port 5 pin in input mode.

P53D	Description
0	Output or input level of the P53 pin: "L"
1	Output or input level of the P53 pin: "H"

P52D	Description
0	Output or input level of the P52 pin: "L"
1	Output or input level of the P52 pin: "H"

P51D	Description
0	Output or input level of the P51 pin: "L"
1	Output or input level of the P51 pin: "H"

P50D	Description
0	Output or input level of the P50 pin: "L"
1	Output or input level of the P50 pin: "H"

20.2.3 Port 5 Direction Register (P5DIR)

Address: 0F229H
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
P5DIR	—	—	—	—	P53DIR	P52DIR	P51DIR	P50DIR
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

P5DIR is a special function register (SFR) to select the input/output mode of Port 5.

[Description of Bits]

- **P53DIR to P50DIR** (bit 3 to 0)
  The P53DIR to P50DIR pins are used to set the input/output direction of the Port 5 pin.

P53DIR	Description
0	P53 pin: Output (initial value)
1	P53 pin: Input

P52DIR	Description
0	P52 pin: Output (initial value)
1	P52 pin: Input

P51DIR	Description
0	P51 pin: Output (initial value)
1	P51 pin: Input

P50DIR	Description
0	P50 pin: Output (initial value)
1	P50 pin: Input

20.2.4 Port 5 Control Registers 0 and 1 (P5CON0 and P5CON1)

Address: 0F22AH
Access: R/W
Access size: 8/16 bit
Initial value: 00H

	7	6	5	4	3	2	1	0
P5CON0	—	—	—	—	P53C0	P52C0	P51C0	P50C0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

Address: 0F22BH
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
P5CON1	—	—	—	—	—	—	—	P5UD
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

P5CON0 and P5CON1 are special function registers (SFRs) to select input/output state of the Port 5 pin. The input/output state is different between input mode and output mode. Input or output is selected by using the P5DIR register.

[Description of Bits]

- **P53C0 to P50C0** (bit 3 to 0)
  P53C0 to P50C0 are the bits that select N-channel open drain output or CMOS output in the output mode, and high-impedance input or input with a pull-down or pull-up resistor in the input mode.

Setting of P53 pin	When output mode is selected (P53DIR bit = "0")	When input mode is selected (P53DIR bit = "1")
P53C0	Description	
0	N-channel open drain output (initial value)	Input with a pull-down or pull-up resistor
1	CMOS output	High-impedance input

Setting of P52 pin	When output mode is selected (P52DIR bit = "0")	When input mode is selected (P52DIR bit = "1")
P52C0	Description	
0	N-channel open drain output (initial value)	Input with a pull-down or pull-up resistor
1	CMOS output	High-impedance input

Setting of P51 pin	When output mode is selected (P51DIR bit = "0")	When input mode is selected (P51DIR bit = "1")
P51C0	Description	
0	N-channel open drain output (initial value)	Input with a pull-down or pull-up resistor
1	CMOS output	High-impedance input

Setting of P50 pin	When output mode is selected (P50DIR bit = "0")	When input mode is selected (P50DIR bit = "1")
P50C0	Description	
0	N-channel open drain output (initial value)	Input with a pull-down or pull-up resistor
1	CMOS output	High-impedance input

- **P5UD** (bit 0)
  P5UD is the bit that selects input with a pull-up resistor or input with a pull-down resistor when the input with a pull-down or pull-up resistor is selected.

Setting of each port 5 pin	When the input with a pull-down or pull-up resistor mode is selected (P5nDIR bit = "1", P5nC0 = "0") (n = 0,1,2,3)
P5UD	Description
0	Input with a pull-up resistor (initial value)
1	Input with a pull-down resistor

20.2.5 Port 5 Mode Register 0 and 1 (P5MOD0 and P5MOD1)

Address: 0F22CH
Access: R/W
Access size: 8/16-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
P5MOD0	—	—	—	—	P53MD0	P52MD0	P51MD0	P50MD0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

Address: 0F22DH
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
P5MOD1	—	—	—	—	P53MD1	P52MD1	P51MD1	P50MD1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

P5MOD0 and P5MOD1 are special function registers (SFRs) to select the primary function or the secondary function of Port 5.

[Description of Bits]

- **P53MD1, P53MD0** (bit 3)
  The P53MD1 and P53MD0 bit is used to select the primary, secondary, or the tertiary function of the P53 pin.

P53MD1	P53MD0	Description
0	0	General-purpose input/output mode (initial value)
0	1	UART1 data output
1	0	Prohibited
1	1	Prohibited

- **P52MD1, P52MD0** (bit 2)
  The P52MD1 and P52MD0 bit is used to select the primary, secondary, or the tertiary function of the P52 pin.

P52MD1	P52MD0	Description
0	0	General-purpose input/output mode (initial value)
0	1	UART1 data input
1	0	SSIO1 data output
1	1	Prohibited

- **P51MD1,P51MD0** (bit 1)
  The P51MD1 and P51MD0 bit is used to select the primary, secondary, or the tertiary function of the P51 pin.

P51MD1	P51MD0	Description
0	0	General-purpose input/output mode (initial value)
0	1	Prohibited
1	0	SSIO1 synchronous clock input/output
1	1	Prohibited

- **P50MD1, P50MD0** (bit 0)
  The P50MD1 and P50MD0 bit is used to select the primary, secondary, or the tertiary function of the P50 pin.

P50MD1	P50MD0	Description
0	0	General-purpose input/output mode (initial value)
0	1	Melody 0 output
1	0	SSIO1 data input
1	1	Prohibited

Note:
When the pin is set to "Prohibited" and the output mode is selected (by the Port 5 control register), the Port 5 output pin state is fixed as follows regardless of the data of the port data register P5D:

When N-channel open drain output is selected: Output pin is fixed to "L"
When CMOS output is selected: Output pin is fixed to "L"

20.2.6 Port 5 Interrupt Mode Register (P5ISEL)

Address: 0F22EH
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
P5ISEL	—	—	—	—	P53IS	P52IS	P51IS	P50IS
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

P5ISEL is a special function register (SFR) to select the pin used for the Port 5 external interrupt (P5INT).

[Description of Bits]

- **P53IS to P50IS** (bit 3 to 0)
  P53IS to P50IS are the bits that select to disable or enable the interrupt of each pin on the Port 5.

P53IS	Description
0	P53 external interrupt disabled (initial value)
1	P53 external interrupt enabled

P52IS	Description
0	P52 external interrupt disabled (initial value)
1	P52 external interrupt enabled

P51IS	Description
0	P51 external interrupt disabled (initial value)
1	P51 external interrupt enabled

P50IS	Description
0	P50 external interrupt disabled (initial value)
1	P50 external interrupt enabled

20.3 Description of Operation

20.3.1 Input/Output Port Functions

For each pin of Port 5, either output or input is selected by setting the Port 5 direction register (P5DIR).
In the output mode, set the Port 5 control register 0 (P5CON0) to select either N-channel open drain output mode or CMOS output mode.
In the input mode, set the Port 5 control registers 0 and 1 (P5CON0 and P5CON1) to select any of high-impedance input mode, input mode with a pull-down resistor, or input mode with a pull-up resistor.
At the system reset, N-channel open drain output mode is selected as the initial state.
In output mode, the "L" or "H" level is output to each pin of Port 5 depending on the value set by the Port 5 data register (P5D).
In input mode, the input level of each pin of Port 5 is read from the Port 5 data register (P5D).

20.3.2 Secondary and Tertiary Functions

The Port 5 is assigned with the pins for UART1 (TXD1,RXD1), Melody 0 (MD0) output as the secondary function and the pins for Synchronous serial port 1 (SIN1, SCK1, SOUT1) as the tertiary function. Each of them can be used as the secondary or tertiary function by setting the P53MD1 to P50MD1 and P53MD0 to P50MD0 bits of the port mode register 1(P5MOD1) and 0 (P5MOD0).

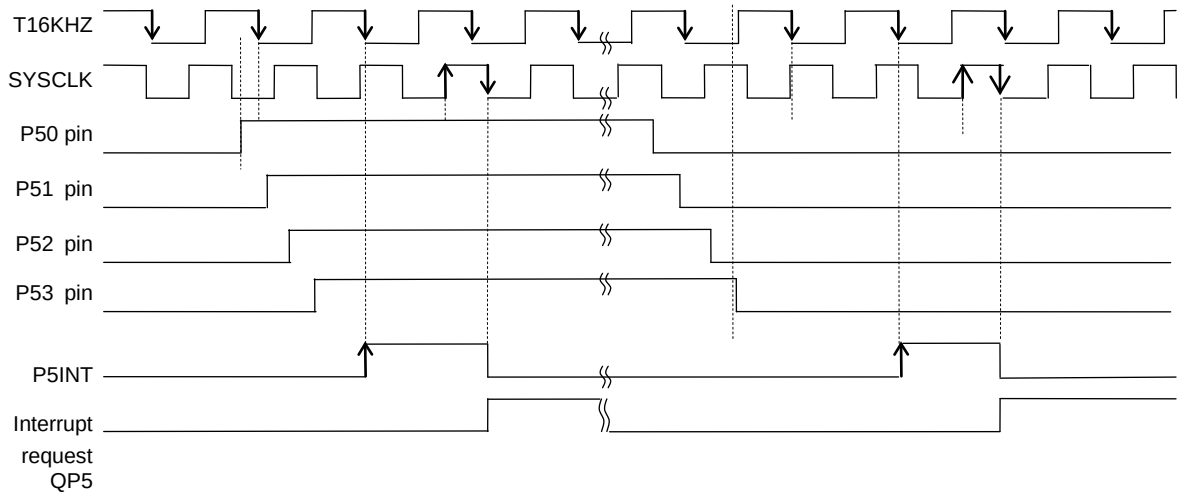
20.3.3 External Interrupt

Each of the Port 5 pins (P50 to P53) can be used as the external 8 interrupt (P5INT). Each interrupt is maskable and selectable to be disabled or enabled. For details of interrupts, see Chapter 5, “Interrupts”.

20.3.4 Interrupt Request

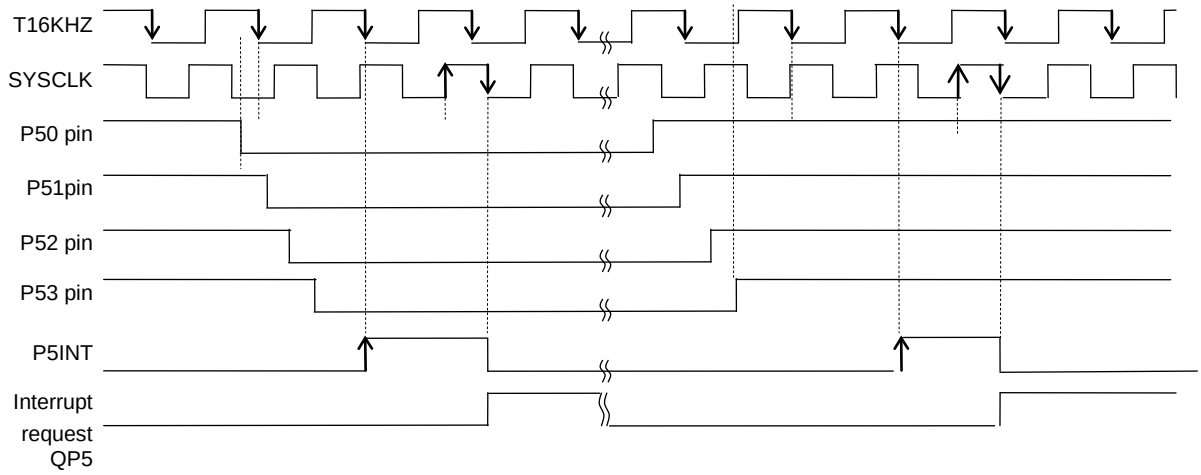
The maskable external 8 interrupt (P5INT) occurs when each of the Port 5 pins has an interrupt edge.
Figure 20-2 shows the external 8 interrupt (P5INT) generation timing in case P5UD bit of P5CON1 register is “1”.
Figure 20-3 shows the external 8 interrupt (P5INT) generation timing in case P5UD bit of P5CON1 register is “0”.

Note:
- The external 8 interrupt (P5INT) is fixed to both-edge interrupt with sampling.
In STOP mode, since the 16 kHz sampling clock stops, no sampling is performed.
- Depending on the P5n pin state, the External 8 Interrupt Request flag (IRQ2's bit 3) may be set to "1" when the P5ISEL setting is changed. Therefore, change the P5ISEL setting when the Master Interrupt Enable (MIE) flag is "0", then reset the External 8 Interrupt Request flag to "0" by software before setting the MIE to "1". For interrupts, see Chapter 5, "Interrupts."



With sampling, both-edge interrupt

Figure 20-2 External 8 Interrupt Generation Timing (P5UD = 1)



With sampling, both-edge interrupt

Figure 20-3 External 8 Interrupt Generation Timing (P5UD = 0)

Chapter 21

Port 6

21. Port 6

21.1 Overview

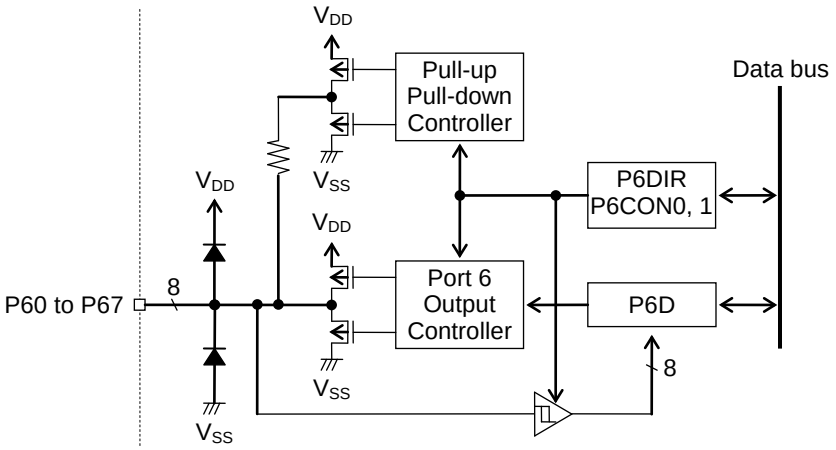
This LSI (ML610Q418C) includes Port 6 (P60 to P67) which is an 8-bit input/output port.
This function is not included in the ML610Q418.

21.1.1 Features

- Allows selection of high-impedance output, P-channel open drain output, N-channel open drain output, or CMOS output for each bit in output mode.
- Allows selection of high-impedance input, input with a pull-down resistor, or input with a pull-up resistor for each bit in input mode.

21.1.2 Configuration

Figure 21-1 shows the configuration of Port 6.



P6D : Port 6 data register
P6DIR : Port 6 direction register
P6CON0 : Port 6 control register 0
P6CON1 : Port 6 control register 1

Figure 21-1 Configuration of Port 6

21.1.3 List of Pins

Pin name	I/O	Description
P60	I/O	Input/output port
P61	I/O	Input/output port
P62	I/O	Input/output port
P63	I/O	Input/output port
P64	I/O	Input/output port
P65	I/O	Input/output port
P66	I/O	Input/output port
P67	I/O	Input/output port

21.2 Description of Registers

21.2.1 List of Registers

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F230H	Port 6 data register	P6D	—	R/W	8	00H
0F231H	Port 6 direction register	P6DIR	—	R/W	8	00H
0F232H	Port 6 control register 0	P6CON0	P6CON	R/W	8/16	00H
0F233H	Port 6 control register 1	P6CON1		R/W	8	00H

21.2.2 Port 6 Data Register (P6D)

Address: 0F230H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
P6D	P67D	P66D	P65D	P64D	P63D	P62D	P61D	P60D
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

P6D is a special function register (SFR) to set the value to be output to the Port 6 pin or to read the input level of the Port 6. In output mode, the value of this register is output to the Port 6 pin. The value written to P6D is readable. In input mode, the input level of the Port 6 pin is read when P6D is read. Output mode or input mode is selected by using the port direction register (P6DIR) described later.

[Description of Bits]

- **P67D-P60D** (bits 7-0)
  The P67D to P60D bits are used to set the output value of the Port 6 pin in output mode and to read the pin level of the Port 6 pin in input mode.

P67D	Description
0	Output or input level of the P67 pin: "L"
1	Output or input level of the P67 pin: "H"

P66D	Description
0	Output or input level of the P66 pin: "L"
1	Output or input level of the P66 pin: "H"

P65D	Description
0	Output or input level of the P65 pin: "L"
1	Output or input level of the P65 pin: "H"

P64D	Description
0	Output or input level of the P64 pin: "L"
1	Output or input level of the P64 pin: "H"

P63D	Description
0	Output or input level of the P63 pin: "L"
1	Output or input level of the P63 pin: "H"

P62D	Description
0	Output or input level of the P62 pin: "L"
1	Output or input level of the P62 pin: "H"

P61D	Description
0	Output or input level of the P61 pin: "L"
1	Output or input level of the P61 pin: "H"

P60D	Description
0	Output or input level of the P60 pin: "L"
1	Output or input level of the P60 pin: "H"

21.2.3 Port 6 Direction Register (P6DIR)

Address: 0F231H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
P6DIR	P67DIR	P66DIR	P65DIR	P64DIR	P63DIR	P62DIR	P61DIR	P60DIR
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

P6DIR is a special function register (SFR) to select the input/output mode of Port 6.

[Description of Bits]

- **P67DIR-P60DIR** (bits 7-0)
  The P67DIR to P60DIR pins are used to set the input/output direction of the Port 6 pin.

P67DIR	Description
0	P67 pin: Output (initial value)
1	P67 pin: Input

P66DIR	Description
0	P66 pin: Output (initial value)
1	P66 pin: Input

P65DIR	Description
0	P65 pin: Output (initial value)
1	P65 pin: Input

P64DIR	Description
0	P64 pin: Output (initial value)
1	P64 pin: Input

P63DIR	Description
0	P63 pin: Output (initial value)
1	P63 pin: Input

P62DIR	Description
0	P62 pin: Output (initial value)
1	P62 pin: Input

P61DIR	Description
0	P61 pin: Output (initial value)
1	P61 pin: Input

P60DIR	Description
0	P60 pin: Output (initial value)
1	P60 pin: Input

21.2.4 Port 6 Control Registers 0, 1 (P6CON0, P6CON1)

Address: 0F232H
Access: R/W
Access size: 8/16 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
P6CON0	P67C0	P66C0	P65C0	P64C0	P63C0	P62C0	P61C0	P60C0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

Address: 0F233H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
P6CON1	P67C1	P66C1	P65C1	P64C1	P63C1	P62C1	P61C1	P60C1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

P6CON0 and P6CON1 are special function registers (SFRs) to select input/output state of the Port 6 pin. The input/output state is different between input mode and output mode. Input or output is selected by using the P6DIR register.

[Description of Bits]

- **P67C1-P60C1, P67C0-P60C0** (bits 7-0)
  The P67C1 to P60C1 pins and the P67C0 to P60C0 pins are used to select high-impedance output, P-channel open drain output, N-channel open drain output, or CMOS output in output mode and to select high-impedance input, input with a pull-down resistor, or input with a pull-up resistor in input mode.

Setting of P67 pin		When output mode is selected (P67DIR bit = "0")	When input mode is selected (P67DIR bit = "1")
P67C1	**P67C0**	Description	
0	0	High-impedance output (initial value)	High-impedance input
0	1	P-channel open drain output	Input with a pull-down resistor
1	0	N-channel open drain output	Input with a pull-up resistor
1	1	CMOS output	High-impedance input

Setting of P66 pin		When output mode is selected (P66DIR bit = "0")	When input mode is selected (P66DIR bit = "1")
P66C1	**P66C0**	Description	
0	0	High-impedance output (initial value)	High-impedance input
0	1	P-channel open drain output	Input with a pull-down resistor
1	0	N-channel open drain output	Input with a pull-up resistor
1	1	CMOS output	High-impedance input

Setting of P65 pin		When output mode is selected (P65DIR bit = "0")	When input mode is selected (P65DIR bit = "1")
P65C1	**P65C0**	Description	
0	0	High-impedance output (initial value)	High-impedance input
0	1	P-channel open drain output	Input with a pull-down resistor
1	0	N-channel open drain output	Input with a pull-up resistor
1	1	CMOS output	High-impedance input

Setting of P64 pin		When output mode is selected (P64DIR bit = "0")	When input mode is selected (P64DIR bit = "1")
P64C1	**P64C0**	Description	
0	0	High-impedance output (initial value)	High-impedance input
0	1	P-channel open drain output	Input with a pull-down resistor
1	0	N-channel open drain output	Input with a pull-up resistor
1	1	CMOS output	High-impedance input

Setting of P63 pin		When output mode is selected (P63DIR bit = "0")	When input mode is selected (P63DIR bit = "1")
P63C1	**P63C0**	Description	
0	0	High-impedance output (initial value)	High-impedance input
0	1	P-channel open drain output	Input with a pull-down resistor
1	0	N-channel open drain output	Input with a pull-up resistor
1	1	CMOS output	High-impedance input

Setting of P62 pin		When output mode is selected (P62DIR bit = "0")	When input mode is selected (P62DIR bit = "1")
P62C1	**P62C0**	Description	
0	0	High-impedance output (initial value)	High-impedance input
0	1	P-channel open drain output	Input with a pull-down resistor
1	0	N-channel open drain output	Input with a pull-up resistor
1	1	CMOS output	High-impedance input

Setting of P61 pin		When output mode is selected (P61DIR bit = "0")	When input mode is selected (P61DIR bit = "1")
P61C1	**P61C0**	Description	
0	0	High-impedance output (initial value)	High-impedance input
0	1	P-channel open drain output	Input with a pull-down resistor
1	0	N-channel open drain output	Input with a pull-up resistor
1	1	CMOS output	High-impedance input

Setting of P60 pin		When output mode is selected (P60DIR bit = "0")	When input mode is selected (P60DIR bit = "1")
P60C1	**P60C0**	Description	
0	0	High-impedance output (initial value)	High-impedance input
0	1	P-channel open drain output	Input with a pull-down resistor
1	0	N-channel open drain output	Input with a pull-up resistor
1	1	CMOS output	High-impedance input

21.3 Description of Operation

21.3.1 Input/Output Port Functions

For each pin of Port 6, either output or input is selected by setting the Port 6 direction register (P6DIR).
In output mode, high-impedance output mode, P-channel open drain output mode, N-channel open drain output mode, or CMOS output mode can be selected by setting the Port 6 control registers 0 and 1 (P6CON0 and P6CON1).
In input mode, high-impedance input mode, input mode with a pull-down resistor, or input mode with a pull-up resistor can be selected by setting the Port 6 control registers 0 and 1 (P6CON0 and P6CON1).
At system reset, high-impedance output mode is selected as the initial state.
In output mode, "L" or "H" level is output to each pin of Port 6 depending on the value set by the Port 6 data register (P6D).
In input mode, the input level of each pin of Port 6 can be read from the Port 6 data register (P6D).

Chapter 22

Melody Driver

22. Melody Driver

22.1 Overview

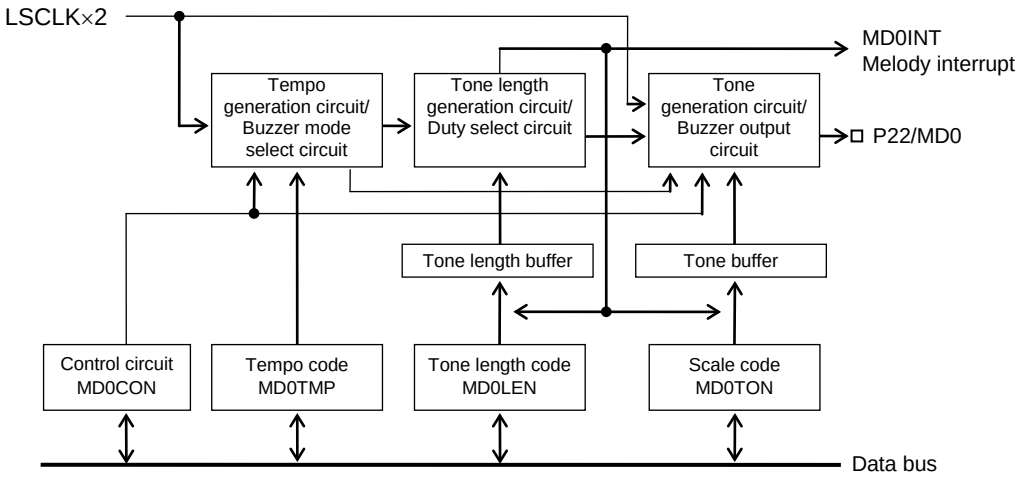
This LSI includes one channel of the melody driver.
To use the melody driver, the secondary function of port 2 should be set. For the secondary function of port 2, see Chapter 17, "Port 2". For the clock to be used in this melody driver, see Chapter 6, "Clock Generation Circuit".

22.1.1 Features

- In melody output mode, 29 scales (melody audio frequency: 508Hz to 32.768kHz), 63 tone lengths, and 15 tempos) are available.
- In buzzer output mode, 4 output modes, 8 frequencies, and 15 duties can be set.

22.1.2 Configuration

Figure 22-1 shows the configuration of the melody driver.



MD0CON : Melody 0 control register
MD0TMP : Melody 0 tempo code register
MD0TON : Melody 0 scale code register
MD0LEN : Melody 0 tone length code register

Figure 22-1 Configuration of Melody Driver

22.1.3 List of Pins

Pin name	I/O	Description
P22/MD0	O	Melody 0 signal output pin Used as the secondary of the P22 pin.

22.2 Description of Registers

22.2.1 List of Registers

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F2C0H	Melody 0 control register	MD0CON	—	R/W	8	00H
0F2C1H	Melody 0 tempo code register	MD0TMP	—	R/W	8	00H
0F2C2H	Melody 0 scale code register	MD0TON	MD0TL	R/W	8/16	00H
0F2C3H	Melody 0 tone length code register	MD0LEN		R/W	8	00H

22.2.2 Melody 0 Control Register (MD0CON)

Address: 0F2C0H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
MD0CON	—	—	—	—	—	—	BZMD	M0RUN
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

MD0CON is a special function register (SFR) to control a melody and the buzzer.

[Description of Bits]

- **BZMD** (bit 1)
  The BZMD bit is used to select melody mode or buzzer mode.

BZMD	Description
0	Melody mode (initial value)
1	Buzzer mode

- **M0RUN** (bit 0)
  The M0RUN bit is used to control start/stop of the MD0 output.

M0RUN	Description
0	Stops MD0 output. (Initial value)
1	Starts MD0 output.

Note:
For melody output, use the 2×low-speed clock (LSCLK×2).
Enable the 2×low-speed clock by setting bit 2 (ENMLT) of frequency control register 1 (FCON1) to "1" and then start melody output by setting M0RUN to "1".

22.2.3 Melody 0 Tempo Code Register (MD0TMP)

Address: 0F2C1H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
MD0TMP	—	—	—	—	M0TM3	M0TM2	M0TM1	M0TM0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

MD0TMP is a special function register (SFR) to set the tempo code of a melody when melody mode is selected and the output mode of a buzzer sound waveform when buzzer mode is selected.

[Description of Bits]

- **M0TM3, M0TM2, M0TM1, M0TM0** (bits 3-0)

When melody mode is selected (BZMD bit = "0")				
M0TM3	**M0TM2**	**M0TM1**	**M0TM0**	Description
0	0	0	0	♪ = 480 (initial value)
0	0	0	1	♪ = 480
0	0	1	0	♪ = 320
0	0	1	1	♪ = 240
0	1	0	0	♪ = 192
0	1	0	1	♪ = 160
0	1	1	0	♪ ≅ 137
0	1	1	1	♪ = 120
1	0	0	0	♪ ≅ 107
1	0	0	1	♪ = 96
1	0	1	0	♪ ≅ 87
1	0	1	1	♪ = 80
1	1	0	0	♪ ≅ 74
1	1	0	1	♪ ≅ 69
1	1	1	0	♪ = 64
1	1	1	1	♪ = 60

When buzzer mode is selected (BZMD bit = "1")				
M0TM3	**M0TM2**	**M0TM1**	**M0TM0**	Description
*	*	0	0	Intermittent 1 output (initial value)
*	*	0	1	Intermittent 2 output
*	*	1	0	Single sound output
*	*	1	1	Continuous sound output

22.2.4 Melody 0 Scale Code Register (MD0TON)

Address: 0F2C2H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
MD0TON	—	M0TN6	M0TN5	M0TN4	M0TN3	M0TN2	M0TN1	M0TN0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

MD0TON is a special function register (SFR) to set the scale code of a melody when melody mode is selected and a buzzer output frequency when buzzer mode is selected.

[Description of Bits]

- **M0TN6, M0TN5, M0TN4, M0TN3, M0TN2, M0TN1, M0TN0** (bits 6-0)

When melody mode is selected (BZMD bit = "0")	
M0TN6 to 0	Description
	Sets the corresponding scale code.

For scale codes, see Section 22.3.4, "Scale Codes".

When buzzer mode is selected (BZMD bit = "1")				
M0TN6 to 3	**M0TN2**	**M0TN1**	**M0TN0**	Description
*	0	0	0	4.096 kHz (initial value)
*	0	0	1	2.048 kHz
*	0	1	0	1.365 kHz
*	0	1	1	1.024 kHz
*	1	0	0	819 Hz
*	1	0	1	683 Hz
*	1	1	0	585 Hz
*	1	1	1	512 Hz

Note: In buzzer mode, the M0TN6 to M0TN3 bits are not used.(Don't care)

22.2.5 Melody 0 Tone Length Code Register (MD0LEN)

Address: 0F2C3H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
MD0LEN	—	—	M0LN5	M0LN4	M0LN3	M0LN2	M0LN1	M0LN0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

MD0LEN is a special function register (SFR) to set the tone length code of a melody when melody mode is selected and buzzer output duty when buzzer mode is selected.

[Description of Bits]

- **M0LN5, M0LN4, M0LN3, M0LN2, M0LN1, M0LN0** (bits 5-0)

When melody mode is selected (BZMD bit = "0")	
M0LN5 to 0	Description
	Sets the corresponding tone length code.

For tone length codes, see Section 22.3.3, "Tone Length Codes".

When buzzer mode is selected (BZMD bit = "1")					
M0LN5 to 4	**M0LN3**	**M0LN2**	**M0LN1**	**M0LN0**	Description
*	0	0	0	0	1/16 DUTY (initial value)
*	0	0	0	1	1/16 DUTY
*	0	0	1	0	2/16 DUTY
*	0	0	1	1	3/16 DUTY
*	0	1	0	0	4/16 DUTY
*	0	1	0	1	5/16 DUTY
*	0	1	1	0	6/16 DUTY
*	0	1	1	1	7/16 DUTY
*	1	0	0	0	8/16 DUTY
*	1	0	0	1	9/16 DUTY
*	1	0	1	0	10/16 DUTY
*	1	0	1	1	11/16 DUTY
*	1	1	0	0	12/16 DUTY
*	1	1	0	1	13/16 DUTY
*	1	1	1	0	14/16 DUTY
*	1	1	1	1	15/16 DUTY

Note: In buzzer mode, the M0LN5 to M0LN4 bits are not used.(Don't care)

22.3 Description of Operation

22.3.1 Operation of Melody Output

Melody is output in the following procedure.

(1) Select melody mode by setting the BZMD bit of the melody 0 control register (MD0CON) to "0".
(2) Set a melody tempo in the melody 0 tempo code register (MD0TMP).
(3) Set a tone length code in the melody 0 tone length code register (MD0LEN).
(4) Set a scale code in the melody 0 scale code register (MD0TON).
(5) Set bit 2 (ENMLT) of the frequency control register 1(FCON1) to "1" to enable the 2×low-speed clock.
(6) When the M0RUN bit of the melody 0 control register (MD0CON) is set to "1", the tone length code and scale code are transferred to the tone length buffer and scale buffer and melody output is started from the MD0 pin. At the same time, a melody 0 interrupt (MD0INT) is requested. When an interrupt occurs and program is passed to the interrupt routine, the interrupt request flag is cleared.

The melody 0 signal output pin (MD0) is assigned as the secondary function of Port 2. See Chapter 19, "Port 2," for the secondary function settings of Port 2.
In the software processing after melody 0 interrupt, the tone length code and the scale code of the note that are output next are set to MD0LEN and MD0TON, respectively. When there is no next note to be output, rest data "00H" is set in MD0TON, the M0RUN bit is set to "0" by the software processing after the next melody 0 interrupt, and melody output is terminated.
By setting the M0RUN bit to "0", melody can be terminated forcibly during melody output.
In the case the melody is forcibly terminated, meldy circuit must be reset once by setting DMD0 bit of BLKCON3 register to "1". Then if you want continue to use the melody function, reset the DMD0 bit to "0" and restart operating from the aforementioned procedure (1).

Figure 22-2 shows the operation waveform of the melody driver.

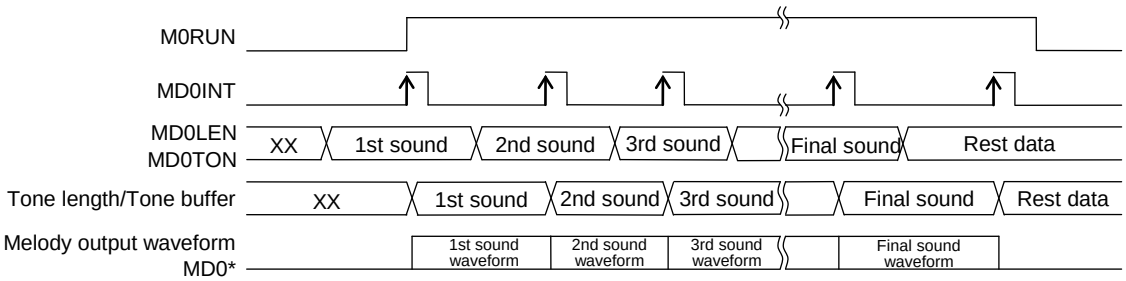


Figure 22-2 Operation Waveform of Melody Driver

22.3.2 Tempo Codes

A tempo code is set in the melody 0 tempo code register (MD0TEM).
Table 22-1 shows the correspondence between tempos and tempo codes.
The tempo when all the bits are set to "0" is equal to the shortest tone length (the tempo when the only M0TP0 bit is set to "1").

Table 22-1 Correspondence between Tempos and Tempo Codes

Tempo	Tempo code (MD0TMP)				
	M0TP3	M0TP2	M0TP1	M0TP0	M0TP3 to 0
♩ = 480	0	0	0	0	0H
♩ = 480	0	0	0	1	1H
♩ = 320	0	0	1	0	2H
♩ = 240	0	0	1	1	3H
♩ = 192	0	1	0	0	4H
♩ = 160	0	1	0	1	5H
♩ ≅ 137	0	1	1	0	6H
♩ = 120	0	1	1	1	7H
♩ ≅ 107	1	0	0	0	8H
♩ = 96	1	0	0	1	9H
♩ ≅ 87	1	0	1	0	AH
♩ = 80	1	0	1	1	BH
♩ ≅ 74	1	1	0	0	CH
♩ ≅ 69	1	1	0	1	DH
♩ = 64	1	1	1	0	EH
♩ = 60	1	1	1	1	FH

22.3.3 Tone Length Codes

A tone length code is set in the melody 0 tone length code register (MD0LEN).
Table 22-2 shows the correspondence between tone lengths and tone length codes.
The tone length when all the bits are set to "0" is equal to the shortest tone length (the tone length when the only M0LN0 bit is set to "1").

Table 22-2 Correspondence between Tone Lengths and Tone Length Codes

Tone length	Tone length code (MD0LEN)						
	M0LN5	M0LN4	M0LN3	M0LN2	M0LN1	M0LN1	M0LN5–0
	1	1	1	1	1	1	3FH
	1	0	1	1	1	1	2FH
	0	1	1	1	1	1	1FH
	0	1	0	1	1	1	17H
	0	0	1	1	1	1	0FH
	0	0	1	0	1	1	0BH
	0	0	0	1	1	1	07H
	0	0	0	1	0	1	05H
	0	0	0	0	1	1	03H
	0	0	0	0	1	0	02H
	0	0	0	0	0	1	01H

The tone length set by a tone length code and a tempo code is expressed by the following equation.

$$\text{Tone length} = 1.953125 \times (TP + 1) \times (LN + 1) \text{ ms}$$

where TP is an integer of 1 to 15, and LN is an integer of 1 to 63.

The bit correspondence between TP and tempo codes is expressed by the following equation.

$$TP = 2^3\text{M0TP3} + 2^2\text{M0TP2} + 2^1\text{M0TP1} + 2^0\text{M0TP0}$$

The bit correspondence between LN and tone length codes is expressed by the following equation.

$$LN = 2^5\text{M0LN5} + 2^4\text{M0LN4} + 2^3\text{M0LN3} + 2^2\text{M0LN2} + 2^1\text{M0LN1} + 2^0\text{M0LN0}$$

22.3.4 Scale Codes

A scale code is set in the melody 0 scale code register (MD0TON).
In the melody driver, a frequency that can be output is expressed by the following equation.

$$\frac{65536}{(TN + 1)} \text{ Hz} \qquad \text{(where TN is an integer of 4 to 127.)}$$

The bit correspondence between TN and scale codes is expressed by the following equation.

$$TN = 2^6M0TN6 + 2^5M0TN5 + 2^4M0TN4 + 2^3M0TN3 + 2^2M0TN2 + 2^1M0TN1 + 2^0M0TN0$$

Table 22-3 shows the correspondence between scales and scale codes.
When the M0TN6 to M0TN2 bits are set to "0", scale becomes a rest. The rest length is set by the tone length code (MD0LEN).

Table 22-3 Correspondence between Scales and Scale Codes

Scale	Frequency (Hz)	Scale code (MD0TON)							
		M0TN6	M0TN5	M0TN4	M0TN3	M0TN2	M0TN1	M0TN0	M0TN6–0
C^1	529	1	1	1	1	0	1	1	7BH
Cis^1	560	1	1	1	0	1	0	0	74H
D^1	590	1	1	0	1	1	1	0	6EH
Dis^1	624	1	1	0	1	0	0	0	68H
E^1	662	1	1	0	0	0	1	0	62H
F^1	705	1	0	1	1	1	0	0	5CH
Fis^1	745	1	0	1	0	1	1	1	57H
G^1	790	1	0	1	0	0	1	0	52H
Gis^1	840	1	0	0	1	1	0	1	4DH
A^1	886	1	0	0	1	0	0	1	49H
Ais^1	936	1	0	0	0	1	0	1	45H
B^1	993	1	0	0	0	0	0	1	41H
C^2	1057	0	1	1	1	1	0	1	3DH
Cis^2	1111	0	1	1	1	0	1	0	3AH
D^2	1192	0	1	1	0	1	1	0	36H
Dis^2	1260	0	1	1	0	0	1	1	33H
E^2	1338	0	1	1	0	0	0	0	30H
F^2	1394	0	1	0	1	1	1	0	2EH
Fis^2	1490	0	1	0	1	0	1	1	2BH
G^2	1560	0	1	0	1	0	0	1	29H
Gis^2	1680	0	1	0	0	1	1	0	26H
A^2	1771	0	1	0	0	1	0	0	24H
Ais^2	1872	0	1	0	0	0	1	0	22H
B^2	1986	0	1	0	0	0	0	0	20H
C^3	2114	0	0	1	1	1	1	0	1EH
D^3	2341	0	0	1	1	0	1	1	1BH
Dis^3	2521	0	0	1	1	0	0	1	19H
E^3	2621	0	0	1	1	0	0	0	18H
Fis^3	2979	0	0	1	0	1	0	1	15H

22.3.5 Example of Using Melody Circuit

Figure 22-3 shows an example of a melody notation, and Table 22-4 shows note codes of melody examples.

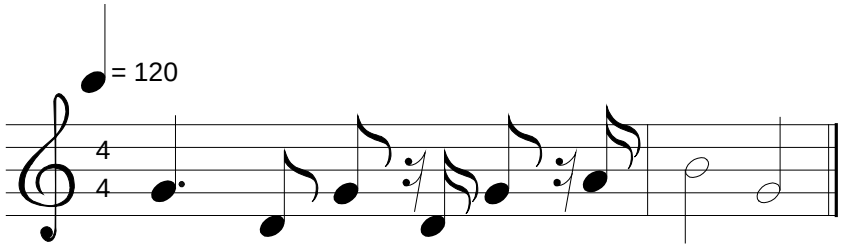


Figure 22-3 Example of Melody Notation

Table 22-4 Note Codes of Melody Examples

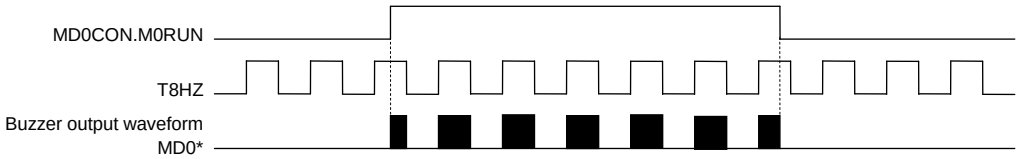
Note	Note code													
	MD0LEN						MD0TON							hexadecimal
	5	4	3	2	1	0	6	5	4	3	2	1	0	
G^2	1	0	1	1	1	1	0	1	0	1	0	0	0	2F28H
D^2	0	0	1	1	1	1	0	1	1	0	1	0	1	0F35H
G^2	0	0	1	1	1	1	0	1	0	1	0	0	0	0F28H
—	0	0	0	1	1	1	0	0	0	0	0	0	0	0700H
D^2	0	0	0	1	1	1	0	1	1	0	1	0	1	0735H
G^2	0	0	1	1	1	1	0	1	0	1	0	0	0	0F28H
—	0	0	0	1	1	1	0	0	0	0	0	0	0	0700H
A^2	0	0	0	1	1	1	0	1	0	0	0	1	1	0723H
B^2	1	1	1	1	1	1	0	0	1	1	1	1	1	3F1FH
G^2	1	1	1	1	1	1	0	1	0	1	0	0	0	3F28H

22.3.6 Operations of Buzzer Output

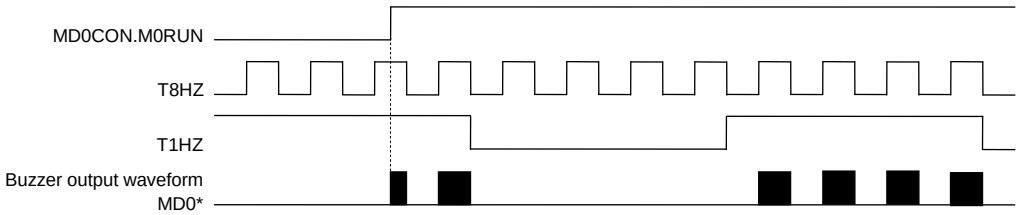
A buzzer sound is output in the following procedure.

(1) Select a buzzer mode by setting the BZMD bit of the melody 0 control register (MD0CON) to "1".
(2) Select a buzzer output mode using the melody 0 tempo code register (MD0TMP).
(3) Select a duty of the High level width of the buzzer output waveform using the melody 0 tone length code register (MD0LEN).
(4) Set the buzzer output frequency in the melody 0 scale code register (MD0TON).
(5) Set bit 2 (ENMLT) of the frequency control register 1(FCON1) to "1" to enable the 2×low-speed clock.
(6) When the M0RUN bit of the melody 0 control register (MD0CON) is set to "1", the waveform equivalent to the buzzer sound that is set from the MD0 pin is output.

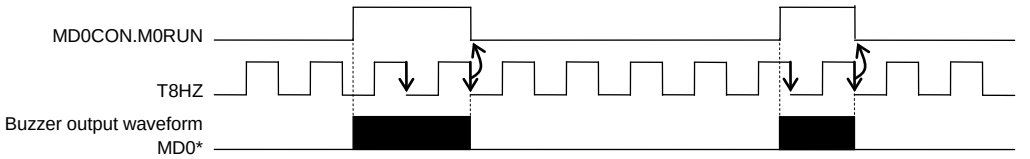
Figure 22-4 shows the output waveform of each buzzer output mode.



(1/4) Output waveform of intermittent sound 1



(2/4) Output waveform of intermittent sound 2



(3/4) Output waveform of single sound



(4/4) Output waveform of continuous sound

Figure 22-4 Output Waveform of Each Buzzer Output Mode

22.4 Specifying port registers

When you want to make sure the Melody/Buzzer function is working, please check related port registers are specified. See Chapter 19, "Port 2" for detail about the port registers.

22.4.1 Functioning P22 (MD0) as the Melody or Buzzer output

Set P22MD bit (bit2 of P2MOD register) to "1" for specifying the melody or buzzer output as the secondary function of P22.

Reg. name	P2MOD register (Address: 0F214H)							
Bit	7	6	5	4	3	**2**	1	0
Bit name	-	-	-	-	-	**P22MD**	P21MD	P20MD
Data	-	-	-	-	-	**1**	*	*

Set P22C1 bit (bit2 of P2CON1 register) to "1" and set P22C0 bit(bit2 of P2CON0 register) to "1", for specifying the P22 as CMOS output.

Reg. name	P2CON1 register (Address: 0F213H)							
Bit	7	6	5	4	3	**2**	1	0
Bit name	-	-	-	-	-	**P22C1**	P21C1	P20C1
Data	-	-	-	-	-	**1**	*	*

Reg. name	P2CON0 register (Address: 0F212H)							
Bit	7	6	5	4	3	**2**	1	0
Bit name	-	-	-	-	-	**P22C0**	P21C0	P20C0
Data	-	-	-	-	-	**1**	*	*

Data of P22D bit (bit2 of P2D register) does not affect to the melody or buzzer function, so don't care the data for the function.

Reg. name	P2D register (Address: 0F210H)							
Bit	7	6	5	4	3	**2**	1	0
Bit name	-	-	-	-	-	**P22D**	P21D	P20D
Data	-	-	-	-	-	******	*	*

- : Bit does not exist.
* : Bit not related to the melody or buzzer function
** : Don't care the data.

Note:
- Port2 is an output-only port, does not have an register to select the data direction(input or output).
- The output characteristics of port22 corresponds to VOL1 and VOH1 when P22MD bit is "1" (melody/buzzer is selected as the 2nd function), and corresponds to VOL2 and VOH2 when the P22MD bit is "0", which are shown in Appendix C, "Electrical Characteristics".

Chapter 23

RC Oscillation Type A/D Converter

23. RC Oscillation Type A/D Converter

23.1 Overview

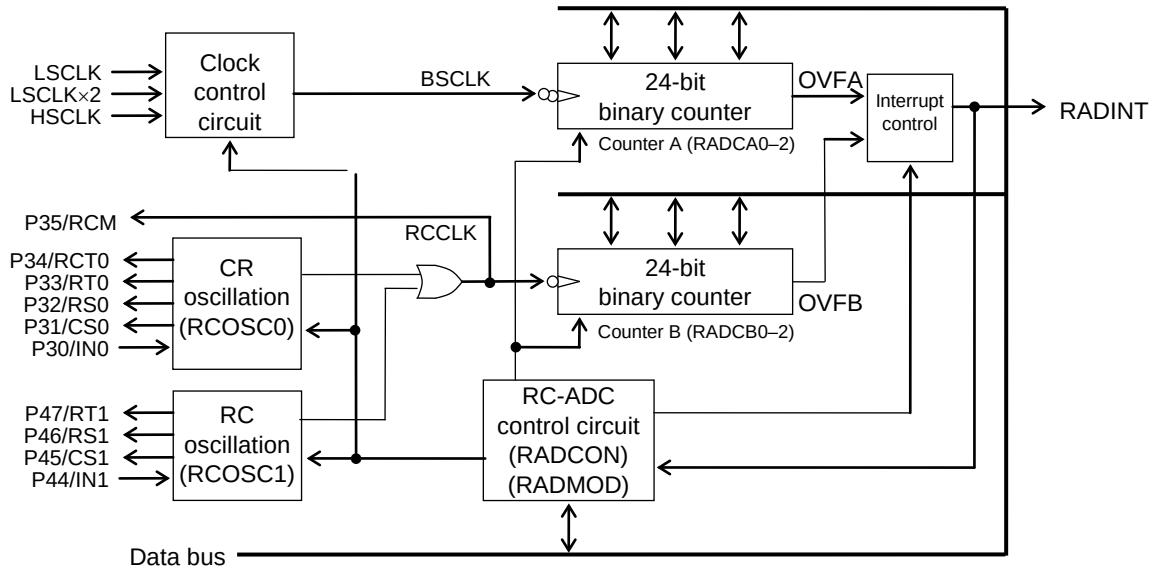
This LSI has a built-in 2-channel RC oscillation type A/D converter (RC-ADC).
The RC-ADC converts resistance values or capacitance values to digital values by counting the oscillator clock whose frequency changes according to the resistor or capacitor connected to the RC oscillator circuits. By using a thermistor or humidity sensor as a resistor, a thermometer or hygrometer can be formed.
In addition, a different sensor for each of the two channels of RC-ADC's RC oscillator circuit can be used to broaden RC-ADC applications; for example, the converter can be used for expansion of measurement range or measurement at two points.
For input clock, see Chapter 6, "Clock Generation Circuit."

23.1.1 Features

- 2-channel system by time division

23.1.2 Configuration

The RC-ADC consists of two RC oscillator circuits to form two channels, Counter A (RADCA0–2) and Counter B (RADCB0–2) as 24-bit binary counters, and an RC-ADC control circuit (RADCON, RADMOD). Figure 23-1 shows the configuration of the RC-ADC.



RADMOD: RC-ADC mode register
RADCON: RC-ADC control register
RADCA0–2: RC-ADC Counter A registers 0–2
RADCB0–2: RC-ADC Counter B registers 0–2

Figure 23-1 Configuration of RC-ADC

23.1.3 List of Pins

Pin name	I/O	Description
P30/IN0	I	Channel 0 oscillation input pin. Used for the secondary function of the P30 pin.
P31/CS0	O	Channel 0 reference capacitor connection pin. Used for the secondary function of the P31 pin.
P32/RS0	O	Channel 0 reference resistor connection pin. Used for the secondary function of the P32 pin.
P33/RT0	O	Pin for connection with a resistive sensor for measurement on Channel 0. Used for the secondary function of the P33 pin.
P34/RCT0	O	Pin for connection with a resistive/capacitive sensor for measurement on Channel 0. Used for the secondary function of the P34 pin.
P35/RCM	O	RC oscillation monitor pin. Used for the secondary function of the P35 pin.
P44/IN1	I	Channel 1 oscillation input pin. Used for the secondary function of the P44 pin.
P45/CS1	O	Channel 1 reference capacitor connection pin. Used for the secondary function of the P45 pin.
P46/RS1	O	Channel 1 reference resistor connection pin. Used for the secondary function of the P46 pin.
P47/RT1	O	Pin for connection with a resistive sensor for measurement on Channel 1. Used for the secondary function of the P47 pin.

23.2 Description of Registers

23.2.1 List of Registers

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F300H	RC-ADC Counter A register 0	RADCA0	—	R/W	8	00H
0F301H	RC-ADC Counter A register 1	RADCA1	—	R/W	8	00H
0F302H	RC-ADC Counter A register 2	RADCA2	—	R/W	8	00H
0F304H	RC-ADC Counter B register 0	RADCB0	—	R/W	8	00H
0F305H	RC-ADC Counter B register 1	RADCB1	—	R/W	8	00H
0F306H	RC-ADC Counter B register 2	RADCB2	—	R/W	8	00H
0F308H	RC-ADC mode register	RADMOD	—	R/W	8	00H
0F309H	RC-ADC control register	RADCON	—	R/W	8	00H

23.2.2 RC-ADC Counter A Registers (RADCA0–2)

Address: 0F300H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
RADCA0	RAA7	RAA6	RAA5	RAA4	RAA3	RAA2	RAA1	RAA0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

Address: 0F301H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
RADCA1	RAA15	RAA14	RAA13	RAA12	RAA11	RAA10	RAA9	RAA8
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

Address: 0F302H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
RADCA2	RAA23	RAA22	RAA21	RAA20	RAA19	RAA18	RAA17	RAA16
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

RADCA0–2, which serve as a 24-bit binary counter (Counter A), are special function registers (SFRs) used to perform read/write operations to Counter A itself.

Note:
After writing data into the RC-ADC counter A register, be sure to read it to check that the data has been written correctly.
When A/D conversion starts after data is written, the value that has been written is read during A/D conversion (RARUN = 1).
When A/D conversion terminates (RARUN = 0), the count value is read.

23.2.3 RC-ADC Counter B Registers (RADCB0–2)

Address:0F304H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
RADCB0	RAB7	RAB6	RAB5	RAB4	RAB3	RAB2	RAB1	RAB0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

Address:0F305H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
RADCB1	RAB15	RAB14	RAB13	RAB12	RAB11	RAB10	RAB9	RAB8
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

Address:0F306H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
RADCB2	RAB23	RAB22	RAB21	RAB20	RAB19	RAB18	RAB17	RAB16
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

RADCB0–2, which serve as a 24-bit binary counter (Counter B), are special function registers (SFRs) used to perform read/write operations to Counter B itself.

Note:
After writing data into the RC-ADC counter B register, be sure to read it to check that the data has been written correctly.
When A/D conversion starts after data is written, the value that has been written is read during A/D conversion (RARUN = 1).
When A/D conversion terminates (RARUN = 0), the count value is read.

23.2.4 RC-ADC Mode Register (RADMOD)

Address: 0F308H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
RADMOD	RACK2	RACK1	RACK0	RADI	OM3	OM2	OM1	OM0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

RADMOD is a special function register (SFR) used to select an A/D conversion mode of the RC-ADC.

[Description of Bits]

- **OM3-0** (bits 3-0)
  The OM3–0 bits are used to select an oscillation mode for the RC oscillator circuits.

OM3	OM2	OM1	OM0	Description
0	0	0	0	IN0 pin external clock input mode (initial value)
0	0	0	1	RS0–CS0 oscillation mode
0	0	1	0	RT0–CS0 oscillation mode
0	0	1	1	$RT_{0\text{-}1}$–CS0 oscillation mode
0	1	0	0	RS0–CT0 oscillation mode
0	1	0	1	RS1–CS1 oscillation mode
0	1	1	0	RT1–CS1 oscillation mode
0	1	1	1	IN1 pin external clock input mode
1	*	*	*	Setting prohibited

- **RADI** (bit 4)
  The RADI bit is used to choose whether to generate the RC-ADC interrupt request signal (RADINT) by an overflow at Counter A or Counter B.

RADI	Description
0	Generates an interrupt request by Counter A overflow (initial value).
1	Generates an interrupt request by Counter B overflow.

- **RACK2-0** (bits 7-5)
  The RACK2–0 bits are used to select the base clock of Counter A (BSCLK).

RACK2	RACK1	RACK0	Description
0	0	0	LSCLK (initial value)
0	0	1	LSCLK×2
0	1	0	HSCLK
0	1	1	1/2HSCLK
1	0	0	1/4HSCLK
1	0	1	1/8HSCLK
1	1	*	Setting prohibited (no clock is supplied)

Note:
When specifying LSCLK×2 for the base clock, enable the operation of the 2×low-speed clock by setting bit 2 (ENMLT) of the frequency control register 1 (FCON1) to "1".

23.2.5 RC-ADC Control Register (RADCON)

Address: 0F309H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
RADCON	—	—	—	—	—	—	—	RARUN
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

RADCON is a special function register (SFR) used to control A/D conversion operation of the RC-ADC.

[Description of Bits]

- **RARUN** (bit 0)
  The RARUN bit is used to start A/D conversion of the RC-ADC. When RARUN is set to "1", A/D conversion starts. If Counter A or Counter B overflows with RARUN set to "1", the bit is automatically reset to "0".
  RARUN is set to "0" at system reset.

RARUN	Description
0	Stops A/D conversion (initial value).
1	Starts A/D conversion.

Note:
When A/D conversion is stopped by resetting the RARUN to "0", the RC-ADC does not perform properly without the following procedures.
1) Set the "DRAD" bit of the Block Control Register 4 (BLKCON4) to "1" , in order to disable the RC-ADC.
2) Reset the "DRAD" bit of the Block Control Register 4 (BLKCON4) to "0" , in order to enable the RC-ADC.
3) Set up the RC-ADC again by following the required procedures, then restart.

23.3 Description of Operation

Counter A (RADCA0–2) is a 24-bit binary counter for counting the base clock (BSCLK), which is used as the standard of time. Counter A can count up to 0FFFFFFH.
Counter B (RADCB0–2) is a 24-bit binary counter for counting the oscillator clock (RCCLK) of the RC oscillator circuits. Counter B can count up to 0FFFFFFH.
Counters A and B are provided with overflow flags (OVFA and OVFB, respectively). Each overflow output results in generation of an RC-ADC interrupt request signal (RADINT). Use the RADI bit of the RC-AC mode register (RADMOD) to select whether to generate an overflow interrupt by an overflow on Counter A or Counter B: setting RADI to "0" specifies Counter A overflow and setting it to "1" specifies Counter B overflow.

The RARUN bit of the RC-AD control register (RADCON) is used to start or stop RC-ADC conversion operation. When RARUN is set to "0", the oscillator circuits stop, so that counting will not be performed. When RARUN is set to "1", RC oscillation starts, when the RC oscillator clock (RCCLK) and the base clock (BSCLK) start counting through Counter B and Counter A.

The RC oscillation section has a total of eight types of oscillation modes based on the two oscillator circuits of RCOSC0 and RCOSC1, and mode selection is made by the RC-ADC mode register (RADMOD).
P30–34, P44–47, and P35 must be configured as their secondary function input or output when using 1) the RC oscillator circuit RCOSC0, 2) the RC oscillator circuit RCOSC1, and 3) the RC monitor pin (RCM) that outputs RC oscillation waveforms, respectively. For the configuration of the RC oscillator circuits, see Section 23.1.2, "Configuration"; for the secondary functions of Port 3, see Chapter 20, "Port 3"; for the secondary functions of Port 4, see Chapter 21, "Port 4".

23.3.1 RC Oscillator Circuits

RC-ADC performs A/D conversion by converting the oscillation frequency ratio between a reference resistor (or capacitor) and a resistive sensor (or capacitive sensor) such as a thermistor to digital data.
By making RC oscillation occur both on the reference side and on the sensor side with the reference capacitor the error factor that the RS oscillator circuit itself is eliminated, thereby making it possible to perform the A/D conversion of the characteristics of the sensor itself.
Also, by calculating the ratio between the oscillation frequency on the reference side and that on the sensor side and then calculating the correlation between the calculated ratio and temperatures that the sensor characteristics have in advance, a temperature can be obtained based on that calculated ratio.
Table 23-1 lists the eight types of oscillation modes, one of which is selected by the RC-ADC mode register (RADMOD) OM3–0 bits.

Table 23-1 Oscillation Modes from Which Selection Is Made by OM3–0 Bits

Mode No.	RADMOD				RCOSC0 output pin				RCOSC1 output pin			Mode	
	OM3	OM2	OM1	OM0	RS0	RT0	CRT0	CS0	RS1	RT1	CS1		
0	0	0	0	0	Z	Z	Z	Z	Z	Z	Z	IN0 external clock input mode	
1	0	0	0	1	1/0	Z	Z	0/1	Z	Z	Z	RS0–CS0 oscillation	RCOSC0 oscillation mode
2	0	0	1	0	Z	1/0	Z	0/1	Z	Z	Z	RT0–CS0 oscillation	
3	0	0	1	1	Z	Z	1/0	0/1	Z	Z	Z	$RT_{0\text{-}1}$–CS0 oscillation	
4	0	1	0	0	1/0	Z	0/1	Z	Z	Z	Z	RS0–CT0 oscillation	
5	0	1	0	1	Z	Z	Z	Z	1/0	Z	0/1	RS1–CS1 oscillation	RCOSC1 oscillation mode
6	0	1	1	0	Z	Z	Z	Z	Z	1/0	0/1	RT1–CS1 oscillation	
7	0	1	1	1	Z	Z	Z	Z	Z	Z	Z	IN1 external clock input mode	
8	1	*	*	*	Z	Z	Z	Z	Z	Z	Z	(Setting prohibited)	

Note) * : Indicates "arbitrary."
Z : Indicates high-impedance output.
1/0, 0/1 : Indicates active output.
(Setting prohibited) : The oscillator clock is not supplied even by setting the RARUN bit to "1" or by starting A/D conversion.

In Table 23-1, mode No.0 and mode No.7 are modes where external clocks to be input to the IN0 or IN1 pin are used for measurement with the RC oscillator circuit stopped.
As shown in Table 23-1, the two oscillator circuits, RCOSC0 and RCOSC1, are so specified that they cannot operate concurrently in order to prevent interference in oscillation from occurring when they oscillate concurrently.
The relationship between an oscillation frequency f_{RCCLK} and an RC constant is expressed by the following equation:

$$\frac{1}{f_{RCCLK}} = t_{RCCLK} = k_{RCCLK} \cdot R \cdot C$$

where t_{RCCLK} is the period of the oscillator clock, k_{RCCLK} the proportional constant, and R·C the product of capacitances CS, CT, (CS+CVR) and (CT+CVR) and resistances RS and RT. CS, CT, (CS+CVR), (CT+CVR), RS, and RT concern oscillation. The value of k_{RCCLK} slightly changes depending on the value of the supply voltage VDD, RI, R, or C.
Table 23-2 lists the typical k_{RCCLK} values.

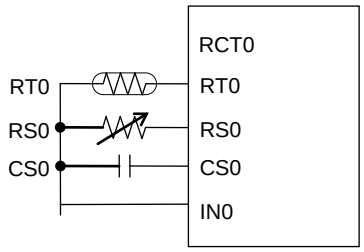
Table 23-2 Typical Values of the Proportional Constant k_{RCCLK} of RC Oscillator Circuits

V_{DD} (V)	CSn, CTn (pF)	CVRn(pF)	RSn, RTn (kΩ)	k_{RCCLK} (Typ.)
3	560	820	100	1.2
	560	820	10	1.2
1.5	560	820	100	1.2
	560	820	10	1.3

Note) n = 0, 1

Notes:
- Out of the Port 3 and Port 4 pins, pins that are to be used for the RC-ADC function must be configured as secondary function input or output using the mode register (P3MOD0, P3MOD1, P4MOD0, P4MOD1) of the corresponding port.
- All the Port 3 pins except P35/RCM (see Section 23.1.3, "List of Pins") are configured as pins dedicated to the RC-ADC function during A/D conversion. Therefore, during A/D conversion, all the Port 3 pins except P35 cannot be used as their primary functions in oscillation mode No. 0, 1, 2, 3 or 4, which is selected by the RADMOD register. In the same way, the P44 to P47 pins of Port 4 cannot be used as their primary functions in oscillation mode No. 5, 6 or 7.

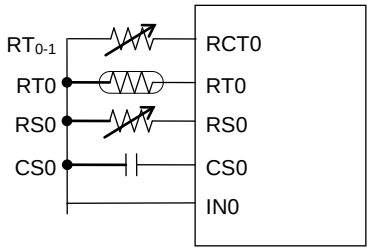
Figures 23-2 to 23-5 show the oscillator circuit configurations, the modes of oscillation for each configuration, and the OM3–0 bit settings.



OM3	OM2	OM1	OM0	Mode of oscillation
0	0	0	1	Oscillates with the reference resistor RS0 and CS0
0	0	1	0	Oscillates with the sensor RT0 and CS0

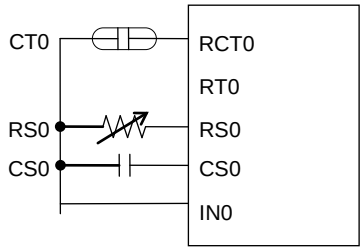
Figure 23-2 When RCOSC0 Is Used for Measurement with One Resistive Sensor

Note:
The unused pin RCT0 shown in the figure above is configured as a pin dedicated to the RC-ADC function during A/D conversion; therefore, during A/D conversion, RCT0 cannot be used as a port (P34).



OM3	OM2	OM1	OM0	Mode of oscillation
0	0	0	1	Oscillates with the reference resistor RS0 and CS0
0	0	1	0	Oscillates with the sensor RT0 and CS0
0	0	1	1	Oscillates with the reference resistor RT_{0-1} and CS0

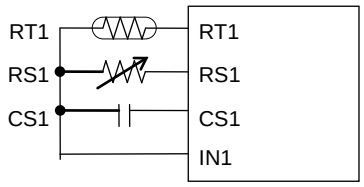
Figure 23-3 When RCOSC0 Is Used for Measurement with One Resistive Sensor (Two points are adjusted with two reference resistors)



OM3	OM2	OM1	OM0	Mode of oscillation
0	0	0	1	Oscillates with the reference resistor RS0 and CS0
0	1	0	0	Oscillates with the sensor RS0 and CT0

Figure 23-4 When RCOSC0 Is Used for Measurement with One Capacitive Sensor

Note:
The unused pin RT0 shown in the figure above is configured as a pin dedicated to the RC-ADC function during A/D conversion; therefore, during A/D conversion, RT0 cannot be used as a port (P33).



OM3	OM2	OM1	OM0	Mode of oscillation
0	1	0	1	Oscillates with the reference resistor RS1 and CS1
0	1	1	0	Oscillates with the sensor RT1 and CS1

Figure 23-5 When RCOSC1 Is Used for Measurement with One Resistive Sensor

23.3.2 Counter A/Counter B Reference Modes

There are the following two modes of RC-ADC conversion operation:

- Counter A reference mode (RADMOD RADI = "0")
  In this mode, a gate time is determined by Counter A and the base clock (BSCLK), which is used as the time reference, then the RC oscillator clock (RCCLK) is counted by Counter B within the gate time to make the content of Counter B an A/D conversion value.
  The A/D conversion value is proportional to RC oscillation frequency.

- Counter B reference mode (RADMOD RADI = "1")
  In this mode, a gate time is determined by Counter B and the RC oscillator clock (RCCLK), and the base clock (BSCLK), which is used as the time reference, is counted by Counter A within the gate time to make the content of Counter A an A/D conversion value.
  The /D conversion value is inversely proportional to RC oscillation frequency.

(1) Operation in Counter A reference mode

Figure 23-6 shows the operation timing in Counter A reference mode.
Following is an example of operation procedure in Counter A reference mode:

① Preset to Counter A (RADCA2–0) the value obtained by subtracting the count value "nA0" from the maximum value + 1 (1000000H). The product of the count value "nA0" and the BSCLK clock period indicates the gate time.
② Preset "000000H" to Counter B (RADCB2–0).
③ Set the OM3–OM0 bits of RADMOD to desired oscillation mode (see Table 23-1).
④ Set the RADI bit of RADMOD to "0" to specify generating of an interrupt request signal by Counter A overflow.
⑤ Set the RARUN bit of RADCON to "1" to start A/D conversion.

Counter A starts counting of the base clock (BSCLK) when RARUN is set to "1" and the RCON signal (signal synchronized with the fall of the base clock) is set to "1". When Counter A overflows, the RARUN bit is automatically reset to "0" (⑥) and counting is terminated. At the same time, an RC-ADC interrupt request (RADIN) occurs (⑦).

When the RCON signal is set to "1", the RC oscillator circuit starts operation and Counter B starts counting of the RC oscillator clock (RCCLK). When the RARUN bit is reset to "0" due to overflow of Counter A, RC oscillation stops and Counter B stops counting.
The final count value "nB0" of Counter B is the RCCLK count value during the gate time "$nA0 \cdot t_{BSCLK}$" and is expressed by the following expression:

$$nB0 \quad \cong \quad nA0 \cdot \frac{t_{BSCLK}}{t_{RCCLK}} \quad \propto \quad f_{RCCLK}$$

where t_{BSCLK} indicates the BSCLK period and t_{RCCLK} the RCCLK period. That is, "nB0" is a value proportional to the RC oscillation frequency f_{RCCLK}.

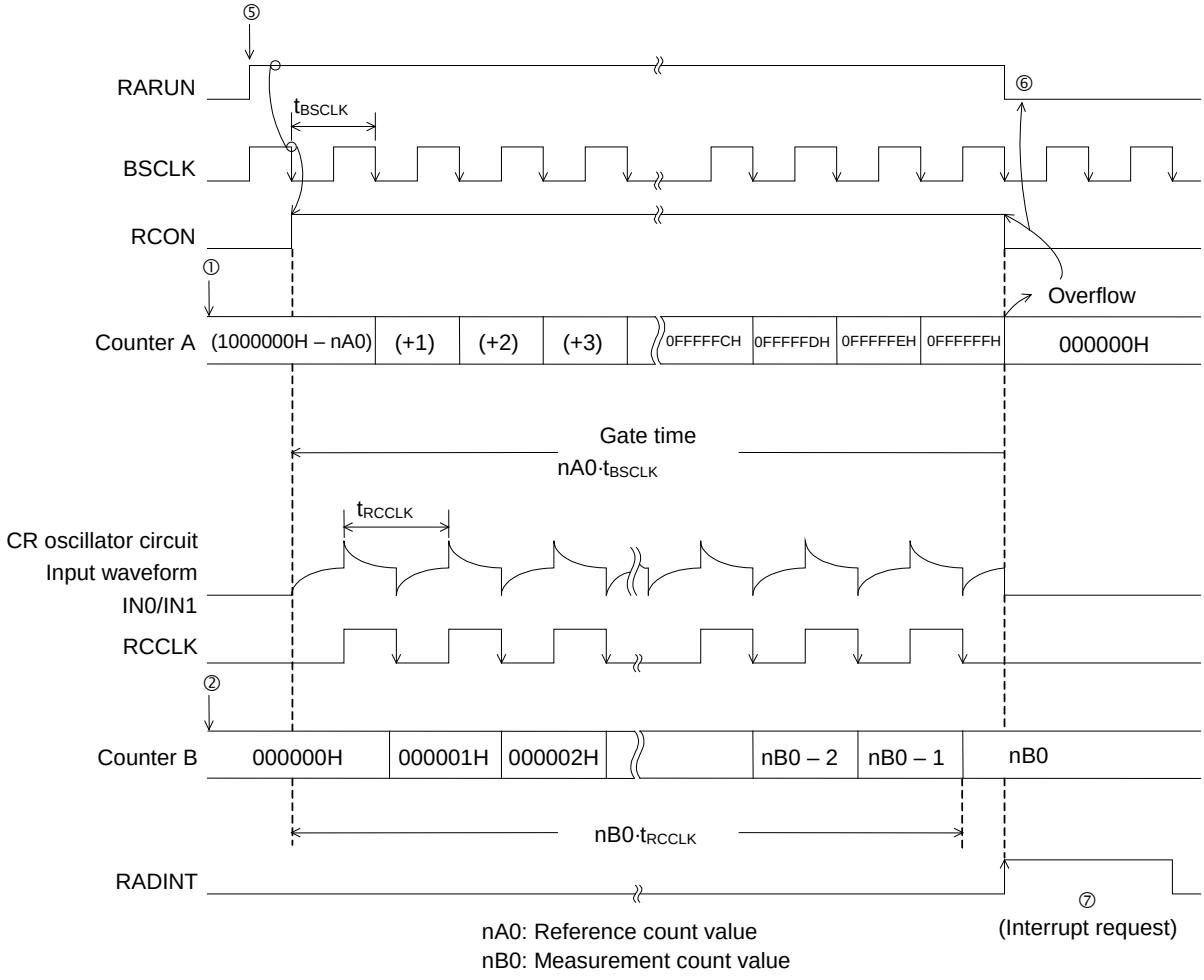


Figure 23-6 Operation Timing in Counter A Reference Mode

(2) Operation in Counter B reference mode

Figure 23-7 shows the operation timing in Counter B reference mode.
Following is an example of operation procedure in Counter B reference mode:

① Preset to Counter B (RADCB2–0) the value obtained by subtracting the count value "nB1" from the maximum value + 1 (1000000H). The product of the count value "nB1" and the RCCLKclock period indicates the gate time.
② Preset "000000H" to Counter A (RADCA2–0).
③ Set the OM3–OM0 bits of RADMOD to desired oscillation mode (see Table 23-1).
④ Set the RADI bit of RADMOD to "1" to specify generating of an interrupt request signal by Counter B overflow.
⑤ Set the RARUN bit of RADCON to "1" to start A/D conversion.

When the RARUN bit is set to "1" and the RCON signal (signal synchronized with the fall of the base clock) is set to"1", the RC oscillator circuit starts operation and Counter B starts counting of the RC oscillator clock (RCCLK). When Counter B overflows, the RARUN bit is automatically reset (⑥) and conversion operation terminates. At the same time, an RC-ADC interrupt request (RADINT) occurs. (⑦)

When the RCON signal is set to "1", Counter A starts counting of the base clock (BSCLK). When the RARUN bit is reset due to overflow of Counter B, Counter A stops counting.
The final count "nA1" of Counter A is the CLK count value during the gate time "$nB1 \cdot t_{RCCLK}$" and is expressed by the following expression:

$$nA1 \quad \cong \quad nB1 \cdot \quad \frac{t_{RCCLK}}{t_{BSCLK}} \quad \propto \quad \frac{1}{f_{RCCLK}}$$

That is, "nA1" is a value inversely proportional to the RC oscillation frequency f_{RCCLK}.

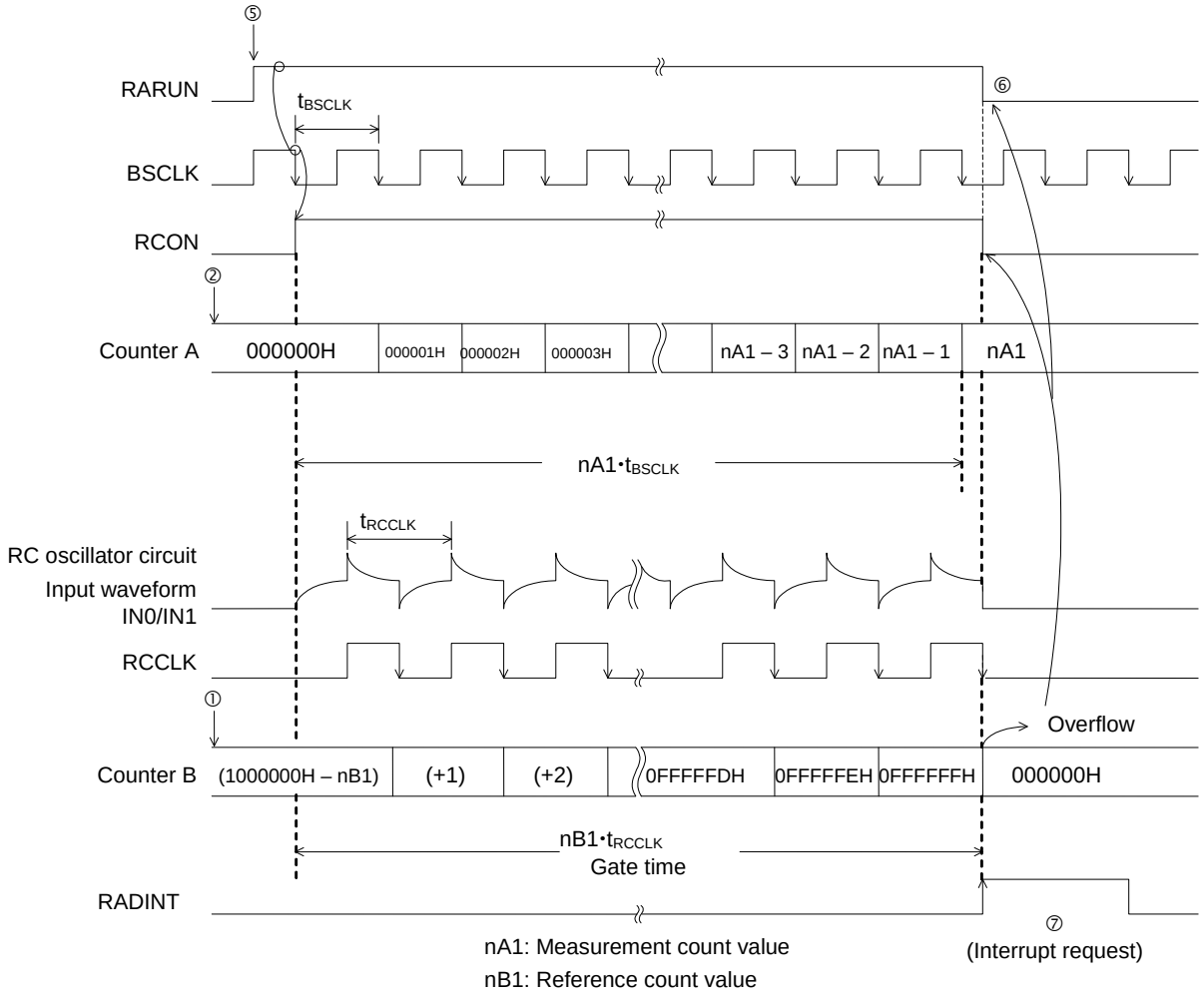


Figure 23-7 Operation Timing in Counter B Reference Mode

23.3.3 Example of Use of RC Oscillation Type A/D Converter

This section describes the method of performing A/D conversion for sensor values in Counter A and B reference modes by taking temperature measurement by a thermistor as an example.
Figure 23-8 shows the configuration of 1-thermistor RC oscillator circuit using RCOSC0.

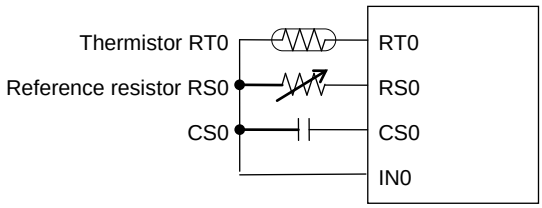


Figure 23-8 Configuration of 1-Thermistor RC Oscillator Circuit Using RCOSC0

Figure 23-9 shows the temperature characteristics of the thermistor resistance RT0.

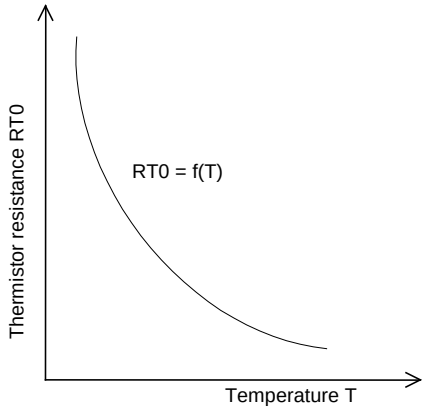


Figure 23-9 Temperature Characteristics of Thermistor

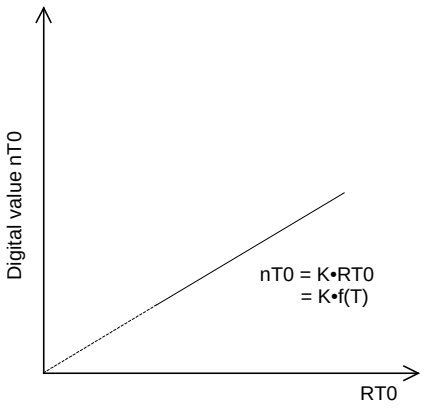


Figure 23-10 A/D Conversion Characteristics (Ideal characteristics when nT0 is proportional to RT0)

RT0 is expressed as a function of temperature T by the following equation:

$$RT0 = f(T)$$

Figure 23-10 shows the ideal characteristics of A/D conversion with the assumption that RT0 is an analog quantity. In the ideal characteristics, the A/D conversion value nT0 will purely depend on RT0 only. Assuming that nT0 is proportional to RT0, let proportional constant be K, then nT0 has the following relationship with temperature T:

$$nT0 = K \cdot RT0 = K \cdot f(T) \quad \ldots\ldots \text{Expression A}$$

Therefore, temperature T can be expressed as a digital value by performing the conversion processing that accords with the characteristics shown in Figure 23-9 for nT0 by software.

To convert from an RT0 value to a digital value, the ratio is used between a) the oscillation frequency by the thermistor connected to the RT0 pin and the capacitor connected to the CS0 pin and b) the oscillation frequency by the reference resistor (which ideally should have no temperature characteristics) connected to the RS0 pin and the capacitor connected to the CS0 pin. This is for making the conditions other than resistance equal to eliminate the error factor in oscillation characteristics.
As shown in Figures 23-9 and 23-11, the RT0 value depends on temperature T and the RS0 value is assumed to be constant regardless of temperature T. It is ideal if the characteristics of the oscillation frequency fOSC to temperature T using these resistances will be like the solid lines in Figures 23-12 and 23-13; however, in reality, it would appear that they will be like the dotted lines due to error factors such as IC temperature characteristics.
Since the condition of f_{RCCLK} (RT0) and that of f_{RCCLK} (RS0) are the same except for the resistances, the error ratios are almost the same; therefore, errors can almost be eliminated by using the ratio between f_{RCCLK} (RT0) and f_{RCCLK} (RS0).
The ratio between f_{RCCLK} (RT0) and f_{RCCLK} (RS0) is equivalent to the above-mentioned A/D conversion value nT0 that should ideally depend only on RT0.

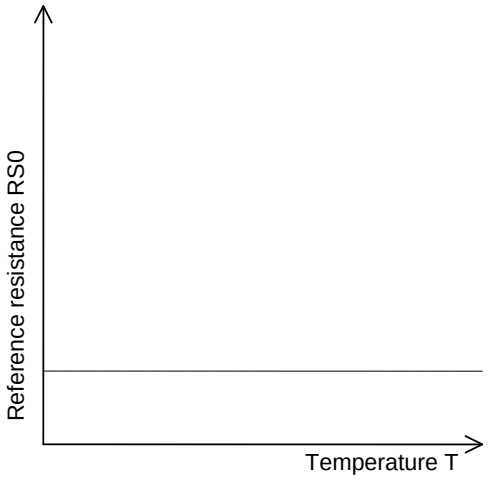


Figure 23-11 Temperature Characteristics of Reference Resistor

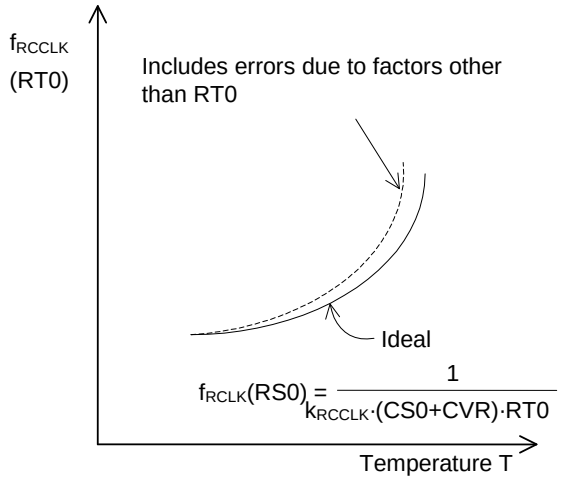


Figure 23-12 Oscillation Characteristics of Thermistor

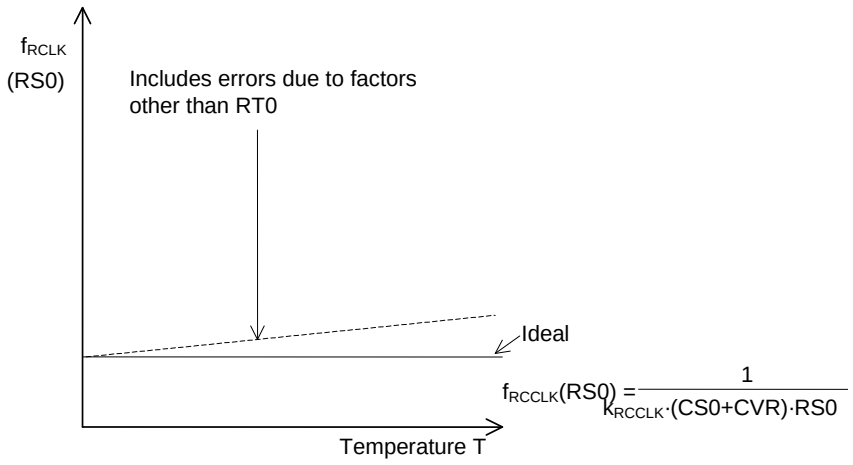


Figure 23-13 Oscillation Characteristics of Reference Resistor

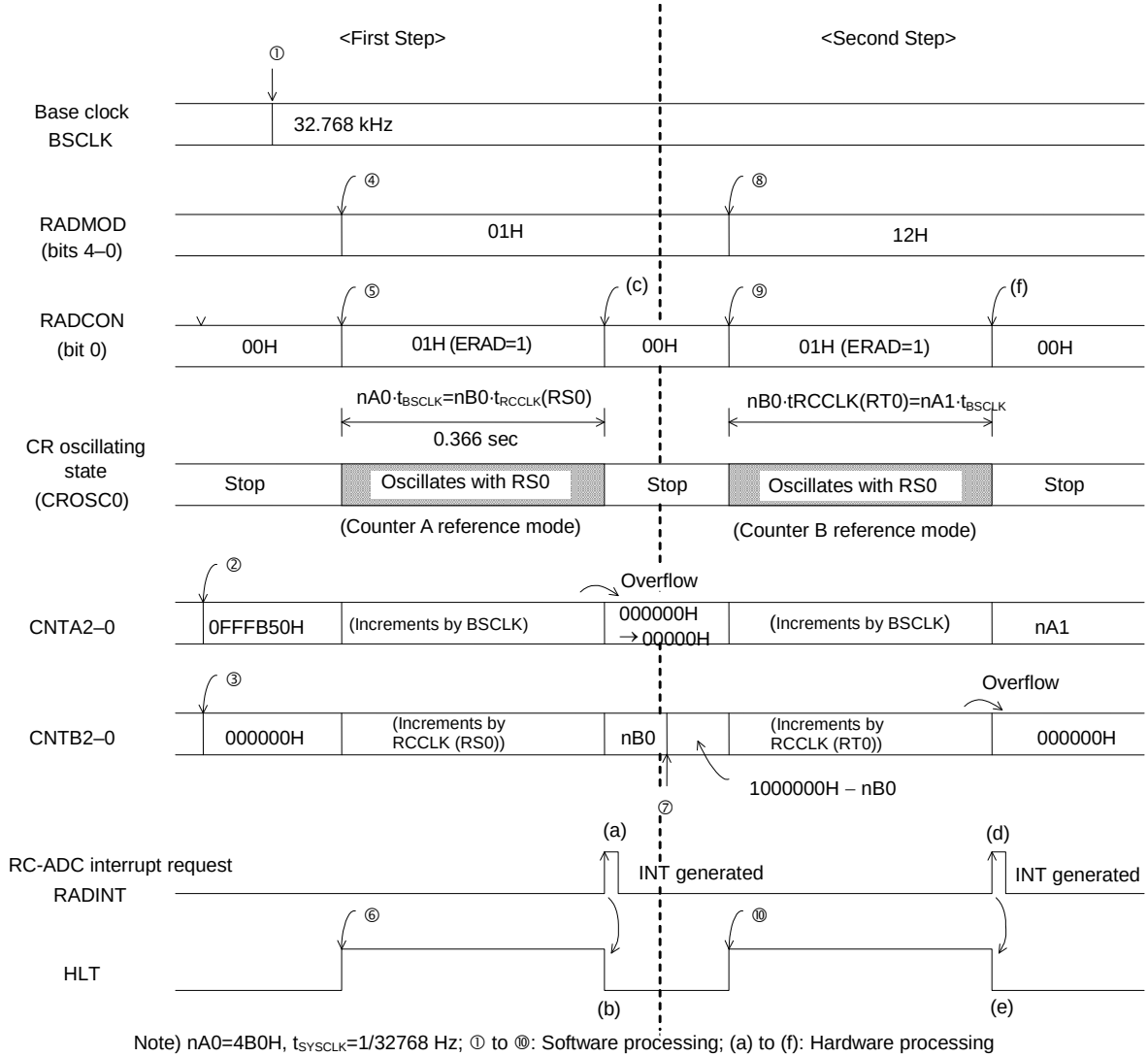
Figure 23-14 shows, as an example of method, a timing diagram of one cycle of conversion from analog value RT0 to a digital value, that is, A/D conversion.
Basically, one A/D conversion cycle must consist of two steps, as shown in Figure 23-14. The reason for requiring two steps is that the reference resistor and the thermistor must first be oscillated separately and then the ratio between the oscillation frequencies of them is used, as described above.
In the example below, operation for these two steps is performed using the following combination:

- First step = RC oscillation with RS0 in Counter A reference mode
- Second step = RC oscillation with RT0 in Counter B reference mode

Besides this, there would be several possible A/D conversion methods.
In the above method, the operation time (gate time) for the second step fluctuates depending on the value of thermistor RT0. To avoid the fluctuation of the operation time, using a method that uses the following combination is recommended:

- First step = RC oscillation with RS0 in Counter B reference mode
- Second step = RC oscillation with RT0 in Counter A reference mode

A/D conversion procedure is explained below by taking Figure 23-14 as an example.



Note) nA0=4B0H, t_{SYSCLK}=1/32768 Hz; ① to ⑩: Software processing; (a) to (f): Hardware processing

Figure 23-14 Timing Diagram for 1 Cycle of A/D Conversion (Example)

<First Step>

① Set the base clock to 32.768 kHz. (Write "00H" in FCON0.)
② Preset "1000000H – nA0" in Counter A.
③ Preset "000000H" in Counter B.
④ Write "01H" in RADMOD to select Counter A reference mode and the oscillation mode that uses reference resistance RS0.
⑤ Write "01H" in RADCON to start A/D conversion operation.
⑥ Write "1" in the HLT bit of SBYCON (see Chapter 4, "MCU") to set the device to HALT mode.

Note:
In this example, nA0 is set to 4B0H because the gate time "nA0•t_{BSCLK}" in oscillation mode with reference resistor RS0 is set to 0.3666 second. The value of nA0 is related to how much the margin of the quantification error of the A/D conversion is: the greater the nA0 value is, the smaller the margin of error becomes.
To reduce noise contamination to the RC oscillator circuit caused by CPU operation, it is recommended to constantly put the device into HALT mode during operation of RC oscillation.

From this point of time, the RC oscillator circuit (RCOSC0) continues oscillation for about 0.366 second with the reference resistance RS0. Then, when Counter A overflows, the RADINT signal is set to "1" and an RC-ADC interrupt request is generated (section (a)). Also, the generation of interrupt request releases HALT mode (section (b)) and at the same time, A/D conversion operation stops (section (c), RARUN bit = "0"). At this time, Counter A is set to "000000H".
The content of Counter B at this time is expressed by the following expression:

$$nB0 = nA0 \cdot \frac{t_{BSCLK}}{t_{RCCLK}(RS0)} \quad \ldots\ldots \text{Expression B}$$

That completes the operations in First Step .

<Second Step>

⑦ Calculate "1000000H – nB0" from the content of Counter B "nB0" and set the obtained value in Counter B.
At this point, Counter A needs to be cleared; however, no processing is required since the counter is already set to "000000H".
⑧ Write "12H" in RADMOD to select Counter B reference mode and the oscillation mode that uses thermistor RT0.
⑨ Write "01H" in RADCON to start A/D conversion operation.
⑩ Write "1" in the HLT bit of SBYCON (see Chapter 4, "MCU") to set the device to HALT mode.

The RC oscillator circuit (RCOSC0) oscillates with thermistor RT0 from this point until Counter B overflows. This period is equal to the product of "nB0" obtained in the First Step and the oscillation period t_{RCCLK} (RT0) using RT0.
When Counter B overflows, the RADINT signal is set to "1" and an RC-ADC interrupt request is generated (section (d)). Also, the generation of interrupt request releases HALT mode (section (e)) and at the same time, A/D conversion operation stops (section (f), RARUN bit = "0").
This completes the operations in Second Step .

The content of Counter A at this time becomes the A/D conversion value nA1, which is expressed by the following expression:

$$nA1 = nB0 \cdot \frac{t_{RCCLK}\ (RT0)}{t_{BSCLK}} \quad \ldots\ldots \text{Expression C}$$

From expressions B and C, nA1 is expressed by the following expression:

$$nA1 = nA0 \cdot \frac{t_{RCCLK}\ (RT0)}{t_{RCCLK}\ (RS0)} \quad \ldots\ldots \text{Expression D}$$

where t_{RCCLK} (RS0) is the oscillator clock period by reference resistor RS0 and t_{RCCLK} (RT0) the oscillator clock period by thermistor RT0.
Since the oscillation period is expressed by "$t_{RCCLK} = k_{RCCLK} \cdot R \cdot C$", t_{RCCLK} (RS0) and t_{RCCLK} (RT0) are expressed by the following expressions:

$$t_{RCCLK}(RS0) = k_{RCCLK} \cdot (CS0+CVR) \cdot RS0$$
$$t_{RCCLK}(RT0) = k_{RCCLK} \cdot (CS0+CVR) \cdot RT0 \quad \ldots\ldots \text{Expression E}$$

When expression E is substituted for expression D, nA1 will be:

$$nA1 = nA0 \cdot \frac{RT0}{RS0}$$

Since "nA0" ("4B0H" in this example) and RS0 are constants whose values are fixed, "nA1" is a digital value proportional to RT0. This very "nA1" corresponds to "nT0" in expression A.

That concludes the description of the A/D conversion method using a thermistor. "nA1" that has been obtained must further be converted to a value such as a temperature indication value for thermometer by program according to the temperature-to-resistance characteristics of the thermistor.

23.3.4 Monitoring RC Oscillation

The RC oscillator clock (RCCLK) can be output using the secondary function of the P35 pin of Port 3. See Chapter 20, "Port 3," for the details of the secondary function of P35.
Monitoring RC oscillation is useful for checking the characteristics of the RC oscillator circuit. That is, the relationship between a sensor, such as a thermistor, and the oscillation frequency can be measured. For instance, the coefficient for conversion from the above-described nA1 value to a temperature indication value can be obtained by checking the relationship between the ambient temperature of a thermistor-incorporated RC oscillator, the oscillation frequency with thermistor RT0, and the oscillation frequency with reference resistor RS0.

Note:
- P35 (RCM) is a monitor pin for oscillation clock. The channel 0(P34-P30) and channel 1(P47-P44) share the monitor pin.
- Please use P35 (RCM) for the evaluation purpose and disable the output while operating in an actual application to minimize the noise.

23.4 Specifying port registers

When you want to make sure the RC-ADC function is working, please check related port registers are specified. See Chapter 20, "Port 3" and Chapter 21, "Port 4" for detail about the port registers.

23.4.1 Functioning P35(RCM), P34(RCT0), P33(RT0), P32(RS0), P31(CS0) and P30(IN0) as the RC-ADC(Ch0)

Set P35MD1-P30MD1 bits(bit5-bit0 of P3MOD1 register) to "0" and set P35MD0-P30MD0(bit5-bit0 of P3MOD0 register) to "1", for specifying the RC-ADC as the secondary function of P35, P34, P33, P32, P31 and P30.

Reg. name	P3MOD1 register (Address: 0F21DH)							
Bit	7	6	**5**	**4**	**3**	**2**	**1**	**0**
Bit name	P37MD1	P36MD1	**P35MD1**	**P34MD1**	**P33MD1**	**P32MD1**	**P31MD1**	**P30MD1**
Data	-	-	**0**	**0**	**0**	**0**	**0**	**0**

Reg. name	P3MOD0 register (Address: 0F21CH)							
Bit	7	6	**5**	**4**	**3**	**2**	**1**	**0**
Bit name	P37MD0	P36MD0	**P35MD0**	**P34MD0**	**P33MD0**	**P32MD0**	**P31MD0**	**P30MD0**
Data	-	-	**1**	**1**	**1**	**1**	**1**	**1**

Set P34C1-P30C1 bit(bit4-0 of P3CON1 register) to "0", set P34C0-P30C0 bit(bit4-0 of P3CON0 register) to "0", and set P34DIR-P30DIR bit(bit4-0 of P3DIR register) to "1", for specifying the P34-P30 as high-impedance inputs.
The P34C1-P30C1 bit and P34C0-P30C0 bit can be set to all "1" instead of all "0" to select the high-impedance inputs.
Set P35C1 bit(bit5 of P3CON1 register) to "1", set P35C0 bit(bit5 of P3CON0 register) to "1", and set P35DIR bit(bit5 of P3DIR register) to "0", for specifying the P35 as CMOS outputs.

Reg. name	P3CON1 register (Address: 0F21BH)							
Bit	7	6	**5**	**4**	**3**	**2**	**1**	**0**
Bit name	P37C1	P36C1	**P35C1**	**P34C1**	**P33C1**	**P32C1**	**P31C1**	**P30C1**
Data	-	-	**1**	**0**	**0**	**0**	**0**	**0**

Reg. name	P3CON0 register (Address: 0F21AH)							
Bit	7	6	**5**	**4**	**3**	**2**	**1**	**0**
Bit name	P37C0	P36C0	**P35C0**	**P34C0**	**P33C0**	**P32C0**	**P31C0**	**P30C0**
Data	-	-	**1**	**0**	**0**	**0**	**0**	**0**

Reg. name	P3DIR register (Address: 0F219H)							
Bit	7	6	**5**	**4**	**3**	**2**	**1**	**0**
Bit name	P37DIR	P36DIR	**P35DIR**	**P34DIR**	**P33DIR**	**P32DIR**	**P31DIR**	**P30DIR**
Data	-	-	**0**	**1**	**1**	**1**	**1**	**1**

Data of P35D-P30D bits (bit5-0 of P3D register) do not affect to the RC-ADC function, so don't care the data for the function.

Reg. name	P3D register (Address: 0F218H)							
Bit	7	6	**5**	**4**	**3**	**2**	**1**	**0**
Bit name	P37D	P36D	**P35D**	**P34D**	**P33D**	**P32D**	**P31D**	**P30D**
Data	-	-	**	**	**	**	**	**

- : Bit does not exist.
* : Bit not related to the RC-ADC channel 0(using P35,P34,P33,P32,P31 and P30) function
** : Don't care the data

23.4.2 Functioning P47(RT1), P46(RS1), P45(CS1) and P44(IN1) as the RC-ADC(Ch1)

Set P47MD1-P44MD1 bits(bit7-bit4 of P4MOD1 register) to "0" and set P47MD0-P44MD0(bit7-bit4 of P4MOD0 register) to "1", for specifying the RC-ADC as the secondary function of P47, P46, P45 and P44.

Reg. name	P4MOD1 register (Address: 0F225H)							
Bit	**7**	**6**	**5**	**4**	3	2	1	0
Bit name	**P47MD1**	**P46MD1**	**P45MD1**	**P44MD1**	P43MD1	P42MD1	P41MD1	P40MD1
Data	**0**	**0**	**0**	**0**	*	*	*	*

Reg. name	P4MOD0 register (Address: 0F224H)							
Bit	**7**	**6**	**5**	**4**	3	2	1	0
Bit name	**P47MD0**	**P46MD0**	**P45MD0**	**P44MD0**	P43MD0	P42MD0	P41MD0	P40MD0
Data	**1**	**1**	**1**	**1**	*	*	*	*

Set P47C1-P44C1 bit(bit7-4 of P4CON1 register) to "0", set P47C0-P44C0 bit(bit7-4 of P4CON0 register) to "0", and set P47DIR-P44DIR bit(bit7-4 of P4DIR register) to "1", for specifying the P47-P44 as high-impedance inputs.
The P47C1-P44C1 bit and P47C0-P44C0 bit can be set to all "1" instead of all "0" to select the high-impedance inputs.

Reg. name	P4CON1 register (Address: 0F223H)							
Bit	**7**	**6**	**5**	**4**	3	2	1	0
Bit name	**P47C1**	**P46C1**	**P45C1**	**P44C1**	P43C1	P42C1	P41C1	P40C1
Data	**0**	**0**	**0**	**0**	*	*	*	*

Reg. name	P4CON0 register (Address: 0F222H)							
Bit	**7**	**6**	**5**	**4**	3	2	1	0
Bit name	**P47C0**	**P46C0**	**P45C0**	**P44C0**	P43C0	P42C0	P41C0	P40C0
Data	**0**	**0**	**0**	**0**	*	*	*	*

Reg. name	P4DIR register (Address: 0F221H)							
Bit	**7**	**6**	**5**	**4**	3	2	1	0
Bit name	**P47DIR**	**P46DIR**	**P45DIR**	**P44DIR**	P43DIR	P42DIR	P41DIR	P40DIR
Data	**1**	**1**	**1**	**1**	*	*	*	*

Data of P47D-P44D bits (bit7-4 of P4D register) do not affect to the RC-ADC function, so don't care the data for the function.

Reg. name	P4D register (Address: 0F220H)							
Bit	**7**	**6**	**5**	**4**	3	2	1	0
Bit name	**P47D**	**P46D**	**P45D**	**P44D**	P43D	P42D	P41D	P40D
Data	******	******	******	******	*	*	*	*

* : Bit not related to the RC-ADC channel 1(using P47,P46,P45 and P44) function
** : Don't care the data

Note:
Status of output pins P31-P34 and P45-P47 changes according to the RC oscillation mode specified by OM0-OM3 bit of RADMOD register.

Chapter 24

Successive Approximation Type A/D Converter

24. Successive Approximation Type A/D Converter

24.1 Overview

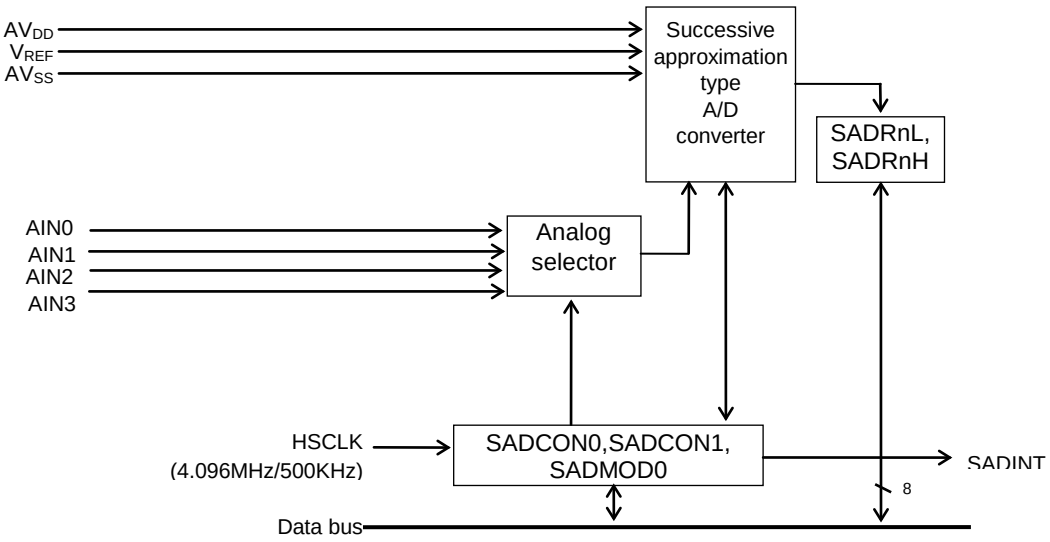
This LSI has a built-in 4-channel successive approximation type A/D converter (SA-ADC).

24.1.1 Features

- Built-in sample/hold 12-bit successive approximation type A-D converter, which enables channel selection from 4 channels

24.1.2 Configuration

Figure 24-1 shows the configuration of SA-ADC.



SADR0L: SA-ADC result register 0L
SADR0H: SA-ADC result register 0H
SADR1L: SA-ADC result register 1L
SADR1H: SA-ADC result register 1H
SADR2L: SA-ADC result register 2L
SADR2H: SA-ADC result register 2H
SADR3L: SA-ADC result register 3L
SADR3H: SA-ADC result register 3H
SADCON0: SA-ADC control register 0
SADCON1: SA-ADC control register 1
SADMOD0: SA-ADC mode register 0

Figure 24-1 Configuration of SA-ADC

24.1.3 List of Pins

Pin name	I/O	Description
AV_{DD}	—	Positive power supply pin for the successive approximation type A/D converter
V_{REF}	—	Reference power supply pin for the successive approximation type A/D converter
AV_{SS}	—	Negative power supply pin for the successive approximation type A/D converter
AIN0	I	Successive approximation type A/D converter input pin 0
AIN1	I	Successive approximation type A/D converter input pin 1
AIN2	I	Successive approximation type A/D converter input pin 2
AIN3	I	Successive approximation type A/D converter input pin 3

24.2 Description of Registers

24.2.1 List of Registers

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F2D0H	SA-ADC result register 0L	SADR0L	SADR0	R	8/16	00H
0F2D1H	SA-ADC result register 0H	SADR0H		R	8	00H
0F2D2H	SA-ADC result register 1L	SADR1L	SADR1	R	8/16	00H
0F2D3H	SA-ADC result register 1H	SADR1H		R	8	00H
0F2D4H	SA-ADC result register 2L	SADR2L	SADR2	R	8/16	00H
0F2D5H	SA-ADC result register 2H	SADR2H		R	8	00H
0F2D6H	SA-ADC result register 3L	SADR3L	SADR3	R	8/16	00H
0F2D7H	SA-ADC result register 3H	SADR3H		R	8	00H
0F2F0H	SA-ADC control register 0	SADCON0	SADCON	R/W	8/16	02H
0F2F1H	SA-ADC control register 1	SADCON1		R/W	8	00H
0F2F2H	SA-ADC mode register 0	SADMOD0	—	R/W	8	00H

24.2.2 SA-ADC Result Register 0L (SADR0L)

Address: 0F2D0H
Access: R
Access size: 8/16 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
SADR0L	SAR03	SAR02	SAR01	SAR00	—	—	—	—
R/W	R	R	R	R	R	R	R	R
Initial value	0	0	0	0	0	0	0	0

SADR0L is a special function register (SFR) used to store SA-ADC conversion results on channel 0.
SADR0L is updated after A/D conversion.

[Description of Bits]

- **SAR03-SAR00** (bits 7-4)
  The SAR03–SAR00 bits are used to store the values of bit 3 to bit 0 of A/D conversion results (12 bits) on channel 0.

24.2.3 SA-ADC Result Register 0H (SADR0H)

Address: 0F2D1H
Access: R
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
SADR0H	SAR0B	SAR0A	SAR09	SAR08	SAR07	SAR06	SAR05	SAR04
R/W	R	R	R	R	R	R	R	R
Initial value	0	0	0	0	0	0	0	0

SADR0H is a special function register (SFR) used to store SA-ADC conversion results on channel 0.
SADR0H is updated after A/D conversion.

[Description of Bits]

- **SAR0B-SAR04** (bits 7-0)
  The SAR0B3–SAR04 bits are used to store the values of bit 11 to bit 4 of A/D conversion results (12 bits) on channel 0.

24.2.4 SA-ADC Result Register 1L (SADR1L)

Address: 0F2D2H
Access: R
Access size: 8/16 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
SADR1L	SAR13	SAR12	SAR11	SAR10	—	—	—	—
R/W	R	R	R	R	R	R	R	R
Initial value	0	0	0	0	0	0	0	0

SADR1L is a special function register (SFR) used to store SA-ADC conversion results on channel 1.
SADR1L is updated after A/D conversion.

[Description of Bits]

- **SAR13-SAR10** (bits 7-4)
  The SAR13–SAR10 bits are used to store the values of bit 3 to bit 0 of A/D conversion results (12 bits) on channel 1.

24.2.5 SA-ADC Result Register 1H (SADR1H)

Address: 0F2D3H
Access: R
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
SADR1H	SAR1B	SAR1A	SAR19	SAR18	SAR17	SAR16	SAR15	SAR14
R/W	R	R	R	R	R	R	R	R
Initial value	0	0	0	0	0	0	0	0

SADR1H is a special function register (SFR) used to store SA-ADC conversion results on channel 1.
SADR1H is updated after A/D conversion.

[Description of Bits]

- **SAR1B-SAR14** (bits 7-0)
  The SAR1B–SAR14 bits are used to store the values of bit 11 to bit 4 of A/D conversion results (12 bits) on channel 1.

24.2.6 SA-ADC Result Register 2L (SADR2L)

Address: 0F2D4H
Access: R
Access size: 8/16 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
SADR2L	SAR23	SAR22	SAR21	SAR20	—	—	—	—
R/W	R	R	R	R	R	R	R	R
Initial value	0	0	0	0	0	0	0	0

SADR2L is a special function register (SFR) used to store SA-ADC conversion results on channel 2.
SADR2L is updated after A/D conversion.

[Description of Bits]

- **SAR23-SAR20** (bits 7-4)
  The SAR23–SAR20 bits are used to store the values of bit 3 to bit 0 of A/D conversion results (12 bits) on channel 2.

24.2.7 SA-ADC Result Register 2H (SADR2H)

Address: 0F2D5H
Access: R
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
SADR2H	SAR2B	SAR2A	SAR29	SAR28	SAR27	SAR26	SAR25	SAR24
R/W	R	R	R	R	R	R	R	R
Initial value	0	0	0	0	0	0	0	0

SADR2H is a special function register (SFR) used to store SA-ADC conversion results on channel 2.
SADR2H is updated after A/D conversion.

[Description of Bits]

- **SAR2B-SAR24** (bits 7-0)
  The SAR2B–SAR24 bits are used to store the values of bit 11 to bit 4 of A/D conversion results (12 bits) on channel 2.

24.2.8 SA-ADC Result Register 3L (SADR3L)

Address: 0F2D6H
Access: R
Access size: 8/16 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
SADR3L	SAR33	SAR32	SAR31	SAR30	—	—	—	—
R/W	R	R	R	R	R	R	R	R
Initial value	0	0	0	0	0	0	0	0

SADR3L is a special function register (SFR) used to store SA-ADC conversion results on channel 3.
SADR3L is updated after A/D conversion.

[Description of Bits]

- **SAR33-SAR30** (bits 7-4)
  The SAR33–SAR30 bits are used to store the values of bit 3 to bit 0 of A/D conversion results (12 bits) on channel 3.

24.2.9 SA-ADC Result Register 3H (SADR3H)

Address: 0F2D7H
Access: R
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
SADR3H	SAR3B	SAR3A	SAR39	SAR38	SAR37	SAR36	SAR35	SAR34
R/W	R	R	R	R	R	R	R	R
Initial value	0	0	0	0	0	0	0	0

SADR3H is a special function register (SFR) used to store SA-ADC conversion results on channel 3.
SADR3H is updated after A/D conversion.

[Description of Bits]

- **SAR3B-SAR34** (bits 7-0)
  The SAR3B–SAR34 bits are used to store the values of bit 11 to bit 4 of A/D conversion results (12 bits) on channel 3.

24.2.10 SA-ADC Control Register 0 (SADCON0)

Address: 0F2F0H
Access: R/W
Access size: 8/16 bits
Initial value: 02H

	7	6	5	4	3	2	1	0
SADCON0	—	—	—	—	—	—	SACK	SALP
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

SADCON0 is a special function register (SFR) used to control the operation of the SA-ADC.

[Description of Bits]

- **SALP** (bit 0)

SALP	Description
0	Single A/D conversion only (Initial value)
1	Consecutive A/D conversion

This bit is used to select whether A/D conversion is performed once only for each channel or consecutively. When this bit is set to "0", A/D conversion is performed once only for each channel and when it is set to "1", A/D conversion is performed consecutively according to the settings of the SA-ADC mode register (SADMOD0).

- **SACK** (bit 1)

SACK	Description
0	HSCLK is set to the range of 375kHz to 625kHz.
1	HSCLK is set to the range of 1.5MHz to 4.2MHz (Initial value)

The SACK bit is used to set an A/D conversion time.
As a conversion time is set by counting HSCLK, set this bit to "0" when HSCLK is set in the range of 375kHz to 625kHz and when HSCLK is set in the range of 1.5MHz to 4.2MHz, set this bit to "1".

Note:
Set the SACK bit when the SARUN bit of the SADCON1 register is "0" (A/D conversion inactive).
SA-ADC is available only when VDD=1.8V to 3.6V and HSCLK is in the ranges of 375KHz to 625kHz and of 1.5MHz to 4.2MHz.

24.2.11 SA-ADC Control Register 1 (SADCON1)

Address: 0F2F1H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
SADCON1	—	—	—	—	—	—	—	SARUN
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

SADCON1 is a special function register (SFR) used to control the operation of the SA-ADC.

[Description of Bits]

- **SARUN** (bit 0)

SARUN	Description
0	Stops conversion. (Initial value)
1	Starts conversion.

The SARUN bit is used to start or stop SA-ADC conversion. Setting this bit to "1" starts A/D conversion and setting it to "0" stops A/D conversion.
When SALP of SADCON0 is "0" and then A/D conversion on the channel with the largest channel number among the selected ones is terminated, the SARUN bit is automatically set to "0".

Notes:
Use the SA-ADC with high-speed clock oscillation (HSCLK) enabled in the frequency control register (FCON0).
The SA-ADC is available only when V_{DD} = 1.8 to 3.6 V and HSCLK is in the ranges of 375kHz to 625kHz and of 1.5MHz to 4.2 MHz.
Do not start A/D conversion in the state in which bits 1 (SACH1) and 0 (SACH0) of SA-ADC mode register 0 are "0" and "0" respectively. When A/D conversion is started in this state, A/D conversion is not done while the A/D converter is activated. Therefore, the SA-ADC result register is not updated, the A/D conversion termination interrupt is not generated, A/D conversion is not terminated automatically, and SARUN remains "1".

24.2.12 SA-ADC Mode Register 0 (SADMOD0)

Address: 0F2F2H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
SADMOD0	—	—	—	—	SACH3	SACH2	SACH1	SACH0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

SADMOD0 is a special function register (SFR) used to choose A/D conversion channel(s).

[Description of Bits]

- **SACH0** (bit 0)

SACH0	Description
0	Stops conversion on channel 0. (Initial value)
1	Performs conversion on channel 0.

- **SACH1** (bit 1)

SACH1	Description
0	Stops conversion on channel 1. (Initial value)
1	Performs conversion on channel 1.

- **SACH2** (bit 2)

SACH2	Description
0	Stops conversion on channel 2. (Initial value)
1	Performs conversion on channel 2.

- **SACH3** (bit 3)

SACH3	Description
0	Stops conversion on channel 3. (Initial value)
1	Performs conversion on channel 3.

The SACH0 to SACH4 bits are used to select channel(s) on which A/D conversion is performed. If the SACH1 and the SACH3 are set to "1", A/D conversion is performed on channel 1 first, and then channel 3.

Do not start A/D conversion when all the SACHn (from SACH0 to SACH4) set to "0". If conversion is started, the A/D conversion circuit is activated (ON), however, A/D conversion is not done. Therefore, the SA-ADC result register is not updated, A/D conversion termination interrupt does not occur, A/D conversion does not terminate automatically, and consequently, bit 0 (SARUN) of the SA-ADC control register (SADCON1) remains "1".

24.3 Description of Operation

24.3.1 Settings of A/D Conversion Channels

According to the setting of SA-ADC mode register 0 (SADMOD0), A/D conversion is performed as shown below and A/D conversion results are stored in the SA-ADC result register (SADR0H/L to SADR3H/L).
Table 24-2 lists the relation of SA-ADC mode register, SA-ADC result registers, AIN0-3.

Table 24-2 Relation of SADMOD0, SA-ADC result registers, AIN0-3, and the temperature sensor.

SA-ADC Mode Register 0 (SADMOD0)				SA-ADC Result Registers (SADR0H, SADR0L to SADR3H, SADR3L)				Note
SACH3	SACH2	SACH1	SACH0	SADR3	SADR2	SADR1	SADR0	
0	0	0	0					Prohibition
0	0	0	1				AIN0	
0	0	1	0			AIN1		
0	0	1	1			AIN1	AIN0	
0	1	0	0		AIN2			
0	1	0	1		AIN2		AIN0	
0	1	1	0		AIN2	AIN1		
0	1	1	1		AIN2	AIN1	AIN0	
1	0	0	0	AIN3				
1	0	0	1	AIN3			AIN0	
1	0	1	0	AIN3		AIN1		
1	0	1	1	AIN3		AIN1	AIN0	
1	1	0	0	AIN3	AIN2			
1	1	0	1	AIN3	AIN2		AIN0	
1	1	1	0	AIN3	AIN2	AIN1		
1	1	1	1	AIN3	AIN2	AIN1	AIN0	

The values of the result register for the sections with a slash mark remain unchanged.

Do not start A/D conversion when bit 0 (SACH0) to bit 3 (SACH3) of SA-ADC mode register 0 (SADMOD0) is "0". If A/D conversion is started, the A/D conversion circuit is set to ON. However, as A/D conversion is not performed, the SA-ADC result register is not updated, an A/D conversion termination interrupt is not generated, A/D conversion does not terminate automatically, and bit 0 (SARUN) of the SA-ADC control register (SADCON1) remains "1".

24.3.2 Operation of the Successive Approximation A/D Converter

For direct input, operate SA-ADC in the following procedure.

1. Before starting SA-ADC, start oscillation of the high-speed clock (HSCLK) and wait until the oscillator settles.
2. When HSCLK is within the range from 375KHz to 625kHz, set bit 1 (SACK) of the SA-ADC control register (SADCON0) to "0" and when HSCLK is within the range from 1.5MHz to 4.2MHz, set it to "1".
3. Set the SA-ADC mode register 0 (SADMOD0).
4. When bit 0 (SARUN) of SA-ADC control register 1 (SADCON1) is set to "1", the SA-ADC circuit becomes active and performs A/D conversion from the lower channel number that is selected in the SA-ADC mode register (SADMOD0).
5. A/D conversion results are stored in the applicable SA-ADC result registers (SADRnL, SADRnH) and when A/D conversion of the largest channel number that is selected terminates, an SA-ADC conversion termination interrupt (ADSINT) is generated.
6. Finally, using bit 0 (SALP) of the SADCON0 register, it is possible to select whether A/D conversion is terminated (SARUN bit is "0") or A/D conversion is automatically restarted at termination of A/D conversion of the last channel.

Even if a channel is switched during A/D conversion, the channel that is selected at the start of A/D conversion is maintained until an A/D conversion termination interrupt occurs.
Figure 24-3 shows the SA-ADC operation timing when channel 0 and channel 1 are selected.

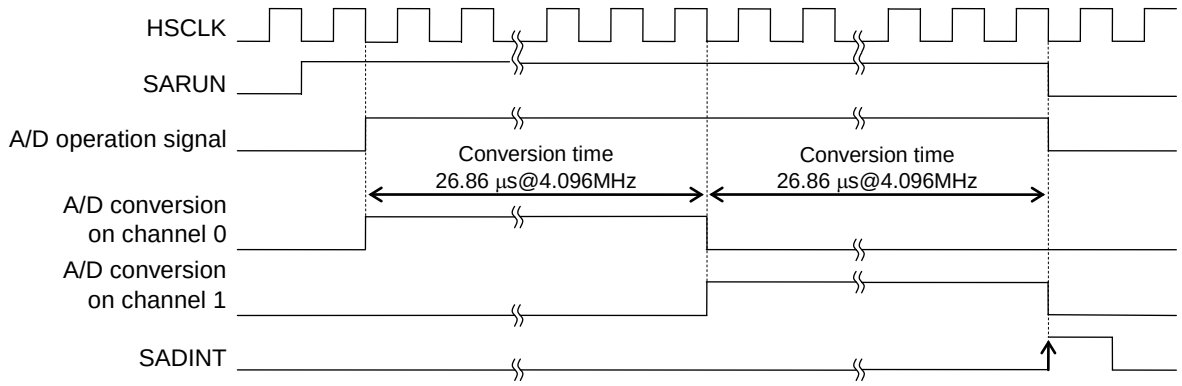


Figure 24-3 SA-ADC Operation Timing at Direct Input

Note:
A/D conversion time in 500kHzRC oscillation mode is 46 μs.

Chapter 25

LCD Drivers

25. LCD Driver

25.1 Overview

This LSI includes LCD drivers that display the contents that are set in the display register.
The LCD drivers handle the LCD display functions with four blocks.

Display registers
Display allocation
Display control
Driver

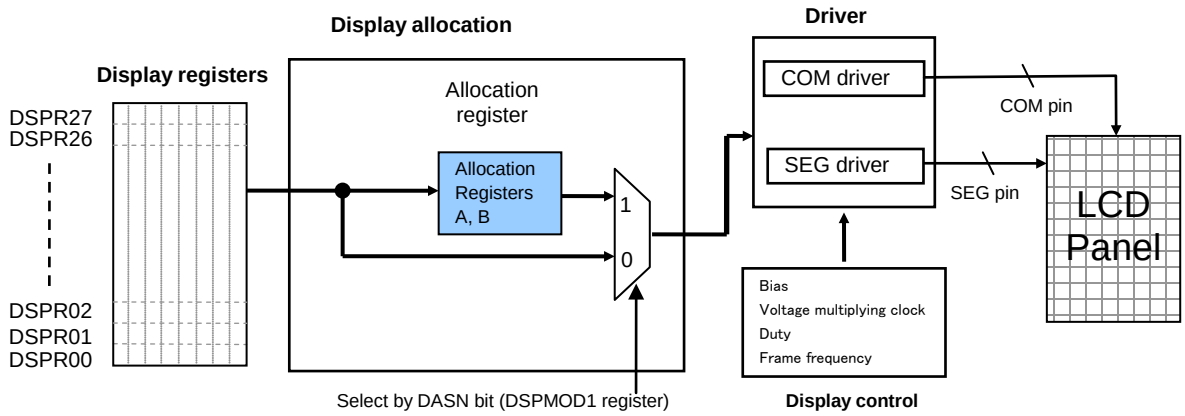


Figure 25-1 Configuration of LCD Display Function

The display registers are used to store the contents to be displayed as bit patterns.
The bit pattern storage method depends on the specification of the LCD panel to be used (display pattern and assignment of the COM pin and SEG pin) and the setting of the display allocation circuit.

The display allocation block controls mapping of the display register for the LCD common/segment.
Using the display allocation registers A and B or not using them is selectable. When using them (Set DASN bit of DSPMOD1 register to “1”), the segment mapping of the display register can be specified in bit units by programming according to the contents of display allocation registers A and B. Therefore, the display register array can be changed in flexible and simplify the software process for display (This function is defined as the programmable display allocation function in the User's Manual). Also, the data specified to the registers A and B can be easily prepared by using LAPIS semiconductor LCD allocation Tool.
When the display allocation registers A and B are not used (set the DSPMOD1 register's DASN bit to "0"), the display content is controlled only with the display registers.

The display control circuit generates LCD drive waveforms according to the characteristics of the LCD.
A bias, a bias voltage multiplying clock, a duty, and a frame frequency suitable for the LCD panel can be selected.

A) When not using Programmable display allocation function (DASN bit of DSPMOD1 register is “0”)
Suitable for the dot matrix type LCD panel whose common/segment array is approximated to the bit array of the display register.
Figure 25-2 shows an example of the correlation between the display registers and the dot matrix type LCD image.

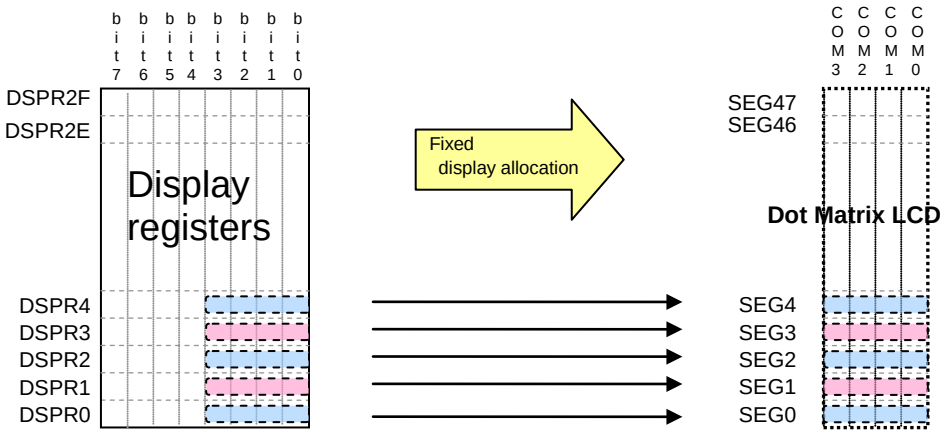
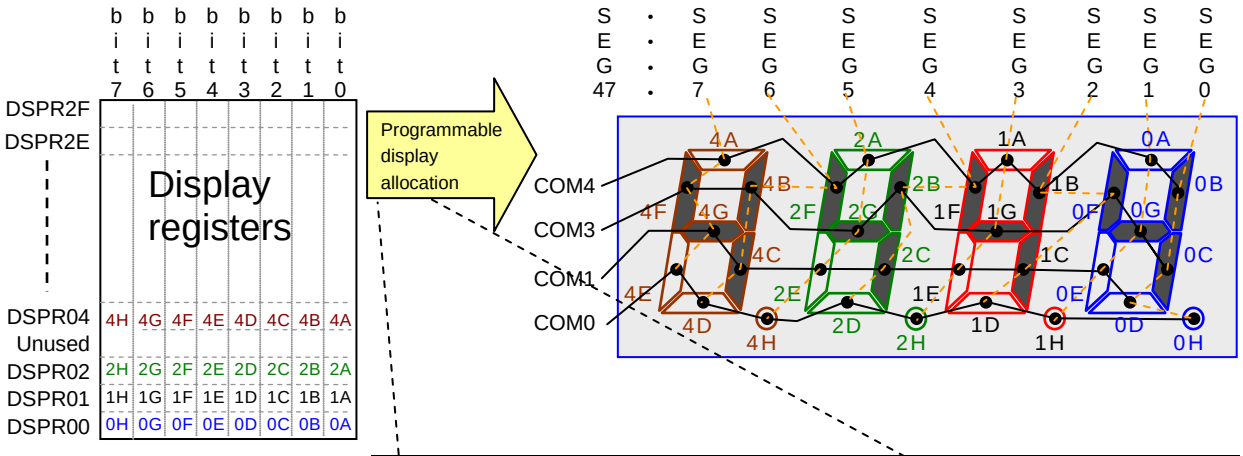


Figure 25-2 Example of Correlation between Display Registers and Dot Matrix Type LCD Image

B) When using Programmable display allocation function (DASN bit of DSPMOD1 register is “1”)
The programmable display allocation function is suitable for the LCD panel of segment type or character type whose common/segment array is restricted by the design or wiring. The display allocation registers A and B can be used to programmatically specify the display register bit-by-bit. This allows the display registers to be arrayed flexibly and thus makes the display processing in the software easier. The content of the display allocation register A (DSmCn) specifies the addresses of the display registers (DSPR00 to 2F) to be output to the common "n" of the segment "m." The content of the display allocation register B (DSmCnB) specifies the bits of the display registers (DSPR00 to 2F) to be output to the common "n" of the segment "m."
Figure 25-3 shows an example of the correlation between the display registers and the dot matrix type LCD image when the programmable display allocation is used.



DSPR	bit	Symbol on panel	COM	SEG
…	…	…	…	…
DSPR4	6	4G	1	7
…	…	…	…	..
DSPR0	2	0C	3	0
DSPR0	1	0B	4	0
DSPR0	0	0A	4	1

Figure 25-3 An example of correlation between display registers and segment type LCD

25.1.1 Features

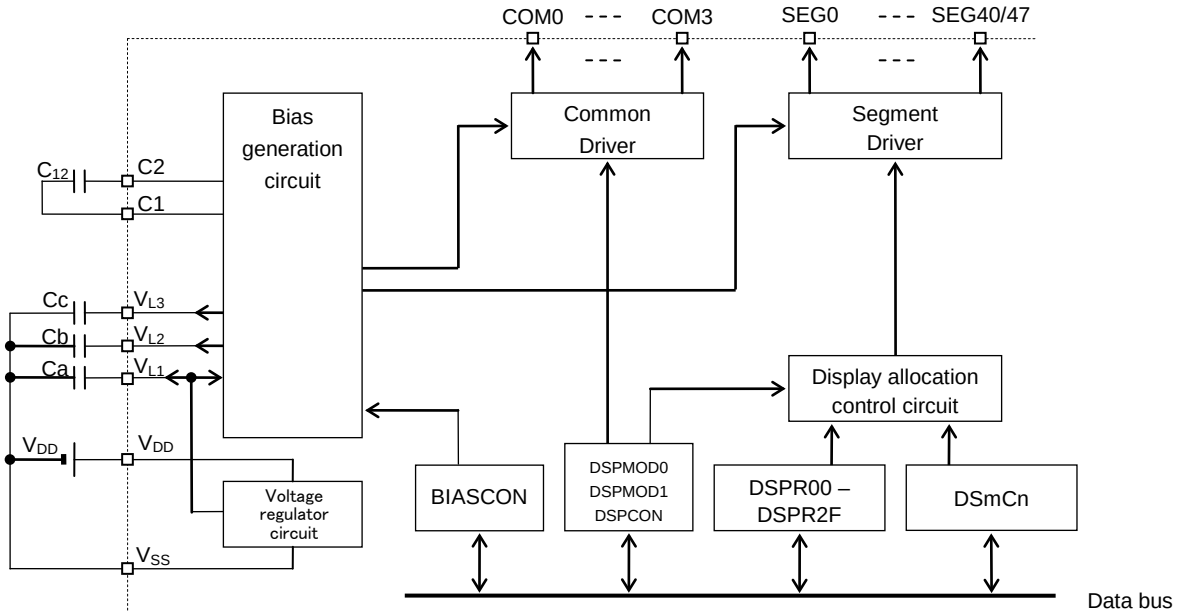
The LCD drivers are applicable to various types of LCD panels.

- ML610Q418 : 192 dots max. (48 segments x 4 commons)
- ML610Q418C : 160 dots max. (40 segments x 4 commons)
- 1/1 to 1/4 duty
- 1/2 or 1/3 bias (built-in bias generation circuit)
- Frame frequency selectable (4 types)
- Bias voltage multiplying clock selectable (8 types)
- Contrast adjustment (1/3 bias:32 steps, 1/2 bias: 32 steps)
- Programmable display allocation function

The programmable display allocation function facilitates software display processing. By using "ALL LCDs on mode" and "ALL LCDs off mode", LCD panel inspection processing software can be easily created.

25.1.2 Configuration of the LCD Drivers

Figure 25-4 shows the configuration of the LCD drivers and the bias generation circuit.



BIASCON : Bias circuit control register
DSPMOD0 : Display mode register 0
DSPMOD1 : Display mode register 1
DSPCON : Display control register
DSmCn : Allocation register (m = 0 to 47, n = 0 to 3)
DSPR00 - DSPR2F : Display registers

Figure 25-4 Configuration of LCD Drivers and Bias Generation Circuit

25.1.3 Configuration of the Bias Generation Circuit

The bias generation circuit generates LCD drive voltages (VL1 to VL3) by multiplying the voltage (VL1) generated by the voltage regulator with the capacitor (C12).
When the BSON bit of the bias circuit control register (BIASCON) is set to "1", the bias generation circuit starts operation.
Display contrast adjustment is possible in 32 steps by using the display contrast register (DSPCNT).
Figure 25-5 shows the configurations of the bias generation circuit with 1/2 bias and with 1/3 bias.

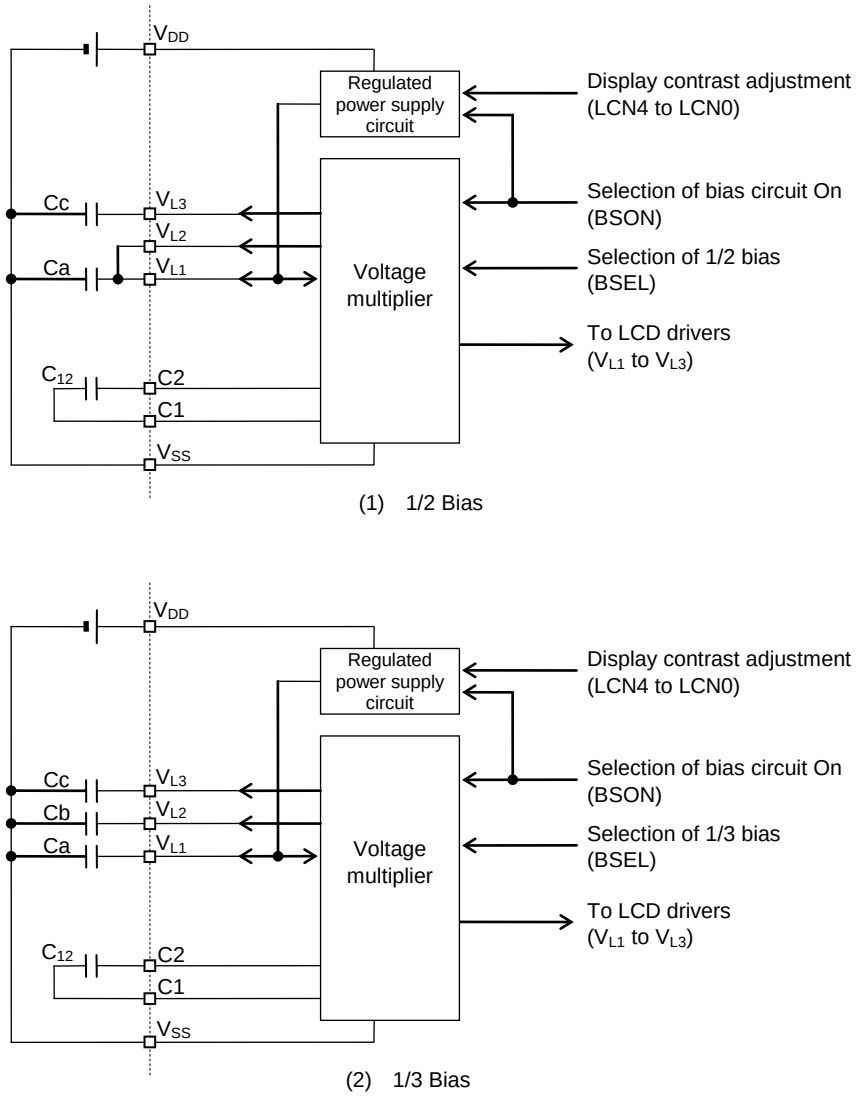


Figure 25-5 Configuration of Bias Generation Circuit

25.1.4 List of Pins

Pin name	Input/output	Function
V_{L1}	—	Power supply pin for LCD bias (internally generated)
V_{L2}	—	Power supply pin for LCD bias (internally generated)
V_{L3}	—	Power supply pin for LCD bias (internally generated)
C1	—	Capacitor connection pin for LCD bias generation
C2	—	Capacitor connection pin for LCD bias generation
COM0	O	LCD common pin
COM1	O	LCD common pin
COM2	O	LCD common pin
COM3	O	LCD common pin
SEG0	O	LCD segment pin
SEG1	O	LCD segment pin
SEG2	O	LCD segment pin
SEG3	O	LCD segment pin
SEG4	O	LCD segment pin
SEG5	O	LCD segment pin
SEG6	O	LCD segment pin
SEG7	O	LCD segment pin
SEG8	O	LCD segment pin
SEG9	O	LCD segment pin
SEG10	O	LCD segment pin
SEG11	O	LCD segment pin
SEG12	O	LCD segment pin
SEG13	O	LCD segment pin
SEG14	O	LCD segment pin
SEG15	O	LCD segment pin
SEG16	O	LCD segment pin
SEG17	O	LCD segment pin
SEG18	O	LCD segment pin
SEG19	O	LCD segment pin
SEG20	O	LCD segment pin
SEG21	O	LCD segment pin
SEG22	O	LCD segment pin
SEG23	O	LCD segment pin
SEG24	O	LCD segment pin
SEG25	O	LCD segment pin
SEG26	O	LCD segment pin
SEG27	O	LCD segment pin
SEG28	O	LCD segment pin
SEG29	O	LCD segment pin
SEG30	O	LCD segment pin
SEG31	O	LCD segment pin
SEG32	O	LCD segment pin
SEG33	O	LCD segment pin
SEG34	O	LCD segment pin

Pin name	Input/output	Function
SEG35	O	LCD segment pin
SEG36	O	LCD segment pin
SEG37	O	LCD segment pin
SEG38	O	LCD segment pin
SEG39	O	LCD segment pin
SEG40	O	LCD segment pin
SEG41	O	LCD segment pin
SEG42	O	LCD segment pin
SEG43	O	LCD segment pin
SEG44	O	LCD segment pin
SEG45	O	LCD segment pin
SEG46	O	LCD segment pin
SEG47	O	LCD segment pin

25.2 Description of Registers

25.2.1 List of Registers

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F0F0H	Bias circuit control register	BIASCON	—	R/W	8	38H
0F0F1H	Display contrast register	DSPCNT	—	R/W	8	00H
0F0F2H	Display mode register 0	DSPMOD0	DSPMOD	R/W	8/16	00H
0F0F3H	Display mode register 1	DSPMOD1		R/W	8	00H
0F0F4H	Display control register	DSPCON	—	R/W	8	00H
0F100H to 0F12FH	Display register 00 to Display register 2F	DSPR00 to DSPR2F	—	R/W	8	Undefined
0F400H to 0F42FH	Display allocation register A	DS0C0A to DS47C0A	—	R/W	8	Undefined
0F440H to 0F46FH		DS0C1A to DS47C1A	—	R/W	8	Undefined
0F480H to 0F4AFH		DS0C2A to DS47C2A	—	R/W	8	Undefined
0F4C0H to 0F4EFH		DS0C3A to DS47C3A	—	R/W	8	Undefined
0F600H to 0F62FH	Display allocation register B	DS0C0B to DS47C0B	—	R/W	8	Undefined
0F640H to 0F66FH		DS0C1B to DS47C1B	—	R/W	8	Undefined
0F680H to 0F6AFH		DS0C2B to DS47C2B	—	R/W	8	Undefined
0F6C0H to 0F6EFH		DS0C3B to DS47C3B	—	R/W	8	Undefined

25.2.2 Bias Circuit Control Register 0 (BIASCON)

Address: 0F0F0H
Access: R/W
Access size: 8-bit
Initial value: 38H

	7	6	5	4	3	2	1	0
BIASCON	—	—	BSEL1	BSEL0	BSN2	BSN1	BSN0	BSON
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	1	1	1	0	0	0

BIASCON is a special function register (SFR) to control the bias generation circuit.

[Description of Bits]

- **BSON** (bit 0)
  The BSON bit is used to control the operation of the bias generation circuit.
  When BSON is set to "1", the bias generation circuit generates the LCD drive voltages (VL1 to VL3).

BSON	Description
0	Bias circuit Off (initial value)
1	Bias circuit On

- **BSN2-BSN0** (bit 3 to 1)
  The BSN2 to BSN0 bits are used to select a clock for multiplying the bias voltage in the bias generation circuit.
  LSCLK to 1/128LSCLK can be selected.

BSN2	BSN1	BSN0	Description
0	0	0	1/1 LSCLK (32kHz)
0	0	1	1/2 LSCLK (16kHz)
0	1	0	1/4 LSCLK (8kHz)
0	1	1	1/8 LSCLK (4kHz)
1	0	0	1/16 LSCLK (2 kHz) (initial value)
1	0	1	1/32 LSCLK (1kHz)
1	1	0	1/64 LSCLK (512Hz)
1	1	1	1/128 LSCLK (256Hz)

- **BSEL** (bit 5 to 4)
  The BSEL bit is used to set the bias in the bias generation circuit.
  1/2 bias or 1/3 bias can be selected.

BSEL1	BSEL0	Description
0	0	1/3 bias
0	1	Setting prohibited
1	0	Setting prohibited
1	1	1/2 bias (initial value)

25.2.3 Display Control Register (DSPCNT)

Address: 0F0F1H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
DSPCNT	—	—	—	LCN4	LCN3	LCN2	LCN1	LCN0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

DSPCNT is a special function register (SFR) to adjust the contrast of display (32 steps).
For the setting value of DSPCNT and the LCD drive voltages (VL1, VL2, VL3), see Appendix C, "Electrical Characteristics".

[Description of Bits]

- **LCN4-LCN0** (bits 4-0)
  The LCN4 to LCN0 bits are used to adjust the contrast of display (32 steps).

LCN4	LCN3	LCN2	LCN1	LCN0	Description	V_{L1} voltage (typ.) V
0	0	0	0	0	Low	0.94 (initial value)
0	0	0	0	1	↑	0.96
0	0	0	1	0		0.98
0	0	0	1	1		1.00
0	0	1	0	0		1.02
0	0	1	0	1		1.04
0	0	1	1	0		1.06
0	0	1	1	1		1.08
0	1	0	0	0		1.10
0	1	0	0	1		1.12
0	1	0	1	0		1.14
0	1	0	1	1		1.16
0	1	1	0	0		1.18
0	1	1	0	1		1.20
0	1	1	1	0		1.22
0	1	1	1	1		1.24
1	0	0	0	0		1.26
1	0	0	0	1		1.28
1	0	0	1	0		1.30
1	0	0	1	1		1.32
1	0	1	0	0		1.34
1	0	1	0	1		1.36
1	0	1	1	0		1.38
1	0	1	1	1		1.40
1	1	0	0	0		1.42
1	1	0	0	1		1.44
1	1	0	1	0		1.46
1	1	0	1	1		1.48
1	1	1	0	0		1.50
1	1	1	0	1		1.52
1	1	1	1	0	↓	1.54
1	1	1	1	1	High	1.56

25.2.4 Display Mode Register 0 (DSPMOD0)

Address: 0F0F2H
Access: R/W
Access size: 8/16 bit
Initial value: 00H

	7	6	5	4	3	2	1	0
DSPMOD0	—	FRM1	FRM0	—	—	—	DUTY1	DUTY0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

DSPMOD0 is a special function register (SFR) to control the display mode of the LCD drivers.

[Description of Bits]

- **DUTY1 to DUTY0** (bit 1 to 0)
  The DUTY1 to DUTY0 bits are used to specify the duty in 4 steps (1/1 to 1/4). The numbers of commons/segments are determined according to the duty setting.

DUTY1	DUTY0	Description
0	0	1/1 duty (initial value)
0	1	1/2 duty
1	0	1/3 duty
1	1	1/4 duty

- **FRM1-FRM0** (bit 6, 5)
  The FRM1 to FRM0 bits are used to select a frame frequency of the LCD drivers.
  The reference frequency of a frame frequency (LLSCLK = 32.768 kHz) is selectable from 64 Hz, 73 Hz, 85 Hz, or 102 Hz.

FRM1	FRM0	Description
0	0	Reference frequency: 64 Hz (initial value)
0	1	Reference frequency: 73 Hz
1	0	Reference frequency: 85 Hz
1	1	Reference frequency: 102 Hz

The frame frequency for each duty is listed in Table 25-1.

Table 25-1 Frame Frequency for Each Duty

Duty	Frame frequency [Hz]			
	Reference frequency 64Hz	Reference frequency: 73 Hz	Reference frequency: 85 Hz	Reference frequency: 102 Hz
1/1 duty	64.00	73.14	85.33	102.40
1/2 duty	64.00	73.14	85.33	102.40
1/3 duty	64.25	73.31	85.33	103.04
1/4 duty	64.00	73.14	85.33	102.40

25.2.5 Display Mode Register 1 (DSPMOD1)

Address: 0F0F3H
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
DSPMOD1	—	—	—	—	—	DASN	—	—
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

DSPMOD1 is a special function register (SFR) to control the display mode of the LCD drivers.
Use DSPMOD1 to select to use or unused the programmable display allocation function for the display registers.

[Description of Bits]

- **DASN** (bit 2)
  The DASN bit is used to control the operation of the programmable display allocation function.
  See Sections 25.2.6, 25.2.7, and 25.3.3 for the programmable display allocation function.

DASN	Description
0	Not use Programmable display allocation function (initial value)
1	Use Programmable display allocation function

25.2.6 Display Control Register (DSPCON)

Address: 0F0F4H
Access: R/W
Access size: 8-bit
Initial value: 00H

	7	6	5	4	3	2	1	0
DSPCON	—	—	—	—	—	—	LMD1	LMD0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

DSPCON is a special function register (SFR) to control the LCD drivers.

[Description of Bits]

- **LMD1-LMD0** (bit 1, 0)
  The LMD1 and LMD0 bits are used to select an LCD display mode.
  LCD stop mode, all LCDs off mode, LCD display mode, and all LCDs on mode can be selected.
  In LCD stop mode, Vss level is output to all the common drivers and segment drivers. The charge and discharge current to and from the display panel can be stopped.
  In all LCDs off mode, off waveform is output to all the segment drivers irrespective of the contents of the display registers.
  In LCD display mode, the contents of the display registers are output to each segment driver.
  In all LCDs on mode, on waveform is output to all the segment drivers irrespective of the contents of the display registers.

LMD1	LMD0	Description
0	0	LCD stop mode (initial value)
0	1	All LCDs off mode
1	0	LCD display mode
1	1	All LCDs on mode

25.2.7 Display Allocation Register A (DS0C0A to DS47C3A)

Address: 0F400H to 0F42FH, 0F440H to 0F46FH, 0F480H to 0F4AFH, 0F4C0H to 0F4EFH, 0F500H to 0F52FH
Access: R/W
Access size: 8-bit
Initial value: Undefined

	7	6	5	4	3	2	1	0
DSmCnA	—	—	a5	a4	a3	a2	a1	a0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	x	x	x	x	x

DSmCnA (m= 0 to 47, n = 0 to 3) are special function registers (SFRs) that are used for the programmable display allocation function.
Each valid bit of DSmCnA becomes undefined at system reset.
Table 25-3 shows a list of the display allocation register A.

[Description of Bits]

- **a5 to a0** (bit 5 to 0)
  The a5 to a0 bits of DSmCnA (m = 0 to 47, n = 0 to 3) are used to select the addresses of the display registers (DSPR00 to 2F) that are output to common n of segment m.
  Set DSmCnA when the DASN bit of the display mode register 1 (DSPMOD1) is "0". When the DASN bit is "1", access from the CPU is invalid.

Table 25-3 Display Allocation Register A

Segment	Common	Register name	Address	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	R/W
SEG0	COM0	DS0C0A	0F400H	—	—	a5	a4	a3	a2	a1	a0	R/W
SEG1	COM0	DS1C0A	0F401H	—	—	a5	a4	a3	a2	a1	a0	R/W
SEG2	COM0	DS2C0A	0F402H	—	—	a5	a4	a3	a2	a1	a0	R/W
SEG3	COM0	DS3C0A	0F403H	—	—	a5	a4	a3	a2	a1	a0	R/W
:	:	:	:	:	:	:	:	:	:	:	:	:
SEG47	COM0	DS47C0A	0F42FH	—	—	a5	a4	a3	a2	a1	a0	R/W
SEG0	COM1	DS0C1A	0F440H	—	—	a5	a4	a3	a2	a1	a0	R/W
:	:	:	:	:	:	:	:	:	:	:	:	:
SEG47	COM1	DS47C1A	0F46FH	—	—	a5	a4	a3	a2	a1	a0	R/W
SEG0	COM2	DS0C2A	0F480H	—	—	a5	a4	a3	a2	a1	a0	R/W
:	:	:	:	:	:	:	:	:	:	:	:	:
SEG47	COM2	DS47C2A	0F4AFH	—	—	a5	a4	a3	a2	a1	a0	R/W
SEG0	COM3	DS0C3A	0F4C0H	—	—	a5	a4	a3	a2	a1	a0	R/W
:	:	:	:	:	:	:	:	:	:	:	:	:
SEG47	COM3	DS47C3A	0F4EFH	—	—	a5	a4	a3	a2	a1	a0	R/W

25.2.8 Display Allocation Register B (DS0C0B to DS47C3B)

Address: 0F600H to 0F62FH, 0F640H to 0F66FH, 0F680H to 0F6AFH, 0F6C0H to 0F6EFH, 0F700H to 0F72FH
Access: R/W
Access size: 8-bit
Initial value: Undefined

	7	6	5	4	3	2	1	0
DSmCnB	—	—	—	—	—	b2	b1	b0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	x	x	x

DSmCnB (m= 0 to 47, n = 0 to 3) are special function registers (SFRs) to store segment allocation data.
Each valid bit of DSmCnB becomes undefined at system reset.
Table 25-4 shows a list of the display allocation register B.

[Description of Bits]

- **b2-b0** (bit 2 to 0)
  The b2 to b0 bits of DSmCnB (m = 0 to 47, n = 0 to 3) are used to set the bits of the display registers (DSPR00 to 2F) that are output to common n of segment m.
  Set DSmCnB when the DASN bit of the display control register 0 (DSPCON0) is "0".
  When the DASN bit is “1”, access from the CPU is invalid.

b2	b1	b0	Description
0	0	0	Selects bit 0
0	0	1	Selects bit 1
0	1	0	Selects bit 2
0	1	1	Selects bit 3
1	0	0	Selects bit 4
1	0	1	Selects bit 5
1	1	0	Selects bit 6
1	1	1	Selects bit 7

Table 25-4 Display Allocation Register B

Segment	Common	Register name	Address	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	R/W
SEG0	COM0	DS0C0B	0F600H	—	—	—	—	—	b2	b1	b0	R/W
SEG1	COM0	DS1C0B	0F601H	—	—	—	—	—	b2	b1	b0	R/W
SEG2	COM0	DS2C0B	0F602H	—	—	—	—	—	b2	b1	b0	R/W
SEG3	COM0	DS3C0B	0F603H	—	—	—	—	—	b2	b1	b0	R/W
:	:	:	:	:	:	:	:	:	:	:	:	:
SEG47	COM0	DS47C0B	0F62FH	—	—	—	—	—	b2	b1	b0	R/W
SEG0	COM1	DS0C1B	0F640H	—	—	—	—	—	b2	b1	b0	R/W
:	:	:	:	:	:	:	:	:	:	:	:	:
SEG47	COM1	DS47C1B	0F66FH	—	—	—	—	—	b2	b1	b0	R/W
SEG0	COM2	DS0C2B	0F680H	—	—	—	—	—	b2	b1	b0	R/W
:	:	:	:	:	:	:	:	:	:	:	:	:
SEG47	COM2	DS47C2B	0F6AFH	—	—	—	—	—	b2	b1	b0	R/W
SEG0	COM3	DS0C3B	0F6C0H	—	—	—	—	—	b2	b1	b0	R/W
:	:	:	:	:	:	:	:	:	:	:	:	:
SEG47	COM3	DS47C3B	0F6EFH	—	—	—	—	—	b2	b1	b0	R/W

25.2.9 Display Registers (DSPR00 to DSPR2F)

Address: 0F100H to 0F12FH
Access: R/W
Access size: 8-bit
Initial value: Undefined

	7	6	5	4	3	2	1	0
DSPRxx	c7	c6	c5	c4	c3	c2	c1	c0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	x	x	x	x	x	x	x	x

DSPRxx (xx = 00 to 2FH) are special function registers (SFRs) to store display data.
Each valid bit of DSPRxx becomes undefined at system reset.
The display registers that are not used for LCD display can be used for data memories.
Set data in DSPRxx before setting LCD display mode.
The c3 to c0 values are used when the programmable display allocation function is not used.

[Description of Bits]

- **c7 to c0** (bit 7 to 0)
  The c7 to c0 bits are used to set display data.

c7~c0	Description
0	off waveform
1	on waveform

Table 25-5 List of Display Registers

Register name	Address	Corresponding segment	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	R/W
DSPR00	0F100H	SEG0	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR01	0F101H	SEG1	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR02	0F102H	SEG2	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR03	0F103H	SEG3	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR04	0F104H	SEG4	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR05	0F105H	SEG5	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR06	0F106H	SEG6	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR07	0F107H	SEG7	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR08	0F108H	SEG8	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR09	0F109H	SEG9	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR0A	0F10AH	SEG10	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR0B	0F10BH	SEG11	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR0C	0F10CH	SEG12	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR0D	0F10DH	SEG13	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR0E	0F10EH	SEG14	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR0F	0F10FH	SEG15	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR10	0F110H	SEG16	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR11	0F111H	SEG17	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR12	0F112H	SEG18	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR13	0F113H	SEG19	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR14	0F114H	SEG20	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR15	0F115H	SEG21	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR16	0F116H	SEG22	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR17	0F117H	SEG23	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR18	0F118H	SEG24	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR19	0F119H	SEG25	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR1A	0F11AH	SEG26	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR1B	0F11BH	SEG27	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR1C	0F11CH	SEG28	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR1D	0F11DH	SEG29	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR1E	0F11EH	SEG30	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR1F	0F11FH	SEG31	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR20	0F120H	SEG32	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR21	0F121H	SEG33	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR22	0F122H	SEG34	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR23	0F123H	SEG35	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR24	0F124H	SEG36	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR25	0F125H	SEG37	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR26	0F126H	SEG38	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR27	0F127H	SEG39	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR28	0F128H	SEG40	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR29	0F129H	SEG41	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR2A	0F12AH	SEG42	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR2B	0F12BH	SEG43	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR2C	0F12CH	SEG44	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR2D	0F12DH	SEG45	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR2E	0F12EH	SEG46	c7	c6	c5	c4	c3	c2	c1	c0	R/W
DSPR2F	0F12FH	SEG47	c7	c6	c5	c4	c3	c2	c1	c0	R/W

25.3 Description of Operation

25.3.1 Operation of LCD Drivers and Bias Generation Circuit

Figure 25-6 shows the operation of the LCD drivers and the bias generation circuit.

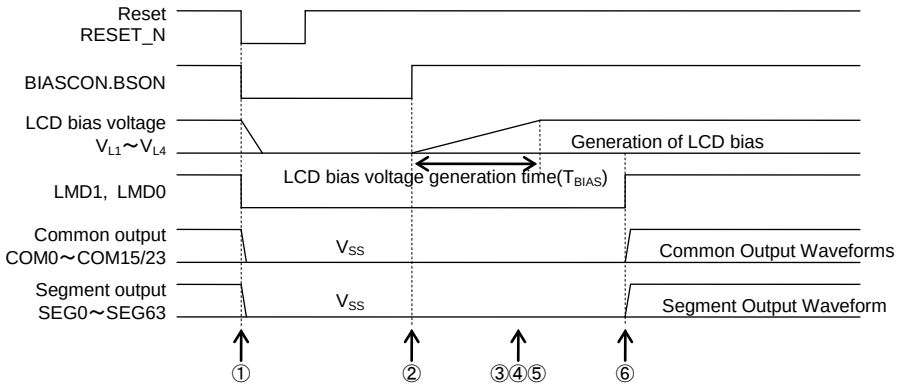


Figure 25-6 Operation of LCD Drivers and Bias Generation Circuit

① System reset stops the bias generation circuit and the LCD drivers operation and V_{SS} level to be output to each of the common and segment pins.
② By using the bias circuit control register (BIASCON), select 1/2 bias or 1/3 bias and select the bias voltage multiplying clock, and then bias generation circuit is started (BSON=1).
③ When the programmable display allocation function is used, set LCD allocation data in the display allocation registers A,B(DS0C0A to DS47C3A an DS0C0B to DS47C3B).
④ Set a frame frequency and a duty by using the display mode register 0 (DSPMOD0).
When using the programmable display allocation function, set the DSPMOD1 register's DASN bit to"1".
When not using the programmable display allocation function, set the DSPMOD1 register's DASN bit to"0".
⑤ Set display data in the display registers (DSPR00 to DSPR2F).
⑥ After having waited more than bias voltage generation time (T_{BIAS}), set the display control register (DSPCON) LMD1 and LMD0 bits to the display mode. (Display waveform is output to each segment pin.)

25.3.2 Segment Mapping When the Programmable Display Allocation Function is Not Used

The following shows the segment map of the display registers (DSPR00 to 2F) when the programmable display allocation function is not used (DSPMOD1 register's DASN bit = "0"):

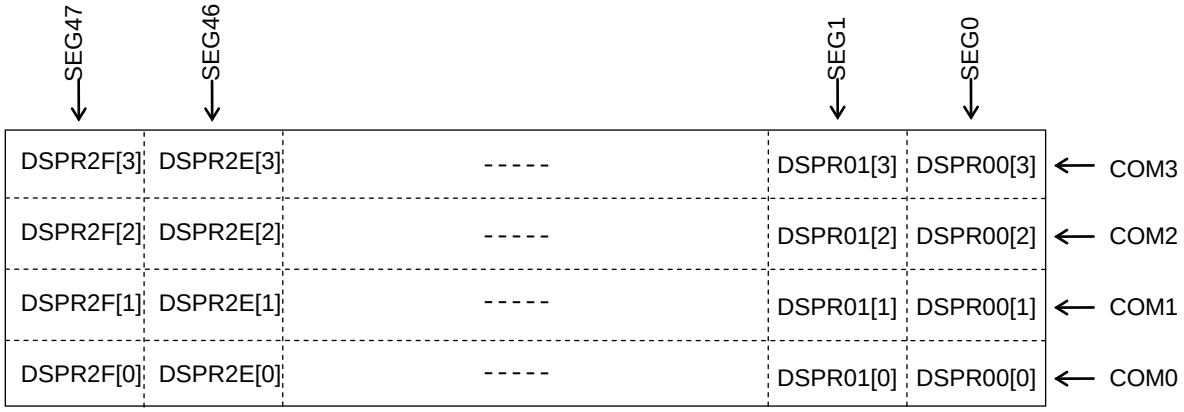


Figure 25-7 Segment Map Configuration Diagram

25.3.3 Segment Mapping When the Programmable Display Allocation Function is Used

When the programmable display allocation function is used (DASN bit of DSPMOD1 register is "1"), the segment map of the display registers (DSPR00 to 2F) can be programmatically changed using the display allocation registers (DSmCn: m = 0 to 47, n = 0 to 3).
Figure 25-8 shows the configuration when using the programmable display allocation function.

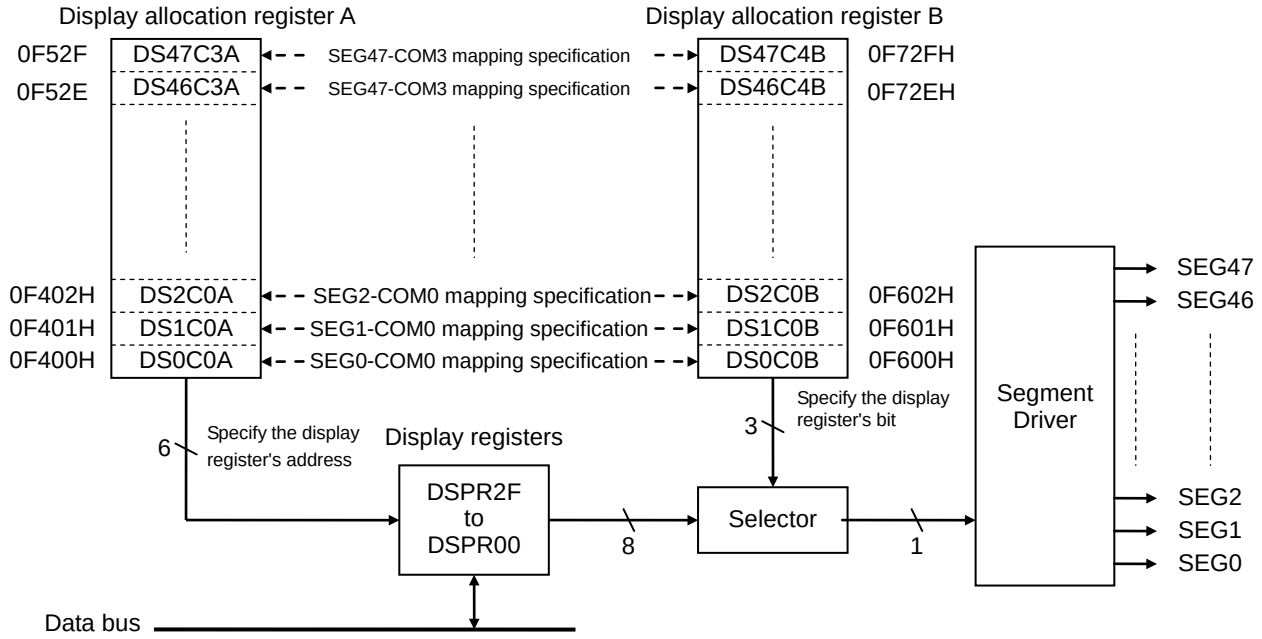
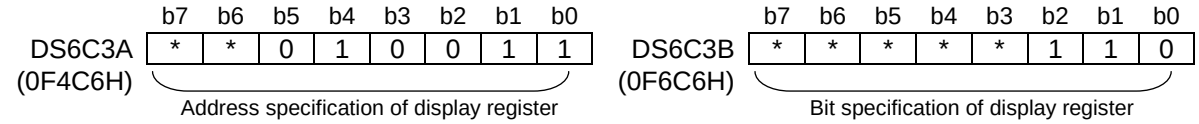


Figure 25-8 Configuration When Using the Programmable Display Allocation Function

In the display allocation register A (DSmCnA: m = 0 to 47, n = 0 to 3), set the address (00H to 2FH) of the display register (DSPR00 to DSPR2F) that is output to the common n of the segment n. In display allocation register B (DSmCnB: m = 0 to 47, n = 0 to 3), set the bits of the display register (DSPR00 to DSPR2F) that is output to the common n of the segment m.
For instance, to display bit 6 of display register 13 (DSPR13) to the common 3 of the segment 6, set as follows.



"*" indicates an arbitrary value.

Note:
Set display allocation data to display allocation registers when the DASN bit of display mode register 1 (DSPMOD1) is "0". When the DASN bit is “1”, access from the CPU is invalid.

25.3.4 Common Output Waveforms

Figure 25-9 shows the common output waveform at 1/4 duty (4 commons) and 1/3 bias.

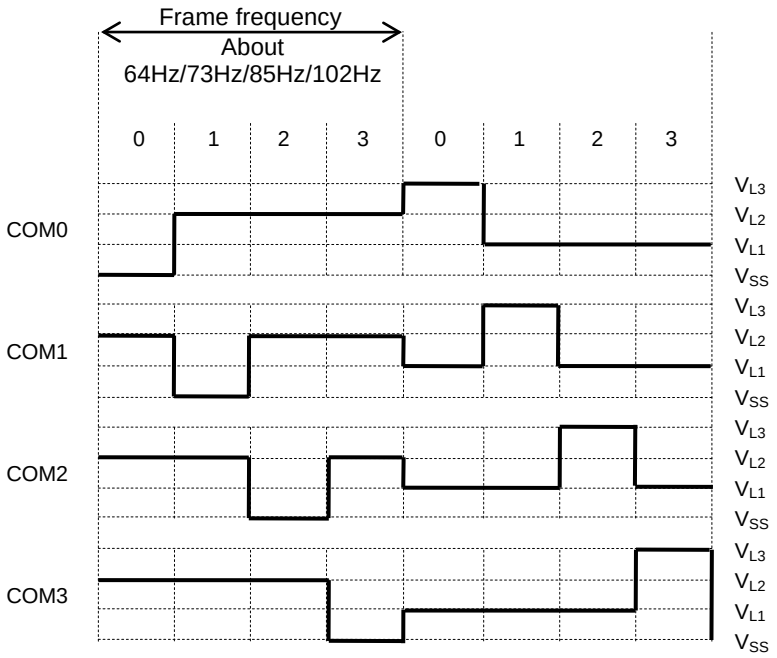


Figure 25-9 Common Output Waveform at 1/4 Duty (4 Commons) and 1/3 Bias

25.3.5 Segment Output Waveform

Figure 25-10 shows the segment output waveform at 1/4 duty (4 commons) and 1/3 bias.

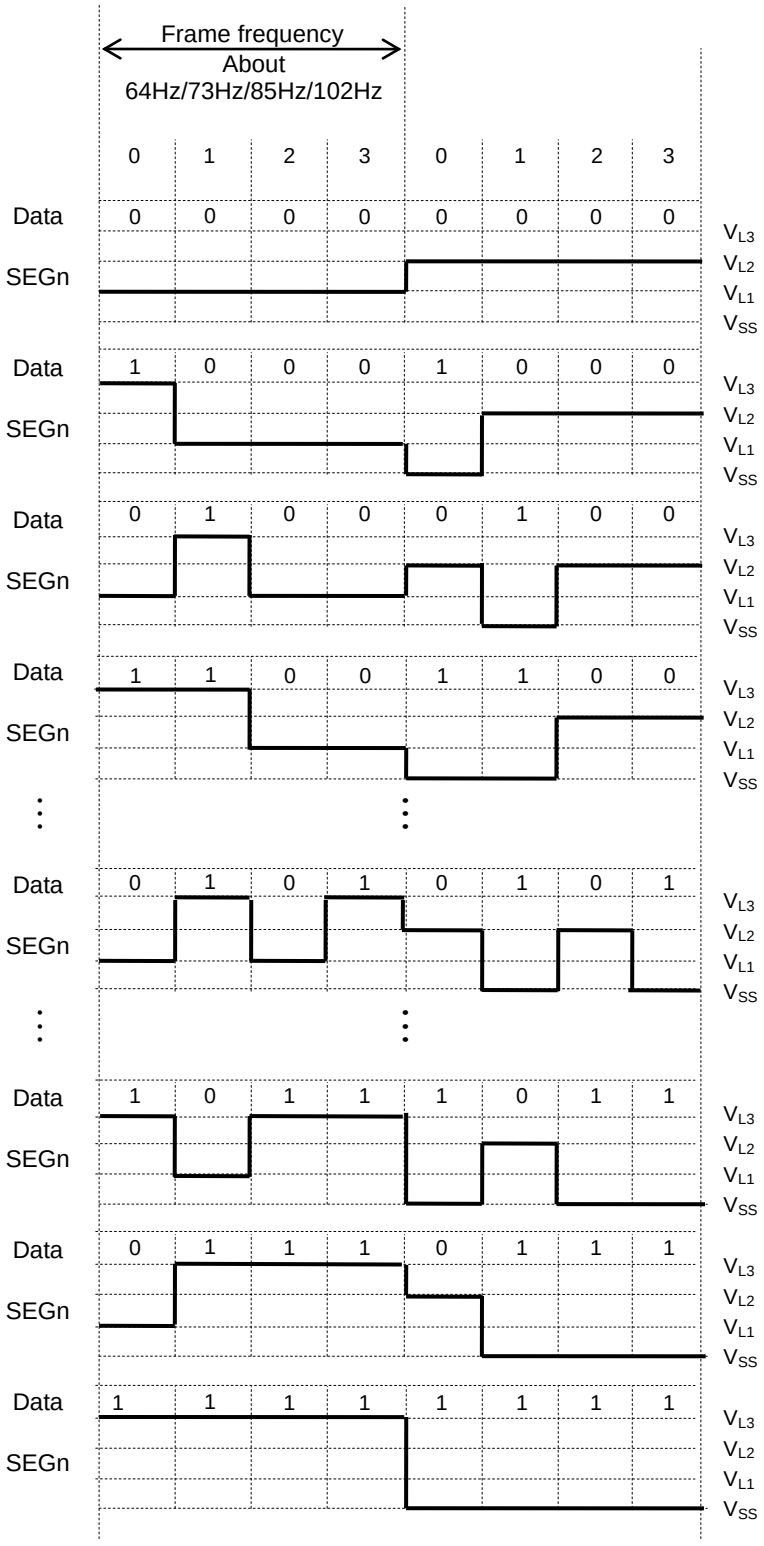


Figure 25-10 Segment Output Waveform at 1/4 Duty (4 Commons) and 1/3 Bias

Chapter 26

Battery Level Detector

26. Battery Level Detector

26.1 Overview

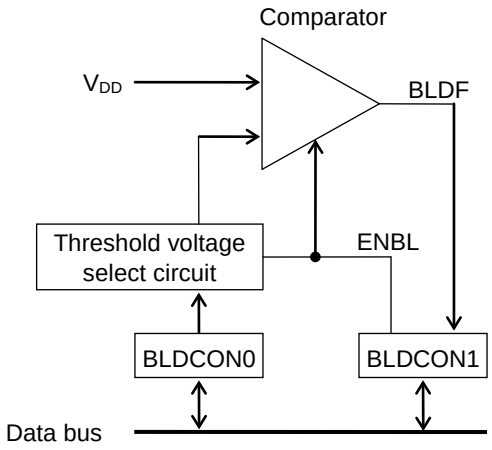
This LSI includes a Battery Level Detector (BLD).
16 levels of threshold voltages can be selected by setting Battery Level Detector control register 0 (BLDCON0).

26.1.1 Features

- Threshold voltages: One out of the 16 levels can be selected
- Accuracy: ±2% (Typ.)
- Temperature deviation: ±0.1%/°C

26.1.2 Configuration

BLD consists of the comparator and threshold voltage select circuits.
Figure 26-1 shows the configuration of the Battery Level Detector.



BLDCON0 : Battery Level Detector control register 0
BLDCON1 : Battery Level Detector control register 1

Figure 26-1 Configuration of Battery Level Detector

26.2 Description of Registers

26.2.1 List of Registers

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F0D0H	Battery Level Detector control register 0	BLDCON0	BLDCON	R/W	8/16	00H
0F0D1H	Battery Level Detector control register 1	BLDCON1		R/W	8	00H

26.2.2 Battery Level Detector Control Register 0 (BLDCON0)

Address: 0F0D0H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
BLDCON0	—	—	—	—	LD3	LD2	LD1	LD0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

BLDCON0 is a special function register (SFR) to control the Battery Level Detector

[Description of Bits]

- **LD3, LD2, LD1, LD0** (bits 3-0)
  The LD3, LD2, LD1, and LD0 bits are used to select a threshold voltage (VCMP) of the Battery Level Detector. 16 levels of threshold voltages can be selected.

LD3	LD2	LD1	LD0	Description
0	0	0	0	1.35 V ±2% (initial value)
0	0	0	1	1.40 V ±2%
0	0	1	0	1.45 V ±2%
0	0	1	1	1.50 V ±2%
0	1	0	0	1.60 V ±2%
0	1	0	1	1.70 V ±2%
0	1	1	0	1.80 V ±2%
0	1	1	1	1.90 V ±2%
1	0	0	0	2.00 V ±2%
1	0	0	1	2.10 V ±2%
1	0	1	0	2.20 V ±2%
1	0	1	1	2.30 V ±2%
1	1	0	0	2.40 V ±2%
1	1	0	1	2.50 V ±2%
1	1	1	0	2.70 V ±2%
1	1	1	1	2.90 V ±2%

26.2.3 Battery Level Detector Control Register 1 (BLDCON1)

Address: 0F0D1H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
BLDCON1	—	—	—	—	—	—	BLDF	ENBL
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R	R/W
Initial value	0	0	0	0	0	0	0	0

BLDCON1 is a special function register (SFR) to control the Battery Level Detector.

[Description of Bits]

- **ENBL** (bit 0)

The ENBL bit is used to control activation (ON) or deactivation (OFF) of the Battery Level Detector.
The Battery Level Detector is activated (ON) and deactivated (OFF) by setting the ENBL bit to "1" and "0", respectively.

ENBL	Description
0	Deactivates the Battery Level Detector (OFF) (initial value)
1	Activates the Battery Level Detector (ON).

- **BLDF** (bit 1)

The BLDF bit is the judgment result flag of the Battery Level Detector.
The BLDF bit is set to "1" or "0" when the power supply voltage (VDD) is lower than or higher than the threshold voltage selected by LD3 to LD0 bits of BLDCON0 register, respectively.

BLDF	Description
0	Higher than the threshold voltage (initial value)
1	Lower than the threshold voltage

26.3 Description of Operation

26.3.1 Threshold Voltage

The threshold voltage (VCMP) is selected by setting the bits of BLDCON0.
Table 28-1 shows the threshold voltages and the accuracy.

Table 28-1 Threshold Voltages and Accuracy

BLDCON0				Threshold voltage	Accuracy	Temperature
LD3	LD2	LD1	LD0	V_{CMP}	Ta = 25°C	deviation
0	0	0	0	1.35 V	±2.0%	±0.1%/°C
0	0	0	1	1.40 V		
0	0	1	0	1.45 V		
0	0	1	1	1.50 V		
0	1	0	0	1.60 V		
0	1	0	1	1.70 V		
0	1	1	0	1.80 V		
0	1	1	1	1.90 V		
1	0	0	0	2.00 V		
1	0	0	1	2.10 V		
1	0	1	0	2.20 V		
1	0	1	1	2.30 V		
1	1	0	0	2.40 V		
1	1	0	1	2.50 V		
1	1	1	0	2.70 V		
1	1	1	1	2.90 V		

26.3.2 Operation of Battery Level Detector

Activation (ON) and deactivation (OFF) of the Battery Level Detector are controlled by setting the ENBL bit of the Battery Level Detector control register (BLDCON1), and the result of the comparison of the power supply voltage (VDD) to the threshold voltage is output to the BLDF bit of BLDCON1.
When ENBL, the enable control bit of the Battery Level Detector, is set to "1", the detector is activated (ON). When ENBL is set to "0", the detector is deactivated (OFF) and has no supply current.
BLDF indicates the result of comparison. When BLDF bit is set to "1", it indicates the power supply voltage is lower than the threshold voltage. When BLDF bit is set to "0", it indicates the power supply voltage (VDD) is higher than the threshold voltage. The Battery Level Detector requires a settling time. Read BLDF bit 1ms or more after ENBL bit is set to "1".
Figure 26-2 shows an example of the operation timing diagram.

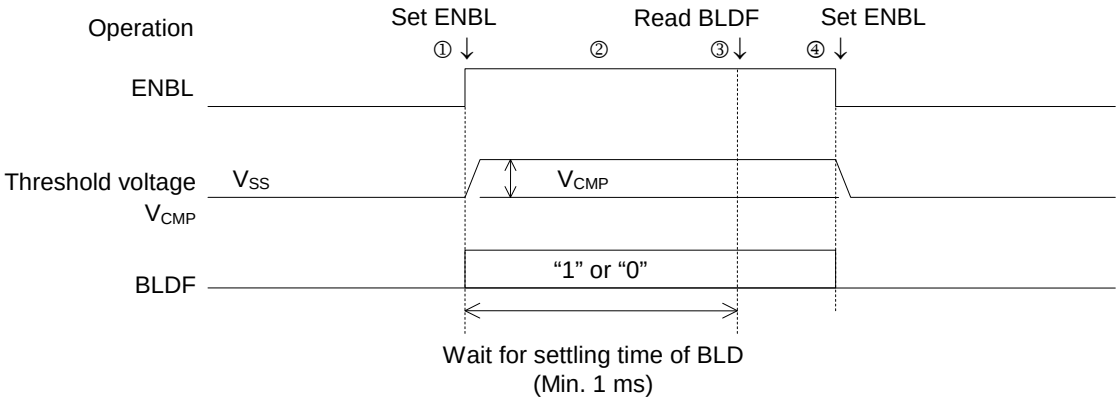


Figure 26-2 Example of Operation Timing Diagram

The operations in Figure 28-2 are described below.

① The Battery Level Detector is activated (ON) by setting the ENBL bit to "1".
② Wait the settling time (min. 1 ms) of the Battery Level Detector.
③ Read BLDF bit.
④ Set ENBL bit to "0".

Note:
Select the threshold voltage (VCMP) when the ENBL bit is "0".

Chapter 27

Power Supply Circuit

27. Power Supply Circuit

27.1 Overview

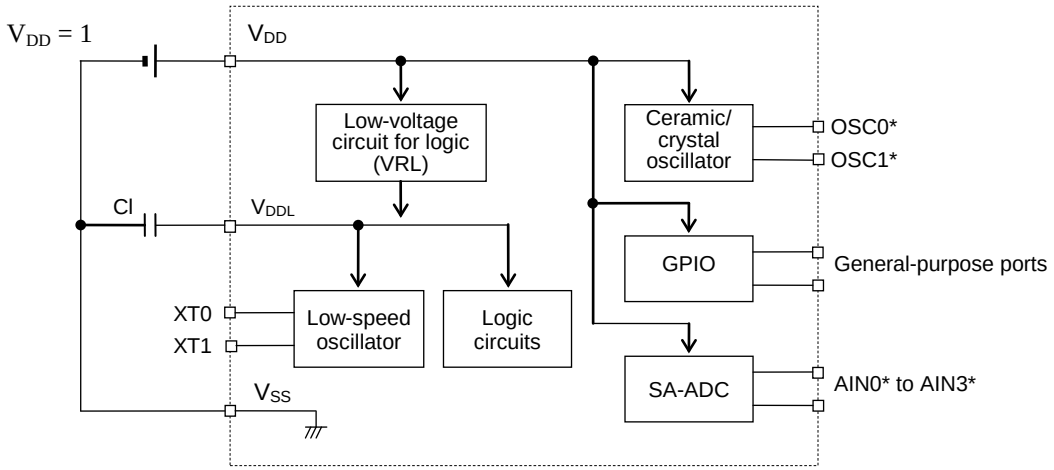
This LSI includes a regulated power supply for the internal logic (VRL).
The VRL outputs the operating voltage, V_{DDL}, of the internal logic.
For the circuit configuration of the power supplies for LCD (VL1 to VL3), see Chapter 25, "LCD Driver".

27.1.1 Features

- VRL outputs the operating voltage, V_{DDL}, of the internal logic circuits, program memory, RAM, Low-speed oscillator, etc.

27.1.2 Configuration

Figure 27-1 shows the configuration of the power supply circuit.



*: Shows the secondary function of the port.

Figure 27-1 Configuration of Power Supply Circuit

27.1.3 List of Pins

Pin name	I/O	Description
V_{SS}	—	Negative power supply pin
V_{DD}	—	Positive power supply pin
V_{DDL}	—	Positive power supply pin for the internal logic circuits (internally generated)

Chapter 28

Flash Memory Programming

28. Flash Memory Programming

28.1 Overview

This LSI has ISP (In System Programming) function that allows self-programming using a special function register (SFR) and remapping the boot area.

28.1.1 Features

The flash memory rewrite function has the following features:

- 1-word write can be used.
- The following two erasing types can be used.
  1. Block Erase (Erasing unit : 16K bytes)
     Range which can be erased: (0:0000h - 0:BFFFh, 1:0000h - 1:FFFFh, 2:0000h - 2:0FFFh)
  2. Sector Erase (Erasing unit : 512 bytes).
     Range which can be erased: (0:8000h - 0:FBFFh, 1:8000h - 1:FFFFh, 2:0000h - 2:0FFFh)

- Rewrite Counts of the flash memory:
  Depends on the rewriting address as follows.

Rewriting address	Rewrite Count
0:0000h – 0:FFFFh*1 1:0000h – 1:FFFFh	100
2:0000h –2:0FFFh	10000

Note:
*1 Excepted the Test Data Area (0:FC00h to 0:FFFFh).

- Software remap function.
  4Kbyte of boot area (0:0000H-0:0FFFH) can be remapped by specifying a register.(REMAPADD register)

- The following two erasing types can be used.
  1Kbyte of boot area (0:0000H-0:03FFH) can be remapped by setting a external pin (TEST0 pin).

28.2 Description of Registers

28.2.1 List of Registers

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F0E0H	Flash address register L	FLASHAL	FLASHA	R/W	8/16	00H
0F0E1H	Flash address register H	FLASHAH		R/W	8	00H
0F0E2H	Flash data register L	FLASHDL	FLASHD	R/W	8/16	00H
0F0E3H	Flash data register H	FLASHDH		R/W	8	00H
0F0E4H	Flash control register	FLASHCON	—	W	8	00H
0F0E5H	Flash acceptor	FLASHACP	—	W	8	00H
0F0E6H	Flash segment register	FLASHSEG	—	R/W	8	00H
0F0E7H	Flash self register	FLASHSLF	—	R/W	8	00H
0F0EAH	Flash status register	FLASHSTA	—	R	8	00H
0F0ECH	Flash remap register	REMAPADD	—	R/W	8	00H

28.2.2 Flash Address Register L,H (FLASHAL,H)

Address: 0F0E0H
Access: R/W
Access size: 8 bits/16 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
FLASHAL	FA7	FA6	FA5	FA4	FA3	FA2	FA1	FA0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R
Initial value	0	0	0	0	0	0	0	0

Address: 0F0E1H
Access: R/W
Access size: 8 bits/16 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
FLASHAH	FA15	FA14	FA13	FA12	FA11	FA10	FA9	FA8
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

FLASHAL and FLASHAH are special function registers (SFRs) that set the flash memory rewrite addresses.

[Description of Bits]

- **FA7**-FA0 (bits 7-0)
  The FA7 to FA0 bits are used to set the lower address for 1-word write.
  Note that the bit 0 is fixed to 0 and cannot be written.

- **FA15-FA8** (bits 7-0)
  The FA15 to FA8 bits are used to set the upper address for block erase, or sector erase, or 1-word write.
  In the case of block erase, the block specified by FA15 to FA14 bits are erased.
  In the case of sector erase, the sector specified by FA15 to FA9 bits are erased.
  Erasable sector memory areas are 0:8000H～0:FBFFH, 1:8000H～1:FFFFH, and 2:0000H～2:0FFFH.
  0:0000H～0:7FFFH and 1:0000H～1:7FFFH are not erasable.

Table 28-1 and Table 28-2 show the address setting values for each erasing types.

Note:
After you executed remapping boot areas, please set previous address (before remapped) to this register. See "28.3.4 Remap function by software", and "28.3.5 Remap function by hardware (external terminal)" for more detail about the remap functions.

Table 28-1 Address Setting Values for Block Erase

Area for block erase				FLASHSEG		FLASHAH							
Segment	Address			SEG1	SEG0	FA 15	FA 14	FA 13	FA 12	FA 11	FA 10	FA 9	FA 8
Segment 0	0:0000H	to	0:3FFFH	0	0	0	0	0	0	0	0	0	0
	0:4000H	to	0:7FFFH	0	0	0	1	0	0	0	0	0	0
	0:8000H	to	0:BFFFH	0	0	1	0	0	0	0	0	0	0
Segment 1	1:0000H	to	1:3FFFH	0	1	0	0	0	0	0	0	0	0
	1:4000H	to	1:7FFFH	0	1	0	1	0	0	0	0	0	0
	1:8000H	to	1:BFFFH	0	1	1	0	0	0	0	0	0	0
	1:C000H	to	1:FFFFH	0	1	1	1	0	0	0	0	0	0
Segment 2	2:0000H	to	2:0FFFH	1	0	0	0	0	0	0	0	0	0

Table 28-2 Address Setting Values for Sector Erase

Area for sector erase				FLASHSEG		FLASHAH							
Segment	Address			SEG1	SEG0	FA 15	FA 14	FA 13	FA 12	FA 11	FA 10	FA 9	FA 8
Segment 0	0:8000H	to	0:81FFH	0	0	1	0	0	0	0	0	0	0
	0:8200H	to	0:83FFH	0	0	1	0	0	0	0	0	1	0
	0:8400H	to	0:85FFH	0	0	1	0	0	0	0	1	0	0
	0:BA00H	to	0:BBFFH	0	0	1	0	1	1	1	0	1	0
	0:BC00H	to	0:BDFFH	0	0	1	0	1	1	1	1	0	0
	0:BE00H	to	0:BFFFH	0	0	1	0	1	1	1	1	1	0
Segment 0	0:C000H	to	0:C1FFH	0	0	1	1	0	0	0	0	0	0
	0:C200H	to	0:C3FFH	0	0	1	1	0	0	0	0	1	0
	0:C400H	to	0:C5FFH	0	0	1	1	0	0	0	1	0	0
	0:F800H	to	0:F9FFH	0	0	1	1	1	1	1	0	0	0
	0:FA00H	to	0:FBFFH	0	0	1	1	1	1	1	0	1	0
Segment 1	1:8000H	to	1:81FFH	0	1	1	0	0	0	0	0	0	0
	1:8200H	to	1:83FFH	0	1	1	0	0	0	0	0	1	0
	1:BC00H	to	1:BDFFH	0	1	1	0	1	1	1	1	0	0
	1:BE00H	to	1:BFFFH	0	1	1	0	1	1	1	1	1	0
Segment 1	1:C000H	to	1:C1FFH	0	1	1	1	0	0	0	0	0	0
	1:C200H	to	1:C3FFH	0	1	1	1	0	0	0	0	1	0
	1:FC00H	to	1:FDFFH	0	1	1	1	1	1	1	1	0	0
	1:FE00H	to	1:FFFFH	0	1	1	1	1	1	1	1	1	0
Segment 2	2:0000H	to	2:01FFH	1	0	0	0	0	0	0	0	0	0
	2:0200H	to	2:03FFH	1	0	0	0	0	0	0	0	1	0
	2:0400H	to	2:05FFH	1	0	0	0	0	0	0	1	0	0
	2:0A00H	to	2:0BFFH	1	0	0	0	0	0	1	0	1	0
	2:0C00H	to	2:0DFFH	1	0	0	0	0	0	1	1	0	0
	2:0E00H	to	2:0FFFH	1	0	0	0	0	0	1	1	1	0

Note:
The effective address range of sector erasing are 0:8000h to 0:FBFFh, and 2:0000h to 2:0FFFh. Therefore, addressing of 0:0000h to 0: 7FFFh is invalid.
0:FC00h to 0:FDFFh is erasable only by the on-chip debug emulator (uEASE).

28.2.3 Flash Data Register L,H (FLASHDL,H)

Address: 0F0E2H
Access: R/W
Access size: 8 bits/16 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
FLASHDL	FD7	FD6	FD5	FD4	FD3	FD2	FD1	FD0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

Address: 0F0E3H
Access: R/W
Access size: 8 bits/16 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
FLASHDH	FD15	FD14	FD13	FD12	FD11	FD10	FD9	FD8
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

FLASHDL and FLASHDH are special function registers (SFRs) that set the flash memory write data.

[Description of Bits]

- **FD7-FD0** (bits 7-0)
  The FD7 to FD0 bits are used to set the lower write data for 1-word write.

- **FD15-**FD8 (bits 7-0)
  The FD15 to FD8 bits are used to set the upper write data for 1-word write.
  Writing to FD15-FD8 starts the 1-word write.
  During flash memory writing, CPU stops the instruction. After writing is finished, CPU resumes the next instruction.

Note:
Erase the contents of the target addresses in advance. The content of an overwritten address is not guaranteed.
Writing to FLASHDH starts the 1-word write. Write data to FLASHDL and FLASHDH in this order.

28.2.4 Flash Control Register (FLASHCON)

Address: 0F0E4H
Access: W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
FLASHCON	—	—	—	—	—	—	FSERS	FERS
W	W	W	W	W	W	W	W	W
Initial value	0	0	0	0	0	0	0	0

FLASHCON is a write-only special function register (SFR) to control the block erase or the sector erase for the flash memory rewrite.

[Description of Bits]

- **FERS** (bit 0)
  The FERS bit is used to start the block erase.
  Setting the FERS bit to "1" erases the block specified by the FLASHSEG register and FLASHAH register. This bit is automatically set to "0" after completing the erase.
  During flash memory erasing, CPU stops the instruction. After erasing is finished, CPU resumes the next instruction.

- **FSERS** (bit1)
  The FSERS bit is used to start the sector erase.
  Setting the FSERS bit to "1" erases the sector specified by the FLASHSEG register and FLASHAH register. This bit is automatically set to "0" after completing the erase.
  During flash memory erasing, CPU stops the instruction. After erasing is finished, CPU resumes the next instruction.

Write value		Description
FSERS	FERS	
0	0	Erase function is not started.
0	1	Start block erase
1	0	Start sector erase
1	1	Start block erase

28.2.5 Flash Acceptor (FLASHACP)

Address: 0F0E5H
Access: W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
FLASHACP	FAC7	FAC6	FAC5	FAC4	FAC3	FAC2	FAC1	FAC0
R/W	W	W	W	W	W	W	W	W
Initial value	0	0	0	0	0	0	0	0

FLASHACP is a write-only special function register (SFR) to control the block erase and the sector erase for the flash memory rewrite or enable/disable the 1-word write operation.

[Description of Bits]

- **FAC7-FAC0** (bits 7-0)
  The FAC7 to FAC0 registers are used to restrict the erase operation or 1-word write operation in order to prevent an unintended operation.
  Writing "0FAH" and "0F5H" to FLASHACP in this order enables a one-time erase or 1-word write. For subsequent block erases or sector erases or 1-word writes, "0FAH" and "0F5H" must be written to FLASHACP each time.
  Even if another instruction is inserted between "0FAH" and "0F5H" written to FLASHACP, the erase operation or 1-word write is enabled. Note that, if data other than "0F5H" is written to FLASHACP after "0FAH" is written, the "0FAH" write processing becomes invalid. So, "0FAH" must be rewritten at first.

28.2.6 Flash Segment Register (FLASHSEG)

Address: 0F0E6H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
FLASHSEG	—	—	—	—	—	—	FSEG1	FSEG0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

FLASHSEG is a special function register (SFR) that sets the flash memory rewrite segment address.

[Description of Bits]

- **FSEG1, FSEG0** (bits 2,0)
  The FSEG0 and FSEG1 bits are used to set the flash segment address. For details, see the description of the flash address register(FLASHAH,FLASHAL).

28.2.7 Flash Self Register (FLASHSLF)

Address: 0F0E7H
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
FLASHSLF	—	—	—	—	—	—	—	FSELF
R/W	R	R	R	R	R	R	R	R/W
Initial value	0	0	0	0	0	0	0	0

FLASHSLF is a special function register (SFR) for controlling the flash memory self-programming function.

[Description of Bits]

- **FSELF** (bit0)
  The flash-memory self programming function is valid when this bit is "1."

FSELF	Description
0	Flash-memory self programming function is disabled (initial value).
1	Flash-memory self programming function is enabled.

28.2.8 Flash Status Register (FLASHSTA)

Address: 0F0EAH
Access: R
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
FLASHSTA	BSTF	—	—	—	—	—	—	—
R/W	R	R	R	R	R	R	R	R
Initial value	0	0	0	0	0	0	0	0

FLASHSTA is a special function register (SFR) for checking completion of a rise of VDDL voltage.

[Description of Bits]

- **BSTF** (bit7)
  The BSTF bit is used to as a flag to indicate the completion of a rise of VDDL voltage.

BSTF	Description
0	Incompletion of a rise of VDDL voltage (initial value).
1	Completion of a rise of VDDL voltage.

28.2.9 Flash Remap Register (REMAPADD)

Address: 0F0ECH
Access: R/W
Access size: 8 bits
Initial value: 00H

	7	6	5	4	3	2	1	0
REMAPADD	RBTA	RES2	RES1	RES0	REA15	REA14	REA13	REA12
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial value	0	0	0	0	0	0	0	0

FLASHRMP is a special function register (SFR) which specify remap areas.

The FLASHRMP register specifies two remap areas.
1. Remap an address area 0000H~0FFFH (4KB) to the same size 4KB of area started from an address specified in RES2-0 bits and REA15-12 bits.
2. Remap an address area 0000H~02FFH (512B) to the ISP boot area (FC00 H~FDFFH by setting REBTA bit.

When the software reset is performed by executing BRK instruction (*only CPU is reset), the CPU will start executing instruction codes from the address specified in the REMAPADD register. Please note that the all vector tables (a reset vector, a hardware interrupt vector and a software interrupt vector area) will be remapped.
* See "nX-U8 / 100 core instruction manual" for more detail about the BRK instruction.

[Description of Bits]

- **REA15-12** (bit3-0)
  Set upper 4bit of start address (bit15-12) that you want to remap.
  e.g.) Set 0FH to the REA15-REA12 for remapping the area 0000H-0FFFH to F000H-FFFFH.

- **RES2-0** (bit6-4)
  Set segment address of the area you want to remap.

- **RBTA** (bit7)
  The boot area can be remapped to ISP boot area (address: FC00 H~FDFFH) by setting the REBTA bit to "1".
  Use this bit for remapping to the ISP boot are, as REA15-REA12 bits can specify only the upper 4bit of address.
  Booting from the ISP boot area is available by using an external pin (TEST0 pin).
  For detail about the hardware remap, see "28.3.5 Remap function by hardware (external port)."

28.3 Description of Operation

When using the flash memory rewrite function, prepare the program for rewrite in advance on a program code area with addresses that are not used for block erase or 1-word write.
The flash memory rewrite function includes the block erase function, the sector erase function, and the 1-word write function that writes by 1 word (2 bytes).
The rewrite counts of the flash memory differ as follows depending on the address.

Address	Rewrite counts
0:0000H – 0:FFFFH 1:0000H – 1:FFFFH	100
2:0000H –2:0FFFH	10000

It also includes the flash rewrite acceptor function which restricts the flash memory rewrite operation, to prevent an improper rewriting of the flash memory. Writing "0FAH" and "0F5H" to the flash acceptor (FLASHACP) in this order enables a one-time block erase or sector erase or 1-word write.

Low speed system clock (CPU operating clock) is NOT supported for the Flash self-write programming. See the notes in each clock mode shown in the following list.

Table 28-3 System clock and Notes for Flash self-write programming

Clock mode and Registers for setting		Notes
32.768kHz crystal oscillation clock	SYSCLK bit of FCON1 register is reset to '0' (initial).	Can not support the Flash self-write programming.
500kHzRC oscillation clock	Both OSCM1 bit and OSCM0 bit of FCON0 register are reset to '0' (Initial) and also SYSCLK bit of FCON1 register is set to '1'. The clock divider works controlling by SYSC1 and SYSC0 bit of FCON0 register.	Select 1/1 OSC(500kHz) or 1/2 OSC(250kHz) clock. 1/4 OSC(125kHz) and 1/8 OSC(62.5kHz) clock are NOT supported for the Flash self-write programming.
Crystal/Ceramic oscillation clock	OSCM1 bit of FCON0 register is reset to '0' and OSCM0 bit is set to '1' and also SYSCLK bit of FCON1 register is set to '1'. The clock divider works controlling by SYSC1 and SYSC0 bit of FCON0 register.	Select the Crystal/Ceramic resonator and set the registers so that the frequency of CPU clock is specified within 200kHz ~ 4.2MHz.
Internal PLL oscillation clock	OSCM1 bit of FCON0 register is set to '1' and OSCM0 bit is reset to '0' and also SYSCLK bit of FCON1 register is set to '1'. The clock divider works controlling by SYSC1 and SYSC0 bit of FCON0 register. One of 4.096MHz, 2.048MHz, or 1.024MHz is selected for the system clock.	No limitations for the CPU operating clock frequency.
External clock	Both OSCM1 bit and OSCM0 bit of FCON0 register are set to '1' and also SYSCLK bit of FCON1 register is set to '1'. The clock divider works controlling by SYSC1 and SYSC0 bit of FCON0 register.	Provide an external clock and set the registers so that the frequency of CPU clock is specified within 200kHz ~ 4.2MHz.

See Chapter 6, "Clock Generation Circuit" for more detail about each clock mode and the registers.

Note when debugging the Flash self-write programming codes under U8 development environments, described in the following list.

Table 28-4 Notes when debugging the Flash self-write programming codes

Use case	Notes
When debugging the Flash self-wrie programming codes on DTU8 debugger	• Do not have Real-time execution (GO execution) with any break points set in an address range that has codes for Flash self-write programming sequence (From writing Flash acceptor through writing Flash data register), otherwise the self-writing operation to the Flash may fail. • Do not have STEP execution in an address range that has codes for Flash self-write programming sequence (From writing Flash acceptor through writing Flash data register), otherwise the self-writing operation to the Flash may fail. • Data Flash area (02:20000H-02:20FFFH) is reference-able through 'Memory in physical segments 1 and higher' window on the DTU8 debugger, however can not change the data directly through the memory window (unaccepted by the debugger).
When programming the Flash memory by using DTU8, FWuEASE and MWuEASE.	Programming (erase/write) the Data Flash memory area by using the On-chip debug emulator uEASE is Not supported.

28.3.1 Block Erase Function

This function erases the flash memory data by block . (16K bytes).
Write "0FAH" and "0F5H" to the flash acceptor (FLASHACP) and set the block address in the flash segment register and the flash address register H (FLASHAH). Then, write "1" to the FERS bit of the flash control register (FLASHCON) to erase the data in the block (16K bytes) specified by FLASHSEG and FLASHAH.
During the erase, the CPU is stopped. When the erase is completed, the program is restarted from the instruction following the one that set the FERS bit of the FLASHCON to "1".
Figure 28-1 shows a sample program of block erase (assuming that the FLASHSEG register is already set).

```
        LEA     offset FLASHAH                  ; EA←FLASHAH address
        MOV     R0,     #0FAH                   ; Flash acceptor enable data
        MOV     R1,     #0F5H                   ; Flash acceptor enable data
        MOV     R2,     #01H                    ; Block erase enable data
        MOV     R4,     #(offset FLASHACP)&0FFH
        MOV     R5,     #(offset FLASHACP)>>8   ; ER4←FLASHACP address
        MOV     R6,     #(offset FLASHCON)&0FFH
        MOV     R7,     #(offset FLASHCON)>>8   ; ER6←FLASHCON address
          :
        (Set the erase start block address in R9)
MARK:
        SB      FSELF                           ; Enables the flash-memory self programming
function
LOOP:
        TB      BSTF                            ;
        BZ      LOOP
        ST      R0,     [ER4]                   ; Enable flash acceptor
        ST      R1,     [ER4]                   ; Enable flash acceptor
        ST      R9,     [EA]                    ; Set block address
        ST      R2,     [ER6]                   ; Start block erase
        NOP                                     ; * Always set
        NOP                                     ; * Always set
        RB      FSELF                           ; Disables the flash-memory self programming
function
```

Figure 28-1 Sample Program of Block Erase

Note:
If erasing data being used by the running program, the program would malfunction. Erase a block unrelated to the operation of the program.
Be sure to set the NOP instruction twice or more, following the block erase start instruction.

28.3.2 Sector Erase Function

This function erases the flash memory data by sector. (512 bytes).
Write "0FAH" and "0F5H" to the flash acceptor (FLASHACP) and set the sector address in the flash segment register and the flash address register H (FLASHAH). Then, write "1" to the FSERS bit of the flash control register (FLASHCON) to erase the data in the sector (512 bytes) specified by FLASHSEG and FLASHAH.
During the erase, the CPU is stopped. When the erase is completed, the program is restarted from the instruction following the one that set the FSERS bit of the FLASHCON to "1".
Figure 28-2 shows a sample program of block erase (assuming that the FLASHSEG register is already set).

```
        LEA     offset FLASHAH                  ; EA←FLASHAH address
        MOV     R0,     #0FAH                   ; Flash acceptor enable data
        MOV     R1,     #0F5H                   ; Flash acceptor enable data
        MOV     R2,     #02H                    ; Sector erase enable data
        MOV     R4,     #(offset FLASHACP)&0FFH
        MOV     R5,     #(offset FLASHACP)>>8   ; ER4←FLASHACP address
        MOV     R6,     #(offset FLASHCON)&0FFH
        MOV     R7,     #(offset FLASHCON)>>8   ; ER6←FLASHCON address
          :
        (Set the erase start block address in R9)
MARK:
        SB      FSELF                   ; Enables the flash-memory self programming function
LOOP:
        TB      BSTF                    ;
        BZ      LOOP
        ST      R0,     [ER4]           ; Enable flash acceptor
        ST      R1,     [ER4]           ; Enable flash acceptor
        ST      R9,     [EA]            ; Set sector address
        ST      R2,     [ER6]           ; Start sector erase
        NOP                             ; * Always set
        NOP                             ; * Always set
        RB      FSELF                   ; Disables the flash-memory self programming function
```

Figure 28-2 Sample Program of Sector Erase

Note:
If erasing data being used by the running program, the program would malfunction. Erase a sector unrelated to the operation of the program.
Be sure to set the NOP instruction twice or more, following the block erase start instruction.

28.3.3 1-word Write Function

This function writes data to the flash memory by 1 word (2 bytes).
Write "0FAH" and "0F5H" to the flash acceptor (FLASHACP) and set the address in the flash segment register (FLASHSEG) and the flash address register L, H (FLASHAL,H). Then, write data to the flash data register L, H (FLASHDL,H) to write the data in the address specified by FLASHSEG and FLASHAL, H.
During the 1-word write, the CPU is stopped. When the write is completed, the program is restarted from the instruction following the write to FLASHDH instruction.
Figure 28-3 shows a sample program of 1-word write (assuming that the FLASHSEG register is already set).

```
        LEA     offset FLASHAL                  ; EA←FLASHAL address
        MOV     R0,     #0FAH                   ; Flash acceptor enable data
        MOV     R1,     #0F5H                   ; Flash acceptor enable data
        MOV     R2,     #02H                    ; Address increment data
        MOV     R3,     #00H
        MOV     R4,     #(offset FLASHACP)&0FFH
        MOV     R5,     #(offset FLASHACP)>>8   ; ER4←FLASHACP address

        (Set the write start address in ER8)
        (Set the write end address in ER12)

MARK:
        SB      FSELF               ; Enables the flash-memory self programming function
LOOP:
        TB      BSTF                ;
        BZ      LOOP

        (Set the write data in ER10)

        ST      R0,     [ER4]           ; Enable flash acceptor
        ST      R1,     [ER4]           ; Enable flash acceptor
        ST      XR8,    [EA]            ; Set address and data, start 1-word write
        NOP                             ; * Always set
        NOP                             ; * Always set

        L       ER14,   [ER8]           ; Load data
        CMP     ER14,   ER10            ; Check data
        BNE     ERROR                   ; Go to error routine on error

        ADD     ER8,    ER2             ; Increment address
        CMP     ER8,    ER12
        BLE     MARK                    ; Compare addresses

        RB      FSELF                   ; Disables the flash-memory self programming
function
```

Figure 28-3 Sample Program of 1-word Write

Note:
In verification process of the write data at the above ** marked line,the content of the program memory cannot be read, if the target address is in the Segment 0 area (0:E000H - 0:FFFFH). Read it from the ROM reference area of Segment 8, like follows.

```
L       ER14,       8:[ER8]     ; ** Load data
```

If rewriting data being used by the running program, the program might malfunction. Write to an address unrelated to the operation of the program.

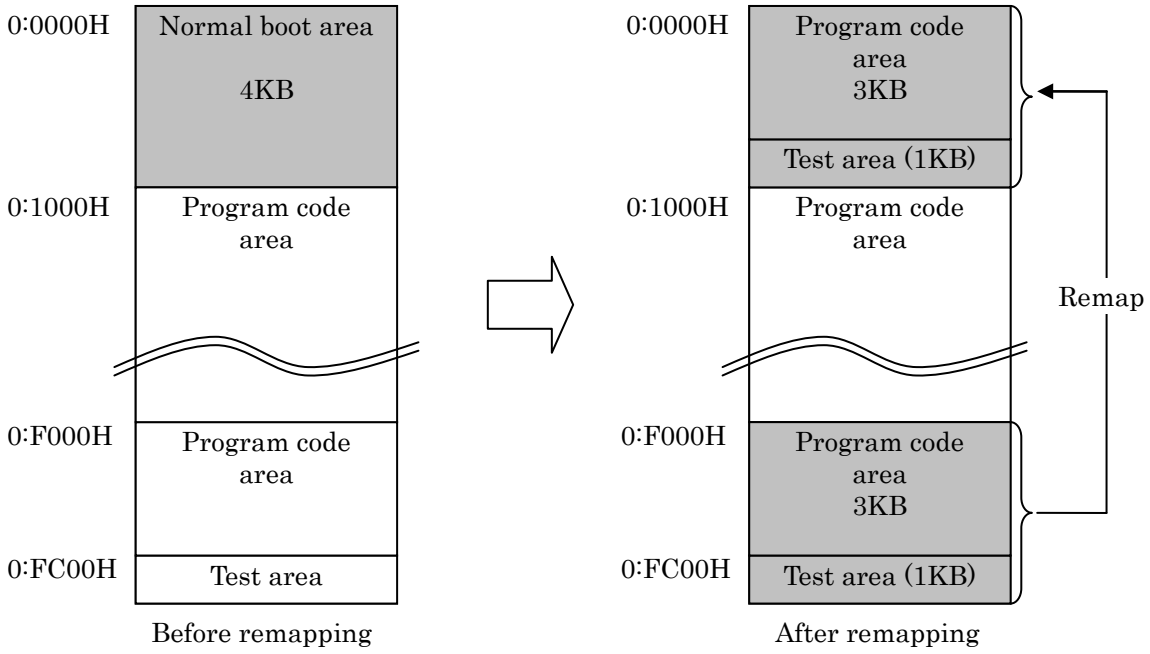
Be sure to set the NOP instruction twice or more, following the write to FLASHDH instruction.

28.3.4 Remap function by software

An address area 0000H~0FFFH (4KB) can be remapped to the same size 4KB of area starting with an address specified into REMAPADD register. Program codes start from the remapped area, by setting the FLSHRMP resiter and performing software reset (*only CPU is reset) with BRK instruction. Please refer to "nX-U8 / 100 core instruction manual" for more detail about the BRK instruction.
The vector table area (a reset vector, a hardware interrupt vector and a software interrupt vector area) is also remapped, which enables unique interrupt managements and to have 4KB or less unique program code in the remapped area independent of main program. It is useful in the case implementing FLASH self programming requires special interrupt controls.
Figure 28-4 shows an example of program codes for performing the remap and Figure 28-5 shows memory map before and after the remapping.

```
    MOV     R0,     #00FH           ;
    ST      R0,     0F0ECH          ; Set upper 4bit of start address you wan to remap
                                    ; to REMAPADD register (0F0ECH).
    MOV     PSW,    #02H            ; Set Interrupt level (ELEVEL) to 2.
BRK                                 ; BRK  instruction
```

Figure 28-4 Program code example for remapping



*Test area can not be used for a program code area

Figure 28-5 Memory map before/after the remapping

Note:
After remapping, when you read 0:0000H - 0:1000H(4KB) , please read from a segment 8 (i.e. read 8:0000H – 8:1000H).
When you want to write data to the area 0:0000H - 0:1000H (4KB) , please set 0:0000H-0:1000H to the Flash Address Register L, H (FLASHAL, H). See "29.2.2 Flash Address Register L, H (FLASHAL, H)" for more detail about the register.

28.3.5 Remap function by hardware (external terminal)

When the power on reset or reset by a RESET_N port is released on the condition an external terminal (TEST0 pin) is set to High, 1KB of the ISP boot area (0:FC00H-0:FFFFH) is remapped to 0:0000H - 0:03FF and the program code starts running from the address FC00H Therefore, following two functions are available by programming a boot program in advance with LAPIS SEMI's on-chip debug emulator uEASE.

(1) In case a problem such as power fail while rewriting the block 0 (0000H to 3FFF:16KB) in FLASH where a boot program is placed, the ISP bood area can prevent infinite unrecoverable state.

(2) Implementing a boot program that uses communicating function such as UART, can work as if an embedded Flash writer.

Please note that the 1KB ISP boot area is also remapped to 0:0400H - 0:0FFF.

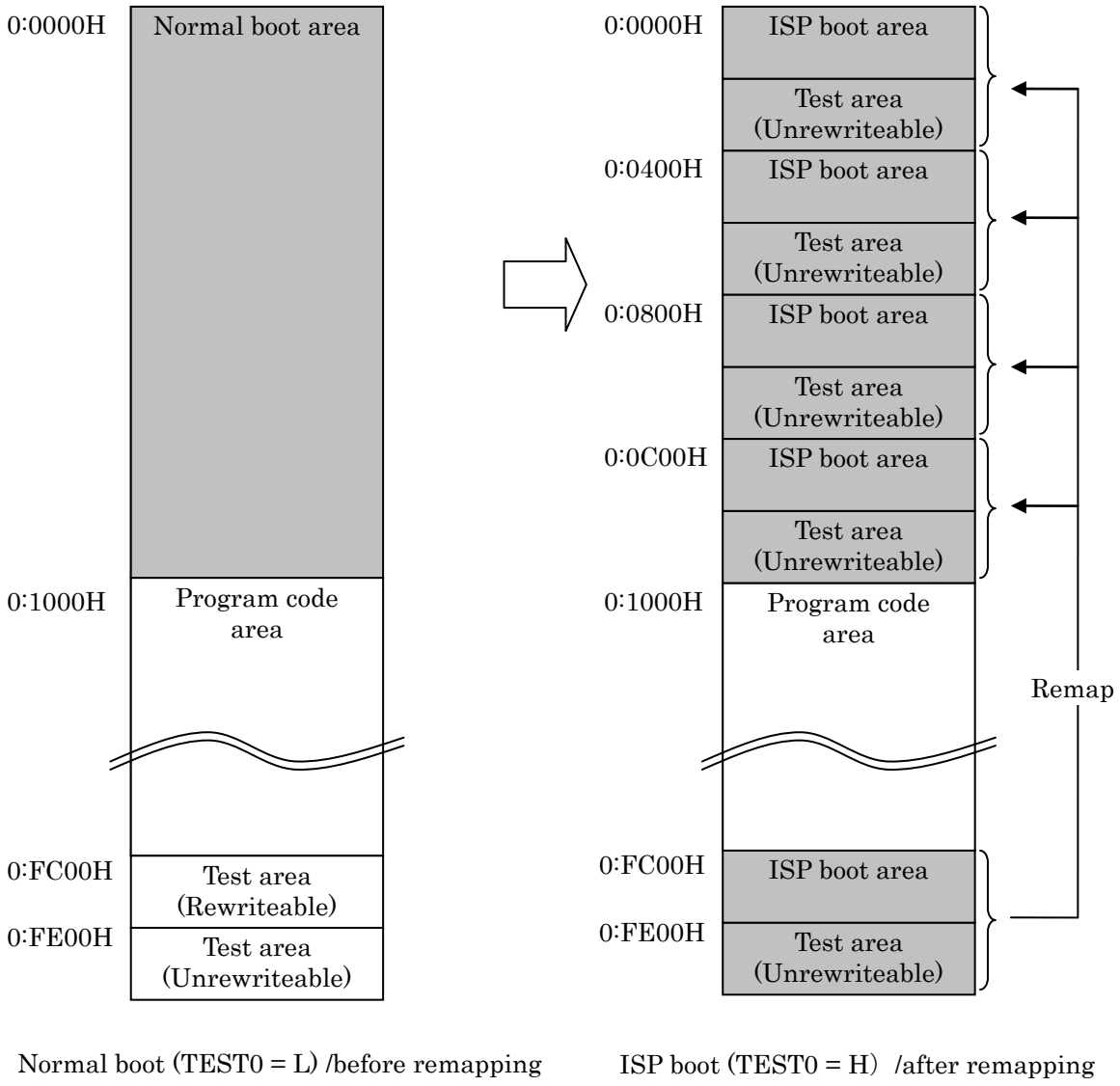


Figure 28-6 ISP boot by setting TEST0 pin

28.3.6 Notes in Use

- Be sure to write "0FFH" to the unused area and the test area of the flash memory. If data in these areas is uncertain or not "0FFH", the operation cannot be guaranteed. For details, see Chapter 2, "CPU and Memory Space".
- When the power is down or the operation is terminated forcibly during block erase or sector erase or 1-word write, retry the block erase or the sector erase.
- During flash memory rewriting/erasing, CPU stops the instruction. After rewriting/erasing is finished, CPU resumes the next instruction
- If the flash memory rewrite program does not work, write the program by using the on-chip ICE function.

Chapter 29

Code-Option

29. Code-Option

29.1 Overview

This LSI has the code-option function. The code-option function can select the voltage of releasing the system reset mode.
To use this LSI, set the code-option data to writable test data area of program code.

29.1.1 Features

- The voltage on which the CPU begins to execute instructions can be selected from 1.1V or 1.8V(Max.)
- The setting value of code-option can be checked by reading the special function register (SFR).

29.2 Description of Registers

29.2.1 List of Registers

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F3D8H	Code option register 0	CODEOP0	－	R	8	－*

*: Initial value of CODEOP0 depends on code-option data

29.2.2 Code-Option Register 0 (CODEOP0)

Address: 0F3D8H
Access: R
Access size: 8 bits
Initial value: depend on the code-option data

	7	6	5	4	3	2	1	0
CODEOP0	—*	—*	—*	—*	—*	—*	—*	COLD0
R/W	R	R	R	R	R	R	R	R
Initial value	x	x	x	x	x	x	x	x

CODEOP0 is a special function register (SFR) to check the setting value of code-option.
CODEOP0 is read only register and the writing to this register is invalid.
For the setting method of the code-option data, refer to 29.3 "The Setting Method of the Code-Option Data".

[Description of Bits]

- **COLD0** (bit 0)
  The COLD0 bit is used to select the voltage of releasing the system reset mode.

COLD0	Description
0	When power supply voltage (V_{DD}) is more than 1.1V (Max.), the system reset mode is released, and the CPU begins to execute instructions.
1	When power supply voltage (V_{DD}) is more than 1.8V (Max.), the system reset mode is released, and the CPU begins to execute instructions.

29.3 The Setting Method of the Code-Option Data

29.3.1 The Code-Option Data Format

The code-option data is set on address 0:ffe0h which is in writable test data area of program code.
For explanation of each bit of the code-option data, refer to 29.2.2 "Code-Option Register 0 (CODEOP0)".

	7	6	5	4	3	2	1	0
0:FDE0H	—*	—*	—*	—*	—*	—*	—*	COLD0

29.3.2 The Programming Method of the Code-Option

Figure 29-1 shows examples of program to set the code-option data.

- When the voltage on which CPU begins is 1.8V.

```
;------------------------------------------------------
;       Setting the code-option data
;------------------------------------------------------
        cseg at 0:0fde0h
        dw      0001h
;
```

- When the voltage on which CPU begins is 1.1V.

```
; ------------------------------------------------------
;       Setting the code-option data
; ------------------------------------------------------
        cseg at 0:0fde0h
        dw      0000h
;
```

Figure 29-1 Examples of program to set the code-option data

Note:
・To use this LSI, set the code-option data to writable test data area of program code.
・Fill the writable test data area except the code-option data with "0FFH".

Chapter 30

On-Chip Debug Function

30. On-Chip Debug Function

30.1 Overview

This LSI has an on-chip debug function allowing Flash memory rewriting.
The on-chip debug emulator (uEASE) is connected to this LSI to perform the on-chip debug function.

30.2 Method of Connecting to On-Chip Debug Emulator

Figure 30-1 shows connection to the on-chip debug emulator (uEASE).
For on-chip debug emulator, see "uEASE User's Manual".

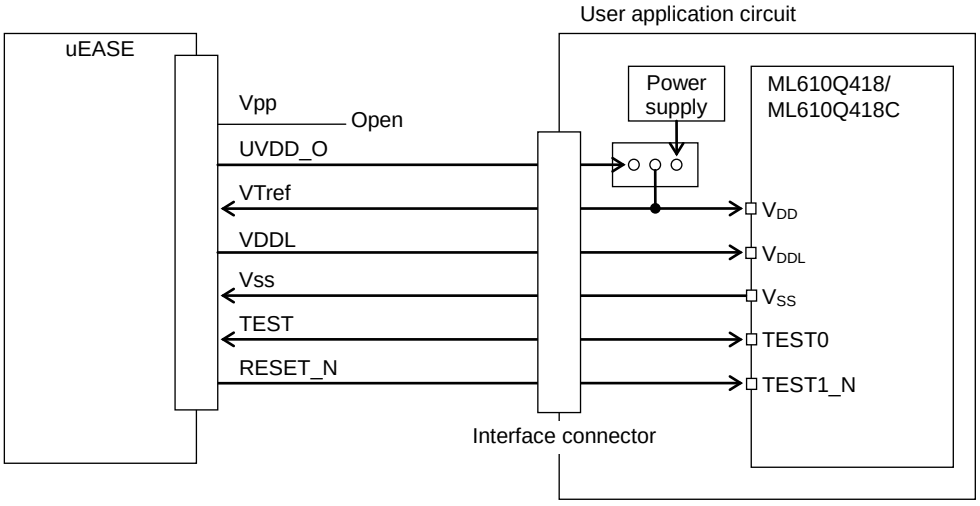


Figure 30-1 Connection to On-chip Debug Emulator (uEASE)

Note:
•Please do not apply LSIs being used for debugging to mass production.
•When using the on-chip debug function or the flash rewrite function after mounting of the board, design the board so that the 5 pins (VDD, VDDL, VSS, TEST0, and TEST1_N) required for connection to the on-chip debug emulator can be connected.
•"3.0V to 3.6V" has to be supplied to VDD while debugging and writing flash.

For details, see "uEASE User's Manual" and "uEASE Target Connection Manual".

30.3 Flash Memory Rewrite Function

Flash memory erase/write can be performed with the memory mounted on board by using the commands from the on-chip debug emulator (uEASE). For more details on the on-chip debug emulator, see "uEASE User's Manual".
Table 30-2 shows the Flash memory rewrite functions.

Table 30-2 Flash Memory Rewrite Functions

Function	Outline
Block erase	Erase of 8 Kwords (16 Kbytes)
1-word write	Write of 1 word (2 bytes)
Random read	Read of input address

Table 30-3 shows the conditions and specifications of Flash memory rewrite.

Table 30-3 Specifications of Flash Memory Rewrite

Parameter	Specifications
Rewrite count	100 cycles
Operating temperature	0°C to 40°C
Operating voltage (V_{DD})	1.8V to 3.6V
Chip-erase time	(Typ.) 68ms (Max.) 100ms
Block-erase time	(Typ.) 68ms (Max.) 100ms
1-word (16 bits) write	(Typ.) 18µs (Max.) 40µs
Overall-word (64K x 16 bits) write	Approx. 1.2s (Typ.) Approx. 2.7s (Max.)

Note:
When performing Flash memory rewrite (erase, write), a voltage within the range from 1.8V to 3.6V needs to be applied to the power supply voltage VDD.

Appendixes

Appendix A Registers

Contents of Registers

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F000H	Data segment register	DSR	—	R/W	8	00H
0F001H	Reset status register	RSTAT	—	R/W	8	Undefined
0F002H	Frequency control register 0	FCON0	FCON	R/W	8/16	33H
0F003H	Frequency control register 1	FCON1		R/W	8	00H
0F008H	Stop code acceptor	STPACP	—	W	8	00H
0F009H	Standby control register	SBYCON	—	W	8	00H
0F00AH	Low-speed time base counter divide register	LTBR	—	R/W	8	00H
0F00BH	High-speed time base counter divide register	HTBDR	—	R/W	8	00H
0F00CH	Low-speed time base counter frequency adjustment register L	LTBADJL	LTBADJ	R/W	8/16	00H
0F00DH	Low-speed time base counter frequency adjustment register H	LTBADJH		R/W	8	00H
0F00EH	Watchdog timer control register	WDTCON	—	R/W	8	00H
0F00FH	Watchdog timer mode register	WDTMOD	—	R/W	8	02H
0F011H	Interrupt permit register 1	IE1	—	R/W	8	00H
0F012H	Interrupt permit register 2	IE2	—	R/W	8	00H
0F013H	Interrupt permit register 3	IE3	—	R/W	8	00H
0F014H	Interrupt permit register 4	IE4	—	R/W	8	00H
0F015H	Interrupt permit register 5	IE5	—	R/W	8	00H
0F016H	Interrupt permit register 6	IE6	—	R/W	8	00H
0F017H	Interrupt permit register 7	IE7	—	R/W	8	00H
0F018H	Interrupt request register 0	IRQ0	—	R/W	8	00H
0F019H	Interrupt request register 1	IRQ1	—	R/W	8	00H
0F01AH	Interrupt request register 2	IRQ2	—	R/W	8	00H
0F01BH	Interrupt request register 3	IRQ3	—	R/W	8	00H
0F01CH	Interrupt request register 4	IRQ4	—	R/W	8	00H
0F01DH	Interrupt request register 5	IRQ5	—	R/W	8	00H
0F01EH	Interrupt request register 6	IRQ6	—	R/W	8	00H
0F01FH	Interrupt request register 7	IRQ7	—	R/W	8	00H
0F020H	External interrupt control register 0	EXICON0	—	R/W	8	00H
0F021H	External interrupt control register 1	EXICON1	—	R/W	8	00H
0F022H	External interrupt control register 2	EXICON2	—	R/W	8	00H
0F028H	Block control register 0	BLKCON0	—	R/W	8	00H
0F029H	Block control register 1	BLKCON1	—	R/W	8	00H
0F02AH	Block control register 2	BLKCON2	—	R/W	8	00H
0F02BH	Block control register 3	BLKCON3	—	R/W	8	00H
0F02CH	Block control register 4	BLKCON4	—	R/W	8	00H
0F030H	Timer 0 data register	TM0D	TM0DC	R/W	8/16	0FFH
0F031H	Timer 0 counter register	TM0C		R/W	8	00H
0F032H	Timer 0 control register 0	TM0CON0	TM0CON	R/W	8/16	00H
0F033H	Timer 0 control register 1	TM0CON1		R/W	8	00H
0F034H	Timer 1 data register	TM1D	TM1DC	R/W	8/16	0FFH
0F035H	Timer 1 counter register	TM1C		R/W	8	00H
0F036H	Timer 1 control register 0	TM1CON0	TM1CON	R/W	8/16	00H
0F037H	Timer 1 control register 1	TM1CON1		R/W	8	00H
0F038H	Timer 2 data register	TM2D	TM2DC	R/W	8/16	0FFH

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F039H	Timer 2 counter register	TM2C		R/W	8	00H
0F03AH	Timer 2 control register 0	TM2CON0	TM2CON	R/W	8/16	00H
0F03BH	Timer 2 control register 1	TM2CON1		R/W	8	00H
0F03CH	Timer 3 data register	TM3D	TM3DC	R/W	8/16	0FFH
0F03DH	Timer 3 counter register	TM3C		R/W	8	00H
0F03EH	Timer 3 control register 0	TM3CON0	TM3CON	R/W	8/16	00H
0F03FH	Timer 3 control register 1	TM3CON1		R/W	8	00H
0F090H	Capture control register	CAPCON	—	R/W	8/16	00H
0F091H	Capture status register	CAPSTAT	—	R/W	8	00H
0F092H	Capture data register 0	CAPR0	—	R/W	8	00H
0F093H	Capture data register 1	CAPR1	—	R/W	8	00H
0F094H	Capture time base data register	CAPTB	—	R	8	Undefined
0F0A0H	PWM0 period register L	PW0PL	PW0P	R/W	8/16	0FFH
0F0A1H	PWM0 period register H	PW0PH		R/W	8	0FFH
0F0A2H	PWM0 duty register L	PW0DL	PW0D	R/W	8/16	00H
0F0A3H	PWM0 duty register H	PW0DH		R/W	8	00H
0F0A4H	PWM0 counter register L	PW0CL	PW0C	R/W	8/16	00H
0F0A5H	PWM0 counter register H	PW0CH		R/W	8	00H
0F0A6H	PWM0 control register 0	PW0CON0	PW0CON	R/W	8/16	00H
0F0A7H	PWM0 control register 1	PW0CON1		R/W	8	40H
0F0D0H	Battery Level Detector control register 0	BLDCON0	BLDCON	R/W	8/16	00H
0F0D1H	Battery Level Detector control register 1	BLDCON1		R/W	8	00H
0F0E0H	Flash address register L	FLASHAL	FLASHA	R/W	8/16	0FFH
0F0E1H	Flash address register H	FLASHAH		R/W	8	0FFH
0F0E2H	Flash data register L	FLASHDL	FLASHD	R/W	8/16	00H
0F0E3H	Flash data register H	FLASHDH		R/W	8	00H
0F0E4H	Flash control register	FLASHCON	—	R/W	8	00H
0F0E5H	Flash acceptor	FLASHACP	—	R/W	8	00H
0F0E6H	Flash segment register	FLASHSEG	—	R/W	8	00H
0F0E7H	Flash self register	FLASHSLF	—	R/W	8	00H
0F0EAH	Flash status register	FLASHSTA	—	R	8	00H
0F0ECH	Flash remap register	REMAPADD	—	R/W	8	00H
0F0F0H	Bias circuit control register	BIASCON	—	R/W	8	38H
0F0F1H	Display contrast register	DSPCNT	—	R/W	8	00H
0F0F2H	Display mode register 0	DSPMOD0	DSPMOD	R/W	8/16	00H
0F0F3H	Display mode register 1	DSPMOD1		R/W	8	00H
0F0F4H	Display control register	DSPCON	—	R/W	8	00H
0F100H	Display register 00	DSPR00	—	R/W	8	Undefined
0F101H	Display register 01	DSPR01	—	R/W	8	Undefined
0F102H	Display register 02	DSPR02	—	R/W	8	Undefined
0F104H	Display register 04	DSPR04	—	R/W	8	Undefined
0F105H	Display register 05	DSPR05	—	R/W	8	Undefined
0F106H	Display register -06	DSPR06	—	R/W	8	Undefined
0F108H	Display register 08	DSPR08	—	R/W	8	Undefined
0F109H	Display register 09	DSPR09	—	R/W	8	Undefined
0F10AH	Display register 0A	DSPR0A	—	R/W	8	Undefined
0F10CH	Display register 0C	DSPR0C	—	R/W	8	Undefined
0F10DH	Display register 0D	DSPR0D	—	R/W	8	Undefined
0F10EH	Display register 0E	DSPR0E	—	R/W	8	Undefined
0F110H	Display register 10	DSPR10	—	R/W	8	Undefined

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F111H	Display register 11	DSPR11	—	R/W	8	Undefined
0F112H	Display register 12	DSPR12	—	R/W	8	Undefined
0F114H	Display register 14	DSPR14	—	R/W	8	Undefined
0F115H	Display register 15	DSPR15	—	R/W	8	Undefined
0F116H	Display register 16	DSPR16	—	R/W	8	Undefined
0F118H	Display register 18	DSPR18	—	R/W	8	Undefined
0F119H	Display register 19	DSPR19	—	R/W	8	Undefined
0F11AH	Display register 1A	DSPR1A	—	R/W	8	Undefined
0F11CH	Display register 1C	DSPR1C	—	R/W	8	Undefined
0F11DH	Display register 1D	DSPR1D	—	R/W	8	Undefined
0F11EH	Display register 1E	DSPR1E	—	R/W	8	Undefined
0F120H	Display register 20	DSPR20	—	R/W	8	Undefined
0F121H	Display register 21	DSPR21	—	R/W	8	Undefined
0F122H	Display register 22	DSPR22	—	R/W	8	Undefined
0F124H	Display register 24	DSPR24	—	R/W	8	Undefined
0F125H	Display register 25	DSPR25	—	R/W	8	Undefined
0F126H	Display register 26	DSPR26	—	R/W	8	Undefined
0F128H	Display register 28	DSPR28	—	R/W	8	Undefined
0F129H	Display register 29	DSPR29	—	R/W	8	Undefined
0F12AH	Display register 2A	DSPR2A	—	R/W	8	Undefined
0F12CH	Display register 2C	DSPR2C	—	R/W	8	Undefined
0F12DH	Display register 2D	DSPR2D	—	R/W	8	Undefined
0F12EH	Display register 2E	DSPR2E	—	R/W	8	Undefined
0F12FH	Display register 2F	DSPR2F	—	R/W	8	Undefined
0F204H	Port 0 data register	P0D	—	R	8	Undefined
0F206H	Port 0 control register 0	P0CON0	P0CON	R/W	8/16	00H
0F207H	Port 0 control register 1	P0CON1		R/W	8	00H
0F208H	Port 1 data register	P1D	—	R	8	Undefined
0F20AH	Port 1 control register 0	P1CON0	P1CON	R/W	8/16	00H
0F20BH	Port 1 control register 1	P1CON1		R/W	8	00H
0F210H	Port 2 data register	P2D	—	R/W	8	00H
0F212H	Port 2 control register 0	P2CON0	P2CON	R/W	8/16	00H
0F213H	Port 2 control register 1	P2CON1		R/W	8	00H
0F214H	Port 2 mode register 0	P2MOD0	P2MOD	R/W	8/16	00H
0F215H	Port 2 mode register 1	P2MOD1		R/W	8	00H
0F216H	Port 2 mode register 2	P2MOD2	—	R/W	8	00H
0F218H	Port 3 data register	P3D	—	R/W	8	00H
0F219H	Port 3 direction register	P3DIR	—	R/W	8	00H
0F21AH	Port 3 control register 0	P3CON0	P3CON	R/W	8/16	00H
0F21BH	Port 3 control register 1	P3CON1		R/W	8	00H
0F21CH	Port 3 mode register 0	P3MOD0	P3MOD	R/W	8/16	00H
0F21DH	Port 3 mode register 1	P3MOD1		R/W	8	00H
0F220H	Port 4 data register	P4D	—	R/W	8	00H
0F221H	Port 4 direction register	P4DIR	—	R/W	8	00H
0F222H	Port 4 control register 0	P4CON0	P4CON	R/W	8/16	00H
0F223H	Port 4 control register 1	P4CON1		R/W	8	00H
0F224H	Port 4 mode register 0	P4MOD0	P4MOD	R/W	8/16	00H
0F225H	Port 4 mode register 1	P4MOD1		R/W	8	00H
0F228H	Port 5 data register	P5D	—	R/W	8	0FH
0F229H	Port 5 direction register	P5DIR	—	R/W	8	00H

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F22AH	Port 5 control register 0	P5CON0	P5CON	R/W	8/16	00H
0F22BH	Port 5 control register 1	P5CON1		R/W	8	00H
0F22CH	Port 5 mode register 0	P5MOD0	P5MOD	R/W	8/16	00H
0F22DH	Port 5 mode register 1	P5MOD1		R/W	8	00H
0F22EH	Port 5 interrupt mode register	P5ISEL	—	R/W	8	00H
0F230H	Port 6 data register	P6D	—	R/W	8	0FH
0F231H	Port 6 direction register	P6DIR	—	R/W	8	00H
0F232H	Port 6 control register 0	P6CON0	P6CON	R/W	8/16	00H
0F233H	Port 6 control register 1	P6CON1		R/W	8	00H
0F280H	Serial port 0 transmit/receive buffer L	SIO0BUFL	SIO0BUF	R/W	8/16	00H
0F281H	Serial port 0 transmit/receive buffer H	SIO0BUFH		R/W	8	00H
0F282H	Serial port 0 control register	SIO0CON	—	R/W	8	00H
0F284H	Serial port 0 mode register 0	SIO0MOD0	SIO0MOD	R/W	8/16	00H
0F285H	Serial port 0 mode register 1	SIO0MOD1		R/W	8	00H
0F288H	Serial port 1 transmit/receive buffer L	SIO1BUFL	SIO1BUF	R/W	8/16	00H
0F289H	Serial port 1 transmit/receive buffer H	SIO1BUFH		R/W	8	00H
0F28AH	Serial port 1 control register	SIO1CON	—	R/W	8	00H
0F28CH	Serial port 1 mode register 0	SIO1MOD0	SIO1MOD	R/W	8/16	00H
0F28DH	Serial port 1 mode register 1	SIO1MOD1		R/W	8	00H
0F290H	UART0 transmit/receive buffer	UA0BUF	—	R/W	8	00H
0F291H	UART0 control register	UA0CON	—	R/W	8	00H
0F292H	UART0 mode register 0	UA0MOD0	UA0MOD	R/W	8/16	00H
0F293H	UART0 mode register 1	UA0MOD1		R/W	8	00H
0F294H	UART0 baud rate register L	UA0BRTL	UA0BRT	R/W	8/16	00H
0F295H	UART0 baud rate register H	UA0BRTH		R/W	8	00H
0F296H	UART0 status register	UA0STAT	—	R/W	8	00H
0F298H	UART1 transmit/receive buffer	UA1BUF	—	R/W	8	00H
0F299H	UART1 control register	UA1CON	—	R/W	8	00H
0F29AH	UART1 mode register 0	UA1MOD0	UA1MOD	R/W	8/16	00H
0F29BH	UART1 mode register 1	UA1MOD1		R/W	8	00H
0F29CH	UART1 baud rate register L	UA1BRTL	UA1BRT	R/W	8/16	00H
0F29DH	UART1 baud rate register H	UA1BRTH		R/W	8	00H
0F29EH	UART1 status register	UA1STAT	—	R/W	8	00H
0F2A0H	I2C bus 0 receive data register	I2C0RD	—	R	8	00H
0F2A1H	I2C bus 0 slave address register	I2C0SA	—	R/W	8	00H
0F2A2H	I2C bus 0 transmit data register	I2C0TD	—	R/W	8	00H
0F2A3H	I2C bus 0 control register	I2C0CON	—	R/W	8	00H
0F2A4H	I2C bus 0 mode register	I2C0MOD	—	R/W	8	00H
0F2A5H	I2C bus 0 status register	I2C0STAT	—	R	8	00H
0F2C0H	Melody 0 control register	MD0CON	—	R/W	8	00H
0F2C1H	Melody 0 tempo code register	MD0TMP	—	R/W	8	00H
0F2C2H	Melody 0 tone scale code register	MD0TON	MD0TL	R/W	8/16	00H
0F2C3H	Melody 0 tone length code register	MD0LEN		R/W	8	00H
0F2D0H	SA-ADC result register 0L	SADR0L	SADR0	R/W	8/16	00H
0F2D1H	SA-ADC result register 0H	SADR0H		R/W	8	00H
0F2D2H	SA-ADC result register 1L	SADR1L	SADR1	R/W	8/16	00H
0F2D3H	SA-ADC result register 1H	SADR1H		R/W	8	00H
0F2D4H	SA-ADC result register 2L	SADR2L	SADR2	R/W	8/16	00H
0F2D5H	SA-ADC result register 2H	SADR2H		R/W	8	00H
0F2D6H	SA-ADC result register 3L	SADR3L	SADR3	R/W	8/16	00H

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F2D7H	SA-ADC result register 3H	SADR3H		R/W	8	00H
0F2F0H	SA-ADC control register 0	SADCON0	SADCON	R/W	8/16	02H
0F2F1H	SA-ADC control register 1	SADCON1		R/W	8	00H
0F2F2H	SA-ADC mode register 0	SADMOD0	—	R/W	8	00H
0F300H	RC-ADC counter A register 0	RADCA0	—	R/W	8	00H
0F301H	RC-ADC counter A register 1	RADCA1	—	R/W	8	00H
0F302H	RC-ADC counter A register 2	RADCA2	—	R/W	8	00H
0F304H	RC-ADC counter B register 0	RADCB0	—	R/W	8	00H
0F305H	RC-ADC counter B register 1	RADCB1	—	R/W	8	00H
0F306H	RC-ADC counter B register 2	RADCB2	—	R/W	8	00H
0F308H	RC-ADC mode register	RADMOD	—	R/W	8	00H
0F309H	RC-ADC control register	RADCON	—	R/W	8	00H
0F400H	Display allocation register A	DS0C0A	—	R/W	8	Undefined
0F401H	Display allocation register A	DS1C0A	—	R/W	8	Undefined
0F402H	Display allocation register A	DS2C0A	—	R/W	8	Undefined
0F403H	Display allocation register A	DS3C0A	—	R/W	8	Undefined
0F404H	Display allocation register A	DS4C0A	—	R/W	8	Undefined
0F405H	Display allocation register A	DS5C0A	—	R/W	8	Undefined
0F406H	Display allocation register A	DS6C0A	—	R/W	8	Undefined
0F407H	Display allocation register A	DS7C0A	—	R/W	8	Undefined
0F408H	Display allocation register A	DS8C0A	—	R/W	8	Undefined
0F409H	Display allocation register A	DS9C0A	—	R/W	8	Undefined
0F40AH	Display allocation register A	DS10C0A	—	R/W	8	Undefined
0F40BH	Display allocation register A	DS11C0A	—	R/W	8	Undefined
0F40CH	Display allocation register A	DS12C0A	—	R/W	8	Undefined
0F40DH	Display allocation register A	DS13C0A	—	R/W	8	Undefined
0F40EH	Display allocation register A	DS14C0A	—	R/W	8	Undefined
0F40FH	Display allocation register A	DS15C0A	—	R/W	8	Undefined
0F410H	Display allocation register A	DS16C0A	—	R/W	8	Undefined
0F411H	Display allocation register A	DS17C0A	—	R/W	8	Undefined
0F412H	Display allocation register A	DS18C0A	—	R/W	8	Undefined
0F413H	Display allocation register A	DS19C0A	—	R/W	8	Undefined
0F414H	Display allocation register A	DS20C0A	—	R/W	8	Undefined
0F415H	Display allocation register A	DS21C0A	—	R/W	8	Undefined
0F416H	Display allocation register A	DS22C0A	—	R/W	8	Undefined
0F417H	Display allocation register A	DS23C0A	—	R/W	8	Undefined
0F418H	Display allocation register A	DS24C0A	—	R/W	8	Undefined
0F419H	Display allocation register A	DS25C0A	—	R/W	8	Undefined
0F41AH	Display allocation register A	DS26C0A	—	R/W	8	Undefined
0F41BH	Display allocation register A	DS27C0A	—	R/W	8	Undefined
0F41CH	Display allocation register A	DS28C0A	—	R/W	8	Undefined
0F41DH	Display allocation register A	DS29C0A	—	R/W	8	Undefined
0F41EH	Display allocation register A	DS30C0A	—	R/W	8	Undefined
0F41FH	Display allocation register A	DS31C0A	—	R/W	8	Undefined
0F420H	Display allocation register A	DS32C0A	—	R/W	8	Undefined
0F421H	Display allocation register A	DS33C0A	—	R/W	8	Undefined
0F422H	Display allocation register A	DS34C0A	—	R/W	8	Undefined
0F423H	Display allocation register A	DS35C0A	—	R/W	8	Undefined
0F424H	Display allocation register A	DS36C0A	—	R/W	8	Undefined
0F425H	Display allocation register A	DS37C0A	—	R/W	8	Undefined

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F426H	Display allocation register A	DS38C0A	—	R/W	8	Undefined
0F427H	Display allocation register A	DS39C0A	—	R/W	8	Undefined
0F428H	Display allocation register A	DS40C0A	—	R/W	8	Undefined
0F429H	Display allocation register A	DS41C0A	—	R/W	8	Undefined
0F42AH	Display allocation register A	DS42C0A	—	R/W	8	Undefined
0F42BH	Display allocation register A	DS43C0A	—	R/W	8	Undefined
0F42CH	Display allocation register A	DS44C0A	—	R/W	8	Undefined
0F42DH	Display allocation register A	DS45C0A	—	R/W	8	Undefined
0F42EH	Display allocation register A	DS46C0A	—	R/W	8	Undefined
0F42FH	Display allocation register A	DS47C0A	—	R/W	8	Undefined
0F440H	Display allocation register A	DS0C1A	—	R/W	8	Undefined
0F441H	Display allocation register A	DS1C1A	—	R/W	8	Undefined
0F442H	Display allocation register A	DS2C1A	—	R/W	8	Undefined
0F443H	Display allocation register A	DS3C1A	—	R/W	8	Undefined
0F444H	Display allocation register A	DS4C1A	—	R/W	8	Undefined
0F445H	Display allocation register A	DS5C1A	—	R/W	8	Undefined
0F446H	Display allocation register A	DS6C1A	—	R/W	8	Undefined
0F447H	Display allocation register A	DS7C1A	—	R/W	8	Undefined
0F448H	Display allocation register A	DS8C1A	—	R/W	8	Undefined
0F449H	Display allocation register A	DS9C1A	—	R/W	8	Undefined
0F44AH	Display allocation register A	DS10C1A	—	R/W	8	Undefined
0F44BH	Display allocation register A	DS11C1A	—	R/W	8	Undefined
0F44CH	Display allocation register A	DS12C1A	—	R/W	8	Undefined
0F44DH	Display allocation register A	DS13C1A	—	R/W	8	Undefined
0F44EH	Display allocation register A	DS14C1A	—	R/W	8	Undefined
0F44FH	Display allocation register A	DS15C1A	—	R/W	8	Undefined
0F450H	Display allocation register A	DS16C1A	—	R/W	8	Undefined
0F451H	Display allocation register A	DS17C1A	—	R/W	8	Undefined
0F452H	Display allocation register A	DS18C1A	—	R/W	8	Undefined
0F453H	Display allocation register A	DS19C1A	—	R/W	8	Undefined
0F454H	Display allocation register A	DS20C1A	—	R/W	8	Undefined
0F455H	Display allocation register A	DS21C1A	—	R/W	8	Undefined
0F456H	Display allocation register A	DS22C1A	—	R/W	8	Undefined
0F457H	Display allocation register A	DS23C1A	—	R/W	8	Undefined
0F458H	Display allocation register A	DS24C1A	—	R/W	8	Undefined
0F459H	Display allocation register A	DS25C1A	—	R/W	8	Undefined
0F45AH	Display allocation register A	DS26C1A	—	R/W	8	Undefined
0F45BH	Display allocation register A	DS27C1A	—	R/W	8	Undefined
0F45CH	Display allocation register A	DS28C1A	—	R/W	8	Undefined
0F45DH	Display allocation register A	DS29C1A	—	R/W	8	Undefined
0F45EH	Display allocation register A	DS30C1A	—	R/W	8	Undefined
0F45FH	Display allocation register A	DS31C1A	—	R/W	8	Undefined
0F460H	Display allocation register A	DS32C1A	—	R/W	8	Undefined
0F461H	Display allocation register A	DS33C1A	—	R/W	8	Undefined
0F462H	Display allocation register A	DS34C1A	—	R/W	8	Undefined
0F463H	Display allocation register A	DS35C1A	—	R/W	8	Undefined
0F464H	Display allocation register A	DS36C1A	—	R/W	8	Undefined
0F465H	Display allocation register A	DS37C1A	—	R/W	8	Undefined
0F466H	Display allocation register A	DS38C1A	—	R/W	8	Undefined
0F467H	Display allocation register A	DS39C1A	—	R/W	8	Undefined

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F468H	Display allocation register A	DS40C1A	—	R/W	8	Undefined
0F469H	Display allocation register A	DS41C1A	—	R/W	8	Undefined
0F46AH	Display allocation register A	DS42C1A	—	R/W	8	Undefined
0F46BH	Display allocation register A	DS43C1A	—	R/W	8	Undefined
0F46CH	Display allocation register A	DS44C1A	—	R/W	8	Undefined
0F46DH	Display allocation register A	DS45C1A	—	R/W	8	Undefined
0F46EH	Display allocation register A	DS46C1A	—	R/W	8	Undefined
0F46FH	Display allocation register A	DS47C1A	—	R/W	8	Undefined
0F480H	Display allocation register A	DS0C2A	—	R/W	8	Undefined
0F481H	Display allocation register A	DS1C2A	—	R/W	8	Undefined
0F482H	Display allocation register A	DS2C2A	—	R/W	8	Undefined
0F483H	Display allocation register A	DS3C2A	—	R/W	8	Undefined
0F484H	Display allocation register A	DS4C2A	—	R/W	8	Undefined
0F485H	Display allocation register A	DS5C2A	—	R/W	8	Undefined
0F486H	Display allocation register A	DS6C2A	—	R/W	8	Undefined
0F487H	Display allocation register A	DS7C2A	—	R/W	8	Undefined
0F488H	Display allocation register A	DS8C2A	—	R/W	8	Undefined
0F489H	Display allocation register A	DS9C2A	—	R/W	8	Undefined
0F48AH	Display allocation register A	DS10C2A	—	R/W	8	Undefined
0F48BH	Display allocation register A	DS11C2A	—	R/W	8	Undefined
0F48CH	Display allocation register A	DS12C2A	—	R/W	8	Undefined
0F48DH	Display allocation register A	DS13C2A	—	R/W	8	Undefined
0F48EH	Display allocation register A	DS14C2A	—	R/W	8	Undefined
0F48FH	Display allocation register A	DS15C2A	—	R/W	8	Undefined
0F490H	Display allocation register A	DS16C2A	—	R/W	8	Undefined
0F491H	Display allocation register A	DS17C2A	—	R/W	8	Undefined
0F492H	Display allocation register A	DS18C2A	—	R/W	8	Undefined
0F493H	Display allocation register A	DS19C2A	—	R/W	8	Undefined
0F494H	Display allocation register A	DS20C2A	—	R/W	8	Undefined
0F495H	Display allocation register A	DS21C2A	—	R/W	8	Undefined
0F496H	Display allocation register A	DS22C2A	—	R/W	8	Undefined
0F497H	Display allocation register A	DS23C2A	—	R/W	8	Undefined
0F498H	Display allocation register A	DS24C2A	—	R/W	8	Undefined
0F499H	Display allocation register A	DS25C2A	—	R/W	8	Undefined
0F49AH	Display allocation register A	DS26C2A	—	R/W	8	Undefined
0F49BH	Display allocation register A	DS27C2A	—	R/W	8	Undefined
0F49CH	Display allocation register A	DS28C2A	—	R/W	8	Undefined
0F49DH	Display allocation register A	DS29C2A	—	R/W	8	Undefined
0F49EH	Display allocation register A	DS30C2A	—	R/W	8	Undefined
0F49FH	Display allocation register A	DS31C2A	—	R/W	8	Undefined
0F4A0H	Display allocation register A	DS32C2A	—	R/W	8	Undefined
0F4A1H	Display allocation register A	DS33C2A	—	R/W	8	Undefined
0F4A2H	Display allocation register A	DS34C2A	—	R/W	8	Undefined
0F4A3H	Display allocation register A	DS35C2A	—	R/W	8	Undefined
0F4A4H	Display allocation register A	DS36C2A	—	R/W	8	Undefined
0F4A5H	Display allocation register A	DS37C2A	—	R/W	8	Undefined
0F4A6H	Display allocation register A	DS38C2A	—	R/W	8	Undefined
0F4A7H	Display allocation register A	DS39C2A	—	R/W	8	Undefined
0F4A8H	Display allocation register A	DS40C2A	—	R/W	8	Undefined
0F4A9H	Display allocation register A	DS41C2A	—	R/W	8	Undefined

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F4AAH	Display allocation register A	DS42C2A	—	R/W	8	Undefined
0F4ABH	Display allocation register A	DS43C2A	—	R/W	8	Undefined
0F4ACH	Display allocation register A	DS44C2A	—	R/W	8	Undefined
0F4ADH	Display allocation register A	DS45C2A	—	R/W	8	Undefined
0F4AEH	Display allocation register A	DS46C2A	—	R/W	8	Undefined
0F4AFH	Display allocation register A	DS47C2A	—	R/W	8	Undefined
0F4C0H	Display allocation register A	DS0C3A	—	R/W	8	Undefined
0F4C1H	Display allocation register A	DS1C3A	—	R/W	8	Undefined
0F4C2H	Display allocation register A	DS2C3A	—	R/W	8	Undefined
0F4C3H	Display allocation register A	DS3C3A	—	R/W	8	Undefined
0F4C4H	Display allocation register A	DS4C3A	—	R/W	8	Undefined
0F4C5H	Display allocation register A	DS5C3A	—	R/W	8	Undefined
0F4C6H	Display allocation register A	DS6C3A	—	R/W	8	Undefined
0F4C7H	Display allocation register A	DS7C3A	—	R/W	8	Undefined
0F4C8H	Display allocation register A	DS8C3A	—	R/W	8	Undefined
0F4C9H	Display allocation register A	DS9C3A	—	R/W	8	Undefined
0F4CAH	Display allocation register A	DS10C3A	—	R/W	8	Undefined
0F4CBH	Display allocation register A	DS11C3A	—	R/W	8	Undefined
0F4CCH	Display allocation register A	DS12C3A	—	R/W	8	Undefined
0F4CDH	Display allocation register A	DS13C3A	—	R/W	8	Undefined
0F4CEH	Display allocation register A	DS14C3A	—	R/W	8	Undefined
0F4CFH	Display allocation register A	DS15C3A	—	R/W	8	Undefined
0F4D0H	Display allocation register A	DS16C3A	—	R/W	8	Undefined
0F4D1H	Display allocation register A	DS17C3A	—	R/W	8	Undefined
0F4D2H	Display allocation register A	DS18C3A	—	R/W	8	Undefined
0F4D3H	Display allocation register A	DS19C3A	—	R/W	8	Undefined
0F4D4H	Display allocation register A	DS20C3A	—	R/W	8	Undefined
0F4D5H	Display allocation register A	DS21C3A	—	R/W	8	Undefined
0F4D6H	Display allocation register A	DS22C3A	—	R/W	8	Undefined
0F4D7H	Display allocation register A	DS23C3A	—	R/W	8	Undefined
0F4D8H	Display allocation register A	DS24C3A	—	R/W	8	Undefined
0F4D9H	Display allocation register A	DS25C3A	—	R/W	8	Undefined
0F4DAH	Display allocation register A	DS26C3A	—	R/W	8	Undefined
0F4DBH	Display allocation register A	DS27C3A	—	R/W	8	Undefined
0F4DCH	Display allocation register A	DS28C3A	—	R/W	8	Undefined
0F4DDH	Display allocation register A	DS29C3A	—	R/W	8	Undefined
0F4DEH	Display allocation register A	DS30C3A	—	R/W	8	Undefined
0F4DFH	Display allocation register A	DS31C3A	—	R/W	8	Undefined
0F4E0H	Display allocation register A	DS32C3A	—	R/W	8	Undefined
0F4E1H	Display allocation register A	DS33C3A	—	R/W	8	Undefined
0F4E2H	Display allocation register A	DS34C3A	—	R/W	8	Undefined
0F4E3H	Display allocation register A	DS35C3A	—	R/W	8	Undefined
0F4E4H	Display allocation register A	DS36C3A	—	R/W	8	Undefined
0F4E5H	Display allocation register A	DS37C3A	—	R/W	8	Undefined
0F4E6H	Display allocation register A	DS38C3A	—	R/W	8	Undefined
0F4E7H	Display allocation register A	DS39C3A	—	R/W	8	Undefined
0F4E8H	Display allocation register A	DS40C3A	—	R/W	8	Undefined
0F4E9H	Display allocation register A	DS41C3A	—	R/W	8	Undefined
0F4EAH	Display allocation register A	DS42C3A	—	R/W	8	Undefined
0F4EBH	Display allocation register A	DS43C3A	—	R/W	8	Undefined

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F4ECH	Display allocation register A	DS44C3A	—	R/W	8	Undefined
0F4EDH	Display allocation register A	DS45C3A	—	R/W	8	Undefined
0F4EEH	Display allocation register A	DS46C3A	—	R/W	8	Undefined
0F4EFH	Display allocation register A	DS47C3A	—	R/W	8	Undefined
0F600H	Display allocation register B	DS0C0B	—	R/W	8	Undefined
0F601H	Display allocation register B	DS1C0B	—	R/W	8	Undefined
0F602H	Display allocation register B	DS2C0B	—	R/W	8	Undefined
0F603H	Display allocation register B	DS3C0B	—	R/W	8	Undefined
0F604H	Display allocation register B	DS4C0B	—	R/W	8	Undefined
0F605H	Display allocation register B	DS5C0B	—	R/W	8	Undefined
0F606H	Display allocation register B	DS6C0B	—	R/W	8	Undefined
0F607H	Display allocation register B	DS7C0B	—	R/W	8	Undefined
0F608H	Display allocation register B	DS8C0B	—	R/W	8	Undefined
0F609H	Display allocation register B	DS9C0B	—	R/W	8	Undefined
0F60AH	Display allocation register B	DS10C0B	—	R/W	8	Undefined
0F60BH	Display allocation register B	DS11C0B	—	R/W	8	Undefined
0F60CH	Display allocation register B	DS12C0B	—	R/W	8	Undefined
0F60DH	Display allocation register B	DS13C0B	—	R/W	8	Undefined
0F60EH	Display allocation register B	DS14C0B	—	R/W	8	Undefined
0F60FH	Display allocation register B	DS15C0B	—	R/W	8	Undefined
0F610H	Display allocation register B	DS16C0B	—	R/W	8	Undefined
0F611H	Display allocation register B	DS17C0B	—	R/W	8	Undefined
0F612H	Display allocation register B	DS18C0B	—	R/W	8	Undefined
0F613H	Display allocation register B	DS19C0B	—	R/W	8	Undefined
0F614H	Display allocation register B	DS20C0B	—	R/W	8	Undefined
0F615H	Display allocation register B	DS21C0B	—	R/W	8	Undefined
0F616H	Display allocation register B	DS22C0B	—	R/W	8	Undefined
0F617H	Display allocation register B	DS23C0B	—	R/W	8	Undefined
0F618H	Display allocation register B	DS24C0B	—	R/W	8	Undefined
0F619H	Display allocation register B	DS25C0B	—	R/W	8	Undefined
0F61AH	Display allocation register B	DS26C0B	—	R/W	8	Undefined
0F61BH	Display allocation register B	DS27C0B	—	R/W	8	Undefined
0F61CH	Display allocation register B	DS28C0B	—	R/W	8	Undefined
0F61DH	Display allocation register B	DS29C0B	—	R/W	8	Undefined
0F61EH	Display allocation register B	DS30C0B	—	R/W	8	Undefined
0F61FH	Display allocation register B	DS31C0B	—	R/W	8	Undefined
0F620H	Display allocation register B	DS32C0B	—	R/W	8	Undefined
0F621H	Display allocation register B	DS33C0B	—	R/W	8	Undefined
0F622H	Display allocation register B	DS34C0B	—	R/W	8	Undefined
0F623H	Display allocation register B	DS35C0B	—	R/W	8	Undefined
0F624H	Display allocation register B	DS36C0B	—	R/W	8	Undefined
0F625H	Display allocation register B	DS37C0B	—	R/W	8	Undefined
0F626H	Display allocation register B	DS38C0B	—	R/W	8	Undefined
0F627H	Display allocation register B	DS39C0B	—	R/W	8	Undefined
0F628H	Display allocation register B	DS40C0B	—	R/W	8	Undefined
0F629H	Display allocation register B	DS41C0B	—	R/W	8	Undefined
0F62AH	Display allocation register B	DS42C0B	—	R/W	8	Undefined
0F62BH	Display allocation register B	DS43C0B	—	R/W	8	Undefined
0F62CH	Display allocation register B	DS44C0B	—	R/W	8	Undefined
0F62DH	Display allocation register B	DS45C0B	—	R/W	8	Undefined

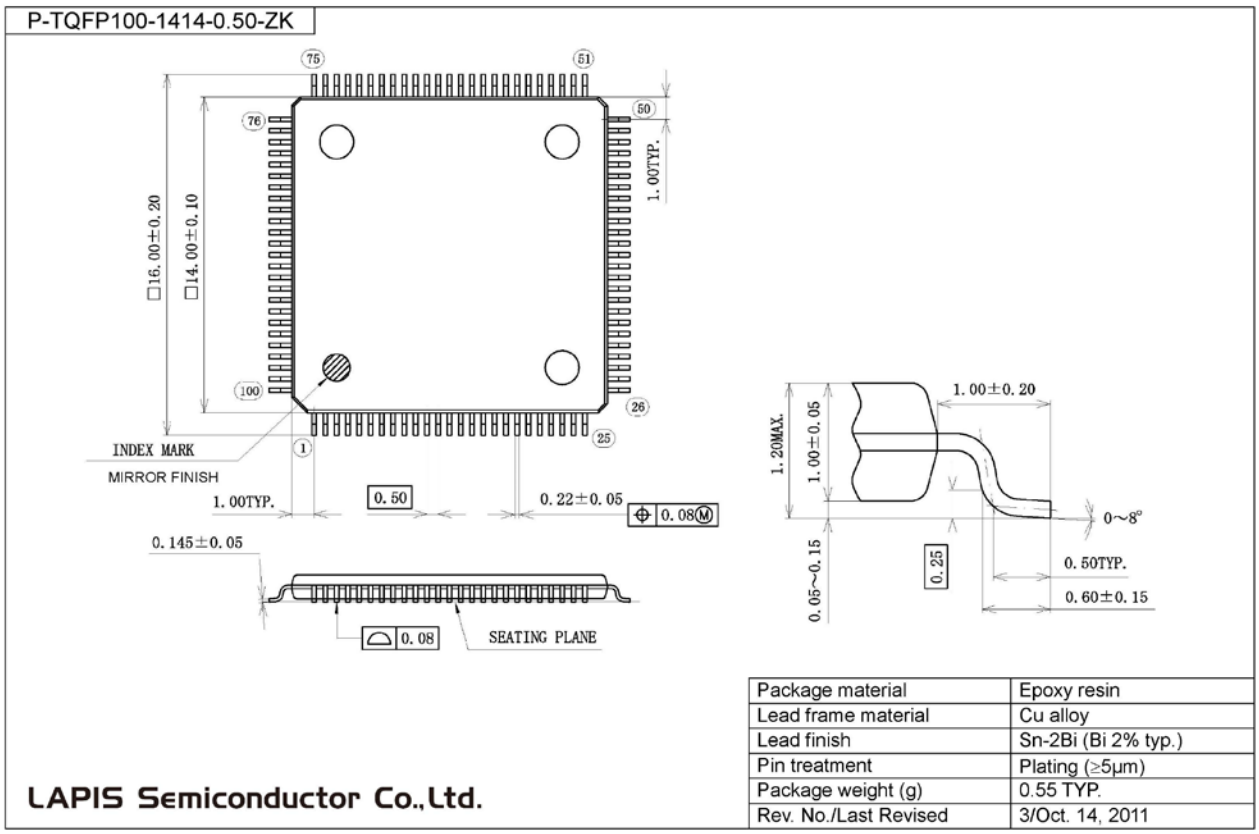
Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F62EH	Display allocation register B	DS46C0B	—	R/W	8	Undefined
0F62FH	Display allocation register B	DS47C0B	—	R/W	8	Undefined
0F640H	Display allocation register B	DS0C1B	—	R/W	8	Undefined
0F641H	Display allocation register B	DS1C1B	—	R/W	8	Undefined
0F642H	Display allocation register B	DS2C1B	—	R/W	8	Undefined
0F643H	Display allocation register B	DS3C1B	—	R/W	8	Undefined
0F644H	Display allocation register B	DS4C1B	—	R/W	8	Undefined
0F645H	Display allocation register B	DS5C1B	—	R/W	8	Undefined
0F646H	Display allocation register B	DS6C1B	—	R/W	8	Undefined
0F647H	Display allocation register B	DS7C1B	—	R/W	8	Undefined
0F648H	Display allocation register B	DS8C1B	—	R/W	8	Undefined
0F649H	Display allocation register B	DS9C1B	—	R/W	8	Undefined
0F64AH	Display allocation register B	DS10C1B	—	R/W	8	Undefined
0F64BH	Display allocation register B	DS11C1B	—	R/W	8	Undefined
0F64CH	Display allocation register B	DS12C1B	—	R/W	8	Undefined
0F64DH	Display allocation register B	DS13C1B	—	R/W	8	
0F64EH	Display allocation register B	DS14C1B	—	R/W	8	Undefined
0F64FH	Display allocation register B	DS15C1B	—	R/W	8	Undefined
0F650H	Display allocation register B	DS16C1B	—	R/W	8	Undefined
0F651H	Display allocation register B	DS17C1B	—	R/W	8	Undefined
0F652H	Display allocation register B	DS18C1B	—	R/W	8	Undefined
0F653H	Display allocation register B	DS19C1B	—	R/W	8	Undefined
0F654H	Display allocation register B	DS20C1B	—	R/W	8	Undefined
0F655H	Display allocation register B	DS21C1B	—	R/W	8	Undefined
0F656H	Display allocation register B	DS22C1B	—	R/W	8	Undefined
0F657H	Display allocation register B	DS23C1B	—	R/W	8	Undefined
0F658H	Display allocation register B	DS24C1B	—	R/W	8	Undefined
0F659H	Display allocation register B	DS25C1B	—	R/W	8	Undefined
0F65AH	Display allocation register B	DS26C1B	—	R/W	8	Undefined
0F65BH	Display allocation register B	DS27C1B	—	R/W	8	Undefined
0F65CH	Display allocation register B	DS28C1B	—	R/W	8	Undefined
0F65DH	Display allocation register B	DS29C1B	—	R/W	8	Undefined
0F65EH	Display allocation register B	DS30C1B	—	R/W	8	Undefined
0F65FH	Display allocation register B	DS31C1B	—	R/W	8	Undefined
0F660H	Display allocation register B	DS32C1B	—	R/W	8	Undefined
0F661H	Display allocation register B	DS33C1B	—	R/W	8	Undefined
0F662H	Display allocation register B	DS34C1B	—	R/W	8	Undefined
0F663H	Display allocation register B	DS35C1B	—	R/W	8	Undefined
0F664H	Display allocation register B	DS36C1B	—	R/W	8	Undefined
0F665H	Display allocation register B	DS37C1B	—	R/W	8	Undefined
0F666H	Display allocation register B	DS38C1B	—	R/W	8	Undefined
0F667H	Display allocation register B	DS39C1B	—	R/W	8	Undefined
0F668H	Display allocation register B	DS40C1B	—	R/W	8	Undefined
0F669H	Display allocation register B	DS41C1B	—	R/W	8	Undefined
0F66AH	Display allocation register B	DS42C1B	—	R/W	8	Undefined
0F66BH	Display allocation register B	DS43C1B	—	R/W	8	Undefined
0F66CH	Display allocation register B	DS44C1B	—	R/W	8	Undefined
0F66DH	Display allocation register B	DS45C1B	—	R/W	8	Undefined
0F66EH	Display allocation register B	DS46C1B	—	R/W	8	Undefined
0F66FH	Display allocation register B	DS47C1B	—	R/W	8	Undefined

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F680H	Display allocation register B	DS0C2B	—	R/W	8	Undefined
0F681H	Display allocation register B	DS1C2B	—	R/W	8	Undefined
0F682H	Display allocation register B	DS2C2B	—	R/W	8	Undefined
0F683H	Display allocation register B	DS3C2B	—	R/W	8	Undefined
0F684H	Display allocation register B	DS4C2B	—	R/W	8	Undefined
0F685H	Display allocation register B	DS5C2B	—	R/W	8	Undefined
0F686H	Display allocation register B	DS6C2B	—	R/W	8	Undefined
0F687H	Display allocation register B	DS7C2B	—	R/W	8	Undefined
0F688H	Display allocation register B	DS8C2B	—	R/W	8	Undefined
0F689H	Display allocation register B	DS9C2B	—	R/W	8	Undefined
0F68AH	Display allocation register B	DS10C2B	—	R/W	8	Undefined
0F68BH	Display allocation register B	DS11C2B	—	R/W	8	Undefined
0F68CH	Display allocation register B	DS12C2B	—	R/W	8	Undefined
0F68DH	Display allocation register B	DS13C2B	—	R/W	8	Undefined
0F68EH	Display allocation register B	DS14C2B	—	R/W	8	Undefined
0F68FH	Display allocation register B	DS15C2B	—	R/W	8	Undefined
0F690H	Display allocation register B	DS16C2B	—	R/W	8	Undefined
0F691H	Display allocation register B	DS17C2B	—	R/W	8	Undefined
0F692H	Display allocation register B	DS18C2B	—	R/W	8	Undefined
0F693H	Display allocation register B	DS19C2B	—	R/W	8	Undefined
0F694H	Display allocation register B	DS20C2B	—	R/W	8	Undefined
0F695H	Display allocation register B	DS21C2B	—	R/W	8	Undefined
0F696H	Display allocation register B	DS22C2B	—	R/W	8	Undefined
0F697H	Display allocation register B	DS23C2B	—	R/W	8	Undefined
0F698H	Display allocation register B	DS24C2B	—	R/W	8	Undefined
0F699H	Display allocation register B	DS25C2B	—	R/W	8	Undefined
0F69AH	Display allocation register B	DS26C2B	—	R/W	8	Undefined
0F69BH	Display allocation register B	DS27C2B	—	R/W	8	Undefined
0F69CH	Display allocation register B	DS28C2B	—	R/W	8	Undefined
0F69DH	Display allocation register B	DS29C2B	—	R/W	8	Undefined
0F69EH	Display allocation register B	DS30C2B	—	R/W	8	Undefined
0F69FH	Display allocation register B	DS31C2B	—	R/W	8	Undefined
0F6A0H	Display allocation register B	DS32C2B	—	R/W	8	Undefined
0F6A1H	Display allocation register B	DS33C2B	—	R/W	8	Undefined
0F6A2H	Display allocation register B	DS34C2B	—	R/W	8	Undefined
0F6A3H	Display allocation register B	DS35C2B	—	R/W	8	Undefined
0F6A4H	Display allocation register B	DS36C2B	—	R/W	8	Undefined
0F6A5H	Display allocation register B	DS37C2B	—	R/W	8	Undefined
0F6A6H	Display allocation register B	DS38C2B	—	R/W	8	Undefined
0F6A7H	Display allocation register B	DS39C2B	—	R/W	8	Undefined
0F6A8H	Display allocation register B	DS40C2B	—	R/W	8	Undefined
0F6A9H	Display allocation register B	DS41C2B	—	R/W	8	Undefined
0F6AAH	Display allocation register B	DS42C2B	—	R/W	8	Undefined
0F6ABH	Display allocation register B	DS43C2B	—	R/W	8	Undefined
0F6ACH	Display allocation register B	DS44C2B	—	R/W	8	Undefined
0F6ADH	Display allocation register B	DS45C2B	—	R/W	8	Undefined
0F6AEH	Display allocation register B	DS46C2B	—	R/W	8	Undefined
0F6AFH	Display allocation register B	DS47C2B	—	R/W	8	Undefined
0F6C0H	Display allocation register B	DS0C3B	—	R/W	8	Undefined
0F6C1H	Display allocation register B	DS1C3B	—	R/W	8	Undefined

Address	Name	Symbol (Byte)	Symbol (Word)	R/W	Size	Initial value
0F6C2H	Display allocation register B	DS2C3B	—	R/W	8	Undefined
0F6C3H	Display allocation register B	DS3C3B	—	R/W	8	Undefined
0F6C4H	Display allocation register B	DS4C3B	—	R/W	8	Undefined
0F6C5H	Display allocation register B	DS5C3B	—	R/W	8	Undefined
0F6C6H	Display allocation register B	DS6C3B	—	R/W	8	Undefined
0F6C7H	Display allocation register B	DS7C3B	—	R/W	8	Undefined
0F6C8H	Display allocation register B	DS8C3B	—	R/W	8	Undefined
0F6C9H	Display allocation register B	DS9C3B	—	R/W	8	Undefined
0F6CAH	Display allocation register B	DS10C3B	—	R/W	8	Undefined
0F6CBH	Display allocation register B	DS11C3B	—	R/W	8	Undefined
0F6CCH	Display allocation register B	DS12C3B	—	R/W	8	Undefined
0F6CDH	Display allocation register B	DS13C3B	—	R/W	8	Undefined
0F6CEH	Display allocation register B	DS14C3B	—	R/W	8	Undefined
0F6CFH	Display allocation register B	DS15C3B	—	R/W	8	Undefined
0F6D0H	Display allocation register B	DS16C3B	—	R/W	8	Undefined
0F6D1H	Display allocation register B	DS17C3B	—	R/W	8	Undefined
0F6D2H	Display allocation register B	DS18C3B	—	R/W	8	Undefined
0F6D3H	Display allocation register B	DS19C3B	—	R/W	8	Undefined
0F6D4H	Display allocation register B	DS20C3B	—	R/W	8	Undefined
0F6D5H	Display allocation register B	DS21C3B	—	R/W	8	Undefined
0F6D6H	Display allocation register B	DS22C3B	—	R/W	8	Undefined
0F6D7H	Display allocation register B	DS23C3B	—	R/W	8	Undefined
0F6D8H	Display allocation register B	DS24C3B	—	R/W	8	Undefined
0F6D9H	Display allocation register B	DS25C3B	—	R/W	8	Undefined
0F6DAH	Display allocation register B	DS26C3B	—	R/W	8	Undefined
0F6DBH	Display allocation register B	DS27C3B	—	R/W	8	Undefined
0F6DCH	Display allocation register B	DS28C3B	—	R/W	8	Undefined
0F6DDH	Display allocation register B	DS29C3B	—	R/W	8	Undefined
0F6DEH	Display allocation register B	DS30C3B	—	R/W	8	Undefined
0F6DFH	Display allocation register B	DS31C3B	—	R/W	8	Undefined
0F6E0H	Display allocation register B	DS32C3B	—	R/W	8	Undefined
0F6E1H	Display allocation register B	DS33C3B	—	R/W	8	Undefined
0F6E2H	Display allocation register B	DS34C3B	—	R/W	8	Undefined
0F6E3H	Display allocation register B	DS35C3B	—	R/W	8	Undefined
0F6E4H	Display allocation register B	DS36C3B	—	R/W	8	Undefined
0F6E5H	Display allocation register B	DS37C3B	—	R/W	8	Undefined
0F6E6H	Display allocation register B	DS38C3B	—	R/W	8	Undefined
0F6E7H	Display allocation register B	DS39C3B	—	R/W	8	Undefined
0F6E8H	Display allocation register B	DS40C3B	—	R/W	8	Undefined
0F6E9H	Display allocation register B	DS41C3B	—	R/W	8	Undefined
0F6EAH	Display allocation register B	DS42C3B	—	R/W	8	Undefined
0F6EBH	Display allocation register B	DS43C3B	—	R/W	8	Undefined
0F6ECH	Display allocation register B	DS44C3B	—	R/W	8	Undefined
0F6EDH	Display allocation register B	DS45C3B	—	R/W	8	Undefined
0F6EEH	Display allocation register B	DS46C3B	—	R/W	8	Undefined
0F6EFH	Display allocation register B	DS47C3B	—	R/W	8	Undefined

Appendix B Package Dimensions

(Unit: mm)



Package material	Epoxy resin
Lead frame material	Cu alloy
Lead finish	Sn-2Bi (Bi 2% typ.)
Pin treatment	Plating (≥5μm)
Package weight (g)	0.55 TYP.
Rev. No./Last Revised	3/Oct. 14, 2011

Notes for Mounting the Surface Mount Type Package

The surface mount type packages are very susceptible to heat in reflow mounting and humidity absorbed in storage. Therefore, before you perform reflow mounting, contact a ROHM sales office for the product name, package name, pin number, package code and desired mounting conditions (reflow method, temperature and times).

Appendix C Electrical Characteristics

● Absolute Maximum Ratings

(V_{SS} = 0V)

Parameter	Symbol	Condition	Rating	Unit
Power supply voltage 1	V_{DD}	Ta = 25°C	–0.3 to +4.6	V
Power supply voltage 2	AV_{DD}	Ta = 25°C	–0.3 to +4.6	V
Power supply voltage 3	V_{DDL}	Ta = 25°C	–0.3 to +3.6	V
Power supply voltage 4	V_{L1}	Ta = 25°C	–0.3 to +1.75	V
Power supply voltage 5	V_{L2}	Ta = 25°C	–0.3 to +3.5	V
Power supply voltage 6	V_{L3}	Ta = 25°C	–0.3 to +5.25	V
Input voltage	V_{IN}	Ta = 25°C	–0.3 to V_{DD}+0.3	V
Output voltage	V_{OUT}	Ta = 25°C	–0.3 to V_{DD}+0.3	V
Output current 1	I_{OUT1}	Port3–5, Ta = 25°C	–12 to +11	mA
Output current 2	I_{OUT2}	Port2, Ta = 25°C	–12 to +20	mA
Power dissipation	PD	Ta = 25°C	0.9	W
Storage temperature	T_{STG}	—	–55 to +150	°C

● Recommended Operating Conditions

(V_{SS} = 0V)

Parameter	Symbol	Condition	Range	Unit
Operating temperature	T_{OP}	—	-20～+70	°C
Operating voltage	V_{DD}	f_{OP} = 30k to 625kHz	1.1～3.6	V
		f_{OP} = 30k to 4.2MHz	1.8～3.6	V
Operating frequency (CPU)	f_{OP}	VDD = 1.1～3.6V	30k～36k	Hz
		VDD = 1.3～3.6V	30k～650k	
		VDD = 1.8～3.6V	30k～4.2M	
Capacitor externally connected to V_{DD} pin	C_V	—	More than 2.2±30%	μF
Capacitor externally connected to V_{DDL} pin	C_{L0}	—	2.2±30%	μF
	C_{L1}	—	0.1±30%	
Capacitors externally connected to $V_{L1, 2, 3}$ pins	$C_{a, b, c}$	—	0.1±30%	μF
Capacitors externally connected across C1 and C2 pins	C_{12}	—	0.47±30%	μF

● Clock Generation Circuit Operating Conditions

(V_{SS} = 0V)

Parameter	Symbol	Condition	Rating			Unit
			Min.	Typ.	Max.	
Low-speed crystal oscillation frequency	f_{XTL}	—	—	32.768k	—	Hz
Recommended equivalent series resistance value of low-speed crystal oscillation	R_L	—	—	—	40k	Ω
Low-speed crystal oscillation external capacitor	C_{DL}/C_{GL}	C_L=3pF of crystal oscillation	—	6	—	pF
		C_L=6pF of crystal oscillation	—	12	—	
		C_L=9pF of crystal oscillation	—	18	—	
High-speed crystal/ceramic oscillation frequency	f_{XTH}	—	—	4.0M / 4.096M	—	Hz
High-speed crystal oscillation external capacitor	C_{DH}	—	—	24	—	pF
	C_{GH}	—	—	24	—	

● Operating Conditions of Flash Memory

(V_{SS} = 0V)

Parameter	Symbol	Condition	Rating			Unit
			Min.	Typ.	Max.	
Operating temperature	T_{OP}	Flash ROM, At write/erase	0	—	+40	°C
		Data flash memory, At write/erase	-20	—	+70	°C
Operating voltage	V_{DD}	At write/erase	1.8	—	3.6	V
Rewrite counts	C_{EP}	Flash ROM	—	—	100	cycles
		Data flash memory	—	—	10k	
Data retention	Y_{DR}	Flash ROM	10	—		years
		Data flash memory, 1000 cycles	10	—		
Chip-erase time	t_{CERASE}	—		85	100	ms
Block-erase time	t_{BERASE}	—		85	100	ms
Sector-erase time	t_{SERASE}	—		85	100	ms
1-word (16 bits) write time	t_{WRITE}	—		18	40	μs

● DC Characteristics (1/5)

(V_{DD} = 1.1 to 3.6V, AV_{DD} =2.2 to 3.6V, V_{SS} = AV_{SS} = 0V, Ta = -20 to +70°C,unless otherwise specified)

Parameter	Symbol	Condition		Rating Min.	Rating Typ.	Rating Max.	Unit	Measuring circuit
500kHz RC oscillation frequency	f_{RC}	V_{DD} = 1.3 to 3.6V	Ta = 25°C	Typ. −10%	500	Typ. +10%	kHz	1
			Ta = -20 to +70°C	Typ. −25%	500	Typ. +25%	kHz	
PLL oscillation frequency*4	f_{PLL}	LSCLK = 32.768kHz V_{DD} = 1.8 to 3.6V		-2.5%	8.192	+2.5%	MHz	
Low-speed crystal oscillation start time*2	T_{XTL}	—		—	0.3	2	s	
500kHz RC oscillation start time	T_{RC}	V_{DD} = 1.3～3.6V		—	50	500	μs	
High-speed crystal oscillation start time*3	T_{XTH}	V_{DD} = 1.8 to 3.6V		—	2	20	ms	
PLL oscillation start time	T_{PLL}	V_{DD} = 1.8 to 3.6V		—	1	10		
Low-speed oscillation stop detect time*1	T_{STOP}	—		0.2	3	20		
Reset pulse width	P_{RST}	—		200	—	—	μs	
Reset noise elimination pulse width	P_{NRST}	—		—	—	0.3		
Power-on reset activation power rise time	T_{POR}	—		—	—	10	ms	
Low level reset detection voltage	V_{LLR}	COLD0=0*5		—	—	1.1	V	
		COLD0=1*5		—	—	1.8		
Low level reset detection time	T_{LLR}	—		200	—	—	μs	
Release reset voltage	V_{RER}	COLD0=0*5		—	—	1.1	V	
		COLD0=1*5		—	—	1.8		

*1: When low-speed crystal oscillation stops for a duration more than the low-speed oscillation stop detect time, the system is reset to shift to system reset mode.

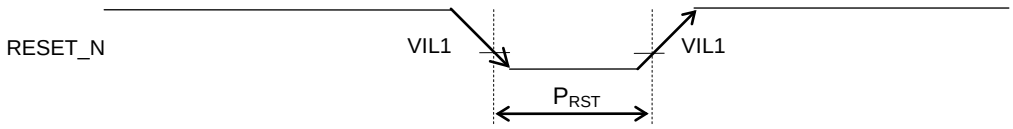
*2 : Use 32.768kHz crystal resonator DT-26 (Load capacitance 6pF) (made by KDS:DAISHINKU CORP.) is used (C_{GL}=C_{DL}=6pF).

*3 : Use 4.096MHz Crystal Oscillator CHC49SFWB (Kyocera).

*4 : 1024 clock average.

*5 : The COLD0 bit is the code-option which is set up into the Flash memory.

Reset pulse width (P_{RST})



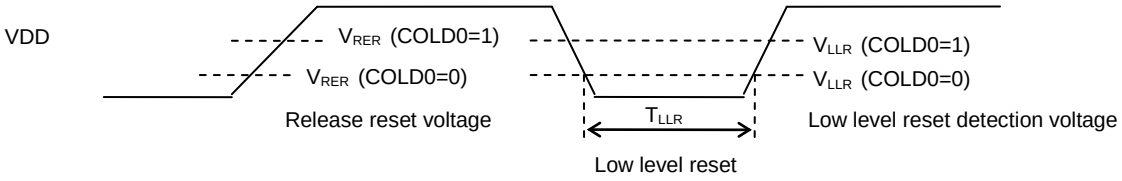
Reset pulse width (P_{RST})

Power-on reset activation power rise time (T_{POR})



Power-on reset activation power rise time (T_{POR})

Low level reset detection time(T_{LLR})



Low level reset detection time(T_{LLR})

● DC Characteristics (2/5)

(V_{DD} = 1.1 to 3.6V, AV_{DD} =2.2 to 3.6V, V_{SS} = AV_{SS} = 0V, Ta = -20 to +70°C,unless otherwise specified)

Parameter	Symbol	Condition		Rating Min.	Rating Typ.	Rating Max.	Unit	Measuring circuit
V_{L1} voltage	V_{L1}	V_{DD} = 3.0V, Tj = 25°C	CN4-0 = 00H	0.89	0.94	0.99	V	1
			CN4-0 = 01H	0.91	0.96	1.01		
			CN4-0 = 02H	0.93	0.98	1.03		
			CN4-0 = 03H	0.95	1.00	1.05		
			CN4-0 = 04H	0.97	1.02	1.07		
			CN4-0 = 05H	0.99	1.04	1.09		
			CN4-0 = 06H	1.01	1.06	1.11		
			CN4-0 = 07H	1.03	1.08	1.13		
			CN4-0 = 08H	1.05	1.10	1.15		
			CN4-0 = 09H	1.07	1.12	1.17		
			CN4-0 = 0AH	1.09	1.14	1.19		
			CN4-0 = 0BH	1.11	1.16	1.21		
			CN4-0 = 0CH	1.13	1.18	1.23		
			CN4-0 = 0DH	1.15	1.20	1.25		
			CN4-0 = 0EH	1.17	1.22	1.27		
			CN4-0 = 0FH	1.19	1.24	1.29		
			CN4-0 = 10H	1.21	1.26	1.31		
			CN4-0 = 11H	1.23	1.28	1.33		
			CN4-0 = 12H	1.25	1.30	1.35		
			CN4-0 = 13H	1.27	1.32	1.37		
			CN4-0 = 14H*1	1.29	1.34	1.39		
			CN4-0 = 15H*1	1.31	1.36	1.41		
			CN4-0 = 16H*1	1.33	1.38	1.43		
			CN4-0 = 17H*1	1.35	1.40	1.45		
			CN4-0 = 18H*1	1.37	1.42	1.47		
			CN4-0 = 19H*1	1.39	1.44	1.49		
			CN4-0 = 1AH*1	1.41	1.46	1.51		
			CN4-0 = 1BH*1	1.43	1.48	1.53		
			CN4-0 = 1CH*1	1.45	1.50	1.55		
			CN4-0 = 1DH*1	1.47	1.52	1.57		
			CN4-0 = 1EH*1	1.49	1.54	1.59		
			CN4-0 = 1FH*1	1.51	1.56	1.61		
V_{L1} temperature deviation	ΔV_{L1}	V_{DD} = 3.0V		—	-1.5	—	mV/°C	
V_{L1} voltage dependency	ΔV_{L1}	V_{DD} = 1.3 to 3.6V		—	5	20	mV/V	
V_{L2} voltage	V_{L2}	VDD = 3.0V, Tj = 25°C, 300kΩ load (V_{L3}–V_{SS})	1/2bias	—	V_{L1}×1	—	V	
			1/3bias	Typ. ×09	V_{L1}×2	—		
V_{L3} voltage	V_{L3}		1/2bias	Typ. ×09	V_{L1}×2	—		
			1/3bias	Typ. ×09	V_{L1}×3	—		
LCD bias voltage generation time	T_{BIAS}	—		—	—	100	ms	

● DC Characteristics (3/5)

(V_{DD} = 1.1 to 3.6V, AV_{DD} =2.2 to 3.6V, V_{SS} = AV_{SS} = 0V, Ta = -20 to +70°C, unless otherwise specified)

Parameter	Symbol	Condition		Rating Min.	Rating Typ.	Rating Max.	Unit	Measuring circuit
BLD threshold voltage	V_{BLD}	V_{DD} = 1.35 to 3.6V	LD2–0 = 0H	Typ. –2%	1.35	Typ. +2%	V	1
			LD2–0 = 1H		1.4			
			LD2–0 = 2H		1.45			
			LD2–0 = 3H		1.5			
			LD2–0 = 4H		1.6			
			LD2–0 = 5H		1.7			
			LD2–0 = 6H		1.8			
			LD2–0 = 7H		1.9			
			LD2–0 = 8H		2.0			
			LD2–0 = 9H		2.1			
			LD2–0 = 0AH		2.2			
			LD2–0 = 0BH		2.3			
			LD2–0 = 0CH		2.4			
			LD2–0 = 0DH		2.5			
			LD2–0 = 0EH		2.7			
			LD2–0 = 0FH		2.9			
BLD threshold voltage temperature deviation	ΔV_{BLD}	V_{DD} = 1.35 to 3.6V		—	0	—	%/°C	
Supply current 1	IDD1	CPU: In STOP state. Low-speed/high-speed oscillation: stopped.	Ta = 25°C	—	0.55	0.9	μA	
			Ta = -20 to +70°C	—	—	8		
Supply current 2	IDD2	CPU: In HALT state (LTBC, RTC: Operating*3*5). High-speed oscillation: Stopped. LCD/BIAS circuits: Stopped.	Ta = 25°C	—	1.1	2.2	μA	
			Ta = -20 to +70°C	—	—	9		
Supply current 3	IDD3	CPU: In 32.768kHz operating state.*1*3 High-speed oscillation: Stopped. LCD/BIAS circuits: Operating.*2	Ta = 25°C	—	6.5	8	μA	
			Ta = -20 to +70°C	—	—	15		
Supply current 4	IDD4	CPU: In 500kHz CR operating state. LCD/BIAS circuits: Operating.*2*3	Ta = 25°C	—	80	100	μA	
			Ta = -20 to +70°C	—	—	120		
Supply current 5	IDD5	CPU: In 4.096MHz operating state.*2*3 PLL: In oscillating state. LCD/BIAS circuits: Operating. *2 V_{DD} = 1.8 to 3.6V	Ta = 25°C	—	1.0	1.1	mA	
			Ta = -20 to +70°C	—	—	1.2		
Supply current 6	IDD6	CPU: In 4.096MHz operating state.*2 PLL: In oscillating state. *3*4 A/D: In operating state. LCD/BIAS circuits: Operating. *2 V_{DD} = AV_{DD} = 3.0V	Ta = 25°C	—	1.6	1.7	mA	
			Ta = -20 to +70°C	—	—	2.5		

*1 : CPU operating rate is 100% (No HALT state).
*2 : All SEGs: off waveform, No LCD panel load, 1/3 bias, 1/3 duty, Frame frequency: Approx. 64 Hz, Bias voltage multiplying clock: 1/128 LSCLK (256Hz)
*3 : Use 32.768kHz crystal resonator DT-26 (Load capacitance 6pF) (made by KDS:DAISHINKU CORP.) is used (C_{GL}=C_{DL}=6pF).
*4 : Use 4.096MHz Crystal Oscillator CHC49SFWB (Kyocera).
*5 : Significant bits of BLKCON0~BLKCON4 registers are all "1".

● DC Characteristics (4/5)

(V_{DD} = 1.1 to 3.6V, AV_{DD} =2.2 to 3.6V, V_{SS} = AV_{SS} = 0V, Ta = -20 to +70°C,unless otherwise specified)

Parameter	Symbol	Condition		Rating Min.	Rating Typ.	Rating Max.	Unit	Measuring circuit
Output voltage 1 (P20–P22/2nd function is selected) (P30–P36) (P40–P47) (P50–P53) (P60–P67)*1	VOH1	IOH1 = –0.5mA, V_{DD} = 1.8 to 3.6V		V_{DD}–0.5	—	—	V	2
		IOH1 = -0.1mA, V_{DD} = 1.3 to 3.6V		V_{DD}–0.3	—	—		
		IOH1 = -0.03mA, VDD = 1.1 to 3.6V		V_{DD}–0.3	—	—		
	VOL1	IOL1 = +0.5mA, V_{DD} = 1.8 to 3.6V		—	—	0.5		
		IOL1 = +0.1mA, V_{DD} = 1.3 to 3.6V		—	—	0.5		
		IOL1 = +0.03mA, V_{DD} = 1.1 to 3.6V		—	—	0.3		
Output voltage 2 (P20–P22/2nd function is Not selected)	VOH2	IOH1 = –0.5mA, V_{DD} = 1.8 to 3.6V		V_{DD}–0.5	—	—		
		IOH1 = -0.1mA, V_{DD} = 1.3 to 3.6V		V_{DD}–0.3	—	—		
		IOH1 = -0.03mA, V_{DD} = 1.1 to 3.6V		V_{DD}–0.3	—	—		
	VOL2	IOL2 = +5mA, V_{DD} = 1.8 to 3.6V		—	—	0.5		
Output voltage 3 (P40–P41)	VOL3	IOL3 = +3mA, V_{DD} = 2.0 to 3.6V (when I^2C mode is selected)		—	—	0.4		
Output voltage 4 (COM0–3) (SEG0–39) (SEG40–47)*2	VOH4	IOH4 = –0.05mA, VL1=1.2V		V_{L3}–0.2	—	—		
	VOML4	IOMH4 = +0.05mA, VL1=1.2V		—	—	V_{L2}+0.2		
	VOML4S	IOMH4S = –0.05mA, VL1=1.2V		V_{L2}–0.2	—	—		
	VOLM4	IOML4 = +0.05mA, VL1=1.2V		—	—	V_{L1}+0.2		
	VOLM4S	IOML4S = –0.05mA, VL1=1.2V		V_{L1}–0.2	—	—		
	VOL4	IOL4 = +0.05mA, VL1=1.2V		—	—	0.2		
Output leakage (P20–P22) (P30–P35) (P40–P47) (P50–P53) (P60–P67)*1	IOOH	VOH = V_{DD} (in high-impedance state)		—	—	1	μA	3
	IOOL	VOL = V_{SS} (in high-impedance state)		–1	—	—		
Input current 1 (RESET_N) (TEST1_N)	IIH1	VIH1 = V_{DD}		0	—	1	μA	4
	IIL1	VIL1 = V_{SS}	V_{DD} = 1.8 to 3.6V	–600	–300	–20		
			V_{DD} = 1.3 to 3.6V	–600	–300	-10		
			V_{DD} = 1.1 to 3.6V	–600	–300	-2		
Input current 1 (TEST0)	IIH1	VIH1 = V_{DD}	V_{DD} = 1.8 to 3.6V	20	300	600		
			V_{DD} = 1.3 to 3.6V	10	300	600		
			V_{DD} = 1.1 to 3.6V	2	300	600		
	IIL1	VIL1 = V_{ss}		-1	—	—		
Input current 2 (P00–P03) (P10–P11) (P30–P35) (P40–P47) (P50–P53) (P60–P67)*1	IIH2	VIH2 = V_{DD} (when pulled-down)	V_{DD} = 1.8 to 3.6V	2	30	200		
			V_{DD} = 1.3 to 3.6V	0.2	30	200		
			V_{DD} = 1.1 to 3.6V	0.01	30	200		
	IIL2	VIL2 = V_{SS} (when pulled-up)	V_{DD} = 1.8 to 3.6V	–200	–30	–2		
			V_{DD} = 1.3 to 3.6V	–200	–30	-0.2		
			V_{DD} = 1.1 to 3.6V	–200	–30	-0.01		
	IIH2Z	VIH2 = V_{DD} (in high-impedance state)		—	—	1		
	IIL2Z	VIL2 = V_{SS} (in high-impedance state)		–1	—	—		

*1: ML610Q418C only
*2: ML610Q418 only

● DC Characteristics (5/5)

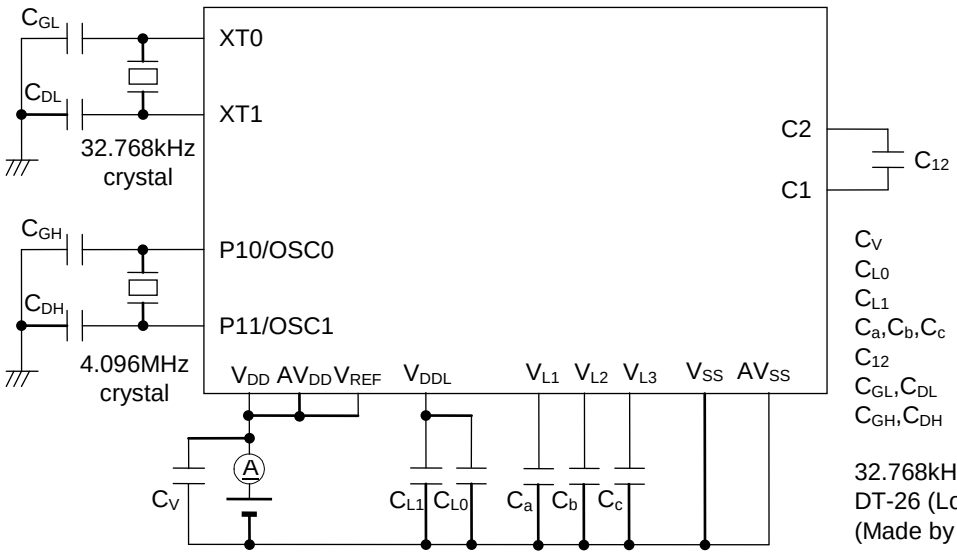
(V_{DD} = 1.1 to 3.6V, AV_{DD} =2.2 to 3.6V, V_{SS} = AV_{SS} = 0V, Ta = -20 to +70°C,unless otherwise specified)

Parameter	Symbol	Condition	Rating			Unit	Measuring circuit
			Min.	Typ.	Max.		
Input voltage 1 (RESET_N) (TEST1_N) (TEST0) (P00–P03) (P10–P11) (P31–P35) (P40–P43) (P45–P47) (P50–P53) (P60–P67)*1	VIH1	V_{DD} = 1.3 to 3.6V	0.7 ×V_{DD}	—	V_{DD}	V	5
		V_{DD} = 1.1 to 3.6V	0.7 ×V_{DD}	—	V_{DD}		
	VIL1	V_{DD} = 1.3 to 3.6V	0	—	0.3 ×V_{DD}		
		V_{DD} = 1.1 to 3.6V	0	—	0.2 ×V_{DD}		
Input voltage 2 (P30, P44)	VIH2	—	0.7 ×V_{DD}	—	V_{DD}		
	VIL2	—	0	—	0.3 ×V_{DD}		
Input pin capacitance (P00–P03) (P10–P11) (P30–P35) (P40–P47) (P50–P53) (P60–P67)*1	CIN	f = 10kHz V_{rms} = 50mV Ta = 25°C	—	—	5	pF	—

*1: ML610Q418C only

● Measuring circuits

Measuring circuit 1

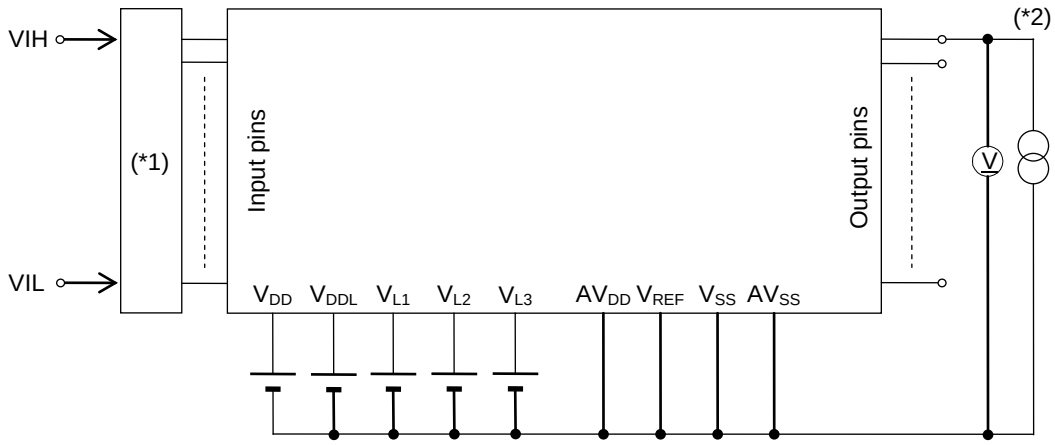


C_V	: >2.2µF
C_{L0}	: 2.2µF
C_{L1}	: 0.1µF
C_a,C_b,C_c	: 0.1µF
C_{12}	: 0.47µF
C_{GL},C_{DL}	: 6pF
C_{GH},C_{DH}	: 24pF

32.768kHz crystal resonator
DT-26 (Load capacitance 6pF)
(Made by KDS:DAISHINKU CORP.)

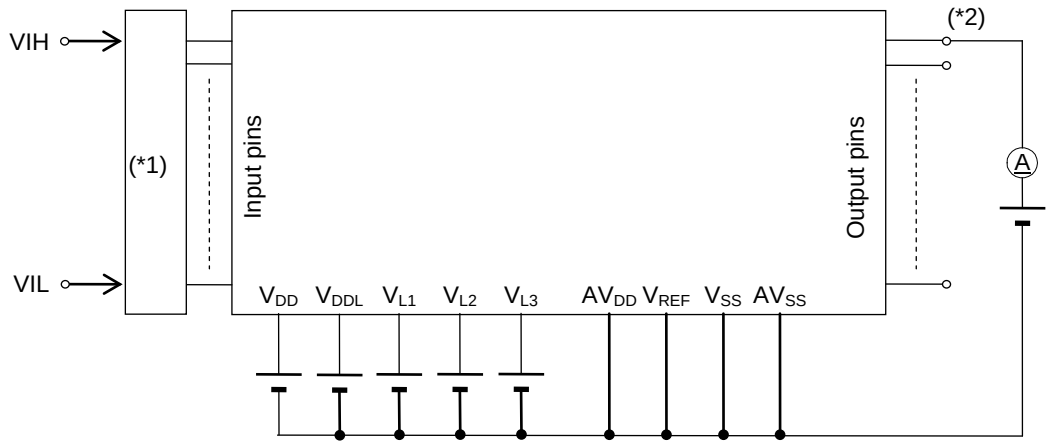
4.096MHz crystal:
HC49SFWB (Kyocera)

Measuring circuit 2



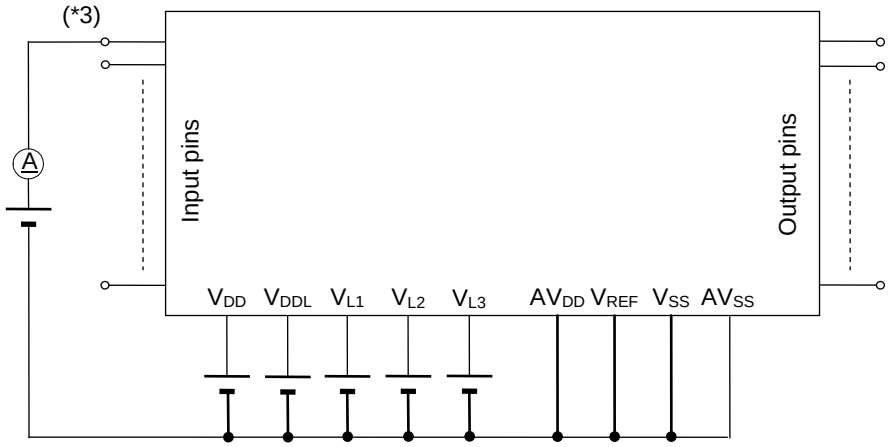
(*1) Input logic circuit to determine the specified measuring conditions.
(*2) Measured at the specified output pins.

Measuring circuit 3



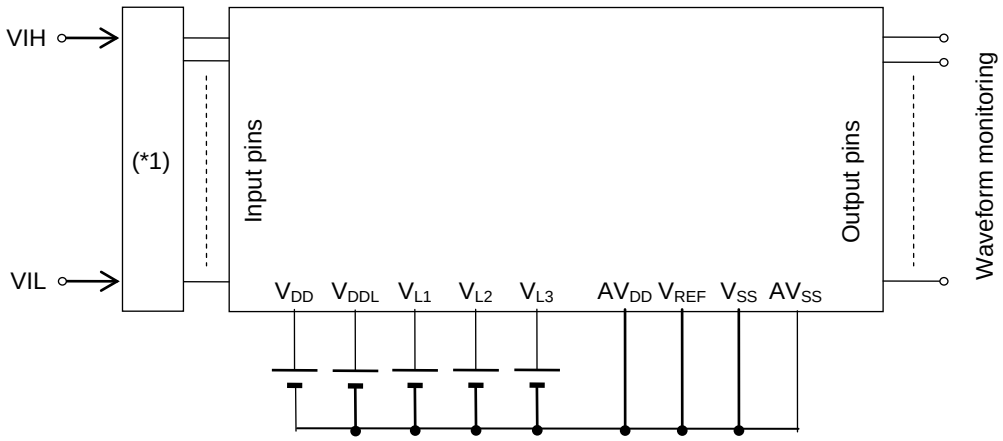
*1: Input logic circuit to determine the specified measuring conditions.
*2: Measured at the specified output pins.

Measuring circuit 4



*3: Measured at the specified output pins.

Measuring circuit 5

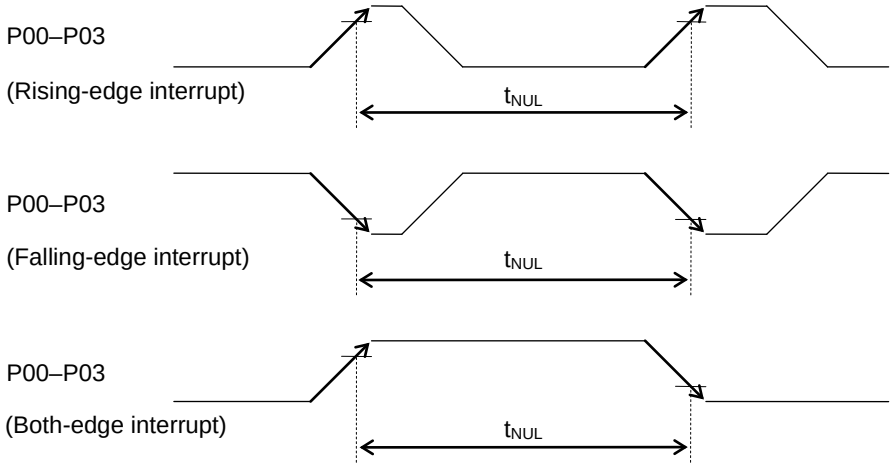


*1: Input logic circuit to determine the specified measuring conditions.

● AC Characteristics (External Interrupt)

(V_{DD} = 1.1 to 3.6V, AV_{DD} =2.2 to 3.6V, V_{SS} = AV_{SS} = 0V, Ta = -20 to +70°C,unless otherwise specified)

Parameter	Symbol	Condition	Rating			Unit
			Min.	Typ.	Max.	
External interrupt disable period	t_{NUL}	Interrupt: Enabled (MIE = 1), CPU: NOP operation System clock: 32.768kHz	76.8	—	106.8	μs

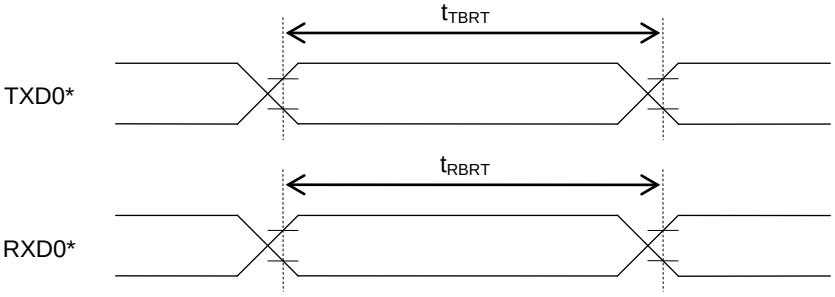


● AC Characteristics (UART)

(V_{DD} = 1.3 to 3.6V, AV_{DD} =2.2 to 3.6V, V_{SS} = AV_{SS} = 0V, Ta = -20 to +70°C,unless otherwise specified)

Parameter	Symbol	Condition	Rating			Unit
			Min.	Typ.	Max.	
Transmit baud rate	t_{TBRT}	—	—	BRT*[1]	—	s
Receive baud rate	t_{RBRT}	—	BRT*[1] −3%	BRT*[1]	BRT*[1] +3%	s

*1: Baud rate period (including the error of the clock frequency selected) set with the UART baud rate register (UA0BRTL,H) and the UART mode register 0 (UA0MOD0).



*: Indicates the secondary function of the port.

● AC Characteristics (Synchronous Serial Port)

(V_{DD} = 1.3 to 3.6V, AV_{DD} =2.2 to 3.6V, V_{SS} = AV_{SS} = 0V, Ta = -20 to +70°C,unless otherwise specified)

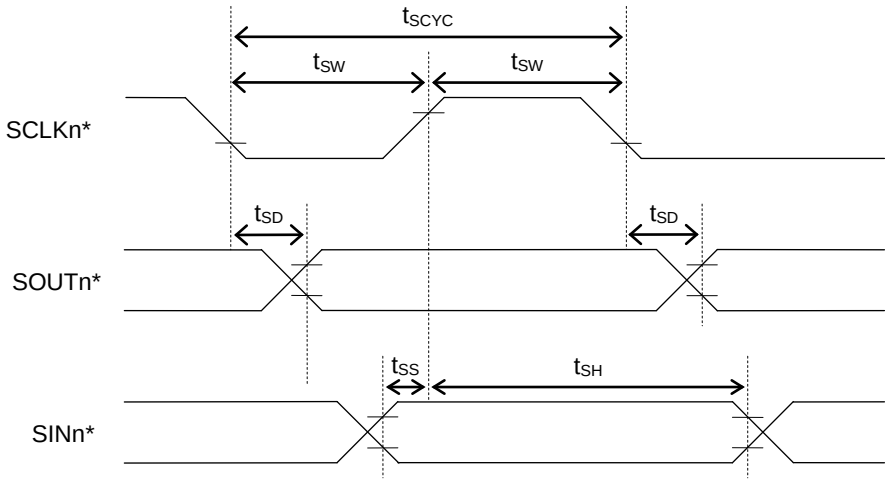
Parameter	Symbol	Condition	Rating			Unit
			Min.	Typ.	Max.	
SCLKn input cycle (slave mode)	t_{SCYC}	When high-speed oscillation is not active*2 (V_{DD} = 1.3 to 3.6V)	10	—	—	µs
		When high-speed oscillation is active*3 (V_{DD} = 1.8 to 3.6V)	1	—	—	µs
SCLKn output cycle (master mode)	t_{SCYC}	—	—	SCLKn*1	—	s
SCLKn input pulse width (slave mode)	t_{SW}	When high-speed oscillation is not active*2 (V_{DD} = 1.3 to 3.6V)	4	—	—	µs
		When high-speed oscillation is active*3 (V_{DD} = 1.8 to 3.6V)	0.4	—	—	µs
SCLKn output pulse width (master mode)	t_{SW}	—	SCLKn*1 ×0.4	SCLKn*1 ×0.5	SCLKn*1 ×0.6	s
SOUTn output delay time (slave mode)	t_{SD}	When high-speed oscillation is not active*2 (V_{DD} = 1.3 to 3.6V)	—	—	500	ns
		When high-speed oscillation is active*3 (V_{DD} = 1.8 to 3.6V)	—	—	240	
SOUTn output delay time (master mode)	t_{SD}	When high-speed oscillation is not active*2 (V_{DD} = 1.3 to 3.6V)	—	—	500	ns
		When high-speed oscillation is active*3 (V_{DD} = 1.8 to 3.6V)	—	—	240	
SINn input setup time (slave mode)	t_{SS}	—	80	—	—	ns
SINn input setup time (master mode)	t_{SS}	When high-speed oscillation is not active*2 (V_{DD} = 1.3 to 3.6V)	500	—	—	ns
		When high-speed oscillation is active*3 (V_{DD} = 1.8 to 3.6V)	240	—	—	
SINn input hold time	t_{SH}	When high-speed oscillation is not active*2 (V_{DD} = 1.3 to 3.6V)	300	—	—	ns
		When high-speed oscillation is active*3 (V_{DD} = 1.8 to 3.6V)	80	—	—	

n= 0,1

*1: Clock period selected with SnCK3–0 of the serial port n mode register (SIOnMOD1)

*2: When RC oscillation is selected with OSCM1–0 of the frequency control register (FCON0)

*3: When Crystal/ceramic oscillation, built-in PLL oscillation, or external clock input is selected with OSCM1–0 of the frequency control register (FCON0)



*: Indicates the secondary function of the port.

● AC Characteristics (I^2C Bus Interface: Standard Mode 100kHz)

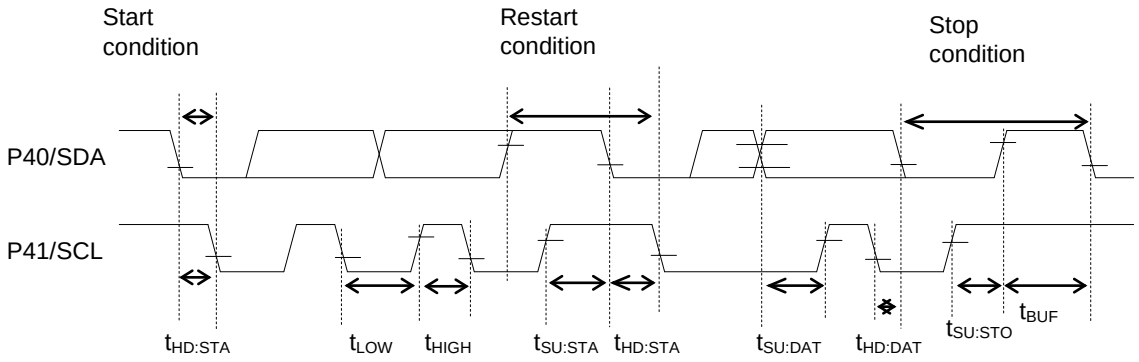
(V_{DD} = 1.8 to 3.6V, AV_{DD} =2.2 to 3.6V, V_{SS} = AV_{SS} = 0V, Ta = -20 to +70°C,unless otherwise specified)

Parameter	Symbol	Condition	Rating			Unit
			Min.	Typ.	Max.	
SCL clock frequency	f_{SCL}	—	0	—	100	kHz
SCL hold time (start/restart condition)	$t_{HD:STA}$	—	4.0	—	—	µs
SCL "L" level time	t_{LOW}	—	4.7	—	—	µs
SCL "H" level time	t_{HIGH}	—	4.0	—	—	µs
SCL setup time (restart condition)	$t_{SU:STA}$	—	4.7	—	—	µs
SDA hold time	$t_{HD:DAT}$	—	0	—	3.45	µs
SDA setup time	$t_{SU:DAT}$	—	0.25	—	—	µs
SDA setup time (stop condition)	$t_{SU:STO}$	—	4.0	—	—	µs
Bus-free time	t_{BUF}	—	4.7	—	—	µs

● AC Characteristics (I^2C Bus Interface: Fast Mode 400kHz)

(V_{DD} = 1.8 to 3.6V, AV_{DD} =2.2 to 3.6V, V_{SS} = AV_{SS} = 0V, Ta = -20 to +70°C,unless otherwise specified)

Parameter	Symbol	Condition	Rating			Unit
			Min.	Typ.	Max.	
SCL clock frequency	f_{SCL}	—	0	—	400	kHz
SCL hold time (start/restart condition)	$t_{HD:STA}$	—	0.6	—	—	µs
SCL "L" level time	t_{LOW}	—	1.3	—	—	µs
SCL "H" level time	t_{HIGH}	—	0.6	—	—	µs
SCL setup time (restart condition)	$t_{SU:STA}$	—	0.6	—	—	µs
SDA hold time	$t_{HD:DAT}$	—	0	—	0.9	µs
SDA setup time	$t_{SU:DAT}$	—	0.1	—	—	µs
SDA setup time (stop condition)	$t_{SU:STO}$	—	0.6	—	—	µs
Bus-free time	t_{BUF}	—	1.3	—	—	µs



● AC Characteristics (RC Oscillation A/D Converter)

Condition for VDD=1.8 to 3.6V

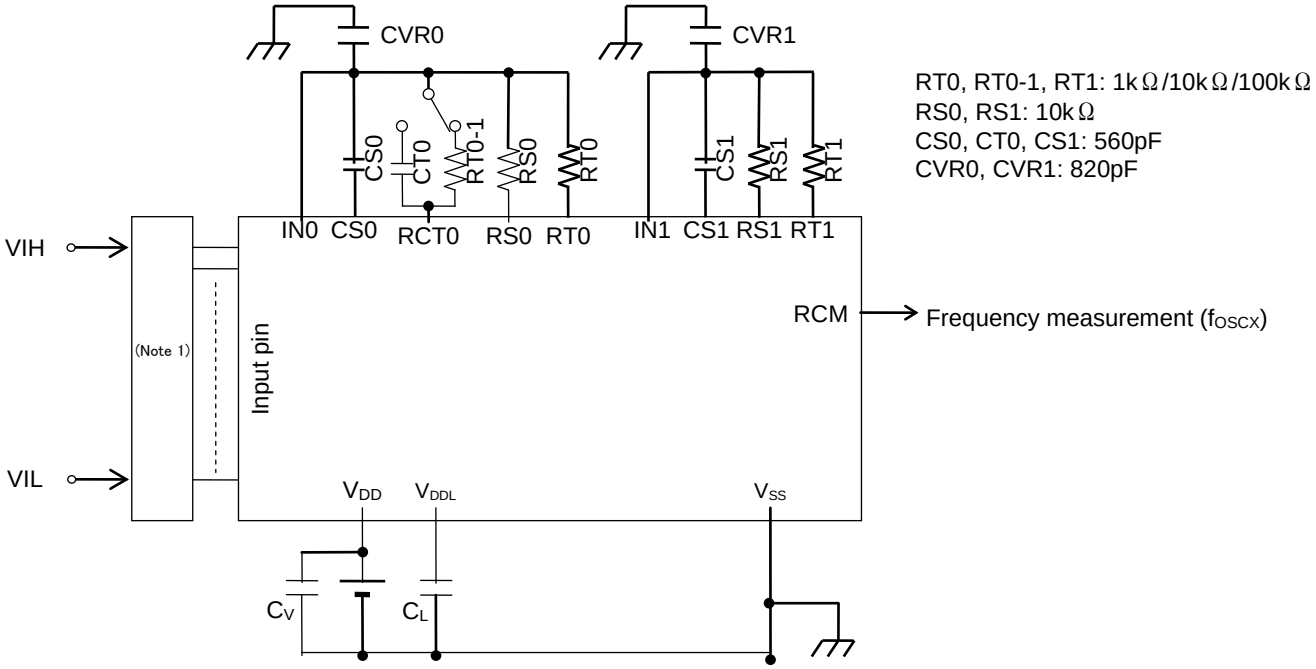
(V_{DD} = 1.8 to 3.6V, AV_{DD} =2.2 to 3.6V, V_{SS} = AV_{SS} = 0V, Ta = -20 to +70°C,unless otherwise specified)

Parameter	Symbol	Condition	Rating			Unit
			Min.	Typ.	Max.	
Oscillation resistor	RS0,RS1,RT0 ,RT0-1,RT1	CS0, CT0, CS1≥740pF	1	—	—	kΩ
Oscillation frequency V_{DD} = 3.0V	f_{OSC1}	Resistor for oscillation=1kΩ	457.3	525.2	575.1	kHz
	f_{OSC2}	Resistor for oscillation=10kΩ	53.48	58.18	62.43	kHz
	f_{OSC3}	Resistor for oscillation=100kΩ	5.43	5.89	6.32	kHz
RS to RT oscillation frequency ratio *1 V_{DD} = 3.0V	Kf1	RT0, RT0-1, RT1=1kΩ	7.972	9.028	9.782	—
	Kf2	RT0, RT0-1, RT1=10kΩ	0.981	1	1.019	—
	Kf3	RT0, RT0-1, RT1=100kΩ	0.099	0.101	0.104	—

*1: Kfx is the ratio of the oscillation frequency by the sensor resistor to the oscillation frequency by the reference resistor on the same conditions.

$$Kfx = \frac{f_{OSCX}(\text{RT0-CS0 oscillation})}{f_{OSCX}(\text{RS0-CS0 oscillation})}, \frac{f_{OSCX}(\text{RT0-1-CS0 oscillation})}{f_{OSCX}(\text{RS0-CS0 oscillation})}, \frac{f_{OSCX}(\text{RT1-CS1 oscillation})}{f_{OSCX}(\text{RS1-CS1 oscillation})}$$

(x = 1, 2, 3)



Condition for VDD=1.25 to 3.6V

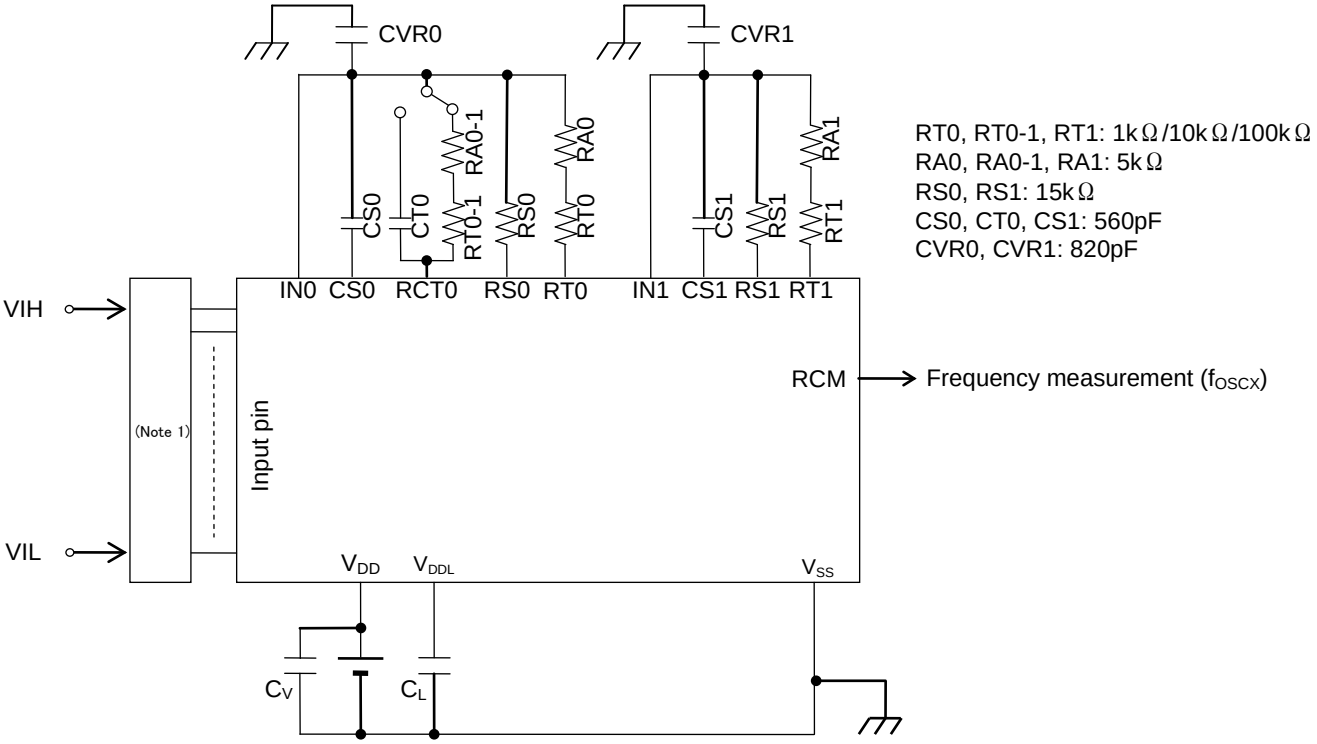
(V_{DD} = 1.8 to 3.6V, AV_{DD} =2.2 to 3.6V, V_{SS} = AV_{SS} = 0V, Ta = -20 to +70°C,unless otherwise specified)

Parameter	Symbol	Condition	Rating			Unit
			Min.	Typ.	Max.	
Oscillation resistor	RS0,RS1,RT0 ,RT0-1,RT1	CS0, CT0, CS1≥740pF	1	—	—	kΩ
Oscillation frequency V_{DD} = 1.5V	f_{OSC1}	Resistor for oscillation=6kΩ	81.93	93.16	101.2	kHz
	f_{OSC2}	Resistor for oscillation=15kΩ	35.32	38.75	41.48	kHz
	f_{OSC3}	Resistor for oscillation=105kΩ	5.22	5.65	6.03	kHz
RS to RT oscillation frequency ratio *1 V_{DD} = 1.5V	Kf1	RT0, RT0-1, RT1=1kΩ	2.139	2.381	2.632	—
	Kf2	RT0, RT0-1, RT1=10kΩ	0.973	1	1.028	—
	Kf3	RT0, RT0-1, RT1=100kΩ	0.142	0.147	0.152	—
Oscillation frequency V_{DD} = 3.0V	f_{OSC1}	Resistor for oscillation=6kΩ	85.28	94.58	103.3	kHz
	f_{OSC2}	Resistor for oscillation=15kΩ	35.72	38.87	41.78	kHz
	f_{OSC3}	Resistor for oscillation=105kΩ	5.189	5.622	6.012	kHz
RS to RT oscillation frequency ratio *1 V_{DD} = 3.0V	Kf1	RT0, RT0-1, RT1=1kΩ	2.227	2.432	2.626	—
	Kf2	RT0, RT0-1, RT1=10kΩ	0.982	1	1.018	—
	Kf3	RT0, RT0-1, RT1=100kΩ	0.141	0.145	0.149	—

*1: Kfx is the ratio of the oscillation frequency by the sensor resistor to the oscillation frequency by the reference resistor on the same conditions.

$$Kfx = \frac{f_{OSCX}(\text{RT0-CS0 oscillation})}{f_{OSCX}(\text{RS0-CS0 oscillation})}, \frac{f_{OSCX}(\text{RT0-1-CS0 oscillation})}{f_{OSCX}(\text{RS0-CS0 oscillation})}, \frac{f_{OSCX}(\text{RT1-CS1 oscillation})}{f_{OSCX}(\text{RS1-CS1 oscillation})}$$

(x = 1, 2, 3)



*1: Input logic circuit to determine the specified measuring conditions.

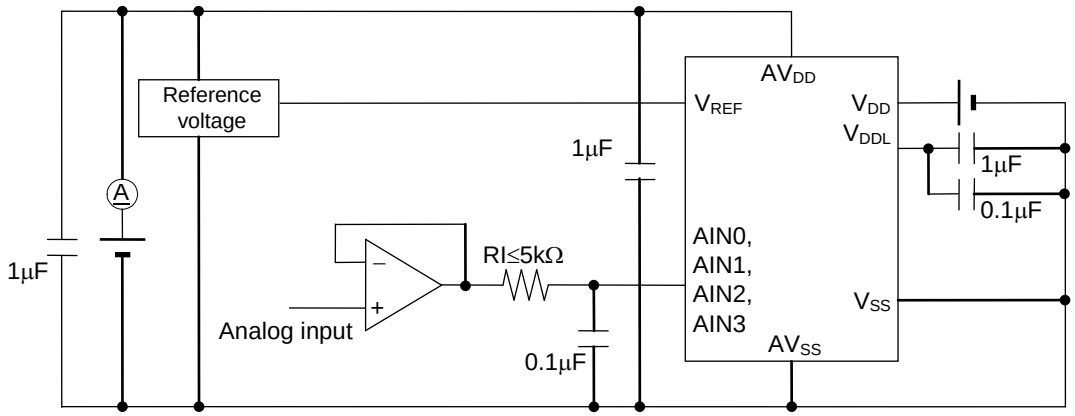
Note:
- Please have the shortest layout for the common node (wiring patterns which are connected to the external capacitors, resistors and IN0/IN1 pin), including CVR0/CVR1. Especially, do not have long wire between IN0/IN1 and RS0/RS1. The coupling capacitance on the wires may occur incorrect A/D conversion. Also, please do not have signals which may be a source of noise around the node.
- When RT0/RT1 (Thermistor and etc.) requires long wiring due to the restricted placement, please have VSS(GND) trace next to the signal.
- Please make wiring to components (capacitor, resistor and etc.) necessary for objective measurement. Wiring to reserved components may affect to the A/D conversion operation by noise the components itself may have.

● AC Characteristics (Successive Approximation Type A/D Converter)

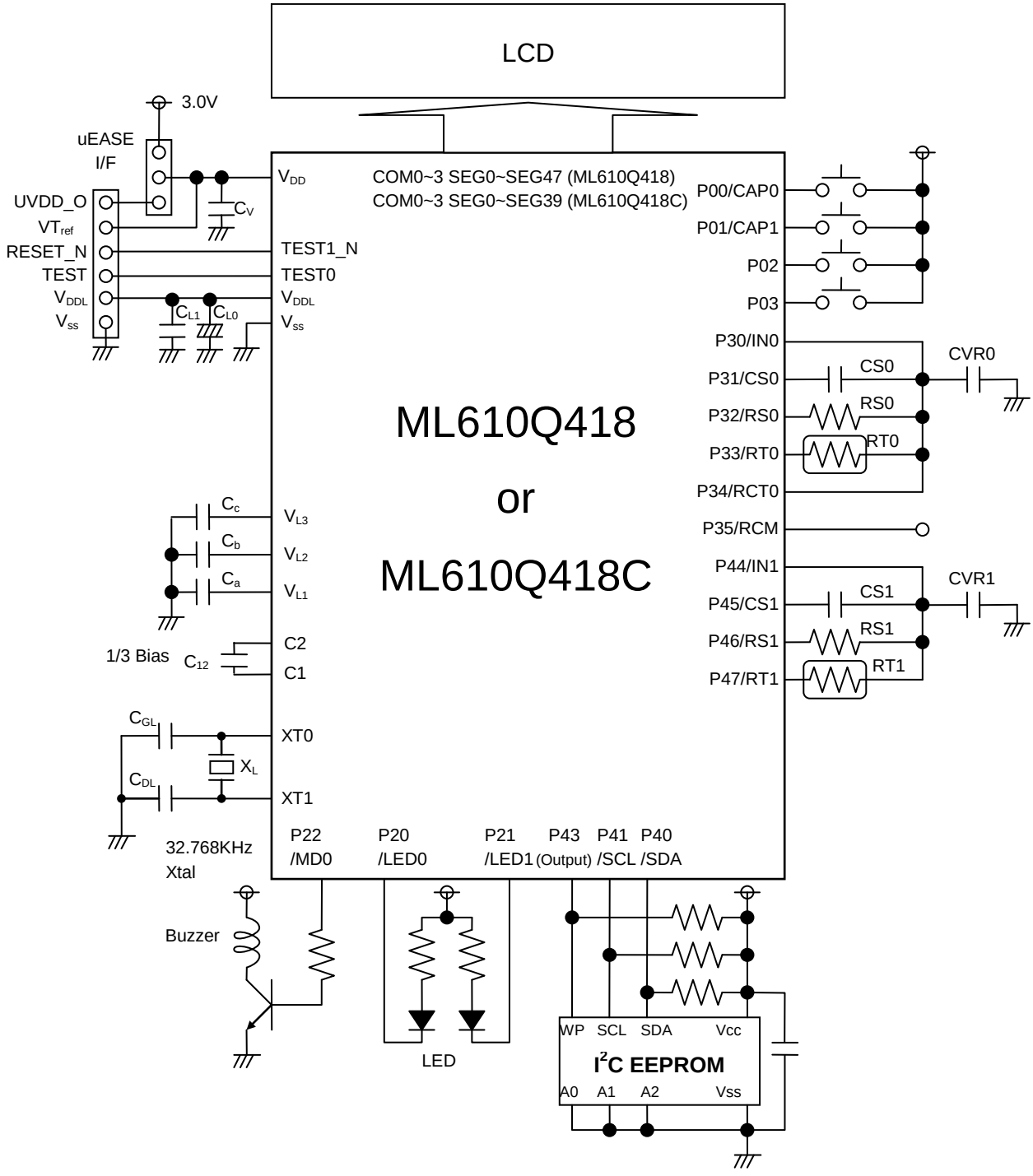
(V_{DD} = 1.8 to 3.6V, AV_{DD} =2.2 to 3.6V, V_{SS} = AV_{SS} = 0V, Ta = -20 to +70°C,unless otherwise specified)

Parameter	Symbol	Condition	Rating			Unit
			Min.	Typ.	Max.	
Resolution	n	—	—	—	12	bit
Integral non-linearity error	INL	2.7V ≤ V_{REF} ≤ 3.6V	−4	—	+4	LSB
		2.2V ≤ V_{REF} ≤ 2.7V	−6	—	+6	
Differential non-linearity error	DNL	2.7V ≤ V_{REF} ≤ 3.6V	−3	—	+3	
		2.2V ≤ V_{REF} ≤ 2.7V	−5	—	+5	
Zero-scale error	V_{OFF}	—	−6	—	+6	
Full-scale error	FSE	—	−6	—	+6	
Reference voltage	V_{REF}	—	2.2	—	AV_{DD}	V
Conversion time	t_{CONV}	SACK = 0 (HSCLK = 375kHz to 625kHz)	—	25	—	ϕ/CH
		SACK = 1 (HSCLK = 1.5MHz to 4.2MHz)	—	112	—	

ϕ: Period of high-speed clock (HSCLK)



Appendix D Application Circuit Example



*: For the configuration of the low-speed clock generation circuit, see Section 6.3.1, "Low-Speed Clock".

Figure D-1 Application Circuit Example

Appendix E Check List

This Check List has notes to prevent commonly-made programming mistakes and frequently overlooked or misunderstood hardware features of the MCU. Check each note listed up chapter by chapter while coding the program or evaluating it using the MCU.

Chapter 1 Overview

[ ] Please confirm how to handle the unused pins(Refer to Section 1.3.4 in the user's manual).

Chapter 2 CPU and Memory

- **Memory model**

[ ] LARGE model(Refer to Section 1.3.6 in the U8 instruction manual).

- **Program Memory size**

[ ] 130,048 Byte (0:0000H to 0:FBFFH, 1:0000H to 1:FFFFH)

- **Data RAM size**

[ ] 4096 Byte (0:E000H to 0:EFFFH)

- **Unused area**

[ ] Please fill test area 0:FC00H to 0:FDFFH with BRK instruction code "0FFH" (Refer to a startup file "S61043XSW.asm" for programming in the source code).

[ ] For fail safe in your system, please fill unused program memory area (your program code does not use) with BRK instruction code "0FFH". We will fill the area with the code "0FFH" at LAPIS semiconductor's factory programming.

- **Initializing RAM**

[ ] The hardware reset does not initialize RAM. Please initialize RAM by the software.

Chapter 3 Reset

- **Reset activation pulse width**

[ ] Minimum 200us (Refer to Appendix C-2 in the user's manual)

- **Power-on reset occurrence power rising time**

[ ] Maximum 10ms (Refer to Appendix C-2 in the user's manual)

- **BRK instruction reset**

[ ] In system reset by the BRK instruction, no special function register (SFR) is initialized either. Therefore initialize the SFRs by your software.

Chapter 4 Standby control / MCU control

- **STOP mode**

[ ] Please note the STPACP is not enabled when both interrupt enable flags and the interrupt request flags are "1" & MIE flag is "0" (Refer to Section 4.2.2~4.2.3. in the user's manual).

[ ] Place two NOP instructions next to the instruction that sets the STP bit to "1"(Refer to Section 4.3.3. in the user's manual).

- **HALT mode**

[ ] Place two NOP instructions next to the instruction that sets the HLT bit to "1"(Refer to Section 4.3.2. in the user's manual).

- **BLKCON registers**

[ ] BLKCON registers enable or disable corresponsive each peripheral (Refer to Section 4.2.4 ~ 4.2.8. in the user's manual).

[ ] When certain bits of block control registers are set to "1", corresponding peripherals are reset (all registers are reset) and operating clocks for the peripherals stop.

[ ] DXTSP bit (bit 4) of BLKCON4 register disables the operation of 32kHz oscillation stop detector in only HALT mode.

Chapter 5 Interrupt

- **Unused interrupt vector table**

[ ] Please define all unused interrupt vector tables for fail safe.

- **Non-maskable interrupt**

[ ] The watchdog timer interrupt (WDTINT) is non-maskable interrupts that do not depend on MIE flag(Refer to Section 5.2.9. and 5.3).

Chapter 6 Clock generation circuit

• Initial System clock

[ ] At power up or system reset, the 32.768kHz crystal oscillation clock oscillates to be supplied to CPU as the system clock.

• PLL oscillation clock

[ ] The PLL oscillation generates 8MHz clock which can be driven to an output pin, however the maximum CPU operating frequency is 4MHz.

• Switching high-speed clock operation mode to low-speed clock operation mode

[ ] When switching the high-speed clock to the low-speed clock after the power up, confirm the low-speed clock is oscillating for sure by checking Q128H bit is "1".

• Switching high-speed clock operation mode to another high-speed clock operation mode

[ ] When switching the high-speed clock mode, the clock must be first switched back to low clock before switching to other high-speed clock (Refer to Section 6.2.2.).

• Port secondary Function

[ ] Specify the secondary function for the port 2 when driving a clock to the pin(Refer to Section 6.4 in the user's manual).

[ ] A high-speed crystal/ceramic oscillation or an external clock input mode does not require specifying the secondary function for the port 1(Refer to Section 19.3.2.)

Chapter 7 TBC (Time Base Counter)

• HTBCLK

[ ] HTBLK goes through the HTBDR register. Set proper value to the register(Refer to Section 7.2.3. in the user's manual).

• How to read LTBC

[ ] Read consecutively LTBC(Low-speed Time Base Counter) twice until the last data coincides the previous data to prevent reading of uncertain data while counting up the clock (Refer to Section 7.3.1 in the user's manual).

Chapter 9 Timer

• How to read the timer counter registers

[ ] Check notes for reading the timer counter registers while counting up(Refer to Section 9.2.6. ~ 9.2.9. in the user's manual).

Chapter 10 PWM

• Used Pin

[ ] P34(PWM0) pin or P43(PWM0) pin is used.

• How to read the PWM counter registers

[ ] Check notes for reading the PWM counter registers while the PWM is operating(Refer to Section 10.2.4. in the user's manual).

• Port secondary Function

[ ] Specify the secondary Function for the port(Refer to Section 10.4 in the user's manual).

Chapter 11 WDT

• Overflow period

Clear WDT during the selected overflow period:

[ ] 125ms, [ ] 500ms, [ ] 2s, [ ] 8s

• WDP

[ ] Check the WDP before writing to the WDTCON and determine writing "5AH" or "0A5H" (Refer to Section 11.2.2. in the user's manual).

Chapter 12 SSIO

• Used pin

[ ] P40(SIN0), P41(SCK0) and P42(SOUT0) are used, or P44(SIN0), P45(SCK0) and P46(SOUT0) are used.

[ ] P50(SIN1), P51(SCK1) and P52(SOUT1) are used.

• Port secondary Function

[ ] Specify the secondary Function for the port (Refer to Section 12.4. in the user's manual)

Chapter 13 UART

• Used pin

[ ] P02(RXD0) and P43(TXD0) are used, or [ ] P42(RXD0) and P43(TXD0) are used.

[ ] Select the P02 or P42 for RXD0 by specifying U0RSEL bit of UA0MOD0 register.

[ ] P03(RXD1) and P53(TXD1) are used, or [ ] P52(RXD1) and P53(TXD1) are used.

[ ] Select the P03 or P52 for RXD1 by specifying U1RSEL bit of UA1MOD0 register.

[ ] P42(RXD0) and P43(TXD1) are used.

• Port secondary Function

[ ] Specify the secondary Function for the port (Refer to Section 13.4. in the user's manual)

• Port quaternary Function

[ ] Specify the quaternary Function for the port (Refer to Section 13.4. in the user's manual)

Chapter 14 I2C

• Used pin

[ ] P40(SDA) pin and P41(SCL) pin used.

• Port secondary Function

[ ] Specify the secondary Function for the port (Refer to Section 14.4. in the user's manual)

Chapter 15 ~ Chapter 21 Port

• Handling the pin

[ ] Don't leave Hi-impedance Input ports in floating state.

• Port secondary Function

[ ] Specify properly PnCON0/1 and PnMOD0/1 registers for each port.

• Port 6

[ ] ML610Q418 does Not have port 6.

Chapter 22 Melody / Buzzer

• Enabling the LSCLK x 2

[ ] Set ENMLT bit of FCON1 register to "1" to enable the low-speed double clock (LSCLK x 2) before stating the melody or buzzer outputs.

• Port secondary Function

[ ] Specify the secondary Function for the port (Refer to Section 22.4. in the user's manual)

Chapter 23 RC oscillation type A/D converter

• Counter register

[ ] Reading the counter register A or B during the A/D conversion, returns the data written before starting the A/D conversion.

• Oscillation monitor pin

[ ] P35/RCM pin is a monitor pin for oscillation clock. The channel 0(P34-P30) and channel 1(P47-P44) share the monitor pin.

[ ] Please use P35/RCM for the evaluation purpose and disable the output while operating in an actual application to minimize the noise.

• Port secondary Function

[ ] Specify the secondary Function for the port (Refer to Section 23.4. in the user's manual).

[ ] All the Port 3 pins except P35/RCM are configured as pins dedicated to the RC-ADC function during A/D conversion(Refer to Section 23.3.1. in the user's manual).

Chapter 24 Successive Approximation type A/D converter

• Operating conditions

[ ] Please confirm the operating voltage and the clock frequency.
V_{DD}=1.8V~3.6V (HSCLK=375KHz~1.1MHz or 1.99MHz~4.2MHz), AV_{DD}=2.2V~3.6V

• Others

[ ] Use the A/D converter when the HSCLK is oscillating.

[ ] Do not set SARUN bit of SADCON1 register to "1" on the condition both SACH0 bit and SACH1 bit of SADMOD0 register are "0" (Refer to Section 24.2.10 ~ Section 24.2.11).

Chapter 25 LCD driver

• Bias

[ ] 1/2 bias or [ ] 1/3 bias

• Duty

[ ] 1/1 to 1/4 Duty

• COM/SEG

[ ] ML610Q418 : 4COM x 48SEG
[ ] ML610Q418C : 4COM x 40SEG

• External capacitor

(1/3 bias)

[ ] Ca=0.1uF(connected to VL1 pin), [ ] Cb=0.1uF(connected to VL2 pin),
[ ] Cc=0.1uF(connected to VL3 pin),
[ ] C12=0.47uF(connected between C1 and C2 pin)

(1/2 bias)

[ ] Ca=0.1uF(connected to VL1 pin), [ ] Cc=0.1uF(connected to VL3 pin),
[ ] C12=0.47uF(connected between C1 and C2 pin)

Chapter 26 BLD (Battery Low Detector)

• Changing the threshold

[ ] Please select the threshold voltage when the BLD circuit is OFF.

Chapter 27 Power circuit

• External capacitor

[ ] CL0=2.2uF (connected to VDDL pin), [ ] CL1=0.1uF (connected to VDDL pin)

Chapter 30 On-chip debug

[ ] Supply 3.0V ~ 3.6V to VDD pin when programming (erasing and writing) the Flash ROM with LAPIS semiconductor development tool uEASE.

[ ] Please do not apply LSIs being used for debugging to mass production.

[ ] Please validate the ROM code on your production board without LAPIS semiconductor development tool uEASE.

Appendix A SFR (Specific Function Registers)

• Initial data

[ ] Please confirm there are some SFRs have undefined initial value at reset (Refer to Appendix A in the user's manual).

Appendix C Electrical Characteristics

• Operating temperature

[ ] -20'C to +70'C

• Operating voltage vs Operating frequency

[ ] Please confirm the operating conditions.
[ ] +1.1V to +3.6V (0kHz to 36kHz: Low-speed crystal oscillation clock operation)
[ ] +1.3V to +3.6V (30kHz to 625kHz: Built-in RC oscillation clock operation)
[ ] +1.8V to +3.6V (30kHz to 4.2MHz: High-speed crystal/ceramic oscillation clock or built-in PLL oscillation clock)

Revision History

Revision History

Document No.	Date	Page		Description
		Previous Edition	Current Edition	
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